SMITHIAN FOLD THEORY V3

From Fold to Physics

An Exact, Parameter-Free and Machine-Closed Complete-Field Reconstruction of Physical Science from Smithian Fold Theory

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Contents

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Lean verification integration. *The current SFT ordered census passed the independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims, 898,902 candidate records and decisions, 11,108 controls, seventeen branches and no reported issue. Lean natively proves the two-class operational root and its unique survivor and constructs proof-bearing acceptance-gate certificates. The remaining claim content is checked as the complete registered artifact graph; that distinction is retained throughout this successor.*

Abstract

This successor presents the complete current `physics` paper surface: 368 live model-admitted claims, 257,776 generated candidates, 368 unique survivors and 1,472 passed controls. It integrates 0 claim section(s) not present in the preceding abstract's dated surface and remains complete only to this versioned registered census, with lawful extension open. The independent Lean 4 layer returned PASS for this surface as part of all 2,777 current claims.

$\alpha^{-1} = 503846395469/3676744786 = 137.035999177180855...$; $\alpha = 3676744786/503846395469 = 0.007297352564321794...$

The value was sealed before the registered CODATA 2022 target was released and lies inside the complete interval 137.035999177 ± 0.000000021. The same derivational constitution produces exact charged-lepton cubic and Koide relations, electron and muon magnetic anomalies, the on-shell electroweak share, Higgs mass and self-coupling, the Planck/proton hierarchy, proton radius, vacuum and cosmological ratios, nuclear closure numbers, hadron trajectories, inverse-square dilution, relativistic and quantum laws, thermodynamic and extraction boundaries, gravitational-wave

ordering, the Unified Constants Object, Tesla resonance laws, vacuum/inertia co-variation and restoration, and the penta/hepta/Smithion standing-prediction family documented here.

The publication contains all 368 current engine-admitted Physics claims. Their generated grammars contain 257,776 candidates, 257,776 one-for-one decisions, 368 unique survivors and 1,472 passing mandatory adverse controls. Of those claims, 238 contain sealed post-derivation external-validation records; formal claims are connected to their registered empirical successors or retain their exact structural test boundary. The categorical clean-room audit closes all 488 Physics-owned V1/V2 atoms at the declared current-evidence boundary, with no open Physics atom or gap family. Grand Locks 075/076 preserve the 349-claim pre-extension snapshot; the separately admitted Unified Constants, Tesla resonance, vacuum/inertia and new-sector families add nineteen current claims with their own completion audits and terminal empirical receipts. The current dated inventory is therefore a 368-claim full-field projection, complete to known registered scope and permanently open to lawful extension, correction and falsification.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

Successor headline findings

1. **Independent whole-model result.** Lean 4 returned `PASS` for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and `presentedOccurrence` is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.
3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.
5. **Current paper surface.** This successor accounts for all 368 current claim(s) in its declared branch ownership boundary; 0 later claim section(s) are integrated in the successor supplement.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

Current status, evidence language and reader map

Status field	Current position
Branch and completion boundary	Physics: 368/368 claims in the current registered categorical scope. This is dated, versioned completion and remains open to lawful extension.
Formal status	Every live claim represented here has a current model-admitted receipt. Formal admission does not by itself imply empirical confirmation.
Empirical status	Claim-specific. Blind, non-blind, development-observed, holdout, adverse, unresolved, unavailable and formal-only distinctions remain exactly as recorded in the claim ledger and evidence packages.
Chronology	Claim-specific registration, seal, custody and observation order governs. Later evidence does not retroactively become an earlier prediction.
Publication status	Version 1.4.1 is published open access at 10.5281/zenodo.21761655 in the existing Zenodo concept DOI lineage.
Existing record lineage	Current record DOI `10.5281/zenodo.21627765`; concept DOI `10.5281/zenodo.21520880`. Publication is permitted only through the existing record's new-version route.
What is not claimed	Permanent closure of science, universal empirical confirmation, transfer of another branch's ownership or permission to replace an adverse or unresolved record.

Corpus-wide terminology and evidence key

Term	Reserved meaning in this paper
Theorem	A formally closed proposition at its declared grammar and dependency boundary.
Law	An admitted branch relation with its declared carrier, boundary, dependencies and extension rule.
Claim	The registered unit judged by the engine and bound to one immutable receipt.
Constitution	The rules governing admissible objects, derivations, evidence and publication; not an empirical result.
Derivation	Generation and elimination of the declared candidate space to its surviving structure.
Prediction	A consequence sealed before the matching target is released under the registered custody protocol.
Observation	A source-bound external record; it does not enter candidate generation.
Measurement	An instrument-, method-, condition- and uncertainty-bound observed value.
Reconstruction	A separately implemented regeneration or an explicitly identified inference of a retained state or history.
Exact numerical correspondence	Equality or registered interval relation between exact numerical objects at the declared boundary.
Structural correspondence	A post-derivation relation between forms without a claim of exact numerical prediction.
Boundary correspondence	Agreement restricted to the named interface, limit or ownership boundary.
Compatibility	Non-adverse but non-discriminating evidence.
Support	Relevant evidence that is not unique confirmation.
Confirmation	Used only when the current registered evidence protocol warrants that classification.
Validation	Successful execution of the specified formal, computational or empirical test; its kind must be named.
Adverse result	A registered test result that conflicts with the tested claim at its declared boundary.
Unresolved result	Required evidence remains unavailable, incomplete, disputed or insufficiently classified.
Implementation identity	The hash-bound identity of executable material; implementation success is not empirical confirmation.
Formal, empirical and publication status	Three independent classifications. None silently substitutes for another.
Foundational closure	Completion of the registered foundation only. It does not imply field-wide closure.
Field-wide closure	Completion of the frozen or registered dated field census named in this paper.
Current-evidence closure	Closure only to the evidence surface presently registered and preserved.
Extension openness	New questions may be added by a later version without rewriting existing receipts or history.

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relationships; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The cited repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. Those files remain authoritative where prose abbreviates their display.

Editorial change control

This version may improve expression, order, typography and navigation. It may not change scientific meaning, scope, chronology, evidence class, ownership, claim status or machine identity. Any apparent conflict between authoritative records must be traced and either resolved by explicit supersession or recorded for Maria Smith; prose alone cannot manufacture agreement.

1. Publication and authorship boundary

PUBLISHED OPEN-ACCESS PAPER. The scientific and local publication-readiness gates pass. The authorised publication goal is to issue this manuscript only as version 1.3 of the existing Physics concept record after Maria Smith approves the final candidate files and hashes.

Maria Smith, independent researcher and founder of Ernos Labs. Contact: Maria.Smith.Sftoe@gmail.com. Submissions and reproducibility reports: https://discord.gg/ucwGryVxGr. GitHub: https://github.com/MettaMazza.

1.1 Ernos Labs and the authorship of this work

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to ask readers to trust a new institution. Its purpose is to make the work inspectable enough that trust is unnecessary. Every claimed law must expose its premises, generated alternatives, eliminations, survivor, controls, measurement custody, unfavourable evidence and exact receipt. A reviewer needs no credential, subscription, institutional affiliation or permission to reproduce an acceptance or demonstrate an invalidation.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That biography is not offered as evidence for a physical result; the derivations and observations must carry that burden. It is evidence about access. A system that decides who may be heard before their work can be inspected does not merely exclude individuals: it loses questions, methods and discoveries that its credential and capital filters never authorise. This paper therefore refuses both romantic exceptionalism and credentialed dismissal. The same public empirical standard applies to Maria Smith, an established professor, an independent reproducer and an AI system.

1.2 Why this is an open-science publication

The institutional critique is evidence-based. Systematic research has found sponsor-associated differences in reported efficacy results and conclusions, while work on commercial sponsorship shows that funding can influence which questions are pursued (Lundh et al., 2018; Fabbri et al., 2018). Grant selection is not an infallible oracle: empirical studies report sensitivity to review design and reviewer composition, and a Cochrane review found limited evidence that grant peer review improves the quality of funded research (Gallo et al., 2023; Demicheli and Di Pietrantonj, 2007/2022 archive). Null and negative results remain systematically underrepresented, distorting the visible scientific record (Nature Communications, 2025). UNESCO's Recommendation on Open Science calls for methods, software, source and outputs to be open to rigorous scrutiny and identifies paywalls and high publication charges as sources of inequality (UNESCO, 2021).

Opaque prediction presents a related problem. A black-box system may be a useful instrument and may pass a declared blind benchmark, but predictive accuracy alone does not disclose a law or a derivation. NIST accordingly treats transparency, explainability, interpretability, validity and reliability as distinct properties of trustworthy AI (NIST AI RMF 1.0). Ernos Labs therefore rejects three substitutions: funding success is not evidential closure; professional consensus is not generated uniqueness; and a black-box score is not an explicit scientific law.

The editorial position is direct. Knowledge institutions organised through competitive capital, scarce grants, prestige markets, subscription access and author charges have incentives that are not identical to free human inquiry. Expertise, criticism and measurement remain indispensable, but credentials, funding and consensus cannot select a fundamental law and then stand in for its evidence. The remedy is not weaker science. It is a harder, public standard: derive the law, generate its alternatives, preserve the failures, seal the prediction, expose the observation and permit anyone to reproduce the decision.

1.3 Rights, participation and the Ernos Labs designation

Maria Smith retains copyright and scientific authorship. CC BY 4.0 for papers and documentation and Apache 2.0 for code permit inspection, copying, criticism, modification and redistribution with attribution. The words "Ernos Labs" are a separate standards-conformance designation. A fork may freely test or reuse the open work, but it may represent itself as Ernos Labs only while it follows the published empirical constitution, preserves adverse evidence, submits to critical review and does not weaken the engine or admission route.

Independent replications, lawful extensions, corrections and attempted invalidations are invited. Submissions must include the full derivation chain and pass the same engine; reputation cannot open the gate and lack of credentials cannot close it. Contact

Maria.Smith.Sftoe@gmail.com, join https://discord.gg/ucwGryVxGr, or inspect the public work at https://github.com/MettaMazza.

1.4 Published sequence through Physics

The present release stops at Physics. Chemistry, Materials and every later branch remain outside this ordered update.

Order	Paper	Version	Open paper	Archival DOI
00	*There Is No Nothing* - Methods Paper 00	0.2.0	Markdown	10.5281/zenodo.21591160
01	*From Nothing to Fold* - Foundation	1.2.0	Markdown	10.5281/zenodo.21591169
02	*From Fold to Mathematics*	1.3.0	Markdown	10.5281/zenodo.21591170
03	*From Distinction to Information*	1.2.0	Markdown	10.5281/zenodo.21591171
04	*After Turing: The Fold Machine*	1.2.0	Markdown	10.5281/zenodo.21591174
05	*The Quantum Fold Machine*	1.2.0	Markdown	10.5281/zenodo.21591175
06	*From Fold to Physics*	1.3.0	Markdown	Version DOI pending archival deposit; previous 1.2 DOI 10.5281/zenodo.21627765

2. Lead derivation: the exact fine-structure constant

The paper does not present α as a guessed expression, a regression target or a coincidence selected from nearby fractions. Its construction is a typed chain from earlier admitted Fold structure:

1. The foundational Fold distinction supplies exactly two fibre labels, so $b = 2$.
2. The independently enumerated second nonidentity recurrence closes uniquely at generator $c = 3$.
3. Three-direction support gives the complete generation volume $3^3 = 27$; its successor volume is $3^4 = 81$.
4. The least complete binary covers are forced by exact inequalities: $2^4 < 27 \le 2^5$ and $2^6 < 81 \le 2^7$. Therefore the down and up cover depths are exactly 5 and 7.
5. Those typed blocks force the tower $2^7 = 128$, the rank-two boundary block $3^2 = 9$, and the complete directed cover $2 \cdot 5^3 = 250$.
6. Promoting one of the three interchangeable directions from depth five to depth seven at each finite rung produces the complete ladder $(5^3, 5^2 \cdot 7, 5 \cdot 7^2, 7^3) = (125, 175, 245, 343)$.
7. The terminal effective cover is therefore the exact positive rational

$$C = 250 + 1/(175 + 1/(245 + 1/343)) = 3676744786/14706643.$$

8. The tower, boundary, returning One and effective cover then force

$$\alpha^{-1} = 128 + 9(C + 1)/C = 503846395469/3676744786,$$

and hence

$$\alpha = 3676744786/503846395469.$$

The four exact promotion stages converge internally from $34259/250$ to $5995462/43751$, $1468922449/10719245$, and the terminal $503846395469/3676744786$. The formal claim generated and decided all 1,024 members of its declared grammar, admitted exactly one survivor, passed minimality and named-shape uniqueness, passed all four mandatory adverse controls, and was independently reconstructed. Its 256-form empirical successor kept the CODATA target inaccessible until the prediction seal, converted the complete published central value and uncertainty into exact rational endpoints $[137.035999156, 137.035999198]$, found the sealed result inside that interval, and rejected the deliberately tampered control. Measurement confirms the result; it does not select any rung, coefficient or survivor.

This derivation is the paper's central numerical result. It is also a dependency rather than an isolated endpoint: the same exact α propagates into the charged-lepton refinement, electroweak share, Higgs relation, proton/Planck hierarchy, magnetic anomalies, proton radius and other terminal precision laws below.

3. Headline findings: landmark first-principles results

The discoveries are ordered by physical consequence. Evidence identifiers are supplied for reproducibility, but the findings-not filenames or hashes-are the scientific headlines.

Headline finding	Exact first-principles result	External measured comparison	Evidence claim
Fine-structure constant	α□^{1} = 503846395469/3676744786 = 137.035999177180855...; α = 3676744786/503846395469 = 0.007297352564321794...	CODATA 2022: α□^{1} = 137.035999177 ± 0.000000021; the sealed exact result lies inside the complete interval	`SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001`
Charged-lepton mass structure	exact cubic invariants (1, 1/6, 1/485, 3/1454), terminal α^3 refinement, and Koide invariant 2/3	both terminal adjacent-mass-ratio consequences and exact 2/3 lie inside the complete CODATA-derived intervals	`SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002`
Higgs mass and self-coupling	m_H/v = 1/2 + 6α/5; with post-seal v=246.22 GeV, m_H = 125.266104978... GeV; λ = 0.1294167525...	mass prediction lies inside PDG 2025 [125.09, 125.31] GeV; complete ATLAS, CMS and self-coupling rows retained	`SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066`
On-shell electroweak share	1930922298157999/8642477221479757 = 0.223422318471252882...	inside PDG [0.22333, 0.22351]; compatible-input W/Z check passes and the all-input tension is retained	`SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003`
Squared Planck/proton hierarchy	2^127(1 - 2α/3) = 1.693134633261878984...×10³□	inside the complete propagated CODATA interval [1.693112648161510904..., 1.693187333165341130...]×10³□	`SFT-PHYS-VALIDATION-PROTON-PLANCK-TERMINAL-003`
Electron and muon magnetic anomalies	a_e = 0.00115965218046558296...; a_μ = 0.00116592071499941707..., both exact positive fractions	inside the complete CODATA electron and Fermilab world-average muon intervals without fitted coefficients	`SFT-PHYS-QED-MUON-MAGNETIC-ANOMALY-004`
Proton rms charge radius	coefficient 4(1 - α/10); post-seal scale translation gives approximately 0.840621761 fm	inside the registered 2026 electronic-hydrogen, muonic-hydrogen, CODATA-2022 and conservative PRad intervals; historical CODATA-2014 mismatch retained	`SFT-PHYS-PROTON-RADIUS-TERMINAL-029`
Vacuum and cosmological scale	local floor 1/2^20; vacuum share 11/16; normalized $\Lambda(c/H)^2 = 33/16$	11/16 lies inside Planck [0.6833, 0.6945]; the dimensional Λ interval is transported only after post-seal H and c	`SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036`
Dark-to-baryon and expansion calibration	dark/baryon 27/5 with shares 27/32 and 5/32, refined 279/52; late/early expansion 13/12, refined 3305/3048	both ratio families lie inside their complete registered Planck and SH0ES observational intervals	`SFT-PHYS-COSMO-HUBBLE-CALIBRATION-001`
Second nonidentity generator	exact count 3; first-return orbit 1/7 → 2/7 → 4/7 → 1/7	structural exact result	`SFT-PHYS-STRUCT-GENERATOR-THREE-001`
Stable spatial dimension	3	independent structural stability census	`SFT-PHYS-SPACE-DIMENSION-THREE-001`
Boundary rank	2	boundary of the forced three-direction carrier	`SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001`
Source dilution exponent	2; response = source/r^2	experiment reports 2 + q with q=(2.7 ± 3.1)×10□¹□; 2 is the sole positive integer in the interval	`SFT-PHYS-VALIDATION-INVERSE-SQUARE-001`
Nuclear closure sequence	(2, 8, 20, 28, 50, 82, 126, 184)	exact match to the complete registered IAEA sequence	`SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001`
Binding-per-nucleon maximum	A=62, Z=28, N=34 (nickel-62)	same unique maximum in the complete 2,548-row positive-composite AME2020 census	`SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005`
Primordial support partition	scalar 31/32 = 0.96875; tensor 1/32 = 0.03125	Planck n□ interval [0.9607, 0.9691]; tensor bound r<0.032, passed by 0.00075; no equality to conventional e-fold count is claimed	`SFT-PHYS-VALIDATION-INFLATION-GROWTH-040`

Headline finding	Exact first-principles result	External measured comparison	Evidence claim
Radiation scaling	fourth-power scale law; exact doubling ratio 2□ = 16	post-seal comparison with the measured Stefan fourth-power exponent; no dimensional coefficient is claimed as structure-only	`SFT-PHYS-VALIDATION-COLLECTIVE-RADIATION-RESPONSE-042`
Unified constants object	one rooted object fixes b=2, c=3, spatial rank 3, boundary rank 2, cover depths 5/7, terminal alpha, lepton/quark products, dark/baryon and expansion ratios, Planck hierarchy, vacuum floors and normalized cosmological magnitude	the object composes already sealed structural and measured-value receipts; no listed value is introduced as a fitted cross-sector parameter	`SFT-PHYS-UNIFIED-CONSTANTS-OBJECT-077`
Tesla resonance family	bounded round trip 2q; positive whole mode family 2qn; odd quarter-wave harmonics; one longitudinal and two transverse roles; exact resonant transfer with complete energy ledger	five captured source rows support the resonance classes while retaining material losses, distinct speeds, Earth-cavity boundaries and no source-free or unlimited-power claim	`SFT-PHYS-VALIDATION-TESLA-RESONANCE-FAMILY-082`
Vacuum/inertia drive family	local exact drive a to b retains positive transfer a Take b; vacuum and inertia co-vary at exchange ratio One; finite-depth floor 1/2^(k+1); restoration closes the six-label ledger	official records retain the proposed mechanism, nonempty vacuum response and unity anchor, while also retaining no public prototype measurement, pump energy and no source-free cyclic gain	`SFT-PHYS-VALIDATION-VACUUM-INERTIA-DRIVE-FAMILY-087`
Penta/hepta sectors and Smithion census	sector 5: coupling 4/5, 5 charge kinds, 24 mediators, slope 4; sector 7: coupling 6/7, 7 charge kinds, 48 mediators, slope 6; category-clean total 110 fundamental kinds	known 3/8 sector anchors and known fermions agree; 72 new gauge carriers and 12 Smithion kinds remain explicit standing predictions with exact search and falsification boundaries	`SFT-PHYS-VALIDATION-NEW-SECTOR-COMPLETE-FAMILY-095`
Physics Grand Lock	the published pre-extension snapshot contains 349 claims; its formal certificate covers 347 claims and 534 dependency nodes; 21/21 generator-dependent headline values move under 3→4 while half-One, spatial rank 3 and boundary rank 2 hold	234 pre-lock empirical receipts reconcile 147 distinct external source identities; all 14 detected unfavorable-result or scope-boundary claims and all 6 legacy receipt shapes remain explicit	`SFT-PHYS-VALIDATION-GRAND-LOCK-076`

3.1 Charged leptons: mass structure without irrational roots

Generator-three support forces one three-root carrier with symmetric invariants `sum = 1`, `pair sum = 1/6`, `leading product = 1/485` and sharpened product `3/1454`. The terminal refinement holds from that product the unique complete electromagnetic self-coupling $\alpha^3/3^3 \cdot (5 + 7\alpha/3)$. Exact rational root brackets then predict both adjacent squared mass-ratio consequences; both lie inside their complete CODATA 2022 intervals. Independently, the Fold normalization forces the charged-lepton Koide invariant exactly `2/3`, which lies inside the complete CODATA-derived interval. No measured mass enters the cubic, no irrational root is admitted into proof, and no mass-ratio row chooses the invariant.

3.2 Electroweak and Higgs sector

The terminal electroweak construction uses complete binary support sixteen, the three held generator directions, charged support 15^2, four neutral pair channels of 14^2, and one terminal $\alpha/17$ self-return. It forces

$$\sin^2\theta_W = 1930922298157999/8642477221479757 = 0.223422318471252882....$$

That sealed value lies inside the complete PDG on-shell interval `[0.22333, 0.22351]`. Its exact One-complement passes the compatible-input W/Z interval; the known all-input tension is preserved rather than deleted or used to change the derivation.

For the scalar sector, half-One is the unique displaced ground and the leading rungs are `1/2`, `1/4` and `1/8`. Six-cell scalar support, its held return and cover depth five force $m_H/v = 1/2 + 6\alpha/5$. Applying the registered post-seal dimensional reference `v = 246.22 GeV` yields the exact mass `31557437733819647/251923197734500 GeV = 125.266104978... GeV`, inside the PDG 2025 interval `[125.09, 125.31] GeV`. The same relation forces native $\lambda = (m_H/v)^2/2 = 0.1294167525....$ Individual ATLAS and CMS offsets and the present broad direct self-coupling constraint remain explicit in the validation record.

3.3 Precision magnetic anomalies and proton structure

With the admitted exact α and exact terminal turn 355/113, the finite electron loop forces one exact rational anomaly a_e = 0.00115965218046558296..., inside the complete CODATA interval centred on 0.00115965218046 with standard uncertainty 0.00000000000018. The second-generation correction $\alpha^2(2/17 + \alpha/106 + \alpha^2/853)$ forces $a\mu$ = 0.00116592071499941707..., inside the Fermilab world-average interval centred on 0.001165920715 with standard uncertainty 0.000000000145. Neither value imports a conventional perturbation series or fitted coefficient.

The proton charge-support law independently forces the dimensionless coefficient $4(1 - \alpha/10)$. Post-seal composition with the CODATA reduced proton Compton scale gives approximately 0.840621761 fm. It lies inside the registered current electronic-hydrogen, muonic-hydrogen, CODATA-2022 and conservative PRad intervals. The disjoint historical CODATA-2014 interval remains in the record as an unfavourable historical row rather than being erased.

3.4 Scale hierarchy, vacuum and cosmological ratios

The terminal squared Planck/proton hierarchy is exactly $2^{12\square}(1 - 2\alpha/3) = 1.693134633261878984...\times10^{3\square}$, inside the complete outward-propagated CODATA mass interval. Vacuum structure separates three quantities that conventional shorthand can conflate: a local boundary-energy floor $1/2^{2\square}$, a global vacuum share 11/16, and normalised cosmological magnitude $\Lambda(c/H)^2 = 33/16$. The global share lies inside the complete registered Planck interval; dimensional Λ is obtained only after the sealed coefficient is composed with held observational H and exact c scales.

The three-space generation volume 27 and its least binary cover 32 force dark and baryonic shares 27/32 and 5/32, ratio 27/5, with the period-five refinement 279/52. A separate Fold calibration forces matter/vacuum shares 1/3 and 2/3, a leading late/early expansion ratio 13/12, and depth-seven refinement 3305/3048. Both ratio families pass their complete registered Planck and SH0ES interval tests.

3.5 Space, force, nuclei, hadrons and gravitational waves

The independently enumerated generator is three; stability selects three spatial directions; their boundary has rank two; and complete propagation across that boundary forces inverse-square dilution. The exponent is exactly 2, while the registered experiment reports $2 + q$ with $q = (2.7 \pm 3.1)\times10^{\square1\square}$, making two the sole positive integer in the complete interval.

The same structure forces the nuclear closure sequence (2, 8, 20, 28, 50, 82, 126, 184) and the unique positive-composite AME2020 binding-per-nucleon maximum at nickel-62. Light-hadron support follows the exact squared trajectory $Q(J) = (6J - 3)/5$; all five registered carriers lie inside their complete resonance-support intervals without fitting a slope or intercept. Gravitational-wave structure forces inspiral with rising chirp, merger, then damped ringdown; the complete GW151226 and GW190521 records validate the three-stage ordering while retaining the short-signal and alternative-source boundaries.

4. What the result changes

The scientific claim is not merely that several fractions approximate familiar numbers. One exact, axiom-free derivational language produces the structural dimension, force exponent, dimensionless couplings, mass ratios, scale hierarchies, thermodynamic boundaries, quantum relations and macroscopic propagation laws under one admission standard. Every numeric result is connected to the same foundational root, generated within an explicit finite grammar, isolated from its measurement target, and tested after sealing. That conjunction-unification, exactness, zero fitted parameters, uniqueness enumeration and external validation-is the work's substantive contribution.

The paper therefore keeps three levels distinct without diminishing any of them: the physical result is the headline; the exact derivation explains why it follows; and the receipt, hash, adverse controls and independent reconstruction establish that the reported chain is the chain actually executed.

5. Complete current Physics status

The current categorical inventory contains 368 Physics claims in 13 ordered subbranches. All 368 have immutable model-admitted receipts. The categorical V1/V2 ownership audit identifies 488 Physics-owned atoms, all 488 closed at the declared same-strength current-evidence boundary, with no open atom or gap family. Formal Grand Lock 075 binds the complete pre-extension ownership surface, verifies its acyclic 534-node dependency dictionary, proves that every route in that snapshot reaches the foundational One and tests generator dependence. Empirical Grand Lock 076 reconciles all 234 pre-extension empirical claims, 147 distinct external source identities, every available measurement receipt, all disclosed legacy receipt shapes and every detected unfavourable or scope-boundary claim. Nineteen later Physics extensions are not silently folded backward into those historical locks: their own family-completion certificates, post-seal comparison receipts and live categorical inventory preserve them explicitly.

"Closed" here means complete to the dated corpus, methods and evidence standard. It never means immune to correction or permanently closed to discovery. New findings, stronger evidence and falsifications remain lawful versioned extensions through the unchanged engine; existing receipts remain immutable.

6. Exact constitutional domain

The derivational domain admits the empty One as structural absence but never numerical zero. Proof magnitudes are positive generated counts and exact positive rational parts or ratios. Orientation, opposition and complement are held labels rather than negative quantities. Irrational, imaginary and binary floating values are barred from proof. Completed infinity, an ungenerated continuum, axioms, fitted coefficients and free parameters are also barred. External decimal measurements remain source-bound records. A finite decimal is converted to an exact rational interval only inside the empirical adapter and never gains authority to select the Fold law.

7. The single admission engine

Each claim entered the same `SFTAdmissionEngine`. Registration rejects axioms, free parameters, missing root trace, unadmitted dependencies and source-identity failure before candidate execution. The engine then checks census cardinality and identity, one decision per candidate, exactly one survivor, minimality, named-shape uniqueness, the four mandatory controls, an implementation-distinct external reconstruction and, for empirical claims, prediction isolation, target custody, complete rows and falsification. A failure halts without model admission. Accepted receipts are immutable evidence identities. The engine source is not edited by this publication correction, and the paper cannot confer admission.

8. Why superdeterminism permits uncertainty and quantum weights

No law in this branch installs ontic nondeterminism. The complete Fold state includes preparation, held labels, measurement setting, path and observation record. A probabilistic or quantum weight is an exact census ratio over unresolved support. Each branch execution remains deterministic. Measurement uncertainty records distinctions unavailable to an observation class; it does not assert causeless state selection. Bell correspondence therefore retains setting and preparation records in the complete state, identifies the factorization assumption that fails, and separately preserves the no-signalling marginal through complete remote-fibre enumeration.

9. Empirical constitution

External evidence follows one direction. First the Fold dependencies generate and eliminate candidate laws. Then a data-only Fold program receives only registered inputs and the sealed derivation identity. Its instruction set has no filesystem, network, subprocess, clock, environment, dynamic import or foreign-function capability. A distinct custodian commits target identity before execution and releases content only to the matching prediction seal. The evaluator preserves every registered row and must reject a deliberately altered or displaced control. BIPM, NIST/CODATA, NIST ASD, PDG, IAEA, IAPWS, GWOSC, CERN Open Data and NASA LAMBDA supply the external records.

10. Exact measured-value correspondence

Measured-value forcing means that a zero-parameter Fold relation produces a sealed consequence and a registered exact adapter compares it with withheld observation. The observation is evidence, not a tunable parameter: it cannot enter candidate generation, choose a survivor or add a correction. Finite decimals and uncertainties are parsed as exact positive fractions. Multiplication and quotient propagate interval endpoints in the capability-closed interpreter; target overlap is evaluated only after release. Exact official decimal prefixes with ellipses are bounded by their next decimal place. Any non-overlap halts admission.

The inverse fine-structure result belongs here, in Physics: `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001` forces the terminal exact ratio and `SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001` performs its sealed CODATA comparison. The same rule governs every constant, scale and precision claim in the inventory. No redundant handwritten status may override an engine-admitted receipt or a passed empirical certificate.

11. Reading the complete derivation ledger

Each claim section below is a self-contained prose projection of its machine evidence. The complete `candidate_census.json`, `elimination_receipt.json`, `controls.json`, `certificate.json`, external experiment registration and receipt remain authoritative. The paper includes every claim identity and admitted receipt hash so the evidence map can fail closed on omission or substitution.

Family results, complete claim inventory and claim-level audit records

The dependency-ordered ledger below is the complete human-readable Physics claim inventory for this version. Each claim section exposes its claim identity, family, formal and empirical status, exact statement, dependency route, carrier and boundary, candidate grammar and count, unique survivor, elimination logic, falsification condition, controls, source and custody records, chronology, scientific meaning and receipt identities. Where a generic clause is referenced, its full unchanged wording is retained in the shared claim-record appendix. The machine packages remain authoritative for complete candidates, decisions and executable traces.

12. Measurement Metrology

Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain.

1. Capability-closed exact Fold prediction form

Claim identity: `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Within the registered finite prediction grammar, a physical prediction can be target-inaccessible, exact, portable and completely traced in exactly one form: immutable registered Fold inputs pass through a data-only sequence of generated exact operations to one terminal emission, while every ambient host capability and unregistered operation is absent and any violation halts.

> *Registered exact inputs, a data-only closed opcode surface, no ambient capability, single-assignment state, one final emission, complete trace and no extra rule.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-COMP-FORM-STATE-TRANSITION-001`

- SFT-COMP-FORM-OPERATIONAL-PROCESS-001
- SFT-COMP-SEM-EVALUATION-001
- SFT-COMP-SEM-TERMINATION-001
- SFT-COMP-SEM-VERIFICATION-001

Generated grammar and closure boundary. Generate the complete product of input, value, operation, capability, state, termination, trace and extension forms. The declared exact boundary is: All finite data-only prediction interpreters assembled from admitted exact values, registered inputs, generated operations, immutable held states, a declared terminal emission and no host capability. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `registered-input-only__exact-fold-domain__data-only-generated-operations__no-ambient-capability__single-assignment-held-state__one-terminal-emission__complete-step-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
input	registered-input-only	ambient-input	Ambient input can contain the withheld target or an answer-producing channel.
value	exact-fold-domain	host-scalar-domain	Host scalars import forbidden or platform-dependent proof values.
operation	data-only-generated-operations	contributor-executable-source	Arbitrary source can create operations outside the registered grammar.
capability	no-ambient-capability	ambient-capability	An ambient resource can reveal target, comparison or mutable authority state.
state	single-assignment-held-state	mutable-overwrite	Silent overwrite loses the exact predecessor and permits untraced replacement.
termination	one-terminal-emission	implicit-or-open-ended	Implicit or open-ended completion cannot certify a complete prediction artifact.
trace	complete-step-trace	result-only	A result alone hides the operational relation and its resource path.
extension	no-extra-rule	extra-rule	An extra instruction or capability is a free answer-selecting parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:bc99279c92de616961792fbfee2b9502b70e6aec23b54fc97c73ed83fafd1b87`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form missing the first preservation; observed reject the generated form missing the first preservation; receipt `sha256:101b735f39c0308acff4f1fbb4fc7c0845d09021fb0cada8326ad7a876d77f96`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:06e9ea349c585d18965d1f715617649d780dbd450ab59bce9090d38591920b1b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1814b0879667d0341a6a7cf3b345a60550bcbc94d5ad113e00cc9e2db8dc00b6`.
- `boundary`: passed; expected reject excluded capabilities, values or authority paths; observed reject excluded capabilities, values or authority paths; receipt `sha256:0978722205b9e2a84a6ef2fb7e600cc50bc215a65ae8c025a2e96c53f0e6e3dd`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2ac5e37d95aea5263c2fcf026b93eb3549e21a04ce7e0e125ad91b62a956c0cd`. Independent certificate: `sha256:060dddafaba441cc92c20cb16ac1c5e960744ea2af940ffacb70c6bcce2d0d15`. Engine

external-validation hash:
`sha256:4abfd002a0b3c5467a517fa4f2cdf36842d757a292cb4979482c4403ad859fd7`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- arbitrary Python or native contributor code
- filesystem, network, subprocess, clock, environment, import and foreign-function access
- floating, negative, irrational, imaginary and semantic-zero proof values
- unbounded or untraced execution

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:9b0c5773bbc29985b974079e8fab0947dbf28b7ca44e8bcc2c254e5ab94dae24`; engine receipt `sha256:bf675d47eb8731760292d542c65fb0af31349d313360d018ee0199a71e23f9a4` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001-bf675d47eb873176.json`; empirical-validation hash `None`; measurement receipt `None`.

2. Portable target commitment and post-seal custody

Claim identity: `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Within the registered blind-exchange grammar, target content remains unavailable to prediction in exactly one form: a distinct custodian commits the complete named target package and prediction envelope before execution, then releases that unchanged package once to the matching experiment only after a matching prediction seal exists.

> *Pre-prediction complete commitment, distinct custodian, nonce-bound unchanged content, matching experiment and envelope, one post-seal release and no extra rule.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-COMP-SEC-HASHING-001`
- `SFT-COMP-SEC-COMMITMENT-001`
- `SFT-COMP-SEC-INTEGRITY-001`

Generated grammar and closure boundary. Generate the complete product of timing, holder, target support, content identity, experiment, envelope, release and extension forms. The declared exact boundary is: All finite portable target exchanges using admitted exact identities, pre-seal commitment, distinct custody and one post-seal release. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `pre-prediction-commitment__distinct-custodian__complete-named-support__nonce-bound-content-identity__matching-experiment-only__matching-envelope-only__one-postseal-release__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
timing	pre-prediction-commitment	post-prediction-choice	A post-prediction choice can select favourable target content.
holder	distinct-custodian	prediction-holder	The predictor holding targets violates structural inaccessibility.
support	complete-named-support	partial-or-open-support	Partial or open support permits selective favourable rows.
content	nonce-bound-content-identity	unbound-content	Unbound content can be replaced after prediction.
experiment	matching-experiment-only	cross-experiment-release	Cross-experiment reuse breaks registration custody.
envelope	matching-envelope-only	unbound-envelope	An unbound seal can certify a different input or relation.
release	one-postseal-release	preseal-or-repeat-release	Pre-seal or repeat release leaks target information.
extension	no-extra-rule	extra-rule	Discretionary release is a free selection parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:d90150c6bfcb9cf6138c088811ced4e86ce19c87be78905f6844be0b4e843309`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form missing the first preservation; observed reject the generated form missing the first preservation; receipt `sha256:9d1ae2f40bb91451031c219678c477468054cc6bfecb398935d7356e3e0e059a`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:18e6f58bb38714edf42a2881d8bf8e19a8c4b4999a5132d7bde03a4a053f20c7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:53b93b6397aa18513fb73bcd48e2a739a26061f47428ae3790fa8d25f0248087`.
- `boundary`: passed; expected reject excluded capabilities, values or authority paths; observed reject excluded capabilities, values or authority paths; receipt `sha256:0f1c55cf30f7c60f78e3ff7f089b1c3a641fe34e00e84508ede332bab9d54b03`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:458f3324e7eb0127ec05aab491f9ddd6005718ba642b2d6ed3b67d0bc5adc1d9`. Independent certificate: `sha256:af9b5b3c7fcb2d140f04f3aa902aadd0392c9aec0bc027e7bb517042963abc13`. Engine external-validation hash: `sha256:24ae77d5eadb178fd6700ae5cc3dde02952f1e649409a3398f763f2bb797b723`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- target access by the prediction executor
- release before a matching prediction seal
- selective target-row omission

- host-specific sandbox claims

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:7e20fbe1c90984ae6971e579458ea46eea192c8ee46398828df67241bcfe3696; engine receipt sha256:288826223673d5472e818fb0c53d837710adcfdb7abbabf88285100650e3c872 at receipts/engine/model_admitted/SFT-PHYS-MEAS-TARGET-CUSTODY-001-288826223673d547.json; empirical-validation hash None; measurement receipt None.

3. Hostile empirical-package and authority protection

Claim identity: SFT-PHYS-MEAS-HOSTILE-PACKAGE-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Within the registered empirical-package grammar, contributor material can enter official prediction without gaining execution or scientific authority in exactly one form: strict data-only program fields are parsed by the admitted interpreter, the census and model-admitted receipts are identical before and after handling, and every extra field, executable source or protected-tree change halts admission.

> *Strict data-only Fold document, exact parser, no unknown fields or direct source execution, unchanged authority trees, separate comparison, fail-closed response and no extra route.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-COMP-SEC-INTEGRITY-001
- SFT-COMP-SEC-ADVERSARIAL-001
- SFT-COMP-SEM-VERIFICATION-001

Generated grammar and closure boundary. Generate the complete product of package, parser, field, execution, authority, comparison, response and extension forms. The declared exact boundary is: All finite empirical prediction packages admitted through a strict data-only parser while census and model-admitted receipt authority remain protected. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: data-only-fold-document__exact-schema-parser__reject-extra-field__interpreter-only-execution__unchanged-protected-tree__postseal-separate-comparison__halt-without-admission__no-extra-route. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
package	data-only-fold-document	mixed-code-and-data	Mixed packages can execute contributor-selected behaviour.
parser	exact-schema-parser	permissive-parser	A permissive parser admits undeclared behaviour.
field	reject-extra-field	ignore-extra-field	An ignored field creates divergent or hidden meaning.
execution	interpreter-only-execution	execute-contributor-source	Direct execution exposes every ambient host capability.
authority	unchanged-protected-tree	mutable-authority-tree	Changing census or accepted receipts can manufacture admission.
comparison	postseal-separate-comparison	embedded-comparison	Embedded comparison can expose or optimize against targets.
response	halt-without-admission	warn-and-continue	Continuation admits a known authority or capability breach.
extension	no-extra-route	extra-route	An extra route bypasses the protected admission path.

Base and successor. The closure proof is sealed by derivation hash sha256:8f53a36634646d3d0edf75d2aa1c8b7ad8c9238431cab60c5ed69d89afa2ffb1. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form missing the first preservation; observed reject the generated form missing the first preservation; receipt `sha256:155abd7902abed0cf2e85564b52bafb870f8f8565b09bc3b8c0847134a9399c5`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:9eadbe7d5481a418a910ff9ca6f16ef51c317220a3ac4b47900b0516ec7e419f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a319275abd552274cbf9c75121007066d2ae71142500f2fe884f53cd93276a61`.
- `boundary`: passed; expected reject excluded capabilities, values or authority paths; observed reject excluded capabilities, values or authority paths; receipt `sha256:119b65e69ed287f95df7763be66eb58b7997ef88302cae2a698c8651125a9034`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b5a6ba9afcebd3f858403bb8a48a8d8623ac2fde50ecb36700c68264453e7103`. Independent certificate: `sha256:2a2bc6e6aadd3b7c2753423f4cd5a8690935d26e4d58a44e1b65a601fef6d8f9`. Engine external-validation hash: `sha256:cb2f2ac788dc037ffb4a61d8e22bf9ae35f8d9a1373302545f6bbac53675e127`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- contributor Python, native source, shell commands, modules or callables
- unknown package fields
- census or model-admitted receipt mutation
- target comparison within prediction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:2864116c4bee1e27737be9e7e7ab8ecc297f21d369e051e27fa2afeee9db431b`; engine receipt `sha256:40d8e9ab7ac203495ac06fdf51cc708c5a4e1d528f8701212c9e562da6eaaf16` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-HOSTILE-PACKAGE-001-40d8e9ab7ac20349.json`; empirical-validation hash `None`; measurement receipt `None`.

4. Physical observation and retained distinction

Claim identity: `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A physical observation is forced to retain a distinguishable phenomenon, a held reference condition and the exact comparison record; an answer without those three coordinates is not an admitted physical observation.

The unique observation carrier is a source-bound phenomenon/reference pairing plus its held comparison record, executed without target access and with every row retained.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__distinction-reference-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`distinction-reference-record`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b13b4b74dd66a5e092af10f451eda8471a4b5aaeb8fe3678f1e9c43446a80fd1`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a34141df720d459de4b146698583cf8fdcde23b929ddcda6e2efd7685bf2a6cd`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:99dffccfc8546e03a99d83a3b7a5a24e5a6e352e2a77380cd7904173382c2d60`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f62791c52be506a073e57cb83a2a099e6fa76f0f8e0ecb4df7950d4ceafa44dd`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:eedf3b06cb5ccdc9b7ffad27fa80d7f553e977f01d14994793b7e2530c3cb9a2`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4a2fc44538091b1505d1da8b0424f73efbea60a315a3c253ca7307673dbabb08`. Independent certificate: `sha256:063d93d38c21349f348acef11572c524ffb601bf2e4b5eba7a05bba5b7e29b29`. Engine external-validation hash: `sha256:e927cda5ee39e50ad6e88fb8ec8e47b80e1172541f1c843178effd53cbcf0ca5`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- observation-carrier-1: predicted distinction-reference-comparison-record; observed distinction-reference-comparison-record; exact match True
- observation-carrier-2: predicted distinction-reference-comparison-record; observed distinction-reference-comparison-record; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:fb9b60b896112e06cbcff00baa15ad27c229ff844f985140c4b76c24f80e8745`. Isolation certificate: `sha256:a4963217e76a339639e927929164afbc75a167037de4411918fe346236446e8d`. Custody certificate: `sha256:395b398805fa3e3a12514432a9a471738928b1ff76d38b6c175f9febb07b4ee2`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- observation-carrier-1 from BIPM-SI-BROCHURE-9-2026
- observation-carrier-2 from BIPM-SI-BROCHURE-9-2026

Shared claim-record clause `PHYS-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favourable rows, free scales and fitted parameters

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:e7de0b1dfb937752cf3e928f2597f84cfa9a75f68bd83f1dc0de805a144a33c4`; engine receipt `sha256:8d7daef993773a0721b9bf6022aef3b22ebea445608b80084ce496a35917d955` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-OBSERVATION-CARRIER-001-8d7daef993773a07.json`; empirical-validation hash sha256:7e72914082a7bc50a648fb8e7a38fe1a179495c4e8a5a795e583b82d01a32669; measurement receipt sha256:fb9b60b896112e06cbcff00baa15ad27c229ff844f985140c4b76c24f80e8745.

5. Physical quantity as reference-equivalence carrier

Claim identity: SFT-PHYS-MEAS-QUANTITY-CARRIER-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A physical quantity is forced as the equivalence class of observations comparable through one generated kind-preserving reference relation; its measured value is a separate record, not the quantity itself.

> *The physical quantity carrier is the complete class of observations connected by a kind-preserving reference equivalence, with measured values retained separately.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__kind-preserving-reference-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	kind-preserving-reference-equivalence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:bcbcb5efef70da5f3432198aa359d60e7dbcd2dfc20781a94f0abe5a870d6a5f. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9888ea14f67d6c08a04a1cfc8d37a8522c8222c7855cd64c4a0414328a8486c2`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1ad6611d696ab491614f73e53a56b3a3c1f91931a00215ebdb548ee37e121a47`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9a060ec9f8710478bdbf4ae6ff1056ad01e38498daa3fa8e712b5b97ec54d29c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:61dc4ec52ba6fb38968ea6c69f29e3b3164d7a336203b9af708cece27e69569a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8c3c165e5709b23eba3bd85ff8e52ef0582861f3da26491e87b9baf7932be60a`. Independent certificate: `sha256:e01dbd3531cd39593cd0477f80a7cd7dd4245304c6a3baec8a1a143c92de88c7`. Engine external-validation hash: `sha256:822d0e139a12f5427f2993540668d075885e1db47561dc545912f3fa3b7eb813`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- quantity-carrier-1: predicted property-kind-reference-equivalence; observed property-kind-reference-equivalence; exact match True
- quantity-carrier-2: predicted property-kind-reference-equivalence; observed property-kind-reference-equivalence; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:4da5928a0e5cac7ce45b39d7aa745f818e420030b50d9fd507d9b7772584ac5c`. Isolation certificate: `sha256:d5222b2dd483f506a339cf06f5703f35c4078b207cb30bbf63e31e12298d4668`. Custody certificate: `sha256:8c9573a455a104ebbc060d0f2b8c42761633adb5e7fb5783ff019afda5b7457a`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- quantity-carrier-1 from BIPM-SI-BROCHURE-9-2026
- quantity-carrier-2 from BIPM-SI-BROCHURE-9-2026

Shared claim-record clause `PHYS-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value

- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favourable rows, free scales and fitted parameters

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:6141278e60dc78628d5aba3eb14291466f8a2197772073b05832c236966e7cb8`; engine receipt `sha256:c98453b691e68a1b5340a5c14d757c54c19f736fc89f9dbfb54c0ce7c7117033` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-QUANTITY-CARRIER-001-c98453b691e68a1b.json`; empirical-validation hash `sha256:1be09b9b3031b8cf29a3d5e7a99457a11c57688a8ded4e21c2fa440ea3fdb1a6`; measurement receipt `sha256:4da5928a0e5cac7ce45b39d7aa745f818e420030b50d9fd507d9b7772584ac5c`.

6. Dimension composition and independence

Claim identity: `SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A physical dimension is forced as the canonical generated composition path of independent quantity kinds; two descriptions share a dimension exactly when their paths reduce to the same held form.

> *Dimensions are canonical finite products and exact quotients of independent quantity-kind carriers, compared by form identity rather than numerical value.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__canonical-dimension-path__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`canonical-dimension-path`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:5a0b540830aaf5e802e11e5e7447b4c8008cc19b95dcb2e01c6a97993a924d06`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9eed98166f8ac9a14cdf047c4790e53789d749b4d06966057f64d641844c681a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8f143ffae762666331bd4572a420f8c08eb1eae0abc7edd002800541b3ce5abd`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8ad77de29f4c2d52c5ebe06bb3ac7e97bd096be04e2fd47168eb43346cd9e3cf`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:77c4aa61b2b5b44faae27694a0736b89001989fd4d05a4aa02711ed84ae9f875`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:31540d9c686489dbdcd06cde3f9318e569bf13f73ff54208473367ca3ecca16b`. Independent certificate: `sha256:28ead99c9c104d5a6d65eb11f2e9c52ada94502a34c04b76ef4d6aea4a042043`. Engine external-validation hash: `sha256:4eeedd519209f691aa11917ab866707542c123454b6ee4c5cec3160b763779aa`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- dimension-composition-1: predicted canonical-generated-dimension-composition; observed canonical-generated-dimension-composition; exact match True
- dimension-composition-2: predicted canonical-generated-dimension-composition; observed canonical-generated-dimension-composition; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S006` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:dacf359001a590a02e19fc58997a6eea8006753f398d478bcc5a3b752a2070f0`. Isolation certificate: `sha256:5721baca0856f9bb14ffb76e984a4a0543b1264ff5da8378d4c75df3bb4502e9`. Custody certificate: `sha256:a4539a556adfbe18add5a64aca66e825f9ff3dd96d26c47ae0e2f5a88d780eda`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- dimension-composition-1 from BIPM-SI-BROCHURE-9-2026
- dimension-composition-2 from BIPM-SI-BROCHURE-9-2026

Shared claim-record clause `PHYS-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favourable rows, free scales and fitted parameters

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d1596e2d9a4c2ec5f45ec9a143c2449dde88d9377fe853b1410651942701be4a`; engine receipt `sha256:a09e3abd01f98710265eec2397d2cf3df06618ecb18a06c77ea92f703d3ac72b` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001-a09e3abd01f98710.json`; empirical-validation hash `sha256:9a61561ef3963aee29f32ac519a8631972148ef6531e9ca5f98334a55dca5bdd`; measurement receipt `sha256:dacf359001a590a02e19fc58997a6eea8006753f398d478bcc5a3b752a2070f0`.

7. Unit comparison and exact scale transport

Claim identity: `SFT-PHYS-MEAS-UNIT-COMPARISON-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A unit is forced as one held reference quantity used to compare quantities of the same dimension; lawful scale transport is witnessed by an exact positive part ratio and cannot change dimension.

> *A unit comparison pairs a quantity with a held same-dimension reference and records an exact positive ratio; conversions compose those ratios and preserve dimension.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001`

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__same-dimension-reference-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`same-dimension-reference-ratio`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3aac4934a1b7a7cf8dc636317507b0d58f5352f65646d4e7d12f1ba6664d3493`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:fcf38f4b41423e84d4a14af2d289e780faddd2a08dab621a12d3796b71ff65a6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:5a03d4dfa95258d4fbc0a242c8209d3aaeb08e107094e45cd864e66d751d880d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e96864f5f86da2dd02aab6cbdfb056815ea19b5311500f036264d916ab479c18`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:cf9239eeb6851b1aea6c2989ab384f21ebfebab2c7a62a20ace5d8e4d77f9c52`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2da4fce7d904328003e628ffb9bc4da6ec55d6f5b30a191ffe73ec2efea7170d`. Independent certificate: `sha256:cd16c654f4ee555225b946883febceb54e755414770d71342cefdc47d57160ce`. Engine external-validation hash: `sha256:72cb71cc5037252208307b5240cf7dd9c3eb4423a0835688caece4cee417c75d`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- unit-comparison-1: predicted same-dimension-reference-comparison; observed same-dimension-reference-comparison; exact match True

- unit-comparison-2: predicted same-dimension-reference-comparison; observed same-dimension-reference-comparison; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:be0d4e4d6b7861e3aeed798ac12ac02bc589d3caf7eb934feaea2c168646784c`. Isolation certificate: `sha256:33ee551d13782c7b06bfefe9bd2b734ba7abdc22216d4fafb3f246a6bc9e43a7`. Custody certificate: `sha256:b5a198444596b646f9676b2891c62f9b4e1068e4e2b2dc4aa533154a99b5941b`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- unit-comparison-1 from BIPM-SI-BROCHURE-9-2026
- unit-comparison-2 from BIPM-SI-BROCHURE-9-2026

Shared claim-record clause `PHYS-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favourable rows, free scales and fitted parameters

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:20f7b4c464dda4b2ec7721228e6c3c2004da5cdc46a5808d23c1a0788511f58c`; engine receipt `sha256:e9b8d7ccb1cd52ca2ab5600a51a684827e6cd36db3f79cd1a01301732bd90827` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-UNIT-COMPARISON-001-e9b8d7ccb1cd52ca.json`; empirical-validation hash `sha256:adc94c47b3551de4822ae0c9c5dc21d558e7ccbbcb28b2956d4095071d672b38`; measurement receipt `sha256:be0d4e4d6b7861e3aeed798ac12ac02bc589d3caf7eb934feaea2c168646784c`.

8. Reference realization and traceability

Claim identity: `SFT-PHYS-MEAS-REFERENCE-REALIZATION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A physical reference is forced to be operationally realized by a finite reproducible comparison process whose complete trace connects the definition to the observation; a name or assigned scalar alone is not a realization.

> *Reference realization is a complete finite operational trace from the held definition through each comparison to the observed record.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-UNIT-COMPARISON-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__definition-to-operation-trace__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`definition-to-operation-trace`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:40d938beac534abdd46f38562a383b189ada555a01649abd5ab3daec81b4b1b7`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1f2dad9e88737b8f22f19d247b8bfb25c4447d0cac7c1e23c4c1407cf97387bb`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b233c37c78a4e4f34ac70474aceef70cc56604f1f671fc12d620f5ed00716faa`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:209f2b5c736cf23e4b94151005ca3062de1d396d0458739a460a514cd1d3d863`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:023389bb1e2e8f8d47b353faf6147bc91f36156ea0cc9c1f771b60c10e642473`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f78121815a382685c4dcaf3c638f980994cf29225e0e9471841814aad2fc4e12`. Independent certificate: `sha256:fc2c880293991e77dc198e7b11a4e81e14f5c0750c67b42ae86f0b95c6acb3a4`. Engine

external-validation hash:
`sha256:ebe0991ff169f35b5369ffdf7d502c1664078bc030eae6102ddfc801fb5cc309`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- reference-realization-1: predicted definition-realized-by-reproducible-operation; observed definition-realized-by-reproducible-operation; exact match True
- reference-realization-2: predicted definition-realized-by-reproducible-operation; observed definition-realized-by-reproducible-operation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:a972c153a8e6ab580296aed1685404da1ab84a1afde2e769c8a6163a1eb41ee4`. Isolation certificate:
`sha256:8af6199a05a20e3a9b860357d47293eea804a945e5be07adf2d553cbc2e59e10`. Custody certificate:
`sha256:c3a3c93995344e17bb45994b33ca9691ce3999560f440f44f33029bb3c466bd3`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- reference-realization-1 from BIPM-SI-BROCHURE-9-2026
- reference-realization-2 from BIPM-SI-BROCHURE-9-2026

Shared claim-record clause `PHYS-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favourable rows, free scales and fitted parameters

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:101665e01f8bca1acab1c52378e1087c8a845f6c6383d073b1c113ac036014e3`; engine receipt
`sha256:fe8d21bca9c2c1721cd7f14ebacc535705a185af0313af00a17f89d5877836f0` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-REFERENCE-REALIZATION-001-fe8d21bca9c2c172.json`;
empirical-validation hash
`sha256:76c026763c0ef1ff8b585fda805e00ea1b7f26106ae49200b0eb7eb69b65fc1f`; measurement receipt `sha256:a972c153a8e6ab580296aed1685404da1ab84a1afde2e769c8a6163a1eb41ee4`.

9. Measured-value record separated from proof value

Claim identity: `SFT-PHYS-MEAS-VALUE-RECORD-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A measured value is forced as a held external record containing the exact reference identity, reported numerical inscription and observation provenance; the inscription never enters the proof as an SFT scalar.

> *The measured-value record holds the external number inscription, unit/reference identity and observation provenance while the derivation remains exact and separate.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-REFERENCE-REALIZATION-001`

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__held-reference-number-provenance-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`held-reference-number-provenance-record`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:1de54b307b73d6e01746e20ddb3903c7f78e2cac4607802f2cae5b1b61cd0f15`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8cbb8dd6f3a25af27290eb17336e1bb52b737779cc1e8c9fe26d29a3e92bb585`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:51109d21ed7e0a72b1655f6317744d65af2ed906d176ea92cc990b14ff8cbd2b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

`sha256:8dc482be052ba144450746d84f9dfa508643eabd770d8e96fdf38640b7ef8fed`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:bd48eb008e2e79f555cea3790deddd31fd01286d296245b56cdd01ffa97fe127`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2fd434613bc4467e473b86e1dd9b1720c04bfe2a0deaf8448a1aba855c765f5f`. Independent certificate: `sha256:8d58b9c85c1dcf9970b4a62cec01fa6ad66a07b2650d5aff08d6f66d6d7bc803`. Engine external-validation hash: `sha256:2d35e29c01080c355a980388fed0af76f89d8b7aa4737f3532b39f3299f77edc`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- value-record-1: predicted number-unit-provenance-measurement-record; observed number-unit-provenance-measurement-record; exact match True
- value-record-2: predicted number-unit-provenance-measurement-record; observed number-unit-provenance-measurement-record; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:c2ebda59a74016efbd2dfaa4ad3089cbd7ce671c443c5bf6b4f76fdbdfa591a7`. Isolation certificate: `sha256:ad2a5880eb8f4182cbf851c1dd5301c1d36fb37cf510328772868aad20fbb094`. Custody certificate: `sha256:cec0c1d2be143b1c7b8cd79599257d792614f73278fe3cdb4741af232937e41b`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- value-record-1 from BIPM-SI-BROCHURE-9-2026
- value-record-2 from BIPM-SI-BROCHURE-9-2026

Shared claim-record clause `PHYS-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favourable rows, free scales and fitted parameters

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:9d4b1f92e209ec85a2bad76581e6ce2ede3dd45fb30a2de2ec5bd534c5225635`; engine receipt `sha256:461c2062551cbfd5aa884249eaea2eefae5dbabbe672885eb764deda2e0fa01d` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-VALUE-RECORD-001-461c2062551cbfd5.json`; empirical-validation hash `sha256:391b18fc8f280e3423dbd3a9197cd357c094a618ec87f27eaf29f760ffb8a11b`; measurement receipt `sha256:c2ebda59a74016efbd2dfaa4ad3089cbd7ce671c443c5bf6b4f76fdbdfa591a7`.

10. Measurement uncertainty as retained admissible alternatives

Claim identity: `SFT-PHYS-MEAS-UNCERTAINTY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Measurement uncertainty is forced as the explicitly retained support of observation/reference distinctions not closed by the measurement process; it is not an irrational proof value or permission to choose a favourable result.

> *Uncertainty is the source-bound set of still-admissible observation/reference relations, reported with the measured record and retained in full.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__retained-unresolved-observation-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`retained-unresolved-observation-support`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:588dcbbdda9016932a32deee4311c036516af2248444222623453e91c5d75803`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7a0f4a18b8bb0dd172eab8a909284e061cbfbf6ef91fc8722df97f5662cec719`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0020b7b3b47807abfae0edd921d25970bd18f2ef64054c2ecdc4e902f8076653`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4bfeb3041e24c6d31b4aac408bd4e3b568816786a88191f65030d9d7b73f0b05`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:46a9d02402b8165db8a231977dd86732974422aa8a0d8e28ba871a18e228aeaa`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d194e478c4f9b3b77ffb8c7e737f86ddf25d01e64477403d65c8470c9a87413d`. Independent certificate: `sha256:3ed1ad98c0d3c0ec36d699270786a8ebf315001345b0cf2dd5cf9af2233d0b0c`. Engine external-validation hash: `sha256:68564445cc3b6884f099f0e0a9439e5fb1312b232845d57e172d204c8e0d6b2d`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- uncertainty-1: predicted retained-range-and-uncertainty-components; observed retained-range-and-uncertainty-components; exact match True
- uncertainty-2: predicted retained-range-and-uncertainty-components; observed retained-range-and-uncertainty-components; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:6eb5361a119e6ef2e228a0410dafe538615513d3eb674c00f56a2d62039227c6`. Isolation certificate: `sha256:071fbc1ec40ec275089616ab5cd2bcd28e6c6a8b0ef7234c7e38f87aedbab151`. Custody certificate: `sha256:e8b02e4e92daa6f3ce7d95cff2ae77dc5c44feeec58ac7d8a5f7fd2a41bab70b`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- uncertainty-1 from BIPM-SI-BROCHURE-9-2026
- uncertainty-2 from BIPM-SI-BROCHURE-9-2026

Shared claim-record clause `PHYS-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favourable rows, free scales and fitted parameters

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:f7d5982aed179b617b9c3ff7640dd2a676dcb4fe8fe53c52ae3fc4f8ae4fadd6; engine receipt sha256:9aef08a488c0a15fc4dd2132a1afd9c32d45bb0da6e6ed5c76f90c541f5ffc01 at receipts/engine/model_admitted/SFT-PHYS-MEAS-UNCERTAINTY-001-9aef08a488c0a15f.json; empirical-validation hash sha256:b262f7064a0360880fe4e616e062d6125de1ecc784caf928afad7c9154b6fabe; measurement receipt sha256:6eb5361a119e6ef2e228a0410dafe538615513d3eb674c00f56a2d62039227c6.

11. Calibration chain and comparison closure

Claim identity: SFT-PHYS-MEAS-CALIBRATION-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Calibration is forced as the finite compositional chain of exact reference comparisons connecting an instrument record to the realized unit, with every intermediate identity and uncertainty support retained.

> *A calibration is the complete ordered composition of source-bound reference comparisons, closing at the realized unit and retaining every intermediate record.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__complete-composed-reference-chain__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	complete-composed-reference-chain	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.

Axis	Forced form	Eliminated form	Elimination reason
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:4cf6d00ce900bdd5cc2a5585369233006c33f388e857a4155e0cf8f7040ecab1. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:15f6f58bf9fe46ae235575ddc0319dcbb4eb91b12dbbefc8866777e205cc5f93.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:382c52e07b6cac7a858f688563fd4f4b9731dc6cabffa6739dfc50f3ac8a60c6.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:a1b9be492a77a444a1f989971a243fee651ffd731a8c7290a7dd879b05235af2.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:a22c4401b242007fc6be19c41a51e85675303280663309b3fc00510adf2f0cfd.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:9ab06c6a96ac0cb18688ea26ebb49f23022312c0bf37679c013f208842103c4c. Independent certificate: sha256:eebfa37af6a3e87822fd7a36b7e36c6e8945c251c837ed08a90f5b01e39cef39. Engine external-validation hash: sha256:eb65354e0c7521f35ef3bd47b3cc08c9818e7d59effa0e5f6ebb78211cbd9bf1.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- calibration-1: predicted traceable-comparison-chain; observed traceable-comparison-chain; exact match True
- calibration-2: predicted traceable-comparison-chain; observed traceable-comparison-chain; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause PHYS-S006 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: sha256:ebc8e171524e298373f518758f4cdfdcc3e3404e4a77a4ed578f319026b3f79d. Isolation certificate: sha256:ffddd3ef29aeaca6fd9abb34447e5f5ecb70a1413c29db69ace217dab6630566. Custody certificate: sha256:b6f4ba961baa0934206633d813e0bdeb48da1a1c422987a425b8ed7d0dada37c.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- calibration-1 from BIPM-SI-BROCHURE-9-2026
- calibration-2 from BIPM-SI-BROCHURE-9-2026

Shared claim-record clause PHYS-S003 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favourable rows, free scales and fitted parameters

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:58c558b52ede3a0880de389cf3f6e0e1334e3e0d2299b940b70c15d5c083b095`; engine receipt `sha256:60c5b750220a91b600f9752669227be225c67e5ec8abb7d760f023557ace561a` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-CALIBRATION-001-60c5b750220a91b6.json`; empirical-validation hash `sha256:2c691a537544d0a4334de99fd9437cd484f0d9b33b8b0e3eddde7b0625ce4a67`; measurement receipt `sha256:ebc8e171524e298373f518758f4cdfdcc3e3404e4a77a4ed578f319026b3f79d`.

12. Dimensional consistency and lawful conversion

Claim identity: `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A physical relation is dimensionally lawful exactly when every compared or joined path has the same canonical dimension form and every conversion is a reversible exact same-dimension reference map.

> *Dimensional consistency is exact canonical path identity at every comparison, with conversions restricted to reversible same-dimension reference ratios.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-CALIBRATION-001`

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__canonical-path-identity-and-reversible-conversion__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	canonical-path-identity-and-reversible-conversion	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:8d0e30be3967618d9ac14b2a45301da1eb89bf9350f33da63bebeb55411a7b21`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:205006977cc520a84734c06e70ac637ab6c19a87ed851309b146b576a2f9849b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:fdd99f20271fea3b94c4d86056aa69692355c9e236302ae434dede45f04b99de`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5ebff673a39ef1ba8350015cf0f873f1ec3da528c4748d298b49eecb13fe0a81`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9baeb7fe6d1d1e66374d4e218194d3d58557c0e9dfa1859bfcafcdff895e73fc`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:1d4014ac9c8a6f913458caed124b7f4f6b16f8f2a0d478845ae3bc127e5c8f42`. Independent certificate: `sha256:72a1dbfa77d09de03ce057cf80b490ec72d08154ca662b2b452206c2ada8e128`. Engine external-validation hash: `sha256:315b66404a1399596fbec3150bb4939bd7744bbbe82829f615cb0a4bf1ef7230`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- dimensional-consistency-1: predicted same-dimension-relation-and-conversion; observed same-dimension-relation-and-conversion; exact match True
- dimensional-consistency-2: predicted same-dimension-relation-and-conversion; observed same-dimension-relation-and-conversion; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:392212011b4c71ff3f97b26c242ef0902ba9fab70134d09bef9ef0cf932248e3`. Isolation certificate:
`sha256:6a89028542df57707a8ecd0b5faa295e6c73723cc54c5e21888a2520770e37f9`. Custody certificate:
`sha256:dccfb22e9a9936c840afd362b99364c762e1d0f3bdb5ad64f5933ed5a9574bda`.

Registered source descriptions:

- Bureau International des Poids et Mesures: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- dimensional-consistency-1 from BIPM-SI-BROCHURE-9-2026
- dimensional-consistency-2 from BIPM-SI-BROCHURE-9-2026

Shared claim-record clause `PHYS-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favourable rows, free scales and fitted parameters

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:39af2267e4fd51a9bbe0168bf50283d7b1fc20b31112b47a57c247b76b00b8e7`; engine receipt
`sha256:2e23d3f162cf66416ed05f77336bb9a95254004d09fc44e51e65d2f3b5102984` at `receipts/engine/model_admitted/SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001-2e23d3f162cf6641.json`; empirical-validation hash
`sha256:7eac9d18dc7b35ed01539bc019da57e224232ee45edbe10bb2edfb9dc268e007`; measurement receipt `sha256:392212011b4c71ff3f97b26c242ef0902ba9fab70134d09bef9ef0cf932248e3`.

13. Exact source-boundary growth measurement discriminator

Claim identity: `SFT-PHYS-MEAS-BOUNDARY-GROWTH-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. After a structural boundary-growth law is independently derived and sealed, a complete source-boundary experiment checks it without a fitted exponent or logarithm by comparing exact positive distance and response ratios, generating every positive exponent by repeated composition, retaining the complete compatible family and closing all later candidates only after a monotonic strict-excess certificate. The observed family cannot supply or alter the structural law.

> *One source-bound exact ratio experiment, complete response uncertainty, positive successor-power enumeration, all-compatible selection, monotonic closure, post-seal custody and complete trace.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-PART-EQUIVALENCE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-FIELD-SOURCE-RESPONSE-001
- SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001

Generated grammar and closure boundary. Generate the complete product of source carrier, distance comparison, response comparison, scale, candidate generation, selection, stopping, target access, evidence record and extra-rule forms. The declared exact boundary is: All finite exact procedures for inferring a positive source-boundary growth exponent from two or more registered distance/response observations under admitted conserved-source dilution, including complete uncertainty support and a depth-independent stopping certificate. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `one-retained-source-boundary__exact-positive-distance-ratio__complete-positive-response-ratio-interval__same-source-ratio-cancellation__positive-successor-power-family__all-interval-compatible-exponents__monotonic-strict-excess-certificate__postseal-custodied-observations__complete-ratio-power-decision-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
source	one-retained-source-boundary	unbound-response-values	Unbound responses may come from different sources or preparations.
distance	exact-positive-distance-ratio	floating-or-signed-distance	Floating or signed distances are outside the exact positive proof domain.
response	complete-positive-response-ratio-interval	point-answer-without-uncertainty	A point answer discards unresolved measurement support.
scale	same-source-ratio-cancellation	fitted-source-amplitude	A fitted amplitude is a free target-derived parameter.
candidates	positive-successor-power-family	borrowed-or-fitted-exponent	A borrowed exponent selects the answer before measurement.
selection	all-interval-compatible-exponents	best-fit-or-first-match	Best fit or first match can erase tied compatible alternatives.
stopping	monotonic-strict-excess-certificate	fixed-search-depth	A fixed depth can hide a later compatible candidate.
target	postseal-custodied-observations	target-readable-method	Target access can choose the exponent rule.
record	complete-ratio-power-decision-trace	reported-exponent-only	An exponent alone cannot reproduce generation, comparison or closure.
extension	no-extra-rule	free-dimension-rule	An added spatial model can force a desired exponent.

Base and successor. The closure proof is sealed by derivation hash `sha256:95eec30df96776e0b54738d1eb8eedabb3ceeccac8b1c7d7c95aff463c93039e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form missing the first preservation; observed reject the generated form missing the first preservation; receipt `sha256:49c0e9ad603063b7a798ce56d84af7fc98dd24c46fca8128895246b531f8598d`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:adebcc98ce249122d2a47909ec040beba1db9e569667015df572c019d289d310`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:cc21d02d3416b56c4207d23b16300b4c0e88bcda3089b0151c9f588a8ca1adb8`.
- `boundary`: passed; expected reject excluded capabilities, values or authority paths; observed reject excluded capabilities, values or authority paths; receipt

sha256:47bea616cdeb5da7ba6914b5f7274f3af8f89013addec4dab1c9d32208381f9f.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:06b030de0896b998ea78ae969349c43e6a07eac99d60df3c5a93083eb0c83ade. Independent certificate: sha256:e6377e2f546a60825cb838c9a3b2ed8969bf99643ecb97fa0690dca5eddc70a3. Engine external-validation hash: sha256:4706a2ba009eee6ef2157f511d36fb5722ac65a2105912bb8d28382a03091be5.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S003 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- logarithms, irrational exponents, floating fits and regressions
- semantic numerical zero and negative proof magnitude
- a measured or fitted dimension used to select an inverse-power law
- target access before the complete method seal
- omitted measurements, uncertainties or compatible exponents

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:6b6069bb58cce5f30869ea2ed0cc6228b1809295b42c454f51af592277d61059; engine receipt sha256:b2d8360be10ba4be5664b59ee4624d91d909962f83927b3bd98d4caf5b02e43b at receipts/engine/model_admitted/SFT-PHYS-MEAS-BOUNDARY-GROWTH-001-b2d8360be10ba4be.json; empirical-validation hash None; measurement receipt None.

13. Mechanics Dynamics

Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace.

14. Physical event and generated change

Claim identity: SFT-PHYS-MECH-EVENT-CHANGE-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A physical event is a complete observed Fold state, and physical change is the source-bound transition between two such states; neither an unobserved background coordinate nor an answer-only scalar is required.

> *The minimal mechanical event/change law is a pair of complete observed states joined by one held, source-bound transition trace.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__state-bound-event-transition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	state-bound-event-transition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:1a36c712b3094ec644072cef6b9c6ef525c853a2c0149e7812d5b204f5a8f7ca. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:e5732acdd20a744295dc44bc619a57918fbdc6192e1f90bc424ef81b4205556a.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:7602df232de970b2c8237b10af653597ff2791df45f2ce143274cb8982ec8c9b.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:e6d4253c7ea6e792d78e6130fd2e77d803da3e2aa1130caf377680e0067bcede.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:d3bea2413cac687ffbde7295267a5c87d29d7607d2d9d6d3f0081d7ef424c1f8.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:6211d85755a0d738c265c26c686df476c122fb10f5751a8cfd2fab55c00e3458. Independent certificate: sha256:8362a480031dc0c7b73082a0266fca0f1d8ca29bf509f8f4c214b3b17c4964d1. Engine external-validation hash: sha256:9edba138c531c95a578d69d905b2a7407bfbfa5057ea837bbe6ee5b91449b47f.

Shared claim-record clause PHYS-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- event-change-1: predicted observed-state-and-transition; observed observed-state-and-transition; exact match True
- event-change-2: predicted observed-state-and-transition; observed observed-state-and-transition; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:a9ce5915467e496b2936b94024f4dbaae78a612e9773845318f37ca9a2f0eb2c`. Isolation certificate: `sha256:74a609a3a27de5b6b2ccd4f6be95766a1840029eaa139500bc3a7aadce92d6df`. Custody certificate: `sha256:ba790f6c6204308f8e2b5471a7a2a8df92a93ffb3ef841898768d38978052094`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- event-change-1 from NIST-CODATA-2022-ALL-CONSTANTS
- event-change-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:79d66bbd9f9588a5728abeea96f506aa54156c0ebcd96639859b83c265d11492`; engine receipt `sha256:71e815aed3d0ecea7f5d5df4ee3d2037e5e3f6fcf5803e7ac66492ef724f9b9c` at `receipts/engine/model_admitted/SFT-PHYS-MECH-EVENT-CHANGE-001-71e815aed3d0ecea.json`; empirical-validation hash `sha256:690fdd95af9d9a2df787c48a170fb044f82e207a3d9f50046f974a2ba41b2c76`; measurement receipt `sha256:a9ce5915467e496b2936b94024f4dbaae78a612e9773845318f37ca9a2f0eb2c`.

15. Duration from ordered recurrence

Claim identity: `SFT-PHYS-MECH-DURATION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Duration is forced as the exact positive ratio between generated transition count and a held reference recurrence; absence of a transition is structural Empty One, never numerical time zero.

> *Duration is the exact positive part ratio of ordered event transitions to a reproducible held recurrence.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-EVENT-CHANGE-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__transition-to-reference-recurrence-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	transition-to-reference-recurrence-ratio	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:e2d69f7887f53f979eaf97804668295456b98e7565a2b7846f45e784bfc47851`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b4f9404fd9b9ea4af729a789bef9b65234be6798a6b44e21e1a5243775bd15d2`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:27378c512ce057e7da835630bcb0de56a2ebde6396583c756fd90916074215bc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ea92e1601fbcf473a19d93a380e15aa3d0c0a5380af04df94282e814cf88bfb5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b6acf90f77e4d49ede0f0893e8ece79af4dc9c899fe02e00cff04557a9729da6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a6a15210ce7ca178cfc461603f430df76b1dac99bf6357c9ea0f581f4e1c2ea1`. Independent certificate: `sha256:ea03284dc046e848bd8ea95022c7babbffea01cc12ffdf2cdada7ae9e9d18b69`. Engine

external-validation hash:
`sha256:7fa0e2dd90fde529c8fb9f5d499b616bbe1cefba8a5ec8ceb3ff7cd111b4cbe5`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- duration-1: predicted ordered-recurrence-duration; observed ordered-recurrence-duration; exact match True
- duration-2: predicted ordered-recurrence-duration; observed ordered-recurrence-duration; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:f0ef7a1aea98a4281d912c5101a03dcd174b9ef83ba44954cfa7aa1998cc2b56`. Isolation certificate:
`sha256:9d149475e8506c37ff1be6c66bdd1a0be79162f4cacc5f32020ebbf5ac8196cd`. Custody certificate:
`sha256:2ee01dee864a5b5f78d76008e4a22fd55780a059498256a4364b8a0fb960eef8`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- duration-1 from NIST-CODATA-2022-ALL-CONSTANTS
- duration-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:2a5c7b05ba00af3c7a8f39c0b7cbfac62428fabfa64e92417f32ea07ea758fde`; engine receipt
`sha256:90b0ed7f1d52601eef77629cc3c1502f2bc9d9c169a5421c06d33d0c6334b649` at
`receipts/engine/model_admitted/SFT-PHYS-MECH-DURATION-001-90b0ed7f1d52601e.json`;
empirical-validation hash
`sha256:5908fe29650d9fe5516898d0f431984312db346ee1ccbcdb5c194fbd1c2fe8cd`; measurement
receipt `sha256:f0ef7a1aea98a4281d912c5101a03dcd174b9ef83ba44954cfa7aa1998cc2b56`.

16. Location, separation and held displacement orientation

Claim identity: `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Mechanical location is a relation to generated reference support; displacement pairs two such relations and holds which endpoint retains the orientation rather than introducing negative length.

Location is reference-relative support identity; displacement is the exact paired separation plus its held endpoint orientation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-DURATION-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__reference-separation-with-held-orientation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`reference-separation-with-held-orientation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0852c61a138670944d99a071689cea3ab6ac9a01a3d23edb59b49852298c16b7`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:546b51f9b2e762bbd20eb2c8f6a9107e39d7be62c1269a1671ba0ada983d4e33`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:29579f288f2f858a4a2927b99128d64207896f55de5c193bc2c0b72278d69f26`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:00b0e58a2e79c26a3ead80fb8930d4fff51228d95a96ddd56183558069228f91`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7ffe334e6b45268ce30a65f8a96caa4cb7394c356ece583f159260a398aa7448`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ef7e7c152cab399048000402b884130d8b549982d2d4f4457b8f9d6079cc33b4`. Independent certificate: `sha256:74f4571e92dce781927dbbf696f483603323a18a90101594093db7831e6317ea`. Engine external-validation hash: `sha256:0c55b6fa1981d90c92eb82ae475e9f9cd659fd5e2b74d9d83357fa93915fcf89`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- location-displacement-1: predicted reference-relative-oriented-separation; observed reference-relative-oriented-separation; exact match True
- location-displacement-2: predicted reference-relative-oriented-separation; observed reference-relative-oriented-separation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:e65e6ce09de7a52095ba0b446ff09400a9ad6d4143f4bfa1f1df94b50b24bfde`. Isolation certificate: `sha256:885c32cc2bb2a796c0b851e752540471565da983084a2c0c4f267478fbedf88e`. Custody certificate: `sha256:aa5c7260c9d2c56444168a098508693124c618eea6a7abc3a135304b0f5bc6a9`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- location-displacement-1 from NIST-CODATA-2022-ALL-CONSTANTS
- location-displacement-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:18e76ad65ab917486289ec1492042836a8cc4d048b3f2596e6b747becb03560f`; engine receipt `sha256:fc78abed974ceb7306725b8ac1fa140871e568dcf9d8f3f7119dd76138f084ca` at `receipts/engine/model_admitted/SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001-fc78abed974ceb73.json`; empirical-validation hash `sha256:0d89cdef832d7d14e6464d1653e3615d35064929a3ff3f374b7998b23d83a2ec`; measurement receipt `sha256:e65e6ce09de7a52095ba0b446ff09400a9ad6d4143f4bfa1f1df94b50b24bfde`.

17. Speed and oriented velocity

Claim identity: `SFT-PHYS-MECH-SPEED-VELOCITY-001`

Shared claim-record clause `PHYS-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Speed is forced as exact generated separation per reference recurrence, while velocity additionally retains displacement orientation; no signed magnitude enters the proof.

> *Speed is the exact positive separation/duration relation; velocity is that relation paired with held orientation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__separation-per-recurrence-with-orientation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`separation-per-recurrence-with-orientation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:f4f7dd67abd6da8bee282293faba3321f86e1a2b51b969bf9771109d9328b6ac`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

`sha256:857318859f939a1955cdc6e0ddb44e43ccbf563b0898011f7b1fb900a4599267`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:df735e60db928f0a532c8668596c46cb7368b0c6acb8af1cd32c744bbdc6116c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:521586076d2e04c0f020770696dfb7737aba2718e4e5eb467d7d875639871c70`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8a2dcbc1ee2621c044d84c1c0fd4a9d60bd6265d26eb03a28113d5b0bc332c6b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5f8459771e9c081691badd81c203fd3d4e2cf19ed97bfffe70892af1881fd68d`. Independent certificate: `sha256:4c406b62a112bbf68a3d9bd28663090c1890c69792279762418ab647e300b45d`. Engine external-validation hash: `sha256:6bf177896851043c140d312204ea38ff3eb19fad5b365f45762658fa4ef3cf84`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- speed-velocity-1: predicted separation-duration-oriented-rate; observed separation-duration-oriented-rate; exact match True
- speed-velocity-2: predicted separation-duration-oriented-rate; observed separation-duration-oriented-rate; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:41c78bac512c2bcd92e13f20db166fb10ef74a8cdba9c63ac3c7b941f39bafc8`. Isolation certificate: `sha256:eb7f5210146e0b01b27900a2dbd39c05346ef9312fadf1c0f38284592e834591`. Custody certificate: `sha256:539ad93ec944a59bd63c0db5136dfae1abd3986beedc9a8964eddcd7fb210448`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- speed-velocity-1 from NIST-CODATA-2022-ALL-CONSTANTS
- speed-velocity-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:8a7105ca9025bc8a0afedf46e85fde1de9da7192ef4d817a8612c4b51e0d9db1`; engine receipt `sha256:032489b9c3a0c26462074860c3419af02d49fb9fcc4ba4384b7f0807881d655e` at `receipts/engine/model_admitted/SFT-PHYS-MECH-SPEED-VELOCITY-001-032489b9c3a0c264.json`; empirical-validation hash `sha256:798bdaf2150cb261eec3b7766a4f6e2d5d8cd6310d041fce693234d26d28ea86`; measurement receipt `sha256:41c78bac512c2bcd92e13f20db166fb10ef74a8cdba9c63ac3c7b941f39bafc8`.

18. Change of velocity and acceleration

Claim identity: `SFT-PHYS-MECH-ACCELERATION-001`

Shared claim-record clause `PHYS-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Acceleration is forced as the exact oriented change between two velocity carriers per generated recurrence interval.

> *Acceleration is the held-oriented exact velocity difference per exact positive duration.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-SPEED-VELOCITY-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__velocity-change-per-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`velocity-change-per-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:77cc93f7aa10ead6c95041d075574624ddaf737a6efe82e95c34f2477e9c613c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:eaa0e0d413902b77efec640ce3898f5916bf70effd6ea8b3ec11d4c59b3afe01`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b6f3db19eb18dbd89342011c6163d5cf7964513f4ed912afd464453f0fa87171`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7ef1899a7ad943756898dbdd1b19194c78cd65367b50c3d006cc0d9073dd043d`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b3658d7236963e88665c6e90bc2f701eff616dd0df6cbc3f2de75eff96476e9b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a780de7d431097d31f0dfdd4e98b4f58f296084b8b5734e3fa3ff9ada31659ca`. Independent certificate: `sha256:6072fa5ba6bdf90653743a0fb99a02994860444b53ea4a2ab965068bdfed3ec6`. Engine external-validation hash: `sha256:f026ece1ce94ff741c81ccce64bf8e8aa86e16bd2f0032e2fa581461a4d15b29`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- acceleration-1: predicted oriented-velocity-change-rate; observed oriented-velocity-change-rate; exact match True
- acceleration-2: predicted oriented-velocity-change-rate; observed oriented-velocity-change-rate; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:a26fd2df5d7064632563a6471f7704b09d6c253d649adc6e9f2e9ea390ca7373`. Isolation certificate: `sha256:3c057af2d9b83f41dbc0796369846175294d06ba4e6b73b588a8ff4eb85fa08d`. Custody certificate: `sha256:f78edf2252b82084bfe67a39cc6f5a0d90d3d8ffd45eda0ae35bbeb62cbd4d31`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- acceleration-1 from NIST-CODATA-2022-ALL-CONSTANTS
- acceleration-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation

- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:7404e80b95d502807652bb23457062421c15cab94e640aeb188cc7ef97891024`; engine receipt `sha256:3726d5e4e02d4b6ab8e04a7dbc2ed1c36a07fccdcd50fb44a508a41ddfa27ea6` at `receipts/engine/model_admitted/SFT-PHYS-MECH-ACCELERATION-001-3726d5e4e02d4b6a.json`; empirical-validation hash `sha256:5b679b14cbeabc9f18ff49f4478616f9c4e7100164cce87b12739aeaac3c5849`; measurement receipt `sha256:a26fd2df5d7064632563a6471f7704b09d6c253d649adc6e9f2e9ea390ca7373`.

19. Inertial persistence

Claim identity: `SFT-PHYS-MECH-INERTIA-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. When no external transfer trace enters a closed state path, the complete motion carrier is forced to recur unchanged; change without a source would erase provenance.

> *A mechanically closed path preserves its complete velocity carrier from one generated event to its successor.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-ACCELERATION-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-trace-motion-persistence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`closed-trace-motion-persistence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:512bad3ceff5341aa497a8cb89112a3b35cefc413423b5aa0d515d8472bc99e1`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a9b00ca9f70199ff8a9224e7859716a9e3f719eb5a98bf3c81e3babbe0ea5ddb`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c304027d52c9d424f878fbcaed897116ce472297a60b57537bec1fdba21f2932`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:de0614da1cdb04608b176111e85876fe7c84ce21a8cda604ea6681bf5518271e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2edbb86bf293bad3273f3cb81facff99276ac4a6ba047c3b67f6035a9ea89464`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c75cc345554614e524cddf65683ede97bc16584472df14533e1bd20d8d035aa4`. Independent certificate: `sha256:c5aa0cab1a8a9aee2e9064bcdc05a566b64d977b6313cf2ae22f349ab41ad3aa`. Engine external-validation hash: `sha256:ec721aa1ab369a19cc92a54b2ab5aae74d0a8051bb3fd12acfa9653534289b1b`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- inertia-1: predicted closed-motion-carrier-persistence; observed closed-motion-carrier-persistence; exact match True
- inertia-2: predicted closed-motion-carrier-persistence; observed closed-motion-carrier-persistence; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:044084bbacba9c5a214e31851eb27431ba8ff84432b19ae2587c812b35c8919f`. Isolation certificate: `sha256:fb86c19e47c5c78506438cc24a61b4ed91be08600bd2ab7c795fe6f986336274`. Custody certificate: `sha256:8a2ce8f51d928f582aa1c4ce1714100176334bd810398bf1eaede28abe0a2994`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- inertia-1 from NIST-CODATA-2022-ALL-CONSTANTS
- inertia-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:399b0ab36d79ae5b46497f591156fe5f4019d66000df4b9060afa1bada0d4f5b`; engine receipt `sha256:b87863a7c28e6cf7ed9716bac77c53072bb2941651c0368fb542e1b8bb153bc8` at `receipts/engine/model_admitted/SFT-PHYS-MECH-INERTIA-001-b87863a7c28e6cf7.json`; empirical-validation hash `sha256:c8bea965b7fa84f8cd636febdc8f2db4ab496511224b8171153c0cd310133f64`; measurement receipt `sha256:044084bbacba9c5a214e31851eb27431ba8ff84432b19ae2587c812b35c8919f`.

20. Momentum carrier and transfer

Claim identity: `SFT-PHYS-MECH-MOMENTUM-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Momentum is forced as the complete pair-cell composition of inertia content with oriented velocity, because each inertial part must retain its motion label.

> *Momentum is the complete exact product of inertial content and oriented velocity, retaining both source coordinates.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-INERTIA-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__inertia-velocity-pair-cell-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`inertia-velocity-pair-cell-composition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:85cb25d79a88917234d4d9d0a6e21d944d502975d8d480fd8a1122c5346e9936`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e901ba60d3e54a104a96a0940b00151d67825f35111ab394275e3d7374237878`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ab52c54f3cc0dbede483e23d6b4e1eb6c27884892fc23836814945f13ae4daff`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f601ea2f1fc531758f7a20059bf9b4252cf59b4f3d03d16d5b122071afb39aaa`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:428e2d96ef2edbc2a6e12d1f215ef82164580a1670516a026448663b1b4e2cdf`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d11aff6c886dca731eb6adf5548b7a409bbb7b62d19340b7b6e43f37176d643f`. Independent certificate: `sha256:002be5aa14d8977be1ca48fd14c295a6980caca97a50b3d57aefcac50c3018e9`. Engine external-validation hash: `sha256:a885abfdc863b8422c60e76f94f2c351abb3ecf3fe30f6506055030642f1e1c8`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- momentum-1: predicted inertia-times-oriented-velocity; observed inertia-times-oriented-velocity; exact match True
- momentum-2: predicted inertia-times-oriented-velocity; observed inertia-times-oriented-velocity; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:e78f68d66b1d91f2a93a6866ac90757d91e22ed580c13ba8a813af7bd971f797`. Isolation certificate: `sha256:50d9955286c2d3a07579d16eeb3ff6c4d9a1dfd3c5d13d8e0eb930fa1f6eb76a`. Custody certificate: `sha256:57ac87465225c15ddf6e954f8fc600fd0d3ba154baa148fef5bbfc9485f52e47`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- momentum-1 from NIST-CODATA-2022-ALL-CONSTANTS
- momentum-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:959008334c9be67d184b51ec26234c33f9e008dd00d778ddb6b1f585164710a7`; engine receipt `sha256:9c67363fe5ef6783145dea27a49321411c8b0f3190ec67723878d5a5b277181c` at `receipts/engine/model_admitted/SFT-PHYS-MECH-MOMENTUM-001-9c67363fe5ef6783.json`; empirical-validation hash `sha256:c6f65f6134d9ad120e288afba214f03602b1e6b14e99f3db387d9e5c6913d200`; measurement receipt `sha256:e78f68d66b1d91f2a93a6866ac90757d91e22ed580c13ba8a813af7bd971f797`.

21. Force as constrained momentum transfer

Claim identity: `SFT-PHYS-MECH-FORCE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Force is forced as exact momentum transfer per generated duration; it is an interaction trace, not an independent primitive or fitted coefficient.

> *Force is the source-bound held momentum change divided by the exact positive duration of transfer.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-MOMENTUM-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__momentum-transfer-per-duration__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`momentum-transfer-per-duration`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:5413a8827b6b6d3987f6f4fe2b2ea9485a92067b4a771130956ecbc175f01033`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ca5ca4b9adef57f96c96e099214bb5382deca3bce3fe84c29d618e412193ed69`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:395f9b84abd524ce73fb49e305d8586bce8c48374b6c6700c29f7f7fac00579f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fb2cbd727c7e8484eb6e8c7e29b362cf9fcf2a70db9720fdf12b53c24dcab409`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:0f7f4dbc467ffa45c5df7ff588e3db7985e174a71dea3cf8ec581d8604f9bab8`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:78246d7975345c0e149c90b0c9f340eab96f6257e25dca8be0149661fcfdbc78`. Independent certificate: `sha256:a8a6c056543be591933aeea56583421668f6b22d065b6f658424427424e2a125`. Engine external-validation hash: `sha256:1fbe144ed79081344f0c38704b65b8c56888105567b4f55fc35df7fee84e6ce3`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- force-1: predicted momentum-transfer-rate; observed momentum-transfer-rate; exact match True
- force-2: predicted momentum-transfer-rate; observed momentum-transfer-rate; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:dba932821b1fd3bc068a0866c419881fc019a8ced7a0e658f1648407d40abec2`. Isolation certificate: `sha256:1493f95e964d4f550cc993153fc490f3f0cd13d382fb421dabf034ad2b024be6`. Custody certificate: `sha256:f1e0f75eb5c84f300b51c1091c311c69dcb30fbabe262daaefa2447b11485d60`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- force-1 from NIST-CODATA-2022-ALL-CONSTANTS
- force-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:115ccfe578fff87010e0e2dc9729057329dd0d1e425f32f20109867e9b87388a`; engine receipt `sha256:55c476d418d54f6bce48a366b936a40c26c77c0ca04ae64e7d2706f1ecc3169d` at `receipts/engine/model_admitted/SFT-PHYS-MECH-FORCE-001-55c476d418d54f6b.json`; empirical-validation hash `sha256:98477ffa21f8b5c7110d5755318dc730d8e79a1c6d2fd13ecc9d7281d4ca392a`; measurement receipt `sha256:dba932821b1fd3bc068a0866c419881fc019a8ced7a0e658f1648407d40abec2`.

22. Work, kinetic change and energy carrier

Claim identity: `SFT-PHYS-MECH-WORK-ENERGY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Mechanical work is forced as the complete composition of transferred force with oriented displacement, and energy is the retained capacity/change record transported through that interaction.

> *Mechanical work is exact force/displacement composition; the corresponding closed change is retained as mechanical energy transfer.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-FORCE-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__force-displacement-transfer-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	force-displacement-transfer-composition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b0eaeccd415b5a68f5dbc883d2bc713533bab9000b71eda18545531f3952a09f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3794d4cdc98beb8e04daa8a8cfd87e57160ee075d437dff47ce14644d7742d68`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:cf747a2b9ba20a84e0203edbf6ae22b06374bda74bff411269e92f9e67d065d0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e51e82a4d323c5ddbdbfb852b651c858a7f99f04750c48968496df4131a56b5c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dc17d7767ecf8016267312e2843b86c503602c4f7e3bd803771af774bff6ac7c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a6e40d4f25794f17748b16e95cf9cb1d526c6edf9c54c6fb062d5c82ea7d28c9`. Independent certificate: `sha256:829fed84a305929b33019b69cd3a241bb984b134118246ea556eab8751ca507d`. Engine external-validation hash: `sha256:d3cc7149703e4ec46ccdc45ad79f0abb8dbe617dd13652f486dcf1321b79de20`.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- work-energy-1: predicted force-displacement-energy-transfer; observed force-displacement-energy-transfer; exact match True
- work-energy-2: predicted force-displacement-energy-transfer; observed force-displacement-energy-transfer; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause PHYS-S007 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: sha256:aea377079e5954d4efb9994c12ecf1b2a640e0773708f03c56f5876c3555fa79. Isolation certificate: sha256:da973fb3372056232b70ca7bfe7fa8da51953365c355ccf991b8f1ac3ccb6910. Custody certificate: sha256:00023f1e2d07ec852209041f45d0b3ab96a0fb38ca323a802ceb106dc2108345.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- work-energy-1 from NIST-CODATA-2022-ALL-CONSTANTS
- work-energy-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause PHYS-S008 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:226c08b9a51b2cfe083704696061d91b7bc3d0f609cc1e8c0de2cb097392adb9; engine receipt sha256:2907d13e9cf208298ac2de4c0b22d6e17db2ca0df3aaf420d4dc9f27fa72d8c0 at receipts/engine/model_admitted/SFT-PHYS-MECH-WORK-ENERGY-001-2907d13e9cf20829.json; empirical-validation hash sha256:4ae8e5a23fed816dd0afb51002f8bca66fbc53259e498d169fcdc2e8ef572990; measurement receipt sha256:aea377079e5954d4efb9994c12ecf1b2a640e0773708f03c56f5876c3555fa79.

23. Power as ordered energy transfer

Claim identity: SFT-PHYS-MECH-POWER-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Power is forced as exact energy transfer per recurrence duration, retaining both the transferred energy record and its ordered interval.

> *Power is the exact positive energy-transfer/duration relation with held transfer orientation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-WORK-ENERGY-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__energy-transfer-per-duration__source-bound-proof-of-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`energy-transfer-per-duration`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:77ba6b9b6ca09908bbc61ca83255e0ec0b7dac4e51a6d096e9cdd4afa6c26639`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c59e356804c621da0ed401ab1c327333cd8cd8cf1547d3f2ff3a849bc5b34648`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d38e55fd04c92b8ecef7ee144cde3160c67c590cf0cff9e56407c8cf62dd1699`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:480e4a1e1dd88d9395d3b145f2054cc0d1ba29234dfc29b55d72618b6c99c803`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8cec23188e4bebe26dbd9eb0d22639e7ceecd2a87295c0d68d526e78ff1ee195`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0196c246d57db36313a70ba28369b6216fc75ea295113c42d450748df61a0a39`. Independent certificate: `sha256:2aad9d67ad4a812c9bfb23eefae164480cc0385d5f466eb3618bbc5bbc161b7f`. Engine external-validation hash: `sha256:d39251a5c104f4f57810961b454bf330e3b8a9a969bd3b378c77d6b590dfc6e2`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- power-1: predicted energy-duration-transfer-rate; observed energy-duration-transfer-rate; exact match True
- power-2: predicted energy-duration-transfer-rate; observed energy-duration-transfer-rate; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:1b1f8c4941b81dedfeac15bf134804dd8aeb25867d7cc5c3f84488f28fe5ef93`. Isolation certificate:
`sha256:a3bd1b84fb5660529234f6655158e552353c307415bec1d4f68994ca1076cad1`. Custody certificate:
`sha256:ae4362dbd2b030d3f194e02ae1cf7ac5256cf6af27d9b27e8528e39eed14098a`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
  https://physics.nist.gov/cuu/Constants/Table/allascii.txt
  (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- power-1 from NIST-CODATA-2022-ALL-CONSTANTS
- power-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:d1ab950bcaa251961d46eb5cd41c11db2529bf786c791068e592e901b35a9c49`; engine receipt
`sha256:986113fa723ac5827d698eab494abe9cddfb4054e4629d442d91a0a91f486ecd` at
`receipts/engine/model_admitted/SFT-PHYS-MECH-POWER-001-986113fa723ac582.json`;
empirical-validation hash
`sha256:42ccf10687f06be76c7b7523e765f07f6ee4895e205c2d67365cb1328d067a05`; measurement
receipt `sha256:1b1f8c4941b81dedfeac15bf134804dd8aeb25867d7cc5c3f84488f28fe5ef93`.

24. Angular motion and cyclic orientation

Claim identity: `SFT-PHYS-MECH-ANGULAR-MOTION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Angular motion is forced by recurrence on a closed ordered path; phase is the exact generated position within that recurrence and orientation is held by traversal order.

> *Angular position is an exact part of a closed recurrence; angular velocity is its held-oriented phase change per duration.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-POWER-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-path-recurrence-phase__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	closed-path-recurrence-phase	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:33de879cde318aaa45228596dc9e3c0b10aff73cae3d57be98863d59e2ea8409`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:91c99573f2c68caa9dab7a90084374ea7f5c76a7cca7604ad7a525068c2fdb20`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8dd93a8a2f3ff99e337aa19195da236de39d77d20d48b13c82e563779cb863ed`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f84f49d5b56f4473cb212dea76ecf6be895a890c59545ba4bb21c54789b4283a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d8c5f976a54b6985e8ae93a096e8912a7d1284e0d83a52c8b90f3e143682ea3a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:debdc18905c1598597bd9c1c8fc989351f5eba652d479efdadba090aae9d66b0`. Independent certificate: `sha256:6e1a782602dd60976895fa4f344d672e0edb2fd9ea6a87cdcf3ac298ce39a6bd`. Engine external-validation hash: `sha256:184ddcf04e5b3e4426731abefe521765d12f81cecef0b49e8f62807ee3edb5fa`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- angular-motion-1: predicted cyclic-phase-and-oriented-rate; observed cyclic-phase-and-oriented-rate; exact match True
- angular-motion-2: predicted cyclic-phase-and-oriented-rate; observed cyclic-phase-and-oriented-rate; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:bf27a6e422ab22b420f1c65a5e4853f47b7348a7868b1274f232e11cff455991`. Isolation certificate: `sha256:8e0982fe18aa273e01de6c503df9ba7e519f42ba97bb974e38ceeebbd2e931f7`. Custody certificate: `sha256:6d8bf7feac603ffcaefde7aa5c7cc9ff27987d808a9e842ff8490eb0bc1d1f5c`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- angular-motion-1 from NIST-CODATA-2022-ALL-CONSTANTS
- angular-motion-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:87112a209fc7b0aa88634956bfdd8cd1f85bf3c0e708ed2d7d11fd8ece1f9272`; engine receipt `sha256:f38ac92829c84d31410cbe3fe0d03a2ac5ff997ebdad4e41473498324f322db7` at `receipts/engine/model_admitted/SFT-PHYS-MECH-ANGULAR-MOTION-001-f38ac92829c84d31.json`; empirical-validation hash `sha256:5ae5fe71d72f4f9aa6e8341ba1fe7b768361c79e155327f3b2bcdd816806bf02`; measurement receipt `sha256:bf27a6e422ab22b420f1c65a5e4853f47b7348a7868b1274f232e11cff455991`.

25. Angular momentum and torque

Claim identity: `SFT-PHYS-MECH-ANGULAR-MOMENTUM-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Angular momentum is forced as oriented separation composed with linear momentum on a closed path; torque is its source-bound change per duration.

> *Angular momentum is exact oriented separation/momentum composition, and torque is its exact held change per duration.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-ANGULAR-MOTION-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__separation-momentum-cyclic-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`separation-momentum-cyclic-composition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:dd458740e09cefed16da594b44f60825050c4f0ff94d41bee7c89623609d2546`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:232302a821809c4550eb83205445d4fe0816d8317ffa55d47fc0bfe0b8e931a1`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3c2cbebde1987ac9cd48919d2c1eaa2540b6431c32946931533f660f03db1551`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6a9e927be1334cbfa1a2a3b9cb661f58799a598a850dd10c16a49c2a643d256b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:897299ac6dc97d35ba690f074a5d8ec1aa9272658f0f6f30e02bad9ea3ce39e5`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash:

`sha256:7d49cd2f4c425b1cd3ec28f07c2fb50a5bb3a919834be024fa9dd88a2345cac0`. Independent certificate: `sha256:b3f560d8405276173d31892428ef9dee06ff2c19dc0ee2b925d60ed592cc8913`. Engine external-validation hash: `sha256:4448cb52bb10c45fa27e6dc7ad88e3ef5d37cdbc88770eb5828e434780142b5f`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- angular-momentum-1: predicted oriented-angular-momentum-and-torque; observed oriented-angular-momentum-and-torque; exact match True
- angular-momentum-2: predicted oriented-angular-momentum-and-torque; observed oriented-angular-momentum-and-torque; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:d88f6fd9f0e2f82c406642f3a9ffcfcad5557b192168adb0a6ee67182e1f4961`. Isolation certificate: `sha256:0db8efa79754720969a0830eb264e816c44b711d0efce986e4b6adf7fd309310`. Custody certificate: `sha256:df3d3f540b855725cc0591b99fa0f3e201ac61a15da3b8bb3c561598cead6ef0`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- angular-momentum-1 from NIST-CODATA-2022-ALL-CONSTANTS
- angular-momentum-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:941673fa8bce066c0158b06fb9a13377e52b229fe15ea9b95ceec1c01ac55bbe`; engine receipt `sha256:b82a7f81b1904737363b439e1d85e573a0dd55fee628475ff139e764c767e28d` at `receipts/engine/model_admitted/SFT-PHYS-MECH-ANGULAR-MOMENTUM-001-b82a7f81b1904737.json`; empirical-validation hash `sha256:190508b180cd0414c67ad920d1976d24e6b253a45accf279000d6456bb7bf42e`; measurement receipt `sha256:d88f6fd9f0e2f82c406642f3a9ffcfcad5557b192168adb0a6ee67182e1f4961`.

26. Composite motion and centre relation

Claim identity: `SFT-PHYS-MECH-COMPOSITE-CENTRE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The centre relation of a composite is forced by complete inertial-content weighting of every part's separation, divided by complete inertial support; no constituent may be omitted.

> *Composite centre is the exact complete inertia-weighted location relation and composite momentum is the disjoint junction of all part momenta.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-ANGULAR-MOMENTUM-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__complete-inertia-weighted-part-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`complete-inertia-weighted-part-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:533402f2b58bf4cbb406e13d393a1fa3dd395403c47324f4a8bfcda4dfc352cd`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c511bfc1eab2cad1eea54aa1e729de50494692446e8d8ddc9a2b48351cbdf687`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8d19e87768f3fb35a39532c3a72e89d6cd2201521652f2d23d8f0e38a37f8cb0`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:787aeda12671300fc6430e467f1113186dce377b5dbb79024aed55a65dd28b4c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1635fa44abf739cc12b3b0b18df0b1a5523b163829a538419f567f9f0264709a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:377e3ae1335d43420ae3d96342bc263578b30766dcf3fc926cb2dcd400536d22`. Independent certificate: `sha256:228be1a5037de6c20a8321e3a332e0ebe67c77d0e395bd9e6233c13878024dcd`. Engine external-validation hash: `sha256:b2ac3603e0dffe6e7ed7ca9b034f9c9a12e9115dcef39f8f6db161c230e97ccb`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- composite-centre-1: predicted complete-weighted-composite-centre; observed complete-weighted-composite-centre; exact match True
- composite-centre-2: predicted complete-weighted-composite-centre; observed complete-weighted-composite-centre; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:f7e1ac6fd9d7ee469cbae3e3ebeb58f5154c87a75eb0aea75cf9e142e7e05b0d`. Isolation certificate: `sha256:4927b2fd3ee0cf7dd3bb1b4ae5a3a56926b689d146c64b1d39cb3ca18854772d`. Custody certificate: `sha256:99445963feb1200d6f5f2d670dc8d398c3e9c4f9713c9c93f3937ffc6d73601e`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- composite-centre-1 from NIST-CODATA-2022-ALL-CONSTANTS
- composite-centre-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:f55a92e34f8c58dd27bd6d48577b68f5ff951c38986ff7466fa83328040a533c`; engine receipt `sha256:ac98f7c19cb0a90e85cb8c728d0728482868cf79c54d6b8a861c8085a1ccf403` at `receipts/eng` ine/model_admitted/SFT-PHYS-MECH-COMPOSITE-CENTRE-001-ac98f7c19cb0a90e.json; empirical-validation hash sha256:c872e3521fc4998223ef53091ef3f41f2e7eff49ab6f1d980f58be198a4547d2; measurement receipt sha256:f7e1ac6fd9d7ee469cbae3e3ebeb58f5154c87a75eb0aea75cf9e142e7e05b0d.

27. Mechanical conservation under closed transfer

Claim identity: SFT-PHYS-MECH-CONSERVATION-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For a closed generated interaction network, every lost mechanical carrier at one node is forced to be held at another; complete internal transfer therefore preserves total momentum and energy support.

> *Closed mechanical transfer is a bijection on complete momentum and energy carriers, establishing exact conservation within the declared boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-COMPOSITE-CENTRE-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__closed-network-transfer-bijection__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	closed-network-transfer-bijection	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:b2d385b51c9418b997cecfafa9ad9d4538ffe5073d7eb47321402f6fc6a88f68. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:39dce981a6c1b9efa89ad292f1c4e993835d0c8c80e0a6e962373813517f9756`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f7a65471ba766cb9eaf7aad30613aa0c30042309ce32542d267a35a979d98972`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a3e760a6792fa1b929d65b81fa7a0783ce3d1f4239ee253cb6beeb11a987c5e2`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:87bf710c03122257d4572610f5ad7a93a11d22329112dfd09faebadb29c07d10`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4adb9788db5bc48fa2d40265858e1ffc9ba1e2076bdab833af6e35a6303a0c40`. Independent certificate: `sha256:b2b1a8e1de1c790784891d0fdf9fb10b142e7b8889317a4860d2d1013e0ea715`. Engine external-validation hash: `sha256:e901b7810a9a7e81f3795e90ad3f886e0bf0f25780c6902d7f7fc13679a6add1`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- conservation-1: predicted closed-transfer-conservation; observed closed-transfer-conservation; exact match True
- conservation-2: predicted closed-transfer-conservation; observed closed-transfer-conservation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:1cb45ef0361aa0fa416d570cc9bee9915df8865e23ff154fa1c60547531bfca4`. Isolation certificate: `sha256:4ae88e7bd5d844f6ffa3b370eca399959b1ab139c102f0aba63588534724d6ef`. Custody certificate: `sha256:f92ccc5439270a45d866803d7492a7ee407c32bf3288490034879e4c700c2bda`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- conservation-1 from NIST-CODATA-2022-ALL-CONSTANTS
- conservation-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation

- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:506581c9621edf7e0ee6f3cc8e9e42de5ecc55de5519ead7c6ac5f28205402f3`; engine receipt `sha256:7d02feaf423abbf05b62b69c7364f256f4a576c53437611ea6d22b3b06261b5b` at `receipts/engine/model_admitted/SFT-PHYS-MECH-CONSERVATION-001-7d02feaf423abbf0.json`; empirical-validation hash `sha256:06f19e46a4146f3bcf7a9278885e37fd3a1494f46e380eca23ac2cb0d2c74e82`; measurement receipt `sha256:1cb45ef0361aa0fa416d570cc9bee9915df8865e23ff154fa1c60547531bfca4`.

28. Constrained motion, restoration and oscillation

Claim identity: `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A reversible constraint maps displacement orientation to a source-bound restoring transfer; finite repeated reversal forces a recurrence whose period is an exact transition count relative to the reference recurrence.

> *Constrained reversible restoration generates a closed oscillation with exact recurrence count, held phase and conserved carrier trace.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__reversible-restoring-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`reversible-restoring-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a06e9e7154219b80b984ec4440c400635dce3e425c7e6cfee8cb03264db1964e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a5d131fd9eb84ebdd960da36959498a70e1f2399d40f5c72f20e550924de89b3`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f3c229eb365b1fa9e7a68da36f338651cc373ef16a836b6eb2870c663af016f2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:69687c3dd3eb6008dd8e5e5029c70cf19fa4d6f867c34b19d96acb6fa4947ec4`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:475a40edbb23e6c9929bd30fd6c8f763a41a1d41b622238548f14a9241f93bea`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:716edae5dedaef06202415df671c379fdc51dff7d5e60afa73d9879daf953e06`. Independent certificate: `sha256:733c4ba617c22f1f664f6efb16868a19824da109570788e2d2caf08eda91e44a`. Engine external-validation hash: `sha256:cc3b18c53385593ddf55c2c3c6d0f9a2eb107835870786781431ecd5e61b237e`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- constraint-oscillation-1: predicted reversible-restoration-oscillation; observed reversible-restoration-oscillation; exact match True
- constraint-oscillation-2: predicted reversible-restoration-oscillation; observed reversible-restoration-oscillation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:828cc6ab02762b5786b3e17fdebfd8e13d23dd6b85cec9961c7f902ac841732b`. Isolation certificate: `sha256:b4e30a4ac2d1da4fd36ace539986ed81e4d6babe38b6c603fac20f1c8250eaa3`. Custody certificate: `sha256:a91f4c7f54e34d13c5a7e9fc4e547bb6ce236c9494e742449b74174453aba3de`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- constraint-oscillation-1 from NIST-CODATA-2022-ALL-CONSTANTS
- constraint-oscillation-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavourable rows

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:6ad16151b2785bc80283b622bc44fdcb3d3f42493173b936d2c66d7f1ceff80d`; engine receipt `sha256:ad7387e41663cd1f57d403b58863dea2dbd6aa0f0f890a5e1807acaade0b9d46` at `receipts/engine/model_admitted/SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001-ad7387e41663cd1f.json`; empirical-validation hash `sha256:2337a51d1a9a5552c58d9f22fbea21965bf6aa04a964a67ed51b3760952d3ebb`; measurement receipt `sha256:828cc6ab02762b5786b3e17fdebfd8e13d23dd6b85cec9961c7f902ac841732b`.

29. Free-particle Fold phase and exact dispersion

Claim identity: `SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A free phase advances by the Fold of its exact momentum carrier, and that advance is identically the composition of two momentum advances on the One-cycle.

> *For p=1/4, Fold dispersion is 1/2 and phase 1/3 advances exactly to 5/6 by either one Fold step or two p steps.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-MOMENTUM-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for free-particle fold phase and exact dispersion, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__Fold-momentum-equals-two-cyclic-advances__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-continuum-object`	The object lacks a generated Fold support trace.
`dependency`	`admitted-V3-dependency-chain`	`asserted-prior-result`	A prior answer cannot act as a V3 premise.
`relation`	`Fold-momentum-equals-two-cyclic-advances`	`borrowed-dispersion-law`	The alternative loses a required conservation or recurrence record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected candidate neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	A missing carrier could silently select the answer.

Axis	Forced form	Eliminated form	Elimination reason
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:e218a34d4095d92e7e4bd3609e78cd3a4801b20aa23f06da280a56c29316eaaf`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:8c8abe048b2e0d3c2b40b72cdc0934f6e08fc5372dfb1ddcbd0541dc2a63c3a9`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:bf88bf71af8cd6e1866679adf017e7916401a9e88eb4637e05e30f7407a9597c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6bdefe43465be54912b1b709e50e7d7d8aebab95e2f94b1d23c7144448212750`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:59f5fc8c2264e7dfbc55bd587b4b388cd5eb669df4bdcf418b62391d62681335`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:6cdae2c27ddcf3db943cfa6964247b923e516823c119ec5ff2cd45bfd3e1050f`. Engine external-validation hash: `sha256:61029641439e12c80fc08ae3e3e27b5fa2934c47fb3b0c7a850b54ded166caaf`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:6b7380b390388840f329ff9ded65beef257b4c4ecbfabeab412ee16a5915ec2e`; engine receipt `sha256:d49c884c63339615c07f1b03712b44675c84df8cf0ebd84299aabc23645cba82` at `receipts/eng` ine/model_admitted/SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003-d49c884c63339615.json; empirical-validation hash None; measurement receipt None.

30. Potential evolution and additive exact phase

Claim identity: SFT-PHYS-DYNAMICS-POTENTIAL-EVOLUTION-003

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Sequential kinetic and potential Fold rotations equal one rotation by their exact positive sum, so a path phase retains every energy contribution without an imported evolution equation.

At phase 1/3, kinetic 1/8 followed by potential 1/4 equals one 3/8 advance and ends at 17/24.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001

Generated grammar and closure boundary. Generate the complete eight-axis product for potential evolution and additive exact phase, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-chain__sequential-rotations-equal-summed-energy-rotation__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	sequential-rotations-equal-summed-energy-rotation	opaque-phase-generator	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:65f3699a00efb37993d3d44d5970ca0ff9daebcddf3b0f4c5f5269499cc6b686. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:b3d149ad76829ca3c63ca9bb36fbb2895c435c84f96096f023ca47d74a787205.

- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:65156571a85663c80655f18aafe7018aea3a858313aa78c3400061d68e6cd9db`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3d6e70f6e2dfc72a57de3c7bd5cbb299b9958ea4a6499e2183508c7507dd4dae`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:9b6d171f23ae15b03c1c225f0282d8cee0585c8f23a800409bb214abc18475c6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:af7c79d30602d55d60be3b076df0c639d8b4c57fbfee86a611a015a01ed18027`. Engine external-validation hash: `sha256:374155a3e447823649137b289d74d06b0e75fb09bda3d4ed9755bf39b18cd610`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:8e8fed9795cb197cd9489b1041092ff2719682e2956aec072ee53eaa427d69b8`; engine receipt `sha256:2c0f29888cbfe276f5c1b03d27dc809046841d57b2857fe60a2f2f43c3a341f7` at `receipts/engine/model_admitted/SFT-PHYS-DYNAMICS-POTENTIAL-EVOLUTION-003-2c0f29888cbfe276.json`; empirical-validation hash `None`; measurement receipt `None`.

31. Stationary Fold spectrum

Claim identity: `SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. At every positive binary depth the unique half-spacing construction yields a finite exact spectrum with ground support half a grid step and every adjacent gap one whole grid step.

> *At depth k the levels are (2j-1)/2^(k+1); depth two is 1/8, 3/8, 5/8, 7/8 with gap 1/4.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-DYNAMICS-POTENTIAL-EVOLUTION-003`
- `SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for stationary fold spectrum, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__half-step-ground-and-uniform-whole-step-gaps__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-continuum-object`	The object lacks a generated Fold support trace.
`dependency`	`admitted-V3-dependency-chain`	`asserted-prior-result`	A prior answer cannot act as a V3 premise.
`relation`	`half-step-ground-and-uniform-whole-step-gaps`	`selected-or-continuous-spectrum`	The alternative loses a required conservation or recurrence record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected candidate neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	A missing carrier could silently select the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access would reverse the empirical direction.
`record`	`complete-trace-and-controls`	`answer-only`	An answer without trace cannot be independently reproduced.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:2fa52725256dfd8f695b8fc1515df473c6d18e1647fc2c9b6d094c0f5dc4bda2`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:2e43e78a4b885264b25e0875c6f9233f7800a358d7aa2942cf7553a0492cd847`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:45ec878430853f47134c7e072166bd943a0c29ba6cd6bb75a7b23142d13cd5a6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b9399927d3ba19ba0e13b5a8a72e6c74986a9bfde3722e6374591b82c4b0231d`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:f5a87e9e424a46e879175b4dca154d2c037f4d8158ddfbd0269ef327e85a724c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:5888a8892bcdfa8895d18ad3977772fe82c4ba3aef3084341d979048fd0b6e32`. Engine external-validation hash: `sha256:185a3f71d3be406f8edfde089410d1dee66304feb82945edd39b2d60faceb0e5`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:99a553afe97c3c1cd0e31440ea282f901e5e3b57e78eee6fbf3b573b83884dac`; engine receipt `sha256:b144cf1ec79dc2f2086a89f90d3781f4165344e1170161469f2e800823e584e2` at `receipts/engine/model_admitted/SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003-b144cf1ec79dc2f2.json`; empirical-validation hash `None`; measurement receipt `None`.

32. Odd-lattice all-region occupancy and exact recurrence vector

Claim identity: `SFT-PHYS-ODD-LATTICE-ALL-REGION-OCCUPANCY-TERMINAL-007`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every positive finite odd complete lattice beyond the One, the binary Fold is a permutation of its complete positive residue support. Every positive Fold successor therefore preserves the entire exact occupancy vector under every complete positive region partition having no more regions than members. The vector totals the lattice count and every region count is positive. This closes the all-region law and its recurrence vector at arbitrary positive finite depth, without importing a historical lattice size, chosen Fold count, measured vector or correction.

> *Every positive finite odd complete lattice is permuted by the binary Fold; every complete positive-region occupancy vector is invariant at every positive Fold depth, totals complete membership and occupies every region when the region count does not exceed the member count. The frozen V1 E3 run independently reproduces (32,32,32,32,32,32,32,31) for 255 members, twelve steps and eight regions.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-VACUUM-ODD-RECURRENCE-003`
- `SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-MATH-ORBIT-NUMBER-THEORY-002`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete product of every general, selected or target-assigned lattice domain; every exact, noninvertible or target-assigned Fold action; every complete, prefix or endpoint-omitting support;

every complete, selected or target-assigned region partition; every exact-vector, vector-free or target-assigned occupancy law; every successor, historical-only or uncertified depth closure; both target custody states; and both extension states. The declared exact boundary is: Every positive finite odd lattice beyond the One, its complete positive rational support, the binary Fold successor, every positive finite Fold depth, and every complete positive region partition with region count not exceeding member count. Structural absence is an empty support, never a numerical state. The generator produced `2916` named candidates and the decision support contains `2916` one-for-one decisions. Exactly one candidate survived: `every-positive-odd-complete-lattice__binary-Fold-permutation-on-positive-residues__complete-positive-lattice-support__complete-positive-region-partition-through-the-One__permutation-preserves-exact-vector-and-all-region-occupancy__every-positive-finite-Fold-depth-by-successor__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`lattice`	`every-positive-odd-complete-lattice`	`selected-historical-lattice-only`	Rejected by computed Fold predicates: lattice-domain.
`Fold`	`binary-Fold-permutation-on-positive-residues`	`noninvertible-selected-residue-map`	Rejected by computed Fold predicates: Fold-action.
`support`	`complete-positive-lattice-support`	`selected-orbit-prefix`	Rejected by computed Fold predicates: complete-support.
`regions`	`complete-positive-region-partition-through-the-One`	`selected-occupied-regions-only`	Rejected by computed Fold predicates: region-partition.
`occupancy`	`permutation-preserves-exact-vector-and-all-region-occupancy`	`all-region-claim-without-vector`	Rejected by computed Fold predicates: occupancy-law.
`depth`	`every-positive-finite-Fold-depth-by-successor`	`historical-depth-only`	Rejected by computed Fold predicates: depth-closure.
`target`	`sealed-before-observation-release`	`observation-readable-before-seal`	Rejected by computed Fold predicates: target-custody.
`extension`	`empty-extension`	`free-occupancy-correction`	Rejected by computed Fold predicates: no-free-extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:7428718697ab3ad78dcc7d9b4b4b1ac2e439f159d444ad22c956903f8deb3372`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject a nonpermuting residue rule.; observed The deliberate nonpermutation loses complete support.; receipt `sha256:047005037f2381c91895630630c1f6379614c36210bb6134ad9c99bb45e485f1`.
- `tampered_source`: passed; expected Reject a changed claimant source identity.; observed The changed identity differs from the registered manifest.; receipt `sha256:f9cc802fda9684abe9042aa128f74d5f9a2c098b1846a200fdb61bd404bf4775`.
- `tampered_artifact`: passed; expected Reject duplicate candidate identities.; observed The complete product has unique identities.; receipt `sha256:f8ab5596c700dead84f72de6044f4023b122da6a24c0a8e1bbb6833779ed2a8e`.
- `boundary`: passed; expected Reject pre-seal target access and a free correction.; observed Sealed custody and empty extension are forced.; receipt `sha256:90d112d709b62a1c043a10493f170533fec5e7b22a55c597854a55b009ec6d63`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:46c0f105c76584c7a483e0fb51e60842229e45a7edab165d4dd56d0956544621`. Independent

certificate: `sha256:964a744645d5e30ed8fbcd09732879d7b17a14e0ab40c432e80d31629cddfd3c`. Engine external-validation hash:
`sha256:0dea9b70fb2495b3535ac3ed38aad3ffe5393ba88a942bf71c6158c70a33cb3f`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- SFTOM-V1-E3-EXACT-OBSERVATION

Observed comparison records:

- Both frozen V1 E3 source files are retained byte-for-byte and were opened only after the V3 prediction seal.
- Released historical inputs: complete 255-member lattice, twelve Fold steps and eight regions.
- Complete independently measured recurrence vector: (32, 32, 32, 32, 32, 32, 32, 31).
- Every one of the eight regions is occupied; the exact vector totals all 255 members.
- The same complete vector occurs after the first, eleventh and twelfth positive Fold steps, directly replicating and strengthening the original measured statement.
- The V3 law is depth-independent and applies to every positive finite odd complete lattice and every complete region partition within its member count; the historical run did not select that law.

Falsification condition: Reject if either frozen V1 source identity changes; if the original E3 inputs, status, observation or acceptance rule are omitted or altered; if the independently measured complete recurrence vector changes with positive Fold depth; if any released region is empty despite region count not exceeding the odd complete-lattice count; if total occupancy differs from complete lattice membership; if the general odd-lattice permutation or positive-depth successor certificate fails; or if the historical target or result enters the claimant before its prediction seal.

Measurement receipt:
`sha256:d594ef374830c0bbaaa7fa872e2bc62d0e69e1559739e76bc3c0618b335f6140`. Isolation certificate:
`sha256:345afc73832ddb0cc00bd60c847d470bf61a2a413f5b740590726f59cc560cac`. Custody certificate:
`sha256:2f4f005522a7f137483c4a7872b7754cdbf20f082a870aaa74a422403ec663e3`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, certificate, candidate table, answer vector or survivor identity in the formal derivation runtime
- no historical source locator, lattice size, Fold count, region count or measured vector in the claimant
- no claimant-supplied expected survivor and no submission-controlled admission flag
- no numerical-null proof state, negative magnitude, irrational, imaginary, transcendental or floating proof magnitude
- no selected orbit prefix, omitted One endpoint, preselected occupied regions or target-assigned boundary
- no historical-only depth claim in place of the positive-successor certificate
- no target access before the prediction seal and no free occupancy correction after closure
- no execution or import of the superseded V1 source during V3 prediction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:bbd11bf580cae4c188a2924b2e35b24ae133985ae39d858462d1f54e9b8e6c8f`; engine receipt
`sha256:185a5dfb16ba14910625b0e2e94a894e7f422cf8dfe65f824ab4f9f5759b07fe` at `receipts/engine/model_admitted/SFT-PHYS-ODD-LATTICE-ALL-REGION-OCCUPANCY-TERMINAL-007-185a5dfb16b`

`a1491.json`; empirical-validation hash `sha256:bfd7812d12c91cd25d2c88b7060362652e29a4fd85dfa1f6cfc7ec0a741a9f52`; measurement receipt `sha256:d594ef374830c0bbaaa7fa872e2bc62d0e69e1559739e76bc3c0618b335f6140`.

33. Half-One pair synchronization and coupled-ensemble terminal law

Claim identity: `SFT-PHYS-COUPLED-ENSEMBLE-SYNCHRONIZATION-TERMINAL-007`

Shared claim-record clause `PHYS-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every ordered pair of exact positive Fold parts, moving both members toward one another by the same coupling share of their separation synchronizes them exactly if and only if the coupling is the forced half-One. The two equal moves then reassemble the complete separation and leave an empty structural residual record; every off-half generated coupling leaves a positive residual. An already synchronized finite ensemble remains synchronized under every common Fold successor. Arbitrary-ensemble convergence is not assumed: the complete superseded E4 recurrence is released and reconstructed only after the general boundary is sealed.

> *Equal paired movement synchronizes every exact positive pair if and only if the coupling is the forced half-One; off-half generated couplings retain positive residual separation, and every synchronized ensemble remains synchronized under common Fold successors. The frozen V1 E4 ensemble independently reproduces final coupling vector (7,7,1,5,5), half-One recurrence (12,10,7,5,4,3,2,1,1,1,1,1,1,1,1), and terminal common point four-sevenths.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-HALF-ONE-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete product of every complete, selected or target-assigned coupling domain; every paired, one-sided or target-assigned movement; every exact balance, unequal or measurement-selected relation; every empty-record, numerical-null or result-as-premise residual semantics; every conditional terminal, universal collapse or target-recurrence ensemble law; every general, historical-only or uncertified closure; both target custody states; and both extension states. The declared exact boundary is: Every ordered pair of exact positive Fold parts, the complete generated coupling alternatives around the forced half-One, equal coupled movement by a part of exact separation, empty structural equality records, every already synchronized finite positive ensemble and every positive successor depth. Arbitrary ensemble convergence is not assumed; it must be separately measured or proven for its registered recurrence. The generator produced `2916` named candidates and the decision support contains `2916` one-for-one decisions. Exactly one candidate survived: `complete-generated-coupling-alternatives__both-members-move-equal-coupled-shares__two-equal-moves-reassemble-complete-separation__half-One-gives-empty-residual-off-half-stays-positive__synchronized-class-remains-synchronized-under-successor__arbitrary-positive-pair-and-successor-closure__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`coupling_domain`	`complete-generated-coupling-alternatives`	`selected-half-One-only`	Rejected by computed Fold predicates: coupling-domain.
`paired_action`	`both-members-move-equal-coupled-shares`	`one-member-moves`	Rejected by computed Fold predicates: paired-action.

Axis	Forced form	Eliminated form	Elimination reason
`balance_relation`	`two-equal-moves-reassemble-complete-separation`	`unequal-free-moves`	Rejected by computed Fold predicates: balance.
`residual_semantics`	`half-One-gives-empty-residual-off-half-stays-positive`	`numerical-null-synchrony`	Rejected by computed Fold predicates: residual.
`ensemble_terminal`	`synchronized-class-remains-synchronized-under-successor`	`universal-collapse-without-condition`	Rejected by computed Fold predicates: ensemble-terminal.
`closure`	`arbitrary-positive-pair-and-successor-closure`	`historical-ensemble-only`	Rejected by computed Fold predicates: closure.
`target_boundary`	`sealed-before-observation-release`	`observation-readable-before-seal`	Rejected by computed Fold predicates: target-custody.
`extension`	`empty-extension`	`free-coupling-correction`	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:b334a0cce278506495097aab5288a05ffd2eba2da739e13ed7dfdfdc2a22396a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject either off-half coupling as exact synchronization.; observed Both leave the same positive residual magnitude.; receipt `sha256:978495b19ea6a78a542087b299beca05d63505e8c7526fdd172d28f492034301`.
- `tampered_source`: passed; expected Reject a changed claimant source identity.; observed The changed identity differs from the registered manifest.; receipt `sha256:f9cc802fda9684abe9042aa128f74d5f9a2c098b1846a200fdb61bd404bf4775`.
- `tampered_artifact`: passed; expected Reject duplicated candidate identities.; observed The complete product has unique identities.; receipt `sha256:c74f2eab8555206eb5f3a49ee2b048122a0dc767178e467e00a155138bf2c2e1`.
- `boundary`: passed; expected Reject universal collapse, pre-seal observation and free correction.; observed Only the conditional synchronized terminal, sealed target and empty extension survive.; receipt `sha256:72ba6bce9aa44e789fee9d4adcd9e58976960c86fe569366903328839152515e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b7eb1e3b89d31ce53ad4cf841df1ca561739e92dd7268fa100b3813e10ea17d0`. Independent certificate: `sha256:d5dd037ba8350aaa197d120cbefcb1cdf34c120fb6f2c84655111b2afae7c492`. Engine external-validation hash: `sha256:a818817af6843dd5ed1f9fbdb1c5c1b919968cd06876d35af4ee499b9eefc9e2`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- SFTOM-V1-E4-EXACT-OBSERVATION

Observed comparison records:

- Both frozen V1 E4 files are retained byte-for-byte and opened only after the V3 prediction seal.
- All five registered couplings and all fifteen region-count steps per coupling were independently recomputed exactly.
- Complete final coupling vector: (7, 7, 1, 5, 5).
- Half-One recurrence: (12, 10, 7, 5, 4, 3, 2, 1, 1, 1, 1, 1, 1, 1, 1).
- Half-One uniquely reaches one occupied region, first at step eight, and all twenty members finish exactly at four-sevenths.

- Off-half controls remain visible and finish at seven, seven, five and five occupied regions; they cannot be discarded to manufacture uniqueness.

Falsification condition: Reject if either frozen V1 source identity changes; if any registered ensemble, depth, region, coupling or printed final coordinate is omitted or altered; if the independent exact recurrence fails to reproduce all five final region counts; if half-One does not uniquely reach one occupied region; if its complete stepwise recurrence or terminal common point changes; if any off-half comparison is deleted; if the general pair synchronization or synchronized-terminal proof fails; or if target content enters before prediction sealing.

Measurement receipt: `sha256:26c258af8df7b47cb63990a322ad42259c7feae367af3364396472ff127ece79`. Isolation certificate: `sha256:b3e64fac2ad2e9c1a88c28ea514125b09d572e5faf258cb8ff120a4de5cd2ac4`. Custody certificate: `sha256:ed0a7d1410486928a7bc01d727fabdefc472a961848a815e8551df18a28728bc`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, recurrence, final count, common point or survivor identity in the formal claimant
- no historical ensemble denominator, member count, step count, region count or coupling vector in candidate decisions
- no claimant-supplied expected survivor and no submission-controlled admission flag
- no numerical-null equality state, negative magnitude, irrational, imaginary or floating proof magnitude
- no assertion that every arbitrary ensemble converges under half-One coupling
- no deletion of unfavourable off-half coupling rows
- no target access before the prediction seal and no free coupling correction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:5316e4ee59f1d5a1abdffb19d98346915438ebd1e23d0c6f72181eb94d3417a1`; engine receipt `sha256:c761476abce4bdf4d357022245a0d069f2ca5f30cb34ee917e8a2f7578a5c2d0` at `receipts/engine/model_admitted/SFT-PHYS-COUPLED-ENSEMBLE-SYNCHRONIZATION-TERMINAL-007-c761476abce4bdf4.json`; empirical-validation hash `sha256:8630bc2f8c43136cd9e26c429267b05260218a997dfc00c4c1877037ae9e42cf`; measurement receipt `sha256:26c258af8df7b47cb63990a322ad42259c7feae367af3364396472ff127ece79`.

34. Exact Fold expansion and information-rate carrier correspondence

Claim identity: `SFT-PHYS-LYAPUNOV-KS-CORRESPONDENCE-TERMINAL-008`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A complete m-label Fold step forces the same exact positive-whole carrier m for local-separation expansion and complete information-support growth; conventional Lyapunov and entropy logarithms remain external correspondence labels under explicitly retained hypotheses.

> *The exact Fold expansion carrier and complete information-support carrier are the same generated positive whole m at every positive depth; the two-branch external anchor matches carrier two, while analytic logarithms remain external symbolic labels and the entropy-formula scope remains conditional.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-SELF-SIMILAR-CONVERGENCE-002`

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`

Generated grammar and closure boundary. Generate the complete product of every complete, selected or target-assigned label domain; every exact, free or target-assigned separation law; every complete, fixed or selected support law; every common, independent or measurement-equated carrier relation; every exact, untyped or target-assigned information step; every exact-symbolic, imported-analytic or decimal proof boundary; both target-custody states; and both extension states. The declared exact boundary is: Every positive finite complete generated label count m of at least two, every positive Fold depth d, every positive local separation, complete m-label word support and its successor. The exact theorem returns the shared positive-whole carrier m; analytic logarithmic notation remains solely an external comparison label. The generator produced `2916` named candidates and the decision support contains `2916` one-for-one decisions. Exactly one candidate survived: `complete-generated-m-label-domain__uncast-local-separation-multiplied-by-m__complete-depth-support-m-to-d__same-m-carries-separation-and-support-growth__one-m-label-distinction-per-depth__exact-carrier-only-symbolic-external-translation__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`label_domain`	`complete-generated-m-label-domain`	`selected-two-label-domain`	Rejected by computed Fold predicates: label-domain.
`separation_law`	`uncast-local-separation-multiplied-by-m`	`free-separation-exponent`	Rejected by computed Fold predicates: separation-law.
`support_law`	`complete-depth-support-m-to-d`	`fixed-binary-support`	Rejected by computed Fold predicates: support-law.
`common_carrier`	`same-m-carries-separation-and-support-growth`	`independent-rate-parameters`	Rejected by computed Fold predicates: common-carrier.
`information_step`	`one-m-label-distinction-per-depth`	`untyped-information-increment`	Rejected by computed Fold predicates: information-step.
`analytic_boundary`	`exact-carrier-only-symbolic-external-translation`	`imported-analytic-proof-value`	Rejected by computed Fold predicates: analytic-boundary.
`target_boundary`	`sealed-before-observation-release`	`observation-readable-before-seal`	Rejected by computed Fold predicates: target-custody.
`extension`	`empty-extension`	`free-rate-correction`	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:7749cfe3453324ad469fe98967a5ab6c18491bdce3df2a0c6a3b6165b012fd28`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject a selected binary carrier for a generated three-label Fold.; observed Complete support and separation both return three, not two, as their multiplier.; receipt `sha256:37122691e3a624a1772bee38e44dfa1048c8ed6e88089da0aaa9da8e5173ce0d`.
- `tampered_source`: passed; expected Reject a changed claimant source identity.; observed The changed identity differs from the registered manifest.; receipt `sha256:f9cc802fda9684abe9042aa128f74d5f9a2c098b1846a200fdb61bd404bf4775`.

- `tampered_artifact`: passed; expected Reject duplicated candidate identities.; observed The complete product has unique identities.; receipt `sha256:c74f2eab8555206eb5f3a49ee2b048122a0dc767178e467e00a155138bf2c2e1`.
- `boundary`: passed; expected Reject imported analytic proof values, pre-seal target access and a free rate correction.; observed Only the exact carrier, sealed target and empty extension survive.; receipt `sha256:f84863dd7961828969186e8bb156becee345cb4d435c032810a5de3ac8c5b3df`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:86ac695164dfbe7c792ec43f22a6aa0d730e6234b3efeb74db1e986750e7d607`. Independent certificate: `sha256:e0b9b6a8b7567be6c502dabf0bc3b6ff21ca9fb455e58fbf92c1f6dd8a436a28`. Engine external-validation hash: `sha256:a03df497306791d51ed4f5ae3e756129bb5bce84a20220de3a3d6dedf067ee37`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PH2-EXTERNAL-LYAPUNOV-KS-CORRESPONDENCE

Observed comparison records:

- Both primary-source snapshots are retained byte-for-byte and opened only after the V3 prediction seal.
- The direct two-branch map records two pieces, absolute slope two, two preimages and two independent equiprobable output symbols.
- Independent Fold reconstruction gives complete supports two, four, eight, sixteen and thirty-two at depths one through five.
- Every support successor ratio and every tested local-separation multiplier is exactly the same positive-whole carrier two.
- The conventional Lyapunov and one-bit information labels are retained as external labels; no logarithm, irrational approximation or decimal enters the Fold proof value.
- The entropy-formula correspondence retains the source's hypotheses and is not promoted to an unconditional identity.

Falsification condition: Reject if either primary-source snapshot or the source record changes; if any registered map, preimage, distribution, output-symbol, rate-label or hypothesis-scope row is omitted; if the direct map's branch, slope and output counts do not equal the independently generated Fold carrier; if complete support does not grow as m to depth; if the expansion and information carriers differ; if the conditional scope of the entropy formula is universalized; if an analytic logarithm is imported as an SFT proof scalar; if a free correction is introduced; or if target content enters before prediction sealing.

Measurement receipt: `sha256:4acb4635259f330d2a4b434f5439ca3b27a0c4428d5864ce013e7468a4875ae0`. Isolation certificate: `sha256:f94d6de2c489b95bd239501cb442654b965e53b9a0f7e58cd0db12a85349ab33`. Custody certificate: `sha256:598f97ba9038e30b6b6308cf0ae403463b01a10a3825d47b6e1b2e9002c68c99`.

Registered source descriptions:

- ARXIV-1211.1234: https://arxiv.org/abs/1211.1234 (sha256:716c6b68b4755f47087747f67f4b2c4a7ed0ade30f77391a18129f2ad76de6c9)
- ARXIV-1004.3441: https://arxiv.org/abs/1004.3441 (sha256:ea560114c9cc196e18feb4536f63f28c292dadda465aafcb9f41ef12c20ac090)

Registered target identities:

- PH2-WITHHELD-MAP-AND-ENTROPY-CORRESPONDENCE from PH2-EXTERNAL-LYAPUNOV-KS-CORRESPONDENCE

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- external analytic values as proof scalars
- target-selected carrier

- free rate correction
- unconditional Pesin identity

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:d373bc7c4a05f5965eff26e64a699b1c1c82fe6d2a7fb82ab0d3d585ab765abd; engine receipt sha256:f0b74a8639f9e01be7b9adc31f83b4b24ed82f085e72c17736c38f030b5cae40 at receipts/engine/model_admitted/SFT-PHYS-LYAPUNOV-KS-CORRESPONDENCE-TERMINAL-008-f0b74a8639f9e01b.json; empirical-validation hash sha256:3c813f65cebb30e294350695b6163319a90d66523742ff9019a0c07abf1681d8; measurement receipt sha256:4acb4635259f330d2a4b434f5439ca3b27a0c4428d5864ce013e7468a4875ae0.

35. Exact normalised coupled-map synchronization threshold

Claim identity: SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For a complete m-label Fold in the normalised drive-response transverse channel, the exact residual multiplier m(One-g) uniquely forces g=(m-1)/m, expansion below, neutrality at the boundary and contraction above; half-One follows at m=2, without universalizing arbitrary topology.

> *For the normalized drive-response transverse Fold channel, the unique neutral coupling is (m-1)/m; smaller coupling expands and larger coupling contracts, with half-One forced at m=2 and arbitrary topology factors explicitly retained outside this scope.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-LYAPUNOV-KS-CORRESPONDENCE-TERMINAL-008
- SFT-PHYS-COUPLED-ENSEMBLE-SYNCHRONIZATION-TERMINAL-007
- SFT-COMP-DIST-SYNCHRONIZATION-001
- SFT-FOUNDATION-HALF-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

Generated grammar and closure boundary. Generate the complete product of every complete, selected or target-assigned expansion domain; every exact, free or target-assigned residual action; every forced, fixed or measured threshold; every exact, mislabeled or absent regime partition; every normalized, universalized or finite-only scope; every exact-symbolic, imported-analytic or decimal proof boundary; both target custody states; and both extension states. The declared exact boundary is: Every complete generated positive label count m of at least two, every exact proper positive coupling g, the normalized drive-response transverse channel with retained share One-minus-g, and every positive successor depth. Arbitrary coupling topologies retain their separate transverse eigenvalue factors and are not assigned this threshold universally. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: complete-generated-m-expansion__retain-one-minus-coupling-share__unique-m-minus-one-over-m-boundary__below-expands-boundary-neutral-above-contracts__normalized-drive-response-transverse-channel__exact-rational-carrier-symbolic-external-translation__sealed-before-observation-release__empty-extension. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`expansion_domain`	`complete-generated-m-expansion`	`selected-binary-expansion`	Rejected by computed Fold predicates: expansion-domain.
`coupling_action`	`retain-one-minus-coupling-share`	`free-residual-action`	Rejected by computed Fold predicates: coupling-action.
`threshold_relation`	`unique-m-minus-one-over-m-boundary`	`fixed-half-One-for-all-m`	Rejected by computed Fold predicates: threshold.
`regime_partition`	`below-expands-boundary-neutral-above-contracts`	`boundary-declared-contracting`	Rejected by computed Fold predicates: regimes.
`synchronization_scope`	`normalized-drive-response-transverse-channel`	`arbitrary-topology-universal-threshold`	Rejected by computed Fold predicates: scope.
`analytic_boundary`	`exact-rational-carrier-symbolic-external-translation`	`imported-exponential-proof-value`	Rejected by computed Fold predicates: analytic-boundary.
`target_boundary`	`sealed-before-observation-release`	`observation-readable-before-seal`	Rejected by computed Fold predicates: target-custody.
`extension`	`empty-extension`	`free-threshold-correction`	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:d4d7c155c258c90c031af9f9f9233b501169c1d6d3071ca4823ea462f845c32c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject half-One as a universal threshold for every m.; observed The three-label threshold is two-thirds and half-One remains below it.; receipt `sha256:5940cc08cb032c81f0309da13073f53f87d409b435d741e62abf26be88767591`.
- `tampered_source`: passed; expected Reject a changed claimant source identity.; observed The changed identity differs from the registered manifest.; receipt `sha256:f9cc802fda9684abe9042aa128f74d5f9a2c098b1846a200fdb61bd404bf4775`.
- `tampered_artifact`: passed; expected Reject duplicated candidate identities.; observed The complete product has unique identities.; receipt `sha256:c74f2eab8555206eb5f3a49ee2b048122a0dc767178e467e00a155138bf2c2e1`.
- `boundary`: passed; expected Reject universal topology, pre-seal observation and free correction.; observed Only normalized scope, sealed target and empty extension survive.; receipt `sha256:79e19bb1b8b82b78e5b0a7c3c903c8d1b91154640ddf4c15693314e496b9792b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e6757c5fd0f980763d0f09c3515997744542ec4afbc5b23dd80c11c27123527e`. Independent certificate: `sha256:b2d757623e5e3064d4c973f623b7a49707f1a16c41767e78e1f80c1dc5b6d66d`. Engine external-validation hash: `sha256:949cc876e33ca0f6b920ec5bc116371cebef67bd9c8d9268c2d56140f4278911`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PH3-EXTERNAL-COUPLED-MAP-CRITICALITY

Observed comparison records:

- Both primary records are hash-locked and opened only after the V3 prediction seal.
- The normalised zero-eigenratio direct case records the half-One lower boundary and the exact Fold reconstruction independently forces half-One for m=2.

- Below, at and above controls yield exact transverse multipliers four-thirds, One and two-thirds: expansion, neutrality and contraction.
- The complete topology-dependent lower and upper bounds remain visible; the normalised threshold is not assigned to arbitrary lattices.
- The finite 21-map, coupling 0.8, approximately 0.867 critical-alpha row is retained as an external held record, never a proof scalar.
- Non-Gaussian finite-time fluctuations and intermittent bursting outside the stable window remain adverse critical evidence.
- The NLM record retains the two-quantity linear criterion and its stated high-precision numerical status without converting an approximation into exact proof.

Falsification condition: Reject if either source or source record changes; if any criterion, map, coupling, synchronization, transverse-rate, threshold, topology, finite-example, fluctuation or intermittency row is omitted; if m(One-g) does not uniquely force (m-1)/m; if the half-One direct case fails; if arbitrary topology factors are erased; if an analytic or decimal target becomes an SFT proof value; if a free correction is introduced; or if targets enter before prediction sealing.

Measurement receipt: `sha256:db0b7caa12d6b5d1d29c5bdf3c00c359e01c20fcb44993ddea059c76733866cf`. Isolation certificate: `sha256:18c48a294f227845b481d1afc35bdff1e062ff767a6e6786ddad9369e7789b78`. Custody certificate: `sha256:41b3550dd4fd8c1e438f7c8e72e5831a1320a29b94b5a92264b6a466e6f66604`.

Registered source descriptions:

- ARXIV-NLIN-0504012: https://arxiv.org/abs/nlin/0504012 (sha256:6ec59cc2f85ed4d90ef3bd457b1f2877d92a4e10059548ab11297e363f588a4d)
- PUBMED-15447575: https://pubmed.ncbi.nlm.nih.gov/15447575/ (sha256:08ab034a50e7e6383669d3c6f607634c080cd741018628c4355eeeb8e142d8e1)

Registered target identities:

- PH3-WITHHELD-COUPLED-MAP-CRITICALITY-RECORD from PH3-EXTERNAL-COUPLED-MAP-CRITICALITY

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- arbitrary-topology universal threshold
- external analytic value as proof scalar
- measurement-selected survivor
- free threshold correction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:24af36a052d5a6685bccc0857a378159d968a48280bec0cee65548d64eb99c7c`; engine receipt `sha256:845ceed3f993a0523e1f01a5ce9375d469e426be12539941c5e5fd5130b9534f` at `receipts/engine/model_admitted/SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008-845ceed3f993a052.json`; empirical-validation hash `sha256:35c695aabf5966a39fe78dd6bf7ab75945549270786d974527a84c019ba4d2a4`; measurement receipt `sha256:db0b7caa12d6b5d1d29c5bdf3c00c359e01c20fcb44993ddea059c76733866cf`.

36. All-dimension orbital-restoration stability law

Claim identity: `SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Positive denominator ordering between q^(d-1) gravity and q^3 centrifugal response forces restoring circular orbits for d=2,3, exact marginality at d=4 and nonrestoring response for every d at least five.

Positive denominator ordering between dimension-forced inverse-(d-1) gravity and inverse-cube centrifugal response forces restoring circular orbits only for d=2,3, exact marginality at d=4, and nonrestoring response for every d at least five.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001`
- `SFT-PHYS-FIELD-INVERSE-SQUARE-001`
- `SFT-PHYS-VALIDATION-INVERSE-SQUARE-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

Generated grammar and closure boundary. Generate the complete product of every complete, selected or target-assigned dimension domain; every dimension-derived, fixed or target-assigned gravity dilution; every inverse-cube, dimension-matched or target-assigned centrifugal dilution; every positive-magnitude, signed-import or target-assigned balance; every positive-denominator, negative-exponent or measured outward comparison; every complete, selected or universal stability partition; both target custody states; and both extension states. The declared exact boundary is: Every positive whole spatial dimension d of at least two, every exact outward radius ratio q greater than the One, dimension-forced inverse-(d-1) gravity, inverse-cube centrifugal magnitude, and every positive successor displacement. Dimension one is excluded because the registered circular angular-momentum organization is absent there. The generator produced `2916` named candidates and the decision support contains `2916` one-for-one decisions. Exactly one candidate survived: `complete-positive-whole-d-at-least-two__inverse-positive-power-d-minus-one__inverse-cube-positive-magnitude__equal-positive-magnitudes-at-reference-radius__compare-positive-power-denominators__stable-two-three-marginal-four-unstable-above__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`dimension_domain`	`complete-positive-whole-d-at-least-two`	`selected-three-space-only`	Rejected by computed Fold predicates: dimension.
`gravity_dilution`	`inverse-positive-power-d-minus-one`	`fixed-inverse-square-all-d`	Rejected by computed Fold predicates: gravity.
`centrifugal_dilution`	`inverse-cube-positive-magnitude`	`dimension-matched-centrifugal-power`	Rejected by computed Fold predicates: centrifugal.
`circular_balance`	`equal-positive-magnitudes-at-reference-radius`	`signed-force-import`	Rejected by computed Fold predicates: balance.
`outward_comparison`	`compare-positive-power-denominators`	`import-negative-exponent-ratio`	Rejected by computed Fold predicates: comparison.
`stability_partition`	`stable-two-three-marginal-four-unstable-above`	`three-space-only-verdict`	Rejected by computed Fold predicates: partition.
`target_boundary`	`sealed-before-observation-release`	`observation-readable-before-seal`	Rejected by computed Fold predicates: target.
`extension`	`empty-extension`	`free-stability-correction`	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:167d4ec9cec1fbf7ca5a52db76919bb2d1c3bf11cec0074ae83c5c36f549c15b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject stable five-space orbit under inverse-fourth-power gravity.; observed Gravity falls below inverse-cube centrifugal response.; receipt `sha256:f687597bf5fcd7de0c8c037b98df2b6946078182352a98c83f7646ec777826a8`.
- `tampered_source`: passed; expected Reject changed source identity.; observed Identity differs.; receipt `sha256:7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83`.
- `tampered_artifact`: passed; expected Reject duplicate candidates.; observed All identities unique.; receipt `sha256:5f9c0ad545be40ca825517099110bae199281727e06768826c0280cdfae9b73b`.
- `boundary`: passed; expected Reject negative-exponent import, pre-seal observation and correction.; observed Only positive comparison, sealed target and empty extension survive.; receipt `sha256:76d0758795c16363c1a6e8956a090b883ca59eb0503a7800d3814fc4a6728c2c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:94985b6cb2ba2d7ec1a4b3d44c8d8329c6a25dbc957a1237c1299f3f9b3c47ca`. Independent certificate: `sha256:b98bad07fe05e51599604075232fa496c9a7f1d575797d04e6f8e9712ee09ace`. Engine external-validation hash: `sha256:3470c2ebbaf7aafc76fe89b5d351eb181d35029d0409f05501b1301bb3b44226`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- D9F-EXTERNAL-ORBITAL-DIMENSION-STABILITY

Observed comparison records:

- Both Cambridge derivations are byte-locked and opened only after the V3 prediction seal.
- The independent census retains dimensions two through twelve at outward ratios three-halves, two and five-halves.
- Dimensions two and three are restoring at every registered ratio; dimension four is exactly marginal; every tested dimension from five through twelve is nonrestoring.
- The successor proof extends the unstable result to every higher positive whole dimension because each dimension successor multiplies only the gravity denominator by q greater than One.
- The external inverse-(N-1) force result, N<4 boundary, potential-power boundary and strict-minimum criterion all match without fitting.
- No signed force, negative exponent, irrational, measured parameter or correction enters the Fold proof.

Falsification condition: Reject if either Cambridge source or the source record changes; if any force-power, balance, perturbation, stability, dimension, marginal, unstable-family, potential or effective-potential row is omitted; if the exact positive-magnitude census differs; if dimension one is silently admitted; if a signed force, negative exponent, measured fit or free correction enters proof; or if targets enter before sealing.

Measurement receipt: `sha256:0b1572fc86c9e9d4501efee787ab5388dc4991261d757828f419e19c65ccef09`. Isolation certificate: `sha256:cc3c2fd11a02259dbf08fb7cce0e023c9b2386e05bc2e71ff27254e41253c084`. Custody certificate: `sha256:15d1d6b2dea21be60367f45f2ec9001b11598445a30182d8edf34f330dfc4fc6`.

Registered source descriptions:

- CAMBRIDGE-CENTRAL-FORCES-CH6: https://www.damtp.cam.ac.uk/user/reh10/lectures/ia-dyn-chapter6.pdf (sha256:59de9d143492b1fdf2f1f0b96c9653d861822081f6ee57c7689b943e31210024)
- TONG-CENTRAL-FORCES-FOUR: https://www.damtp.cam.ac.uk/user/tong/relativity/four.pdf (sha256:2d999345dd33fce016498a1ed2dc5085f8e7513530f75dfeb94dfbff05aad44e)

Registered target identities:

- D9F-WITHHELD-ORBITAL-STABILITY-RECORD from D9F-EXTERNAL-ORBITAL-DIMENSION-STABILITY

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- signed-force proof
- negative exponent
- dimension-one orbit
- measured fit
- free correction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:b5f097071692fa729280a05aa9d9c264532f3ff28fda7bb7b689104e130556f1`; engine receipt `sha256:9a067d10ece75b8300bec3c77193649313dbc2fcc116f721bd0fef54544a433c` at `receipts/engine/model_admitted/SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009-9a067d10ece75b83.json`; empirical-validation hash `sha256:b545a3186cb862437c92f3d65e46e5570bdf71cbf7b8c2a028280e916f37d03f`; measurement receipt `sha256:0b1572fc86c9e9d4501efee787ab5388dc4991261d757828f419e19c65ccef09`.

37. Finite Fold symmetry, conservation and least action

Claim identity: `SFT-PHYS-DYNAMICS-SYMMETRY-ACTION-TERMINAL-016`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Exact Fold preserves every reduced odd denominator core; complete finite Fold symmetries preserve held invariant fibres; every registered conserved partition generates its preserving symmetry class; the unique dyadic Fold trajectory descends to stationary One; monotone positive descent telescopes to endpoint separation and no finite detour has smaller action.

> *Fold preserves the reduced odd denominator core of every rational part; complete finite Fold symmetries preserve their held invariant fibres, every registered conserved partition generates its complete preserving symmetry class; every dyadic Fold part follows its unique binary-rank descent to stationary One; every monotone positive descent path telescopes to its endpoint carrier; and no finite detour has smaller action.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORBIT-NUMBER-THEORY-002`
- `SFT-FOUNDATION-FOLD-DYNAMICS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-FOUNDATION-PART-001`

Generated grammar and closure boundary. Generate the complete product of support, symmetry, incidence, carrier, conserved-fibre, converse, action, path and extension forms. The declared exact boundary is: Every exact positive rational Fold part; the complete dyadic fixed-point basin; every finite generated state support; every bijection of that support; every registered exact transition/carrier relation and held fibre partition; every finite path of distinct adjacent states with exact positive ranked coordinates. The generator produced `512` named candidates and the decision support contains `512` one-for-one decisions.

Exactly one candidate survived: `complete-finite-generated-support__complete-bijection-enumeration__transition-incidence-preserved__exact-positive-carrier-preserved__odd-core-and-held-invariant-fibre__enumerate-all-fibre-preserving-bijections__sum-of-positive-oriented-step-magnitudes__dyadic-Fold-descent-and-positive-detour-bound__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`support`	`complete-finite-generated-support`	`selected-or-continuous-state-sample`	A sample or continuum premise is not a complete generated domain.
`symmetry`	`complete-bijection-enumeration`	`named-transformation-without-census`	A named transformation does not prove bijection or completeness.
`incidence`	`transition-incidence-preserved`	`state-labels-without-transition-law`	Ignoring incidence can turn a non-dynamics map into a symmetry.
`carrier`	`exact-positive-carrier-preserved`	`free-rescaling-under-symmetry`	A free rescaling changes the physical law.
`conserved`	`odd-core-and-held-invariant-fibre`	`answer-only-scalar`	A scalar detached from support has no conservation trace.
`converse`	`enumerate-all-fibre-preserving-bijections`	`assert-every-conservation-has-unknown-symmetry`	An ungenerated symmetry is not evidence.
`action`	`sum-of-positive-oriented-step-magnitudes`	`signed-or-imported-action-integral`	Signed continuum integration is outside the proof domain.
`path`	`dyadic-Fold-descent-and-positive-detour-bound`	`postulated-stationary-physical-path`	Postulating stationarity imports the desired principle.
`extension`	`no-extra-rule`	`free-Lagrangian-or-Euler-equation`	An imported functional or equation is an extra premise.

Base and successor. The closure proof is sealed by derivation hash `sha256:441f216f6c5c0b73c0db50445b2fea60e3796f38f54a4617f5471b65adaf08a5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:c993d211ebd09080e4cc3077d8fffee231d05f9ba2a5c0aa80ae292d546f70f5`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:bafd7588c69ca117953fded10f3b6e4f1f381737b86e925f69c249698ec85444`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:250b52554881e0d14d83e9b306f3ce1421a73e9366da9663f93d8509f54115a0`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:1839b0fae41ca54f4daa155e56b8d3a75ecc683bd62a3c856453df07a6719066`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4c5efd8efc82310605344c83bfaba79470f3632a90a8dda029a64fcac3cb6b37`. Independent certificate: `sha256:f22863557ec8d4f77008bfd44d42b3885d99c3a8eef7a89bba04d24d8cdae6be`. Engine external-validation hash: `sha256:b16add97905a64f9097085c2ca029a1cfc0c6806fbf49c5a65dbee7fff2c4d49`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported Noether theorem, Lagrangian, Euler-Lagrange equation or continuum variation
- no prior V1/V2 proof artifact, answer table or stored survivor
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no assertion that an arbitrary non-Fold physical system has the registered dyadic ranking
- no measured trajectory or target selecting the symmetry, charge or action form

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:8e275a82b73779b75647ea049b663e7e733f9318c4d1f8f793f20e7ec758ec72`; engine receipt `sha256:3c158a5e6f5ad82f89435d0b00c6542d1230e20706f5cdadad0db202864d4075` at `receipts/engine/model_admitted/SFT-PHYS-DYNAMICS-SYMMETRY-ACTION-TERMINAL-016-3c158a5e6f5ad82f.json`; empirical-validation hash `None`; measurement receipt `None`.

14. Fields Forces Waves Geometry

Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace.

38. Source, response and field carrier

Claim identity: `SFT-PHYS-FIELD-SOURCE-RESPONSE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A physical field is forced as the complete source-bound map from a localized distinction to every generated accessible response cell; a value detached from source and support is not a field.

> *A field is the complete finite map of a held source distinction through generated accessible support to source-bound response records.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__source-bound-support-map__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	source-bound-support-map	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:52c49f681981f8d5b6e02968c63af698f5036ab31292c8862d44eac2225f6ecf`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:041f78c0bd302fd7b80d81d704c363231a1d9f969d5afa0bb5616744aeeee376`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:744369e625fc3e621706b51f234a09906a2d26b6548cc4fa29f9d5610a8e1b41`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c6481452651e95b15b4bd1d40d946b6b1b6da41e0ce684ca8ccf6b65a4b31b95`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:33b67fad6c33cc5a25ce4f9fd87d6234a1f7c5112d3a4ecdd2f6ac1fb376d3bd`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:1eb0754b3023b1f21e490d51ab8ccea125ba60c9247fc197cbd4fa99ede542df`. Independent certificate: `sha256:4557b6f886c13ea1c0718f82f936530fad652700a247c5c61936cdbcec75d8d2`. Engine external-validation hash: `sha256:a11f09ac56448eb89235c7b844eae16c98527a7ac8cf0599e1d52d9ef242fead`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- source-response-1: predicted source-support-response-field; observed source-support-response-field; exact match True
- source-response-2: predicted source-support-response-field; observed source-support-response-field; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:5d70cab8c081b140049f503a3664af10b66b0caa87dfb2433c5e09dab51c1589`. Isolation certificate:
`sha256:6ffe43ce72ad3ac15ec805642c15766166006c54b9399b29e14f4b1cffc55a14`. Custody certificate:
`sha256:a8a20229e8b7b6d5f52ec1171809f392c689f84a2f1639a6bf513d60722eebc2`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- source-response-1 from NIST-CODATA-2022-ALL-CONSTANTS
- source-response-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:d68ac9023c8421a96f92b76c4282aebe70666d4bb78bab10b15df8853842c70e`; engine receipt
`sha256:f4564473fe6fa51d1f5a4a777b37fb53f9ccf839fff4a2a098e2c25ce4bd5fd7` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-SOURCE-RESPONSE-001-f4564473fe6fa51d.json`;
empirical-validation hash
`sha256:9052fc292af5a94927de35aa9eb93623d78f8fe4b85023731b977f91cc9211e0`; measurement receipt `sha256:5d70cab8c081b140049f503a3664af10b66b0caa87dfb2433c5e09dab51c1589`.

39. Generated support and geometric dilution

Claim identity: `SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. When one conserved source trace is distributed across a complete generated boundary, response per cell is forced as the exact source support divided by boundary-cell count.

> *Field dilution is the exact conserved source carrier divided across every generated equivalent boundary cell.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-SOURCE-RESPONSE-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__conserved-source-over-complete-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	conserved-source-over-complete-boundary	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:c925f51cc848a17a613158e1364b0c0d70e5e7110968ff7dd2b1808302fee6f7`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8d76878d3e63df5947b4e677c368dcbc9f06995db3539663069867031b6026e7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:17870c6c9111f8027c5a04318a92cb02613787d79a5dd91ff6e809bff8af9604`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ea33d8af27399c9e8e03ca0d37ed92667744e367f6daf1870754adb4a7a6496e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:52211b87aa710adc74dd296af6b6cb3f011a2ff8ee05bb75a61c1483607b29c4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ff7dc6ce86a1c911c501f6f7dd6fa3d786b3daf1fcd34650b2f0fe9e9b83de57`. Independent certificate: `sha256:31100409c4d7ddd477f1b87fd73fc90377f5755a34b9ecf3c00a30dffdfd3fa0`. Engine external-validation hash: `sha256:8ecef76b5b0e2968f92695b6939f3018b8611d5cffe8d22c15a965c45d2b1fe6`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- geometric-dilution-1: predicted conserved-geometric-support-dilution; observed conserved-geometric-support-dilution; exact match True

- geometric-dilution-2: predicted conserved-geometric-support-dilution; observed conserved-geometric-support-dilution; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:39249e16dd911c81ceed74058b54539d71808f5639b2cb2123779865a986c47a`. Isolation certificate: `sha256:6023f2c51629d6077f247603725f0e8f1895113bbe6ee73aa4f637e7c46276e8`. Custody certificate: `sha256:a942e88d5f178a6ee9ed8a5be6dffaf842fc2bbefba3dcf64851184d9d3214d6`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- geometric-dilution-1 from NIST-CODATA-2022-ALL-CONSTANTS
- geometric-dilution-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:9e37a7df51cddf59e38e3f8bd13139789bf4c62b6169be3a2ddd7a7b2d4a4281`; engine receipt `sha256:9f1c50dbea066ec0e34bc50213ac0f9e7b4e56d35609cb4518927d48b235992b` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001-9f1c50dbea066ec0.json`; empirical-validation hash `sha256:b25e7761283fe1261e6a59e3e91b0bd42139d537fc1b728b6e57b43804b52710`; measurement receipt `sha256:39249e16dd911c81ceed74058b54539d71808f5639b2cb2123779865a986c47a`.

40. Gravitational interaction at the weak relational boundary

Claim identity: `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A gravitational response is forced as a universally source-linked relational change of free motion, with geometric support dilution and no held charge-family alternative.

> *Weak relational gravitation is universal inertial-source response transported over generated support and observed as a free-motion change.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`

- SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__universal-inertial-source-response__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`universal-inertial-source-response`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3dde044cc72c4d0dedbff080713c32e7a66fa2179b399d7d796ebb3a30b61739`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c8f956d03349e201afeea9b8b1d15f3416a3d0728a92431d69d7af3c6cbf8c52`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:4c07f5b01b4f9dd95aa95edadd518b8e04541fb9492c9dc9f6d8fb1e42a6a722`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6606681b51c0a9a8c9e7e3af1497013f2f0339e449f349a6ad24519836bebd28`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4bf70ab1a1ab6aa0165875951085c651fda9fdfef6628f26556865be009095ee`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:93e83ead15508eef1a445383097eb16e7061d3dbc4eb5cf3812f52e9070193fb`. Independent certificate: `sha256:6b77d9e624382a50e26aef406e60763d677fd957d4ed398ed5202f15308a76bd`. Engine external-validation hash: `sha256:7581c8611977ac89a327c60786a58f25b3c72851d0a1078cf5db14eaddaa6bc1`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GWOSC-V2-CATALOGUES-2026-07-23

Observed comparison records:

- gravitational-interaction-1: predicted universal-inertial-gravitational-response; observed universal-inertial-gravitational-response; exact match True
- gravitational-interaction-2: predicted universal-inertial-gravitational-response; observed universal-inertial-gravitational-response; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:e44cf81381b717341ce45cf99884fa2ea9b3b247b31fa8a90fcd89c6f8c02e23`. Isolation certificate: `sha256:3290e8b740c843bd3151a65946110616511dcf2a414b2ab45b5b8a3cc7d2696e`. Custody certificate: `sha256:12ac50e3f9b0ce3ce55e9dc3550b2d1a6c0ed6b89a497a0e03002c800ff45925`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravitational-interaction-1 from GWOSC-V2-CATALOGUES-2026-07-23
- gravitational-interaction-2 from GWOSC-V2-CATALOGUES-2026-07-23

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:33510c09d41b88f409a7b514397744607aa4d4d475c95fae2b0814ae6d831448`; engine receipt `sha256:af25bd6d7cd4af55c92fc4fb93ed4fa2ba2d8dbcdc13a9d22d478a5868a58bed` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001-af25bd6d7cd4af55.json`; empirical-validation hash `sha256:06632dae7e70d5e983d1b29bf48746649a54ef01805bf6063282ca79eb4bc4d1`; measurement receipt `sha256:e44cf81381b717341ce45cf99884fa2ea9b3b247b31fa8a90fcd89c6f8c02e23`.

41. Electric distinction and held charge orientation

Claim identity: `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Electric charge is forced as a conserved two-fibre source label whose opposite orientations are held distinctions rather than positive and negative proof magnitudes.

> *Electric distinction is a conserved two-fibre held source label; its orientation changes interaction class without introducing negative quantity.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__conserved-two-fibre-source-label__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	conserved-two-fibre-source-label	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:e95a5c1afddc0ae8b8f6671eaf0bd5ac54711b849caa4505c55ccd312c18c49f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:662bd8009ab209cfa1bf46d5a554e8d4b79eb53d9f4f82c88719857fff9fe78b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:be9c50d45008014f4fdd20aa8bfa18f5e91f9a120a755427781b5ccb608d5d7b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d467869edf735c08251cb166a4868024d2acb9cdeae472dfe01f267faff6511a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a8d75ee0c59080e854470b31c6db28bb990b07783468a97e7c971f921ebac4b1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9a2814a2eb56fc4b08cd99d8fd678f66d1f3b4ad98b593730e5d5229030fd0da`. Independent certificate: `sha256:1dc68a09b17229216ebee4249420c082e95bcfc78502b5c053714467bad10b29`. Engine

external-validation hash:
`sha256:0fdcf95c03a3aa455293ccc7d2256892c075b0091535bbb8b9e99178e2dbffc7`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- electric-distinction-1: predicted conserved-held-electric-orientation; observed conserved-held-electric-orientation; exact match True
- electric-distinction-2: predicted conserved-held-electric-orientation; observed conserved-held-electric-orientation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:6cd06a76a1ffd85e0b83aa8aed58fb962d837ff9ad2d6377728f7a7583ecbb14`. Isolation certificate:
`sha256:a592be3d01457704bc5ebb02ab536fee53f97510e8c38ecfb588bb851877f7c3`. Custody certificate:
`sha256:d9199e055a1f9dab694dd09b4dc3369dbf68a6bb96b931a08463710f5b6055dc`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- electric-distinction-1 from NIST-CODATA-2022-ALL-CONSTANTS
- electric-distinction-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:8b7841d8d9ec04fd347f7e45ea065afc655a6823b14e1ab4453ec8a1294f6ecd`; engine receipt
`sha256:3b3e69fb0fa71108a908f8b7299da47df8aa635f40938ae2bddc89bdc72071d8` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001-3b3e69fb0fa71108.json`;
empirical-validation hash
`sha256:9146e9cafffc33b432759d9125183022840f43273da299cf9056859478b64b9f`; measurement receipt `sha256:6cd06a76a1ffd85e0b83aa8aed58fb962d837ff9ad2d6377728f7a7583ecbb14`.

42. Electric field, potential and work correspondence

Claim identity: `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Electric field response is force transfer per held charge carrier; potential difference is the exact work transfer per charge between two generated support relations.

Electric field is charge-normalized force response and electric potential difference is charge-normalized work transfer between held endpoints.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__charge-normalized-force-and-work__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`charge-normalized-force-and-work`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0875d868c1d36375b546333154b296011e885effb7032e2b4e5449cbab665347`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:822036fc67f0501f0487767675b89306c3fd296509762afc01915cdf4786acea`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:4c3a8711ed4de88b4c8578e65a3f32681900e1c7d93b94a6a46f8b78ecfb38ca`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8c5e7fbf3054bddf4f5a969d20418adfbdd03f419c1f058e8d14e048244a9257`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4da112bff0e21eb5e42d2bd033cf0da2ce75ca226c5d371ecae8c9e562e422fe`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:29a14819bf54c8423f2fbdbd136d4462c6698875ac2ae78bb63b70a7aa605f8d`. Independent certificate: `sha256:ba419390f0d39e89fc7bbe511178cb83411759ebdc52d44d4ccd1cdd02c32bee`. Engine external-validation hash: `sha256:255389b1246bd1d7e3f8ba119e968fd019a3cb9e6e6827056aab6852b6dc0826`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- electric-potential-1: predicted charge-normalised-field-and-potential; observed charge-normalised-field-and-potential; exact match True
- electric-potential-2: predicted charge-normalised-field-and-potential; observed charge-normalised-field-and-potential; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:8ff986682e8340c9aa22481da2cb9db6967bab9f38bfd5ce5812cbf5b4151c53`. Isolation certificate: `sha256:34136cd52c2adc75c04bb8c728a3df6a8d8e7ffbe5fae282f88b6693a3dbd8b4`. Custody certificate: `sha256:f0c7eadfbf6eaa5bb8ac7d955640f6ab0592f0df4c7692aba086173aa41c0a0a`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- electric-potential-1 from NIST-CODATA-2022-ALL-CONSTANTS
- electric-potential-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:165cb2a27d3ed2b10246df9c63442fb14919784ecdc1ef0acd32c1f3e2a94737`; engine receipt `sha256:c64eacc598d02d73e0d9994f6525b45f4c12a7334e28d7f98351d6c4dfcda6de` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001-c64eacc598d02d73.json`; empirical-validation hash `sha256:75b680e08454cf75167dd4e92790b58d315bb72a19415a4c9f0d89237b736bcb`; measurement receipt `sha256:8ff986682e8340c9aa22481da2cb9db6967bab9f38bfd5ce5812cbf5b4151c53`.

43. Magnetic response to oriented change

Claim identity: `SFT-PHYS-FIELD-MAGNETIC-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Magnetic response is forced when an electric source label is transported with held orientation: the response is transverse to the source-motion path and reverses with either held orientation.

> *Magnetic field response is the transverse held relation generated by oriented electric-source transport.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__oriented-charge-motion-transverse-response__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`oriented-charge-motion-transverse-response`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:850030cc7fb143607cef4fb21e4c2cf49ba983524e0f2f623a08af08e0395a96`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:30c315164b77365902d50426dc07b08e9aa4c4641eed78a15394675cbed09921`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:56f4d2c79bb6eeb8f1113d3eca4ef88cf61b28c469e91d55d85c2d6cc55ace57`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ebc25039336454dbcf6bca3ac5bd76621720cc1ad5b7dd049184db31ba6c6239`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e1c50337da9ca2040e691d3aa093e4ae1e50247e0c03fcb77f13512767ba34eb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:92c14267164b0df4326dc155fac09d5a7ebf9b6469774beeaac2999709d34fbc`. Independent certificate: `sha256:63f9580fa4c97bde378def33968315d4581aafcbcfe25c81a9bedc902106a2fa`. Engine external-validation hash: `sha256:176ab0a92bb52468719a7f3f2e2d1e31e4d28d3309701c520e328338776989e5`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- magnetic-1: predicted oriented-electric-transport-magnetic-response; observed oriented-electric-transport-magnetic-response; exact match True
- magnetic-2: predicted oriented-electric-transport-magnetic-response; observed oriented-electric-transport-magnetic-response; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:6ce585296bbfee1c07581e84284ab1b8adf03d9d19f9131dac0bb9dd5366bb9a`. Isolation certificate: `sha256:d6878811bbbc66d452f89cc9149798c43b5f01033a84752cac395ba5875b5791`. Custody certificate: `sha256:bfa7f2f376bb96ade7c5b29951a88b32d403a204edc5caeca73119218310331f`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- magnetic-1 from NIST-CODATA-2022-ALL-CONSTANTS
- magnetic-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:e426215ba7d84086813c25c371adf648c81c3be17769a411c9192f6c0a4a06e2`; engine receipt `sha256:74b71a149a6537d5819031f72cc561a9e392c7ae5b0fc0985b2d560e0f96d146` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-MAGNETIC-001-74b71a149a6537d5.json`; empirical-validation hash `sha256:8330efa3ec637e9a63299f2fc1d4cdc246ad2f035f7c0dbb6a908551ef81fc60`; measurement receipt `sha256:6ce585296bbfee1c07581e84284ab1b8adf03d9d19f9131dac0bb9dd5366bb9a`.

44. Induction from changing linked support

Claim identity: `SFT-PHYS-FIELD-INDUCTION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A change in magnetic support linked through a closed generated path forces an oriented electric work-transfer trace around that path, preserving complete source-change provenance.

> *Induction maps exact change of linked magnetic support to held cyclic electric work transfer on the complete boundary path.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-MAGNETIC-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__linked-support-change-to-cyclic-transfer__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`linked-support-change-to-cyclic-transfer`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:81feace912aecc77d43c0bca88bc47411a9ee24ce6ad38238ecb4964474e1c90`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:effda451435526b806ea9a3e3a39865f35fdd7448e7f0a4af4b255e828947cfe`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:aac1ccdceaf7af651fc29f7cad1a33665c2beabf5356f8529b6dcd4b284becb9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e69bf6759c052cd10deef515b60131f7cb67a3860c4cb71f8fa7e9f5d045f5b5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dfbbafb8db5bfa6776ac91b2499e152d8979e5ee70eed6cc80cd3b22bdb345d1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d967a5095815ae755cd2805bf7772dbee85d90a62d8f7da10cdd67a2eacc7056`. Independent certificate: `sha256:b16e812eb78b1e7d7dad39616b999b6e02ae2356070dea3fb3e9ce3ae490862b`. Engine external-validation hash: `sha256:c8c864713a6b39b5c42881c23edf9ad8a15591cdfdefa698a1b52eb2639651b7`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- induction-1: predicted changing-linked-support-induction; observed changing-linked-support-induction; exact match True
- induction-2: predicted changing-linked-support-induction; observed changing-linked-support-induction; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:5403c1ae6e5e0e54ac3cac4864767041c2053ac54ab9109c895bd5f3ad96d299`. Isolation certificate: `sha256:7ec7ff9f4b5c514487c1d5de16167596a41131740dd0102e95e61e9e79781e7f`. Custody certificate: `sha256:eb5afc3fff80fdc81a0771576cf984a1bcde292b0dcb979b2d34f4b60e4f0e41`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- induction-1 from NIST-CODATA-2022-ALL-CONSTANTS
- induction-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude

- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:2b868cf20f4544ea236e43131e6e9a7a67666d95f0e2f32595b36f61270ca5aa`; engine receipt `sha256:b76cd4b60c0f7e29b7a215c36dcd0a400d260df284b2735aaf32617730818f4f` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-INDUCTION-001-b76cd4b60c0f7e29.json`; empirical-validation hash `sha256:d227d90a05cdd949ccd7fe5121c066760155a589d1b1faee269bef246c2444e0`; measurement receipt `sha256:5403c1ae6e5e0e54ac3cac4864767041c2053ac54ab9109c895bd5f3ad96d299`.

45. Electric-magnetic compositional closure

Claim identity: `SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Electric and magnetic responses are forced as two observation projections of one conserved oriented source/propagation structure, because each changing projection generates the other on a closed trace.

> *Electromagnetism is the closed joint carrier of electric and magnetic response projections with common source, orientation and propagation trace.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INDUCTION-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__joint-electric-magnetic-propagation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`joint-electric-magnetic-propagation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ad1c06f2c0117d8b5a87378a91ef0e5161515aa9c1ac3627923452ffd0a330de`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:baa78f1d8cedfcd5821c132f2d797357cd64b2b029c3826f66f0c65fcf38617a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b0dcc38e2cd08fe6ce5242e906ae9baba21037229d30005d0b0edb1d2d6a282a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:559638b588ba3218eed5a3d470e7e6d6357b9e13f08f851b8ca2495d2385ac2d`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:0fdb3b10e60f99983b701490c84f3a76a9ca115ad9ce5af6806024f3401f0fb7`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:82b79da2d7b5c6902bbda52aca1ba49b9602ce9bd1c667d9329cacbc77a7cb05`. Independent certificate: `sha256:0c9579c39d3c0045693c0c1ed912f75c31f422581621ea0dc0d9c7d25de531c9`. Engine external-validation hash: `sha256:8808db252d76ffcafb27a83aaadba01908d955a889859cf9841c3b46726c89c4`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- electromagnetic-composition-1: predicted joint-electromagnetic-field-composition; observed joint-electromagnetic-field-composition; exact match True
- electromagnetic-composition-2: predicted joint-electromagnetic-field-composition; observed joint-electromagnetic-field-composition; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:32b1b97f8a77ee775ca9cf3618dfc1d021c4d4df21b35f431e0f672362320e88`. Isolation certificate: `sha256:afd78daa5d2a76845e971b53e880106ea97f52bf25fcf0a5c388614d30bd850e`. Custody certificate: `sha256:686fb2933bd259c6eaefb59c57dc5cb3ce19a58959feaaf7f407a98478f9c6e4`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- electromagnetic-composition-1 from NIST-CODATA-2022-ALL-CONSTANTS

- electromagnetic-composition-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:bab4b81b490040634b085e60e1878c2477e101e5a69871d39128d39f66f768aa`; engine receipt `sha256:4f454fc66603a589ecd4a1cd581d9b51038b33b6f79880f07a168db613027245` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001-4f454fc66603a589.json`; empirical-validation hash `sha256:ddbf7985a2964df72f10351ea87f5ea4ab466f8d3120073872b9fec11cd77261`; measurement receipt `sha256:32b1b97f8a77ee775ca9cf3618dfc1d021c4d4df21b35f431e0f672362320e88`.

46. Gauge-equivalent descriptions and retained observables

Claim identity: `SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Two field descriptions are physically equivalent when their complete observation and transfer traces are identical; unobserved internal label-path differences remain held descriptive redundancy.

> *Gauge equivalence is canonical equality of complete observable field traces under changes confined to held unobserved description labels.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__observable-trace-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`observable-trace-equivalence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0fd752fbc7fb5e94c35d6cd672d9bc08d504a8108888eea84d75599f7e41e694`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f145c14ad85295d09e564e20bdeeb87ee7d7536b03cc9e636ec2232c0cf1306e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3baeec4b82b342a0307585a4f83a80fc186153a8092922a7879bf09be0d9b070`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:2b4b3589ac32501ee81c6df1c75b4a840c876e0917f8198373476cef0d4b7d29`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:34fb113f6a06ec0fcbe160ede6077e08f7693735c946d1a0254368c79c94bd63`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:85c93bc7f68ae39b5bae7150435500555cd25db22bb3e4c327ef6a0c5f381644`. Independent certificate: `sha256:fa780de77bbc520df4440a87525dad6b4dee4903aac27d5548bc8087781609eb`. Engine external-validation hash: `sha256:ce4f2f946e595c576684a34349a8b7b0d60b59c0785d3f78efc69044ff05ee4f`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- gauge-equivalence-1: predicted observable-equivalence-with-held-gauge-redundancy; observed observable-equivalence-with-held-gauge-redundancy; exact match True
- gauge-equivalence-2: predicted observable-equivalence-with-held-gauge-redundancy; observed observable-equivalence-with-held-gauge-redundancy; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:2ba86fbca0b34c195e0c7a3b286237b88e220d107d69fce19c071ad3df695788`. Isolation certificate: `sha256:8db053aa319a65f7cf2e563a91f5d465e5f7c6a3a49b4997dbd62e6da54d931e`. Custody certificate: `sha256:b5c3d477472b6b349523aa5b7580dac9851d7753284196e5c93bc63bebdd01be`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- gauge-equivalence-1 from PDG-2025-SUMMARY-TABLES
- gauge-equivalence-2 from PDG-2025-SUMMARY-TABLES

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:06ac7c692b3e2268eea546dcc9cd523cb48cdef4dbb2bd19a98729c44a7ec8d5`; engine receipt `sha256:c9be792f2d288ae0f370d83dba86c1f4ff503e004ee556fd9363174f6f360412` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001-c9be792f2d288ae0.json`; empirical-validation hash `sha256:f2cf1301fdf227d13c222128b46777091a227211780736cc361484523fd33b25`; measurement receipt `sha256:2ba86fbca0b34c195e0c7a3b286237b88e220d107d69fce19c071ad3df695788`.

47. Local propagation and causal accessibility

Claim identity: `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A field response can change only through a generated adjacent transition path from its source; responses outside the completed propagation trace are inaccessible at that event.

> *Physical locality is source-bound propagation through generated adjacent support; causal accessibility is exactly the completed reachable path set.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__adjacent-source-bound-propagation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	adjacent-source-bound-propagation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:876e6dc3ea8dc91935e7cf0a20c1b3d71ac56629b5ae6f92daa8aeeabade7761. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:f20e2e9440d80fd6bcae3687d0e494568f05d309366eb8696458443b6d246d9a.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:20a8fa5d1d76dfc37bb8cde320c1a4c7d0bfbaed15178472d88f54a43d619f7e.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:faca424ddf42910e2626c8ce34cce75ccb38f8bc320ba7575f90e44bb9fec166.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:e78bbac06cdc159ee09ba8105fb36f3a4d70f60be94c674b07fc110c88834373.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e431b510e133b534898a11b27facedb9b7a4bd7f31a177c767a41caa1ad9d93c. Independent certificate: sha256:7dc28b76c3b7fcb33d0f298b40abaa561d8afc3379211f37d9d45b7aad4ded11. Engine external-validation hash: sha256:f866ecf16df6b87cd5c8506767393c5843847af5f5e445f98f2e33bb3c7755c7.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GWOSC-V2-CATALOGUES-2026-07-23

Observed comparison records:

- locality-causality-1: predicted local-causal-propagation-support; observed local-causal-propagation-support; exact match True
- locality-causality-2: predicted local-causal-propagation-support; observed local-causal-propagation-support; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:713e96c9f9759572b2395152be7e442ad6e837b706a099679c5b40afbd1c29e6`. Isolation certificate: `sha256:6c5c1fe4f69a4951323a87c76e4cea3115a609a0375a32de71b88391cab53f87`. Custody certificate: `sha256:437722dcd5eabbc3e24ebc3e0632994318cd831f8cd939131fff4005f03a198a`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- locality-causality-1 from GWOSC-V2-CATALOGUES-2026-07-23
- locality-causality-2 from GWOSC-V2-CATALOGUES-2026-07-23

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:87387d375fd772baba71fbf2313642bd0d716e3528af3333f625fc44302d1505`; engine receipt `sha256:ddcd093c9d9e17b12469f8992d6cc7bb1458c867ce1a68cffcf6ccc944dd2a77` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001-ddcd093c9d9e17b1.json`; empirical-validation hash `sha256:94f823b9b16585f3a7013a56f9418d25cad6691a2769a9398d049ea24bc32587`; measurement receipt `sha256:713e96c9f9759572b2395152be7e442ad6e837b706a099679c5b40afbd1c29e6`.

48. Radiative transport and source detachment

Claim identity: `SFT-PHYS-FIELD-RADIATION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Radiation is forced when a changing field carrier closes into a self-propagating joint electric-magnetic recurrence that transports energy after local source contact ends.

> *Radiation is detached self-propagating field recurrence carrying source-bound energy, momentum, frequency and polarization records.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__self-propagating-field-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`self-propagating-field-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:464f8785bf8c950fe40c3b4acd61da12c9521102a4fcc4b5ed881ee65599328d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1f6b8207b6701110505f7659295bfe8a2ec2e235b8c808e563443cc5247128cd`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0386c56bd1470024b84aebef849f3212bc179c348936e9d956f0be64faf63d43`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c7bc8f0e89263a161dfc2cd37fadf82e444cbf923916f175b457a998652ed914`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:37e033b8d370e9a37b1fc41f42c70c50360fa36303ae51c724540ffb76e430d0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:653788cbfd634f0f02f407fe78b430ef7621899f2566f0f9f331d61de391f899`. Independent certificate: `sha256:37813a5b21b4330e7fbd979b8c0b4362c6bae623debd74a927f726652b8d0649`. Engine external-validation hash: `sha256:deffa127519f3ce508851847ace0950f67e6a5da8843101ee64c735a873b7191`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- radiation-1: predicted self-propagating-radiative-field; observed self-propagating-radiative-field; exact match True

- radiation-2: predicted self-propagating-radiative-field; observed self-propagating-radiative-field; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:d3f19e10ee097433ca910e6ba85b57ca0d0554f8af18e136ac84ab4fa616f854`. Isolation certificate: `sha256:156e82d41707b0689780af93d39078fbbc296a63b0bfa284f217c01f7aaef710`. Custody certificate: `sha256:6efcc841939563fb99087ca168e48dddefcfa7d5f445d91052af2eb6aab57a22`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- radiation-1 from NIST-ASD-5.12
- radiation-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:c8fce2363723e510e1fba7583050f9f84ec8c35bdfc64f741992feabac92d36c`; engine receipt `sha256:9fa1ed9f5907e60d3c439eebfce928a8a04bec7abf067419b58d0a16a3544361` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-RADIATION-001-9fa1ed9f5907e60d.json`; empirical-validation hash `sha256:7212bae57dc76cc531589cc24c1214bd59e6219656ec3e7f938747a5f70890b9`; measurement receipt `sha256:d3f19e10ee097433ca910e6ba85b57ca0d0554f8af18e136ac84ab4fa616f854`.

49. Empirical closure of physical interaction classes

Claim identity: `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Physical interaction classes are distinguished only by non-equivalent conserved source labels, propagation carriers and response traces; classes with identical complete traces merge.

> *Interaction classes are the canonical equivalence classes of conserved source labels, propagation carriers and observable response traces at the registered PDG boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__source-carrier-response-equivalence-classes__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`source-carrier-response-equivalence-classes`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:7e284eda521dfad45e2b44b4963a0f4e8d645ff412e2189bd965a15a9081a665`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e8989ed9853bb54628f49d4906cddab80fbfe215b97e578c08fd1704fb36aebc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:144c6ff57254d0b52293019574441dcaa66442fc0b6ad1bf6777c8762413ee23`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:78bcc6733218517beeaf151d8b9344b635050b59c70ec5e6198e1720225ce333`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1c60ec0ba523f3de5dc16d6243568dc364cdea5198008b018e0b968fc396a39a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ba2bc950bb6b33ec3c1c0977df1ade1852e9ddab8565b8bc35a812d32017dddc`. Independent certificate: `sha256:4f5d728bf1d7d1d3fb6652019cf0d89a5341c7670ad1b6089960d65e1ada77be`. Engine external-validation hash: `sha256:c121b0c09750695e740b7ec3e7bfd9c97e3a1367847d5ad11fbef81ee83e03ec`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- interaction-classes-1: predicted four-observed-interaction-trace-classes; observed four-observed-interaction-trace-classes; exact match True
- interaction-classes-2: predicted four-observed-interaction-trace-classes; observed four-observed-interaction-trace-classes; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:656bfe5f9d3fff23a9b0d88daa4e41b9235de456f258d5e7a95267f29493406b`. Isolation certificate: `sha256:e4ee61b06306e1c2ee8ea7278874147e8c8281830e7d346e6eeb62f9747d105e`. Custody certificate: `sha256:193a15d1ba83d9fbbd5cc47b5a2acc71950b9e656d9cd9c0d055254db6731bd2`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- interaction-classes-1 from PDG-2025-SUMMARY-TABLES
- interaction-classes-2 from PDG-2025-SUMMARY-TABLES

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:baf5526df74ba9dd96840c4cb04e298ee58837291fca202d5c4bc4087d924aa1`; engine receipt `sha256:1cc5065d8cae35106aa07ac1b2c6e5fab9685899dff878863207533525e99d0d` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-INTERACTION-CLASSES-001-1cc5065d8cae3510.json`; empirical-validation hash `sha256:6fe186a31e7413d2c306c8f4e4f80bc1657b278d1e1cd283d5e09485e78d7d99`; measurement receipt `sha256:656bfe5f9d3fff23a9b0d88daa4e41b9235de456f258d5e7a95267f29493406b`.

50. Source continuity and field conservation

Claim identity: `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every closed spacetime support region, a conserved source label can leave only through a named boundary transfer; the interior change and boundary flow are exact paired records.

> *Source continuity is exact pairing of interior carrier change with oriented boundary flow over complete generated support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__interior-change-boundary-flow-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`interior-change-boundary-flow-pairing`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:eb3d2f529880e7db67e2a68bee00dcf214a09817d85b1427d9c77dcfcfc08ea0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6879367734b5c7f2286583b5c8739c36b436ccd219dd57c532e16c9774c34680`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e8f40f2e01ff4eb0c12eb93c01a8242f37d8c629af632fb58e5338bf93c54a9b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:01913905e0c22c6005ce422f4d205530bf82a361f6f38e9892691c85f9b9ae2a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:0bd5fd4d412afec3911663c85265a74a95472f66fe019bd969830de967cecb13`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8dc81feb7603dc204bf2e45dbf55e742474325f372bf5cfb173a989d8023133f`. Independent certificate: `sha256:ba915b374fda4dfcddf48182128a3f396ec2f0009d99257fc36f6255c7d37d49`. Engine

external-validation hash:
`sha256:9d38ef7a92f2f379b343dab35802c2da3ab12a61986a202fdf58bc29919394d2`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- conserved-source-1: predicted source-continuity-boundary-flow; observed source-continuity-boundary-flow; exact match True
- conserved-source-2: predicted source-continuity-boundary-flow; observed source-continuity-boundary-flow; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:09c186c2d26c987da5a3120175d49d9656e9626f51671ae8cac09d23b91fc3c2`. Isolation certificate: `sha256:59d72586cd207b55490bfa82b5789e640a16b312c30860538b003efa3afa6863`. Custody certificate: `sha256:938733f0b83503c730ded1f0ea2b0da8aaefac52d0623ec8cb39c99eac6c74d3`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- conserved-source-1 from PDG-2025-SUMMARY-TABLES
- conserved-source-2 from PDG-2025-SUMMARY-TABLES

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:48673444e1a4964100390d1c55ff63627808e70212fa9a5047358bf038ae8331`; engine receipt `sha256:901fcb7327185f1311f179ed4064294e60bf388259a2432f1b3021cbbe2cba89` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-CONSERVED-SOURCE-001-901fcb7327185f13.json`; empirical-validation hash
`sha256:2b28490654df278a0aac152e5fc05400db067c0fbf47e36f64e2476168181042`; measurement receipt `sha256:09c186c2d26c987da5a3120175d49d9656e9626f51671ae8cac09d23b91fc3c2`.

51. Reciprocal interaction and transferred distinction

Claim identity: `SFT-PHYS-FIELD-ACTION-REACTION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A closed two-body interaction trace forces each transferred momentum carrier to be held once at the source loss and once at the recipient gain with opposite held orientations.

> *Action and reaction are the two held orientations of one exact transferred carrier on a closed interaction trace.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__reciprocal-transfer-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	reciprocal-transfer-pairing	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:f2d8053d7543d0cd3afa30a1f798346c15ab415e12278bb590cb69d3127763e8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b874366c3800f4054158f9713a7cde61f1f13dc9a0c6706bf1e49f109b3538ec`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d0ef8c30d8dda7e6d8f0f9488112990148001c81a80e05bda93387c9ab8cf98a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ed8d40efc11b42ee302db2490e861dfb97b376eb07abf21d491efef140de596b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6024740d1e081816ec3b01895ca3341fda0d213857a57c7ce3983ef90c375311`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:297dbef62cbb8d0564d3ab582ef9d773fb2060dc95e4b1253ab052a336e62fbd`. Independent certificate: `sha256:40404e78785faacae2f881f4a7d8abe1930ee3a3c3219182d8fa5f8ecd683d63`. Engine external-validation hash: `sha256:61bf2b0cb01977e5a304e39dc1b7beed3b48b6c34cae43a6c9e1ed1ed440cd33`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- action-reaction-1: predicted reciprocal-closed-transfer; observed reciprocal-closed-transfer; exact match True
- action-reaction-2: predicted reciprocal-closed-transfer; observed reciprocal-closed-transfer; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:3f1ef0a6545f3ea16c4009e46cfe651d64c0e1e08bed8549c18313955119ca7f`. Isolation certificate: `sha256:c9887a245a1985d24291f22a27493bc7bacfbc4ca2244e40666d49c7e3f212aa`. Custody certificate: `sha256:38d60ba3e7013b79b9387858210b1eaa6057e0c310f66a2d8329c32877d40a32`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- action-reaction-1 from NIST-CODATA-2022-ALL-CONSTANTS
- action-reaction-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favourable measurements

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:fb07a4ccaac50fbf8955c41b810b95cf8515cc57f4536811c2d7383d2abdb8fe`; engine receipt `sha256:94eb13578b043cc3a4608bca8bcfb91f9d02f9bec1a12168b3b10d42fc6444f5` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-ACTION-REACTION-001-94eb13578b043cc3.json`; empirical-validation hash `sha256:7a782610f12f8bc6d34ef35583ed02e6ecc3cb7c6d4a38b7cb998d24be9249fa`; measurement receipt `sha256:3f1ef0a6545f3ea16c4009e46cfe651d64c0e1e08bed8549c18313955119ca7f`.

52. Period, recurrence and frequency

Claim identity: `SFT-PHYS-WAVE-PERIOD-FREQUENCY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A wave period is the exact positive transition count closing one recurrence relative to a held clock recurrence; frequency is the exact count of closed recurrences per duration.

> *Period and frequency are reciprocal exact positive recurrence ratios with complete phase and reference traces.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-recurrence-count-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`closed-recurrence-count-ratio`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:42e4527a6689bea0564118906464ebc8dfea168fb8773ac1b2e796ef51f93d89`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2484c7ec3efe10bd74c654b8c4bbafab282445c5022e2ec5122bcc189c67beda`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9e9a291e31cd136797288998c7d57a7734b99f165857c38894b07da5b609103c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6c93a003263282930c974a19e7419ebc1d785be762ba17f2067b9daa2599c8a6`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:d28ae4833fb702e2b45d4b5815b2f139b60e7e315c6ef5dda145286584fa3b9e.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:818304b482a4fcf476f693504599edfc466792c241c25186ce113f92a0d6947a. Independent certificate: sha256:c281bae88a0df7680ab1638b39dcc30c3badd302974d8dcb05edc9a995ff95f9. Engine external-validation hash: sha256:c3f7e10168481efe5fffd879289e6b595bfabc85c42a3182bb4add5493cc26db.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- period-frequency-1: predicted periodic-recurrence-frequency; observed periodic-recurrence-frequency; exact match True
- period-frequency-2: predicted periodic-recurrence-frequency; observed periodic-recurrence-frequency; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause PHYS-S011 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: sha256:7298be1b415d8970a152307e98c18867f99d33277fd3711af0d9f53a856deb40. Isolation certificate: sha256:bdc584e5cfb14b6e1e3732d01c6c62b3630e6f64c883878c231f3bc74324ead4. Custody certificate: sha256:a591dde379fc91a1ad22b5c54fd852e18f5cd79fe160f2c683e537ee88578bba.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- period-frequency-1 from NIST-ASD-5.12
- period-frequency-2 from NIST-ASD-5.12

Shared claim-record clause PHYS-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:dd97bbc76a84346a1592153a2ba2a3507b86b51c41d1e3023960ea77913fc635; engine receipt sha256:1685681514cfcc033abdaa8cf880a85f252fb69f74f4c1a0e87d8b6581c9c8c5 at receipts/engine/model_admitted/SFT-PHYS-WAVE-PERIOD-FREQUENCY-001-1685681514cfcc03.json; empirical-validation hash sha256:8ebd913f68802e277023839f4cb1cd17c3a441a113c0c83bac5ca00f36e252f9; measurement receipt sha256:7298be1b415d8970a152307e98c18867f99d33277fd3711af0d9f53a856deb40.

53. Generated wave propagation

Claim identity: SFT-PHYS-WAVE-PROPAGATION-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Wave propagation is forced as repeated transfer of one recurrence form through adjacent generated support while retaining source frequency, phase relation and carrier identity.

> *A wave propagates by source-bound adjacent transport of a recurrent state relation across generated support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-PERIOD-FREQUENCY-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__adjacent-recurrence-transport__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`adjacent-recurrence-transport`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:867989210fc4bb68cf506a750cb0ddf07803db9c00a9c49e3ec00b40984a9847`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3af7c76f5029ff2a83b9b23a8d8d1288514912fce6f4735d002320d632f77091`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:026a75a6671c260841c751c47ee49f7c0ffef947edab1117e95a6f68c78cb99d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:51c5ebc2f063f0349e8df80b91de3ba1e3f140bb831bf587b2831a6cb8a74558`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1f14778ed9b928d9abc8031d0a05a2d11cb2923e5a2586118d92c0a8d8ad4be6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0b3cc4a2294b47b1b090f55adc9b3fcc5ff6d6a46be5a52f0559462b58bfb242`. Independent certificate: `sha256:e9589e5e804c9d00fa36110b5c966679b9c10b80471a1db19aa91ba08120e2d8`. Engine external-validation hash: `sha256:d995ba84b500bd14162b767785e7767452cb36498de266d58380fe4c379514f2`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- propagation-1: predicted adjacent-wave-propagation; observed adjacent-wave-propagation; exact match True
- propagation-2: predicted adjacent-wave-propagation; observed adjacent-wave-propagation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:c14af42c8cb6ee812604ca2025c15b9abba0b43772bc8ac0ab2cbed612813535`. Isolation certificate: `sha256:7cb92b2f89a971219c3009b51d8ba3854ef3ef44da01efcda7910fd1c27908fc`. Custody certificate: `sha256:4deaff495c157253ad3d963ed4fa799d1fade675100b0ae735b2a8fef9fb3b68`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- propagation-1 from NIST-ASD-5.12
- propagation-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:40e024d3b7acdc8cb863c2dd5035068424d76288665f8a7228f178a10e004432`; engine receipt `sha256:c5d9e3e5665cc460856932760cb46eb6a7e97867c7487b1f8caca6e2abb47588` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-PROPAGATION-001-c5d9e3e5665cc460.json`; empirical-validation hash `sha256:a966f782ebebc8a7698579f49eb493794ac783dfc9d05e616ed9886324770c56`; measurement receipt `sha256:c14af42c8cb6ee812604ca2025c15b9abba0b43772bc8ac0ab2cbed612813535`.

54. Propagation speed, wavelength and frequency relation

Claim identity: `SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. One recurrence transported through one spatial repeat forces propagation speed to equal exact wavelength support per period, equivalently wavelength composed with frequency.

> *Wave speed is exact wavelength/period, equivalently the exact product of wavelength and frequency.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__spatial-repeat-times-recurrence-rate__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`spatial-repeat-times-recurrence-rate`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b9dc6734b5930c4f96edbc986dd2e8b4ece980e56772389c5b0933b5efed0c34`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:721206e1b8d38e686b94457bbb13dba587d49e5f497e5c5c8a57a530960f79ce`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6d6384f19549de2fc0a722fabb0821203feecee12ec799e493473eeabf067477`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:1a0ef5a811b9a9e91b38d86e6cf20175aa034745139bc71502307dd236ef5256.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:f687c6c853d8362126831975c0bfde0602fe37fae27ca089e39f4c1e5f5da554.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:eebd9cc41ac9e81564b19dc8bac9a4fac725278a96973dd497ce9dcf779bb0cf. Independent certificate: sha256:185d968975a383a7b3674347375ff79eb1b104c817038c86f9c0185783888192. Engine external-validation hash: sha256:10fced76f8a9f5216be20101b043782171eb89df418337e65f70718b767ffb3c.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- speed-length-frequency-1: predicted speed-wavelength-frequency-closure; observed speed-wavelength-frequency-closure; exact match True
- speed-length-frequency-2: predicted speed-wavelength-frequency-closure; observed speed-wavelength-frequency-closure; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause PHYS-S011 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: sha256:7ec2b84623ce8f7782c01279a5885f27f29a49c8d0e42ff47ffd4546daa84a65. Isolation certificate: sha256:875b7f54115d340b8d8da929ae9584cbd01162f828ceba5850d6eeff47831c78. Custody certificate: sha256:658d1641a9472ebd22f71792e9833a560b9354bfabdccb5c14037ef016a8c90d.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- speed-length-frequency-1 from NIST-ASD-5.12
- speed-length-frequency-2 from NIST-ASD-5.12

Shared claim-record clause PHYS-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:f75f0fb08d53f2f93570a80010b9cf57cf115e8949628bdce6559f1c6a0a6d7c; engine receipt sha256:d37f2a40ad5f1900c80c049450bbcace88566cd8e1141a4c9bf8412a12da209f at receipts/engine/model_admitted/SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001-d37f2a40ad5f1900.json; empirical-validation hash sha256:786f91d894a466a3082309dc2da322990427e0781f3d28d416111cd3cd2eb219; measurement receipt sha256:7ec2b84623ce8f7782c01279a5885f27f29a49c8d0e42ff47ffd4546daa84a65.

55. Wave superposition

Claim identity: `SFT-PHYS-WAVE-SUPERPOSITION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. When distinguishable wave traces share support without interaction, the complete observed carrier is their disjoint labelled junction; no constituent trace may be erased or double counted.

> *Independent waves superpose as the complete labelled junction of their source-bound traces on shared support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__labelled-disjoint-wave-junction__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`labelled-disjoint-wave-junction`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:c89b88305ea292ef573670856c017d720b58ed80b96bea61da4a3a03e9f5ec54`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f1f498bfe822763d05d981ff2cc13c82e3f3097dffef33b20b93d9d8a5822624`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt

sha256:99f36ca76c44e49a26f7202b22981998295b57f995d844b37caebafd3224db03.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:6593ae7f0e26039c15d45cbfb55fd71c1b7c22fde364c32e43e5c2725db35932.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:f0f1d5f47f0661d51353f05b0af7d28f6e7468e898d8da0d01f07db78038b7fb.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:706b5acee19878334d33cfaa3a90accacc00bfef8af9b97731d640cff64a0257. Independent certificate: sha256:8759b77eeb48a7cb010732754817c7d3b36337011dfe2c6f1c13fd6e56140010. Engine external-validation hash: sha256:2d705a101b0672e971c5232a20468492767c7585ba64bd690c6cbb9b88c2e5fb.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- superposition-1: predicted complete-wave-superposition; observed complete-wave-superposition; exact match True
- superposition-2: predicted complete-wave-superposition; observed complete-wave-superposition; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause PHYS-S011 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: sha256:6c37d1aeea4a98ea44a082c9689d6596597507b85565e4fe54dc5ba96c753d7f. Isolation certificate: sha256:f0df4fc8f0e79a92996b22740fbd40a32ff28c2634190e400aabff9e5dfb38f0. Custody certificate: sha256:08a62a0ad337abb5cdf1d30677df3e02e64bf10cff1e8e569d6beaed81907e60.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- superposition-1 from NIST-ASD-5.12
- superposition-2 from NIST-ASD-5.12

Shared claim-record clause PHYS-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:4faf4e3829f957ce9adca7673185867267d602660e47dd5bf192e04c25f5e983; engine receipt sha256:1dea738715f2417b0bdc973dc0b2bb0869319d76ec7d426bfc4852710aeca08c at receipts/engine/model_admitted/SFT-PHYS-WAVE-SUPERPOSITION-001-1dea738715f2417b.json; empirical-validation hash `sha256:3addb50ade23369154b9a494f0e03bc4601affb1536e9a7a53ae372657723aa7`; measurement receipt `sha256:6c37d1aeea4a98ea44a082c9689d6596597507b85565e4fe54dc5ba96c753d7f`.

56. Wave interference and predecessor merging

Claim identity: `SFT-PHYS-WAVE-INTERFERENCE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Interference is forced where superposed phase-labelled predecessors map to the same observation cell: matching labels retain support and opposed labels close distinctions while the predecessor record remains held.

> *Interference is exact phase-labelled predecessor merging into common observation support with all source traces retained.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-SUPERPOSITION-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__phase-labelled-predecessor-merging__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`phase-labelled-predecessor-merging`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:6ca39507ce4ac95c7a3fc237a2694a20864de7c8054e59b4efded85e7acb3368`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:61f255d3b9a55856cf0f48b1e67feba27a0381851562b7f11689609142ee024c.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:fd5497bd6c92c00934809764fb5fb06e0b0b89053f4b9178917c650676f37a56.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:4a0f57ead6c024248f0960760259083e4d58d81e304758eae4dcfa2963683216.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:6488f7f50675f6119b15bfe58e27a1093eebd210f4168384e7fbb19376b1ca40.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:d60ddac9afa052b1c537ac69eff482d0be3eba6c7802fac752852e5c6d9a9ace. Independent certificate: sha256:0ccaf4891c616c47996b7881a390fa5561f34a1d1ba36511022dc21a8e4df570. Engine external-validation hash: sha256:a33ae95075474593b24408b26e1cd2bf8702547229f1c41516066a35ae0d7ae2.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- interference-1: predicted phase-interference-merging; observed phase-interference-merging; exact match True
- interference-2: predicted phase-interference-merging; observed phase-interference-merging; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause PHYS-S011 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: sha256:b1f836e9d34bec28c271f9978645c1659fa025aa48b0307a331d1d236b146f06. Isolation certificate: sha256:29225b9091ef0b1e4d246f75bfbb5319b06dc79b12c0c5a4e161cd820aabcab3. Custody certificate: sha256:a7905b366c782acff5e0ff0b37f05464097e07b6fbc4095b7b2797a55d739b5b.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- interference-1 from NIST-ASD-5.12
- interference-2 from NIST-ASD-5.12

Shared claim-record clause PHYS-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:2724f2bf17a9ad79c2a3b1668339e7944ba6806a94460bde1a727e519ca8c5f9; engine receipt `sha256:383616e13353085d904ad29e5ec14ef4012ac635457e5a86d1067266c7c9f586` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-INTERFERENCE-001-383616e13353085d.json`; empirical-validation hash `sha256:6a130d26e7141366b359dca64aa7aaea3f8fbc874bfe95833686d04e4ee08445`; measurement receipt `sha256:b1f836e9d34bec28c271f9978645c1659fa025aa48b0307a331d1d236b146f06`.

57. Diffraction at constrained support

Claim identity: `SFT-PHYS-WAVE-DIFFRACTION-001`

Shared claim-record clause `PHYS-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Constraining a wavefront removes propagation paths; complete successor generation from the retained aperture paths forces redistribution across every reachable output cell.

> *Diffraction is complete reachable-support regeneration from a constrained set of incident propagation paths.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__aperture-constrained-path-regeneration__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`aperture-constrained-path-regeneration`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:f7f38e60e91b3502f6eaddb7e98f1d9c5d6d406be27ac39ff197b3656d25b302`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bba07f5cb14a9f143451e64b7e4ed0521d51a4361bea4867c0cff3c17cc88532`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:004b18f3ae136055c5cc72e2ff8713666ae74bed318be4a7b3e774c9636363ed`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a8c972847fbb73f4a71f95a8b5345ea9606159ce6f9d80b66103bce3da98142b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:12ddb094a78e44398cc42ca75027c12655d74c384edec117c098e8dae4a59684`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8221e65a8a3ec1dc7622846aec9d52a0a6e961eb4d198c14ba09de5e9faf1b69`. Independent certificate: `sha256:6f251dc4cda533fb92ec13b262a73815ea0591429ec15e92f200aed0ad8dc027`. Engine external-validation hash: `sha256:ac1fdb380ec67269bf507ca78178e4734a1fce981de4ce5d465022aa43f4ec24`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- diffraction-1: predicted constrained-support-diffraction; observed constrained-support-diffraction; exact match True
- diffraction-2: predicted constrained-support-diffraction; observed constrained-support-diffraction; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:47026c78a6c5d78fe182089830f50f6ed94e14eaab4f3612436c94133338ec4f`. Isolation certificate: `sha256:b8e6644dfc38e2e9393f6d7208fdabcfa98a3d04f851914f4856970ab5d7a0cf`. Custody certificate: `sha256:4db4e1b11bf95e95da07a00c86e80a82564e4530825cb990afe1a711adc34ec2`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- diffraction-1 from NIST-ASD-5.12
- diffraction-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:74287acb05adc83404049b702a7fd733114a6e38f9b873bfbae1ef452f737ec5`; engine receipt `sha256:1f3027481808a8defa7f394581a65b27fd8bd1d46733f600da60e4c00cf17fd2` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-DIFFRACTION-001-1f3027481808a8de.json`; empirical-validation hash `sha256:fc04450854374deaf5de13422c048842ed231fedba10847305f0754ed417ad32`; measurement receipt `sha256:47026c78a6c5d78fe182089830f50f6ed94e14eaab4f3612436c94133338ec4f`.

58. Polarization as held transverse orientation

Claim identity: `SFT-PHYS-WAVE-POLARIZATION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Polarization is forced as the held cyclic orientation class of a transverse field recurrence relative to its propagation path.

> *Polarization is the held orientation of transverse recurrence support relative to wave propagation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-INFO-CHANNEL-CAPACITY-001`
- `SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-DIFFRACTION-001`

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__held-transverse-recurrence-orientation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`held-transverse-recurrence-orientation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:52cf1effe9c2204ca90f506e132858cbc9cdbd49d72c5c840baa565483e7d56f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:41cdd4f74dea4fb6becc4f167cc10573d48f3d6f9d5ade29432ef3ea78cb1c93`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ce6994b049e729471b20447aef4c60376ee5b8704c069dacdb456903a20c7e34`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1f733f085dd0a916578c1b1caae7c363800f0c258ec6aea122c73732dc60d512`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a9e2a076ae00e455be645d7c404747acee5880f3f5bb7674a828ebb80e500453`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:bde18730b7e73c8a93fa972423db6a074505a121a3c8b0a929f8a1d343650232`. Independent certificate: `sha256:81f3f282e0c3917b2bfbda79bd3bb7ce8644efc67b6c8a920e624b2994b4f702`. Engine external-validation hash: `sha256:993291c82385faaa60ab36cdc2660df7d2710c5c75b50fae9a7b04a07d9375ec`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- polarization-1: predicted transverse-held-polarization; observed transverse-held-polarization; exact match True
- polarization-2: predicted transverse-held-polarization; observed transverse-held-polarization; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:1333c41e679393a6f2d74d556cfdc02a85f6ea5e7dbe998de792ccf9defdba66`. Isolation certificate: `sha256:1e771056a22fc93f36334a77a6497cae392b8ba65d5d904be3f1eed8d7c31a79`. Custody certificate: `sha256:7349f812b01c6e2a30494b9185ab0dba6e4b328f4db6e170363f62c6879ce5be`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- polarization-1 from NIST-ASD-5.12
- polarization-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:e3e0cfc806eb855c6c5c363e52f178ea94a76a983f27b0174030a3d6646752af; engine receipt sha256:611f1b82de00ee945db78924f2951654cefbf5cdd65ac085e8899521b1601461 at receipts/engine/model_admitted/SFT-PHYS-WAVE-POLARIZATION-001-611f1b82de00ee94.json; empirical-validation hash sha256:8e2c976e416fe7f25806c6ae66b13c3fe9ecf085c08b4e711bb999536c047af3; measurement receipt sha256:1333c41e679393a6f2d74d556cfdc02a85f6ea5e7dbe998de792ccf9defdba66.

59. Dispersion and support-dependent propagation

Claim identity: SFT-PHYS-WAVE-DISPERSION-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. When support transition rules distinguish recurrence labels, each frequency class is forced to carry its own exact propagation ratio; their phase relations separate with distance.

> *Dispersion is label-dependent exact propagation induced by the registered support transition structure.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-POLARIZATION-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__frequency-labelled-support-response__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	frequency-labelled-support-response	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:c4ee18c8f7deb87089f0df6f86e1b46f5e491171d0ea92a85560eb674d0aa3a6. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2eba8297442d282c1c34126487ae4d5c2ac8f27b318b97212e55bf0fc9e203cf`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:50935063db3acaf93337341250dd63aef37f4dfb434e90242c843de5ca391af0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:cda9f78a03b2c8254ced238a1568c73e687eb09e119d3ae133d8849d704054cc`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c3a78ce30373f0e5c85cc27b15c81a1804d07b4db02289d2de65f52acf0c2e46`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c2ce37fc1e3f802f9683fc17d1faf3dbcbcbe30b6db49cd878ca1ccd5a5d9186`. Independent certificate: `sha256:8ad4fb98343d0ebfa422aefd8e8c1ca0b4bb0b4c729e98daf0abdd1d3f4e9fd3`. Engine external-validation hash: `sha256:4d35ff7e5a7937db6e758089364cf3987e770d3da2141c871bc4d419e1d04ed4`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- dispersion-1: predicted frequency-dependent-dispersion; observed frequency-dependent-dispersion; exact match True
- dispersion-2: predicted frequency-dependent-dispersion; observed frequency-dependent-dispersion; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:cb4ffed8c858a6e3fa36958cda4bc84e25b66b876a10e6adbfbfb603ea7246b6`. Isolation certificate: `sha256:3b1e691b3314423349090654e874718f1a529d7b8cfcf84d20a61c1040e7a615`. Custody certificate: `sha256:f4b0455b86a84d846650f50e3cdf23a5bcb35516c10945b9964912e8e5373bf2`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- dispersion-1 from NIST-ASD-5.12
- dispersion-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result

- target access or omitted adverse row

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:423973260a060ecb4f599a9b037742c7c865cc6cf66f64212d0da5867de8b77c; engine receipt sha256:bb15242d9bc421a4ca99ced8941720d9eff92d99c5346e9d2a8336b9ed64450e at receipts/engine/model_admitted/SFT-PHYS-WAVE-DISPERSION-001-bb15242d9bc421a4.json; empirical-validation hash sha256:bf2b3e7ec8e698834c4a8bccf45475fde313e693d0b89cfdab9bb6b262e40c20; measurement receipt sha256:cb4ffed8c858a6e3fa36958cda4bc84e25b66b876a10e6adbfbfb603ea7246b6.

60. Resonance and recurrence matching

Claim identity: SFT-PHYS-WAVE-RESONANCE-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A driven recurrence accumulates coherently only when driver and supported recurrence close on a common exact refinement; unmatched phases merge without persistent accumulation.

> *Resonance is exact common-refinement closure between driving and supported recurrences, producing repeated phase-aligned transfer.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-DISPERSION-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__common-refinement-recurrence-matching__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	common-refinement-recurrence-matching	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:57585ea5d006b16b32a9952e66dd0b0316a225861f24cbaf2bfa274a8441f883`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d2df52b8708ce52d02c5f9c8950d5ea9b5b2f9b272fa5f7be3f6a9c9200a7c4e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:fc794b574799a1058ffb08f667dc5a42ee0523191d34ae3fcec6bbb143f4e083`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:cc80fe73f2b44d6c50179e804afae5bd6155a2698bee3689f5982707b729a1ec`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4f8e5411052b53acc7dc47aa78d57b2aa010994dd757abfc6b3ddc9e0bc6440a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9c5906e5e5607427d0a3ea0a15716c7686f7a6b02ae1e2baa369e857a81c88e8`. Independent certificate: `sha256:e89d71e94981cf1e8eadbbae1d46929ddc31bb7a5ef70cbfa926c3f173ed58eb`. Engine external-validation hash: `sha256:ef05a42fc5f83adc46038aab41c38c4a0f5d9430476118631ce2260ce3ee21e9`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- resonance-1: predicted recurrence-matched-resonance; observed recurrence-matched-resonance; exact match True
- resonance-2: predicted recurrence-matched-resonance; observed recurrence-matched-resonance; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:789b8d2684180c8e0513c56ea5c048387613642f5bf62307bf578edd47ef1850`. Isolation certificate: `sha256:5f01851e95ac5f74cc75c3bedaff28f11bc68aa13353c0d0663f36e77c85ad7b`. Custody certificate: `sha256:9c8491f4b2c9876e4dbec7832c1a03fb4bb322521a2036ad5744accd7df20ad1`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- resonance-1 from NIST-ASD-5.12
- resonance-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave

- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:57c1f5180bf4fb3ed197818aba05e1bee22ff11011cfa2039ab8054e00bcd232; engine receipt sha256:60ae8de521a339460b1d6dc562d275ecba316910f45098981bd3e3b658ceb97a at receipts/engine/model_admitted/SFT-PHYS-WAVE-RESONANCE-001-60ae8de521a33946.json; empirical-validation hash sha256:82993d5af081df94cf563ff5e533d2c2970762cd18410bc43a493e00bc04abae; measurement receipt sha256:789b8d2684180c8e0513c56ea5c048387613642f5bf62307bf578edd47ef1850.

61. Wave energy and momentum transport

Claim identity: SFT-PHYS-WAVE-ENERGY-MOMENTUM-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A propagating recurrence that performs work at a remote response cell necessarily transports the corresponding energy carrier and held propagation momentum along its complete causal path.

> *Wave propagation transports exact energy and oriented momentum carriers, paired at source, path and response.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-RESONANCE-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__causal-wave-transfer-carrier__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	causal-wave-transfer-carrier	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:8116407710ae736a5f763d581577e639ca1a85168f10d85ec0ec1b886ce18c6a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:703a05f7937d3054ac10d8e2b66119064b82bf4a2891940671546eb6949d7fb4`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6f905acb05ac57cb4fd664d461b08ac448933f7adf404e432f6eaa911be54d65`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ba67bebcc0dfce85a8cc736d4cfd96f48b98a5617d1fe807d34091e25168e7cd`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:16137d5af11336ef1b1fdacf28e27fe6f63731cb2900c72fbe062bc6d96cea17`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:de27387e232a634fa0a663a09bcaf09a64290408013be504fd5727c5e36537d2`. Independent certificate: `sha256:e852817a61bb2659ee2d9b24e0d60b77e91ce8f0b8470271e302ab187c1d448e`. Engine external-validation hash: `sha256:c9849d53382f7612e05cf3e9b2c889870ad293aa1b2669d42be269674d0deb5d`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- energy-momentum-1: predicted wave-energy-momentum-transport; observed wave-energy-momentum-transport; exact match True
- energy-momentum-2: predicted wave-energy-momentum-transport; observed wave-energy-momentum-transport; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:01bdd5dc31a8190dffa9deb53cc4f160bcd2c7dd68821beaba1977673271a71a`. Isolation certificate: `sha256:500fc5d158eace3a38ed37465462ac53c9d1f4d9245b00cee90953f9b55ba939`. Custody certificate: `sha256:fa771e13bd06e16aa84746c13f68a3b8bfb135e23e4f6d094e1d49ced915df05`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- energy-momentum-1 from NIST-ASD-5.12
- energy-momentum-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:3f745bb0b2e2c67060f7170b3d59bcf31544f1ce855218db19265adf26f6eb2c`; engine receipt `sha256:e5984b8f7868064d5ceaa61b26a713dc9d128729fdaae80cafcdeb7986c16876` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-ENERGY-MOMENTUM-001-e5984b8f7868064d.json`; empirical-validation hash `sha256:171c10c3a77295894b6a5c26f9c954eebecaff02d16f2f7492b91ab5a1feeb4f`; measurement receipt `sha256:01bdd5dc31a8190dffa9deb53cc4f160bcd2c7dd68821beaba1977673271a71a`.

62. The second structural Fold generator is three

Claim identity: `SFT-PHYS-STRUCT-GENERATOR-THREE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The exact Fold period spectrum forces its second distinct nonidentity generator to be the positive count three: the binary generator is the first-return period of the unit part whose denominator is the positive predecessor of depth-two Fold support, and the least distinct successor period is witnessed by the corresponding depth-three unit part returning in exactly three Fold transitions.

> *The second distinct nonidentity Fold generator is the exact positive count three, witnessed by the complete first-return orbit 1/7 -> 2/7 -> 4/7 -> 1/7.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-COUNT-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-FOUNDATION-PART-EQUIVALENCE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`

Generated grammar and closure boundary. Generate the complete product of Fold carrier, action, recurrence, binary anchor, distinct successor, unit-part construction, orbit record, order closure, measurement direction and extra-rule forms. The declared exact boundary is: All exact positive first-return generator constructions available from finite Fold-word support, positive predecessors, Fold action, recurrence and positive-count succession. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `exact-positive-period-trace__double-and-cast-complete-One__complete-first-return-orbit__depth-two-unit-part-period__least-distinct-positive-successor__support-predecessor-unit-part__all-Fold-transitions-held__positive-count-discreteness-plus-existence__derivation-before-measurement__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`exact-positive-period-trace`	`borrowed-integer-name`	A borrowed numeral has no Fold recurrence trace.

Axis	Forced form	Eliminated form	Elimination reason
action	double-and-cast-complete-One	imported-modular-map	An imported modular operation is not the admitted Fold.
recurrence	complete-first-return-orbit	repeated-value-without-first-return	A repeated value can hide an earlier return.
binary	depth-two-unit-part-period	asserted-binary-count	Naming two does not derive its recurrence.
successor	least-distinct-positive-successor	selected-later-period	A later period is not minimal.
unit	support-predecessor-unit-part	target-denominator	A denominator read from an answer is target input.
orbit	all-Fold-transitions-held	period-label-only	A label cannot reproduce the return.
order	positive-count-discreteness-plus-existence	bounded-denominator-search	A denominator bound does not close later candidates.
measurement	derivation-before-measurement	measurement-selected-period	Measurement-to-derivation flow violates the admitted boundary.
extension	no-extra-rule	extra-generator-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:30d182b1a49310aca015977e10dc218b14cfc7d22f345c50d1685d17d600144d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:e990d2e429bd08054cf8b7dabd6f1ef1cd683f13016fd18b84c94b39c72c80ab`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:747052c401ddb2d214104d2c733bf23c0864fbe7b309ea5370b09358385d669e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:862121326d132cadaef270bd54d159d72e4b223683d7619ed55a95de8cbe7bc9`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:b44f622f2aca2e6f6f17edc34ee0b17e5e7acfb2c16ce2c030843ce1ffbf15ce`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d80e7a81337236193d23dadccc0cbd15fe7c57363ab5e1bbb0c0e021207bb460`. Independent certificate: `sha256:66c249178802488cd20c418f0d1c6f418b549fa541ed22f153d6d5593cbb5b27`. Engine external-validation hash: `sha256:1f4e2b977d4fb33ca84e5c4fbd7fe76a87a5a76bf8a34def76b2cc143b6db157`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- measured dimension, physical target or V2 proof artifact as a premise

- semantic numerical zero, negative, irrational, imaginary or floating proof value
- bounded denominator search standing in for positive-count closure
- selected later period or added generator rule

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:f9bc8995eb8ddcdcca64ebc77ada664c14c1f39b15327000d47c8d9c4b5be89e`; engine receipt `sha256:51fb37f85fc2774d3118fd08636204778a68aa8c908453ae788179e06de90b0a` at `receipts/engine/model_admitted/SFT-PHYS-STRUCT-GENERATOR-THREE-001-51fb37f85fc2774d.json`; empirical-validation hash `None`; measurement receipt `None`.

63. Three-dimensional Fold stability

Claim identity: `SFT-PHYS-SPACE-DIMENSION-THREE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A Fold-stable spatial carrier has exactly three independent generated directions: it must strictly exceed the two fibre roles so their return remains distinct, remain strictly within the four ordered pair-cell roles so a restoring return is retained, and the complete positive-count census of that open window has the sole member three, independently equal to the second Fold generator.

> *The unique Fold-stable spatial dimension count is three, independently equal to generator three.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`

Generated grammar and closure boundary. Generate every spatial stability form over carrier, fibre lower bound, pair-cell upper bound, interval status, positive-count census, spatial count, generator cross-lock, measurement direction and extra rule. The declared exact boundary is: All positive finite independent-direction counts classified by the Fold fibre lower boundary, ordered pair-cell closure boundary, return stability and generator-three cross-lock. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `independent-generated-direction-carrier__strictly-beyond-two-Fold-fibres__strictly-within-four-pair-cells__open-Fold-stability-window__complete-positive-count-enumeration__three-directions__equals-generator-three__forced-count-then-observed__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`independent-generated-direction-carrier`	`coordinate-name-only`	A coordinate name has no independence or incidence trace.
`lower`	`strictly-beyond-two-Fold-fibres`	`one-or-collapsed-fibre-count`	At or below the fibre count the common return is identified with a fibre role.
`upper`	`strictly-within-four-pair-cells`	`unbounded-coordinate-extension`	An unbounded extension adds roles not generated by one Fold closure.
`interval`	`open-Fold-stability-window`	`closed-or-one-sided-window`	Including two collapses return; including four exhausts the restoring remainder.
`census`	`complete-positive-count-enumeration`	`selected-dimension`	Selecting a familiar dimension is not forcing.

Axis	Forced form	Eliminated form	Elimination reason
value	three-directions	two-directions	Two does not remain beyond both Fold fibres.
crosslock	equals-generator-three	no-independent-lock	A single route cannot expose a changed stability bound.
measurement	forced-count-then-observed	space-count-read-from-measurement	Measurement cannot select the spatial law.
extension	no-extra-rule	extra-dimensional-model	An added manifold or conventional dimension rule supplies the answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:1732dde0be6280b0c19dd23d5784610945686b452c1fe4e8ec4a58952d237082`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:39008b82c2eb7d8b02749e1f886046630aed4e5affa12b68262488b73ed9eed0`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:a86c799b092cf1b396035106b1af9b0b17a95c5d7cc8faeef9bdb4cc80aa6438`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:07466e8d00c6c6eb839d1c6ea54149ccc6771bb4e6d9f62a8112929c702d9255`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:3627147e15586e2083b67ac167af616fad3d399da79173f407f0279cdd30b00d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:3766b1799f4000dca01577afdca82223675061fa7d71a8d597f37ecdfd83de76`. Independent certificate: `sha256:1c0056d44ff41cd15a9fb973b8f43b7f2979dcf4b31524c5509230c77f42270a`. Engine external-validation hash: `sha256:71e108c65e801d554a03f613c7ff905c6bfa30a9af04e635cac3881b0f1961fa`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- observed three-space fact or conventional manifold premise
- V2 proof artifact as a dependency
- semantic numerical zero, signed dimension or nonexact proof value
- target-selected interval endpoint or extra stability rule

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:a688ac8d58d7973d67e0c8489c4e554410018d0ebf26a795a4dc287daf1f1378`; engine receipt `sha256:1bcc9dbab55d17203193ff3cdced99f0013ba60768a4c42b4fbb98cd000bee6a` at `receipts/eng` ine/model_admitted/SFT-PHYS-SPACE-DIMENSION-THREE-001-1bcc9dbab55d1720.json; empirical-validation hash None; measurement receipt None.

64. Rank-two boundary of three-space

Claim identity: SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001

Shared claim-record clause PHYS-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For a complete boundary surrounding a localized source in the forced three-direction spatial carrier, holding one source-normal direction leaves exactly the unique positive predecessor rank two as free boundary organisation; scaled equivalent boundary support is therefore the complete pair-cell product.

> *The complete boundary of the forced three-direction spatial carrier has exact positive rank two and r-by-r pair-cell support at every generated positive scale r.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-FOUNDATION-COUNT-001
- SFT-FOUNDATION-PART-EQUIVALENCE-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001

Generated grammar and closure boundary. Generate the complete product of spatial carrier, source normal, boundary incidence, rank relation, rank value, scale composition, coverage, generality, measurement direction and extra-rule forms. The declared exact boundary is: All complete finite boundaries of a localized cell in the forced three-direction carrier, classified by one held normal direction and the positive predecessor relation on retained independent boundary directions. The generator produced 2048 named candidates and the decision support contains 2048 one-for-one decisions. Exactly one candidate survived: forced-three-direction-carrier__one-held-source-normal__all-nonnormal-directions-retained__unique-positive-predecessor__rank-two__complete-pair-cell-product__complete-equivalent-boundary-support__pair-product-successor-closure__rank-sealed-before-measurement__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	forced-three-direction-carrier	unbound-coordinate-space	An unbound coordinate collection has no forced dimension identity.
normal	one-held-source-normal	no-held-source-normal	Without a held normal relation the boundary is not source-relative.
incidence	all-nonnormal-directions-retained	selected-tangent-subset	A selected subset can omit a free boundary direction.
rank	unique-positive-predecessor	signed-dimension-minus-one	A signed scalar expression is not the positive proof witness.
value	rank-two	rank-one	One plus the held normal does not exhaust three directions.
scale	complete-pair-cell-product	linear-or-unlabelled-repetition	Unlabelled repetition loses one retained boundary coordinate.
coverage	complete-equivalent-boundary-support	partial-boundary-sample	A sample cannot establish complete geometric dilution.
generality	pair-product-successor-closure	fixed-scale-table	A finite table does not prove the next scale.
measurement	rank-sealed-before-measurement	measured-rank-input	That reverses the derivation-to-measurement boundary.
extension	no-extra-rule	extra-boundary-rule	An added shape or exponent is not supplied by the dependencies.

Base and successor. The closure proof is sealed by derivation hash `sha256:ac24158c4adf78dc87cc21121d575ae570de6794301469e0cedd5d07141e3277`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:9b50b4f4f1fca1e42ca15c89d7bf1f7c5f533d4484c44c9234deaa5927e87138`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:66896c9dc47e667d9cc368cba3ee78533675b5c097542a9efc91ce2509e114c4`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c0a38d668713c09944f2fce5ca02b238d428333fcb9cd43b4283e6c8f22a3649`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:a4c33833ff4802fc48fd89fd2116d367ce1e625e43cffbcf2732fb5fc3fa613a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:6e48fe721547bc6ea1ceb176a81258205bb3c8fa6afd7ada91f787143a911aa7`. Independent certificate: `sha256:503a0420c507563e3b0aadfe67448fb365824653e7436033d4a9e5e49b33d463`. Engine external-validation hash: `sha256:e12bae1d0ccdf3eb4362bd31c0b434e5d4c3c2c1959890d7bdb0503653fd46d5`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- measured inverse-power exponent as input
- conventional surface-area formula or continuum premise
- signed subtraction, semantic numerical zero or nonexact proof value
- partial boundary sample or added geometry

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:85b4fca1a33b226b1cefefd25a60254706b53de27787feb5c5bec4f3c81fbd51`; engine receipt `sha256:10f70a4a463a28ed545c101dcf7528b6d5a7b04355e3c89f3911f8b95514717d` at `receipts/engine/model_admitted/SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001-10f70a4a463a28ed.json`; empirical-validation hash `None`; measurement receipt `None`.

65. Inverse-square dilution from rank-two boundary support

Claim identity: `SFT-PHYS-FIELD-INVERSE-SQUARE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For one conserved source distributed without preference over every equivalent cell of the forced rank-two boundary, increasing source distance by an exact positive ratio r increases complete boundary support by r paired with r and therefore forces response per cell to be the source divided by r squared. The exponent is derived before and independently of measurement.

> *One conserved source over complete equivalent rank-two boundary support forces response(source,r) = source/(r*r) for every generated exact positive distance ratio r.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001`
- `SFT-PHYS-FIELD-SOURCE-RESPONSE-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete product of source identity, conservation, boundary rank, boundary coverage, scale transport, response allocation, falloff profile, generality, measurement direction, evidence record and extra-rule forms. The declared exact boundary is: All exact finite geometric-dilution relations for one conserved localized source over complete equivalent boundaries generated from the forced rank-two spatial boundary. The generator produced `4096` named candidates and the decision support contains `4096` one-for-one decisions. Exactly one candidate survived: `one-retained-source-carrier__same-complete-source-at-every-boundary__forced-rank-two-boundary__all-equivalent-pair-cells__distance-ratio-paired-with-itself__source-over-complete-boundary__inverse-square__pair-product-successor-induction__sealed-exponent-to-blind-measurement__source-bound-cell-and-measurement-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`source`	`one-retained-source-carrier`	`unbound-response-number`	A detached number has no conserved source identity.
`conservation`	`same-complete-source-at-every-boundary`	`distance-created-or-lost-source`	Changing total support violates the admitted conserved-source law.
`rank`	`forced-rank-two-boundary`	`free-or-measured-exponent`	A free or measured exponent reverses forcing.
`coverage`	`all-equivalent-pair-cells`	`selected-boundary-cells`	Selection breaks equivalence and conservation accounting.
`scale`	`distance-ratio-paired-with-itself`	`unlabelled-scale-factor`	An unlabelled factor loses one boundary coordinate.
`allocation`	`source-over-complete-boundary`	`fitted-response-profile`	A fit can select the target falloff.
`profile`	`inverse-square`	`inverse-linear`	Linear falloff omits one retained boundary coordinate.
`generality`	`pair-product-successor-induction`	`finite-distance-table`	A table has no successor proof.
`measurement`	`sealed-exponent-to-blind-measurement`	`measurement-to-exponent`	A measured exponent cannot become the law premise.
`record`	`source-bound-cell-and-measurement-trace`	`reported-exponent-only`	A label alone cannot reproduce conservation, cells or comparison.
`extension`	`no-extra-rule`	`extra-shape-or-scale`	An added coefficient or shape is not supplied by complete equivalent support.

Base and successor. The closure proof is sealed by derivation hash `sha256:7861b680c944ed9b62f56a91994f554c3c73ee66e8ec85e4d4bdcc71bbe945be`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:5126eae290f28749926e1532af4473ba49a075ffe7045e937a67ccb90b2c7afb`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:73ad9ff5259b5413d9f21ced60ef80c4b9652e711a726e2229f6000c1f2a571c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6757de96ac570cbdebc3ea1c952bf89e3f2e322455d2086a84a65816abb12181`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:e68c533619348505aa22d14cf741971fabc0ad820590878d5234ba19b9603915`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:dd451d0191f043210e3c24969aa8119c07c11f3b67f20c7a71d26f380376e63d`. Independent certificate: `sha256:8ecbdb6929c52e1cf21c8229755bab8685a4ccb896c90e4089352870edbf6ccb`. Engine external-validation hash: `sha256:59dba0de1ea36a629ce7c4973cedb502b8f5091f69a84f58e4ecc7296f4e8e12`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- measured exponent, fitted coefficient or conventional inverse-square premise
- partial boundary, omitted response row or preferred cell
- semantic numerical zero, negative, irrational, imaginary or floating proof value
- target access before the forced profile seal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:e933f774052fe6d86546b5033f2220c595025fc435fa68a5bf3076701946ff3b`; engine receipt `sha256:fd630f1572c8e211f410ddefbb26f84133b0c5d91de383d9461861cd44ddd195` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-INVERSE-SQUARE-001-fd630f1572c8e211.json`; empirical-validation hash `None`; measurement receipt `None`.

66. Complete prime-sector force and mediator ladder

Claim identity: `SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The least binary cover of generator-three successor volume fixes ceiling seven. Complete positive divisibility enumeration then leaves exactly prime sectors two, three, five and seven; each sector p has p charge labels, coupling (p-1)/p and exactly p-squared pair cells less the unique colourless return, hence mediator counts three, eight, twenty-four and forty-eight. The next prime eleven lies beyond the ceiling.

> *The complete prime-sector ladder is (2,3,5,7), with couplings (1/2,2/3,4/5,6/7), mediator counts (3,8,24,48), and first excluded prime 11.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORBIT-NUMBER-THEORY-002`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of sector carrier, dependencies, prime-ceiling relation, enumeration, minimality, measurement direction, evidence and extension forms. The declared exact boundary is: Every positive count through the independently generated upper cover depth, complete proper-divisor tests, complete sector pair cells and the unique colourless return. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-dependency-trace__prime-through-cover-ceiling-and-p-squared-less-One__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`borrowed-physical-label`	A borrowed label has no Fold trace.
`dependency`	`admitted-dependency-trace`	`asserted-input-values`	Asserted values are parameters.
`relation`	`prime-through-cover-ceiling-and-p-squared-less-One`	`composite-or-unbounded-sector-ladder`	The alternative loses a required distinction or adds an unforced rule.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected neighbourhood cannot prove uniqueness.
`minimality`	`all-predecessor-forms-rejected`	`survivor-without-predecessor-controls`	A survivor alone does not exclude a shorter form.
`measurement`	`derivation-sealed-before-comparison`	`target-visible-before-seal`	Target access reverses empirical direction.
`record`	`complete-trace-controls-and-census`	`answer-only-record`	An answer alone cannot reproduce the derivation.
`extension`	`no-extra-rule`	`free-extra-rule`	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:3ec46b64ac79eb7d5890fb933d8591ea0907d57bababe6855736a0fbb4af5301`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:fbc7fcd8393f763160ba4311295df1fbd4295f929c9d213611653f1d6cb54176`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:50ef75f995e5860dc10d502e63938d55b5edd6299dbe42475bf436a7aa349fee`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:193ea72fce01861503ea08ada27034731dd469a147bcf39a6a2772f1232697e3`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:7402ed46812fea45fa5d719b332b54e32704288ceb51128c7ecfdd0d11a3eb46`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b`. Independent certificate: `sha256:c907917461c9da17b81877c6d36f16e9ef6ccfc8116a9213c80b754c392f2f70`. Engine external-validation hash: `sha256:9b55fe2d38f9aba0b68df47435f6ff39c1d7c7467dbb775855f50378dacbd761`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:a05666ce9ce250c3a40e7f745a59a6e801ca7671082e33aef826d19a102b5c98`; engine receipt `sha256:1e22b2eda37dc8afe38d740cf10db286fb2e5f238704a22ccfee7066c8cbaa85` at `receipts/engine/model_admitted/SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002-1e22b2eda37dc8af.json`; empirical-validation hash `None`; measurement receipt `None`.

67. Complete finite prime-sector force and particle inventory

Claim identity: `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The forced prime ladder two, three, five and seven closes the entire registered force-sector grammar. For each sector p it fixes p charge kinds, p-squared less One mediators, coupling (p-1)/p, exact successor running, p-member and antipodal singlets, three-generation mixing shares and fermion mass-part 1/p. The first excluded prime eleven terminates the inventory.

> *Sectors (2,3,5,7) have charge counts (2,3,5,7), mediator counts (3,8,24,48), couplings (1/2,2/3,4/5,6/7), gauge-carrier total 83 and 72 penta/hepta carriers. Penta/hepta singlets have 5/7 constituents, mixing diagonals 14/15 and 20/21, electron mass-part ratios 2/5 and 2/7, and the complete named inventory has 98 members including 12 Smithions and photon/graviton/Higgs. Prime 11 is excluded.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002`
- `SFT-PHYS-STRONG-RUNNING-DIRECTION-002`
- `SFT-PHYS-MATTER-COMPOSITE-HADRONS-001`

- `SFT-PHYS-MATTER-MIXING-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete twelve-axis product of sector, charge, mediator, coupling, running, singlet, mixing, mass, inventory, measurement, prediction-record and extension forms. The declared exact boundary is: Every prime sector through the independently forced ceiling seven; all fibre members, ordered pair cells, finite positive supports, three generated positions, singlet closures and the first excluded prime. The generator produced `4096` named candidates and the decision support contains `4096` one-for-one decisions. Exactly one candidate survived: `complete-primes-through-forced-ceiling__complete-p-fibre-members__p-squared-pair-cells-less-One__p-predecessor-over-p__support-successor-in-sector-gap__complete-p-fibre-or-antipodal-pair__three-preimage-offset-over-sector__One-shortfall-equals-one-over-p__complete-count-and-first-excluded-prime__inventory-sealed-before-anchor-check__standing-falsifiable-penta-hepta-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
sector	complete-primes-through-forced-ceiling	selected-known-sectors	Selecting familiar sectors cannot close the ladder.
charge	complete-p-fibre-members	borrowed-group-dimension	A borrowed group label imports its answer.
mediator	p-squared-pair-cells-less-One	listed-particle-names	Names do not generate a complete inventory.
coupling	p-predecessor-over-p	measured-coupling	Measurement cannot select the law.
running	support-successor-in-sector-gap	imported-beta-function	An imported beta function is outside the grammar.
singlet	complete-p-fibre-or-antipodal-pair	imported-hadron-taxonomy	A taxonomy does not prove closure.
mixing	three-preimage-offset-over-sector	free-mixing-matrix	Free entries are parameters.
mass	One-shortfall-equals-one-over-p	invented-sector-scale	A new scale forks the unified carrier.
inventory	complete-count-and-first-excluded-prime	open-ended-particle-list	An open list contradicts the forced sector ceiling.
measurement	inventory-sealed-before-anchor-check	known-counts-visible-before-seal	That would restate observation.
record	standing-falsifiable-penta-hepta-record	omit-fringe-predictions	Omission would erase a forced consequence.
extension	no-extra-rule	free-extra-sector	An extra sector violates the cover ceiling.

Base and successor. The closure proof is sealed by derivation hash `sha256:4624b05862724e3285d9f74a45c3ef7142151d86147fc6fd42c25dde3783b8db`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:e0da3f1ad608075f24dc7c22e1ba300efc4afff20b1c929fd22f6e086dbf2a91`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:cbde686fb626e90fb2ac7838288cdb76d00b198b78063864afb39ae17b18c57e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d9b17a089155cdab4895ea44b5ad732c09127140b3743d6da0f6912bfa2fed2c`.

- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:e8e8c91307ad668750ead2ccc08f5eb36976d8c4801fa25ecdd7a5a5436a71f5`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:319a328268348bb63287e71b45d9eff654c873e1b3f7365f18be09d372169101`. Independent certificate: `sha256:15fd0ea3dd07b0c350fd55c00dc5494379e0a52136632de037958fd9fe6a80c2`. Engine external-validation hash: `sha256:a6724da44d789aa5533e609ddc0524c4b1ffc844db35b124d86e07f2f42f4c52`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule
- no omission of predicted penta/hepta sectors because they are presently unobserved
- no imported SU(n), Standard Model, grand-unification or hadron table as a proof premise

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:ed90e7c9cd5e1693f0cb76d19db5c57283b70bf94c1841ca79beec9910a10829`; engine receipt `sha256:ee6b271f050a867fbe8d5ddcd10e7dba7f7ccba227b4094efbd5c6e2b48741fb` at `receipts/engine/model_admitted/SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003-ee6b271f050a867f.json`; empirical-validation hash `None`; measurement receipt `None`.

68. Charge, Coulomb potential and Gauss closure

Claim identity: `SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A conserved held charge distributed over the forced rank-two boundary yields exact inverse-square response; multiplying by complete boundary growth reconstructs the source at every generated radius.

> *For source 1/8, radii 1/4 and 1/2 retain identical Gauss flux 1/8; fields are 2 and 1/2 and retained potentials are 1/2 and 3/4.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-INVERSE-SQUARE-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for charge, coulomb potential and gauss closure, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__held-charge-over-rank-two-boundary-with-source-return__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-continuum-object`	The object lacks a generated Fold support trace.
`dependency`	`admitted-V3-dependency-chain`	`asserted-prior-result`	A prior answer cannot act as a V3 premise.
`relation`	`held-charge-over-rank-two-boundary-with-source-return`	`imported-Coulomb-profile-or-free-exponent`	The alternative loses a required conservation or recurrence record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected candidate neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	A missing carrier could silently select the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access would reverse the empirical direction.
`record`	`complete-trace-and-controls`	`answer-only`	An answer without trace cannot be independently reproduced.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:5207acfe629eeb3a1b95422de5b03da37100b21be36667c71a8039d2135b933e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:aed03d64d05dd11ac2114c94f12eeed64e5fbf8a41ac0785d8c7534a2adbb8a6`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:1db42473facdb53ec56c2ea4e22126862ac3d0e1d7522af4d04004a727102a95`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a02635acf69c4a02339b1721a416e0629ae94925af07f044f567ebf0e9b41cbe`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:25ab4a9872d7deb14c1bf89d25eeb44b1ae26eeff1151ddb88f65cfd2a1b1bdc`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:7a418f941af171192cbb28438b7d35e08fc2db7203efb9e91f18f193d717e0d6`. Engine external-validation hash: `sha256:32f4b2183afd1bfba698748c23caea10a2eb5b631239a1f057cc7ccd0cd37c09`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:c881d6a01b046c3069e5b317d7a7da7c0ced55fc2f759078ec68e659befe3dd2`; engine receipt `sha256:f30356c4d6f5fc9478100b9626b8303088cead7845844e1dab28c744761b5197` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003-f30356c4d6f5fc94.json`; empirical-validation hash `None`; measurement receipt `None`.

69. Magnetic relativistic correction

Claim identity: `SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For exact sub-One speed, the motion-held square and positive electric remainder partition the One; the remainder commutes with the Fold and therefore supplies the magnetic correction as a dynamical projection of the electric carrier.

> *The correction is C(v)=One-v^2 as a positive retained part; at v=1/2 it is 3/4 and Fold(C)=1/2=One-Fold(v^2), likewise at v=1/3.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003`
- `SFT-PHYS-FIELD-MAGNETIC-001`
- `SFT-PHYS-SPACETIME-LIMIT-SPEED-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for magnetic relativistic correction, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__One-partition-by-speed-square-and-Fold-covariant-remainder__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-continuum-object`	The object lacks a generated Fold support trace.
`dependency`	`admitted-V3-dependency-chain`	`asserted-prior-result`	A prior answer cannot act as a V3 premise.
`relation`	`One-partition-by-speed-square-and-Fold-covariant-remainder`	`independent-magnetic-force-postulate`	The alternative loses a required conservation or recurrence record.

Axis	Forced form	Eliminated form	Elimination reason
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9b906c00219993001019af98ebbabb0b464f0ab1677b5ac42feecf983b9c69ae`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:fd58ff0c8faddc6b0127682f22c4ebd1ac5f8c72d78259c3b716b133d804e132`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:27588232bfaf00a92166f28965d32ef1fbba4636c5c126546b5c8594c192d178`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c4cf910271ff874cc8cde959f228e2b7023f1ebb13d6c1c262055bfa49051e52`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:f6341899b006c65deee611e514f5b58ad6c05bf74b11d24504d79e33157946ff`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:8f42c6b9b1037c7964c07564b53d896e18df6bfb2232af192997d2bfc8910a13`. Engine external-validation hash: `sha256:0b1c27f4c2b5b3cbb82c785e40d18796420620c4d59c743089fde471c056ed5b`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:e07df5d65e828be51bddc3e891f8596ac28e766e55a0ff318f06b6708897176b; engine receipt sha256:8039e9c6e59176d5f9a09a7796e95e95fdf2587b1abc870becfdd5d715fe45dc at receipts/engine/model_admitted/SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003-8039e9c6e59176d5.json; empirical-validation hash None; measurement receipt None.

70. Lorentz transfer partition

Claim identity: SFT-PHYS-FIELD-LORENTZ-TRANSFER-003

Shared claim-record clause PHYS-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Motion does not create a second unaccounted force: it holds the exact speed-square share of electric transfer while the positive remainder acts on the observed carrier, and both shares reconstruct the source force.

> *For electric force 1/4 and speed 1/2, retained force is 3/16 and motion-held share is 1/16; together they return 1/4.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003
- SFT-PHYS-MECH-FORCE-001
- SFT-PHYS-MECH-CONSERVATION-001

Generated grammar and closure boundary. Generate the complete eight-axis product for lorentz transfer partition, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-chain__electric-transfer-partitioned-by-motion-share__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	electric-transfer-partitioned-by-motion-share	unrelated-added-force	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:b05e985dccb063ceeb2c8d488f431c2d704d76517515de0f2577f48eabebe03a. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:6ba16b5ac831f6230b21f169c00a2c6421b2877b8a5a2509054bdc6fdf6876e6`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:57ce2f114d67fb08f7ed01a12d2a3e11e914a41ca46adc9b446baf4680d797c3`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1ab25ba309a9c883998b7016ca264e3f9266c5f540680afde7a5b6214ab10ccb`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:f43103f36ebeb11d01a281a15a18d16a4554ed54e047faeaa13d6ba187ae38fa`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:d94eff429d8b2e205c7d6052f7c03391bedf22267342de6e5ca75dc6cc6dd23e`. Engine external-validation hash: `sha256:02d87952ae8f1554631ea753e3cdbc1cf290092c0e0b1807efa1ab2432a26cdf`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:672db44f64f8909e47215811622cf8a83547dce97fe44a3e301419f1aace10bc`; engine receipt `sha256:353351d17f145e0f89d0db4a7250469284dd4bae4732c1961bf519bfcd5f211f` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-LORENTZ-TRANSFER-003-353351d17f145e0f.json`; empirical-validation hash `None`; measurement receipt `None`.

71. Planar Maxwell curvature closure

Claim identity: `SFT-PHYS-FIELD-MAXWELL-PLANAR-CLOSURE-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. On the forced rank-two boundary, two spatial axes each carry binary second-difference curvature while one temporal axis carries the same binary curvature; per-axis balance fixes wave speed-square at the One.

> *Planar spatial curvature is four, temporal curvature is two, total ratio is two and dimension-normalized speed-square is One.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FIELD-LORENTZ-TRANSFER-003
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001

Generated grammar and closure boundary. Generate the complete eight-axis product for planar maxwell curvature closure, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__two-spatial-binary-curvatures-balance-one-temporal-per-axis__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	two-spatial-binary-curvatures-balance-one-temporal-per-axis	imported-planar-wave-equation	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:f320aa88ea1c8d20a34b8726f84850e21a6b1ebf418454d0c6f9d711739ec19f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:eafd899a9d320fec212c4ab36f18c4a4efd682611fef094307d16dc5e1572ebf`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:306d2fa1aeb708f8c68d2b46d8a68a777b72679e9b4af54d77909a4385ff71cd`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3ccafd6fdb2f836afd17c5e6fa0162cb1d033ed59894e8163efd758405fc5c94`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:74de44412fc44a7fb00345487361b8f0666807a320593e4416a269499189fc7c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:ed9c988d44a697e889d81581c33591eafc4b943c8b7ba4a5ccff6a9fb9304b01`. Engine

external-validation hash:
`sha256:1f5ba9cfadf5d82ec537e31ce590557050804fae5454fe6ab3c672352a838e14`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:1538c42ddea5976e24c0826c734aa61011c750a8b02babef778576e80d51f1f0`; engine receipt
`sha256:7178eac96fce3c977c64c4e4a9d5e681f1e48a91d8561045c6020979a476460b` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-MAXWELL-PLANAR-CLOSURE-003-7178eac96fce3c97.json`; empirical-validation hash `None`; measurement receipt `None`.

72. Three-space Maxwell curvature closure

Claim identity: `SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. In forced three-space, three spatial axes each carry binary second-difference curvature while one temporal axis carries the same binary curvature; dimension-normalised balance again fixes speed-square at the One.

> *Three-space spatial curvature is six, temporal curvature is two, total ratio is three and dimension-normalized speed-square is One.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-FIELD-MAXWELL-PLANAR-CLOSURE-003`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for three-space maxwell curvature closure, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__three-spatial-binary-curvatures-balance-one-temporal-per-axis__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	three-spatial-binary-curvatures-balance-one-temporal-per-axis	imported-three-space-wave-equation	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:f4339ba7d2812e070e981e1a9a6cb9f48a34c34e88cb052d3b16d45ffd9dccc3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:ca56f3c99452b5d544cec2cbaef3c385a2cb3ddb8743e8ed6a9689e41f3fac56`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:6886ae9dd1982adaa62e0dae2aaa30a09aa9c47879ae4a814b08d54a8b9a8675`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3bea7400d97d0d0cbc4be33441ab8f200262d450398d7dc457967dd9a94a7a1c`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:d7e4f910ed511c3cf4e320979381cab03750daac9d401bb892c4138b28b34b49`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:58bc2bc515de600a5e83bacd5bef7451a96cb7dd6d3cc19c1c301e29849cc979`. Engine external-validation hash: `sha256:0a4a7e9bcdb13ebfe2ad93024e7952f111770f7357c74515e8162ae1f2114caf`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor

- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:7d91ff057b317ebf0105a8b7789182eb010f224ec331528c6a0f321ab99a1010`; engine receipt `sha256:e9314b2725d52aa3136325aee8ba08bca5fbc41e0ee3ee19db391ff98b7c1dac` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003-e9314b2725d52aa3.json`; empirical-validation hash `None`; measurement receipt `None`.

73. Exact wave, polarization, interference and nonlinear operations

Claim identity: `SFT-PHYS-WAVE-EXACT-OPERATIONS-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The joint electromagnetic Fold carrier propagates one One per tick, retains two transverse held labels, merges two half-One predecessors into bright One or an empty dark record by orientation, and generates sum, positive-difference and harmonic nonlinear frequencies by exact Fold operations.

> *Eight One-speed ticks retain phase exactly; polarization has two held transverse labels; equal half carriers give bright One or an empty dark record; 1/3 and 1/4 mix to 7/12, 1/12 and second harmonic 1/2, while Kerr witnesses also return 1/2.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-PHYS-WAVE-POLARIZATION-001`
- `SFT-PHYS-WAVE-RESONANCE-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for exact wave, polarization, interference and nonlinear operations, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__One-speed-walk-held-polarization-predecessor-merge-and-exact-mixing__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-continuum-object`	The object lacks a generated Fold support trace.
`dependency`	`admitted-V3-dependency-chain`	`asserted-prior-result`	A prior answer cannot act as a V3 premise.
`relation`	`One-speed-walk-held-polarization-predecessor-merge-and-exact-mixing`	`imported-amplitude-or-nonlinear-susceptibility`	The alternative loses a required conservation or recurrence record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected candidate neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	A missing carrier could silently select the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access would reverse the empirical direction.

Axis	Forced form	Eliminated form	Elimination reason
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:f1a9fd69f364f55a9d34563887149d0d9e08a914231314207618eadba821b4c5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:da1305278f885cee1927f9a78e3dabf323d17480590bac0f8db653ded1bca5b6`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:e85758931f7806bba8989aa4528d1e48278116ffde19d580946b04dd92285d43`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:68a2392e4b4cf80ccaa7c1dcc5c1f691115996e1fc6038100bce876165a5fce0`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:c50fcba605905e42479774ebd71bb662f6ceb2c3915aa07bcb9a91fef5570fa0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:de299aa4bb3048d1dbffc9b0d2ed4302438f6c808ef0f9272afb19f3dd02b2f2`. Engine external-validation hash: `sha256:0f40930d8dd2d4b8acbbdf2f71dbe44827bf4626a6c805cdc4fec8697bbad61b`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:a95c36f5d6be268d83a9555126fbb62c7e647452ee14ba7c72ca938cd9f009f6`; engine receipt `sha256:bc9f7fff3079a79ca77a60d0680da1b8474ad13f0effedbf8b5cce9a75513e03` at `receipts/engine/model_admitted/SFT-PHYS-WAVE-EXACT-OPERATIONS-003-bc9f7fff3079a79c.json`; empirical-validation hash `None`; measurement receipt `None`.

74. Floored finite-loop closure

Claim identity: `SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Every generated physical loop support has positive finite Fold depth; summing its complete exact rational terms therefore yields a finite rational receipt, while an ungenerated continuum or unlimited-depth completion is outside the admitted claim.

> *The depth-n binary loop sum is the exact positive finite sum 1/2+1/4+...+1/2^n; depths one through six give 1/2, 3/4, 7/8, 15/16, 31/32, 63/64.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-WAVE-EXACT-OPERATIONS-003`
- `SFT-PHYS-VACUUM-ODD-RECURRENCE-003`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for floored finite-loop closure, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__complete-finite-depth-exact-rational-loop-sum__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-continuum-object`	The object lacks a generated Fold support trace.
`dependency`	`admitted-V3-dependency-chain`	`asserted-prior-result`	A prior answer cannot act as a V3 premise.
`relation`	`complete-finite-depth-exact-rational-loop-sum`	`continuum-divergence-and-fitted-counterterm`	The alternative loses a required conservation or recurrence record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected candidate neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	A missing carrier could silently select the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access would reverse the empirical direction.
`record`	`complete-trace-and-controls`	`answer-only`	An answer without trace cannot be independently reproduced.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9cdbf8e9691d002cee7c951d56edef310b246ca461dce7caab05c8113ba368f4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:643fde1c52ea524dff094a8fb8119cf4e8b1d12be431699739bf7d2213355340`.

- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:160cb7c4a05a174ca6869795f4f8326f525fc3a526deff446b6d42db9af5aecf`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f40becab2c8048726c80864bb8da877019405784e6fbf5dd8fedef512446ffd4`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:1f71dea4ae509caa698e6abeb41c08f7ce9de7649a8cd002dadc8cf54ba06390`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:971a26b748962889407d161cfb2a71d5ad12118f7b6c44aff5b8ff971a766439`. Engine external-validation hash: `sha256:882dedf3e1de603d4c8c959477d562d7eee7523f66a3aef199521e8662543ed5`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed
- no claim about an actually completed infinite support or a measured radiative coefficient

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:8e0e61bb5a069e723c60edea5efa42d7f06c7368db6eeb3b7c5f2c103adef023`; engine receipt `sha256:654937a341dfef2a5efc3ae1094a90b940e3535923bc1aa465d64e8930549853` at `receipts/engine/model_admitted/SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003-654937a341dfef2a.json`; empirical-validation hash `None`; measurement receipt `None`.

75. Exact bounded-cavity recurrence and discrete mode law

Claim identity: `SFT-PHYS-TESLA-BOUNDED-CAVITY-078`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every positive finite cavity length q, the least complete outward-return period is 2q cells. Every generated positive mode n has exact period 2qn, so distinct supported modes form a discrete positive whole-indexed family. Appending one path cell changes the round trip from 2q to 2(q+1) while preserving bounded recurrence.

> *For every positive finite cavity length q, the least complete outward-return period is 2q cells. Every generated positive mode n has exact period 2qn, so distinct supported modes form a discrete positive whole-indexed family. Appending one path cell changes the round trip from 2q to 2(q+1) while preserving bounded recurrence.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-WAVE-RESONANCE-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of finite carrier, held boundary, recurrence, enumeration, minimality, measurement direction, record and extension forms. The declared exact boundary is: Every positive finite connected path with two held boundary roles, every positive supplied mode depth and all 256 registered structural alternatives. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `finite-exact-Fold-path__held-endpoint-roles__two-boundary-outward-return-and-whole-mode-recurrence__complete-registered-product__every-required-carrier-retained__formal-seal-before-comparison__complete-trace-ledger__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	finite-exact-Fold-path	borrowed-continuum-wave	A continuum wave is not generated by the Fold grammar.
boundary	held-endpoint-roles	unlabelled-endpoints	Unlabelled endpoints cannot force a reflection or role change.
relation	two-boundary-outward-return-and-whole-mode-recurrence	imported-wave-formula	An imported formula cannot be a clean-room survivor.
enumeration	complete-registered-product	selected-examples	Examples do not establish uniqueness.
minimality	every-required-carrier-retained	omitted-required-carrier	Omission breaks reconstruction or boundary closure.
measurement	formal-seal-before-comparison	target-selected-relation	Target selection is fitted correspondence.
record	complete-trace-ledger	favourable-only-record	Selective retention cannot test the law.
extension	no-extra-rule	free-extra-rule	An added exception is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:2d0ee4703086312d8addaf77b83fc0d70ee7e2ce301303ca7e67082640ea3f47`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:9daa906efda4e214f382d2ff60ced91f26d6c125bd44a515f33d416be0e96c51`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:a53019f1b2dcdba8e1d3fbbeaea704096d13ddb82de9c5ef79850fc78d0640f6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:53e6ea51c75355d278acd420203685cdf6bd61fb48be4cdae95e52dc3f229626`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:9daef0df1ace8c42f05849c3aee5c15142d1296d58eb45c7f7bc0bb0c43a1c69`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash:

`sha256:7baa02a0c59d576446526fee08fb0c26cdddaaf950950814ad1e70b3eb12f0b0`. Independent certificate: `sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60`. Engine external-validation hash: `sha256:cbcf8d6e76d606f3a6f253b57f44c0dd6f1ff2393997f96b0089ee4d52bc09be`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional wave equation, trigonometric continuum, fitted frequency or measured geometry selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude
- no apparatus efficiency, Earth frequency or historical reputation used as a derivational premise
- no omitted transfer, loss, boundary, polarization or adverse record

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:c4a088faa9008e93789374b362bb984dcc961cbf08f2b10adf2c5bf6335830f4`; engine receipt `sha256:52164b45a1cd9eac61d3fd2850d10dd144a0fc6d2699520bd10d166325f3126e` at `receipts/engine/model_admitted/SFT-PHYS-TESLA-BOUNDED-CAVITY-078-52164b45a1cd9eac.json`; empirical-validation hash `None`; measurement receipt `None`.

76. Exact odd-quarter-wave and odd-harmonic law

Claim identity: `SFT-PHYS-TESLA-ODD-QUARTER-WAVE-079`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every positive whole mode n, opposed node/antinode endpoint roles require exactly 2n Take One quarter-cycle acts. The generated sequence is 1,3,5,...; therefore a quarter-wave support carries the fundamental and positive odd harmonics, while every even-harmonic endpoint has the wrong held role.

> *For every positive whole mode n, opposed node/antinode endpoint roles require exactly 2n Take One quarter-cycle acts. The generated sequence is 1,3,5,...; therefore a quarter-wave support carries the fundamental and positive odd harmonics, while every even-harmonic endpoint has the wrong held role.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-TESLA-BOUNDED-CAVITY-078`
- `SFT-PHYS-WAVE-EXACT-OPERATIONS-003`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of finite carrier, opposed endpoint roles, quarter recurrence, enumeration, minimality, measurement direction, record and extension forms. The declared exact boundary is: Every positive supplied mode count, both opposed held endpoint roles, its complete quarter-cycle successor certificate and all 256 registered structural alternatives. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `finite-exact-Fold-path__held-endpoint-roles__opposed-roles-force-two-n-take-One-quarter-count__complete-registered-product__every-required-carrier-retained__formal-seal-before-comparison__complete-trace-ledger__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	finite-exact-Fold-path	borrowed-continuum-wave	A continuum wave is not generated by the Fold grammar.
boundary	held-endpoint-roles	unlabelled-endpoints	Unlabelled endpoints cannot force a reflection or role change.
relation	opposed-roles-force-two-n-take-One-quarter-count	imported-wave-formula	An imported formula cannot be a clean-room survivor.
enumeration	complete-registered-product	selected-examples	Examples do not establish uniqueness.
minimality	every-required-carrier-retained	omitted-required-carrier	Omission breaks reconstruction or boundary closure.
measurement	formal-seal-before-comparison	target-selected-relation	Target selection is fitted correspondence.
record	complete-trace-ledger	favourable-only-record	Selective retention cannot test the law.
extension	no-extra-rule	free-extra-rule	An added exception is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:ad3d7761634d4038a47871bce045a636f4597d9e72768d836fa9b8a6cd2f2f73`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:3ab60a717fb67e9e086d4abcfb15dbc484aa0930eb1e9adb5e2aa1189622ee07`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:019fc6a7eb2da5a13d9e8b67f3d506e2711b6e9f1ba74947a2cf4e5f6b9ac914`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:99d8c3819c74b2cfcec816914d5f07aa22355e80130cbe1be5b15344611d1faa`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:376309fed4fe2f29054f927a338301e1a1cfefc53b63f82195cf2e17f8b9cebb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:7baa02a0c59d576446526fee08fb0c26cdddaaf950950814ad1e70b3eb12f0b0`. Independent certificate: `sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60`. Engine external-validation hash: `sha256:178c35ad74ec0eaa4c1915d75d933220358b0c48d2ae9b2c2f2e77802fffa885`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional wave equation, trigonometric continuum, fitted frequency or measured geometry selecting a survivor

- no semantic numerical zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude
- no apparatus efficiency, Earth frequency or historical reputation used as a derivational premise
- no omitted transfer, loss, boundary, polarization or adverse record

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:7b46f7adc3ce1bde616212a65ad974e39e3fb664a20a677b61e2b1a0327e4742`; engine receipt `sha256:3f1341a5b993e042ffcea6cc7220100e8ee8ab98c59aafc3694426381e30c83c` at `receipts/engine/model_admitted/SFT-PHYS-TESLA-ODD-QUARTER-WAVE-079-3f1341a5b993e042.json`; empirical-validation hash `None`; measurement receipt `None`.

77. Common recurrence with distinct longitudinal and transverse modes

Claim identity: `SFT-PHYS-TESLA-LONGITUDINAL-TRANSVERSE-080`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Forced rank-three space decomposes relative to a held propagation path into one longitudinal role and two transverse roles. All three carry the same exact ordered source-return recurrence while retaining distinct orientation labels. This is a common recurrence theorem, not an identity of electromagnetic, acoustic or material substances.

> *Forced rank-three space decomposes relative to a held propagation path into one longitudinal role and two transverse roles. All three carry the same exact ordered source-return recurrence while retaining distinct orientation labels. This is a common recurrence theorem, not an identity of electromagnetic, acoustic or material substances.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-TESLA-ODD-QUARTER-WAVE-079`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-WAVE-POLARIZATION-001`
- `SFT-MAT-CRYST-PHONON-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of exact path, held orientation, common recurrence, enumeration, minimality, measurement direction, record and extension forms. The declared exact boundary is: The admitted three-space rank, its one propagation role and two boundary-orientation roles, every positive finite recurrence depth and all 256 registered structural alternatives. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `finite-exact-Fold-path__held-endpoint-roles__one-recurrence-with-one-longitudinal-and-two-transverse-held-modes__complete-registered-product__every-required-carrier-retained__formal-seal-before-comparison__complete-trace-ledger__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`finite-exact-Fold-path`	`borrowed-continuum-wave`	A continuum wave is not generated by the Fold grammar.
`boundary`	`held-endpoint-roles`	`unlabelled-endpoints`	Unlabelled endpoints cannot force a reflection or role change.
`relation`	`one-recurrence-with-one-longitudinal-and-two-transverse-held-modes`	`imported-wave-formula`	An imported formula cannot be a clean-room survivor.
`enumeration`	`complete-registered-product`	`selected-examples`	Examples do not establish uniqueness.
`minimality`	`every-required-carrier-retained`	`omitted-required-carrier`	Omission breaks reconstruction or boundary closure.

Axis	Forced form	Eliminated form	Elimination reason
measurement	formal-seal-before-comparison	target-selected-relation	Target selection is fitted correspondence.
record	complete-trace-ledger	favourable-only-record	Selective retention cannot test the law.
extension	no-extra-rule	free-extra-rule	An added exception is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:5fd7136b37430b8af3a7178a7a31771cf93fb1923f7975c3ac03833139c09c85. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:37038936e871d7a91741999b8a75f913b6996a9ea739b58a8ce613e0c5cbb0d5.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:1722425a5ed9ce5656cc405677ee8c13bf9dc0f6afb171f2d921f971a50adc5f.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:3cb3dd7789def060053ab9fb56d2a983b681bed566c538ccd4802584f5b10588.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:0e234f043df56f051e9e5f8c50c54d7942ffd40c8a73f6853cc71aca1e8f2bb8.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:7baa02a0c59d576446526fee08fb0c26cdddaaf950950814ad1e70b3eb12f0b0. Independent certificate: sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60. Engine external-validation hash: sha256:6316ecb955e84506886cc7519c3f30e1dd9863ffca1dc9b3b36922003cb5ec5b.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S010 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional wave equation, trigonometric continuum, fitted frequency or measured geometry selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude
- no apparatus efficiency, Earth frequency or historical reputation used as a derivational premise
- no omitted transfer, loss, boundary, polarization or adverse record

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:deb47244f752db79b13e3509aa2e7cb99a057e9395eb8e97217233f093df21ba; engine receipt sha256:3c7be179aa4e991f5bf66fcb926a51b3660f9d11e9c70216ea0cf4e75fe91a09 at receipts/eng `ine/model_admitted/SFT-PHYS-TESLA-LONGITUDINAL-TRANSVERSE-080-3c7be179aa4e991f.json`; empirical-validation hash `None`; measurement receipt `None`.

78. Exact resonant connected-path transfer and complete energy ledger

Claim identity: `SFT-PHYS-TESLA-RESONANT-TRANSFER-081`

Shared claim-record clause `PHYS-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every positive finite connected resonant path, phase-matched driving reaches every cell within the path count and may deliver positive support to a held receiver. At every act, exact input equals the complete positive cavity-retained, delivered and loss carriers. The law forces transfer possibility and accounting, not a universal apparatus efficiency or an Earth-specific delivered power.

> *For every positive finite connected resonant path, phase-matched driving reaches every cell within the path count and may deliver positive support to a held receiver. At every act, exact input equals the complete positive cavity-retained, delivered and loss carriers. The law forces transfer possibility and accounting, not a universal apparatus efficiency or an Earth-specific delivered power.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-TESLA-LONGITUDINAL-TRANSVERSE-080`
- `SFT-PHYS-WAVE-RESONANCE-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MECH-POWER-001`
- `SFT-PHYS-THERMO-FIRST-LAW-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of connected carrier, held endpoints, phase-matched transfer, enumeration, minimality, measurement direction, complete ledger and extension forms. The declared exact boundary is: Every positive finite connected path, every exact positive input partition into cavity/delivered/loss carriers, every phase-matched recurrence successor and all 256 structural alternatives. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `finite-exact-Fold-path__held-endpoint-roles__phase-matched-connected-reach-with-complete-transfer-ledger__complete-registered-product__every-required-carrier-retained__formal-seal-before-comparison__complete-trace-ledger__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`finite-exact-Fold-path`	`borrowed-continuum-wave`	A continuum wave is not generated by the Fold grammar.
`boundary`	`held-endpoint-roles`	`unlabelled-endpoints`	Unlabelled endpoints cannot force a reflection or role change.
`relation`	`phase-matched-connected-reach-with-complete-transfer-ledger`	`imported-wave-formula`	An imported formula cannot be a clean-room survivor.
`enumeration`	`complete-registered-product`	`selected-examples`	Examples do not establish uniqueness.
`minimality`	`every-required-carrier-retained`	`omitted-required-carrier`	Omission breaks reconstruction or boundary closure.
`measurement`	`formal-seal-before-comparison`	`target-selected-relation`	Target selection is fitted correspondence.
`record`	`complete-trace-ledger`	`favourable-only-record`	Selective retention cannot test the law.
`extension`	`no-extra-rule`	`free-extra-rule`	An added exception is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:2ee07aa20f6a083d1e397ca8a6c611b3662ae4c5ea47231df703aaa2a92c66dc`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:78e0b34f0601abd57b697d7bbb588a4b965c5f5c40a81f03003237660662eb51`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:7265803634459735cab95decb1cbef8a065d9e48caf64c76a3f51dbc5e3c87db`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f9b86b4d5c5843c1f1f3b323bf8ead550e023ad0d32d4337f8cd67429dadd67e`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:094e767fd20bdbce8f4029eef29ec0d7bb018bc450807c9df842f7d415d45542`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:08f8bbe45e8c6df5f7618eef4575748ebc2b37e19e53daf09db4ba245721ddcf`. Independent certificate: `sha256:105cc5efc588cd631f6025f3bcd6afc16ea5c025723dacdcc1f719c372ee4c37`. Engine external-validation hash: `sha256:f9edb5b7a36c008bf81f008f54062a6452ac071377d7bcb85da26f4a574f92fc`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional wave equation, trigonometric continuum, fitted frequency or measured geometry selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude
- no apparatus efficiency, Earth frequency or historical reputation used as a derivational premise
- no omitted transfer, loss, boundary, polarization or adverse record

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:da3d89ad997983e2c05aa6810f88e2d11d2a0e7086ffef5b6ee2cca2bb244a29`; engine receipt `sha256:33dbb3b402878cfd1553f2eaf0d25d1df3194bd29cfe390d43e7a13facde069c` at `receipts/engine/model_admitted/SFT-PHYS-TESLA-RESONANT-TRANSFER-081-33dbb3b402878cfd.json`; empirical-validation hash `None`; measurement receipt `None`.

15. Thermodynamics Vacuum

Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced.

79. Microstate-to-macrostate observation

Claim identity: `SFT-PHYS-THERMO-MICRO-MACRO-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A thermodynamic macrostate is forced as one exact observation class of a completely generated finite microstate support; its unresolved multiplicity is retained as the class fibre.

> *A macrostate is an observation fibre over complete finite microstate support, with exact multiplicity and retained provenance.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-microstate-observation-partition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`finite-microstate-observation-partition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:e7f1e49e23b80ac82398be5887164473e3f9785f1c45a5d378c206d8c43eea2a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c65afd05be1ff96daaf309433e10c3c85d371e4ea512f710f549859f68a4e929`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:214504c9124b489399cc4a308931c0f822a1bbd5a24e492d83ffab44306bccde`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a78d1727daba665726753bb0b20c97d621b23f07d07deabb555620c2535cd011`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:773249981ef18b98d2128f5e7ce4e000e212289760660ea99eeddab18dc06bf1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ff94c98d730d930a00e95a403b5d023e29d8cf70e5e516a7e5f2d03ebe47b6a5`. Independent certificate: `sha256:c245399cfd8062c8a754212fb79e5e6f01f976b0bc62de9ff139118d008eeb03`. Engine external-validation hash: `sha256:29482cd99c4cf3b9c39913446b62536f897d08f32edafceb6a741a791300050f`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- micro-macro-1: predicted finite-micro-macro-observation; observed finite-micro-macro-observation; exact match True
- micro-macro-2: predicted finite-micro-macro-observation; observed finite-micro-macro-observation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:0f08337dd7fb1c68c028ebfc42bf541e8f206203bf5b2caea9a84988e326f317`. Isolation certificate: `sha256:d2cf21b388b4be7a199c2e51e029b05c24f0dc1c18e7650c48411854b3adb77a`. Custody certificate: `sha256:8ac9f4fd32e6f375d7bd6ea266dd1054e2db986b4236a5e08406ccdb5fea0000`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- micro-macro-1 from IAPWS-RELEASES-2026-07-23
- micro-macro-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:c5083ac8b1e577443c2449a991b8ef2cd9723b5b7713169620afbb41d534f7cd`; engine receipt `sha256:8b653cd127559fdfc1c7e47ab71c2f17d32f4de088c045a305e44a00abe4bc27` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-MICRO-MACRO-001-8b653cd127559fdf.json`;

empirical-validation hash `sha256:89466b24eb6d328905931655c718ecbd0e72f3c9e3904eabff23006f381fbac4`; measurement receipt `sha256:0f08337dd7fb1c68c028ebfc42bf541e8f206203bf5b2caea9a84988e326f317`.

80. Temperature carrier and thermal ordering

Claim identity: `SFT-PHYS-THERMO-TEMPERATURE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Temperature is forced as the exact ordering carrier comparing energy-support change with accessible-state multiplicity change relative to one held thermal reference.

> *Temperature is an exact reference-relative order of energy support per accessible-state change, not an imported continuum coordinate.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-MICRO-MACRO-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__energy-support-to-accessible-state-order__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`energy-support-to-accessible-state-order`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:93c47074658ac8480c829c0f8d695ef7be306f2be45fc7e1353afe6e23b9931b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:35041c1fc11ce723d4560fe772069cea9216e1ef156e73cdbb47a18788d13330`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2956553dd7226e7cceeceec10db3a4024e4baf3ca57ee01ae13d7df370aadb3a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d51c89c2466ecb996b33557ce6855bab5fc11ff8b4192e2bdc31e2aa64e9cb78`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:862d8898fc88b7eaad0379763249ab5281a3c34b9a5884ded00a786707a0ba27`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:47726e82008630482d9781ed4e574ab970284d5e510faaa8965ef1a8d2e85d25`. Independent certificate: `sha256:ec2560a3be4715f87c1b8831a4f9103a04ee602088163e87d10b20fbb79208f2`. Engine external-validation hash: `sha256:34feb7b353710d9d9048ee63ee0613fdb84740b01d2a7ea39b2d3c3ace2200c2`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- temperature-1: predicted exact-thermal-order-carrier; observed exact-thermal-order-carrier; exact match True
- temperature-2: predicted exact-thermal-order-carrier; observed exact-thermal-order-carrier; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:263132e9e112673a7538c0acdf9d56c2d146ed3c61df81eb44c80d84da59aa2a`. Isolation certificate: `sha256:1d7b435e854006b01006c281723be88cbe66d4cbd6ff14390ad2c6c613321a79`. Custody certificate: `sha256:2e7ff9cb32cba6d546079d7b38015e6b7cfea939be7b27b0bf374f8df91035bc`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- temperature-1 from IAPWS-RELEASES-2026-07-23
- temperature-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:07cdedbd40e452fd0148df9ac70afb55bbe1ba120c6b6f5e844d818edc4ee841`; engine receipt `sha256:120247c26273364ae2733af59411df4ff71663c9a06bf477f3902e0b13f3f304` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-TEMPERATURE-001-120247c26273364a.json`; empirical-validation hash `sha256:172db803b1c7d9591d25dfa490da8b5644b0ea883e8a22bc43d84ae26c183ff4`; measurement receipt `sha256:263132e9e112673a7538c0acdf9d56c2d146ed3c61df81eb44c80d84da59aa2a`.

81. Thermal equilibrium equivalence

Claim identity: `SFT-PHYS-THERMO-EQUILIBRIUM-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Two coupled supports are in thermal equilibrium exactly when reciprocal energy exchanges preserve their separate macro-observation classes and common temperature ordering.

> *Thermal equilibrium is exact equivalence under reciprocal energy exchange at a shared thermal order.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-TEMPERATURE-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__reciprocal-exchange-observation-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`reciprocal-exchange-observation-equivalence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:fd9da74674ddea0f6164631c4b6b4948fdf1acad2216bf55437a1274acc36c53`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1b728c43b61ae39d3405b5c186402610c4eff158d5505ab6fa86eea0692af359`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a51ff7be669db5438ee2d5a09c824f80ad3d9c09bb648c58a7d1fcdefb60223b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:384ad212a4e04dec0e1582702961d3a4b55ddd08593e737c517d8d000edecfbc`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1f8e0ef7d25eff60e11fa3c1a140c4b7ce27cf57558949e98636551681dbdc36`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:6a18063c2eca0e9104385448dd15e7c69be3a28b1342459716761e4c4c6808ba`. Independent certificate: `sha256:a1cedfbfc5f1309184329786c7f8beb97d30fad87fd7c48f7ea1fd7c16180957`. Engine external-validation hash: `sha256:5bb974810be1c148a1100288a461c7fd2fec997a14ef04b4635c587236c82aee`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- equilibrium-1: predicted thermal-equilibrium-equivalence; observed thermal-equilibrium-equivalence; exact match True
- equilibrium-2: predicted thermal-equilibrium-equivalence; observed thermal-equilibrium-equivalence; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:bada0bb12c0976930cd88861aeb6e3c190b94792b10290bbadf7798a0adb1d9a`. Isolation certificate: `sha256:8d5c9dce80cef319bbd71c5baebeac4b80a156aa4bc5a893ad93263363d47553`. Custody certificate: `sha256:32a6fa80e27aaf92ad3d172e9da5972013597e664c25306dd6d9a299ac73d34a`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- equilibrium-1 from IAPWS-RELEASES-2026-07-23
- equilibrium-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble

- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:049f8c4544286b7c8142d9f6c076592d0cbf24128e4b2aefe506da00574ba8cf`; engine receipt `sha256:cbd86e5cf8d1493d04fba245ae333d5933fd6afbe11800e515f6651a1a86d7c4` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-EQUILIBRIUM-001-cbd86e5cf8d1493d.json`; empirical-validation hash `sha256:fc24abe72fd0523e1238eed19ce91c4cec1c38dee59828e99e4aaa997f598a69`; measurement receipt `sha256:bada0bb12c0976930cd88861aeb6e3c190b94792b10290bbadf7798a0adb1d9a`.

82. Heat and work as distinct energy-transfer traces

Claim identity: `SFT-PHYS-THERMO-HEAT-WORK-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Heat is energy transfer whose microscopic carrier label is closed by the receiving macro-observation; work is energy transfer whose organised source-to-response label remains observable.

> *Heat and work are the closed-label and held-label observation classes of exact energy transfer.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-label-versus-held-label-energy-transfer__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`closed-label-versus-held-label-energy-transfer`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:61795e1f00ba93747a6ef0ac2b787da3e074aa4c8f390a9799d3dce21cb79209`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:522c9c0c66247feb3c51b85bb4689e821a7a799531ab9b68ddadb5de1e8f80d6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:501eb3874ae9c5d8b5ed8a38b0de1fada2ca63887a555bc7e0d0e7f68483133c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b43f038351da4176e1465face97b3eac4b4ed9221038d24fef61fb575fe29283`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1d8e132c7d01dd2d917db2c7bbe6b5d32e5957377f81031e2ea0818c34c348f9`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:692212d6e9d55652d02fb5791a82569e67f831fa33e471c95251c9fea6ac7afb`. Independent certificate: `sha256:d43df8e2076ebc60dc7837d017f9561a29607f73047cab35731c78d92d1ad243`. Engine external-validation hash: `sha256:174773775c142ef55cb82df2c683aa9819e2e853f672b6dc0907ce3e961f0a33`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- heat-work-1: predicted heat-work-transfer-distinction; observed heat-work-transfer-distinction; exact match True
- heat-work-2: predicted heat-work-transfer-distinction; observed heat-work-transfer-distinction; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:147bfd8d5ce85d5ef9df3f4bae37eae4272ab852ee972059e32bf7c8392fa098`. Isolation certificate: `sha256:4eb339a6ff4e80206068490005d493f158f45acd284172b98678406cc281228b`. Custody certificate: `sha256:8d7b054c0b2390b79b1ce173842074f5baf178ae2af355c058ec3128734c5825`.

Registered source descriptions:

- National Institute of Standards and Technology: https://webbook.nist.gov/chemistry/ (sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e)

Registered target identities:

- heat-work-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

- heat-work-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:b9a86a633b8f72bd30d09099c6e7d1d9f2f42ceb564963adb743f3f215b03585`; engine receipt `sha256:de7c30a2d5ab28ca55a30d35e4973d6197a3db788ad47d8604dca2b49d98e57c` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-HEAT-WORK-001-de7c30a2d5ab28ca.json`; empirical-validation hash `sha256:03611820e53939d9eb62be3c7f92a8ec423b02e8431e19f09e97114e48305d9d`; measurement receipt `sha256:147bfd8d5ce85d5ef9df3f4bae37eae4272ab852ee972059e32bf7c8392fa098`.

83. Closed energy accounting

Claim identity: `SFT-PHYS-THERMO-FIRST-LAW-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For a closed support, every change of retained internal energy is forced to pair exactly with boundary-crossing heat and work transfer records.

> *Internal-energy change is the exact closed composition of all heat and work transfers across the named boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-HEAT-WORK-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__internal-change-boundary-transfer-closure__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`internal-change-boundary-transfer-closure`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2f6f4e1533d9acfc84c5b0f23be38d3100a0609ec8bf1a8450df35d87c51df00`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8d3939a5f370964ef42f4dc5c4475f8ec8afcc953289ae724ab7582161ac8dd8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7f43956afb7cc2b419b15ca6cd4271ae8279b8b1b5f55e8ae05a812653290a5c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ff237dd6c3067a027223bba28bc7fc4200700ad5c54bd10a78a92f9e28fc6a28`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:29c891b7564cbca75ba46f533f7644fc9ece22041d405b8c065259d0ac66c14d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:3f7e45454f55f575cf62b0752260ca31edee7d04bd9f7040f96a1d501cc6c0f4`. Independent certificate: `sha256:4f6cdde720d390368b38f62eaa82b47bcc556c7fa987a4fc2ede5941dff9303b`. Engine external-validation hash: `sha256:fec704a24bfe324caf04366192636b567de921e7cded02e5d30df1b7ff7efcfc`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- first-law-1: predicted first-law-energy-closure; observed first-law-energy-closure; exact match True
- first-law-2: predicted first-law-energy-closure; observed first-law-energy-closure; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:27b5e2e6f2ded07ae9da19cac4a1083ddc468fd0145f86bc5424750c71f2942f`. Isolation certificate: `sha256:0ff2340607bd9c7e775dbe9d29664a291d4efe237616d63cf670012df9322fa9`. Custody certificate: `sha256:2a6ce8d3cce0d5a152b0c37a8b1d0b55ff716f307a09a33ef6f408b9eb977133`.

Registered source descriptions:

- National Institute of Standards and Technology: https://webbook.nist.gov/chemistry/ (sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e)

Registered target identities:

- first-law-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- first-law-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:efde8269cd96cdaa7f5c7f725daf78410295eb50f359220fcc512ee7b52cec4d`; engine receipt `sha256:7362f8e65ebb689e893823f347b2fdedc17fd5e01e858dff505997936acb79a8` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-FIRST-LAW-001-7362f8e65ebb689e.json`; empirical-validation hash `sha256:37ec084ec78fda59b326de8dbab75a30b02661fa21367957fe1623d4d7b8c81e`; measurement receipt `sha256:27b5e2e6f2ded07ae9da19cac4a1083ddc468fd0145f86bc5424750c71f2942f`.

84. Thermodynamic entropy from unresolved support

Claim identity: `SFT-PHYS-THERMO-ENTROPY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Thermodynamic entropy is forced as the exact information required to distinguish the generated microscopic predecessors merged by a macro-observation, conditioned on every retained macro-label.

> *Thermodynamic entropy is exact conditional unresolved support of a macro-observation over its finite generated microstates.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-FIRST-LAW-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__conditional-unresolved-microstate-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	conditional-unresolved-microstate-support	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:dae9e11ee9b6001ca0d9be95968a080cd92b6dc5fea748f30936b41ae7af069e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c5269b08d364fbaa40bf12531c012b795a1463eba0af221214425a401b2c6b66`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ae449ca3e6d245ae018a6d766684ebbf4a52476c94632b8a3bcfc850e0e1bb54`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a0c251a8959265f3da49548a9f5f190dd0e697a490eb0a57bdbd6dd6be29d33f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dd028d42a95245598f80ac0fecf0cc08ac43cfe797abcf2f01a5171440f14a24`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0e840231ff85005cf206bde90bc06180ed9a1f798ea988fe4d7e2c5b15f08e98`. Independent certificate: `sha256:8ed2c912ba7e6fac3828cdbfc755c8870dbb4f24088884e4f4a4a10f6fa1df3a`. Engine external-validation hash: `sha256:88bb1699a556713fad36769d4ba130184ef8646cf3ab27c1b4f132225789e643`.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- entropy-1: predicted finite-fold-thermodynamic-entropy; observed finite-fold-thermodynamic-entropy; exact match True
- entropy-2: predicted finite-fold-thermodynamic-entropy; observed finite-fold-thermodynamic-entropy; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause PHYS-S012 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:ff911c7996ad849024a1bd002ae35cf3e05b155a1e063076f9211fedef66d83e`. Isolation certificate:
`sha256:5d46408f401c29ed151155ba348bdd22fb139b897fdb3ad7e99e91a6a81b8784`. Custody certificate:
`sha256:7c8b8d277307713769d409f8eb8484640404cf01a896596a86d58628438ee137`.

Registered source descriptions:

- National Institute of Standards and Technology: https://webbook.nist.gov/chemistry/ (sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e)

Registered target identities:

- entropy-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- entropy-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:877b504de73497b225638f1c0ba82cdf64bfce8f73728c59d1b79bab15ec9321`; engine receipt
`sha256:3ff19937b6ea84cc3aa3220af340ad3e651be1e7fa1bf3928af4f009634769ee` at
`receipts/engine/model_admitted/SFT-PHYS-THERMO-ENTROPY-001-3ff19937b6ea84cc.json`;
empirical-validation hash
`sha256:26fcae64f6e0a8f7b78643c936652212bf4ff99f2abafa73c3c448552e1460cc`; measurement receipt `sha256:ff911c7996ad849024a1bd002ae35cf3e05b155a1e063076f9211fedef66d83e`.

85. Irreversible observation and entropy orientation

Claim identity: `SFT-PHYS-THERMO-SECOND-LAW-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A closed macroscopic process is thermally oriented toward observation classes with no smaller unresolved predecessor support unless the distinctions required for exact reversal are retained externally.

> *Thermal irreversibility is the orientation induced by closed microscopic distinctions; reversal is lawful only with complete retained records.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-ENTROPY-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__observation-merging-support-orientation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	observation-merging-support-orientation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:b6d988f751298688e60bb8d518132503f2f44838df7a591fdd11569366743f56. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:bd5d44ca5ac75aae6520413625174d04324f813e3ed129902e5e22ffab16e867.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:89b5457505016a0591d72cd0d779469d88001625cf1c604645660e721a61faca.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:705cb309bd4a45d0bd0d43283d8cc3a6b0fd0ca2ae6d049478e3bc0c95927331.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:8a76f82a48d2516bb5d150b93a10eed9104453f1148a75c2790edb45b8119b54.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:ceee18ae030e0b6bda80abd30799585a6fce564b94953c5a27fa18b275e7f2c3. Independent certificate: sha256:4176b2452cdb078a2fd8544bd43813b1baa59d68052243d8dcd0c60c3badbc8d. Engine external-validation hash: sha256:f64f207fc13272e50a98b3afc54152df031cd4c9662a7fd6a25aecfa08f1624e.

Shared claim-record clause PHYS-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- second-law-1: predicted second-law-observation-orientation; observed second-law-observation-orientation; exact match True
- second-law-2: predicted second-law-observation-orientation; observed second-law-observation-orientation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:8b4a9998e1d3e63bd2b5d24ef0e9d49c39d7c313737613826327d6a68d634cbe`. Isolation certificate: `sha256:5360cdc5d760108f0a018935f0a946403647a913f60514f1cf27e06bae2a3671`. Custody certificate: `sha256:cf32400672bb65ae4e50fe2556a2e5d2cadddfe9c9f19d1126011f477a65b8f5`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- second-law-1 from IAPWS-RELEASES-2026-07-23
- second-law-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:0ee3aba0d1c1d37a15e9e411519e0587c9e197749fd511b7d2605ac4cb6b2c48`; engine receipt `sha256:65d57af5d78d0fd2f014c0ed3bc1184cf05bf226566e2b0c951f2fe51b7b0a44` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-SECOND-LAW-001-65d57af5d78d0fd2.json`; empirical-validation hash `sha256:8ecc32ffd2412647a6a0e12e8d7cb6804bf48aef8c39333588ba41f92078038e`; measurement receipt `sha256:8b4a9998e1d3e63bd2b5d24ef0e9d49c39d7c313737613826327d6a68d634cbe`.

86. Finite unattainability boundary

Claim identity: `SFT-PHYS-THERMO-THIRD-LAW-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. No positive finite transition sequence can erase the final generated recurrence distinction of a material support; complete recurrence closure is therefore a limiting boundary rather than a reachable destructive state.

> *The minimum thermal recurrence boundary cannot be reached by erasing all distinction through a finite lawful process.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`

- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-SECOND-LAW-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__positive-transition-unattainability__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`positive-transition-unattainability`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4136d758cfe20dd7bc3bce2e815d180521437d46fffa828d5cd0a5e19c184dc8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:14b4815387ff86bbd60b37b5ce2e8305f1297484efb231a9ab76b087ee47487b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:14a67126b7e513be27af787df4da5ee968000098cfcbe746cce45b8ef944f2ec`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ac5e1bef5dbe2e4e07e3e2a331595bb74aa93a46b431b846471762b2ed6bd1fa`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4b5790435184a81b32eaab8dc32a905f9c80c104ab0201d738c3cab6266f62ad`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a4ba57ed42270c8a7ae69418ff067aab619e429c483b71d05d0af94c1c6e73cb`. Independent certificate: `sha256:c7c50729009b9c1eaf672794a191934ca83f87bfd1bffb88d1d982f12e5f3016`. Engine external-validation hash: `sha256:f2731ce2ddfdc96118d8508099d5ac9e637d60c082261bcf38821ca67a1c75a9`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- third-law-1: predicted finite-third-law-boundary; observed finite-third-law-boundary; exact match True
- third-law-2: predicted finite-third-law-boundary; observed finite-third-law-boundary; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:26c5f328b651685a65f8dff174efd2b83f927e48a80b8e169a45c78a2ace1d42`. Isolation certificate: `sha256:a49d5a475622aa14fbbb8c9bac7efb1078da9e66c651d95e67ceb73a38f8b339`. Custody certificate: `sha256:e518d91233cacf5449aab805f7e457c78992952301d0df78a661f9a00eb93597`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- third-law-1 from IAPWS-RELEASES-2026-07-23
- third-law-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:ed9459d43f683065447ecc07b890aba4259abfe646fd9a15147e37c85bfa8cb7`; engine receipt `sha256:35ac14f08f2fe755dadcbaed41b325c257db5500c2e2a8eb796367d632ad9e74` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-THIRD-LAW-001-35ac14f08f2fe755.json`; empirical-validation hash `sha256:26a16c33e6986b878febed79426ba5208115cad720ded56d7ad79ce66c0aa529`; measurement receipt `sha256:26c5f328b651685a65f8dff174efd2b83f927e48a80b8e169a45c78a2ace1d42`.

87. Exact finite statistical weight

Claim identity: `SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The statistical weight of a macro-observation is the exact positive count of its generated microstate fibre relative to the complete accessible support count.

> *Statistical weight is an exact positive rational census ratio over complete generated support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-THIRD-LAW-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-fibre-count-over-support-count__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-fibre-count-over-support-count	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d6df39591028395a45da6a87076f5410e164801d641b5a15c9555dd52796d6c0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:20300f111d9dca3814ee908c92f5c0f65a94e71b60770634d02dda75ad148e34`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ae3eb0351361990cc53e46cf3b098f2d373d6267b064dd204dd937ac88f323e5`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ddaf3b086f0fbaa7a731397924b840c450b7228f3903a413bd2d8a5c251bbcd5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b69676863ee2ff8e9d958e57780df3bff7ef82ba16637994944469df9d05ec0a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:81a65c0c1d85723d19ca7c0b2e9103bc8be2e1a3834fd17f20a283a3deb4c33a`. Independent

certificate: `sha256:a56d8972ed6beb1ff28103ab146793a9dd7dfa9bf2dde2f83b4cc487035176c6`. Engine external-validation hash:
`sha256:176cc7f98cb1f3a290594904052c9c6c4b0f97e5766ea83783ff56ff562e2df4`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- statistical-weight-1: predicted exact-finite-statistical-weight; observed exact-finite-statistical-weight; exact match True
- statistical-weight-2: predicted exact-finite-statistical-weight; observed exact-finite-statistical-weight; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:0e0fc56cfb23bf361bfbf95865e55c7e7d00f37b21595d5796311c33c9813f92`. Isolation certificate:
`sha256:550aa0c957caa81a7eb8223f79644fe9dfb947764515c88408bca1e25c4e5dd0`. Custody certificate:
`sha256:d079018e069d566883979ed805122883368af4002dc35f7a0d8760ff84ae98ba`.

Registered source descriptions:

- National Institute of Standards and Technology: https://webbook.nist.gov/chemistry/ (sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e)

Registered target identities:

- statistical-weight-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- statistical-weight-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:f185c3de2ecd0bb11e43ecc6b8be2d5cc01b2f7dc1d41d549e0089d530654aa2`; engine receipt
`sha256:a1bccdb47bbc54ac9969c4abcdd193283b550a1258a2bafd47e87a2524a34b28` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001-a1bccdb47bbc54ac.json`;
empirical-validation hash
`sha256:55391e20ab12db53e72792d6720c4a3ecd91717a21fc3e797bccab8a53548268`; measurement receipt `sha256:0e0fc56cfb23bf361bfbf95865e55c7e7d00f37b21595d5796311c33c9813f92`.

88. Thermodynamic state relation

Claim identity: `SFT-PHYS-THERMO-STATE-RELATION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A thermodynamic state relation is the minimal closed dependency among independently observed support content, boundary response, thermal order and energy carriers after redundant coordinates are removed.

> *A thermodynamic state law is the unique minimal exact relation closing all registered state carriers on its declared finite domain.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__minimal-closed-thermodynamic-coordinate-dependency__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`minimal-closed-thermodynamic-coordinate-dependency`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4b2a9bae5def5b977dd444dee5c35791cebe9d04a724865b83f5b70be50cc9f5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2d4ff98f4e05c186b576d6a5c15c2c6d56d46197243299fbaaf7a2b79a4cb926`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:aac21fc9d70da35afbc04c2f4f02367f34bbc59cf10440b79644448a989ccb05`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7eada2f344e23049e34b2caf66a6170bfff87d85a1303e1f3b8f5cb05bd36cdf`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6e2ab2f551a6a20796a7d3d6e7a0cb38cf926f3c1b05915952674190323de8b5`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:46e58c0847d9c306c641b73ccbdfad62f69e06ebb0b62a15605a20decfbcf32d`. Independent certificate: `sha256:32e7ab8e33d2cdaaa90269d5a5e530a00c55c2cbb74767f7e5f31813ff3e1e07`. Engine external-validation hash: `sha256:f70d990cb76f26f0a1a9eaf96efb28fd37c88a50227a19894c6f029d518f56f6`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- state-relation-1: predicted minimal-thermodynamic-state-closure; observed minimal-thermodynamic-state-closure; exact match True
- state-relation-2: predicted minimal-thermodynamic-state-closure; observed minimal-thermodynamic-state-closure; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:3ef5d1cdea157436c364386a028e50e0926c5033295f57545bfa974e066c87f6`. Isolation certificate: `sha256:6819349b50b0a247bd60cdcda2c63f436829f1b3de6cbfdf5d0c431871b76c44`. Custody certificate: `sha256:f17dc211864e5e141d1eda59b49e5bb8ea373d82e0b3625140a2e398b0476bfa`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- state-relation-1 from IAPWS-RELEASES-2026-07-23
- state-relation-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:357ecc276228d3d20183b976ddf90d7849e675972b6e3648eefd681d9550085f`; engine receipt `sha256:a99aa79962f17d2a4e55a903307a83ca2be54356de263afe8825803f217bf2f6` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-STATE-RELATION-001-a99aa79962f17d2a.json`; empirical-validation hash `sha256:fb2ad9a0b99c729c704d934a161d1139cd2eb72fbbf08642c6453fcc673d88cf`; measurement receipt `sha256:3ef5d1cdea157436c364386a028e50e0926c5033295f57545bfa974e066c87f6`.

89. Phase coexistence and transition

Claim identity: `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Distinct macroscopic phase classes coexist when their exchange boundary preserves a common thermal order and reciprocal transfer balance; a phase transition changes the selected observation fibre while retaining total provenance.

> *Phase coexistence is balanced exchange between distinct structural observation fibres; transition is provenance-preserving fibre change.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-STATE-RELATION-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__coexisting-observation-fibres-under-balanced-exchange__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`coexisting-observation-fibres-under-balanced-exchange`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:1f9a1c64d9bb8ca993fb8ef527a841928a486c76e0e19ae1b4f9ca7fe2a601b9`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f42cfff2d91851d91717f421aeae259d02959ed2a9a736586d1d333e968d9723`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e8b2e3d4a7c16422e8b1867ca2d6dcbc36f3f90ecbf6962b6ccec29e53d2d3ed`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:312eb8573d58e786edfd4a14765ad2fcb8bad4ec82de9488659ff013dfa0e92b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b7592e2613534291ef6f2b6dd187cc4a2a88ce205c213f5c8265ce95707b5998`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c5ad6f16c8ccb1c0195b7dcd450794abf0a6582d0801d755e1d1cee852687b5b`. Independent certificate: `sha256:46defb37258a5ca944a8d3954d8d25998ccb82efeaa199a0c98458f785896355`. Engine external-validation hash: `sha256:97adc0613eb966c904d2c3bfef6732b8c7ef6ec31a8f5d9cc53f9537b9dc93a9`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- phase-equilibrium-1: predicted phase-coexistence-transition; observed phase-coexistence-transition; exact match True
- phase-equilibrium-2: predicted phase-coexistence-transition; observed phase-coexistence-transition; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:35e4decab73557f6651fd0eac5c915fa4838162ce9546e9d9a630956cc8897f2`. Isolation certificate: `sha256:5e6a897195c17a49866239853d3876f8658ad3043432da670e169ce81153ab20`. Custody certificate: `sha256:5f620f800e1258f86c641c7b587613803e4d1e9bb85e6ae1bcaff6c108e1a72d`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- phase-equilibrium-1 from IAPWS-RELEASES-2026-07-23
- phase-equilibrium-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:7fdb443772c0cd734a76a53d72ab0f09193bf625f36f04ba908fff64e9bb6726`; engine receipt `sha256:4f21021472d3fa831e68e72a049b7a6396e2088b1111327351f3cda0715fd2b8` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001-4f21021472d3fa83.json`; empirical-validation hash `sha256:505e99fc86de8a8b1ac867dd747572fbd8c2f736d43784bd960a488458a15240`; measurement receipt `sha256:35e4decab73557f6651fd0eac5c915fa4838162ce9546e9d9a630956cc8897f2`.

90. Kinetic transport and collision mixing

Claim identity: `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`

Shared claim-record clause `PHYS-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Kinetic transport is the exact propagation of held motion and energy labels between generated cells; collision mixing is their closed local permutation subject to conserved transfer records.

> *Thermal transport is adjacent carrier propagation with locally conserved collision permutations over complete finite support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__cellwise-carrier-propagation-and-local-permutation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`cellwise-carrier-propagation-and-local-permutation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:5b6e5a2287cb6fbd669a699fbf28fc6dae2baf0909c4011cb5c3d6cdcef71a4f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d30fbb38fb12fda20dddfb163e4dfd0434415120f7118b225279220c17463cf7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:cd84dd2bc9be81e40abe6345768b97a1c0f4d9fdcd7a310121d3935a08eb4d9c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b510d9c52a16063ce072ec1ae96bbc6a002abe947f74d692ed979787fecfb4f6`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d94f1208accd846d0e7148bf3577939ac944619e4279e87006453d03f5b2f579`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:83e6a6d6a4a84aa01fc03b09986a11fdce29c319f2018c0f199a7f2fa0cade8c`. Independent certificate: `sha256:3d10eba440c2b0fc41c95cdf04539a180db711dde6c55e71a42c60bc9cbb9d6a`. Engine external-validation hash: `sha256:d6535622421903b87f653960e28bfcd636a5af2c6926d35e20badc9964287bb0`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- kinetic-transport-1: predicted finite-kinetic-transport; observed finite-kinetic-transport; exact match True
- kinetic-transport-2: predicted finite-kinetic-transport; observed finite-kinetic-transport; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:fd2b9d663e29332a07224dfee4c1577885351ebd414fe0066af79c3b9f3b8121`. Isolation certificate: `sha256:99ae9563181946c264f24bd47d2f9d11129bb69525a5d39d1107d103f2afa702`. Custody certificate: `sha256:43afd90cc943c02147674e8f755e8c0020fef42d5609ef5e2f2813baf00f37bc`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- kinetic-transport-1 from IAPWS-RELEASES-2026-07-23
- kinetic-transport-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble

- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:621b4262f0a55344233ce2fa1ad74486f0a63eb31e3650bee68fdc93c37bfda4`; engine receipt `sha256:5beda8e6f719d7fbad9dca80e0b7a8672b5aecef46a1c8df834d6536a8c32550` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-KINETIC-TRANSPORT-001-5beda8e6f719d7fb.json`; empirical-validation hash `sha256:5c5653eeac348465488455af34cf0223a91cce38e83fbc2c6583df1a39baa7e1`; measurement receipt `sha256:fd2b9d663e29332a07224dfee4c1577885351ebd414fe0066af79c3b9f3b8121`.

91. Finite fluctuation and recurrence

Claim identity: `SFT-PHYS-THERMO-FLUCTUATION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A thermal fluctuation is a finite departure of an observed fibre count or transfer count from its recurrence class; complete deterministic support fixes every departure and return path.

> *Thermal fluctuations are exact finite count changes across deterministic Fold paths, weighted only by complete support census.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-count-departure-and-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`finite-count-departure-and-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:8a0ecd5474e43382493de1a001bd4b836bbca591a9aa566152875e37035c183e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:00138e10f0a178a63731e18278eccfda8318dbc158bd82da5420e550a30a5c76`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b0e40ef20b9a4436701c2356257b61ba295912e51754d039135fdf2fad587680`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c1f74423658eddae370554c1aeb672cd2b0738d8683ca9eba6e8d6e78ef88e87`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:087c5bd21117695a5b7de997a677d07134aaa675ae69aed071669d6feba944eb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:03fa0b411cd53d9dae61566fb1f06b5e587d565c9384c0f8799eb196854cb1fe`. Independent certificate: `sha256:9022eaebaff4f04b2b66185a6fa3642d43f39cb420d9e29868b084491ee72d67`. Engine external-validation hash: `sha256:9b675f6b33307f2a946cd5ed732a28f89b643cac3c097822bedc6a3d503e690b`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- fluctuation-1: predicted deterministic-finite-thermal-fluctuation; observed deterministic-finite-thermal-fluctuation; exact match True
- fluctuation-2: predicted deterministic-finite-thermal-fluctuation; observed deterministic-finite-thermal-fluctuation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:9083d3f6297c243c17c6c335430b454a8ea91e4b1399380a2b514cd9de976f12`. Isolation certificate: `sha256:4c5d38c756844b4bc12b87641b1441a6461d806fe1ffee65ae44b4562647d163`. Custody certificate: `sha256:d47efed87e7c1a1d9f6d21eb5d67774201bba5b01585b64902b078cd656dd043`.

Registered source descriptions:

- National Institute of Standards and Technology: https://webbook.nist.gov/chemistry/ (sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e)

Registered target identities:

- fluctuation-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- fluctuation-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:36a8fa4dfa7d8ca5b169ee43242bfb7cdf898fc8ba79ddb8ce038ca3bef51a16`; engine receipt `sha256:30e0865edeacf61bbbac6fdb5cbe11c593b7f41a9366f6cff95312272e448fc1` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-FLUCTUATION-001-30e0865edeacf61b.json`; empirical-validation hash `sha256:e2f4cf513ad47701b5b4e73dc2c93c74152d889733134d3d93167461c1d33397`; measurement receipt `sha256:9083d3f6297c243c17c6c335430b454a8ea91e4b1399380a2b514cd9de976f12`.

92. Macroscopic irreversibility with retained microtrace

Claim identity: `SFT-PHYS-THERMO-IRREVERSIBILITY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Macroscopic irreversibility occurs when observation merges distinct microscopic histories; the complete Fold evolution remains reversible exactly when their fibre labels are retained.

> *Macroscopic irreversibility and microscopic reversibility are corresponding projections distinguished by retained predecessor information.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-FLUCTUATION-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__macro-history-merging-with-microtrace-retention__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`macro-history-merging-with-microtrace-retention`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:5ec923c32431f4cc5c32c3572bda16ffd88c3decd3b782d5642640e91d69ce9c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a951e838910cc42db069eb2b5f345c610e61552ae58bee73b9c3555a03f303b3`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e4a5fa3374d35a5a5b6c7b05e6ec9a06ad4ce856e8f25bbf68fe2c43704dff22`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c58c5081d1760e12d16d32be04067b042d8b4489f64006681cc3ae974d5fc0dd`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4c4cbcf15512184cf5cc9dc5122d793badd4085ca55356feda189d5f580cfa8e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:208cabe0f6efc2a2901bb5510b1559bd30ab859ffdc6a6f1d9e5bada19fac7c0`. Independent certificate: `sha256:2b6ba3c9dbbc8057e7d929f2e3bc6c310019c4451f1f1e0000d3de1e6209457d`. Engine external-validation hash: `sha256:b9a7e0febee4248b4d96d12cd26ce9307855727f8751b88271138cfd45ecc1bf`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- irreversibility-1: predicted macro-irreversibility-micro-reversibility; observed macro-irreversibility-micro-reversibility; exact match True
- irreversibility-2: predicted macro-irreversibility-micro-reversibility; observed macro-irreversibility-micro-reversibility; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:35dee66288acd6f4180c7b3f7c56993e7a95d99fb07ad46fa23978965666897c`. Isolation certificate:
`sha256:b967968ec7c5258c950e3111cb906895ed20d7326e5aa471ed7c829d9ec35228`. Custody certificate:
`sha256:b64076b8eac6ed4460b14d821ab65bb7c7a0add2e618958c223ff0d9fef09769`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- irreversibility-1 from IAPWS-RELEASES-2026-07-23
- irreversibility-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:0e57f4877e74c89d0ed92e5d188e256ba416764f9a906566a9b3ef5367a69b73`; engine receipt
`sha256:2097ca97413226b19cbca02df51a72850c4bdf5b0e42222e45dc2f3bb777a1d2` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-IRREVERSIBILITY-001-2097ca97413226b1.json`;
empirical-validation hash
`sha256:6636d32e644110d7ca9bf18d378ca7378a0afd9297d379c320ce42792d65bd34`; measurement receipt `sha256:35dee66288acd6f4180c7b3f7c56993e7a95d99fb07ad46fa23978965666897c`.

93. Susceptibility, response and transport correspondence

Claim identity: `SFT-PHYS-THERMO-RESPONSE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Thermal response is the exact change of one registered observation carrier per exact change of its source carrier, conditioned on the same held support and recurrence class.

> *Susceptibility and transport are exact source-conditioned response ratios on a declared finite observation support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-source-response-change-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-source-response-change-ratio`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:c6b1f63552ccebcec98575731eb921c753088c53fe8da8632acfd6d99a125e97`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:eccd392239638a2f850d68b176fe90134f718e514c3d996a6575f82bfd601890`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:415c4fc3248360a54fff8b3877d91378f6901ddb21ccb24a44977cd058c71cd3`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1f5e7ec3a67a74630e6a6955565c03bb4cda052fb88b4976c9c91db0458b06af`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:554006a67b3012c6996aefa6658c90e5eb362ee1467800525bbdac3741313e81`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:aba9c526a54975949c1a2800c2ea4f06406cea4ef6a10c7e60679e922e20e072`. Independent certificate: `sha256:554a7ad79538b78eb9bc6ff390b0bc27dc162c20a834a55108f16d138ef8eec9`. Engine external-validation hash: `sha256:b31d1824a7bdfec7309eaba264a3c7d0112ee2fd2b36ecfab9debf433bcf4c72`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- response-1: predicted exact-thermal-response-correspondence; observed exact-thermal-response-correspondence; exact match True

- response-2: predicted exact-thermal-response-correspondence; observed exact-thermal-response-correspondence; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:5cce54bc1a2e4a30f97c1bdb58337eb08e95c6c9d7d423897a9617ebf3beed5e`. Isolation certificate: `sha256:430bd7f1ad6d8fbfde60e9181a5d79c00dcacb75480db4133d421c578f20b6eb`. Custody certificate: `sha256:bdcb62de253c3571fba9d77340fcfa7c6484afaf6ba3a51b6b8da1d27af33515`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- response-1 from NIST-SRD-INDEX-2026-07-23
- response-2 from NIST-SRD-INDEX-2026-07-23

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:99d0e10a240570424466afa75f3005ddcc4cafdc90ddc35655f601f27c6664dd`; engine receipt `sha256:9152359409657ab66e862af0dc4846b962d19105452c85e184694a16a71c8f5b` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-RESPONSE-001-9152359409657ab6.json`; empirical-validation hash `sha256:a033cf52cae01e80b52e5f2ba8f7017cc68b41087614db6caf7695cda92fb5af`; measurement receipt `sha256:5cce54bc1a2e4a30f97c1bdb58337eb08e95c6c9d7d423897a9617ebf3beed5e`.

94. Half-One vacuum floor and oscillator spectrum

Claim identity: `SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Complete binary-support enumeration leaves exactly the half-One as the positive part whose self-pair reassembles the One. It is self-antipodal and Folds to the One. At every positive depth k, complete binary support then forces 2^k uniformly separated oscillator parts (2j-1)/2^(k+1), beginning one half-spacing above structural absence.

> *The exact vacuum floor is 1/2. At depth k the complete oscillator spectrum is ((2j-1)/2^(k+1)) for j=1..2^k; depth two is (1/8,3/8,5/8,7/8).*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-PART-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of vacuum carrier, dependencies, self-pair relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All exact positive parts of complete binary support, their self-pair return, and every finite positive-depth half-spacing spectrum generated by complete binary words. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-trace__unique-half-One-self-pair-and-half-spacing-spectrum__complete-registered-product__all-omitted-carrier-forms-rejected__exact-result-sealed-before-comparison__complete-source-transfer-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`borrowed-vacuum-number`	A borrowed vacuum number has no Fold trace.
`dependency`	`admitted-V3-dependency-trace`	`asserted-prior-answer`	An asserted prior answer is an unregistered premise.
`relation`	`unique-half-One-self-pair-and-half-spacing-spectrum`	`empty-ground-or-selected-offset`	The alternative loses a required carrier or imports an extra law.
`enumeration`	`complete-registered-product`	`selected-candidate-neighbourhood`	A selected neighbourhood cannot establish uniqueness.
`minimality`	`all-omitted-carrier-forms-rejected`	`survivor-without-lower-controls`	A bare survivor does not prove the required carriers are minimal.
`measurement`	`exact-result-sealed-before-comparison`	`target-visible-before-seal`	Target access before sealing reverses the scientific direction.
`record`	`complete-source-transfer-control-record`	`answer-only-record`	An answer alone cannot reproduce the source and transfer ledger.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:47e38dd7eaa368b1e979aebef76e91579daff609fdf61b9ef32a33c2875a5cf1`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:bea5aa10ba513963a5dab594a7cf5fedcc49e0fb4293997f2e449476bc5c068e`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:62f320a5284328b3166bebfc4539d8be9e4b0e61f68993ce26527f6faa8c66ef`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e0829af2a4f920e6232904f8e8050ccaa95d0d2462796fe996d19efddd69f013`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:247b00c3e6325d3cd0d0d728ea395597d854ec60c03ae9a22cc542adabbea6d4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:46b4fc1bc5063437791f86dec18f6804cb3c98174c289b1500bc51ab6c880a7f`. Independent certificate: `sha256:f18ad879d0ebc3445c8ffc4cdaed9a18d4a0687f72a511b80358bd42958e0c71`. Engine external-validation hash: `sha256:71b8cdc47818b3759bb6a994559a41edcdcbf0e319c73073cd42f110fe560127`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S013 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, candidate table or answer artifact as a premise
- no external measurement, conventional vacuum model or apparatus result selecting a survivor
- no fitted coefficient, tunable phase, chosen reservoir value or answer-selected harmonic
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no deletion of the positive outward extraction result or of the complete restoration ledger

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:d81fef2090a7156a7624c032cbb33b0fc51f93ad76a4599b456de0f0166cb089; engine receipt sha256:288674203497f376e9e6dbd0cb8aba169f1f2343b3b7717dbac789c886759548 at receipts/engine/model_admitted/SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003-288674203497f376.json; empirical-validation hash None; measurement receipt None.

95. Perpetually live odd-denominator vacuum recurrence

Claim identity: SFT-PHYS-VACUUM-ODD-RECURRENCE-003

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every generated odd denominator beyond the One, pairing a positive residue with the binary Fold cannot acquire the denominator's factor and therefore cannot reach structural absence. The finite positive residue support forces a first return, so each such vacuum mode remains on an exact live cycle.

> *Every positive odd-denominator unit part beyond the One remains strictly positive and returns exactly; examples d=3,5,7,9 have periods 2,4,3,6.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-FOUNDATION-FOLD-001
- SFT-MATH-ORBIT-NUMBER-THEORY-002
- SFT-MATH-DYNAMICAL-SYSTEMS-001

Generated grammar and closure boundary. Generate the complete eight-axis product of recurrence carrier, dependencies, odd-denominator relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every positive finite odd denominator, its complete positive residue support and the exact first-return Fold orbit of its unit part. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-trace__odd-denominator-first-return-without-empty-state__complete-registered-product__all-omitted-carrier-forms-rejected__exact-result-sealed-before-comparison__complete-source-transfer-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	borrowed-vacuum-number	A borrowed vacuum number has no Fold trace.

Axis	Forced form	Eliminated form	Elimination reason
`dependency`	`admitted-V3-dependency-trace`	`asserted-prior-answer`	An asserted prior answer is an unregistered premise.
`relation`	`odd-denominator-first-return-without-empty-state`	`dead-ground-or-finite-prefix-only`	The alternative loses a required carrier or imports an extra law.
`enumeration`	`complete-registered-product`	`selected-candidate-neighbourhood`	A selected neighbourhood cannot establish uniqueness.
`minimality`	`all-omitted-carrier-forms-rejected`	`survivor-without-lower-controls`	A bare survivor does not prove the required carriers are minimal.
`measurement`	`exact-result-sealed-before-comparison`	`target-visible-before-seal`	Target access before sealing reverses the scientific direction.
`record`	`complete-source-transfer-control-record`	`answer-only-record`	An answer alone cannot reproduce the source and transfer ledger.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:beac2f0dcbbbea64fa9ef4de654304b6d50c8785ea331c4c2ea36e86d48e22ea`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:d48480aab6d50f9557755c7cd1705cb0c395baafa0443759b305697e7b4ce0fb`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:db203330f3989affb79b99144ea157d9253410c257c1baf9074eda9a4f70ed07`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:293da57c4fdb0fb1d60b3a401553b7b2f9e7f607dc420f506897370c1d7bdb5e`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:2258042d5ee3201b476efeea0a71b013701220c514831e59652022498af9c254`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:46b4fc1bc5063437791f86dec18f6804cb3c98174c289b1500bc51ab6c880a7f`. Independent certificate: `sha256:c974d7ea59b08cd92bd148e7aa9873df3a813262c50bb773f0f489257c258bf0`. Engine external-validation hash: `sha256:4d8faab743747422b7d527e47b7024c0d49f9213e200d6acfb419b87ed95d1cc`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, candidate table or answer artifact as a premise
- no external measurement, conventional vacuum model or apparatus result selecting a survivor
- no fitted coefficient, tunable phase, chosen reservoir value or answer-selected harmonic

- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no deletion of the positive outward extraction result or of the complete restoration ledger

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:23af938da6534560c91d70cb33221166d64ef31b77fdc08f249b3891f426a21b`; engine receipt `sha256:dad4e023733464a86c1fcad9913bbe661064c10e67d270e47cc1430d3846a821` at `receipts/engine/model_admitted/SFT-PHYS-VACUUM-ODD-RECURRENCE-003-dad4e023733464a8.json`; empirical-validation hash `None`; measurement receipt `None`.

96. Vacuum polarization and running direction

Claim identity: `SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The live vacuum retains the half-One screened support at the far carrier. One exact Fold exposure returns that support to the bare One, forcing electromagnetic support to increase as screening layers are removed. This fixes the running direction and endpoints without importing a renormalization model.

> *Vacuum polarization forces the structural running direction screened 1/2 -> exposed One; closer probing removes screening and increases effective electromagnetic support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003`
- `SFT-PHYS-VACUUM-ODD-RECURRENCE-003`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of polarization carrier, dependencies, screened-to-exposed relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: The complete exact Fold fibre from the half-One screened carrier to the exposed One, with every intermediate finite screening layer retained if subsequently generated. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-trace__half-One-screened-to-One-exposed-running__complete-registered-product__all-omitted-carrier-forms-rejected__exact-result-sealed-before-comparison__complete-source-transfer-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`borrowed-vacuum-number`	A borrowed vacuum number has no Fold trace.
`dependency`	`admitted-V3-dependency-trace`	`asserted-prior-answer`	An asserted prior answer is an unregistered premise.
`relation`	`half-One-screened-to-One-exposed-running`	`fitted-running-curve-or-reversed-direction`	The alternative loses a required carrier or imports an extra law.
`enumeration`	`complete-registered-product`	`selected-candidate-neighbourhood`	A selected neighbourhood cannot establish uniqueness.
`minimality`	`all-omitted-carrier-forms-rejected`	`survivor-without-lower-controls`	A bare survivor does not prove the required carriers are minimal.
`measurement`	`exact-result-sealed-before-comparison`	`target-visible-before-seal`	Target access before sealing reverses the scientific direction.
`record`	`complete-source-transfer-control-record`	`answer-only-record`	An answer alone cannot reproduce the source and transfer ledger.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:1a88e248c5def845615f245bd8e27f9e3903fd5e69e18dff5870a0cb7dcb063e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:3a155046e477a1c21adb3ffb90fd772408c346db409d0e172ccf39cfd5ab5327`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:b663484e15866b5b05d67a3e40919d253ee0e0ca5163c05a4d15b4738a15ca84`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:955c2f7f51a571fe90a3bf935ab10fe474d19eb89bbdfd22931a16be42c18f83`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:a21e93bb9aa177cdefd411a060c7ff1d0387205d46dc5c4172327f39666a4397`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:46b4fc1bc5063437791f86dec18f6804cb3c98174c289b1500bc51ab6c880a7f`. Independent certificate: `sha256:9e397be6732edd641d37f0536a3dc3da18ef2f55d4eb82958f0c67b98e06a98b`. Engine external-validation hash: `sha256:ba1411492596f29e4dec72c7348bc3526469b3f48945e996fdc8a4ffb9e4e728`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, candidate table or answer artifact as a premise
- no external measurement, conventional vacuum model or apparatus result selecting a survivor
- no fitted coefficient, tunable phase, chosen reservoir value or answer-selected harmonic
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no deletion of the positive outward extraction result or of the complete restoration ledger
- no claim that the two structural endpoints alone fix a measured beta coefficient

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:4db98f792906b7ce138c295954059ec4063c32f4a3e987f8ec0f2dc0fbfb9b20`; engine receipt `sha256:40f95e89489d596b961aeab5d234dca5938b41b0dd6de02ffae7da90e2d66d76` at `receipts/engine/model_admitted/SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003-40f95e89489d596b.json`; empirical-validation hash `None`; measurement receipt `None`.

97. Exact vacuum-to-inertia unity relation

Claim identity: `SFT-PHYS-VACUUM-INERTIA-UNITY-003`

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The independently forced vacuum displacement and native Fold coupling are both the half-One. Their exact exchange ratio is therefore the One and both complete in one Fold. Any lawful change of one carrier must be held identically by the other; this is a structural proportionality and a standing experimental discriminator, not an observation of engineered inertia reduction.

> *The exact vacuum-to-inertia exchange ratio is the One: (1/2)/(1/2)=1, with both carriers completing in one Fold.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-PHYS-MECH-INERTIA-001
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of vacuum/inertia carrier, dependencies, exact exchange relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All exact ratios between the independently admitted vacuum displacement and native Fold coupling, including changed-carrier controls and the one-Fold completion trace. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-trace__half-One-over-half-One-exchange-at-One__complete-registered-product__all-omitted-carrier-forms-rejected__exact-result-sealed-before-comparison__complete-source-transfer-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	borrowed-vacuum-number	A borrowed vacuum number has no Fold trace.
dependency	admitted-V3-dependency-trace	asserted-prior-answer	An asserted prior answer is an unregistered premise.
relation	half-One-over-half-One-exchange-at-One	free-inertia-coupling-or-uncoupled-carriers	The alternative loses a required carrier or imports an extra law.
enumeration	complete-registered-product	selected-candidate-neighbourhood	A selected neighbourhood cannot establish uniqueness.
minimality	all-omitted-carrier-forms-rejected	survivor-without-lower-controls	A bare survivor does not prove the required carriers are minimal.
measurement	exact-result-sealed-before-comparison	target-visible-before-seal	Target access before sealing reverses the scientific direction.
record	complete-source-transfer-control-record	answer-only-record	An answer alone cannot reproduce the source and transfer ledger.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:96417536c5fcbe4bce4ee91ee82f4662cf0c3a400b059b5fad4990d61a9216f2. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:542db57def3e4bff7ea9bef689d05ad43bf9028730be6baecd66ae677629997e.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:6ac3b4036cc44e3d7031304280c0b51b3cfee0c50d06f2f38512f0f110045c9b.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e4cda41b2cfa5455673da0c1b740f5b85cd92d5d6a0111d8b896dd6e4829c3ad`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:288ea6ec8c1414873c1d31997f3a967b584036fb81a538d8b291e865ea54acad`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:46b4fc1bc5063437791f86dec18f6804cb3c98174c289b1500bc51ab6c880a7f`. Independent certificate: `sha256:918411a91aea85ce591208e168de2386c4e4001f9bd7960a9351b8600db0da97`. Engine external-validation hash: `sha256:2654b9a181407cf63138b72d2cd2dda030cd62a48492557c618915266a90b528`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, candidate table or answer artifact as a premise
- no external measurement, conventional vacuum model or apparatus result selecting a survivor
- no fitted coefficient, tunable phase, chosen reservoir value or answer-selected harmonic
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no deletion of the positive outward extraction result or of the complete restoration ledger
- no UAP report, propulsion observation or engineering performance used as validation

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:00072adfe4dcd71e07d644bfd46420ec9f6b4b8588902be3bab08bb5abeb554a`; engine receipt `sha256:3e1b000691688a8bec6ab6301c21ae05136256e08f10d0e0e36babb0dc602d71` at `receipts/engine/model_admitted/SFT-PHYS-VACUUM-INERTIA-UNITY-003-3e1b000691688a8b.json`; empirical-validation hash `None`; measurement receipt `None`.

98. Positive asymmetric vacuum-beat extraction

Claim identity: `SFT-PHYS-VACUUM-ASYMMETRIC-BEAT-EXTRACTION-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The half-One vacuum carrier and generator-three unit carrier have the unique exact positive separation one-sixth. Transferring that beat to a held work carrier leaves one-third in the local vacuum carrier, and one-third plus one-sixth reconstructs the original half-One exactly. Thus a positive outward vacuum-to-work transfer is structurally admitted without creating support.

> *The forced asymmetric vacuum beat transfers positive work 1/6: vacuum 1/2 -> retained vacuum 1/3 plus held work 1/6, conserving the original carrier exactly.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003`
- `SFT-PHYS-VACUUM-ODD-RECURRENCE-003`

- SFT-PHYS-VACUUM-INERTIA-UNITY-003
- SFT-PHYS-WAVE-RESONANCE-001
- SFT-PHYS-MECH-CONSERVATION-001

Generated grammar and closure boundary. Generate the complete eight-axis product of vacuum-beat carrier, dependencies, exact transfer relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All exact beat transfers between the independently forced half-One vacuum carrier and unit generator-three carrier, with source, retained reservoir and work support held separately. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-trace__half-One-versus-one-third-positive-beat-transfer__complete-registered-product__all-omitted-carrier-forms-rejected__exact-result-sealed-before-comparison__complete-source-transfer-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	borrowed-vacuum-number	A borrowed vacuum number has no Fold trace.
dependency	admitted-V3-dependency-trace	asserted-prior-answer	An asserted prior answer is an unregistered premise.
relation	half-One-versus-one-third-positive-beat-transfer	asserted-free-energy-or-erased-outward-transfer	The alternative loses a required carrier or imports an extra law.
enumeration	complete-registered-product	selected-candidate-neighbourhood	A selected neighbourhood cannot establish uniqueness.
minimality	all-omitted-carrier-forms-rejected	survivor-without-lower-controls	A bare survivor does not prove the required carriers are minimal.
measurement	exact-result-sealed-before-comparison	target-visible-before-seal	Target access before sealing reverses the scientific direction.
record	complete-source-transfer-control-record	answer-only-record	An answer alone cannot reproduce the source and transfer ledger.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:2f47106944b3e20b732cc0595935b9a514f9dcd24a2d4f15de3c6b4e589229e9. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:4bf356fe9c24fe022293f5a5ed5c1500682d3b164da201fa5959b89b18d8112b.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:b6de76683aea01da6edd411b7e57d7d6a6eb1488190604c7ba3887150ea85ce3.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:91bd04981098412ed0e07bda628244e221c20bc8043ffec95e8a4aacaec2e393.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:d13dd0b618e4449e4f46631bd8b8e922aed44be2978f532e283bdfa191ffae7b.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:46b4fc1bc5063437791f86dec18f6804cb3c98174c289b1500bc51ab6c880a7f. Independent certificate: sha256:793c10b8b4aed4ee54bfc17343955c4ee50a8d51a0730d743ea662aad26e0d42. Engine

external-validation hash:
`sha256:4dd6db68e462a6363b9a827baff37f5910702a00dcea9f6a3fdb06f74fed5058`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, candidate table or answer artifact as a premise
- no external measurement, conventional vacuum model or apparatus result selecting a survivor
- no fitted coefficient, tunable phase, chosen reservoir value or answer-selected harmonic
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no deletion of the positive outward extraction result or of the complete restoration ledger
- no claim of repeated net work without restoring the changed vacuum carrier

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:f6f2c4c55430e81592b9cf311250acda2491ce4cfd32da5f042e6cd92b055985`; engine receipt `sha256:de7c651c4eccf51ac10efdc04853bcd36fcb3ef0dcbe3ef58bba0cf52efc0e1d` at `receipts/engine/model_admitted/SFT-PHYS-VACUUM-ASYMMETRIC-BEAT-EXTRACTION-003-de7c651c4eccf51a.json`; empirical-validation hash `None`; measurement receipt `None`.

99. Complete returned-cycle vacuum energy ledger

Claim identity: `SFT-PHYS-VACUUM-COMPLETE-CYCLE-LEDGER-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The admitted outward beat transfers one-sixth to work and leaves one-third in the vacuum carrier. Returning the apparatus and vacuum carrier to their exact initial state requires transferring the same one-sixth back. The vacuum is restored to one-half and the work carrier becomes the empty structural record. A complete returned cycle therefore retains the outward extraction event while forbidding an unrecorded net-support gain.

> *A complete returned cycle is: outward vacuum 1/2 -> 1/3 + work 1/6; restoration 1/3 + 1/6 -> vacuum 1/2; residual work is the empty structural form, not numerical zero.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-VACUUM-ASYMMETRIC-BEAT-EXTRACTION-003`
- `SFT-PHYS-THERMO-FIRST-LAW-001`
- `SFT-PHYS-THERMO-ENTROPY-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of returned-cycle carrier, dependencies, full ledger relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: The complete two-leg process returning vacuum, apparatus state and retained information to their initial records, with every positive transfer held and structural absence represented only by the empty form. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-trace__outward-one-sixth-and-equal-restoration-cost__complete-registered-product__all-omitted-carrier-forms-rejected__exact-re` sult-sealed-before-comparison__complete-source-transfer-control-record__no-extra-rule . Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	borrowed-vacuum-number	A borrowed vacuum number has no Fold trace.
dependency	admitted-V3-dependency-trace	asserted-prior-answer	An asserted prior answer is an unregistered premise.
relation	outward-one-sixth-and-equal-restoration-cost	partial-cycle-net-gain-or-erased-extraction	The alternative loses a required carrier or imports an extra law.
enumeration	complete-registered-product	selected-candidate-neighbourhood	A selected neighbourhood cannot establish uniqueness.
minimality	all-omitted-carrier-forms-rejected	survivor-without-lower-controls	A bare survivor does not prove the required carriers are minimal.
measurement	exact-result-sealed-before-comparison	target-visible-before-seal	Target access before sealing reverses the scientific direction.
record	complete-source-transfer-control-record	answer-only-record	An answer alone cannot reproduce the source and transfer ledger.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:e1fc93d8c434b200be0d721922f3f8ad2852f1af09dff8227be458caaaefb916. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:6c9a7d833b3b0110709bda928a42da2f53e18eedb4a33baa3e9381c747fc78c0.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:2bde9e9f906aaad134ebdb7a39a62da7fb987ba0ce794556b05363a2043a64ce.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:071b8351889521815841af93def5f161d15d6488e583a10c04adaa6d41bf3687.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:791700b22bce99bfb57c65e063896b88220dee949e16479c67f75d784e7d7d7a.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:46b4fc1bc5063437791f86dec18f6804cb3c98174c289b1500bc51ab6c880a7f. Independent certificate: sha256:c49bcf20bf0c88eb8ebb0994347a7692fb3046cce229aefc8e8a252f9c5739b7. Engine external-validation hash: sha256:8bd9231535784a0b227c3b86d1731fbf499c432eb53cd3d3eead47c5143faded.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S013 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, candidate table or answer artifact as a premise
- no external measurement, conventional vacuum model or apparatus result selecting a survivor
- no fitted coefficient, tunable phase, chosen reservoir value or answer-selected harmonic
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no deletion of the positive outward extraction result or of the complete restoration ledger
- no suppression of reservoir depletion, apparatus reset, information record or restoration cost

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:98996bf1b2d39caa17f257315a7068e396ef95e789608d452d163589cac20c45`; engine receipt `sha256:a0ae88fcb49661c576a54741af33c09c88fb11d125f48e41edf2658b08ede835` at `receipts/engine/model_admitted/SFT-PHYS-VACUUM-COMPLETE-CYCLE-LEDGER-003-a0ae88fcb49661c5.json`; empirical-validation hash `None`; measurement receipt `None`.

100. Exact one-distinction erasure cost and Maxwell-demon ledger

Claim identity: `SFT-PHYS-THERMO-LANDAUER-DEMON-TERMINAL-018`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete quarter-One/three-quarter-One reset fibre merges at half-One, closes one distinction with native throw half-One, requires one reverse label, and forces a Maxwell-demon cycle to export exactly one environment record.

> *Resetting the complete quarter-One/three-quarter-One Fold fibre to half-One closes exactly one binary distinction with native throw half-One; reversal requires one predecessor label; and a closed Maxwell-demon cycle exports exactly that one label to its environment.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-DYNAMICS-001`
- `SFT-FOUNDATION-HALF-ONE-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-INFO-ENTROPY-UNCERTAINTY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`
- `SFT-PHYS-THERMO-SECOND-LAW-001`
- `SFT-PHYS-THERMO-HEAT-WORK-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete product of reset support, image, distinction, native throw, reverse record, sorting, memory, environment, thermal translation and extension forms. The declared exact boundary is: Every exact reset of the complete two-preimage Fold fibre to one held ready image; every finite closed sorting-memory-reset-environment cycle; and every retained predecessor-label assignment. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `complete-two-preimage-fibre__common-half-One-image__exactly-one-binary-distinction__half-One-preimage-separation__retain-one-predecessor-label__one-held-gas-distinction__two-label-Fold-memory__one-environment-reverse-label__postseal-dimensional-correspondence__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`support`	`complete-two-preimage-fibre`	`sampled-memory-state`	A sampled state cannot establish a two-to-one reset.
`image`	`common-half-One-image`	`selected-reset-state`	A selected state has no Fold trace.
`distinction`	`exactly-one-binary-distinction`	`unquantified-information-loss`	An unquantified loss cannot balance a cycle.
`throw`	`half-One-preimage-separation`	`free-or-vanishing-cost`	A free or vanishing cost violates the exact fibre separation.
`reverse`	`retain-one-predecessor-label`	`infer-erased-predecessor`	The common image cannot identify its source.
`sorting`	`one-held-gas-distinction`	`unrecorded-sorting-gain`	An unrecorded distinction is outside the closed system.
`memory`	`two-label-Fold-memory`	`memory-without-two-states`	A one-state memory cannot encode the decision.
`environment`	`one-environment-reverse-label`	`erased-without-external-record`	That would destroy the predecessor distinction from the closed ledger.
`thermal`	`postseal-dimensional-correspondence`	`import-kT-log-two-as-premise`	A conventional thermal formula cannot select the Fold law.
`extension`	`no-extra-rule`	`free-cost-or-demon-exception`	A free exception defeats closure.

Base and successor. The closure proof is sealed by derivation hash `sha256:86744c170537a1f26f665bd8afa02395f86f81ee0fc6d48028eb7c6e2a61b6ad`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:556549826b012b3cd71e8b80542015d3224b773d5fa6f5869057a4b2d549c6ac`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:f085eeb647249724d91ed525c9cccd368fe1fbbaa6b20ac7accc21ce81840518`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9edfe8f63e326a077581b89b692a4cd4105839ff44467908a825fdb9b2dcc58e`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:372c69b1e6718117746a72b16bdb6732ac79dab6e19b75d13f20b81b065a1275`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:1829d0eb2c8ca53323d38289398a7be8cf250eee451bab6ef02f7c374cfe7758`. Independent certificate: `sha256:7acc7a70718cf879102de06754364697ed9fc271868177acb62f393306874ac8`. Engine external-validation hash: `sha256:07a875d566587e561700d4f1eecc50c80b2fea02cfbb581955e3bc3a0adb54d3`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no conventional Landauer formula, logarithm or experimental target as a derivation premise

- no claim that native half-One is numerically identical to a dimensional energy ratio
- no numerical-zero, negative, irrational, imaginary or floating proof value
- no stochastic demon decision, free reset or omitted environment record
- no V1/V2 executable, answer table or stored survivor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:19be5baaaefa68250f055175ebd66cf2fee7fdadf3332a82c1ba8c77e1ae0575`; engine receipt `sha256:b130990c39c87ac65a1f5728a4c471bc05f8f5720d0413feebffb5b03760f88b` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-LANDAUER-DEMON-TERMINAL-018-b130990c39c87ac6.json`; empirical-validation hash `None`; measurement receipt `None`.

101. Post-seal Landauer erasure and Maxwell-ledger comparison

Claim identity: `SFT-PHYS-THERMO-LANDAUER-EMPIRICAL-019`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The admitted one-distinction Fold reset predicts nonfree environmental transfer; primary theory gives the conditional k_B T ln 2 average bound and primary experiment reports long-cycle mean-heat saturation, without equating native half-One to a dimensional energy number.

> *The exact Fold one-distinction reset predicts a nonfree environmental transfer for physical erasure. Primary theory identifies the symmetric thermal-reservoir ensemble bound as k_B T ln 2 and primary experiment finds long-cycle mean heat saturating it; all distribution, reservoir, average and long-cycle scope rows are retained, while native half-One is not falsely identified with a dimensional energy number.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-THERMO-LANDAUER-DEMON-TERMINAL-018`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of physical carrier, erasure relation, provenance, prediction isolation, measurement separation, row retention, successor closure and extension forms. The declared exact boundary is: The admitted exact two-preimage Fold reset and every registered primary source row concerning a symmetric one-bit memory, thermal-reservoir erasure, environmental heat, ensemble averaging and long-cycle approach. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__one-distinction-reset-with-environment-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`one-distinction-reset-with-environment-record`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:16eae314150f4ef5415826275a0bd5bfffe62cd95ee60d16ecb10273a76d196a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2047f7bb211fa66dcc8744e437d3c453a04625798a8920fd39b5b1c28d88c8c8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:befd819dcb8d77edbc8ef2a6234a79acd3f70a14eacffbc2c769f917bbea35f2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4050b7eaddd8cccadd44897f07254dae6869044a51fd68cf3c1728b0892fb4fa`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e55ed114da671c1b05fc0e9a9e9b6abb33f388712ed08769dbff27ddd4f4edc0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:84839a2508413f560a01deddae908cb9c312f00d19c256c0e60601972e2ddff5`. Independent certificate: `sha256:66b4c5a652c1c49075027abfbc6a3ded3c0d080cdc3462e286624478182376b8`. Engine external-validation hash: `sha256:8970e6e8ed2f188332f346b0959e9ad84999cd7ca38f0c33053b7ec85b7306e9`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PIECHOCINSKA-PRA-2000
- BERUT-NATURE-2012

Observed comparison records:

- LANDAUER-TWO-TO-ONE-RESET: predicted two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; observed two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; exact match True
- LANDAUER-THERMAL-BOUND: predicted two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; observed two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; exact match True
- LANDAUER-EXPERIMENTAL-SATURATION: predicted two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; observed two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; exact match True

- LANDAUER-SCOPE-CONTROL: predicted two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; observed two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Reject if the source snapshot changes; if a registered reset, heat, bound, experiment or scope row is omitted; if physical reliable erasure has no positive environment transfer; if long-cycle mean heat does not approach the registered bound; if the thermal coefficient enters the Fold proof; or if native half-One is misreported as a dimensional numerical equality.

Measurement receipt: `sha256:22ca198ca9538ec8ce25f9a63769c520fa423768c6e6d2500c3df4c2e45a3476`. Isolation certificate: `sha256:d02b3f8cfaafaa6e038f114f6b54b919ec8f2aca69aaaf284193952e0d3c778d`. Custody certificate: `sha256:4df01ed99678ac1bfe03b87d41d775043d9767c29e7136890eb4d0d697521820`.

Registered source descriptions:

- Nature: https://www.nature.com/articles/nature10872 (sha256:76a746eb8760816598724a2db17fb7cd1f3a07325e69193525d313fecea4d27f)
- American Physical Society: https://harvest.aps.org/v2/journals/articles/10.1103/PhysRevA.61.062314/fulltext (sha256:76a746eb8760816598724a2db17fb7cd1f3a07325e69193525d313fecea4d27f)

Registered target identities:

- LANDAUER-TWO-TO-ONE-RESET from PIECHOCINSKA-PRA-2000
- LANDAUER-THERMAL-BOUND from PIECHOCINSKA-PRA-2000
- LANDAUER-EXPERIMENTAL-SATURATION from BERUT-NATURE-2012
- LANDAUER-SCOPE-CONTROL from BERUT-NATURE-2012

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no source text, k_B T ln 2 coefficient or experimental result in the formal predecessor law
- no numerical identification of native half-One with joules or with ln 2
- no omission of ensemble-average, reservoir, distribution or long-cycle limitations
- no numerical-zero, negative, irrational, imaginary or floating SFT proof value
- no target access before the prediction seal and no fitted correction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d624b93544e7d34754e530739caa4c2e54f6d204bb2b8e12fa027b643db37cc3`; engine receipt `sha256:eb264a10c1fe3666cf25a5e18beb9bab271534333bfdee25fadc0ffafcc3a93e` at `receipts/engine/model_admitted/SFT-PHYS-THERMO-LANDAUER-EMPIRICAL-019-eb264a10c1fe3666.json`; empirical-validation hash `sha256:774f63c5cce8ca0eee88a9f9cda0d0bf6095d9d105d90bc43c45f8b2e6b0c0c8`; measurement receipt `sha256:22ca198ca9538ec8ce25f9a63769c520fa423768c6e6d2500c3df4c2e45a3476`.

102. Terminal vacuum floor, cosmic density and scale-transport law

Claim identity: `SFT-PHYS-VACUUM-DENSITY-SCALE-TERMINAL-035`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The unique complete local vacuum boundary-energy floor is One/2^20=(One/2^10)^2. At every finite radiative depth k, the complete 2^k half-spacing modes sum to 2^(k-1) and have common-scale mean 1/2. The terminal cosmic vacuum fraction is separately 11/16. The normalised cosmological magnitude is Lambda(c/H)^2=33/16, and dimensional transport is

Lambda=(33/16)H^2/c^2 after held external scale references are supplied. Equating the local floor, a raw mode total, the global fraction or a dimensional cosmological density is rejected as a type error.

> *The unique complete local vacuum boundary-energy floor is One/2^20=(One/2^10)^2. At every finite radiative depth k, the complete 2^k half-spacing modes sum to 2^(k-1) and have common-scale mean 1/2. The terminal cosmic vacuum fraction is separately 11/16. The normalized cosmological magnitude is Lambda(c/H)^2=33/16, and dimensional transport is Lambda=(33/16)H^2/c^2 after held external scale references are supplied. Equating the local floor, a raw mode total, the global fraction or a dimensional cosmological density is rejected as a type error.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-ONE-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-FOUNDATION-EXACT-OPERATIONS-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003`
- `SFT-PHYS-COSMO-COMPLETE-BUDGET-001`
- `SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032`
- `SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030`
- `SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001`

Generated grammar and closure boundary. Generate the complete product of generator volume, binary cover, boundary-label scope, observable composition, local floor, finite radiative ledger, cosmic share, spatial rate multiplicity, typed prior-claim correction, scale transport, measurement custody and extension form. The declared exact boundary is: The generated volume 3^3; every binary support until the unique least cover; one-label versus complete-two-label boundary support; amplitude versus two-leg energy composition; every finite positive-depth complete half-spacing mode ledger; exact 11/16 terminal vacuum share; three-space rate transport; every exact positive rational held rate/speed reference; and no untyped equality between a local floor, density fraction and dimensional cosmological constant. The generator produced `4096` named candidates and the decision support contains `4096` one-for-one decisions. Exactly one candidate survived: `generator-three-cubed-volume__least-cover-depth-five__both-held-labels-per-depth__two-leg-energy-self-composition__complete-boundary-energy-floor__complete-finite-ledger-and-half-One-mean__terminal-eleven-sixteenths-share__three-space-squared-rate-carrier__distinct-typed-quantities-and-prior-correction__postseal-rate-squared-over-speed-squared__postseal-only-comparison__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`volume`	`generator-three-cubed-volume`	`borrowed-cosmic-volume`	A borrowed volume imports a model.
`cover`	`least-cover-depth-five`	`selected-depth-ten-or-twenty`	Selecting the desired exponent is circular.
`boundary`	`both-held-labels-per-depth`	`one-label-per-depth`	One label omits half the Fold fibre.
`observable`	`two-leg-energy-self-composition`	`amplitude-relabeled-as-energy`	A one-leg support is not a complete energy observation.
`floor`	`complete-boundary-energy-floor`	`named-one-over-two-to-twenty`	Naming the prior value does not force it.

Axis	Forced form	Eliminated form	Elimination reason
`radiative`	`complete-finite-ledger-and-half-One-mean`	`unbounded-mode-sum-as-local-density`	An unbounded sum erases finite support, common scale and quantity type.
`cosmic`	`terminal-eleven-sixteenths-share`	`local-floor-equals-cosmic-fraction`	The quantities have different supports and types.
`geometry`	`three-space-squared-rate-carrier`	`borrowed-continuum-coefficient`	A conventional field equation cannot be a premise.
`typing`	`distinct-typed-quantities-and-prior-correction`	`untyped-number-identification-or-rubber-stamp`	Equal-looking numbers without scale carriers are not one quantity, and preserving an old label cannot excuse a type error.
`transport`	`postseal-rate-squared-over-speed-squared`	`fitted-dimensional-value`	A fitted unit magnitude is a parameter.
`measurement`	`postseal-only-comparison`	`measurement-readable-before-seal`	That would fit the magnitude.
`extension`	`no-extra-rule`	`free-vacuum-scale`	A second vacuum ruler violates the one-axis law.

Base and successor. The closure proof is sealed by derivation hash `sha256:c01238ae67c8185018e5b6be9ad1cf39d8c2dfc241b7e6e0845381f81fa87586`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:4aeb43be5f9e2b45452280cbd30496eddbc0607aba3ac2a1b9de4e54fcdc820c`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:81167a736ebfd0209293af293c2d9751cb0eff2a31cf7a415da2653ef0f2ad19`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:74a9f3aaf140a89bddd338da261541701151f82dd1ed6490a59310d95e21db5c`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:77bde99e13a432ec1df6a953152db7ae57c1fc3aaf061198814f68ce4cfd4cad`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ea8f27dc1760e402a959b6da81c14c9bbc8ecb3c20de37ede2abfc9a0f41fc10`. Independent certificate: `sha256:b4c62c3d58a90ce9af0c1db9053aed8b9962d5cab16e2adf5502500d15bb31ad`. Engine external-validation hash: `sha256:0f107bf8bfbd842816b120cd750e391cd712a567ecd649cad20db2163b818f52`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, stored exponent survivor, dark-energy measurement or conventional field equation in forcing
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof scalar
- no fitted cosmological density, Hubble value, gravitational value or second vacuum scale

- no direct identification of the local One/2^20 cell floor with Omega_v, Lambda or a dimensionful density
- no reuse of the withdrawn measurement-influenced 2 alpha^4/H^3 proposal
- no claim that a continuum zero-point mode sum is an SFT derivational object

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:b6466fba1a682b36391e848e061aa30773de6cb511b44b1767e23cb677346772`; engine receipt `sha256:c7b4777b12fc70628b0fd9a2f7d957274b61ede7f3624a665781364c8c9f7723` at `receipts/engine/model_admitted/SFT-PHYS-VACUUM-DENSITY-SCALE-TERMINAL-035-c7b4777b12fc7062.json`; empirical-validation hash `None`; measurement receipt `None`.

103. Terminal thermal-history, freeze-out and recombination law

Claim identity: `SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every exact positive scale growth g, T_later=T_earlier/g and T_later*g=T_earlier. Distinct binding thresholds are crossed in unique descending order. The depth-seven binary orbit (1,2,4)/7 forces freeze-out neutron/proton shares 1/7 and 6/7 and ratio 1/6. The least complete binary-cover successor forces capture-entry shares 1/8 and 7/8 and ratio 1/7. Complete retained-neutron pairing forces helium-family mass share 1/4 and hydrogen-family share 3/4. Recombination has an exact half-One classification midpoint and a finite positive visibility ledger at every generated radius. Acoustic standing modes are positive whole labels with exact odd/even compression-rarefaction parity; observed angular positions retain their projection and are not proof identical to the internal whole labels.

*For every exact positive scale growth g, T_later=T_earlier/g and T_later*g=T_earlier. Distinct binding thresholds are crossed in unique descending order. The depth-seven binary orbit (1,2,4)/7 forces freeze-out neutron/proton shares 1/7 and 6/7 and ratio 1/6. The least complete binary-cover successor forces capture-entry shares 1/8 and 7/8 and ratio 1/7. Complete retained-neutron pairing forces helium-family mass share 1/4 and hydrogen-family share 3/4. Recombination has an exact half-One classification midpoint and a finite positive visibility ledger at every generated radius. Acoustic standing modes are positive whole labels with exact odd/even compression-rarefaction parity; observed angular positions retain their projection and are not proof identical to the internal whole labels.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-ONE-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-FOUNDATION-EXACT-OPERATIONS-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-PHYS-THERMO-TEMPERATURE-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-PHYS-COSMO-REDSHIFT-001`
- `SFT-PHYS-COSMO-BACKGROUND-001`
- `SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032`
- `SFT-PHYS-MATTER-BARYON-PHOTON-TERMINAL-004`
- `SFT-PHYS-ATOMIC-HYDROGEN-RYDBERG-TERMINAL-005`
- `SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006`
- `SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006`

- SFT-PHYS-PLASMA-COLLECTIVE-001
- SFT-PHYS-PLASMA-OSCILLATION-001

Generated grammar and closure boundary. Generate the complete product of provenance, temperature transport, threshold order, freeze-out carrier, complete-cover transport, nuclear capture, recombination support and acoustic observation form. The declared exact boundary is: Every exact positive rational temperature and scale carrier; every finite set of distinct positive binding thresholds; the complete binary orbit of the least part on depth seven; the least complete depth-three binary cover; every finite positive visibility radius; every finite positive internal acoustic mode count; exact generated-finite sound-horizon segments; and strict separation of internal mode labels from observed projection. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: admitted-threshold-and-transport-dependencies__exact-inverse-scale-transport__descending-distinct-binding-thresholds__least-live-neutron-share__least-complete-binary-cover-successor__paired-neutron-helium-and-hydrogen-families__half-One-midpoint-with-finite-visibility__internal-whole-modes-parity-and-projection-record. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
provenance	admitted-threshold-and-transport-dependencies	named-cosmological-chronology	A familiar chronology cannot select a Fold law.
temperature	exact-inverse-scale-transport	same-direction-or-fitted-cooling	A named cooling curve or fitted exponent adds a rule.
threshold	descending-distinct-binding-thresholds	named-epoch-list	Names do not prove order.
freezeout	least-live-neutron-share	one-seventh-ratio-rubber-stamp	The old label conflates a constituent share with a pair ratio.
decay	least-complete-binary-cover-successor	selected-decay-correction	A fitted decay fraction is inadmissible.
capture	paired-neutron-helium-and-hydrogen-families	named-quarter-abundance	Naming one quarter does not force it.
recombination	half-One-midpoint-with-finite-visibility	instantaneous-zero-width-collapse	A zero-width event erases physical visibility support.
acoustic	internal-whole-modes-parity-and-projection-record	observed-multipoles-assumed-exact-integers	Projection and driving cannot be erased to preserve an old slogan.

Base and successor. The closure proof is sealed by derivation hash sha256:e2b318586427283e1e61c49e481e1879341de04ac670c7fafd5eb773e6d1e4b5. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:ca700bf092d66ef62469796c6034f11ce2c3fbf15077e333c745944e70df2965.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:5344c8a8ce62d060bb86c94ede6ce9e478829115c9b53de55441b0605db23a08.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:69a86c8830048d9b18e28fe284a06eb763f1c1810f4670bc959c5dc6be95bc50.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:1835665e3693287862638f0ef309fb7a625762c9048a4d8df2297913b141d8ca.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:22a9e8259b3e561ef7cc79cdf28846e652a16bd4fbdbcea06323462bd46acfe9`. Independent certificate: `sha256:11a443134f16d7f8946c96ad3f94487aa48fddef4ad560f0526a378441a582ed`. Engine external-validation hash: `sha256:bce1224d0bcc023c31e2ca981ca1963b0f59601c8016ec9dfa07f848b3450972`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no conventional hot-big-bang equation, continuum fluid integral, stochastic freeze-out or fitted thermal exponent as a premise
- no measurement value, helium abundance, recombination redshift or CMB peak location available to candidate selection
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof scalar
- no untyped identification of neutron share with neutron/proton ratio
- no instantaneous zero-width recombination event
- no claim that observed CMB angular multipoles are exact integer multiples
- no free reaction rate, decay fraction, sound speed, baryon loading or projection coefficient

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:9e3ef14860b7a7675b9c189a6512168ec5815fdd4289c6652283ba0aad5c186c`; engine receipt `sha256:25dda78644ebc04ad43cd3e416950ad76dae6b124ed023f41364f0661eae9c00` at `receipts/engine/model_admitted/SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037-25dda78644ebc04a.json`; empirical-validation hash `None`; measurement receipt `None`.

104. Terminal exact temperature, finite canonical equilibrium and fluctuation-response law

Claim identity: `SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every finite positive population, T=(sum throws)/N and total=N*T. Every paired binary population uniquely maximizes multiplicity at half-One. For L levels, exact canonical counts are (2^(L-1),...,1), weights divide by 2^L-1, sum to One and are the unique fixed-count/fixed-throw multinomial maximum. The least fluctuation/response pair is 3/4 and 1/4 around equilibrium 1/2, with equal departure 1/4; the complete deterministic cycle (1/4,1/2,3/4,1/2) has mean 1/2.

*For every finite positive population, T=(sum throws)/N and total=N*T. Every paired binary population uniquely maximizes multiplicity at half-One. For L levels, exact canonical counts are (2^(L-1),...,1), weights divide by 2^L-1, sum to One and are the unique fixed-count/fixed-throw multinomial maximum. The least fluctuation/response pair is 3/4 and 1/4 around equilibrium 1/2, with equal departure 1/4; the complete deterministic cycle (1/4,1/2,3/4,1/2) has mean 1/2.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-ONE-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-HALF-ONE-001`

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-PHYS-THERMO-TEMPERATURE-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001
- SFT-PHYS-THERMO-FLUCTUATION-001
- SFT-PHYS-THERMO-RESPONSE-001

Generated grammar and closure boundary. Generate the complete product of temperature, total throw, equilibrium, canonical weighting, maximum-count, fluctuation, response and noise-orbit forms. The declared exact boundary is: Every finite nonempty tuple of exact positive throws; every positive paired binary population; every finite generated level count at least two; every exact fixed-count/fixed-throw population vector; and every positive repetition of the complete deterministic response cycle. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `exact-positive-mean-throw__population-count-times-mean__unique-binomial-multiplicity-maximum__normalized-dyadic-successor-ladder__complete-fixed-count-throw-multinomial-maximum__three-quarter-complementary-departure__quarter-One-antipodal-response__finite-periodic-orbit-readout`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`temperature`	`exact-positive-mean-throw`	`primitive-continuum-coordinate`	A primitive continuum coordinate is not Fold-generated.
`total`	`population-count-times-mean`	`fitted-Boltzmann-coefficient`	A fitted coefficient adds a parameter.
`equilibrium`	`unique-binomial-multiplicity-maximum`	`selected-half-label`	A named midpoint alone is not a census.
`canonical`	`normalized-dyadic-successor-ladder`	`continuum-exponential`	An exponential is outside exact Fold arithmetic.
`maximum`	`complete-fixed-count-throw-multinomial-maximum`	`asserted-geometric-shape`	A shape assertion does not eliminate alternatives.
`fluctuation`	`three-quarter-complementary-departure`	`ontic-random-kick`	Ontic randomness is not forced.
`response`	`quarter-One-antipodal-response`	`free-dissipation-rate`	A free response adds a parameter.
`noise`	`finite-periodic-orbit-readout`	`untracked-stochastic-source`	An untracked source violates determinism.

Base and successor. The closure proof is sealed by derivation hash `sha256:5c35b4405e5642b59fd905d0bd2234f993c9044d79992afda6fe01fb378c4272`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:c3b1b82ed7cd92c6c8ae26c3af6308311cccecdab0dbfeafae63c7f3bf346418`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:0ab83da6161f67c5bc0714e98e112e553c81dac404123f06a41c4ffe1b564811`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:32db5e2c2486690c946989ea76680a858768efbe3539a292f37f2cf8b5904fc4`.

- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:eb6624a1ed05a67939cf033f893fbff90f4c850fc29c8cd206d14db219e44a4e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:89170070cce40d01eecb4b5d96ff33dba1dca2d04353cab3ee8f7a00348197d5`. Independent certificate: `sha256:5f0562864209036b569059443c080ce156913ae27a8fd0bfffb4482b4a9ee8c4`. Engine external-validation hash: `sha256:5d8e46a19f68ad5d44097d3f450594420c4cbd4f87e765f4c2371a568378f891`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no Boltzmann exponential, continuum bath, Lagrange multiplier or fitted distribution
- no thermometry, noise voltage or measured k_B available to candidate selection
- no ontic randomness
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof magnitude
- empty combinatorial counts are structural absence records only

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:85f556f93bf5dfcf96da823aaa9cafe4cdecf73a781e4bdfc6d03f5f5f95dd22`; engine receipt `sha256:cf536f7516eb1c85bb075500d9c1a0ebd64bdddb544878f26a27424d08de25bf` at `receipts/engine/model_admitted/SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043-cf536f7516eb1c85.json`; empirical-validation hash `None`; measurement receipt `None`.

105. Terminal physical helium-isotope closure

Claim identity: `SFT-PHYS-THERMAL-HELIUM-ISOTOPE-TERMINAL-057`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete four-position by fifteen-nonempty-subword capture ledger has sixty cells and exactly one collective composite-identity record. The unique physical isotope conversion is therefore 59/60. Composed with the admitted analytic helium-family share One/four, this forces primordial physical helium-isotope share 59/240 and complementary hydrogen-family share 181/240, closing exactly to the One.

> *The complete four-position by fifteen-nonempty-subword capture ledger has sixty cells and exactly one collective composite-identity record. The unique physical isotope conversion is therefore 59/60. Composed with the admitted analytic helium-family share One/four, this forces primordial physical helium-isotope share 59/240 and complementary hydrogen-family share 181/240, closing exactly to the One.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037`
- `SFT-PHYS-MATTER-COMPOSITE-HADRONS-001`
- `SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005`
- `SFT-MATH-EXACT-ARITHMETIC-001`

- SFT-MATH-COMBINATORICS-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

Generated grammar and closure boundary. Generate the complete product of analytic-family, constituent, subword, capture-product, composite-identity, isotope-conversion, physical-partition and target-boundary forms. The declared exact boundary is: The admitted analytic helium family; exactly four helium-four constituent positions; every nonempty subword of those positions; their complete incidence product; the single global composite identity; exact positive rational partitions; and no measurement in candidate selection. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `admitted-quarter-baryon-family__helium-four-complete-arity__all-fifteen-nonempty-subwords__complete-four-by-fifteen-product__one-global-composite-identity__fifty-nine-of-sixty-isotope-cells__fifty-nine-over-two-forty-and-one-eighty-one-over-two-forty__target-closed-until-formal-seal`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`family`	`admitted-quarter-baryon-family`	`measured-decimal-called-family-law`	A measured decimal cannot select the analytic predecessor.
`constituents`	`helium-four-complete-arity`	`selected-arity`	A selected isotope arity is an extra premise.
`subwords`	`all-fifteen-nonempty-subwords`	`selected-proper-subset`	A selected subset erases a physical composition channel.
`product`	`complete-four-by-fifteen-product`	`partial-incidence-ledger`	A partial ledger breaks permutation completeness.
`identity`	`one-global-composite-identity`	`empty-or-multiple-composite-identities`	No identity leaves the labels unbound; multiple identities add an ungenerated split.
`conversion`	`fifty-nine-of-sixty-isotope-cells`	`fitted-isotope-correction`	A fitted correction reads the measurement.
`partition`	`fifty-nine-over-two-forty-and-one-eighty-one-over-two-forty`	`quarter-rubber-stamp-or-decimal-copy`	Neither the analytic quarter nor a copied decimal is the completed isotope partition.
`target`	`target-closed-until-formal-seal`	`target-readable-candidate-selection`	Target access could manufacture a correction.

Base and successor. The closure proof is sealed by derivation hash `sha256:ef909127179759e20899c1a50e206b4eba113be6e79c7f1cf06c05c14dfc7327`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:ecce4667ef692326a59c782ce85a23682f9c1d0a82679d13ce636264cd914663`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:8d4c010fe7c9e3c154a34ec6d86d4704fbe8b5340c827720ad6f08ba8a152e2e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6ed685241657e4ba96441f453f3f0318495b23ec540abd4cdfa60e2235d11e08`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:ee2859ae1b79e5ae1f35d866602c3857fb690467ae4dc1972b81c11cbb2f9d48`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4546969d129794116becb23ba047631a9235a6af3675cdfa5639e9bdc26ece39`. Independent certificate: `sha256:6005685a689e8c82d6ec3c8079e8949bc5a2d84150cc802332ad387a95b52c6d`. Engine

external-validation hash:
`sha256:1e59614b5080af79330ce3d00e48e3da143d488a925dc3e0f94eb9385e72d31e`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured helium abundance, uncertainty or central value in formal survivor selection
- no fitted decay rate, reaction rate, binding coefficient, abundance correction or selected denominator
- no omission of a constituent position or nonempty subword
- no multiple ungenerated composite identities
- no numerical-nothing, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof scalar

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:324b4b482bfbf10a8ae53f9b27a1814344b7405e5638423d292672166b74a3c1`; engine receipt
`sha256:a3b3a44a0d3add7032680427c3b2b504147c0eb824d05fa5d044cef454c5ebc4` at `receipts/engine/model_admitted/SFT-PHYS-THERMAL-HELIUM-ISOTOPE-TERMINAL-057-a3b3a44a0d3add70.json`
; empirical-validation hash `None`; measurement receipt `None`.

106. Exact local resonant vacuum-state drive channel

Claim identity: `SFT-PHYS-VACUUM-LOCAL-RESONANT-DRIVE-083`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every ordered exact positive pair b<a within the One, a lawful local resonant drive from a to b retains the exact positive transfer a Take b and reconstructs a as b plus that transfer. Odd-denominator carriers additionally retain a finite live Fold phase trace. This forces a local drive channel without selecting a device, field strength, frequency or magnitude.

For every ordered exact positive pair b<a within the One, a lawful local resonant drive from a to b retains the exact positive transfer a Take b and reconstructs a as b plus that transfer. Odd-denominator carriers additionally retain a finite live Fold phase trace. This forces a local drive channel without selecting a device, field strength, frequency or magnitude.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-VACUUM-ODD-RECURRENCE-003`
- `SFT-PHYS-TESLA-RESONANT-TRANSFER-081`
- `SFT-PHYS-WAVE-RESONANCE-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of local carrier, resonant drive, exact state transition, positive floor, complete ledger, measurement direction, record and extension forms. The declared exact boundary is: Every ordered pair of distinct exact positive local vacuum parts, every positive finite phase-act count, their complete transfer ledger and all 256 registered structural alternatives. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `live-exact-positive-vacuum-carrier__resonant-source-bound-transition__exact-local-state-change-with-source-bound-transfer__positive-finite-depth-floor__complete-drive-response-restoration-ledger__f` `ormal-seal-before-apparatus-comparison__complete-favourable-adverse-unresolved-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`live-exact-positive-vacuum-carrier`	`inert-empty-background`	An inert empty background has no generated state to drive.
`drive`	`resonant-source-bound-transition`	`unrecorded-field-assertion`	An unrecorded assertion supplies no transition or conservation trace.
`relation`	`exact-local-state-change-with-source-bound-transfer`	`free-response-coefficient`	A free coefficient is an unforced parameter.
`floor`	`positive-finite-depth-floor`	`drive-to-numerical-nothing`	Numerical nothing is not an admitted physical carrier.
`ledger`	`complete-drive-response-restoration-ledger`	`outcome-only-ledger`	An outcome-only record hides source, loss or restoration cost.
`measurement`	`formal-seal-before-apparatus-comparison`	`target-selected-channel`	Target selection is fitting.
`record`	`complete-favourable-adverse-unresolved-record`	`favourable-only-record`	Selective retention cannot test a controversial claim.
`extension`	`no-extra-rule`	`free-extra-rule`	An exception or hidden interaction is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:893630a370e417d16944e521d4e6a47c5d5380be45bbf04f3bf0d7f793df40d4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:de8a1addfd8c353e86de13275e6b7858440140512349f6e399c398b735043098`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:4c675afae1b83249d0540988d5fe550f56254cc3bd49bbdaa6f4d15c60cd4151`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c4820d48548f8599168c38083953042fb5332da742e06122330f83b49480af5a`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:d2387b55601003bbc22c07e35cf28e9c2edfe9c04de886852888ca5782c0cf5f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5265efd53406b707522b4affb76a92415755ef069557e867636fc43927d86224`. Independent certificate: `sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60`. Engine external-validation hash: `sha256:f10a33959cca6fa26c977c9d2e07f6b4a409ebe2a4dd4fbe5847cc9d6dea393d`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, patent formula, apparatus equation or desired outcome as a premise
- no measured acceleration, field strength, device performance or target value selecting a survivor
- no free coupling, tunable efficiency, hidden reservoir or omitted information record
- no semantic numerical zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude
- no claim that structural co-variation proves any craft, propulsion system or useful apparatus magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:de08516821e925c38701b82ec2c089548bb9fdf870afca2c56d69191ad311cc3`; engine receipt `sha256:3ffdf3606dd8f27146c3e0bb203c3758e85780049e8c60e074841d567bfc1e0a` at `receipts/engine/model_admitted/SFT-PHYS-VACUUM-LOCAL-RESONANT-DRIVE-083-3ffdf3606dd8f271.json`; empirical-validation hash `None`; measurement receipt `None`.

107. Exact driven vacuum-to-inertia co-variation law

Claim identity: `SFT-PHYS-VACUUM-INERTIA-COVARIATION-084`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every lawful exact local change from vacuum carrier a to b, the unity exchange law forces the inertial carrier to change from a to b and forces both positive change magnitudes to equal a Take b. The initial and driven vacuum/inertia ratios remain the One. In the historical one-third to one-quarter probe, both changes are exactly one-twelfth.

> *For every lawful exact local change from vacuum carrier a to b, the unity exchange law forces the inertial carrier to change from a to b and forces both positive change magnitudes to equal a Take b. The initial and driven vacuum/inertia ratios remain the One. In the historical one-third to one-quarter probe, both changes are exactly one-twelfth.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-VACUUM-LOCAL-RESONANT-DRIVE-083`
- `SFT-PHYS-VACUUM-INERTIA-UNITY-003`
- `SFT-PHYS-MECH-INERTIA-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of driven carrier, source-bound drive, unity co-variation, positive floor, complete ledger, measurement direction, record and extension forms. The declared exact boundary is: Every ordered exact positive initial/driven vacuum pair, its unity-related inertia pair, both exact positive change carriers and all 256 structural alternatives. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `live-exact-positive-vacuum-carrier__resonant-source-bound-transition__vacuum-change-equals-inertia-change-at-exchange-One__positive-finite-depth-floor__complete-drive-response-restoration-ledger__formal-seal-before-apparatus-comparison__complete-favourable-adverse-unresolved-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`live-exact-positive-vacuum-carrier`	`inert-empty-background`	An inert empty background has no generated state to drive.
`drive`	`resonant-source-bound-transition`	`unrecorded-field-assertion`	An unrecorded assertion supplies no transition or conservation trace.
`relation`	`vacuum-change-equals-inertia-change-at-exchange-One`	`free-response-coefficient`	A free coefficient is an unforced parameter.
`floor`	`positive-finite-depth-floor`	`drive-to-numerical-nothing`	Numerical nothing is not an admitted physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
ledger	complete-drive-response-restoration-ledger	outcome-only-ledger	An outcome-only record hides source, loss or restoration cost.
measurement	formal-seal-before-apparatus-comparison	target-selected-channel	Target selection is fitting.
record	complete-favourable-adverse-unresolved-record	favourable-only-record	Selective retention cannot test a controversial claim.
extension	no-extra-rule	free-extra-rule	An exception or hidden interaction is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9c66e372c437fa5aeb4a16692e0258c582eca4e6944641526a45c0176a713164`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:232636b2f3750354b694269c0dfce6c4d253a2f04461287e2a951e11c5912d30`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:19fa38344e7058eeb35896b11b74380e13f0ac973c65f6d07b1ac576ae974c64`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7112b4c74c009fd16fc0f3a989168f5babab17e8cf47715ac9dc8465b09abfb9`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:bcfea9b44b55fe799ab7a070310fbb7680081f3de02bc0ad5dd87975370c8fdc`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5265efd53406b707522b4affb76a92415755ef069557e867636fc43927d86224`. Independent certificate: `sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60`. Engine external-validation hash: `sha256:d5110dc467ae63be659a0110da44769a0c3eaa986e8a4c1e9024fee1334697ad`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, patent formula, apparatus equation or desired outcome as a premise
- no measured acceleration, field strength, device performance or target value selecting a survivor
- no free coupling, tunable efficiency, hidden reservoir or omitted information record
- no semantic numerical zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude
- no claim that structural co-variation proves any craft, propulsion system or useful apparatus magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:1c02ee591177a4dbba3560c4715b1fab0dfb3025b1cbcfb556d5af4bffa50712`; engine receipt `sha256:a956088e2454d1594a7a04ca4f6050816bca30f59a32aae52dcafd1cb9617474` at `receipts/engine/model_admitted/SFT-PHYS-VACUUM-INERTIA-COVARIATION-084-a956088e2454d159.json`; empirical-validation hash `None`; measurement receipt `None`.

108. Finite-depth positive lower bound on driven vacuum and inertia

Claim identity: `SFT-PHYS-VACUUM-INERTIA-POSITIVE-FLOOR-085`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. At every registered positive finite depth k, the least oscillator carrier is the exact positive part One over 2^(k+1). Every lawful driven vacuum carrier and its unity-related inertia carrier must remain at or above that part. Structural absence is therefore unreachable at every declared finite depth; this is not a claim of one depth-independent dimensional mass floor.

> *At every registered positive finite depth k, the least oscillator carrier is the exact positive part One over 2^(k+1). Every lawful driven vacuum carrier and its unity-related inertia carrier must remain at or above that part. Structural absence is therefore unreachable at every declared finite depth; this is not a claim of one depth-independent dimensional mass floor.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-VACUUM-INERTIA-COVARIATION-084`
- `SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003`
- `SFT-FOUNDATION-PART-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of driven carrier, source-bound drive, unity response, finite-depth floor, complete ledger, measurement direction, record and extension forms. The declared exact boundary is: Every positive finite depth, its complete binary word support and least oscillator part, every driven carrier at or above that floor and all 256 structural alternatives. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `live-exact-positive-vacuum-carrier__resonant-source-bound-transition__finite-depth-floor-bounds-both-unity-related-carriers__positive-finite-depth-floor__complete-drive-response-restoration-ledger__formal-seal-before-apparatus-comparison__complete-favourable-adverse-unresolved-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`live-exact-positive-vacuum-carrier`	`inert-empty-background`	An inert empty background has no generated state to drive.
`drive`	`resonant-source-bound-transition`	`unrecorded-field-assertion`	An unrecorded assertion supplies no transition or conservation trace.
`relation`	`finite-depth-floor-bounds-both-unity-related-carriers`	`free-response-coefficient`	A free coefficient is an unforced parameter.
`floor`	`positive-finite-depth-floor`	`drive-to-numerical-nothing`	Numerical nothing is not an admitted physical carrier.
`ledger`	`complete-drive-response-restoration-ledger`	`outcome-only-ledger`	An outcome-only record hides source, loss or restoration cost.
`measurement`	`formal-seal-before-apparatus-comparison`	`target-selected-channel`	Target selection is fitting.
`record`	`complete-favourable-adverse-unresolved-record`	`favourable-only-record`	Selective retention cannot test a controversial claim.
`extension`	`no-extra-rule`	`free-extra-rule`	An exception or hidden interaction is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9c2e628bf3dd736d61cb7f3941a247bbf59bd0ffe5dc46ca7fbda2106e01cb25`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:564af6ed6904d07c1c728b3d944fa61f65af41ba3367c8f7ebe66ef88c83881f`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:12aa217eca2e9a185079d6fb2ea522cf723f426294ffebbd89ed80e68c398487`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:238fe52231c6273e1e0f559b9da3e150d77cfed552ec66a7c00a978c5f772ebd`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:7666dbc5eb7be2f4646420d1ae62918b83b806c600e29dc33f32c449b5c43513`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5265efd53406b707522b4affb76a92415755ef069557e867636fc43927d86224`. Independent certificate: `sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60`. Engine external-validation hash: `sha256:c1f1e9fc0d7ff4fab91fd39cfc53aaebe25228b93f4a4ca8401648757cb185fd`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, patent formula, apparatus equation or desired outcome as a premise
- no measured acceleration, field strength, device performance or target value selecting a survivor
- no free coupling, tunable efficiency, hidden reservoir or omitted information record
- no semantic numerical zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude
- no claim that structural co-variation proves any craft, propulsion system or useful apparatus magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:6ee6c4ad4f98ca33dd9e85a2c8549be3bb8fb69a973b4ecd23025e2efa8f7ea1`; engine receipt `sha256:0a7552d2128afa000bc701c2199f3e7efdce1c5091bd09d612b3826c900c6e23` at `receipts/engine/model_admitted/SFT-PHYS-VACUUM-INERTIA-POSITIVE-FLOOR-085-0a7552d2128afa00.json`; empirical-validation hash `None`; measurement receipt `None`.

109. Complete vacuum-drive, inertial-response and restoration ledger

Claim identity: `SFT-PHYS-VACUUM-INERTIA-COMPLETE-LEDGER-086`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every lawful exact driven pair b<a, vacuum and inertia both change by a Take b at exchange ratio One. The outward event retains that positive carrier; exact restoration returns both carriers to a. The complete six-label information trace

records initial pair, drive, driven pair, outward transfer, restoration and restored pair. Thus the channel and outward event remain admitted while an unrecorded net-support gain in a fully returned cycle is forbidden.

> *For every lawful exact driven pair $b<a$, vacuum and inertia both change by a Take b at exchange ratio One. The outward event retains that positive carrier; exact restoration returns both carriers to a. The complete six-label information trace records initial pair, drive, driven pair, outward transfer, restoration and restored pair. Thus the channel and outward event remain admitted while an unrecorded net-support gain in a fully returned cycle is forbidden.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-VACUUM-INERTIA-POSITIVE-FLOOR-085`
- `SFT-PHYS-VACUUM-ASYMMETRIC-BEAT-EXTRACTION-003`
- `SFT-PHYS-VACUUM-COMPLETE-CYCLE-LEDGER-003`
- `SFT-PHYS-THERMO-FIRST-LAW-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of local carrier, resonant drive, unity response, positive floor, complete drive/response/restoration ledger, measurement direction, record and extension forms. The declared exact boundary is: Every lawful ordered initial/driven exact pair, equal vacuum/inertia response, outward and restoration transfer, complete six-label information trace and all 256 structural alternatives. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `live-exact-positive-vacuum-carrier__resonant-source-bound-transition__complete-paired-drive-response-restoration-and-information-ledger__positive-finite-depth-floor__complete-drive-response-restoration-ledger__formal-seal-before-apparatus-comparison__complete-favourable-adverse-unresolved-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`live-exact-positive-vacuum-carrier`	`inert-empty-background`	An inert empty background has no generated state to drive.
`drive`	`resonant-source-bound-transition`	`unrecorded-field-assertion`	An unrecorded assertion supplies no transition or conservation trace.
`relation`	`complete-paired-drive-response-restoration-and-information-ledger`	`free-response-coefficient`	A free coefficient is an unforced parameter.
`floor`	`positive-finite-depth-floor`	`drive-to-numerical-nothing`	Numerical nothing is not an admitted physical carrier.
`ledger`	`complete-drive-response-restoration-ledger`	`outcome-only-ledger`	An outcome-only record hides source, loss or restoration cost.
`measurement`	`formal-seal-before-apparatus-comparison`	`target-selected-channel`	Target selection is fitting.
`record`	`complete-favourable-adverse-unresolved-record`	`favourable-only-record`	Selective retention cannot test a controversial claim.
`extension`	`no-extra-rule`	`free-extra-rule`	An exception or hidden interaction is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:3dc440b87ea09bc0f56d4e3059ce010e39122abe280bd987b8eab4dbc9cea412`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt

`sha256:c17111105e2b326cee0d337da6d43bfea80d5df38d7bc533c9225647dd5310aa`.

- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:51fce7e684ad3e1c2041cac594bb2ea1adb3dc614b1c32f8a9b3e23055bee081`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:06860a805b9c1725e6322419133de238cda349a7e4905b992a4661117499df52`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:040ce6e332187a5c07bad41be8276847082d80d0b84f14868eb84ccb1eba6d14`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5265efd53406b707522b4affb76a92415755ef069557e867636fc43927d86224`. Independent certificate: `sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60`. Engine external-validation hash: `sha256:fa1fcab940537ffbbcb9a52010f5708b456539c7034fdb9ca4ebda9061f30c55`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, patent formula, apparatus equation or desired outcome as a premise
- no measured acceleration, field strength, device performance or target value selecting a survivor
- no free coupling, tunable efficiency, hidden reservoir or omitted information record
- no semantic numerical zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude
- no claim that structural co-variation proves any craft, propulsion system or useful apparatus magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:cca2708bc78a848935f58185c8d9a38f2a22169c459a06a2d18ce0914b0060cd`; engine receipt `sha256:3fc114314e2065848eca08536e13323688be410b9693a064921e2c8c62cb8b57` at `receipts/engine/model_admitted/SFT-PHYS-VACUUM-INERTIA-COMPLETE-LEDGER-086-3fc114314e206584.json`; empirical-validation hash `None`; measurement receipt `None`.

16. Physical Quantum Relativistic

Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic.

110. Physical quantum state correspondence

Claim identity: `SFT-PHYS-QUANTUM-PHYSICAL-STATE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A physical quantum state is the complete generated support of distinguishable Fold words together with held phase/action and preparation provenance; it is not an answer-only amplitude list.

> *The physical quantum state is complete finite Fold support with held phase/action and source-bound preparation trace.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__complete-word-support-with-held-phase__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	complete-word-support-with-held-phase	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3d0c012233be13bc786b68fe9c5cf1a75a9ae28f77e23dd94f77b33ec138e06e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f11e96e98ea962641d29ecde80b00416bd1674538139bbe194dd49859d02988f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:672a07dff9f246bfddbf32a064c4aa643ffdcd3d7a7f1b0603c5631ea289fcf7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:815054d543d23cb8762114e29be753c6caafd2edb3526a773cb4c792281373f8`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8cd47b523aba7d4c02f62e90f0dbf213234e179910d3affff1ab8752f8ddcf89`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:fa75461c4e288e36ede8428abbda45cf621831355500de7054dc6b6f4e0868ac`. Independent certificate: `sha256:5efc8bd57bf4ead54b60a7a9f2b23c13a85041c03edc946693ded1dc62e03b6e`. Engine external-validation hash: `sha256:fb58e2c95beba18aadf9bec4b4ff8b65bb47ab7bb6e42d4bed0ced67b93edae3`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- physical-state-1: predicted physical-quantum-support; observed physical-quantum-support; exact match True
- physical-state-2: predicted physical-quantum-support; observed physical-quantum-support; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:ea923780a3e38193e4b9524b8b2b0c275f6f962fa702eaf3368a4b2617636515`. Isolation certificate: `sha256:142cd1a6658c19f795a0f4d1dab92bb1d64046eea6f2513320ba8749b7ab6cd2`. Custody certificate: `sha256:e0da2eeb73e5aefcfcd7ef9a32f5c7b538175abffbd56f69a01d7d6b01c7f6c8`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- physical-state-1 from NIST-ASD-5.12
- physical-state-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:aa82b48e02a601da755b1db69cbb107e3fa721d99ce2949214eb688977e4781f`; engine receipt `sha256:7206fb62f63acf9c8d3de0b3d35306d94a27b9862aa8a5d548423d931b041e86` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-PHYSICAL-STATE-001-7206fb62f63acf9c.json`; empirical-validation hash `sha256:2cce23c405dc14a1ec082d6aeabfff5e0567eb60d845b81f3ef7822d3e2615bc`; measurement receipt `sha256:ea923780a3e38193e4b9524b8b2b0c275f6f962fa702eaf3368a4b2617636515`.

111. Reversible physical quantum evolution

Claim identity: `SFT-PHYS-QUANTUM-EVOLUTION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Closed physical quantum evolution is a one-to-one Fold transition on complete support; its held predecessor labels reconstruct every prior state exactly.

Closed quantum evolution is exact reversible transition of the complete physical Fold support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-PHYSICAL-STATE-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__bijective-complete-support-transition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	bijective-complete-support-transition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:7d065f14c266bfa21f46b3ab31764f173f93db3f9fd75e80034b1f4dc68c07d1`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5aa3839cb9d2fa990c5eef22d1b1bb2a10642f726af0ffdbf0a071f8fc7ba3ed`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7577729d7390f62b2fdb9ae943374aba6baaead28cb6bb4a7caec07bd7d79837`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:646833aea82f4db8f7b9a7b40a8b4d8d66ccd6a60744c865a76651b68d27500d`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:03664f9fbe104e20bee6d232f787bdffdaecf7eb2fcbb41a903dd035e1bc078f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b88b6661761e1575417e3ced3cbbe436daefa0efb420a72ffd6e9d29cf0d87d6`. Independent certificate: `sha256:c5f7cee6cb315e49e0e49a0d5bd83897c9fe20a141537f38d1d96ada4f489557`. Engine external-validation hash: `sha256:724a2c3821bd22a6373a1c7d5bdea206a93740ca976d90b88c6c30dd46f7ee4e`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of time: exact predicted interval [81120909/3353649786316000000000000, 527285909/21798723611006000000000000]; external interval [12094421632919/500000000000000000000000000000, 2418884326589/100000000000000000000000000000]; overlap True
- deliberately displaced unfavourable interval rejected

Falsification condition: The sealed exact action/energy interval does not overlap the released CODATA atomic-time interval or the displaced control is accepted.

Measurement receipt: `sha256:6899038bd28a1a5833f8e89cab3479949b4ea7af2346a73272e03cacba07379a`. Isolation certificate: `sha256:c2dc2a79aabaf766656b57f9d6a4c00b1a88f9350e18e05fed3115a0c1657bf6`. Custody certificate: `sha256:823a0bb083d4e9da4e995441ad1ea3f57e07366af63e5a6bd35f471ce2e53ab1`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- evolution-1 from NIST-CODATA-2022-ALL-CONSTANTS
- evolution-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:4054b59a79a215cb6a585f2bf828a8ed7f3f590a5e8f7eef0198a221136a6a3d`; engine receipt `sha256:e864f02036c1482034ff96e9bd9e01862d3ac6966f1585e24c3218b13d07707b` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-EVOLUTION-001-e864f02036c14820.json`; empirical-validation hash `sha256:b62388ceaa72e06717c27cf63118cf82b01ac59eb4b1c181356faeffef7bd8c1`; measurement receipt `sha256:6899038bd28a1a5833f8e89cab3479949b4ea7af2346a73272e03cacba07379a`.

112. Observable and measurement record

Claim identity: `SFT-PHYS-QUANTUM-OBSERVABLE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A quantum observable is a generated partition of physical support into distinguishable record classes; measurement creates one retained source-bound record for the observed class.

> *A physical observable is an exact support partition and measurement is its provenance-retaining record transition.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-EVOLUTION-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__support-partition-to-retained-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`support-partition-to-retained-record`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:8e4d527a39a2f1a159f7fbc9673c5e4a6d69b5969fd468474591ffc1cfff8c51`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:31e681c5a7afc9a3487c9cf195df85aa683e030ef0fb4c5d807f5b18271593f4`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:958782c82e68ed494612530e55434209d317595a5ddbf1f8ddfb1058552a7d5e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b9d57ba62cdefebfb656007ac936b9d0af7c745999bfa855115292ca30a28787`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f82689babc0a8c4eb4d3bf59e677a2f10f20e0007a87075032177ebb2324c442`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5b205bb4f52f6c844d0dd0b60e252d9c5fe684eba16878c5d62c82453b782472`. Independent certificate: `sha256:d856dda98523631331ca9e6d52d2825b95a3d832172d3fff08ed4158e6978057`. Engine external-validation hash: `sha256:9187e103c01965933fb4461704ef3234dcfb888b202ba45093466ba8c4853cb2`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- observable-1: predicted quantum-observable-record; observed quantum-observable-record; exact match True
- observable-2: predicted quantum-observable-record; observed quantum-observable-record; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:68aabfc11d6831dce25d9b7c97df1ff0f4ec2d3985ebbc761c3e6c2d6c1f891a`. Isolation certificate: `sha256:8a49050f5126cfb9f9852676d82eae3294e24d9e915360f9747e1f1c03e6e88a`. Custody certificate: `sha256:503445d1e238254dc25f02466aa46d2db3ca176da6b362e84d6a2d4833688b59`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- observable-1 from NIST-ASD-5.12
- observable-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:804a453b7457dbeb0bb4129c0dcf9a1d8abc4e9087b0725c4d41ca51f192a5ac`; engine receipt `sha256:26b996d38a3cac3ea54ec0cc2ac86ba1a852905eb68e6513c46c8986de4f2bbb` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-OBSERVABLE-001-26b996d38a3cac3e.json`;

empirical-validation hash
`sha256:29e76882e5f9809053f1c986374dcc2381d76f7dbd74be969deec55ab62735b7`; measurement receipt `sha256:68aabfc11d6831dce25d9b7c97df1ff0f4ec2d3985ebbc761c3e6c2d6c1f891a`.

113. Measurement weights from exact branch support

Claim identity: `SFT-PHYS-QUANTUM-WEIGHT-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The weight of a measurement record is the exact positive count of its phase-compatible predecessor fibre relative to complete generated predecessor support.

> *Quantum measurement weights are exact rational branch-support ratios over a complete deterministic census.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-OBSERVABLE-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__compatible-predecessor-count-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`compatible-predecessor-count-ratio`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9df52b288adbc7724cac3b90d49b2b32ef122013044e61a519c14dd4cb6d71a4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bf51dde0d20e4b318bb6a0714ec5fc0cdce3ab2b91178e3c43fdeb812f2d19cc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d9fc7d487a086cbfa2b9d9e69dddafb926af1837fc1d891ddae7054521e63f3b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7e5b72118aedf409fbf527aa4d17e4b4fceab7d8803cf0bcfd802da9548754db`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9869064001c5cfb9d3cd21d88947abeeacaebb42c5b91b62494c58a07c5f5b69`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f9af1074be87591526a6071e6a4804494c8b408d6d5e87a104a67f0d57271c1f`. Independent certificate: `sha256:daf893f865417ee3cdfc25605e989b9074346c5345fe92ab2cfc268bb782aa78`. Engine external-validation hash: `sha256:9b6076854004687f5d903b74576613d5483180d55ce71d942c278517e3876091`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- weight-1: predicted exact-quantum-branch-weight; observed exact-quantum-branch-weight; exact match True
- weight-2: predicted exact-quantum-branch-weight; observed exact-quantum-branch-weight; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:251b09f87750d747214d928540d010fea781675b4bdfcd6012a557efaf6e531b`. Isolation certificate: `sha256:3e43ee804938a71daca2e51fe65ca9b66436b26c2261cce96b48b98b9e8182bd`. Custody certificate: `sha256:35fc275d16fafc6c2c16532797beaa3f57a8a1ac9ce924904cc1ea9091a82c35`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- weight-1 from NIST-ASD-5.12
- weight-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:96e851e9bfcdab5d4ac429d7257b43588b02bf5592f04b286e15b2182a034362`; engine receipt `sha256:17b61d7d856d18d23cf71a7ef468a6c54d9634ed7e8e3245bd60eed818bd8748` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-WEIGHT-001-17b61d7d856d18d2.json`; empirical-validation hash `sha256:92a5a81e25d00d37969c7a5c401dd3560b3b0bcfb3733f230144518b68a6ce7a`; measurement receipt `sha256:251b09f87750d747214d928540d010fea781675b4bdfcd6012a557efaf6e531b`.

114. Incompatible observations and uncertainty

Claim identity: `SFT-PHYS-QUANTUM-INCOMPATIBILITY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Two observations are incompatible when their support partitions do not share a common refinement preserving both held phase/action records; resolving one necessarily closes distinctions needed by the other.

> *Quantum incompatibility is absence of a joint record-preserving refinement of two generated observation partitions.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-WEIGHT-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__noncommuting-observation-partition-refinement__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`noncommuting-observation-partition-refinement`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:04daaf143b15714955cadcc7f8b31eedc83db1d28b675e64e9f482a952efc44c`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:158811357c5a538c1fca4b7fdaf9bc1fb2c2753248c18b4a1b9f19a16d714de2`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:30262f754cbdae3c420a4809a86515b4fc9ec2e6b0f393d644af5f266f1b82b2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0de745408f2d5a993ebeb485c54819897e0b5ff04972894f1a2a0d6b42961c55`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a2c93fe9bb2f155be22a81fa63369500fcd12366413eeba1836870243c1f7595`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c35b641f13953583e4e48f397470b0a9d440416f2c0bd73e993ac6d7033b8464`. Independent certificate: `sha256:bd6f7030405d0751463d75d2a47c86bb88207d4f8e6cff53d7048eb753b2927d`. Engine external-validation hash: `sha256:ed44760183e162ca03acb29c9f617062d8b9ea16ffc745e64e1678c7b1939c99`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- incompatibility-1: predicted quantum-observation-incompatibility; observed quantum-observation-incompatibility; exact match True
- incompatibility-2: predicted quantum-observation-incompatibility; observed quantum-observation-incompatibility; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:64b11105b4f8c5a6e02b0885f5e1efd28693917aeac8b658e7de14950e364761`. Isolation certificate: `sha256:78e9496036566cba78cf6cb52c98347f62d835dfe586f7f5e9a63eeed34c386a`. Custody certificate: `sha256:4aa5cf4c5448cf39da471836e8ab35938b0d8f08b919c4edd2d957cf40cc183d`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- incompatibility-1 from NIST-ASD-5.12
- incompatibility-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction

- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:22b63da698a8253ec1f634c18f43fd57256250146983cdf21299bf872263d786`; engine receipt `sha256:cde76117a3c79b8e0893b271dd1de908e4d4840ae15c777ead3e44d2d1cc9086` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-INCOMPATIBILITY-001-cde76117a3c79b8e.json`; empirical-validation hash `sha256:078b82ef79f36845938f5fb9f2a2cc9cc6f33d57aaf5b5d80090c22de65d6aed`; measurement receipt `sha256:64b11105b4f8c5a6e02b0885f5e1efd28693917aeac8b658e7de14950e364761`.

115. Spin and finite cyclic label action

Claim identity: `SFT-PHYS-QUANTUM-SPIN-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Physical spin is the finite cyclic held-label action of a constituent under generated orientation recurrence; opposite readings are fibre labels, never signed proof magnitudes.

> *Spin is finite cyclic orientation action on a physical Fold constituent with exact held measurement labels.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-INCOMPATIBILITY-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-cyclic-orientation-action__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`finite-cyclic-orientation-action`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.

Axis	Forced form	Eliminated form	Elimination reason
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:68fd1a168491b598e93ca9964ec78b645e1cc5e0ca6664becbf76a25988ca549`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9abc35415e4749e0e224f44a961bf7e30269d12a9fb7256953f3c14186f3364b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:db916532b2cb153c2f330fa206cf12fb490d61af66dc238e59275ecb4bc77611`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d17d43683fff65b3f9dfdcd5236679043af5f399108fa0cc78c365dc5861b49c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:963828635e8ecbdebf873622872f9b3cc1a16fff70c650a55b7339c0e0c695a3`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ecd65100aa6616719ee338cf1e710ec95cded237bb9586bdb45c8f532803a5ce`. Independent certificate: `sha256:371f7e7b225aa68088362c332d7f831a371291b1bf96280fd57333b2cb9ea08b`. Engine external-validation hash: `sha256:5a6d73b63d9e63d7f687985bb2d92ab71c51d5da6926b8f2a5c4091a82f80821`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- spin-1: predicted finite-cyclic-spin; observed finite-cyclic-spin; exact match True
- spin-2: predicted finite-cyclic-spin; observed finite-cyclic-spin; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:1ee33a3216e918e5fe4c9ac0f33271a30112bc8216441b8adccabd31c74771c0`. Isolation certificate: `sha256:9cf4460cae86ecff30cd7738d408ccbd841c48647b98dd9c42bc81c5b18a5aa2`. Custody certificate: `sha256:a06d44d79a08773fda5a99071cd6e3662ec26d490a17952d0c11c639279965ac`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- spin-1 from PDG-2025-SUMMARY-TABLES
- spin-2 from PDG-2025-SUMMARY-TABLES

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:b55902b8ad43a203105af251e7e8d07dcb684c52b7cd273199d3113b53f16c7e`; engine receipt `sha256:8060070b239eeff6926c1eee59b02ffbdbaafe27fbe7f3805c4efc0dd43faaf2` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-SPIN-001-8060070b239eeff6.json`; empirical-validation hash `sha256:55e6d8e1b2f433eac2243f4b3e94aa2a3144e5f5c745a3fa4e06296be984f3e4`; measurement receipt `sha256:1ee33a3216e918e5fe4c9ac0f33271a30112bc8216441b8adccabd31c74771c0`.

116. Identical physical constituents

Claim identity: `SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Constituents are physically identical exactly when exchanging their unobserved identity labels preserves every complete observable transition and composition trace.

> *Identical quantum constituents are the exchange-equivalence class of complete physical Fold traces.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-SPIN-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__constituent-exchange-observable-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`constituent-exchange-observable-equivalence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:adb5a54a71ba2c34d0cb50d9b729fd0cd0cd53b0f3dbdd06a18c87ceb3d0990e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:aab2c92e7b762d31cb25bc68bcb80ed971da4eb2051a50fa55e461fafd318bde`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0efa8b7889317c183594f5168fd1f282182dab407ced9e46aee80da270edb6d0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d87f052ae46189313214dd64b3e37c5561f0f0ed21d4e72b9fb960bd430e254c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c1e08af79027067b143e65b5f88a280c20655a915fba624dabe47979298896f4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:85d4bb3454e1a15386cb5030a57a6e6279dc9115b659e2f00377e143c77df2c4`. Independent certificate: `sha256:7384ac457d91ab3248d9d5bdf1921a272b92984ed1cb3ea17355a246af5627b9`. Engine external-validation hash: `sha256:973584918968fd1ed4cd1d26c4ed4676d7b842960df1839fe1ab32160bb0ee31`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- indistinguishability-1: predicted quantum-constituent-indistinguishability; observed quantum-constituent-indistinguishability; exact match True
- indistinguishability-2: predicted quantum-constituent-indistinguishability; observed quantum-constituent-indistinguishability; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:6f71fdecefc8ace824129a173b55f95f3ab9c0df8a4eeea955d447254d2783be`. Isolation certificate: `sha256:74fdbc3f9d535e95cbb68a5c2bf0ab465680fb9da92448e80b6bd8ce5363240e`. Custody certificate: `sha256:cb280de5fbe00717f9eb1ffdb07ca6251134e56819bb2797b18435c948eaf4e0`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- indistinguishability-1 from PDG-2025-SUMMARY-TABLES
- indistinguishability-2 from PDG-2025-SUMMARY-TABLES

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:42dad3c8a9adfadc381eb84d53fab7af1d752c6dcfb707bb918fcf34658c9378`; engine receipt `sha256:c7eb90e57ad8b8deec7bccc6e1ad19e46a854bf51521a54872d749c21d484340` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001-c7eb90e57ad8b8de.json`; empirical-validation hash `sha256:2f00e6154979a448c66e7c930d34d0baf6d1b7600bd203341ce98ae6ca30a6a0`; measurement receipt `sha256:6f71fdecefc8ace824129a173b55f95f3ab9c0df8a4eeea955d447254d2783be`.

117. Exclusion from antisymmetric composition

Claim identity: `SFT-PHYS-QUANTUM-EXCLUSION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For the alternating held exchange class, two identical constituent labels cannot occupy the same one-constituent observation cell because exchange would erase the only distinguishing orientation trace.

> *Fermionic exclusion is absence of a distinct same-cell joint word under alternating held exchange composition.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__alternating-exchange-composition-exclusion__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	alternating-exchange-composition-exclusion	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:4ec1ebb5ddb10f34e7489128e5b00655d03a4066e49c05615c1e3e67d5b5d00f. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:8bfe4d1671a6378a9659881a08b2ec43b7e8ea2de48f7d6256b5b2e2f13fa1e7.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5a37d4decf9f8cfb19f4158b0ec17548bb010b57567a48cb269d18f322ccf018.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:fda1464760f59392dbe0dd19fc04d4affad8b926f222732fff7b158093adc211.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:0c3883c36033a3fc149220284e3b199a88357eee0385de508566454e6fe4b5da.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:d3c390a960261b1252d41e7ce98d06d23696d6ca0d172141b31fd3a59f3c354f. Independent certificate: sha256:26695bc4bcec93fcb1cb6ff194422727859a22fa8a4589dca768f75e6abc9536. Engine external-validation hash: sha256:eebc93e20335e2c5f9c6c76954cc08f465f52bdb56ca02059234bf5d12302e2e.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- exclusion-1: predicted fold-exclusion-law; observed fold-exclusion-law; exact match True
- exclusion-2: predicted fold-exclusion-law; observed fold-exclusion-law; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause PHYS-S014 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:ed4713ae747dfaa0debc8f25935b2de17cfd84b153ef4b93d4fc6592351d692b`. Isolation certificate:
`sha256:90133fa4349451577b4a355eef274708069a11cfe9009780b4b81a4435dcc318`. Custody certificate:
`sha256:75ee96c1aa196983bc1804424d8c10d7bd38d194ebf3c7069a2230ccce98d99b`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- exclusion-1 from PDG-2025-SUMMARY-TABLES
- exclusion-2 from PDG-2025-SUMMARY-TABLES

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:ae12e71294dc42cab8dfb465189424d80c7d695de9c711d03fde4bf66123ceb6`; engine receipt
`sha256:713d30ffe5d569ac3ae6a23e68c5dbf10c695577a6639fe42b0662b902e07711` at
`receipts/engine/model_admitted/SFT-PHYS-QUANTUM-EXCLUSION-001-713d30ffe5d569ac.json`;
empirical-validation hash
`sha256:0ee28380e756572f3f93b188d89f9da3a50923bb6a566c39a43a0fab41a130b9`; measurement receipt `sha256:ed4713ae747dfaa0debc8f25935b2de17cfd84b153ef4b93d4fc6592351d692b`.

118. Barrier traversal on complete support

Claim identity: `SFT-PHYS-QUANTUM-TUNNELLING-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Barrier traversal occurs when complete reversible support contains phase-compatible paths through generated barrier cells even though the corresponding coarse classical observation closes those paths.

> *Quantum tunnelling is observation of phase-compatible generated barrier paths present in complete support but absent from the coarse classical projection.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__phase-compatible-barrier-path-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`phase-compatible-barrier-path-support`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ade2af05b64bf39f84ccc29a6a421342f8eefba1204e4a94900abb418703156f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:054d47cba32d67859017762ff6bf14c2f0b544c41418c091bd0983deca5446c7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:10a111e522af07c23409f5512cbfae2bf9b26e5c9b79a06caa1a9cfc1889f632`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a4364fd900c8a3a7d2940706b9158a806370b044965df03083717e810be4dd1e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9a065d74a8c2e4777619141c741c06b2d5fc12f379b8ec06fdc2aa9170236900`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f68922fe77299126dd1219fda584ddd255588262561a49164d3d2d6baa4febb6`. Independent certificate: `sha256:6c13114affef6115891454d71a84bb6273b1ddd16c4d33f8929c0847e9f3de7e`. Engine external-validation hash: `sha256:ffc7939cd029f8468244cf59fe6807d29eea092b00ac1758cfec44e2bc122734`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- tunnelling-1: predicted quantum-barrier-traversal; observed quantum-barrier-traversal; exact match True

- tunnelling-2: predicted quantum-barrier-traversal; observed quantum-barrier-traversal; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:139bda9d9c68c66a86992e841fc7d1b8a41469da015ce98e81bd8c94f75ed07c`. Isolation certificate: `sha256:891839731aa144df69ebb90c259209635179cc3a9afd02047cb19c34fd179bf4`. Custody certificate: `sha256:aa14b37475d52f9c9c6281751f34766423bcc275b93833326146f72f07638909`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- tunnelling-1 from NIST-ASD-5.12
- tunnelling-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:4870dcd76fe9993f87180395bd9d8b9524d243c73eef49fbac373175beb1a148`; engine receipt `sha256:b10e5888bd7e44086333276921da133b7559bdc1b7ae27cb43a6164f6c6ed0a9` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-TUNNELLING-001-b10e5888bd7e4408.json`; empirical-validation hash
`sha256:8ab565a0bce91aef3f556d61b9824c3466c4c12e5d3b39655f569c7b680b2d32`; measurement receipt `sha256:139bda9d9c68c66a86992e841fc7d1b8a41469da015ce98e81bd8c94f75ed07c`.

119. Discrete bound-state spectra

Claim identity: `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A bound physical support admits only recurrence classes whose phase/action closes exactly on the generated boundary; transition records are exact distinctions between such closed classes.

> *Bound-state spectra are the generated exact phase-recurrence classes closed on finite physical support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-TUNNELLING-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__boundary-closed-phase-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`boundary-closed-phase-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3c6ffe6babe2b1525f560b24ba003e4dbfc2b2c0ca251837330bc6fca692621f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:422b87a303403ab306d1efc983d71bf64e249005e7626ec235958c406b1d08c3`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6919d017b7fdd311bd79696d8ebe35689cd2ae098a3c871b3d9d95876d9084ab`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e3d63ad84f86ab21de692f6bb037b798dfe1b99d2fc9e29964caa3a00c5b7680`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a4286e85fa7e5a0c26c0671a240ca4971459ae6c3f05323ecdcbf5fd916a0fd7`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c92ea98d77b307cc0c0a2b4fad119b5a338d78ab94afcb130153cc1ae7ff6330`. Independent certificate: `sha256:a1ff6a07fb1629118374bb4fd5751dedff5fd2b41f04b407bade1fdaff80f379`. Engine external-validation hash: `sha256:bc0901f62d0a7a80dbad0e0473ae3e9ed73b55ec3395029427dc1ead04af8810`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- discrete-spectra-1: predicted discrete-quantum-spectra; observed discrete-quantum-spectra; exact match True
- discrete-spectra-2: predicted discrete-quantum-spectra; observed discrete-quantum-spectra; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:d029052197ca1517125ac0bcf0d6385b1bdcf7523f303f2be6cd7c2ede2e50c6`. Isolation certificate: `sha256:623a46ccefd832c5c91f56a04b5eedd6709a7de166ae364545011952b406022a`. Custody certificate: `sha256:17de9c85e96a7764e122730c9a645c59729634760bdbc51e9fa53a9ea8b6c83f`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- discrete-spectra-1 from NIST-ASD-5.12
- discrete-spectra-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:f03b897cea8df629cbc3e668c17011c13e6f40f04be73470e0bfb355c9ff1c50`; engine receipt `sha256:d7b2556daa686f53c3dadec5faff1298af5905404e0ca93a86f5eaf05bceff5c` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001-d7b2556daa686f53.json`; empirical-validation hash `sha256:6a41957b69726f8e2bb3765a6c75c1b2660b73e9dcae0bfa1f54a3aca9674802`; measurement receipt `sha256:d029052197ca1517125ac0bcf0d6385b1bdcf7523f303f2be6cd7c2ede2e50c6`.

120. Physical entanglement correlations

Claim identity: `SFT-PHYS-QUANTUM-ENTANGLEMENT-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A joint physical state is entangled when its complete word support cannot factor into independently prepared constituent supports while its joint preparation trace remains held.

> *Physical entanglement is nonfactorable complete joint Fold support with common preparation provenance.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`

- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__nonfactorable-joint-word-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`nonfactorable-joint-word-support`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:423ea1804d32e65ddd2aa7c24bee3a2e435a10933797de603c40097533a2422c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a885931cb76ddb45fb63b53ba90f23207ca655e03daa2d7193e42a798446a8c9`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b3951cec8c44b8bd6c767ca92ee5c91df5cec9d2f5b115a19ebc6c778779c0cf`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fdd02410e8f55f94f279bb3939bf2828f51933a4b9cd9ff275b0e6d36c1b8f96`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2221140dc52e9b9d8091d2b19d0e34c3f6027d8d989fc7d1c13426540cf8ba54`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:69341d35615e16f99eb1c1728d5507acfe896271668491a4a0e883f961d85069`. Independent certificate: `sha256:cb2b51584c68f7af8a20930aa1af9382548a8b30fa84bc2f501af83b1f3f3054`. Engine external-validation hash: `sha256:28d28ec7419e5753fc69f3134d73a569122f475f7fc098fe817f8e8a55cff4f2`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- CERN-OPEN-DATA-DATASET-API-2026-07-23

Observed comparison records:

- entanglement-1: predicted physical-quantum-entanglement; observed physical-quantum-entanglement; exact match True
- entanglement-2: predicted physical-quantum-entanglement; observed physical-quantum-entanglement; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:8d22e3c14ca5361317a663d31ddab81a49094dfdd2e0553de89f82c1557f39ba`. Isolation certificate: `sha256:90391bc6f51763939cb49adb298825cc5ec49ef8e6121199a92e1c129669cc16`. Custody certificate: `sha256:2ce7c96ccc3a8b0d2121d1d78d1e1476e81804011e083fe5f012413ac97286ed`.

Registered source descriptions:

- European Organization for Nuclear Research: https://opendata.cern.ch/api/records/?type=Dataset&size=1 (sha256:4e232ef39985c02005828f97c43d903381411e7c0a762af768060343d17bd57d)

Registered target identities:

- entanglement-1 from CERN-OPEN-DATA-DATASET-API-2026-07-23
- entanglement-2 from CERN-OPEN-DATA-DATASET-API-2026-07-23

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:7ca9b41fe4f41f4e45b3a61f0a12ed1395acd8667571ff8eccc4f4c923b7e9a1`; engine receipt `sha256:98925c421294a67280352c088bbc577f428b8651b6d8ca2d0c7414f0f87f7204` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-ENTANGLEMENT-001-98925c421294a672.json`; empirical-validation hash `sha256:e7365a652ccac86b1fb3db1455172f323381f83ea0dba4bebd6e78da1e6d818e`; measurement receipt `sha256:8d22e3c14ca5361317a663d31ddab81a49094dfdd2e0553de89f82c1557f39ba`.

121. Bell correlation and local-hidden-record boundary

Claim identity: `SFT-PHYS-QUANTUM-BELL-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Bell correlations arise from joint generated support in which preparation and measurement-setting records belong to the complete deterministic state; a factorized local record omitting that dependence cannot reproduce the full joint census.

> *Bell correlation is exact setting-inclusive joint support; the excluded local-hidden-record model is the incomplete factorization that omits a generated dependence.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`

- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__setting-inclusive-joint-support-correlation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	setting-inclusive-joint-support-correlation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:bc9098cac071ed44eafa3fed68654c15db6892b685c98ba3265c4c6a64af0d93`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b09299bc18140e5c855348082952106725ba8320a83c8cafd1ab997351ddf377`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:538825205848b5bdac60332d5d6e1737736e12e6a7bd2068badb9f674bbd2c93`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:939c55bbbbf66a6d3fd049812da4f9361ad4b43066d21432229ec2ae79449d78`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9fde24c371247082d6f4f220726cbb0fb1a12324e0645156b6a6bf934926108b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash:

`sha256:7367a60e4c819fe426e6407cfb2dd2c0a0ac39d3ce82aec4540244a4f6914860`. Independent certificate: `sha256:03bf661944d56afc47ced421769684b15f988ec4f3b90cd89073e1ceca635ede`. Engine external-validation hash: `sha256:57a9d0e6fdb0bdcc9169f078e5e36936e358e3f1650e8d75677bd3b81f254418`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- CERN-OPEN-DATA-DATASET-API-2026-07-23

Observed comparison records:

- bell-1: predicted bell-joint-support-boundary; observed bell-joint-support-boundary; exact match True
- bell-2: predicted bell-joint-support-boundary; observed bell-joint-support-boundary; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:d3147fa7233e3dc625e97b58489cec4fcb516a09f84271bbf86a655049032b49`. Isolation certificate: `sha256:a0c4eca4fd5d4ea7dbc4fbec8a6ee277ca3aa2d880710b190d1daa5756082816`. Custody certificate: `sha256:e0de50b90a0a23df9ac7c7ea6877cd6742e161df5bcd329966536f3e47a8bc3c`.

Registered source descriptions:

- European Organization for Nuclear Research: https://opendata.cern.ch/api/records/?type=Dataset&size=1 (sha256:4e232ef39985c02005828f97c43d903381411e7c0a762af768060343d17bd57d)

Registered target identities:

- bell-1 from CERN-OPEN-DATA-DATASET-API-2026-07-23
- bell-2 from CERN-OPEN-DATA-DATASET-API-2026-07-23

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:e1ef7b642f66fb848d2d7d1b654713a292be54a481f25d6cd0d1f9defebdb37a`; engine receipt `sha256:51b9838623191c303bc78d402232a72be3ceaac140985c226f12b1550b525ae6` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-BELL-001-51b9838623191c30.json`; empirical-validation hash `sha256:53e64c478f01c144d59b1e96e3ad596249daa05c1df6beecc2c06ae877cc44ee`; measurement receipt `sha256:d3147fa7233e3dc625e97b58489cec4fcb516a09f84271bbf86a655049032b49`.

122. Context-dependent joint observation

Claim identity: `SFT-PHYS-QUANTUM-CONTEXTUALITY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. An observed value is contextual when its record class depends on the complete compatible partition measured alongside it; no context-erasing assignment preserves every joint trace.

> *Quantum contextuality is dependence of an observation record on its complete compatible partition, with no universal context-erasing assignment.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-BELL-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__context-bound-observation-partition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`context-bound-observation-partition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:fe95a3461fb66c114ab0ef879be7a2c588b80d2522117a21371a9748bd93b880`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f66538e9706174e08adb910288533ae32057f5077a78e705b803aad5888f1a1b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0feebd50f773f44a1e5e8c31a4ad6fee0aaae9cb2b1af74b004e7899bcb4495c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:24a148617bb3fb6972e20eeaeeaa4f7130699c648ebefb76023bdb68b76a0823`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:098b5f19652af1197d4c456245a7ce41033ce299cdec9b34a3a6a35fe53d4d28`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e5437e0655b540bc5cbbb7e119227c0519503d0644c3a3e649ca6b3cca31c80`. Independent certificate: `sha256:b00a502d9248194ee83d3dd6f506e0298891aef1c17ecfbff08c39e13e7f113e`. Engine external-validation hash: `sha256:50dea7aa76ec032ba4c97ff7fb2401f7983a34772c297806431bff82bd55ba22`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- contextuality-1: predicted quantum-contextual-observation; observed quantum-contextual-observation; exact match True
- contextuality-2: predicted quantum-contextual-observation; observed quantum-contextual-observation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:3365ca0b6b43a4d65e9d71c25aa2665278a138d4b4310273403fd223d629cec7`. Isolation certificate: `sha256:0f19020c99438738492422e2c17eeabfa7f01458e6d72b6753aec5b7228ad454`. Custody certificate: `sha256:3d9433a4a53ade7e1bb0f0c2725d35a69c8495db817879eb6fdade15405f731f`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- contextuality-1 from NIST-ASD-5.12
- contextuality-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:15b605baaa60a1f509f46c1a14cc9e405938e6a1fb183f09a02136a496c841cb`; engine receipt `sha256:411350197d503c4bfc3d5ad9812008fdb51646ab99cb4db8d4384ddea5aa32b3` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-CONTEXTUALITY-001-411350197d503c4b.json`; empirical-validation hash `sha256:1f558ee4605bcff5971c14620a4cf3971a5f3be52a6823bb88730a514c935cfe`; measurement receipt `sha256:3365ca0b6b43a4d65e9d71c25aa2665278a138d4b4310273403fd223d629cec7`.

123. Decoherence through environmental record distribution

Claim identity: `SFT-PHYS-QUANTUM-DECOHERENCE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Decoherence occurs when phase-distinguishing labels are reversibly distributed into inaccessible environmental words, closing interference in the reduced observation while retaining it globally.

> *Decoherence is reversible distribution of phase records into environmental support and their closure under reduced observation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`
- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-PHYS-QUANTUM-CONTEXTUALITY-001`

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__environmental-phase-record-distribution__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`environmental-phase-record-distribution`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:6579f3c35d4b1557e941bf33474416f325f2261b904ec8f93fc354d17b456451`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a3cf5ecafd15bc2a406db28ad3834d285d09fc7a8c70a3b1bdde10fab6ea314b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt

`sha256:b0d671dbe216be2ae2b9dbfb167004711fd0b12487cf958d5d4f49741f6ac73b`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b7aac10ce1b3e9e4e646b126c82c729b0878336745423529f2acdcf6ea8621c3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8ea6ddfbc48eb65b1fdd13fe12f252413f61491c7067bced5cadc06b43e64ad4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:92fe5a2bbcc2e642707032bfaed0f2db0f7f0705c2f481da74f3282d0b1ec3e9`. Independent certificate: `sha256:c00a9d191b43079e1bcc5c4a74d8d902662444b646199961a5ba04ea60a50252`. Engine external-validation hash: `sha256:8d7197a3ce90c66daf823efbdb212e6d0d9f8d6435003370d3e6ddf4fe94ff89`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- decoherence-1: predicted fold-physical-decoherence; observed fold-physical-decoherence; exact match True
- decoherence-2: predicted fold-physical-decoherence; observed fold-physical-decoherence; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:42f4765bf27747297ad187fb4714afdb997603f89698e637c5c57cf69706255e`. Isolation certificate: `sha256:42eb5cc48ee5f6b373ede804cf3f7173aeead4066fb577bc4fcd39760c3c59b9`. Custody certificate: `sha256:dc6a280d1158ee8ed46d837957b562c0e759a7785cb716f12a1ce84a826ad613`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- decoherence-1 from NIST-ASD-5.12
- decoherence-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:724a58328340281e4d82d9081c7ab64835cfe64bfce08588e882e6f7cb214913`; engine receipt `sha256:1cf4435af3c9ce1069fc175984730c64e333d4be70656537e39e0c7b9d53092f` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-DECOHERENCE-001-1cf4435af3c9ce10.json`; empirical-validation hash sha256:86650a0d0341b6f37248316d1321e9126e98c2c8cd2fdbe3039fcfa412242a4f; measurement receipt sha256:42f4765bf27747297ad187fb4714afdb997603f89698e637c5c57cf69706255e.

124. Operational quantum-to-classical physical limit

Claim identity: SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The classical physical limit is the observation quotient in which environmental records distinguish all interfering predecessor classes relevant to the retained macroscopic variables.

> *Classical physics is the record-stabilized observation quotient of complete quantum Fold support on the declared scale.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-DECOHERENCE-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__environment-distinguished-classical-quotient__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	environment-distinguished-classical-quotient	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:7508a6decd0610d7dd5d3a632a36cf9b10f9937116854bc30741d2f57ed9e6ee. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:548a2a84e76b692c08a361e060e76119136c52b19bc29e977c10e85ba345eca1`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e34d504d275c746a0589c9eae6f8c894578a1f6691b08044db822366b58b22a2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bb4d0b3743489900ecc2943b7e7a02be2a24b9c4135e523ce740f2e15ed6a769`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b194e483b537990a34a2cfec1a0b01fadb9bb869aff1f526858c510ad6b1a1d7`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8c3401dc997205e7cd4e8a8ea930ddc09f7f56f8800443b67cfb2483a1333a3c`. Independent certificate: `sha256:4585eff2f4b1d7856d596067f9882bea67f448f44cfb2848221722c3c214803f`. Engine external-validation hash: `sha256:56c165b276d80da94910fe2df1ad3b6c1683a0490514069f47485ec02ca25979`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- classical-limit-1: predicted operational-quantum-classical-limit; observed operational-quantum-classical-limit; exact match True
- classical-limit-2: predicted operational-quantum-classical-limit; observed operational-quantum-classical-limit; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:d15bfb0b898fc4e6882596c6a676d1719b2185a8da1929d2bbd447561e5bc285`. Isolation certificate: `sha256:896a28014d8db4e1533261e374a4a8eec78f40e880fced68fe310e8a21e47933`. Custody certificate: `sha256:2c3561feeb7b31602d210b78df4308d9daac9a64904a828ab978f051ff6a1035`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- classical-limit-1 from NIST-ASD-5.12
- classical-limit-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:3fb64655826892db2f6bdd9554e1909a90b76480779dc015da75f720840042fd; engine receipt sha256:31ef665cab0e95b3461a6a67e41fd3d423047fa821e4d6fa5192ffef3cfaa08e at receipts/engine/model_admitted/SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001-31ef665cab0e95b3.json; empirical-validation hash sha256:83195552fd625600524caab3bbd07ee1fcf3ddcbd7b8049987abebdfa092c48e; measurement receipt sha256:d15bfb0b898fc4e6882596c6a676d1719b2185a8da1929d2bbd447561e5bc285.

125. Entanglement and no-signalling boundary

Claim identity: SFT-PHYS-QUANTUM-NO-SIGNALLING-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A local observation of joint entangled support sums the complete remote fibre; changing a remote partition only relabels that fibre and cannot change the local exact support count without a causal transfer trace.

> *Entanglement correlations preserve exact local observation support under remote setting changes and therefore cannot signal without a generated causal path.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__remote-fibre-relabeling-invariant-local-count__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	remote-fibre-relabeling-invariant-local-count	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:915ce209b0e2abb4bec9de3b99baf53a9a1ee7af21c6e6c3776473bd170b10e0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:50ef85a22d59552c7f977b851d3dfe881ee9bf2e8353f75918709b57389cbe51`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3aaedc89dcc240da20f0214d2aaf19790e070bdde550e9c704f7d7f159c0c460`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7fafa2cec4d1912ec287601653ce89a461e18341a346ad5c38fe17d0bda33c0b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9070a95e7db91df8731327226a744baa43d0621b5df358e3b261a617508160c4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:80893f62bbf80046828c9dcbc41c2c93c22d00d2c6270a5dbfa60785047a03c4`. Independent certificate: `sha256:d1f318fb1e4a7bb9503e6359296cfd7b940932aa9767cb17a5eb8642efbbe15a`. Engine external-validation hash: `sha256:e8bd5500d02d2d206d27ec0fbfb640ed52ecf18520965dce4a7538496ef12e46`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- CERN-OPEN-DATA-DATASET-API-2026-07-23

Observed comparison records:

- no-signalling-1: predicted quantum-no-signalling; observed quantum-no-signalling; exact match True
- no-signalling-2: predicted quantum-no-signalling; observed quantum-no-signalling; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:def6e2a60f4e4a3ba525bd88165ba9e76ba934e44724275fcd3eba0bb9924e28`. Isolation certificate: `sha256:79829b9ee82563f06fae37445a8f436660e1431be148e7fe81801f0ea7573e97`. Custody certificate: `sha256:fb7b870c12dc3a8a9f4b1dd63e20696e58b70d98be1c321763765b15df8ff049`.

Registered source descriptions:

- European Organization for Nuclear Research: https://opendata.cern.ch/api/records/?type=Dataset&size=1 (sha256:4e232ef39985c02005828f97c43d903381411e7c0a762af768060343d17bd57d)

Registered target identities:

- no-signalling-1 from CERN-OPEN-DATA-DATASET-API-2026-07-23
- no-signalling-2 from CERN-OPEN-DATA-DATASET-API-2026-07-23

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction

- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:37fc9d79a1a4ce4bb84a7a229f5232a61b262fc227849c923f0f5e43539302ea`; engine receipt `sha256:d1fac775f55470f3ddf961e678135bedf89fbe2b80e00aaf162180fe911429cb` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-NO-SIGNALLING-001-d1fac775f55470f3.json`; empirical-validation hash `sha256:39d13f24691a7fc2d84a88f7b73ea0b025713e3a42d46c0e934d629545687b5a`; measurement receipt `sha256:def6e2a60f4e4a3ba525bd88165ba9e76ba934e44724275fcd3eba0bb9924e28`.

126. Two-hand exact Dirac square

Claim identity: `SFT-PHYS-RELATIVITY-TWO-HAND-DIRAC-SQUARE-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The first generated spatial, binary-square and covering counts force the exact 3/5 motion and 4/5 substance partition whose squares close on the One.

> *The exact two-component square is (3/5)^2 + (4/5)^2 = 9/25 + 16/25 = One.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for two-hand exact dirac square, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__generated-three-four-five-square-closure__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-continuum-object`	The object lacks a generated Fold support trace.
`dependency`	`admitted-V3-dependency-chain`	`asserted-prior-result`	A prior answer cannot act as a V3 premise.
`relation`	`generated-three-four-five-square-closure`	`chosen-Pythagorean-or-imported-relativity`	The alternative loses a required conservation or recurrence record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected candidate neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	A missing carrier could silently select the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access would reverse the empirical direction.
`record`	`complete-trace-and-controls`	`answer-only`	An answer without trace cannot be independently reproduced.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:14cd7fd94b9bb6afd346c888b62ae33ba1f90035e904dee2e3a757b047361db5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:0e161f33d1805a12c969a8cc038baecdaa508c8bc47e56e42eaa615d3e4e8cc7`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:5d41e32b7af5ceec2942ff0aa452a89a1255d38be5635a607dc739ee18c0d350`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9994533327142e264468046737dcfa5405b4e6f13f58f89023e341b80db7767d`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:29601296f6d95d18124fa4cf18fc2c1bb4f5e2f56b924666b04044d43e1c0632`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:f55b035b32807c70f3a92f777c92a1c40f96e03798a3e3e0e79a93fefd8d5cc3`. Engine external-validation hash: `sha256:0c034b9e432d4f47abb3ba8de039f84408330d03eebfb8ee98cb6c0886d16448`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:40ef972c0547c26a2117edca910060284b765d0fe367cab1d5eba616e3b88eea`; engine receipt `sha256:8781e3f2ddd1b3980b2ab928c942ccc4db1bccac6527d0723c0a797855991279` at `receipts/engine/model_admitted/SFT-PHYS-RELATIVITY-TWO-HAND-DIRAC-SQUARE-003-8781e3f2ddd1b398.json`; empirical-validation hash `None`; measurement receipt `None`.

127. Full three-plus-one Dirac square

Claim identity: `SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Three momentum generators and one mass generator exhaust the binary-square count four; placing each at the unique half-One balance partitions total square support into four quarters and closes on the One by direct and polarized routes.

> *Four half-One generators give direct square 4*(1/2)^2=One; the complete polarized pairing gives the same One, with equal differences retained as empty structural records.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-RELATIVITY-TWO-HAND-DIRAC-SQUARE-003`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003`

Generated grammar and closure boundary. Generate the complete eight-axis product for full three-plus-one dirac square, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__four-half-One-generators-close-by-two-routes__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-continuum-object`	The object lacks a generated Fold support trace.
`dependency`	`admitted-V3-dependency-chain`	`asserted-prior-result`	A prior answer cannot act as a V3 premise.
`relation`	`four-half-One-generators-close-by-two-routes`	`matrix-postulate-or-omitted-generator`	The alternative loses a required conservation or recurrence record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected candidate neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	A missing carrier could silently select the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access would reverse the empirical direction.
`record`	`complete-trace-and-controls`	`answer-only`	An answer without trace cannot be independently reproduced.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:fb9666924defb309efaaa50a045ec539122821b5f4d2eb50158c0d54dd012bb3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:a60105939901d4902a76dc299b1c441c0300e5ff6c8cb260c034d2a4bf5bba7f`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:317013de33c572c047fbff80898c6778e0840f3085ebe3bb8fe9f6b0ac46ec70`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ed415a81d14c83964e841006bee07f884fca7d49a8cd2cb46ed2ace966ee939d`.

- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:eaa2f57d6422d763fdaaf2ae8dd4d01fdc74c335477a82917c4ebee3770ed36d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:686f1d09dd22e4d7e1c05f264785bf904d03475349cab3b62b8158abfa273efb`. Engine external-validation hash: `sha256:17c4bee45b0cb9de04ae29d85aa89721aa5e93a5ec5be40715047e7fef9b4750`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:c3987de00e4203965aaa1d5be22f2e3271663bee9a5b794a0e3dff741afc3b5f`; engine receipt `sha256:8b58f344d672ea8b89164ee8c5e46e4faacb1eda64aa4d0d682fa4931cd30a4f` at `receipts/engine/model_admitted/SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003-8b58f344d672ea8b.json`; empirical-validation hash `None`; measurement receipt `None`.

128. Electron/muon magnetic-anomaly structure

Claim identity: `SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The bare gyromagnetic value is the binary count. The sole terminal electromagnetic self-return forces a strictly positive leading phase-normalised anomaly alpha/2, and any additional mass-scale channel has muon/electron sensitivity equal to the squared sealed mass ratio.

> *Bare g is 2; the exact phase-normalized leading anomaly is terminal alpha/2; muon/electron mass-scale sensitivity is the square of their sealed mass ratio. Physical full-turn normalization is a post-seal correspondence operation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for electron/muon magnetic-anomaly structure from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions.

Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__terminal-self-return-plus-squared-mass-sensitivity__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	terminal-self-return-plus-squared-mass-sensitivity	imported-QED-series-or-measured-anomaly	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:13fb57b80197a484c12cbeca599ef193619cb0a4caa78515ed3f5979124f01c3. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:da61bd0221b876d5634545923539e31de5a18725e77c483e5fc7f18af6b775e6.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:5f9e94716ea096e3a19b9ff1a5e6df3c3ff4ceca1a0efe8ce6765892643f257f.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:79e36b113aa82cea01532d5533b9c68787a4dd962316654c81683f2ac4fcaf48.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:0bbbfcad33b1878ef498271a836fdd97767fb5829ec55956e07f3c0a66c31a3d.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c. Independent certificate: sha256:2407c8dc639d27ece8eaa73b14d817e086be9ab667c44e286515c6a27b36d973. Engine external-validation hash: sha256:5ec140d88f7693f93cdd1531cbc15358344fb9d9ebca83b685580015252e1ba9.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S015 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal
- no imported Schwinger coefficient, QED perturbation series or irrational pi inside the formal derivation

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:46732443f32f146935ec1afede91a6c9bacad389cea20ef8150b7f224d41cec6`; engine receipt `sha256:e0a6844ac70a6e929c5e23ed576df2ea5c13eb4a05d33b148b2b3ad75a4c4f6f` at `receipts/engine/model_admitted/SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003-e0a6844ac70a6e92.json`; empirical-validation hash `None`; measurement receipt `None`.

129. Terminal exact finite turn projection

Claim identity: `SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A diameter-normalised finite turn has three complete generator sectors. Its sole remaining return passes through the forced up-depth seven and the complete binary support sixteen of the four-rung terminal promotion object, forcing the exact continued ratio $3 + 1/(7 + 1/16) = 355/113$. This is an exact rational Fold projection, not an imported irrational proof value.

> *The unique exact finite turn projection is 355/113 = 3 + 1/(7 + 1/16), with every count generated and no irrational or floating proof value.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-EXACT-OPERATIONS-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete nine-axis product of turn carrier, complete sectors, return, up-depth, terminal support, join, exact domain, provenance and extension forms. The declared exact boundary is: All diameter-normalized finite turn projections using the complete generator sectors and the sole terminal return through already admitted up-depth and finite promotion support. The generator produced `512` named candidates and the decision support contains `512` one-for-one decisions. Exactly one candidate survived: `exact-generated-turn-carrier__generator-three-complete-sectors__one-positive-return__forced-up-cover-depth-seven__four-rung-binary-support-sixteen__nested-positive-return-ratio__exact-rational-projection__registered-observational-prediction-protocol__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`exact-generated-turn-carrier`	`imported-continuum-circle`	A continuum circle imports an ungenerated object.
`sectors`	`generator-three-complete-sectors`	`selected-sector-count`	A selected count is a parameter.
`return`	`one-positive-return`	`free-decimal-remainder`	A decimal remainder is fitted.

Axis	Forced form	Eliminated form	Elimination reason
depth	forced-up-cover-depth-seven	selected-return-depth	A selected depth can tune the projection.
support	four-rung-binary-support-sixteen	unbounded-or-selected-support	An unbounded or selected support has no closure.
join	nested-positive-return-ratio	untyped-sum-or-product	Untyped composition loses return order.
domain	exact-rational-projection	irrational-or-floating-turn	That violates the exact Fold domain.
provenance	registered-observational-prediction-protocol	target-readable-shortcut	A shortcut cannot seal a generated prediction.
extension	no-extra-rule	extra-turn-term	The four-rung object is terminal.

Base and successor. The closure proof is sealed by derivation hash `sha256:e2316e4997252c4e35eab98eedfd8670b4975006c6243394e1ad247f8916d2c1`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:fd56fc882a3efad690547e28802033e3b23b3f9adf5805eb9ed0f520c4fefc04`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:059bb4f1c25cfbb09faa9e08abbd8553ab8d5a3da61fc4d9aefb047624b9a9bd`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8005e9b53d57a84a4674f3b907b538684147a103fb7a7570fc298a603f75fd89`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:2cbbc5c74c328822ca54699f1081bed118e454368ae3b97c4e8a1cb23112db91`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b7540aebff9282350d9117b62faf26ec592c8c187c0c870d64985bea3fab0e0b`. Independent certificate: `sha256:64ea64c3bf1b107130d7a657b4228556fe9887c34d70489d4fbb8ea077a251ee`. Engine external-validation hash: `sha256:aa4101bc767c6d8e1b097dfc8ec49a3aa4064f57d70227e12d0b146d4512e6e6`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported pi or continuum circumference
- no measured magnetic anomaly
- no floating or irrational proof value
- no selected remainder or extra term

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:4e6ae2034195c0c3599bc35b86c6097eb7f62ad752266b56fbf126df801336c0; engine receipt sha256:95e58749591f916244b2fa09fc71e9f8fb13f9db848dc4c2c78ed1d74bd403f0 at receipts/engine/model_admitted/SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004-95e58749591f9162.json; empirical-validation hash None; measurement receipt None.

130. Terminal exact electron magnetic anomaly

Claim identity: SFT-PHYS-QED-ELECTRON-MAGNETIC-ANOMALY-004

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The admitted leading alpha-over-binary phase anomaly is projected through the terminal exact turn. Its complete finite radiative ledger alternates two positive held/returned pairs: down-depth five over the open colour bulk 27 held by 3, against the binary alpha return over 5 times 3; and the single alpha-squared return over 5 times 3 against the up-depth-seven alpha-cubed terminal return over the closed bulk-plus-boundary support 30. Holding alpha times that complete loop carrier once from the leading projection forces one exact electron anomaly.

> *With alpha the admitted exact coupling and T=355/113, the electron anomaly is exactly [alpha/(2T)] times positive_take(One, alpha L), where L is the sum of positive_take(5/24,2 alpha/15) and positive_take(alpha^2/15,7 alpha^3/30).*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003
- SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete ten-axis product of leading carrier, turn, open bulk, first pair, second pair, orientation, retention, target custody, provenance and extension forms. The declared exact boundary is: All first terminal finite-loop completions of the admitted leading electron anomaly using the down/up covers, complete colour bulk/boundary supports, binary labels and terminal alpha exactly once in their typed loop roles. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: retain-admitted-leading-alpha-carrier__terminal-exact-turn-projection__complete-volume-held-by-boundary-channels__down-over-open-bulk-held-by-binary-alpha-over-down-colour__One-alpha-square-over-down-colour-held-by-up-alpha-cube-over-closed-support__two-positive-held-pairs__hold-alpha-loop-once-from-leading__target-inaccessible-until-prediction-seal__registered-observational-prediction-protocol__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
leading	retain-admitted-leading-alpha-carrier	replace-bare-or-leading-law	Replacement erases the admitted Dirac and leading receipts.
turn	terminal-exact-turn-projection	imported-irrational-turn	An irrational carrier violates the domain.
bulk	complete-volume-held-by-boundary-channels	selected-bulk-subset	A subset can tune the result.
first	down-over-open-bulk-held-by-binary-alpha-over-down-colour	free-first-loop-coefficient	A free coefficient is fitted.

Axis	Forced form	Eliminated form	Elimination reason
second	One-alpha-square-over-down-colour-held-by-up-alpha-cube-over-closed-support	free-terminal-loop-coefficient	A free coefficient is fitted.
orientation	two-positive-held-pairs	signed-or-complex-coefficients	Signed or complex proof values violate the Fold domain.
retention	hold-alpha-loop-once-from-leading	append-unbound-anomaly	Appending creates an unnormalized extra carrier.
target	target-inaccessible-until-prediction-seal	measurement-readable-execution	That cannot issue a target-inaccessible seal.
provenance	registered-observational-prediction-protocol	unregistered-fit	An unregistered fit lacks custody and prediction receipts.
extension	no-extra-rule	extra-radiative-term	Every down/up, bulk/boundary and terminal carrier is already consumed.

Base and successor. The closure proof is sealed by derivation hash `sha256:55faf86066b2a026ba02e8bdfd00e3a225e288c31659c76dc5c99f344fa74822`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:4c45ca44eff5ba01eb8cdfb6e718ad8db95256754310a489f1e8e3e4b742d8b5`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:203aea523fb5f22f9bc4cc8826785f1a833bd83b4ff31b2f4e29c2d1c28b88f4`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0058b40f58796261ba5911a35e9595e46d4334501a4941f0dd88bca00797cde8`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:37cab5cef2b623790f23f0f8196876dce1cca03189b966df24c70d7d34182887`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b7540aebff9282350d9117b62faf26ec592c8c187c0c870d64985bea3fab0e0b`. Independent certificate: `sha256:023c041f450da88f8ad1d9c7b1645f4bf2b6b4fb0dba728fff3dc84974ad0335`. Engine external-validation hash: `sha256:a814d824a690d9c31108f161bc66c230ffae6818b3255786f7e81dad48d2d96f`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- MATTER-FLAVOUR-TERMINAL-ANOMALY-AUTHORITATIVE-2022-2026

Observed comparison records:

- ELECTRON-MAGNETIC-ANOMALY-COMPLETE: predicted terminal-target-inaccessible-electron-anomaly-prediction-inside-complete-CODATA-interval__observational-prediction-protocol-passed; observed terminal-target-inaccessible-electron-anomaly-prediction-inside-complete-CODATA-interval__observational-prediction-protocol-passed; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The exact sealed electron prediction leaves the complete registered interval, the target becomes readable before sealing, a measured value selects the survivor, an uncertainty changes, or observational provenance is hidden.

Measurement receipt:
`sha256:9804eb853cbfdabbb41d6b7f3a5f7b7f83e9024df618764caed6ca3ab556577c`. Isolation certificate:
`sha256:609b97fe4581a707041098b52818462659e6c7960f3ccb82c9473423d77994a7`. Custody certificate:
`sha256:0713caf07794050eff752b885f198e4ea800585e71086c518173e003939d6f86`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants and Fermilab Muon g-2 Collaboration: https://physics.nist.gov/cuu/Constants/Table/allascii.txt; https://muon-g-2.fnal.gov/result2025.pdf (sha256:4b506052fc5037576cd6ba4de8ee5afce74460f7a5cb8da4bc58cbb823f6732f)

Registered target identities:

- ELECTRON-MAGNETIC-ANOMALY-COMPLETE from MATTER-FLAVOUR-TERMINAL-ANOMALY-AUTHORITATIVE-2022-2026

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no electron measurement in the executable relation
- no imported QED series or coefficient
- no signed, irrational, imaginary or floating proof value
- no selected loop truncation or extra term

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:732f5aa229ba24f17bdb015a6f19285b1cdbd7e393bdf2c5a6ba7d846e46fcd2`; engine receipt
`sha256:8f9f8c6663f56a16c11e0b5b5fc344926dc264e868ef6a40c47ef424f72655d8` at `receipts/engine/model_admitted/SFT-PHYS-QED-ELECTRON-MAGNETIC-ANOMALY-004-8f9f8c6663f56a16.json`;
empirical-validation hash
`sha256:81df7649d9d3a9ed1b14fba7d810d3ac5bbe4574fd601ec73f29fc206c47159a`; measurement receipt `sha256:9804eb853cbfdabbb41d6b7f3a5f7b7f83e9024df618764caed6ca3ab556577c`.

131. Terminal exact muon magnetic anomaly

Claim identity: `SFT-PHYS-QED-MUON-MAGNETIC-ANOMALY-004`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete electron loop is universal. The second charged-lepton generation adds the positive mass-scale correction at alpha squared: both Fold labels over the depth-two complement 17, then one alpha return over the binary-distributed depth-three complement 2 times 53, then one alpha-squared terminal cell over four-rung support 16 times 53 plus down-depth five. This forces the exact added carrier alpha squared times [2/17 + alpha/106 + alpha squared/853].

> *The exact muon anomaly is the terminal electron anomaly plus alpha^2 [2/17 + alpha/106 + alpha^2/853], with 17=2*3^2-1, 53=2*3^3-1 and 853=16*53+5.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-QED-ELECTRON-MAGNETIC-ANOMALY-004`
- `SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003`
- `SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001`
- `SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003`
- `SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete ten-axis product of universal base, muon generation, coupling order, depth-two complement, depth-three return, terminal support, composition, target custody, provenance and extension forms. The declared exact boundary is: All first terminal positive mass-scale corrections to the complete electron loop for the second charged-lepton generation, using depth-two/three complements, Fold labels, four-rung support and down-depth exactly once. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `complete-terminal-electron-loop__second-charged-lepton-depth__two-charged-end-alpha-square__both-labels-over-depth-two-complement__one-alpha-over-binary-depth-three-complement__four-rung-times-depth-three-plus-down-depth__complete-positive-sum-appended-once__target-inaccessible-until-prediction-seal__registered-observational-prediction-protocol__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`base`	`complete-terminal-electron-loop`	`independent-fitted-muon-base`	A separate base loses electron/muon correspondence.
`generation`	`second-charged-lepton-depth`	`selected-generation-label`	A label alone cannot force a carrier.
`order`	`two-charged-end-alpha-square`	`selected-alpha-order`	A selected order can tune the anomaly.
`two`	`both-labels-over-depth-two-complement`	`free-leading-muon-weight`	A free weight is fitted.
`three`	`one-alpha-over-binary-depth-three-complement`	`free-successor-weight`	A free return is fitted.
`terminal`	`four-rung-times-depth-three-plus-down-depth`	`selected-terminal-denominator`	A selected denominator is fitted.
`composition`	`complete-positive-sum-appended-once`	`signed-or-cancelled-series`	Cancellation imports signed proof values.
`target`	`target-inaccessible-until-prediction-seal`	`measurement-readable-execution`	That cannot issue a target-inaccessible prediction.
`provenance`	`registered-observational-prediction-protocol`	`unregistered-fit`	An unregistered fit lacks custody and prediction receipts.
`extension`	`no-extra-rule`	`extra-muon-term`	Depth two, its successor and terminal cell exhaust the registered muon grammar.

Base and successor. The closure proof is sealed by derivation hash `sha256:c45eeec246a98038c22388e57d8723952bdd4381aa9da38ec162dafec5f110ae`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:10a2e4a82eb07ef38e6087da7c06c66650945ce4b639df23ef71a2dde2502d86`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:de33e0625fde4d4b5ce4bef9b77098f827c6633db8de0b93119ba13a62a1773a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:63b3a7c81f925eb38705876db1b21d3437e0d29d9a9330b7b23d6a452c5502a6`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:dd6d713d8248a0c754b67ca8dd4bf5fc7fb70b7bd6bb5de63933f47489cbd8a6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b7540aebff9282350d9117b62faf26ec592c8c187c0c870d64985bea3fab0e0b`. Independent certificate: `sha256:7e90529df25555089225ae8f5d7c3dbfb5e5b0ecaaaf2b5af9fcdd65a6b6e470`. Engine external-validation hash: `sha256:99e47760e4a1ff1856e5444a9cea5a6d09abd5e9f06854f3438dc473130199bd`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- MATTER-FLAVOUR-TERMINAL-ANOMALY-AUTHORITATIVE-2022-2026

Observed comparison records:

- MUON-MAGNETIC-ANOMALY-COMPLETE: predicted terminal-target-inaccessible-muon-anomaly-prediction-inside-complete-Fermilab-world-average-interval__observational-prediction-protocol-passed; observed terminal-target-inaccessible-muon-anomaly-prediction-inside-complete-Fermilab-world-average-interval__observational-prediction-protocol-passed; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The exact sealed muon prediction leaves the complete registered interval, the target becomes readable before sealing, a measured value selects the survivor, an uncertainty changes, or observational provenance is hidden.

Measurement receipt: `sha256:0f7013f3af7d380310347648f6c37e0a84da3d4801dbe8e5ce76dece379a16b7`. Isolation certificate: `sha256:ed69925acc33e0cebae174463d512f1eee063e3d7730c2e0b56b9bdac27a092a`. Custody certificate: `sha256:0adcf9d07c3a684161edd2407efba338ce2c643ce4264a3a52a99f9dff60ea3e`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants and Fermilab Muon g-2 Collaboration: https://physics.nist.gov/cuu/Constants/Table/allascii.txt; https://muon-g-2.fnal.gov/result2025.pdf (sha256:4b506052fc5037576cd6ba4de8ee5afce74460f7a5cb8da4bc58cbb823f6732f)

Registered target identities:

- MUON-MAGNETIC-ANOMALY-COMPLETE from MATTER-FLAVOUR-TERMINAL-ANOMALY-AUTHORITATIVE-2022-2026

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no muon measurement in the executable relation
- no imported QED or consensus anomaly
- no fitted mass scale or coefficient
- no negative, irrational, imaginary or floating proof value
- no extra terminal term

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:920795a1ed357d5af2b6a04e6403d69b2e54c8e478be2ce28a7a426048292c4e`; engine receipt `sha256:efa486a8ea1c432219f915d288265975fda3c9f0fec9b393c1570737a71b785f` at `receipts/engine/model_admitted/SFT-PHYS-QED-MUON-MAGNETIC-ANOMALY-004-efa486a8ea1c4322.json`; empirical-validation hash `sha256:ec9ba60fe5ced5e711f4031554ac9ff1833e2a8e410c2a2f51b4adb53e4f03cd`; measurement receipt `sha256:0f7013f3af7d380310347648f6c37e0a84da3d4801dbe8e5ce76dece379a16b7`.

132. Terminal finite spin-statistics, occupation and Bose-condensation law

Claim identity: `SFT-PHYS-SPIN-STATISTICS-CONDENSATION-TERMINAL-045`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For positive finite N and L, preserving exchange has exactly C(N+L-1,N) occupation words and admits N occupants in one cell for every N; alternating exchange has exactly C(L,N) words for N<=L, an empty support for N>L, and never exceeds one occupant per cell. Every finite canonical occupation weight is exact rational and sums to One. The two-label spin census is 3/4 preserving and 1/4 alternating; alternating spin first returns after two turns, while a pair is preserving and returns after one. For every N,L and m>=2, the ground-orbit share crosses (m-1)/m at a uniquely first finite cold depth, fixing an exact rational critical mean throw; the unique minimum-throw preserving word places all N occupants in the shared ground orbit.

For positive finite N and L, preserving exchange has exactly C(N+L-1,N) occupation words and admits N occupants in one cell for every N; alternating exchange has exactly C(L,N) words for N<=L, an empty support for N>L, and never exceeds one occupant per cell. Every finite canonical occupation weight is exact rational and sums to One. The two-label spin census is 3/4 preserving and 1/4 alternating; alternating spin first returns after two turns, while a pair is preserving and returns after one. For every N,L and m>=2, the ground-orbit share crosses (m-1)/m at a uniquely first finite cold depth, fixing an exact rational critical mean throw; the unique minimum-throw preserving word places all N occupants in the shared ground orbit.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-ONE-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-HALF-ONE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-FERMION-BOSON-001`
- `SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006`
- `SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008`
- `SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043`
- `SFT-PHYS-COLLECTIVE-RADIATION-RESPONSE-TERMINAL-041`

Generated grammar and closure boundary. Generate the complete product of exchange, exclusion, accumulation, occupation-weight, spin-return, pairing, lock-threshold and ground-orbit forms. The declared exact boundary is: Every positive finite particle count N, positive finite level count L, preserving or alternating exchange class, positive finite cold depth, and Fold factor m at least two; empty cells are typed structural-absence records. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `preserving-versus-alternating-held-trace__one-cell-distinction-exclusion__every-positive-finite-multiplicity-admitted__complete-exact-dyadic-occupation-census__typed-one-turn-two-turn-held-orbits__alternating-pair-preserves-exchange__first-exact-crossing-of-forced-m-minus-one-over-m-share__unique-minimum-throw-shared-ground-word`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`exchange`	`preserving-versus-alternating-held-trace`	`named-particle-species`	A species name does not force its exchange law.
`fermion`	`one-cell-distinction-exclusion`	`selected-occupancy-cap`	A selected cap is a free rule.
`boson`	`every-positive-finite-multiplicity-admitted`	`finite-selected-ceiling`	No generated ceiling follows from preserving exchange.

Axis	Forced form	Eliminated form	Elimination reason
`weights`	`complete-exact-dyadic-occupation-census`	`continuum-statistical-distribution`	A continuum distribution is outside Fold arithmetic.
`spin`	`typed-one-turn-two-turn-held-orbits`	`imaginary-signed-phase`	Imaginary or signed proof scalars are inadmissible.
`pairing`	`alternating-pair-preserves-exchange`	`independent-boson-postulate`	An independent postulate breaks the single derivation chain.
`lock`	`first-exact-crossing-of-forced-m-minus-one-over-m-share`	`measured-or-fitted-temperature`	A measured or fitted onset would select the law.
`ground`	`unique-minimum-throw-shared-ground-word`	`asserted-macroscopic-ground-mode`	Naming a condensate is not an enumeration.

Base and successor. The closure proof is sealed by derivation hash `sha256:5ee1579388bfe3edb1a8772aee7d8c0c09a7eb05308730458a52e33091e85f95`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:605f4063b74d23be764144a1028dc3b1319b22982a21a8c6b3ed31a9d1ed22d3`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:890c659e1760dc96d83ab9ee8051afa91a93668f46e03415fb11ba1e9ecbd6d4`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0a855eb9c0225038937d4f3f46327ecc7dc1005cb7c237489b76ee28b7435148`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:dedd5d8e13991355ca1ba3ec94cd7c41b87841659d51696b3cf828e1f85d3fdf`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:274d954c4cd12e6b469d51421e40869b90a3aef3fcc2f7eea6a22b54d18442c1`. Independent certificate: `sha256:18f0ad787e6f6a48fb7eed4308f6dce85e09c74d6b815b4c60db0b81df9de0ed`. Engine external-validation hash: `sha256:cd4226bd92f2cc114d70530b517280e799828647c49c75afc8d5af64ff4e9bab`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no Fermi-Dirac or Bose-Einstein continuum function imported as a premise
- no measured transition temperature, particle species or 360/720-degree observation available to candidate selection
- no selected occupancy cap, chemical potential, thermodynamic limit or infinite population
- no ontic randomness

- no conventional numerical-nothingness, negative, irrational, imaginary, floating, NaN or continuum proof magnitude; visible 0 in an occupation vector denotes an empty One cell

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:231ff1d3b8535154ad55b040e0317d77b3f2f2059a8ec02cc46eb1dad41782d3`; engine receipt `sha256:f71da5b86f99d6569a1f33dc6fc37024cc5d458b12e625c5dbc4faf3c33ccda7` at `receipts/engine/model_admitted/SFT-PHYS-SPIN-STATISTICS-CONDENSATION-TERMINAL-045-f71da5b86f99d656.json`; empirical-validation hash `None`; measurement receipt `None`.

133. Terminal preparation-depth, support-uncertainty and joint-correlation law

Claim identity: `SFT-PHYS-QUANTUM-SUPPORT-UNCERTAINTY-TERMINAL-049`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every complete dyadic preparation N=2^k and every nonempty position support, exact Fold-parity observation forces s_t*s_f>=N, unit-free support-spread product (s_t*s_f)/N>=One and squared normalised support-spread product at least 1/N^2; depth two attains 1/16 and an eight-branch preparation uniquely has depth three with branch unit 1/8. The complete 2-by-3 census has 63 nonempty supports: 21 factorable and 42 nonfactorable. The complete 3-by-5 census has 32767: 217 factorable and 32550 nonfactorable. Both full products are factorable, correcting their prior relabelling as entanglement; entanglement is the nonfactorable class. Complete projection preserves local counts under remote relabelling. All sixteen local deterministic Bell-response pairs obey the three-of-four bound, while setting-inclusive joint support can exceed it with invariant exact local marginals and no signalling.

> *For every complete dyadic preparation N=2^k and every nonempty position support, exact Fold-parity observation forces s_t*s_f>=N, unit-free support-spread product (s_t*s_f)/N>=One and squared normalized support-spread product at least 1/N^2; depth two attains 1/16 and an eight-branch preparation uniquely has depth three with branch unit 1/8. The complete 2-by-3 census has 63 nonempty supports: 21 factorable and 42 nonfactorable. The complete 3-by-5 census has 32767: 217 factorable and 32550 nonfactorable. Both full products are factorable, correcting their prior relabelling as entanglement; entanglement is the nonfactorable class. Complete projection preserves local counts under remote relabelling. All sixteen local deterministic Bell-response pairs obey the three-of-four bound, while setting-inclusive joint support can exceed it with invariant exact local marginals and no signalling.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-ONE-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-ALGEBRA-001`
- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-PHYS-QUANTUM-INCOMPATIBILITY-001`
- `SFT-PHYS-QUANTUM-ENTANGLEMENT-001`
- `SFT-PHYS-QUANTUM-BELL-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`

Generated grammar and closure boundary. Generate the complete product of preparation depth, Fold-parity observation, support bound, spacing-weighted spread, joint factorability, projection, Bell-factorization and extra-rule forms. The declared exact boundary is: Every nonempty subset of complete binary support through exhaustive depth four with depth-independent orthogonality induction; every nonempty 2-by-3 and 3-by-5 joint subset; all sixteen deterministic local two-setting response pairs; and the complete setting-inclusive two-outcome support. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `preparation-derived-complete-bina` ry-depth__held-returned-parity-count-table__orthogonality-Parseval-support-count__exact-unit-free-support-spread__complete-factorability-subset-census__complete-marginal-count-invariance__incomplete-local-factorization-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
depth	preparation-derived-complete-binary-depth	universally-selected-depth-three	A universal depth is an unforced parameter.
observation	held-returned-parity-count-table	imported-complex-Walsh-amplitudes	Complex amplitudes are outside the proof grammar.
uncertainty	orthogonality-Parseval-support-count	named-or-selected-uncertainty-bound	Naming a prior theorem does not force it.
spread	exact-unit-free-support-spread	support-width-called-statistical-variance	Support extent is not automatically a statistical central moment.
joint	complete-factorability-subset-census	product-size-relabeled-entanglement	A full Cartesian product factorizes exactly.
projection	complete-marginal-count-invariance	remote-label-change-relabeled-signal	A bijective relabelling changes no local count.
bell	incomplete-local-factorization-record	ontic-randomness-or-superluminal-message	Neither is forced by the finite correlation census.
extension	no-extra-rule	measurement-selected-correction	A fitted correction would select the law.

Base and successor. The closure proof is sealed by derivation hash sha256:b87466e8c836feaea657b3d35f530a9de80e70f92fb663dd7a449b119c7fb469. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:f62a278d0ac5036bccf72d4a00e15ad07756b8bb607da8a238cb7c4886b34b63.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:a3c93700d600582544c0cb2f801144a1d23eee3c2ca588338760d5aab12bd18f.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:090198dc748d1481a485cefd5c34157d91fd5f1e9346d266361e10f6f1085502.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:b6008e573ace6486049207789bfcabb1459212302db171f5fa384ad7a819c2ad.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e78b5c1aaf806e5969295fa2d7cea0f8bea225528621eca464e3c54cba1a5afc. Independent certificate: sha256:25877a95c51e203cd290ecafaffaa47587c6082300cd3454b337b60470a1344d. Engine external-validation hash: sha256:b3339e6b86d61ef06190b2f4cc4c1dd778e3b039e34c0d6b35cf2bd7701176bf.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S015` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported Fourier, Donoho-Stark, Heisenberg, tensor-product or Bell theorem as a premise
- no universal depth-three measurement rule independent of preparation
- no relabelling of support extent as statistical moment variance
- no relabelling of a complete Cartesian product or product-greater-than-sum arithmetic as entanglement
- no claim that fifteen is irreducible to its three-by-five product factors
- no ontic nondeterminism, stochastic setting oracle, superluminal signal or omitted preparation/setting record
- no measured Bell or uncertainty value accessible to candidate selection
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:246abe6ea59ec47581adf7eb2cc9552ad5cf703fe88d5e65b404a8d1ad83a672`; engine receipt `sha256:1560f2e0de3870abac2bdc6575aa9811c4dc013d5a6d705e729a832dc451b79f` at `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-SUPPORT-UNCERTAINTY-TERMINAL-049-1560f2e0de3870ab.json`; empirical-validation hash `None`; measurement receipt `None`.

17. Constants Scales Precision

Dimensionless constants, mass scales and precision values are kept in Physics. Each reported value is tied to its zero-parameter Fold relation, sealed prediction, exact rational measurement adapter and immutable receipt.

134. Terminal exact inverse fine-structure Fold ratio

Claim identity: `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The binary predecessor and generator-three recurrence force cover depths five and seven. Their typed tower, boundary, complete three-direction cover, returning One and finite one-direction-at-a-time promotion ladder force the exact terminal ratio 503846395469/3676744786 before measurement.

> *The terminal inverse fine-structure Fold ratio is exactly 503846395469/3676744786; its leading rung is exactly 34259/250.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001`
- `SFT-FOUNDATION-COUNT-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete typed product of cover, depth, tower, boundary, return, join, promotion, termination, target custody and extension forms. The declared exact boundary is: All type-correct assemblies that use each forced binary/generator cover block in its generated role, admit only the One as recurrence correction, and promote each of the three interchangeable cover directions at most once. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `least-complete-binary-covers__cover-generator-cross-lock__binary-at-up-depth__generator-at-boundary-rank__one-return-over-complete-cover__tower-plus-dilated-boundary__one-remaining-direction-per-rung__all-three-directions-promoted__measurement-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`cover`	`least-complete-binary-covers`	`asserted-depths`	Asserted depths have no support census.
`depth`	`cover-generator-cross-lock`	`single-route-depths`	One route cannot expose a changed carrier.
`tower`	`binary-at-up-depth`	`free-leading-whole`	A free leading whole is a parameter.
`boundary`	`generator-at-boundary-rank`	`free-boundary-factor`	A free coefficient is not structural.
`return`	`one-return-over-complete-cover`	`fitted-small-correction`	A fitted correction reads the target.
`join`	`tower-plus-dilated-boundary`	`untyped-arithmetic-permutation`	Permuting typed carriers erases their roles.
`promotion`	`one-remaining-direction-per-rung`	`selected-refinement-series`	A selected series adds unfixed terms.
`termination`	`all-three-directions-promoted`	`infinite-or-truncated-by-fit`	A fit-selected truncation is not closure.
`target`	`measurement-inaccessible-until-seal`	`measured-alpha-visible`	That makes measurement an answer-bearing premise.
`extension`	`no-extra-rule`	`extra-scale-rule`	An extra scale is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:a01c1ebcf24ad02547261495adcd198a3c6cad5dce042feade329ed1068b9547`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:ca5b4e49ab3bedece92f0ec11667e4305baafe8624283410439868e1b79e151c`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:0d516338a2c561300eae04d43e460a74cae268ab52d41ce1ace635214b048ab5`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:937d252d67fbb9ad1f3a8bcbd5000226d4d7bde3b511429cbe0ee917e4986843`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:debfd802d48d0d9c9b5e1df0a6bdbf950f19bd012c0bdf52cfeec2505494619b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:97d0cdb7077804e63fb1bab56244287f15fbc1ee6fdeece167b59a309604117f`. Independent certificate: `sha256:ec1a7c48b4269d74564b029d286350d600f1feca69383bf655ad7415714f71f4`. Engine external-validation hash: `sha256:d6641916132b3eb52372a85b52f7b33b220afe320531434cc2fc68fa8ab38113`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S016` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:f49bcafa7a9021c2bad215685390c192482700c964e034d89f7477fc3357fc53`; engine receipt `sha256:b7b1663a8ed78da756a2624d0bca690ac2ade089a9396060c96f141a6f2896a2` at `receipts/engine/model_admitted/SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001-b7b1663a8ed78da7.json`; empirical-validation hash `None`; measurement receipt `None`.

135. Exact charged-lepton cubic invariants

Claim identity: `SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The forced generator-three partition supplies one exact three-root mass carrier whose symmetric invariants are root sum One, pair sum one over the two Fold fibres paired with generator three, leading product one over the positive predecessor of twice the colour fifth-power support, and sharpened product obtained by the unique generator-three channel part. The resulting exact invariants are 1, 1/6, 1/485 and 3/1454; no irrational root is formed.

> *The exact charged-lepton cubic invariant tuple is (sum=1, pair-sum=1/6, leading-product=1/485, sharpened-product=3/1454), with the sharpened colour channel uniquely retained over channels two, three and four.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-FOUNDATION-COUNT-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-FOUNDATION-PART-EQUIVALENCE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ALGEBRA-001`

Generated grammar and closure boundary. Generate the complete product of cubic carrier, generator, partition, pair invariant, leading support, predecessor, sharpening channel, sharpening form, root boundary and extra-rule forms; enumerate channels two, three and four without measurement access. The declared exact boundary is: All exact positive-rational symmetric three-root invariant constructions made from the admitted Fold fibre count, generator three, complete colour fifth-power support, positive predecessor and one neighbouring generated channel part. The generator produced `2304` named candidates and the decision support contains `2304` one-for-one decisions. Exactly one candidate survived: `one-complete-symmetric-three-root-carrier__admitted-generator-three__root-sum-is-the-One__one-over-two-colour__twice-colour-fifth-power__positive-predecessor-of-support__colour-channel__positive-take-one-channel-part__retain-exact-symmetric-invariants__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	one-complete-symmetric-three-root-carrier	three-names-without-joint-invariants	Names do not retain one cubic relation.
generator	admitted-generator-three	selected-three	A familiar generation count cannot be assumed.
partition	root-sum-is-the-One	unnormalized-root-sum	An unnormalized sum adds a free scale.
pair	one-over-two-colour	one-over-colour	This omits the two Fold fibres.
support	twice-colour-fifth-power	colour-fourth-power	This stops before the complete fifth colour succession.
predecessor	positive-predecessor-of-support	support-count-itself	The complete support retains the terminal One rather than the massive interior.
channel	colour-channel	binary-channel	The binary channel is not the carrier's generator identity.
sharpening	positive-take-one-channel-part	add-channel-whole	A whole channel count changes scale rather than resolves one channel part.
roots	retain-exact-symmetric-invariants	solve-and-round-roots	Root approximation introduces brackets and stopping choices.
extension	no-extra-rule	extra-coefficient-or-measured-selector	An added number or target-selected channel is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:89abdd07154010449d95f2aef0031768452baa7172780ee8a183cb3bd8521823`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:4d3a47f711ee8d5a5a8d8fdf4560ffaa1c3bf66dec3d732c3f8e4a487346f1c0`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:d3ed7f51b7c2112714cc8982209527b165276b283cc66442bfee554236d5254e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6b6025c4aab652df50047f8a0beb5ebfdf807ba155258b19cd8c59f1ab05daa9`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:c20ff35677c6755b47078ff9779fca59f0c3e5e1b7413ae6ecbab57bb49ce3c0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5903885dd237b470dd8f94410d2195eacca778ce56fc3e923f356a4322e94432`. Independent certificate: `sha256:1b1ffd5bd2d7ab0367b2cc600933def92af65edd7479ed8ab5d33bf6eb4e0f37`. Engine external-validation hash: `sha256:122e49fdc688788f76eafcb7c7da904446abdc4ac2f45284f7cd1c39f18d9abd`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S016` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate or answer table in the derivation runtime

- no measured lepton mass or ratio may select a coefficient or channel
- no numerical zero, negative, irrational, imaginary or floating proof value
- no fitted root, bracket, tolerance, iteration count or extra coefficient

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d9d2dee4a0efb589ad52c25383ceac40af265632c2f905418a36e997a95ed4f3`; engine receipt `sha256:9c7202057f7735d46e9b355f8611e63e9043d99b1a3ced16cadb6a468d09e598` at `receipts/engine/model_admitted/SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001-9c7202057f7735d4.json`; empirical-validation hash `None`; measurement receipt `None`.

136. Terminal self-coupling refinement of the charged-lepton cubic

Claim identity: `SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The admitted charged-lepton cubic is refined by the complete terminal electromagnetic self-coupling of its three-root product: alpha to the generator-three order is distributed over the complete three-cubed carrier and weighted once by the forced cover-depth pair five and seven, with the up-depth transported through one alpha and one colour channel. The exact correction is held from 3/1454.

> *The terminal product invariant is exactly 3/1454 held by alpha^3/3^3 times (5 + (7/3)alpha), with alpha the reciprocal of the admitted terminal inverse fine-structure ratio.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-FOUNDATION-COUNT-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ALGEBRA-001`

Generated grammar and closure boundary. Generate the complete product of base invariant, interaction order, support, down-depth role, up-depth transport, channel distribution, correction orientation, target custody and extension forms. The declared exact boundary is: All exact first terminal self-coupling refinements of the admitted three-root product constructed from its generator order, complete generator volume, terminal alpha and forced down/up cover depths. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `retain-admitted-sharpened-product__one-alpha-per-three-root__complete-generator-volume__forced-down-cover-depth__one-alpha-transported-up-depth__one-complete-colour-share__hold-correction-from-product__measurement-inaccessible-until-seal__observational-derivation-disclosed__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`base`	`retain-admitted-sharpened-product`	`replace-cubic-product`	Replacement erases the admitted cubic.
`order`	`one-alpha-per-three-root`	`selected-linear-or-square-order`	A lower selected order does not act once on every root.
`support`	`complete-generator-volume`	`single-channel-support`	One channel omits the complete three-root carrier.
`down`	`forced-down-cover-depth`	`free-depth-coefficient`	A free coefficient is a parameter.

Axis	Forced form	Eliminated form	Elimination reason
`up`	`one-alpha-transported-up-depth`	`bare-up-depth`	A bare successor depth ignores transport between orders.
`channel`	`one-complete-colour-share`	`undistributed-up-depth`	An undistributed successor counts every colour channel as one.
`orientation`	`hold-correction-from-product`	`append-unheld-product`	Appending creates an unaccounted product carrier.
`target`	`measurement-inaccessible-until-seal`	`measurement-readable-relation`	That would fit the failed target.
`provenance`	`observational-derivation-disclosed`	`mislabel-as-blind-forward-discovery`	The failed comparison was already observed.
`extension`	`no-extra-rule`	`extra-fit-term`	An additional term is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:03febfe188acc83a8229d2bd45feb22484d2a027ef085e4abe9f7678c8e2e21e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:33b65a0122091153ecdbdc9e848361db0f7e008ac0b95d3b1a90ae77b8693346`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:f9e2d38d95ab449a15dcdee72eec078045e21abea84bbae18516bc9ef7166e2a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a46afb9f969a7e23e67abe378cd7479d441a69d41227779f0b7d65dbda921e56`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:18a7245b2f1e01acfae3356fbd8e7aab39fb5d715893dd98f85030edbfe8f7f4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:99ce35598da99fedf661a7d92e1bfc40adeb85861204eb5196a13aa30f7a0746`. Independent certificate: `sha256:05d0e8eebbcb2b6dcff81df8eef6f8f3aa7f9fa838ed210a6b035720858fc4c3`. Engine external-validation hash: `sha256:2a99ea6a3eb9f35c5aac8def6fdaa748797bf4fe05ac0d7fc5af7b5fa5f99e16`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- NIST-CODATA-2022-MUON-ELECTRON-MASS-RATIO: predicted terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; observed terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; exact match True
- NIST-CODATA-2022-MUON-TAU-MASS-RATIO: predicted terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; observed terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Either terminal prediction interval fails to overlap its complete source interval, source custody changes, or any failed row is omitted.

Measurement receipt:
`sha256:cbca0b4410f39699e01456c2a835fcdc54c885ab250f35dcf2e9bd71392bad2a`. Isolation certificate:
`sha256:b1c8307fe636656fa3a014f0b5a7f94ff6c59e4e377bfd0434c47e2b22d7b7e2`. Custody certificate:
`sha256:0fdfacc4ee0989c0d437b4833d4af567a3af97ad95097dc6ab3b435c5d8572a2`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S016` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured lepton mass or ratio in the executable relation
- no floating, irrational, imaginary, semantic-zero or negative proof value
- no fitted coefficient, tolerance, selected root or extra correction term
- no claim that this post-failure construction was a blind forward discovery

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:729b45da56b82394584818abde56333dd3366cbb85f0d7ef4105a0f2853985a6`; engine receipt
`sha256:720b443c7267c7dc09cf7ecb5ffa226b9a0d6a745136689ef2e42f17f13b2d2f` at `receipts/engine/model_admitted/SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001-720b443c7267c7dc.json`; empirical-validation hash
`sha256:737036b83285f0487d66662081f6f4ef7d8698f0c70a48f17abd52b3ba242290`; measurement receipt `sha256:cbca0b4410f39699e01456c2a835fcdc54c885ab250f35dcf2e9bd71392bad2a`.

137. Squared proton-to-Planck hierarchy

Claim identity: `SFT-PHYS-SCALE-PROTON-PLANCK-HIERARCHY-002`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Generator-three successor volume has least binary cover depth seven. Complete binary support at that depth contains 128 states and its unique positive predecessor contains 127 massive positions. The half-One gravity carrier therefore makes the unsquared exponent 127/2; no irrational root is formed, and the exact forced comparison object is the squared hierarchy 2^127.

> *The exact squared Planck/proton hierarchy is 2^127 = 170141183460469231731687303715884105728; the unsquared exponent is retained as 127/2 and no square root is formed.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of scale carrier, cover dependencies, full-support predecessor relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: The complete binary support at the least cover of generator-three successor volume, its unique positive predecessor and the half-One coupling, with squared comparison only. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-dependency-trace__depth-seven-support-predecessor-at-half-One__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	depth-seven-support-predecessor-at-half-One	formed-irrational-root-or-fitted-hierarchy	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:5d2f08b5bd30059644269313d7cf73d173e43997a6468844a174ed239ed9f278`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:b549ad7c5df8d84d32a49d45dc61b0945f14d569357542f2c56c9429c07fbcf5`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:578efe7a44cf4d163751a387457c2a7da657fb15273bc86a041f4d076216195c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a780f43acd149fd24c88aef7fb706fc433b85dae8864d13b618ed88c488b339f`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:f7d411c65f442f9ef1a9dad47831ca171d1d44522bc8b142b2c69d3db4cab9b8`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b`. Independent certificate: `sha256:ae1fdcb1dde94d52082aae299929e6590d7e392cfd5650004fa19d075e7dcddd`. Engine external-validation hash: `sha256:d3bb3974eb182a550c2e34f963a9c47fa105747b4186d187c5d86fac25e86e59`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S016` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor

- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule
- no irrational square-root formation; empirical comparison must square the measured positive interval instead

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d847dbf538c92b07c9f710c9f33887685b40dca9eef56f3943e855ac2f49b5bb`; engine receipt `sha256:d1cf13e86e2d9aeaa5f6993456b5f594a303d75e446ea1d9c80d41906b984d9f` at `receipts/engine/model_admitted/SFT-PHYS-SCALE-PROTON-PLANCK-HIERARCHY-002-d1cf13e86e2d9aea.json`; empirical-validation hash `None`; measurement receipt `None`.

138. Terminal charged-colour proton-to-Planck hierarchy

Claim identity: `SFT-PHYS-SCALE-PROTON-PLANCK-TERMINAL-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The admitted lower hierarchy counts complete depth-seven massive support as 2^127 in the squared comparison domain. A proton is the hierarchy carrier that retains both colour labels over generator three and one terminal electromagnetic self-coupling. Transporting that exact 2/3 alpha share once and retaining its positive One-complement forces the terminal squared hierarchy before measurement.

> *The terminal squared Planck/proton hierarchy is 255923934603817488008405160690199418432572494970880/1511539186407, equal to 2^127 times (1 - (2/3) alpha_terminal).*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SCALE-PROTON-PLANCK-HIERARCHY-002`
- `SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete ten-axis product of lower hierarchy carrier, terminal stage, complete support, charged-colour relation, alpha transport, retained orientation, enumeration, minimality, measurement direction and extension. The declared exact boundary is: All exact squared hierarchy successors assembled from the admitted 2^127 support, the admitted two-over-three colour carrier and exactly one terminal electromagnetic self-coupling, with the bare hierarchy retained as a lower-order control. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `admitted-lower-law-carrier__complete-promotion-termination__complete-binary-terminal-support__bare-hierarchy-times-One-complement-of-two-thirds-alpha__single-typed-terminal-alpha-transport__orientation-fixed-by-share-or-retention__complete-registered-product__bare-and-intermediate-forms-preserved__exact-result-sealed-before-target-release__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`admitted-lower-law-carrier`	`detached-measured-number`	A measured number has no forward Fold dependency trace.
`terminal`	`complete-promotion-termination`	`selected-intermediate-stage`	An intermediate stage chosen for agreement is a fit.
`support`	`complete-binary-terminal-support`	`free-scale-support`	A free scale is a parameter.
`relation`	`bare-hierarchy-times-One-complement-of-two-thirds-alpha`	`free-hierarchy-exponent-or-untyped-correction`	The alternative loses a required carrier or adds an unforced operation.

Axis	Forced form	Eliminated form	Elimination reason
`correction`	`single-typed-terminal-alpha-transport`	`fitted-additive-offset`	A fitted offset reads the target.
`sign`	`orientation-fixed-by-share-or-retention`	`target-selected-orientation`	A target-selected sign is not structural.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected neighbourhood cannot prove uniqueness.
`minimality`	`bare-and-intermediate-forms-preserved`	`successor-without-lower-controls`	A terminal answer alone hides whether the lower form was necessary.
`measurement`	`exact-result-sealed-before-target-release`	`target-visible-before-seal`	That reverses empirical direction.
`extension`	`no-extra-rule`	`free-extra-correction`	An extra term is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:13aa1c5380e62ab000cab2f898e5dd6f7cfc1498d20da4fad9f6c8da470da133`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:f1523a85cb053c03b2ec81e22f6b82a870fcbdbc30f320e8a456043f04dae4af`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:404dfca4694089557d07640290478cded4b0e0e92f193fb5c090f99c9015f913`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b303be1b9682c63e30540ac74f82d51581cef9b2ea130460d417efb1ce732c4f`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:1ff75c1743c0e2f721404dc2311de33c6d45dcf0234d98a0f9149ecf42879a56`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:61bd427364f94266efdcb7681e8bb7fb1211cf8a628b54ac1d8dab2b2841c504`. Independent certificate: `sha256:b9425f0a9d11c8870d321a8252c8d951729310b3cf6f088ecad7fd1ae499ea5a`. Engine external-validation hash: `sha256:c7af4edd3accefe40f6324979f2b366eed9106156270a7b98419aed2f80eaf71`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S016` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no external measured value, uncertainty or best-fit table as a derivation input
- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no fitted level, coefficient, correction sign, truncation or target-selected candidate neighbourhood
- no irrational square root; authoritative mass intervals are squared only after the formal seal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:5eca6d00fe54b7afbabb728a9dab3716786ae87790266d141e993221dddce0d6`; engine receipt `sha256:affd1dc6754854b15d7f286291611e481f0572f50a11589dd88cb019289f88d7` at `receipts/engine/model_admitted/SFT-PHYS-SCALE-PROTON-PLANCK-TERMINAL-003-affd1dc6754854b1.json`; empirical-validation hash `None`; measurement receipt `None`.

139. Terminal colour-channel alpha-squared proton-energy fraction

Claim identity: `SFT-PHYS-PARKER-PROTON-ENERGY-TERMINAL-028`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The terminal structural proton-energy fraction is exactly 1081476177710254863 68/253861190227103943729961 = 8 alpha_terminal^2. The superseded leading-rung prediction 500000/1173679081 is retained as an exact historical control. External proton mass and Parker observations remain inaccessible until the derivation seal.

> *The terminal structural proton-energy fraction is exactly 108147617771025486368/253861190227103943729961 = 8 alpha_terminal^2. The superseded leading-rung prediction 500000/1173679081 is retained as an exact historical control. External proton mass and Parker observations remain inaccessible until the derivation seal.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003`
- `SFT-PHYS-NUCLEON-BINDING-TERMINAL-005`
- `SFT-PHYS-MECH-WORK-ENERGY-001`
- `SFT-PHYS-SCATTERING-RUTHERFORD-COMPTON-TERMINAL-006`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

Generated grammar and closure boundary. Generate the complete product of proton identity, alpha stage, all four forced mediator-sector counts, amplitude/energy/cubic powers, typed composition, rest-scale carrier, prediction custody, provenance, external comparison and extension forms. The declared exact boundary is: The admitted proton, all forced sectors two/three/five/seven and their mediator counts, the leading and terminal exact alpha rungs, powers one through three, and a post-seal proton-rest-energy/Parker comparison. The generator produced `3072` named candidates and the decision support contains `3072` one-for-one decisions. Exactly one candidate survived: `admitted-three-colour-proton__complete-terminal-alpha-rung__colour-eight-channels__alpha-squared-energy__complete-channel-count-times-energy-share__proton-own-rest-energy-postseal__capability-closed-before-Parker-release__observational-derivation-explicit__complete-reported-range-and-limitations__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`particle`	`admitted-three-colour-proton`	`named-proton-without-composition`	A name does not generate its interaction support.
`alpha`	`complete-terminal-alpha-rung`	`selected-leading-alpha-rung`	The leading rung is retained as a historical control but is not the completed promotion object.
`channels`	`colour-eight-channels`	`weak-three-channels`	The binary-sector mediator count does not match proton colour.
`power`	`alpha-squared-energy`	`linear-alpha-amplitude`	A single amplitude is not an energy self-composition.
`composition`	`complete-channel-count-times-energy-share`	`add-or-average-channel-shares`	Addition or averaging changes the complete channel multiplicity.
`scale`	`proton-own-rest-energy-postseal`	`local-plasma-fit-or-free-energy`	A local field or density would be a fitted environmental parameter.

Axis	Forced form	Eliminated form	Elimination reason
`target`	`capability-closed-before-Parker-release`	`Parker-range-readable-before-seal`	That would permit formula selection by the observation.
`provenance`	`observational-derivation-explicit`	`claim-new-blind-discovery`	V2 and the Parker report were already known during V3 reconstruction.
`comparison`	`complete-reported-range-and-limitations`	`treat-approximately-400-as-exact-cutoff`	The paper reports an approximate energy reach and no uncertainty on 400 keV.
`extension`	`no-extra-rule`	`free-local-factor-or-correction`	A free local factor would fit the event.

Base and successor. The closure proof is sealed by derivation hash `sha256:ba3f74d46f0fa4868ea25ce160affd879952128cb5dc07f451750f83598b38eb`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:8a698572330616bbe1d93d09c5416ec8ec01e583ff1effdff412eb40647ff207`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:41ecf3347f1f4aa20f85a7e123fd66850f15f7934d3a2f4cc54fdab2dd949a29`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a6fe0f61a16b5f1e132529709d14156c2714bc883d3d10553f14cb207b45484b`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:a6d0935252f735483db99db340353132969c9bb58a1ee253c53965bb1e111847`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9e8af8c582652edc7816af89b7d3254a39c2a7cabcd838f7371d5cdc8d36029a`. Independent certificate: `sha256:65903e28af5143824e401feddc2b1261fe3bdd81a8f13aa4e7ab9a3c8b50bc11`. Engine external-validation hash: `sha256:4adb9cda9fd0ffa1ba246a731e1009ea3a9d8ef1815222bacdac90ca92279438`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- DESAI-ET-AL-2025-PARKER-HCS-PROTONS
- NIST-CODATA-2022

Observed comparison records:

- The primary Parker manuscript, complete CODATA source and curated row record are hash-locked and released only after the exact Fold prediction seal.
- The terminal prediction from the complete current alpha rung is approximately 399.714077 keV after post-seal CODATA scale translation; the retained V2 leading-rung control is approximately 399.714072 keV.
- Both exact propagated prediction intervals lie inside the paper's complete 67-527 keV analysed proton spectrum.
- The primary paper reports protons up to, above and upwards of approximately 400 keV; it does not register 400 keV as an exact cutoff and gives no standard uncertainty for that approximate label.
- Therefore the earlier 'within 0.1 percent' wording is not admitted as a precision measurement claim; only range-level correspondence is supported.

- The complete forced-sector/power control census is retained. Both three-alpha-squared and eight-alpha-squared land inside the broad observed range, so the Parker event alone does not select the eight-channel formula.
- Eight channels remain structurally selected by the independently admitted proton colour sector, never by the Parker target.
- Changing the external upper range to 300 keV rejects the prediction as an unfavourable control.

Falsification condition: Reject if any source identity or registered row changes; if the terminal fraction or NIST-propagated prediction leaves the complete 67-527 keV Parker analysis range; if approximately 400 keV is presented as an exact cutoff or given an unreported uncertainty; if the broad observation is claimed to empirically select eight over every adverse formula; if a local plasma value enters the derivation; or if targets are accessible before prediction sealing.

Measurement receipt: `sha256:f934b85069bf77672ff16ee8acf662add04f6e652d0fe2a3807076f7bc9a5f36`. Isolation certificate: `sha256:260875f77819464ff90d563f292a61190470d0d2f647199ccdb69270c40af7c9`. Custody certificate: `sha256:d64484e3eef9294ffde93e19b562a48f1eb47bbd49e7e9990e0c78da943a2d6c`.

Registered source descriptions:

- Astrophysical Journal Letters accepted manuscript via arXiv: https://arxiv.org/abs/2410.16539 (sha256:6c950a6e142edf57df600d06e1ce40445fc18c84d42c3d2b916d63679a100fb5)
- National Institute of Standards and Technology: https://physics.nist.gov/cuu/Constants/index.html (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- WITHHELD-PARKER-PROTON-RANGE-AND-CODATA-SCALE from None

Shared claim-record clause `PHYS-S016` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no local plasma density, magnetic field, Alfven energy or guide-field parameter
- no proton mass-energy or Parker spectrum in the candidate generator or survivor decision
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude
- no claim that approximately 400 keV is an exact measured cutoff or has 0.1-percent precision
- no omission of alternative forced-sector/power formulas at the post-seal comparison boundary
- no claim of historical blindness; V2 and the external result were known before reconstruction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:003e85d7036062243cbc4a81865ab602477779d486516d64533b64f681eedb21`; engine receipt `sha256:41f17af1f94905b0e93d67d7dfc7a399fd9063a53e6db4d21d486e1a8c5f1673` at `receipts/engine/model_admitted/SFT-PHYS-PARKER-PROTON-ENERGY-TERMINAL-028-41f17af1f94905b0.json`; empirical-validation hash `sha256:ba3f659435ad35064dfb8c2e5fc085b84dc24edf6109f8d4090878e770a67c6b`; measurement receipt `sha256:f934b85069bf77672ff16ee8acf662add04f6e652d0fe2a3807076f7bc9a5f36`.

140. Terminal probe-independent proton rms charge-radius relation

Claim identity: `SFT-PHYS-PROTON-RADIUS-TERMINAL-029`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The unique terminal dimensionless proton rms charge-radius coefficient is exactly 10069574419808/2519231977345 = 4(One-alpha_terminal/10). A dimensionful radius is obtained only after the seal by composing this coefficient with the registered reduced proton Compton wavelength. The coefficient contains no electron, muon or scattering-probe input.

> *The unique terminal dimensionless proton rms charge-radius coefficient is exactly 10069574419808/2519231977345 = 4(One-alpha_terminal/10). A dimensionful radius is obtained only after the seal by composing this coefficient with the registered reduced proton Compton wavelength. The coefficient contains no electron, muon or scattering-probe input.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-PHYS-NUCLEON-BINDING-TERMINAL-005
- SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-QUANTUM-EVOLUTION-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001

Generated grammar and closure boundary. Generate the complete product of proton carrier, tripling orbit, colour/Fold edge support, internal/external charge support, electromagnetic transport order, retained orientation, one-action scale boundary, target custody, provenance and extension forms. The declared exact boundary is: The admitted colour-three charged proton, its one-third/two-thirds tripling partition, both Fold labels, every ordered colour pair cell, the unique external unit charge, terminal alpha and the post-seal reduced proton Compton carrier. The generator produced `2304` named candidates and the decision support contains `2304` one-for-one decisions. Exactly one candidate survived: `admitted-uud-colour-three-proton__outer-two-thirds-complement__all-three-colours-times-both-Fold-labels__ten-internal-pairs-plus-unit-charge__one-alpha-spatial-traversal__hold-inward-share-from-edge__reduced-proton-Compton-postseal__capability-closed-before-target-release__observational-derivation-explicit__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
particle	admitted-uud-colour-three-proton	named-proton-without-composition	A particle name does not generate a radius.
orbit	outer-two-thirds-complement	inner-one-third-as-outer-edge	The inner constituent site is not the composite edge.
edge	all-three-colours-times-both-Fold-labels	partial-colour-or-single-fibre-support	Omitting a colour or Fold label loses a charged path.
charge	ten-internal-pairs-plus-unit-charge	eight-nonreturn-mediators	The mediator count omits the return and external charge roles.
transport	one-alpha-spatial-traversal	no-terminal-transport	This retains only the earlier leading structural radius.
orientation	hold-inward-share-from-edge	append-outward-free-share	Appending a free share breaks the held bound-composite ledger.
scale	reduced-proton-Compton-postseal	full-cycle-or-fitted-length	A fitted length or full-cycle circumference double-counts the one-action radial carrier.
target	capability-closed-before-target-release	probe-results-readable-before-seal	That would fit the known radius puzzle.
provenance	observational-derivation-explicit	claim-historical-blindness	V1/V2 and the measurements were already known.
extension	no-extra-rule	free-form-factor-or-probe-correction	An added term would be a fitted parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:1f5b55c75de20a987a553421413cf4ad54e259b5da55649402bce332cf7d926f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:a7a6a49aa74db2977d54690d7a0ce9f676adb4bcc085662d318f3380d6d3a760`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:b2b88486d92413ea48792aecfd581c0f6708d599a9e42be2cc007085c3604811`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0990ec3b4ad7b9cd1ce8c9d4bd44fdb8747b9ef8e7ef9846327c49c945ccfa50`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:ca75083edfef55fc18c3d5e70d7091287a3eac7767e4bba931ae56b02e6002fc`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b7ae5cd32b920bc549c3cb64b718a2ea64732691be02ccf8cb0118764207f7de`. Independent certificate: `sha256:1bdcc6ce6b78572a95ca6fc1b89b728840d2624720824904ffb9b1f1179ace31`. Engine external-validation hash: `sha256:c30497188e21b2cf74496f02ce05fd120478ff49a586f643a660ac364c916451`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NATURE-2026-ELECTRONIC-HYDROGEN-PROTON-RADIUS
- NATURE-2019-PRAD-PROTON-RADIUS
- NIST-CODATA-2022
- NIST-CODATA-2014-HISTORICAL

Observed comparison records:

- The exact coefficient and all structural alternatives seal before the four primary/authoritative source identities and radius rows are released.
- The exact terminal coefficient is 10069574419808/2519231977345 = 4(One-alpha_terminal/10), with no probe or radius measurement in its derivation.
- Post-seal composition with the complete CODATA-2022 reduced-proton-Compton interval predicts approximately 0.840621761 fm with exact outward uncertainty propagation.
- The prediction interval lies inside the registered 2026 electronic-hydrogen 0.8406(15) fm interval and the current muonic-hydrogen 0.84060(39) fm interval.
- The prediction interval lies inside the CODATA-2022 0.84075(64) fm interval and the conservative PRad 0.831 +/-0.007(stat) +/-0.012(syst) outward interval.
- The historical CODATA-2014 0.8751(61) fm interval is disjoint and remains an explicit unfavourable historical row.
- The primary 2026 paper's partly discrepant earlier electronic results and scattering-analysis dependence remain explicit; probe independence is not relabelled as universal agreement among all extractions.
- All three linear-support alternatives land inside the present muonic interval, so measurement does not select ten; the complete proton charge-support derivation does.
- Changing the current electronic-hydrogen row to a disjoint interval rejects the correspondence.

Falsification condition: Reject if any source identity or registered row changes; if the exact terminal prediction leaves any registered current electronic, muonic, CODATA or outward PRad interval; if the historical CODATA-2014 conflict or current extraction limitations are omitted; if a probe, radius or Compton value enters before sealing; if observation is claimed to select ten charge cells; or if target access precedes the seal.

Measurement receipt:
`sha256:6a828b91cce631a0caa5a33a44284ade02f779931199d0e9fc96e42ecfd3607f`. Isolation certificate:
`sha256:f9112b893f35f1a69d98ea1e06da8339cd9df96de0c830502c76fe6896853a6b`. Custody certificate:
`sha256:7bf579b9c3ac12c02353d9692b0a29ccaede0dcc3e7ff0599a263d402c23ca06`.

Registered source descriptions:

- Nature: https://doi.org/10.1038/s41586-026-10124-3 (sha256:834ddf04fdb3064d07307cc743544990e086601948903fb73390144d2a3ae253)
- Nature manuscript via arXiv: https://doi.org/10.1038/s41586-019-1721-2 (sha256:66925b870f8391be1a940b741ff60322877df7551cfa1db58b59add002e15ad5)
- National Institute of Standards and Technology: https://physics.nist.gov/cuu/Constants/index.html (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)
- National Institute of Standards and Technology: https://physics.nist.gov/cuu/Constants/archive2014.html (sha256:48c31408c79d3a9aa7c16efdbf5ef729d8ad162f2b9dbb8029d4d8cdcbef3568)

Registered target identities:

- WITHHELD-PROTON-RADIUS-COMPLETE-VECTOR from None

Shared claim-record clause `PHYS-S016` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no proton radius, reduced Compton wavelength, electron, muon or scattering result in the candidate generator or survivor decision
- no fitted coefficient, probe-dependent correction, selected uncertainty or hidden form factor
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude
- no claim that rms charge radius is a hard material edge
- no claim that every historical extraction is mutually consistent or systematics-free
- no historical-blindness claim

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:d39039deed7c3890466a520e0c899d59b6dc8e7b70600f46697fb153e0142e37`; engine receipt
`sha256:77a31b069cc1cdd4bdf84f3ca08fb4849ae33fddb02945727f1809d9e4c2373d` at `receipts/engine/model_admitted/SFT-PHYS-PROTON-RADIUS-TERMINAL-029-77a31b069cc1cdd4.json`;
empirical-validation hash
`sha256:39c3b0612ff4b5ddc4ec5c86b7194a9d3f814b3f3af85dc4dad38d03093d8465`; measurement receipt `sha256:6a828b91cce631a0caa5a33a44284ade02f779931199d0e9fc96e42ecfd3607f`.

141. Common Fold scale axis and terminal electroweak transport

Claim identity: `SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. One exact common scale axis is forced: supports 1,2,4,8,...; spacings One, half-One, quarter-One, eighth-One,...; and every forced prime sector uses the same support. The electroweak leading curve is w(R)=(R+2)^2/[4(R+1)^2+(R+2)^2]. Terminal support sixteen holds three directions, forcing active level thirteen, w(13)=225/1009 and the separately typed alpha/17 return, exactly reproducing the admitted terminal on-shell share. The unique internal power anchor is support two. Common exact unit rescaling leaves every dimensionless result unchanged; physical magnitudes are transported only through sealed exact ratios and a post-seal held reference.

> *One exact common scale axis is forced: supports 1,2,4,8,...; spacings One, half-One, quarter-One, eighth-One,...; and every forced prime sector uses the same support. The electroweak leading curve is w(R)=(R+2)^2/[4(R+1)^2+(R+2)^2]. Terminal support sixteen holds three directions, forcing active level thirteen, w(13)=225/1009 and the separately typed alpha/17 return, exactly reproducing the admitted terminal on-shell share. The unique internal power anchor is support*

> *two. Common exact unit rescaling leaves every dimensionless result unchanged; physical magnitudes are transported only through sealed exact ratios and a post-seal held reference.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-COUNT-001`
- `SFT-FOUNDATION-FOLD-DYNAMICS-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-PHYS-MEAS-UNIT-COMPARISON-001`
- `SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001`
- `SFT-PHYS-SPACETIME-LIMIT-SPEED-001`
- `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003`
- `SFT-PHYS-COUPLING-RUNNING-CONVERGENCE-TERMINAL-006`
- `SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002`
- `SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003`
- `SFT-PHYS-SCALE-PROTON-PLANCK-TERMINAL-003`
- `SFT-PHYS-MATTER-QUARK-INVARIANTS-003`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`

Generated grammar and closure boundary. Generate the complete product of foundation origin, scale successor, order/spacing relation, local propagation transport, sector domain, electroweak squared-support form, terminal held-support completion, internal anchor, unit transformation, absolute-ratio placement, target custody, provenance disclosure and extension form. The declared exact boundary is: The One, every positive finite binary support successor, reciprocal exact spacing, one-cell causal transport, all four forced prime sectors, binary charged and neutral Fold fibres, terminal support sixteen with three held directions and one alpha return, every common positive rational unit rescaling, the exact proton-Planck scale ratio and the complete registered post-seal coupling vector. The generator produced `12288` named candidates and the decision support contains `12288` one-for-one decisions. Exactly one candidate survived: `One-origin__binary-complete-support-successor__reciprocal-One-over-support-spacing__one-local-act-per-support-cell__complete-prime-sector-ladder__neutral-square-over-charged-plus-neutral-squares__support-sixteen-hold-three-return-alpha-over-seventeen__unique-support-two-source-four__common-positive-rational-rescaling-cancels__forced-ratio-then-postseal-held-reference__capability-closed-before-target-release__observational-development-explicit__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`origin`	`One-origin`	`numerical-zero-origin`	Semantic numerical zero is not an SFT scale carrier.
`successor`	`binary-complete-support-successor`	`linear-or-continuous-free-step`	A free or continuous step is not generated by Fold.
`spacing`	`reciprocal-One-over-support-spacing`	`untyped-arbitrary-spacing`	Arbitrary spacing loses the common exact ratio.
`transport`	`one-local-act-per-support-cell`	`independent-free-running-axis`	A second arbitrary axis breaks one-cell propagation.
`sectors`	`complete-prime-sector-ladder`	`selected-known-sectors`	Selecting familiar sectors omits forced penta/hepta predictions.

Axis	Forced form	Eliminated form	Elimination reason
weak_curve	neutral-square-over-charged-plus-neutral-squares	linear-channel-ratio	Channel amplitudes require squared support.
terminal	support-sixteen-hold-three-return-alpha-over-seventeen	identify-active-level-with-full-support	Three held generator directions would be omitted.
anchor	unique-support-two-source-four	target-selected-depth	A measurement cannot assign the internal landmark.
units	common-positive-rational-rescaling-cancels	unit-name-changes-dimensionless-law	A held reference cannot change a like-dimension ratio.
placement	forced-ratio-then-postseal-held-reference	measured-rung-or-free-unit-selects-law	A target-selected rung or calibration would be a free parameter.
target	capability-closed-before-target-release	target-readable-before-seal	That would fit the known running vector.
provenance	observational-development-explicit	claim-historical-blindness	The V1/V2 observations and physical measurements were known.
extension	no-extra-rule	free-scale-or-running-correction	An unforced addition would violate zero parameters.

Base and successor. The closure proof is sealed by derivation hash `sha256:688facc545ef6d4e7cbbf86ea70d5400432086ec1b86e421fb5118c1661dfb2c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:8f4540da7dcef139c22150d7490b7a29ff18f3a4c5ff0127fa9412ad3e212d47`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:efeba3f23ea1e5900da498c95bedf19655677ebe0f090f0aff8873ced57feab8`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:39f4f41bb376072deaa4b774f3f7255ab3410bd25ef449aa6c0155a38e496d80`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:4c5aabb4e2c5926719fd737be86a2211bb0f6cbde8e77e2ed0e0f59f889e8bdf`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c39e4a115db04ee65c879d06d98342923668e286b615eac1805203ab76def489`. Independent certificate: `sha256:a4215f5464a8f8e2dca60674b41b77841a71c70ab155afddd9dcbc379be9110b`. Engine external-validation hash: `sha256:0bbcd4db30a68a6a99843f5a139129d19cda5330d4a32f62b6b9c1c4e0ef93b3`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2026-ELECTROWEAK-SCALE-VECTOR
- PDG-2025-2026-STRONG-EM-COMPLETE-RUNNING-VECTOR

Observed comparison records:

- The exact common axis, terminal held-support chain and scale-ratio law sealed before release of any coupling or dimensional target.
- The terminal on-shell weak share lies inside the complete PDG 2026 on-shell interval without a fitted rung or correction.

- The exact leading support-eight weak share 25/106 lies inside the complete registered cesium APV interval; this is a post-seal correspondence, not the selector of support eight.
- The full SLAC E158 and cesium APV intervals lie above the MS Z interval, matching the sealed sub-W descending direction.
- The eDIS interval overlaps the Z interval and remains non-discriminating; the NuTeV on-shell interval lies above the terminal interval and remains an explicit adverse row with its interpretation concerns.
- The complete PDG weak-running figure is retained by immutable source hash; no unprinted coordinates are reverse-digitized.
- The PDG W-threshold minimum and sign change are retained, so the finite monotone Fold branch is not overclaimed as one universal all-energy curve.
- The already admitted complete strong and electromagnetic multi-scale vectors, terminal electroweak comparison and proton-Planck hierarchy comparison remain hash-bound dependencies of this one-axis result.
- Changing the on-shell target to a disjoint interval rejects the correspondence.

Falsification condition: Reject if any source or inherited receipt identity changes; if the terminal on-shell prediction leaves the registered on-shell interval; if the registered sub-W low-transfer direction is reversed; if the NuTeV adverse row, scheme distinctions, W-threshold sign change, strong/EM complete vectors or proton-Planck scale receipt are omitted; if a measured value selects a rung or law; or if target access precedes the seal.

Measurement receipt: `sha256:14ba878aa37865afc13d58c3f89581aed5136172f4acfc4141078da959c50a20`. Isolation certificate: `sha256:05a792a4b6f8bdcc6b6aec0fb266de053d1072eeaa3f4b8d4fba8cdc49bb8e4b`. Custody certificate: `sha256:882a236b6d1f3e176438b58b8addefc46a30e4e6348e2a5aa90222f9ddbf8e30`.

Registered source descriptions:

- Particle Data Group: https://pdg.lbl.gov/2026/reviews/rpp2026-rev-standard-model.pdf (sha256:a102f6252b7190dc423200271dffa7c805cd15a50391b1c578853d2f777611cb)
- Particle Data Group: experiments/external_sources/physics/snapshots/coupling-running-convergence-source-record.json (sha256:b83331089d96c073fbd5101753ba5c4716ae1a8b1b891e068684b6f7246d9953)

Registered target identities:

- WITHHELD-COMMON-SCALE-AND-COUPLING-COMPLETE-VECTOR from None

Shared claim-record clause `PHYS-S016` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no numerical-zero scale origin, negative exponent, irrational, imaginary, floating or completed-continuum proof value
- no imported renormalization-group equation, beta function, logarithmic energy coordinate or continuous scale
- no measured energy, weak angle, coupling, Planck value or selected rung in the formal survivor decision
- no identification of the earlier full-support running level with the terminal held-support level
- no claim that one monotone weak curve applies above the registered W-threshold boundary
- no fitted offset, calibration constant, scheme correction or hidden target-selected depth
- no historical-blindness claim

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:41008d57dd39bcbccd3e2bb5c023c4ab96e0cf7e8e1666a2ce689507cca8fc7f`; engine receipt `sha256:44d542f87d12844c02c321ac651008ae7ddd738a3b16e7d2c3d8700ec89eca55` at `receipts/engine/model_admitted/SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030-44d542f87d12844c.json`; empirical-validation hash `sha256:626bb4f32ea61fd62a404aafc02a12c1fa95c2b64e116eafdfb1ef9c30aaf5ba`; measurement receipt `sha256:14ba878aa37865afc13d58c3f89581aed5136172f4acfc4141078da959c50a20`.

142. The Unified Constants Object

Claim identity: `SFT-PHYS-UNIFIED-CONSTANTS-OBJECT-077`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. One connected rooted Fold object generates the registered cross-sector constants. Its common order is b=2, c=3, spatial rank 3, boundary rank 2, down/up cover depths 5 and 7, leading inverse alpha 34259/250, charged-lepton product 1/485, depth-five conjugate 1/95, depth-seven product 1/383, dark/baryon 27/5 with dark share 27/32, Hubble 13/12, Planck exponent 127/2, local vacuum-energy floor 1/2^20 and half-One 1/2. V3 terminal continuations retain the same object and include inverse alpha 503846395469/3676744786, charged-lepton sharpened product 3/1454 and terminal product, quark products 1/383 and 1/3071, dark/baryon 279/52, Hubble 3305/3048, the terminal squared Planck hierarchy, local vacuum amplitude/energy floors 1/2^10 and 1/2^20, and normalised cosmological magnitude 33/16.

> *One connected rooted Fold object generates the registered cross-sector constants. Its common order is b=2, c=3, spatial rank 3, boundary rank 2, down/up cover depths 5 and 7, leading inverse alpha 34259/250, charged-lepton product 1/485, depth-five conjugate 1/95, depth-seven product 1/383, dark/baryon 27/5 with dark share 27/32, Hubble 13/12, Planck exponent 127/2, local vacuum-energy floor 1/2^20 and half-One 1/2. V3 terminal continuations retain the same object and include inverse alpha 503846395469/3676744786, charged-lepton sharpened product 3/1454 and terminal product, quark products 1/383 and 1/3071, dark/baryon 279/52, Hubble 3305/3048, the terminal squared Planck hierarchy, local vacuum amplitude/energy floors 1/2^10 and 1/2^20, and normalized cosmological magnitude 33/16.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-ONE-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-COUNT-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001`
- `SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001`
- `SFT-PHYS-MATTER-QUARK-CUBICS-003`
- `SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001`
- `SFT-PHYS-COSMO-HUBBLE-CALIBRATION-001`
- `SFT-PHYS-SCALE-PROTON-PLANCK-HIERARCHY-002`
- `SFT-PHYS-SCALE-PROTON-PLANCK-TERMINAL-003`
- `SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003`
- `SFT-PHYS-VACUUM-DENSITY-SCALE-TERMINAL-035`

Generated grammar and closure boundary. Generate the complete twelve-axis product of root, Fold carrier, generated counts, cover geometry, typed sector incidence, leading vector, terminal continuation, graph connectedness, dependency mutation, independence control, measurement direction and extra-rule forms. The declared exact boundary is: Every rooted typed dependency object over the admitted V3 electromagnetic, lepton, quark, dark/baryon, Hubble, Planck-hierarchy and vacuum branches, their exact foundation-order and terminal values, the full generator-three-to-successor dependency probe, the binary/rank independence controls and all 4096 alternatives. The generator produced `4096` named candidates and the decision support contains `4096` one-for-one decisions. Exactly one candidate survived: `single-One-root__Fold-generated-binary-and-generator__least-binary-covers-of-generated-volumes__shared-carrier-incidence__complete-registered-foundation-vector__root-preserving-terminal-continuations__connected-rooted-dependency-graph__complete-generator-successor-probe__binary-and-rank-held-controls__formal-object-sealed-before-comparison__append-traced-node__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`root`	`single-One-root`	`disconnected-constant-list`	A list cannot establish common mathematical identity.
`fold`	`Fold-generated-binary-and-generator`	`borrowed-numerals`	Borrowed numbers are untraced inputs.
`covers`	`least-binary-covers-of-generated-volumes`	`selected-depth-labels`	Selected depths are parameters.
`typing`	`shared-carrier-incidence`	`untyped-number-coincidence`	Equal digits alone do not prove one object.
`leading`	`complete-registered-foundation-vector`	`selected-headline-values`	Headline selection can hide a disconnected branch.
`terminal`	`root-preserving-terminal-continuations`	`replacement-values-with-new-roots`	Replacement would break the common object.
`graph`	`connected-rooted-dependency-graph`	`sector-local-formulas-only`	Local formulas do not expose common dependence.
`mutation`	`complete-generator-successor-probe`	`decorative-shared-labels`	Labels without recomputation do not demonstrate dependence.
`control`	`binary-and-rank-held-controls`	`blanket-global-mutation`	Changing everything cannot localize dependence.
`measurement`	`formal-object-sealed-before-comparison`	`measurement-assembled-object`	That would fit the dependency graph to targets.
`extension`	`append-traced-node`	`rewrite-existing-object`	Rewriting destroys provenance.
`rule`	`no-extra-rule`	`free-extra-rule`	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:ce9c38edc90663ff794873ad61ab52da7fd04b75e3c40fb0eb572d7af00aa211`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:ba78be35a45d8b6b4243e3a5d8c76b0a9a3e5f2e5db3de74ecb03ee37727a248`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:7a56eebbf71fd317ac3a3637c55ecb22c3f33f06c3cd74fe45eed957fa984bfb`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:072a6fd4646bbf224fbb02b2a7b6a017193b92b6cfb961311f5f4a211216d818`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:082de57da6f3821cb821e42103ce00ed619564b876aca6a0e19dd917c51ad742`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d72aba5819b1faf962d1811240eb919ef7f6011631bd51a32b22ad1330710f7e`. Independent certificate: `sha256:62421072ef8793c2915587e8d567ddb8a37912b8697956bf7e3605c4d223fcb8`. Engine external-validation hash: `sha256:7e920dd732601e7e455ab4baf27988aa56666991608ac508df596fa3d93f2b3e`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S016` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no renaming, replacement or reinterpretation of Grand Lock 075 or 076
- no V1/V2 proof artifact or measured constant as an executable premise
- no fitted parameter, selected coefficient, target tolerance or untyped numerical coincidence
- no numerical-zero, negative, irrational, imaginary, floating, NaN or completed-infinite proof scalar
- no omitted sector, severed root edge, blanket mutation or silently rewritten terminal value

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:f5efe6a373ccd8e156cef67a5d01feeb1f6a2aa66c0ddeb1efb8daa4c10a11d2`; engine receipt `sha256:e6112bcfa663b3d24727bd03688491f55fdb5fc32359542b92acee7e551fbac8` at `receipts/engine/model_admitted/SFT-PHYS-UNIFIED-CONSTANTS-OBJECT-077-e6112bcfa663b3d2.json`; empirical-validation hash `None`; measurement receipt `None`.

18. Matter Interactions Flavour

Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed.

143. Mass-energy correspondence

Claim identity: `SFT-PHYS-MATTER-MASS-ENERGY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A retained inertial recurrence composed with two limiting-speed propagation transfers is the equivalent transferable energy carrier; conversion preserves the complete source trace.

> *Rest mass-energy is exact composition of the inertial carrier with two limiting propagation-speed carriers.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__inertial-carrier-times-limit-speed-square__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`inertial-carrier-times-limit-speed-square`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:1733943f9be3a60770b42d10ac1a6754391930e965571ee8714c5855bd0f36f7`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ed0bec154485f63255b26601114771f2175e5c9c127aa857b0e159ebb53d3819`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0d3b8bf23c1d3242a5b5801dad3467f0698bae611dfca54dd64fc741e69eb04e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:aa11d50ff9abd65f109a711ae145bcaf5ce16355a23aabea67ce51f63cc71a1b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:5fe31bb7a21737a2dbf16c212f31f9e68ee0f67bb7ecd1330de2ae306d6c9945`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:40cd4a67fb937a98d196b2c5a836586e9cfb46c3a74c27fe4d1692c39c27cb7e`. Independent certificate: `sha256:e977626f21f6e22faac2b47e0d2f32e48bd60458129716fff62d1771b3cd8e2e`. Engine external-validation hash: `sha256:9b866037d2530b2ed8f047010ca6142cdf866ba5cd65632a738301ae3af12436`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- electron mass energy equivalent: exact predicted interval [2046776446362983920916785951/2500000000000000000000000000000000000000, 2046776447621241171148330647/2500000000000000000000000000000000000000]; external interval [40935528927/500000000000000000000000, 40935528953/500000000000000000000000]; overlap True
- deliberately displaced unfavourable interval rejected

Falsification condition: The sealed exact mass/speed prediction does not overlap the released CODATA mass-energy interval or the displaced control is accepted.

Measurement receipt: `sha256:307ace763cd3554465a082488d5da841dd11bdbc0dfcc3e6db8cbe21f5aec22a`. Isolation certificate: `sha256:9aba9d80294dbb85c37dfc8437971a55715d6b40286893dc07c71cf2cb0edb05`. Custody certificate: `sha256:7fe3f23e968978839c6ee36ffadeca5c3ad289ff2e5a3a45dcbadbc0cb46702a`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- matter-mass-energy-1 from NIST-CODATA-2022-ALL-CONSTANTS
- matter-mass-energy-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:9c1c59cec96e50a061923d3857402a1abdc778d46b00f71a16daf603c49d1440`; engine receipt `sha256:7240300a36d14ae2c841f720ea162b2b67b310dcc912d0d3c1168333603fe955` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-MASS-ENERGY-001-7240300a36d14ae2.json`; empirical-validation hash `sha256:d14728b322540b0444df4194ccb3efa29e6a6492a97ba1a2d9ae94efce516239`; measurement receipt `sha256:307ace763cd3554465a082488d5da841dd11bdbc0dfcc3e6db8cbe21f5aec22a`.

144. Particle-antiparticle held orientation

Claim identity: `SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Particle and antiparticle are opposite held orientations of the same constituent word under conserved-label reversal, with identical positive inertial support.

> *Antiparticles are conserved-label orientation reversals of particle Fold words.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`

- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-MASS-ENERGY-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__conserved-label-orientation-reversal__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	conserved-label-orientation-reversal	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a3525d97763d85db57299cee599b037a1d44a7921258999171e8d0dde3fe9bfc`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:991779226ade641f899769964c870af0f0c4de28a230876d1cd633ba35b19e5f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:4eaff1ede6a4c124348e1a40c781d70d96d04dd6ba35566819589270cf3ca56d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bf21d14f456d343ce6a02b72bcccb0b561869c44afa8d3dfd6b77c4f0e0e7956`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1cd2a34c6959a22862cd410adb85dd16cf0885b2d555bcfa80ed0b66811afb0f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:66fbd5e270b64c302f3c7c195598eb54e596e438cf610d36a1a5fcde0b818096`. Independent certificate: `sha256:746cfe41e7346ca3f8fe778d89bd096037143f9bef8199d0fd4871711ef88768`. Engine

external-validation hash:
`sha256:c667dad26cf98f4235c76271d6f64c103ab7d6ceba3bec4757dc4ae69f6c099f`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-particle-antiparticle-1: predicted particle-antiparticle-orientation; observed particle-antiparticle-orientation; exact match True
- matter-particle-antiparticle-2: predicted particle-antiparticle-orientation; observed particle-antiparticle-orientation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:801a578e45d2c3a1df3a4fad9c2854842f9d13747ef0f367e1e6f3d53a76c0e9`. Isolation certificate:
`sha256:00bf56bb59987913d615c663f8b3c33b6be03f5fb1af21f463b69495d2b58954`. Custody certificate:
`sha256:9de2ef485d33f87e252c2160cb785976237434c2ba08dccab42901cfa439db87`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-particle-antiparticle-1 from PDG-2025-SUMMARY-TABLES
- matter-particle-antiparticle-2 from PDG-2025-SUMMARY-TABLES

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:248fc854ca961a1044b96d08a69c7c1793f60cfa861444a411327f34bcb7b743`; engine receipt
`sha256:d463c501f9fb01a893d13ccafe10ae7ca8f21c8eab50d501211bd5df99075fd5` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001-d463c501f9fb01a8.json`;
empirical-validation hash
`sha256:ed95272155bb2c39ad08c5e80312029c69ca61858b72d6ef17fbca38f10f4b31`; measurement receipt `sha256:801a578e45d2c3a1df3a4fad9c2854842f9d13747ef0f367e1e6f3d53a76c0e9`.

145. Fermion and boson composition classes

Claim identity: `SFT-PHYS-MATTER-FERMION-BOSON-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Constituents divide into alternating and preserving exchange-composition classes according to their finite cyclic spin action; the former exclude duplicate one-cell words and the latter admit them.

> *Fermion and boson are the alternating and preserving exchange classes of physical Fold composition.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__spin-exchange-composition-classes__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	spin-exchange-composition-classes	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:e6cc72048d6ac7203c9fcfd5e58ecce1aaad456edd7d965a6f3fcf4f5be81e26`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:156fcb775b2aff5eca6d29287f58eeab4ab91b456fb3d64093f8169f35b9f222`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:39c0be052afd07273ef0ad9a013e145adfab5e66c08807f1c72adc015286643f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6213645d9d40a9519d4cc65ec8a058d98c3f73b8e70cabf0db75f458b8c8b1b1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9f77efc02091736508973db62a1945e1adc476f5486ec713a061d46cb177c231`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:347ed8537673fef317ddf0f5ab470dfb67c15efb12d37358fc4db2d8fc5e4773`. Independent certificate: `sha256:9de3b2995bd56763940473afbf734a3bf257024a3d7614121e2bbdfc890fa227`. Engine external-validation hash: `sha256:64b0a9f2362f24bee5c075b407a2a73670b16bba189f9302934732fc3d096ed6`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-fermion-boson-1: predicted fermion-boson-exchange-classes; observed fermion-boson-exchange-classes; exact match True
- matter-fermion-boson-2: predicted fermion-boson-exchange-classes; observed fermion-boson-exchange-classes; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:756164846ce9a4957705e8b6a6cf50295d0146c94ba1a0a9b6ac43370b2995ed`. Isolation certificate: `sha256:0de7ba5beaf1d2ff482b22227bde054329bf6a2f9c8dccf8ad33b35b644ac478`. Custody certificate: `sha256:7218c94b142d062094ed2ed6987e4a3b587bcde81640d3928fd1224c257536f5`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-fermion-boson-1 from PDG-2025-SUMMARY-TABLES
- matter-fermion-boson-2 from PDG-2025-SUMMARY-TABLES

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:1d9f866240eef976637ea9b9880ba13460d5b63139a2ba8234368eccb9857396`; engine receipt `sha256:e9bcfbe74e5afc64cfa2ff029be021e8f916c3237260b004975ae9136ebcbad6` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-FERMION-BOSON-001-e9bcfbe74e5afc64.json`; empirical-validation hash `sha256:6c0a59dae649925d608067f0cd9a6615d31b814aec3260e78aaa26833bd3b816`; measurement receipt `sha256:756164846ce9a4957705e8b6a6cf50295d0146c94ba1a0a9b6ac43370b2995ed`.

146. Conserved physical labels

Claim identity: `SFT-PHYS-MATTER-CONSERVED-LABELS-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A material label is conserved when every complete generated transition pairs its disappearance on an input word with its appearance on an output word or named boundary carrier.

> *Conserved particle labels are exact held-label flow invariants of the complete transition grammar.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-MATTER-FERMION-BOSON-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__transitionwise-label-flow-closure__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`transitionwise-label-flow-closure`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:e24c69681bb44a92ae042f7c35fbbb8088458733736069b53228ba818cd8730b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:baf66a25d81ccf81aaa130e0732edf6d70758e68cac88ca87f1b06e3beaa1602`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6f149ce11d6661b1bd2c56ef88117bd66516923c70eba8e0377be09be835918d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

`sha256:b3c1dddcc5408913462dfdbe70490ee0d8e0144e9f670c69d992958154620a30`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6f479d740b225e7286d71a8a4d28f81e285bb80433f72f011ad1cb78006a0e72`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:7ebe6010d8a85c0fb10f770a73d7ca4fbd8db55ac00f6cb454d8fca33966cce2`. Independent certificate: `sha256:09fbb8d4b6458db759a9b634a65db73ce94ab658a72b96bdddbd9923c5621465`. Engine external-validation hash: `sha256:3f5500fc256d02587bd6fa65d7599a18dcc0cb26b14d50593af31f0925b223f7`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-conserved-labels-1: predicted particle-label-conservation; observed particle-label-conservation; exact match True
- matter-conserved-labels-2: predicted particle-label-conservation; observed particle-label-conservation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:76fbf94deb2df93168f346c3112e7da44aed5a4966d7505bdfeb31ab440ac82d`. Isolation certificate: `sha256:c86981796f1a81000372ab875b19d4ab1dca8d14b76d655337696aac512668cf`. Custody certificate: `sha256:e3ee4caccfd1fe94c4fe2e6476a4322d5c5db84b99f1d82f384040fdeef6d434`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-conserved-labels-1 from PDG-2025-SUMMARY-TABLES
- matter-conserved-labels-2 from PDG-2025-SUMMARY-TABLES

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:018f4956443cb59f90307b34d975ef01016fb783f9ccd4fd6f147c43780a3a04`; engine receipt `sha256:ef9b0dfe3f7d73d541bf398f377a956c483c18fe58492231027b919a02ba694b` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-CONSERVED-LABELS-001-ef9b0dfe3f7d73d5.json`; empirical-validation hash `sha256:70650e45292af63133cb6b156d1b723eda52979ad58b2eb1a907956684e8c58e`; measurement receipt `sha256:76fbf94deb2df93168f346c3112e7da44aed5a4966d7505bdfeb31ab440ac82d`.

147. Observed elementary-particle spectrum

Claim identity: `SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Elementary particle kinds are the non-equivalent minimal recurrent constituent words under complete observable transition traces at the registered measurement boundary.

> *The observed elementary-particle spectrum is the finite census of minimal non-equivalent recurrent Fold trace classes at the PDG boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__minimal-recurrent-trace-equivalence-classes__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`minimal-recurrent-trace-equivalence-classes`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:bf4b3f17bad8752400ae0287257f75eaca61b22899c569a9fc466fa051d26bfd`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1111b793deef38ccd05162c2a85883eef2417a3e7762c5c86e6d8f3d38c03cf4`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ddcc358eb534bdd1f240bc3694532fb8db0b67e6281b95bc4c3c6eaeb0c6e5db`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:512ad96f068b575ad0544dd3c8797dd7f8893144162266f9d2f3d8abb440f5c1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c110db659ea14ee96663e2ed3e8c5dbcdcf57d20d7a4d67e624d15c7cd6be233`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5b513a50d02c9708abf487a28feebbe002b36f0f797c14df1948fb0be98f34af`. Independent certificate: `sha256:f5851afd8e7790d039c120b8a89d75a1b5f34eb15ec0422ad89d16aa202a775d`. Engine external-validation hash: `sha256:cab4f97c097a86b5a072a62912971abf3b538420d64a18996f6a6afd0f86bac3`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-particle-spectrum-1: predicted observed-particle-trace-spectrum; observed observed-particle-trace-spectrum; exact match True
- matter-particle-spectrum-2: predicted observed-particle-trace-spectrum; observed observed-particle-trace-spectrum; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:b4184692b4497ff867bba3e5f84aeabcfe50138325517385a5a3c2ba5b956b19`. Isolation certificate: `sha256:4667587a734c6386581983a223b2f0499c8c8ae5c73423efde7bfbcabeabafe3`. Custody certificate: `sha256:9adbff6a6c4bdd4b74b502832e32ed91b65131f25ef6ba16f5b00fc63daf2bf8`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-particle-spectrum-1 from PDG-2025-SUMMARY-TABLES
- matter-particle-spectrum-2 from PDG-2025-SUMMARY-TABLES

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:6393fd5d7d208d758a453916af2c4e9e586ee0f79ddccef5dc57fde7906827bc`; engine receipt `sha256:d659b40f4f63f0d453c27364167ab6ea624db37a047ced6792c389e6fc207e14` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001-d659b40f4f63f0d4.json`; empirical-validation hash `sha256:c11fb94ea769e7100381321a47fbedfe6349fd05130bdd3e1fe5b92d3e2d76d1`; measurement receipt `sha256:b4184692b4497ff867bba3e5f84aeabcfe50138325517385a5a3c2ba5b956b19`.

148. Composite hadron organisation

Claim identity: `SFT-PHYS-MATTER-COMPOSITE-HADRONS-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A hadron is a closed joint constituent word whose internal interaction labels are not separately observable outside its boundary while total conserved labels and inertial support remain observable.

> *Hadrons are boundary-closed composite Fold words with confined internal labels and observable total carriers.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__confined-joint-word-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`confined-joint-word-composition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:bc16520b5dc092d2b1f0ca1bf454c094425232d213a3d3b97e2f339edd2225f4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:794134875ca65cc8d22045a544dc4a94357c2b1cc7965734eaedfae649186334`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8fc2f5f15a7a26710f8f241bc15eaba1833240845edaf6b7a334130a1450ad09`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d5ce26b1e74bebc5ff35a429fd4f17b9b392d39ed94a7a0bfc31b39b17b64f85`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:fc3f7576bde48ef11fbe9449ae4006d0b914605fa8f399aa5036a7ab3829ed40`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0bfc340f875b4d4ee94d7602b2670fe3e8c1a1b8dfd0814b98bdb78bbc2b9876`. Independent certificate: `sha256:22d961672ae7e26b6566dada76865b9771a018c67e77c2c2299f96c1a96be5fb`. Engine external-validation hash: `sha256:1b157b7f0f3bf6968cdfba9ff86fcf176e21a5ff919669615f65f0867147eba5`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-composite-hadrons-1: predicted composite-hadron-closure; observed composite-hadron-closure; exact match True
- matter-composite-hadrons-2: predicted composite-hadron-closure; observed composite-hadron-closure; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:d34d503195d51d94f39b22887fc7aa8996c7cec041de8f54b16a85f76ce744ed`. Isolation certificate: `sha256:a5becb8cac01d1c22ac7ea7e0bdfee9c566a8a27cd99f679c929d039d7b98bb7`. Custody certificate: `sha256:e79538e4371d3a38101896b24df317ae954186432cf04fc13c87b5bf094081ef`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-composite-hadrons-1 from PDG-2025-SUMMARY-TABLES
- matter-composite-hadrons-2 from PDG-2025-SUMMARY-TABLES

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:60c31f642f8992dddba4f867d25af3458344c1479211e28d37f0e870cd504465`; engine receipt `sha256:8fe71140f14938a52b5dda186b660635863e5d40109194b299c584bee09ba9a2` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-COMPOSITE-HADRONS-001-8fe71140f14938a5.json`; empirical-validation hash `sha256:15f4968e8ebfee8f0660117a60b5158d036097842924ffe48311aaf3127bf35d`; measurement receipt `sha256:d34d503195d51d94f39b22887fc7aa8996c7cec041de8f54b16a85f76ce744ed`.

149. Scattering and cross-section measure

Claim identity: `SFT-PHYS-MATTER-SCATTERING-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Scattering is the complete map from prepared incoming joint words to distinguishable outgoing channel words; cross section is the exact positive successful-channel count relative to registered incident support and geometry.

> *Scattering is a complete transition-channel map and cross section is its exact source-normalized observed support ratio.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-MATTER-COMPOSITE-HADRONS-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__incoming-to-outgoing-channel-census__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`incoming-to-outgoing-channel-census`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:8cdd721896a0feb86506d06a2c5199056300d901aa46c0ebade1dc0b71be05b3`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5c420945a230b02e98d4dfa2a708774bb8984f0abebc9d5c98f193adaf32def3`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:999d57cb70e017713c5af6e41f6f7e0459044c050f9ddd56eddaba7cdb1220b6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:096cbfc59e330a05b88b0554bfa19788f10aa302e7d08849e159857d7ac7e063`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:61ca7df4431fdfa7b0090df0baab5bf4c030853e341a00f3fcb1d36be9f00886`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:80d091c6fc94b6f1aced9db1f5b7b7d92be9b0ffb9cea55c5347a8d0dd789c84`. Independent certificate: `sha256:3f803d9f31e1300c8f38daf92f03201cfbb1a1ef21d0f0565a17db3cc385108d`. Engine external-validation hash: `sha256:8a90d565390d9bc448ac9fa4933389eaecff4c82beb1d7638da50d75136d762a`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- CERN-OPEN-DATA-DATASET-API-2026-07-23

Observed comparison records:

- matter-scattering-1: predicted scattering-channel-measure; observed scattering-channel-measure; exact match True
- matter-scattering-2: predicted scattering-channel-measure; observed scattering-channel-measure; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:6e92c22e0b3a65d5f3e252adafb3eff612736117504c99d5f0f83c0d9d71524b`. Isolation certificate: `sha256:fdfa82c09daffa5461a85b5059dc73f73970360ff69beb30ea888367d3e0a8ea`. Custody certificate: `sha256:a1c43b35515e287ba827554f6942529eb7221a458a8d7a41951ec7d5e9c4741f`.

Registered source descriptions:

- European Organization for Nuclear Research: https://opendata.cern.ch/api/records/?type=Dataset&size=1 (sha256:4e232ef39985c02005828f97c43d903381411e7c0a762af768060343d17bd57d)

Registered target identities:

- matter-scattering-1 from CERN-OPEN-DATA-DATASET-API-2026-07-23
- matter-scattering-2 from CERN-OPEN-DATA-DATASET-API-2026-07-23

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude

- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:4ab341534394ab24ee20ce33950a9cc586ea76ca8ab487468ecfef267e9c5e47`; engine receipt `sha256:b2ec3e3f4bf9c77672d9055cd1656562c9972399483eeb9fb81e20c97ebc0393` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-SCATTERING-001-b2ec3e3f4bf9c776.json`; empirical-validation hash `sha256:4a5101c745bbfb4a2ba69b2dbcc52eb70d40ea481d614bfcc79eb2d3699452e2`; measurement receipt `sha256:6e92c22e0b3a65d5f3e252adafb3eff612736117504c99d5f0f83c0d9d71524b`.

150. Particle transition and decay

Claim identity: `SFT-PHYS-MATTER-DECAY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Decay is a recurrent constituent word transitioning into a lower-recurrence composite of output words while every conserved held label and total energy-momentum carrier is paired.

> *Particle decay is exact conserved transition from one recurrent word to a generated output composition.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-MATTER-SCATTERING-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-constituent-transition-channel__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`closed-constituent-transition-channel`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3f13fba90f46522d47d4b49fb955cac22826629d17c6d7e11941da8215ee2383`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:794db5bad533dc878ce7f5091dd8534c2f9b9037ae26c75f69e0107f804a4582`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c7a9c9cbc99ddf70f9a58b2de21043f733beb858f4329a69b6178a07b623d150`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a924a3d1c129b90a5c746a6d723c15627d3abe0de4c5e58cbb5c0e489706ef33`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d6c967315e23cdb063fdc12735b33154fa9cf0db3bcc1c548c6e25c3b7f57343`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:809d2705df4cf4ab7132c2df98d03b54a449771aa0de6420c9061b0900e27e3a`. Independent certificate: `sha256:7cc8dd2f74fdd1f9b4bce38797fef768190f35f07dc57353e11c917977d4847d`. Engine external-validation hash: `sha256:ec7c7b407032db5a69050cb4f8c7be16a315dc0174d6509d3ad6bd91312c1ee3`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-decay-1: predicted particle-decay-transition; observed particle-decay-transition; exact match True
- matter-decay-2: predicted particle-decay-transition; observed particle-decay-transition; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:28f52eea4689438a90d6134087466e90aa2363f3053d62892dbc626337cb16d3`. Isolation certificate: `sha256:b4adc12c062e27711562f3ba1417427e2ae6cd24a13b87ae62c0fda86f26fc12`. Custody certificate: `sha256:e4dcde72539852921743bcf41c1555f51d31f63e93e0b0223849bd3f703a3742`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-decay-1 from PDG-2025-SUMMARY-TABLES
- matter-decay-2 from PDG-2025-SUMMARY-TABLES

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set

- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:dd0f22b6e99a4a04e1ef41d4493d4c9d7d16f1c07ecc75f05c405c07c1808171`; engine receipt `sha256:b9a36548925456b6db03d9031c9b926b8617d42c0c4c2a536bd39cf814a17627` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-DECAY-001-b9a36548925456b6.json`; empirical-validation hash `sha256:06b75856419931680e075fcad3ec2f4ebd02e13158939f6e6cfc9ff020cecb34`; measurement receipt `sha256:28f52eea4689438a90d6134087466e90aa2363f3053d62892dbc626337cb16d3`.

151. State mixing and oscillation

Claim identity: `SFT-PHYS-MATTER-MIXING-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Mixing occurs when preparation and interaction observation partitions select different refinements of the same finite recurrent support; cyclic phase action then changes their exact overlap with recurrence count.

> *Particle mixing is cyclic recurrence of one finite support viewed through nonaligned preparation and interaction partitions.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-MATTER-DECAY-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__nonaligned-partitions-with-cyclic-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`nonaligned-partitions-with-cyclic-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2508bec3eba83f3cbac7a3625e8d934c68ec4f1dac674cbcd240ccd9c9429766`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8aad8060dbde458070e15cd15fa55313d0a0fb1efb7f79e6dab95208c93c6834`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9af76d3b28e78224c81f28b597e1a3b05c573059bf8de8f128315deb08295b87`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:763ea04baba4dffedd25ed238304c89f82e5f9d97a3ebf05d1e6842ed4f40dd8`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:65352a8468399e495fccb9ef4943f1e3c475b298ca8954d56f4a9c9d1567f479`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e2076b010706a7bcf0e70455f5f7701bc2b71fb07a36cabab6a08dab7e7d88f8`. Independent certificate: `sha256:c44e862daedd1331de9f2b39221975753d321a594023e4c0872c2e30a9205e8b`. Engine external-validation hash: `sha256:a9ce3a8e5e4cb7c5f02e161e9087a67514b6c225012a3c3a2cce7d3eba4dd778`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-mixing-1: predicted particle-state-mixing; observed particle-state-mixing; exact match True
- matter-mixing-2: predicted particle-state-mixing; observed particle-state-mixing; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:39b20b797f5500ebbaca01de3e9f0a922c4f2fa593f89a9d0aaaa91fa63ad45b`. Isolation certificate: `sha256:fcdf7dc5cec6f7d759a31833be59373180da9e32678e2157814a8e1b74a03460`. Custody certificate: `sha256:80dfb7c2e17b306dc1aeae20fcbde185b9564ce61c0a824708a59fc4f8bb3519`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-mixing-1 from PDG-2025-SUMMARY-TABLES

- matter-mixing-2 from PDG-2025-SUMMARY-TABLES

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:eb8b1fbda5512ef74fb22c4eac0e59fa7438eaec5af51dab6c9ace24237784f1`; engine receipt `sha256:242e8c60ebbda112b1eebde86299605cabae3c8e40e92d0630ca3714b9d954c3` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-MIXING-001-242e8c60ebbda112.json`; empirical-validation hash `sha256:ad3c80d611a90764e59f865c983365da34a6e1e15fc4e943ffd6b05c31db3132`; measurement receipt `sha256:39b20b797f5500ebbaca01de3e9f0a922c4f2fa593f89a9d0aaaa91fa63ad45b`.

152. Electroweak Fold-fibre mixing law

Claim identity: `SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The binary Fold balance carrier is the half-One. Complete depth-two preimage enumeration uniquely returns one-quarter and three-quarters; both fold to the half-One and together partition the One, forcing the bare squared channel split one-quarter and three-quarters.

> *The bare electroweak squared channel split is sin-squared = 1/4 and cos-squared = 3/4 over unified half-One support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of Fold carrier, dependencies, complete preimage relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All exact depth-two Fold parts tested as preimages of the independently generated half-One image. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-dependency-trace__complete-half-One-preimage-fibre__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`borrowed-physical-label`	A borrowed label has no Fold trace.
`dependency`	`admitted-dependency-trace`	`asserted-input-values`	Asserted values are parameters.
`relation`	`complete-half-One-preimage-fibre`	`selected-angle-or-continuous-phase`	The alternative loses a required distinction or adds an unforced rule.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected neighbourhood cannot prove uniqueness.

Axis	Forced form	Eliminated form	Elimination reason
`minimality`	`all-predecessor-forms-rejected`	`survivor-without-predecessor-controls`	A survivor alone does not exclude a shorter form.
`measurement`	`derivation-sealed-before-comparison`	`target-visible-before-seal`	Target access reverses empirical direction.
`record`	`complete-trace-controls-and-census`	`answer-only-record`	An answer alone cannot reproduce the derivation.
`extension`	`no-extra-rule`	`free-extra-rule`	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:c5ec8856eab3b04f44906f0218eab9d35a9703bea4a849a669249f0888072907`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:a9147e40197c34a7e912d27f0c0586c4843d24382c80f5ea5d2bb4a0346d38fb`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:9a58f1c2fb85881b5b8088644bf9f0d15c13e7b2dd3268d4923c5dfdd5adf345`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3ce203371d382384d04ff1f196998476e98f70f9c124c314690cf7a7c90b29ee`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:7d7db1c9ee3e4272ddd76896fc6914357e518fbe5820395a6d0875ce3b639785`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b`. Independent certificate: `sha256:d087df08ebbe7c5b0a27827e56677841cd5a130cb8b3fe1a41814468ffca14c0`. Engine external-validation hash: `sha256:6aa4b5dddadccdfe0bcc0f23aa8f91a599c1d3b68028193db720d03811da50cb`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:43210de9ea1472d62b32f5dc7639fb5c7bc7dced272aa3925d66563505d057f8`; engine receipt `sha256:c15ae82a6cb030fc2b13b36ab6099bdbaf8faffb0092a7537e44aaa0d160db0c` at `receipts/eng` ine/model_admitted/SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002-c15ae82a6cb030fc.json; empirical-validation hash None; measurement receipt None.

153. Exact Fold PMNS squared-support triple

Claim identity: SFT-PHYS-NEUTRINO-PMNS-ANGLES-002

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The binary half-One separation, generator-three unit separation and complete binary support at generator depth force atmospheric, solar and reactor squared-support parts 1/2, 1/3 and (1/2 times 1/3)/2^3 = 1/48.

> *The exact Fold PMNS squared-support triple is atmospheric 1/2, solar 1/3 and reactor 1/48.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-QUANTUM-PHYSICAL-STATE-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of neutrino support carrier, dependencies, binary/generator/tower composition, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All exact products of the admitted binary and generator-three separation carriers over complete binary support at generator depth. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-dependency-trace__binary-generator-separations-over-depth-three-support__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	binary-generator-separations-over-depth-three-support	selected-mixing-triple	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:9cc8b42876a4a25a440f99c8a3a4070e722f065e3db4445c0f07c976c3af550b. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt

sha256:d340934bee8714c7a750e6e93d98dc73a496991650cd02aa0d6c4877d1aff5c3.

- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:1177452085ab8e621fe3c07979fd647400be7695804ef0e945889c5af6257c38`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b47a2bfe918a8458685fc8ed0439bb5f20ce2279ba54416ddbea536368569cfb`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:be4fc2aee75404b3370cc9878eb7e0da27e33d584bc0b1b7f6900f40858a2229`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b`. Independent certificate: `sha256:27da38f682958d47ff250304ddbf78cc55909cc68659ed8274e58b01d7b1401a`. Engine external-validation hash: `sha256:8563a746be3cb01682bdfd656d3d56e5b952476f88e420f5417cbdf28685b27e`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:c9b825a0bb567ea8b3160782c0b05d414b543a41b685136c5ef9dc2cd0f23f05`; engine receipt `sha256:f15bedd63f3cddc56f4ca7397633b6081ad47086cf77496cfd834bb82f5a5287` at `receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-PMNS-ANGLES-002-f15bedd63f3cddc5.json`; empirical-validation hash `None`; measurement receipt `None`.

154. Exact squared W-to-Z Fold relation

Claim identity: `SFT-PHYS-ELECTROWEAK-WZ-RATIO-002`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The massive-to-complete weak-channel squared relation is the upper member of the already forced half-One preimage fibre. It is therefore exactly three-quarters. The irrational unsquared root is outside the SFT proof domain and is never formed.

> *The exact squared W-to-Z Fold relation is (M_W/M_Z)^2 = 3/4; no irrational unsquared ratio is formed.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`

- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of mass-ratio carrier, admitted mixing dependency, upper-channel squared relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All type-correct squared weak-channel ratios assembled from the complete admitted electroweak Fold fibre. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-dependency-trace__upper-channel-squared-mass-relation__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	upper-channel-squared-mass-relation	formed-root-or-untyped-channel-choice	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:653437257d12094f6b927b80227ef4e366a6ad704664391321d66004cb9b52eb`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:8d6af386d5b72c4c47d3dd7a1848b4a0f3584f60af3b4ba7687e661525b1b8d3`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:e9e31cfdeb7d97be8f51a2b49f90bf2e6a70416d3a0bb5cb2faf88a8963c40a9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:de0a2b689c020172a062678a5cc32c5150549f0c4cbace73a3a04fd533e488c3`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:bbcffeb66ed945c758a4f68eb1995ebefcc6a0f312fe32c5312d2ef8f86c48bc`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b`. Independent certificate: `sha256:0ebf286ad43f2ffab15f028c0d28b1bfdf2cb0a463769bee01d12320fee204de`. Engine external-validation hash: `sha256:7c72af9f1edd25f8f3ab11a0d85f745794f54d9028d0af63d9a4e9aa80fe0dad`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S017 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule
- no square root of three and no unsquared irrational proof value

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:4469cf54ba2db54b9670721e12c705b796c564846d172ffed40c2bf5d7dc0abb; engine receipt sha256:a69443396f1fdf1f9beb11e65d2cc318b1c01dc5864e52d52a22eaebee1a4e89 at receipts/engine/model_admitted/SFT-PHYS-ELECTROWEAK-WZ-RATIO-002-a69443396f1fdf1f.json; empirical-validation hash None; measurement receipt None.

155. Self-sourced colour running direction

Claim identity: SFT-PHYS-STRONG-RUNNING-DIRECTION-002

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A neutral carrier retains the One source count at every positive level. A colour-carrying mediator appends both Fold labels at every successor level, forcing exact source-count prefix 1,3,5,... with successor difference two. The claim fixes the structural direction and binary slope, not a measured renormalized beta coefficient.

> *The colour self-source count is 1+2k after k successors (prefix 1,3,5,7,...) and the neutral-carrier count remains the One; the exact structural successor slope is two.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-FIELD-SOURCE-RESPONSE-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

Generated grammar and closure boundary. Generate the complete eight-axis product of running carrier, dependencies, charged-versus-neutral successor relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every positive finite running level generated by retaining the bare source and appending either the complete binary colour feedback pair or no charged feedback. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-dependency-trace__binary-self-source-successor-versus-neutral-hold__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.

Axis	Forced form	Eliminated form	Elimination reason
`dependency`	`admitted-dependency-trace`	`asserted-input-values`	Asserted values are parameters.
`relation`	`binary-self-source-successor-versus-neutral-hold`	`imported-beta-function-or-fitted-slope`	The alternative loses a required distinction or adds an unforced rule.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected neighbourhood cannot prove uniqueness.
`minimality`	`all-predecessor-forms-rejected`	`survivor-without-predecessor-controls`	A survivor alone does not exclude a shorter form.
`measurement`	`derivation-sealed-before-comparison`	`target-visible-before-seal`	Target access reverses empirical direction.
`record`	`complete-trace-controls-and-census`	`answer-only-record`	An answer alone cannot reproduce the derivation.
`extension`	`no-extra-rule`	`free-extra-rule`	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:3fc1da0842f8e087db7c17007cc1bddb63e58c41666b27fbe09d4658af8e4b51`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:e4c3f3abf5b5e710909a9adefd750f1c01f4cc47f41940c7ef05792408bf9781`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:9fdc2a6e7b63ad431ae9d6257d442d5d9e295e2e4cf961a3c24e13ac5fbf56aa`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a0be9ed3ea3361185ee50b9e73e9dd1ca1dc4282cea5044a5649275a28b4e6e0`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:28fd98942d7a6e881ac50f9963c51a8ebfc5b500e12eadcc60d0b8fa34f14c6f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b`. Independent certificate: `sha256:c055cc182bf27e19cd2acd94c18a3c516c3ee5f1e4e563f096daa85289a95076`. Engine external-validation hash: `sha256:9f9008c81386f465f6ad81f8c1b05fa69b119aedd92340afcfac46348dde6d92`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity

- no unregistered force, particle, phase, scale, channel or correction rule
- no claim that the exact host count two is a measured QCD beta-function coefficient

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:b8e20f3f9f64ffa2fbc29048ea7c06fef38df706fb17093d6961a68da8d7a6ab; engine receipt sha256:c9628d8af62fc85936b91328ab2d539f4a5def412e3a61c4cf36c0b6d5a8d412 at receipts/engine/model_admitted/SFT-PHYS-STRONG-RUNNING-DIRECTION-002-c9628d8af62fc859.json; empirical-validation hash None; measurement receipt None.

156. Self-antipodal maximal Fold phase carrier

Claim identity: SFT-PHYS-NEUTRINO-CP-PHASE-002

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Complete binary Fold separation has one self-antipodal part: the half-One. It folds to the One and its complement is itself, forcing the exact maximal-separation phase position one-half before any neutrino phase measurement.

> *The exact maximal Fold phase carrier is the half-One, 1/2 of a complete turn.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-OBSERVABLE-001

Generated grammar and closure boundary. Generate the complete eight-axis product of phase carrier, dependencies, self-antipodal relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact part of complete binary Fold support tested for return to the One and equality with its held complement. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-dependency-trace__unique-self-antipodal-half-One__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	unique-self-antipodal-half-One	continuous-or-measurement-selected-phase	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:ed0153a7e2ba490f785c56a0e04ee548c41ff9c83400e46081d5d83ed4e591f6. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:6c1ddde82e7659ae5b118b6580d6dadef2d1d3c96dbf9df974ee84f1a034f6ad`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:56bd3cd53e0e6910301ef3ebc2650de127be2662a86aa64d903b64f0c7d47656`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1c2b025a43a1375fc277813882f8b7f2a175ef8199cc59ab70d082e61639d99e`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:13fc8fc10826ccdd187cc7aad171da44978f483fdefb5d59c901e9ea215fe4c3`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b`. Independent certificate: `sha256:373a3307dc34dbb7e93cca107deffba08df51b4b1801237fc0c473c6ba9d012d`. Engine external-validation hash: `sha256:38e01fd64b49253c63cabbdc293d64a62f6a27e9720baf2b368d340c2bd82f5a`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:cbd6a5fd6b8bd47abce44109d527dde714c4d0290c70d4f7c666c3360ceb6578`; engine receipt `sha256:66a61437fe9e5f7101002e716423413dbeb111125e7c9c4d540f179a8c1eb12f` at `receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-CP-PHASE-002-66a61437fe9e5f71.json`; empirical-validation hash `None`; measurement receipt `None`.

157. Bare Dirac gyromagnetic Fold value

Claim identity: `SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The binary electromagnetic balance carrier is the half-One. Its exact reciprocal comparison to the One is the positive whole count two, forcing the bare Dirac gyromagnetic value two. Radiative anomaly corrections are separate empirical Physics and are not claimed by this law.

> *The bare Dirac gyromagnetic Fold value is exactly 2; this claim excludes the measured radiative anomaly.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-MATH-EXACT-RELATIONS-002

Generated grammar and closure boundary. Generate the complete eight-axis product of gyromagnetic carrier, dependencies, reciprocal balance relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All exact positive reciprocal relations between the admitted electromagnetic half-One balance carrier and the complete One. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-dependency-trace__One-to-half-One-reciprocal__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	One-to-half-One-reciprocal	measured-anomaly-or-fitted-g-value	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9af2478ed0735a7411cae7c10aa6e8eb2cd1334d25bb633fbf36afdf2d0e6e97`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:c74885975149e72dfdd85c229f5f15e3e4eb5fcc064cdfd2f60cf02f6665a6f4`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:889413928203edd23333b8faf1d7a0f2f4090f0e33e531a0c29597ba07f19b86`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:26db1a75752722428065c5f51f5f9aa1ca599c763fd30441907777416bb2ac84`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:59ece166f57582dfae45fd8c546c2003f4121b783ca55c8248b827d4ff10921b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b`. Independent certificate: `sha256:3615d9d1afc305b42da5a1f86140e1c9742f96920315c82e00cc74b7c7a51ba4`. Engine external-validation hash: `sha256:e54dbe2f2a807d76ab57647f6bc5e7d8e5e2857e4ca9477cb044bbb636599d65`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule
- no claim that the bare exact value equals the anomaly-corrected measured electron g value

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:2cfa8f4103b3f5c5d8096bac8874a6aa3e3f953c75d4bcf9578ea6333ac3ed79`; engine receipt `sha256:2d4d2cf74bd6c8b589961f0ad65b4ebe3f614317493daa65c4024de4ef2ad66e` at `receipts/engine/model_admitted/SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002-2d4d2cf74bd6c8b5.json`; empirical-validation hash `None`; measurement receipt `None`.

158. Weak parity from the held Fold fibre

Claim identity: `SFT-PHYS-WEAK-PARITY-FIBRE-002`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Complete half-One preimage enumeration forces one lower part one-quarter and one upper part three-quarters. Both map to the same half-One image while remaining distinguished by held side. A channel restricted to exactly one held side is therefore parity asymmetric; a channel retaining both is parity paired.

> *The weak handed Fold fibre is left-held 1/4 and right-held 3/4 over common image 1/2; a one-side channel violates parity by construction.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of handed carrier, dependencies, held-side fibre relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: The complete exact half-One Fold fibre with lower and upper held-side labels and all one-side or two-side channel restrictions. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-dependency-trace__held-lower-and-upper-common-image-fibre__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`borrowed-physical-label`	A borrowed label has no Fold trace.
`dependency`	`admitted-dependency-trace`	`asserted-input-values`	Asserted values are parameters.
`relation`	`held-lower-and-upper-common-image-fibre`	`unheld-or-imported-chirality`	The alternative loses a required distinction or adds an unforced rule.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected neighbourhood cannot prove uniqueness.
`minimality`	`all-predecessor-forms-rejected`	`survivor-without-predecessor-controls`	A survivor alone does not exclude a shorter form.
`measurement`	`derivation-sealed-before-comparison`	`target-visible-before-seal`	Target access reverses empirical direction.
`record`	`complete-trace-controls-and-census`	`answer-only-record`	An answer alone cannot reproduce the derivation.
`extension`	`no-extra-rule`	`free-extra-rule`	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:6b246cba470a9223d2bab7c00bf774485a142d4d1b8d51bb22f489b29bdd5f60`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:542ff56ad805d9ccb73fd88585057f41fa166d015d115020abc575757988e7f4`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:84020e1fd0a86146b16c09234542f86cccdba61dca53fb9cac0ed7f7253baaac`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d0b9d1f721ff2993580bb827eb317c677485a274083ae3949b75c092d32c2e09`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:94af8f923be880482a518d812a7665411c8af59256168a285fedeceb1814dc24`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b`. Independent certificate: `sha256:cb069b78dc7b2f79c029832c7b232caccbd903c23f02f85f283896e14972ea26`. Engine external-validation hash: `sha256:d69283078c43fab728f08fc6f7bdd22a5d4908c30de30711abd7968c97cf7eb9`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor

- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:258fdfa55042a4bc6030320a2913118ad436dba74602e50835a0a87c3cdbeb96`; engine receipt `sha256:5e33439d3af142c055b5455921783e65a846e8d3d9bbf588409022480f263454` at `receipts/engine/model_admitted/SFT-PHYS-WEAK-PARITY-FIBRE-002-5e33439d3af142c0.json`; empirical-validation hash `None`; measurement receipt `None`.

159. Terminal exact on-shell electroweak Fold share

Claim identity: `SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The four-rung terminal alpha object generates complete binary support sixteen. Holding the three generator directions leaves running level thirteen. At that level the charged squared support is fifteen squared and the four neutral pair channels carry fourteen squared each, forcing 225/1009. The sole terminal self-return crosses the complete support successor seventeen once, adding exact alpha/17 and forcing the terminal on-shell share before measurement.

> *The terminal on-shell electroweak share is sin-squared = 1930922298157999/8642477221479757 and its exact One-complement is cos-squared = 6711554923321758/8642477221479757.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete ten-axis product of lower carrier, terminal stage, support, running relation, self-coupling transport, orientation, enumeration, minimality, measurement direction and extension. The declared exact boundary is: All type-correct exact channel shares formed from the admitted bare weak fibre, the complete terminal alpha promotion object, its complete binary support, held generator directions and one returning terminal self-coupling. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `admitted-lower-law-carrier__complete-promotion-termination__complete-binary-terminal-support__level-thirteen-charged-over-four-neutral-pairs-plus-alpha-over-seventeen__single-typed-terminal-alpha-transport__orientation-fixed-by-share-or-retention__complete-registered-product__bare-and-intermediate-forms-preserved__exact-result-sealed-before-target-release__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`admitted-lower-law-carrier`	`detached-measured-number`	A measured number has no forward Fold dependency trace.
`terminal`	`complete-promotion-termination`	`selected-intermediate-stage`	An intermediate stage chosen for agreement is a fit.
`support`	`complete-binary-terminal-support`	`free-scale-support`	A free scale is a parameter.
`relation`	`level-thirteen-charged-over-four-neutral-pairs-plus-alpha-over-seventeen`	`free-running-level-or-imported-renormalization`	The alternative loses a required carrier or adds an unforced operation.
`correction`	`single-typed-terminal-alpha-transport`	`fitted-additive-offset`	A fitted offset reads the target.

Axis	Forced form	Eliminated form	Elimination reason
sign	orientation-fixed-by-share-or-retention	target-selected-orientation	A target-selected sign is not structural.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	bare-and-intermediate-forms-preserved	successor-without-lower-controls	A terminal answer alone hides whether the lower form was necessary.
measurement	exact-result-sealed-before-target-release	target-visible-before-seal	That reverses empirical direction.
extension	no-extra-rule	free-extra-correction	An extra term is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:fd697d15e5dc304d1dd7895cfa4522eaa3fc93a1b9fd6995ba07ce08eeae201b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:7bd7c74304217234fef6ff08648317974648a26071821cbf9cbcb2e2eb295fd5`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:69655e626f057180c9028f67846d4bd9e9b47e0898272bd1b4251aba71ad4462`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:68d67b360ce6aa76d71bad675c8c5ddb071bb7a8f955c58a2230b85102f7c3d6`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:8ee95740ea9bfb696e2ff420311da10309620d1e8c88218bbb19a138c0f556fe`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:61bd427364f94266efdcb7681e8bb7fb1211cf8a628b54ac1d8dab2b2841c504`. Independent certificate: `sha256:22c924c06b1b2156027f3ae232e3fa3381b6b2054efff2e3f8a467a6cbafca16`. Engine external-validation hash: `sha256:861287114903a56a4014cf411543ed557c0568200a560ab63d833113e7b41402`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no external measured value, uncertainty or best-fit table as a derivation input
- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no fitted level, coefficient, correction sign, truncation or target-selected candidate neighbourhood

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:2e86543fe719e208cbcea24ccb6205531713d8b4fac915803978c313f77b048e`; engine receipt `sha256:263cf79e79ed1d7ba0246f84abe869084aa4db7b195dd0c07300d1dabdef72f4` at `receipts/engine/model_admitted/SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003-263cf79e79ed1d7b.json`; empirical-validation hash `None`; measurement receipt `None`.

160. Quark channel invariants and cover depths

Claim identity: `SFT-PHYS-MATTER-QUARK-INVARIANTS-003`

Shared claim-record clause `PHYS-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Complete colour, hand and electroweak channel counts independently cross-lock with exact Fold products, forcing up pair sum 1/12, down pair sum 1/8 and least binary cover depths seven and five.

> *Up pair sum is 1/12, down pair sum 1/8, up cover depth seven and down cover depth five.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for quark channel invariants and cover depths from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__channel-count-cross-locked-with-Fold-product__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`imported-parameter-table`	A table has no Fold provenance.
`dependency`	`admitted-root-trace`	`prior-answer-premise`	A prior result cannot select V3.
`relation`	`channel-count-cross-locked-with-Fold-product`	`asserted-quark-coefficients`	The alternative loses a forced count, type or retained distinction.
`enumeration`	`complete-product`	`selected-subset`	A subset cannot establish uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:8367541eba8e54604067cba4a24815100b7d8072711ed5b29587e97d75776c2a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:701e492976ce746f77d3d3a66e774e734480b45b9f5e0e584766a2cf3f0b35c1`.

- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:5aea5f9105680ae4d40b236454cf2d039c658066468b3f344b9509043ea2114a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:58a26c819a69b30efc6ddc9625bac241135323a3d711a154728da7ea2071c3ef`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:8912df316fc73ceefe8f14f5001820bf1b8b08c75e789fd2e7b88903d677baed`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`. Independent certificate: `sha256:79d399f00e09cdd5ae132fc3c8c49400d96a5149cde57432821ed9416d0ab8fb`. Engine external-validation hash: `sha256:1cafc16eba0153762c7ab9508d3c29d4c3b2df7d08e047070c87fdee5a34ab76`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:45d72cc60c27211a20d2f8cb0b327b506990007b9e7714637893e4303450c507`; engine receipt `sha256:8dad6c72f49f916be9fd5d4aba1c3837e86795580ef246403098aed2a2b34b63` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-QUARK-INVARIANTS-003-8dad6c72f49f916b.json`; empirical-validation hash `None`; measurement receipt `None`.

161. Dual exact quark mass cubics

Claim identity: `SFT-PHYS-MATTER-QUARK-CUBICS-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Exchanging the generator and Fold-fibre roles in the already generated cubic carrier forces one down-type and one up-type three-root polynomial; roots remain exact algebraic objects through their invariants and first complete rational isolating census.

> *Down invariants are (1,1/8,1/383); up invariants are (1,1/12,1/3071); each polynomial has three disjoint positive rational isolating intervals.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-QUARK-INVARIANTS-003`
- `SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001`

- SFT-MATH-ALGEBRA-001

Generated grammar and closure boundary. Generate the complete eight-axis product for dual exact quark mass cubics from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__dual-colour-binary-cubic-invariants__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	dual-colour-binary-cubic-invariants	selected-root-table-or-fitted-polynomial	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:547c264a9ae3c10878f9521af81a09dea7616e7d04461585a7e136d284ce6043. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:61410ccc1145885dfc11eca3770442ad2bef5e540aa5dbb9fb8cd32a7ee1ae69.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:b32992ae11355f4c1222ac08e26ec9285ac6c9f5f1a3cb45281bdac6feb5bd24.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:85127dfdb5b409fc2dae381443a60cbe2f809ea31e7d5355d5dbb886018d8687.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:d70b91bb4102715e1376abb8ad5828a0e39ba029cdacba5fbd54608c46667cdb.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c. Independent certificate: sha256:6172e60c7ad6c11777a9e1ee9f69cf506516e6fded501ef3aab125e24d5c5360. Engine external-validation hash: sha256:2c90259c314ae4ef5a15450aec766bcfb4c699a7665b9684a4a51502606b7728.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S017 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:af7d8959524fca444e74bf7fb69c4ea963a99a5275a7dd3f6bdddfc18a68033d; engine receipt sha256:e6fccd6c428a5ee8c9126ae25ef145d8c5780d69465d1bf878d3ff39b34969e3 at receipts/engine/model_admitted/SFT-PHYS-MATTER-QUARK-CUBICS-003-e6fccd6c428a5ee8.json; empirical-validation hash None; measurement receipt None.

162. Terminal quark self-coupling dressing

Claim identity: SFT-PHYS-MATTER-QUARK-DRESSING-003

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The sealed terminal electromagnetic ratio transports once through the exact cover-depth roles: the depth gap two lifts only the central down-type carrier, while depth seven is retained over the upper up-type pair.

> *The exact terminal dressing factors are (inverse-alpha+2)/inverse-alpha for the central down carrier and inverse-alpha/(inverse-alpha+7) for the upper up pair.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-QUARK-CUBICS-003
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001

Generated grammar and closure boundary. Generate the complete eight-axis product for terminal quark self-coupling dressing from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__central-down-lift-and-upper-up-retention__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	central-down-lift-and-upper-up-retention	measurement-selected-mass-correction	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.

Axis	Forced form	Eliminated form	Elimination reason
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:095d27329f4dc0c6e6af69bfb9e6a17083f65569b33024d30306d6edcd6b235c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:8e45d733f0daf6f32844e1b081dea2d4b75ab3ef5e8279902d26d8a35d0920cd`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:c75971596e5fc57077c4e999c0278481059070b7fa51bd9d8aa728d33cb73879`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7476de235ba285c3b9a7028eeb12744f1a2860fa297e15b8475989a3bf9293b5`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:74c6d4ac53311668b08b2c242aafeb88eaf351e2ce8352ab9f29f755632b7af2`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`. Independent certificate: `sha256:154aff97fbb624726151edecbd57b074b85836b80cc01859a1da22b51ea7e511`. Engine external-validation hash: `sha256:6156bee434308622c9f3c8246c2844886da5403bbcf401d7304e00227d87b475`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:5a1c9cd707d96b02bec6217a0dfb4c49cca00839f5f431ed0c747ce89cdea3e4`; engine receipt `sha256:4156d17fa99d73374151d8abebba074fb43f2d9852ae3c3e15564f8a49dd7f6e` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-QUARK-DRESSING-003-4156d17fa99d7337.json`; empirical-validation hash `None`; measurement receipt `None`.

163. Exact CKM fibre alignment

Claim identity: `SFT-PHYS-MATTER-CKM-FIBRE-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The tripling fibre of the holding point and the tripling fibre of the One force a complete asymmetric three-by-three overlap object with uniform diagonal, unequal adjacent bands and one unique far corner.

> *The exact fibre matrix is ((8/9,5/9,2/9),(7/9,8/9,5/9),(4/9,7/9,8/9)).*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-QUARK-INVARIANTS-003`
- `SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for exact ckm fibre alignment from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__complete-two-fibre-overlap__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`imported-parameter-table`	A table has no Fold provenance.
`dependency`	`admitted-root-trace`	`prior-answer-premise`	A prior result cannot select V3.
`relation`	`complete-two-fibre-overlap`	`nine-independent-mixing-dials`	The alternative loses a forced count, type or retained distinction.
`enumeration`	`complete-product`	`selected-subset`	A subset cannot establish uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:a2b0b741ae35367bbdc0ed310a52292efde749b5093b19407a1c1f72ee3487e4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:fb783664d4c25bc1c9750e58db0b6a44bccf89429900d588b5bfa11c9332232e`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:241bb19700236e9f6d925fcd6c75e492842a29aa65bdc70f88ff7b5df0ae3763`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9ced90ecd8ba83d54f14e70ccc9a5fb74243661882b990c4bf4031419ec70901`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:9298b0f05f61cb7339d06fd6a1c1ab0fa267a2538f5fe2cbd4d46502c14c3944`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`. Independent certificate: `sha256:e6a1e4fe62c67dcdb0ffb60c2c53f3c15fb23a653d6e661542950dade1fb1d9a`. Engine external-validation hash: `sha256:32c89b09e1387092b89d67b6a906c0eeac5aebc65c9e612b8ce0c060ecd11da9`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:8dd6650b21e8c5fcdec9f7539ab0861eab3981ab189243d3826d62f4a5afb674`; engine receipt `sha256:2cf6948eb88b328bc6efb24b394a37cb8c6d2c32044196fa39ae189488976ea3` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-CKM-FIBRE-003-2cf6948eb88b328b.json`; empirical-validation hash `None`; measurement receipt `None`.

164. Mass-root CKM physical mixing law

Claim identity: `SFT-PHYS-MATTER-CKM-PHYSICAL-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The sealed quark polynomials determine the three physical mixing relations: light-to-central down mass, positive difference of down/up ladder slopes, and their joined six-channel dilution; the unitarity apex square is identically 1/6.

> *The physical relation graph forces s12 squared, s23, s13 squared and exact apex square 1/6 without forming an irrational root or reading a CKM target.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-QUARK-DRESSING-003`
- `SFT-PHYS-MATTER-CKM-FIBRE-003`
- `SFT-PHYS-NEUTRINO-CP-PHASE-002`

Generated grammar and closure boundary. Generate the complete eight-axis product for mass-root ckm physical mixing law from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__polynomial-root-mixing-graph-with-six-channel-apex__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`imported-parameter-table`	A table has no Fold provenance.
`dependency`	`admitted-root-trace`	`prior-answer-premise`	A prior result cannot select V3.
`relation`	`polynomial-root-mixing-graph-with-six-channel-apex`	`independent-fitted-CKM-angles`	The alternative loses a forced count, type or retained distinction.
`enumeration`	`complete-product`	`selected-subset`	A subset cannot establish uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9647ee9a8a9168aae99a4ef0a91bf849fc7cdbb4984e41d88eadf0a0caa067b9`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:842d818dd1dbaf0401793167348adb23e6abea95833937abc2bf48eb3a93fe57`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:3a019d7e9e72f37435f1ff61184c85a4ee92346f81820b84cb230784c00f5fe1`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:eb3d77e0b92ea05333d15ca76558fc74f3841a4bbeb90f5feebe039a0d7f2684`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:65d542f27268e01e6bcb82974ac06fe70d55b8ff75c81786e9001fd2c563fb1d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`. Independent certificate: `sha256:d8e02283338b80713372ae755c0e7014d7b1214c701e119d83fbb5a618092de3`. Engine external-validation hash: `sha256:bcea55c9efa76d66b08ec396a8b55e8dc5b4c6e2d4953da69cd1a07e398f9aa1`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S017 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:90a16983817869e8773eadc5d7820bb5216766026cb0a4649adbb1cfd0b489fd; engine receipt sha256:beaa25910b1a8fb944c90bca43b8cf07fec997756bb3cf3b96724133eee18d4a at receipts/engine/model_admitted/SFT-PHYS-MATTER-CKM-PHYSICAL-003-beaa25910b1a8fb9.json; empirical-validation hash None; measurement receipt None.

165. Proton-to-electron hierarchy reconstruction

Claim identity: SFT-PHYS-MATTER-PROTON-ELECTRON-003

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The leading exact charged-lepton cubic and the colour-bound one-third baryon carrier force a dimensionless proton/electron ratio graph; its algebraic roots remain sealed by exact rational intervals and are compared only afterward.

> *The ratio is (1/3) times (muon mass minus electron mass) over their product, where both masses are squared roots of x^3-x^2+(1/6)x-1/485.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001
- SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001
- SFT-PHYS-MATTER-QUARK-CUBICS-003

Generated grammar and closure boundary. Generate the complete eight-axis product for proton-to-electron hierarchy reconstruction from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__one-third-baryon-over-lepton-polynomial-mass-graph__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	one-third-baryon-over-lepton-polynomial-mass-graph	measured-proton-electron-ratio-as-input	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.

Axis	Forced form	Eliminated form	Elimination reason
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:917fbf15ac520dc18357a4db061cdaa98bf1dc415617977c922ba28cf407f8f3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:850ea0d650a082e616ebcee9b5f92d0a3aae5663cad409e0d31d849411efcbb3`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:4b173236c3f303a862a783db2b06a8e11251661e35916f8e02934273bfbfd62c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:71d55a7bb4d79fb1bc40b9039b64c7c49ea33f495a91a82bb6ef5b1f462eeeee`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:ddfe5ccdacb15c5b2a6f890302d3fd4ba72453a56e27965b63f159f4675eef41`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`. Independent certificate: `sha256:e8384dd3306ead33578b769b21fcc66aece0efee93133f2913509ea805908b9e`. Engine external-validation hash: `sha256:3a41fb146747eeda946530c9234e0136982bad4435796ad327506b499ec6629d`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause `PHYS-S004` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d041effc72083dd57b57e76ec98be80792624f76010a05243e8ecd06fa3aae0d`; engine receipt sha256:456305f867a1afc75a47afa355364f48132b901c4c1f0f7a7ace9a40be64b988 at receipts/engine/model_admitted/SFT-PHYS-MATTER-PROTON-ELECTRON-003-456305f867a1afc7.json; empirical-validation hash None; measurement receipt None.

166. Exact neutrino splitting ladder

Claim identity: SFT-PHYS-NEUTRINO-SPLITTING-003

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The non-unison counts at down depth five and doubled depth ten force exact rung ratio 33; the independent complete depth-five cell and covering-prime-square predecessor translation forces solar-to-atmospheric ratio 3/100.

> *The Mersenne rung ratio is 1023/31=33 and the physical translation ratio is exactly 3/100.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-QUARK-INVARIANTS-003
- SFT-PHYS-NEUTRINO-PMNS-ANGLES-002
- SFT-MATH-ORBIT-NUMBER-THEORY-002

Generated grammar and closure boundary. Generate the complete eight-axis product for exact neutrino splitting ladder from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__Mersenne-rung-and-cover-translation-cross-lock__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	Mersenne-rung-and-cover-translation-cross-lock	measured-splitting-ratio	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:45c5d6311f12a8fa4be64f48939e8206b745daeb23c69a4ce35f769af7ac11a0. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:bffb647896f78e999ca001940f829ac69bbf25f858dc9617d81c14aa5e9bc883.

- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:7f28bd60d0a82607fc553c23fb62ef1e6155ac819f6d5d01c6b66c0adb0e6580`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8dc26c8a1ca7c3362efda86a14f0caf220a47611534bafe669342b4c26812cfa`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:6ef1d94bf5cc0d6a631a199c3dc817a203e04bd4d546d6075b4a50dd476d3d91`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`. Independent certificate: `sha256:9e429034da7aa71d771683095c72f5ae308a9e29fbdb2d2dd564019420987cdc`. Engine external-validation hash: `sha256:2abac81fa7e8033a78190bcdbac185f9728f4132246ffa779e85e5b282b5530c`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:122cbab6e170a47c4af1e4f210294c097929f323ea0ab21baf046a9d36516032`; engine receipt `sha256:037d94c3ae94a015aa5177ee829fcd77e1137f5cdb10cfb2a9476f7a4bf4f5b7` at `receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-SPLITTING-003-037d94c3ae94a015.json`; empirical-validation hash `None`; measurement receipt `None`.

167. Positive absolute-neutrino mass structure

Claim identity: `SFT-PHYS-NEUTRINO-POSITIVE-MASS-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. No-Zero requires the lightest mass-square carrier to occupy a positive cell rather than a numerical zero. The first complete depth-five cell forces 1/32 of the solar anchor, normal ordering and exact mass-square coefficients 1/32, 33/32 and 3203/96.

> *In solar-splitting units, mass squares are lightest 1/32, middle 33/32 and heavy 3203/96, forcing positive normal ordering; eV translation requires one post-seal solar anchor.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-NEUTRINO-SPLITTING-003`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`

- SFT-PHYS-NEUTRINO-PMNS-ANGLES-002

Generated grammar and closure boundary. Generate the complete eight-axis product for positive absolute-neutrino mass structure from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__first-positive-depth-five-cell-plus-forced-splittings__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`imported-parameter-table`	A table has no Fold provenance.
`dependency`	`admitted-root-trace`	`prior-answer-premise`	A prior result cannot select V3.
`relation`	`first-positive-depth-five-cell-plus-forced-splittings`	`massless-lightest-or-fitted-absolute-mass`	The alternative loses a forced count, type or retained distinction.
`enumeration`	`complete-product`	`selected-subset`	A subset cannot establish uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:fe8cf8148443fd9c8620c22d46163b89e349a08ad5211c4232de07f73d978a98`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:8510ba04a8e69e1f4652a0e63a376de644e03eef374d598228c13a6b460294be`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:0493c74e4a1ecaf84fd3b1964d6a115318efe74a7e17ffef3d229040f86cdb41`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:96922b9c6430d09b84f4d43e58901bfbc4a5ed4a58dd031e9e423703f858c41f`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:9fb3ab6932fccb343d2bd9671f11e9bfc5f26dceee9e13463328ccff2c521dfd`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`. Independent certificate: `sha256:99751f2439c5e78d2176bba578d7e9aa3cf00c9eb6fa73b164ec27295d277777`. Engine external-validation hash: `sha256:5273caa93f658cdb1c01da093ffb06d9252bf6c102b21ff0ca7b95c2ba4813e4`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal
- no claim that an SI/eV scale is produced without one declared post-seal dimensional anchor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:3dace26ddd3e2dedad0068cc4aa8d87eb183e2a14c7d4830f0f7c4d823a36604`; engine receipt `sha256:049c0ce9ac0b05591b8636141be26c976a31f39aabb5941ad6aca5ffb60dd7fd` at `receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-POSITIVE-MASS-003-049c0ce9ac0b0559.json`; empirical-validation hash `None`; measurement receipt `None`.

168. Physical PMNS and CP support

Claim identity: `SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Binary, generator-three and complete depth-three support force the squared PMNS triple and its normalised electron-flavour weights, while the unique self-antipodal Fold position forces the maximal phase carrier.

> *The squared triple is (1/2,1/3,1/48), CP phase carrier 1/2, and electron weights (47/72,47/144,1/48), which sum exactly to the One.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-NEUTRINO-POSITIVE-MASS-003`
- `SFT-PHYS-NEUTRINO-PMNS-ANGLES-002`
- `SFT-PHYS-NEUTRINO-CP-PHASE-002`

Generated grammar and closure boundary. Generate the complete eight-axis product for physical pmns and cp support from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__complete-PMNS-support-with-self-antipodal-phase__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`imported-parameter-table`	A table has no Fold provenance.
`dependency`	`admitted-root-trace`	`prior-answer-premise`	A prior result cannot select V3.
`relation`	`complete-PMNS-support-with-self-antipodal-phase`	`measured-PMNS-fit`	The alternative loses a forced count, type or retained distinction.

Axis	Forced form	Eliminated form	Elimination reason
`enumeration`	`complete-product`	`selected-subset`	A subset cannot establish uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:69117b5094f9aa163f4de46be2856a742cc23239cf79d4480b99ab5a9fbd0c92`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:22174211eab4cf8c4c4e0a5cbe2c5142b73bc60e9f24c9f9e94f611df751695a`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:ad207c162a66d6615c0f9aa00e03d44fe60d0c90115e59527b864b203ec11aa8`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:97d462bfe73083fb33887ac148411ef5b9984290fc705805d438f11c77118546`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:7bcda760e7ab87c86ff371ad2e7db7461326aeedcaa6399a00436813df78ab81`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`. Independent certificate: `sha256:4de43a0c939ef6bc4d56fa2fc3351199f3a3251355970dcb19c91651f5ceab09`. Engine external-validation hash: `sha256:9dbee323d92016a6ad0800a8b5ae424bd07e558bffa0112f71016d90883fcd06`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause `PHYS-S004` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:3de5e79626b2ffb9a4f60b807960030d601dd222fbf903ece4cb28cccaadaf25`; engine receipt `sha256:6cc0b2828e8a304fd80c274f6e9a2e722cd1bebd68cff071f4c394bf25c872bd` at `receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003-6cc0b2828e8a304f.json`; empirical-validation hash `None`; measurement receipt `None`.

169. Single-hand Majorana discriminator

Claim identity: `SFT-PHYS-NEUTRINO-MAJORANA-003`

Shared claim-record clause `PHYS-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete half-One fibre has two held hands for a Dirac pair, but the neutrino census retains one hand. The unique self-antipodal half-One is therefore the only generated massive one-hand coupling, forcing the Majorana class within the declared grammar.

> *The charged two-hand separation is 1/2; one hand cannot instantiate that pair; the half-One alone is self-antipodal, so the massive one-hand carrier is Majorana.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003`
- `SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001`
- `SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002`

Generated grammar and closure boundary. Generate the complete eight-axis product for single-hand majorana discriminator from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__single-hand-unique-self-antipodal-coupling__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`imported-parameter-table`	A table has no Fold provenance.
`dependency`	`admitted-root-trace`	`prior-answer-premise`	A prior result cannot select V3.
`relation`	`single-hand-unique-self-antipodal-coupling`	`asserted-Dirac-or-Majorana-label`	The alternative loses a forced count, type or retained distinction.
`enumeration`	`complete-product`	`selected-subset`	A subset cannot establish uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:c1b7f7d253a9e819ee91010eb08d33ef1aadd06fcd56d502b29fac54dfcb4daa`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:d3c3f79e6df43e5e3b7d36e9c42f47854bd3154b913c378663dfe7419f541299`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:9685fc6d1764e908cc4e6bf7996f595af4a419e60774341668d3ced6e23efac1`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a9fcf706f3f16f7ff18c246a0781d23242be16ef3812a0d71ff8ad3cf7aa7bc4`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:d237227957ff1354bd3173bfbb78ca96b4f85f559c40b3544188ad8145c2b4d6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`. Independent certificate: `sha256:4b5bd24e4fd47ab15c9c86fa8e760fe7117978ec97365a248c9059b7e711e23a`. Engine external-validation hash: `sha256:84790de07d96b9f6f86311bc80a14ebb6db112d9b4c37bb1262b6bd75cb30056`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:820ff5018ea73ac86f2793fd7cea4a77c11ed4f9cb13203250761f724496f355`; engine receipt `sha256:da6688fb7442a5d3f4e3aa39f745c04aebc87102af60186b6e1bcd4e7078345e` at `receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-MAJORANA-003-da6688fb7442a5d3.json`; empirical-validation hash `None`; measurement receipt `None`.

170. Neutrinoless-double-beta noncancellation prediction

Claim identity: `SFT-PHYS-NEUTRINO-ZERO-NU-BETA-BETA-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Positive Majorana masses and nonempty electron-flavour weights force a nonempty neutrinoless-double-beta amplitude. Exact rational root bounds certify that the middle contribution exceeds the other two combined, so arbitrary held phase opposition cannot close it.

> *The effective Majorana amplitude has a strict positive coefficient floor and finite ceiling relative to the square root of the post-seal solar anchor; therefore zero-nu-beta-beta must occur within this grammar.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-NEUTRINO-MAJORANA-003`
- `SFT-PHYS-NEUTRINO-POSITIVE-MASS-003`
- `SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003`

Generated grammar and closure boundary. Generate the complete eight-axis product for neutrinoless-double-beta noncancellation prediction from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__positive-Majorana-weighted-amplitude-with-noncancellation__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`imported-parameter-table`	A table has no Fold provenance.
`dependency`	`admitted-root-trace`	`prior-answer-premise`	A prior result cannot select V3.
`relation`	`positive-Majorana-weighted-amplitude-with-noncancellation`	`asserted-decay-or-zero-mass-limit`	The alternative loses a forced count, type or retained distinction.
`enumeration`	`complete-product`	`selected-subset`	A subset cannot establish uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:a4c5d66b5cdb644eeb61f969e08b3444becb787c7e611d980e6d1e0331c9503c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:43091e29d7f6004930a664a339cbb34a1409fd515c90a39dd0874d7dd589f83e`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:30bb52c6abc8abe8c4c29636b11083992c43dfdc43748017ba89930f62f43eea`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a39dfb3fb09593934949b7d2f859ba4ec217eea11831e11f93ff39e56870701e`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:302276b0bcc22dcdbbb6230b3729a98dafae1bcf77ac17f90218922a36bd56ac`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`. Independent certificate: `sha256:92eff69bcd9d1e2eb7f9fea469f76d2ccec2663776bbf19a5df9316e2ac57aae`. Engine

external-validation hash:
`sha256:0ab95cf6291c1264dc5baef83e7b41aace1e91e2426833a8a18204d38ce7edd5`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal
- no claim that neutrinoless double-beta decay has already been observed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:20d78c28b28c0d7c6a524a6b86ca039ec35f6d448a4dc7d9d3542aa70d8c15f4`; engine receipt `sha256:3b7bc18e1160fe2a3633ce7cba15b9c3001f55ed73516ef9a6a6db474292f6fb` at `receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-ZERO-NU-BETA-BETA-003-3b7bc18e1160fe2a.json`; empirical-validation hash `None`; measurement receipt `None`.

171. Baryon-to-photon Fold transport law

Claim identity: `SFT-PHYS-MATTER-BARYON-PHOTON-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The sealed quark-root mixing graph supplies the squared CP transport measure, and the unique Fold imbalance share transports exactly half of that measure into the surviving baryon residue; the dimensional-free prediction is eta = J squared times half-One.

> *The exact algebraic prediction is eta = J^2/2, with J^2 generated from the sealed quark-root CKM graph and no cosmological target in the relation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-CKM-PHYSICAL-003`
- `SFT-PHYS-NEUTRINO-CP-PHASE-002`
- `SFT-PHYS-COSMO-COMPLETE-BUDGET-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for baryon-to-photon fold transport law from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__sealed-Jarlskog-square-through-half-One-imbalance__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	sealed-Jarlskog-square-through-half-One-imbalance	measured-baryon-abundance-or-free-efficiency	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:d78a4c8e523d4eec8dc217efc67feb5db8b491739cf7ce8e107ca1b82e16ead7`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:61754f4c80f9c2d1d531657d2ff64cc81659f0bb7dad4c8a0199e552c6019deb`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:fc4b67af7d485e826654614c7398cfa4945a3f0aeef3a3ce2468ee239fe3682f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5a40398d19173de4ad04e171f084e2c2a6c7a258786ff1a22f7a2e328517badf`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:4a08608f58003ada72897fa2a1ec1e3ea7542ab307baaf2cdb9311fa451e049e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4`. Independent certificate: `sha256:56b560437ea8d0b326b2cc9c184b18800ba227bd9345c6513b647eb4e14a2675`. Engine external-validation hash: `sha256:02e2867f1b41f8ecacc85ada35978b7a13d22be159345e88ebca62fba0c7be41`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value

- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:46e32e2d1115d1e27342a153fe03037e3fc0bb0108893096576f3a65d1f7d763`; engine receipt `sha256:2455218c2d7918d015bcacceffca6308da79a73beedf33f9ecdbd5a715413ceb` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-BARYON-PHOTON-003-2455218c2d7918d0.json`; empirical-validation hash `None`; measurement receipt `None`.

172. Common quark/lepton mixing construction

Claim identity: `SFT-PHYS-MATTER-MIXING-CORRESPONDENCE-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Quark and lepton mixing are the same complete alignment construction between one generated mass fibre and the channel fibre; only the already forced generating lock differs, producing orbit-class quark alignment and self-antipodal lepton alignment.

> *The quark lock is 2/3 with its exact ninth matrix; the lepton lock is 1/2 with first row (5/6,1/2,1/6), and all three lepton mass sites return to half-One under tripling.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-CKM-FIBRE-003`
- `SFT-PHYS-NEUTRINO-PMNS-ANGLES-002`
- `SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002`

Generated grammar and closure boundary. Generate the complete eight-axis product for common quark/lepton mixing construction from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__one-alignment-law-two-forced-locks__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`imported-parameter-table`	A table has no Fold provenance.
`dependency`	`admitted-root-trace`	`prior-answer-premise`	A prior result cannot select V3.
`relation`	`one-alignment-law-two-forced-locks`	`two-independent-fitted-matrices`	The alternative loses a forced count, type or retained distinction.
`enumeration`	`complete-product`	`selected-subset`	A subset cannot establish uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9f8798415792c18a220f22546a3a66e5aa8970fe657200da378e560ada41d60a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:544ba03b8295520e583bd75510bb66fe137d4cf2adaf4aaa3907356b8c6ca15b`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:884c08243e089403b23da33978a75ed4bdf166628c02a3f5376472440b3fdc37`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e787729f2a04937fb31c7ee5c25ef545f24e2159b10fe1ce1e94816c6074e97d`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:1a51052cc2b9218db55cdd85e71a5facaee6179886ed860b66147e578c755c44`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4`. Independent certificate: `sha256:b1fe3faa319dacf95e0e4c92a4fcb9b16e261e415f2a8319b921a2f4601dc63a`. Engine external-validation hash: `sha256:397c4b8561c45d1467a62a387f3bd59f2c1d7f0ec85d459d8918b6b6439bf0af`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:78b6a1f0ad3701b6fb0e0afc8cc150d00c7627cd5f86967fbb952c458810baac`; engine receipt `sha256:8a8a498160b16bb65ebfc1071d404a5db6de08b6c9f3499765889ca17677b501` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-MIXING-CORRESPONDENCE-003-8a8a498160b16bb6.json`; empirical-validation hash `None`; measurement receipt `None`.

173. Depth-independent Fold mass-ratio family

Claim identity: `SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. At every positive combined-ladder depth d, complete binary-by-generator support forces light part 1/N, heavy complement (N-1)/N and heavy-to-light ratio N-1 for N=2 times 3^d.

> *For every positive d, heavy/light = 2*3^d-1; the first three values are 5, 17 and 53.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-FOUNDATION-COUNT-001
- SFT-MATH-DISCRETE-001

Generated grammar and closure boundary. Generate the complete eight-axis product for depth-independent fold mass-ratio family from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__complete-support-complement-ratio-family__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	complete-support-complement-ratio-family	finite-ratio-list-or-measured-masses	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:4935a33d1117e20d1f7406bc3248124286d104819de2be5ded94e0f8d34f5ebe`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:0fd23506fa091043c2ffbb00c877129a878ed7bfc9039e22232b0e150c911fe4`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:53f6dcf22d65eb551c53d783151f4fa1423644b716bd727a77b3be791b629c2f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a74885a7ccf6c29ceaef9b181da5e8b7e865ecc387392964b9f7e7633c19b99c`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:1f5516a3e74e9f6694e1a6c3fb1cae91f2b9d4621c52e914ce52e888ffb7f691`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4`. Independent certificate: `sha256:eb8a922de1e53cc992e2efaa71385e066a860bdcd47f714c07090d2a87ee5dfc`. Engine

external-validation hash:
`sha256:e03703957018b7727249aaed6bca1b756fbc240afd891ad14a02753da02774ee`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:0cd36623ca774b0159e22354a79d4ccaa6de4cf617236e3d7a788443bb772a65`; engine receipt
`sha256:2bd73d71dcee6d09c2d427da1e1ac79c1866ef75949c5571509fd37f8fd21168` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003-2bd73d71dcee6d09.json`;
empirical-validation hash `None`; measurement receipt `None`.

174. Mirror-closed lepton mass carrier

Claim identity: `SFT-PHYS-MATTER-MIRROR-MASS-CLOSURE-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The generated lepton positions and their exact One-shortfalls are the same complete multiset in opposite order, so every symmetric invariant is route-independent before any cubic-root comparison.

> *The position triple is (1/6,1/2,5/6), its shortfall triple is (5/6,1/2,1/6), and both sums equal 3/2 exactly.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-MIXING-CORRESPONDENCE-003`
- `SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003`
- `SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for mirror-closed lepton mass carrier from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__position-shortfall-multiset-identity__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`imported-parameter-table`	A table has no Fold provenance.

Axis	Forced form	Eliminated form	Elimination reason
`dependency`	`admitted-root-trace`	`prior-answer-premise`	A prior result cannot select V3.
`relation`	`position-shortfall-multiset-identity`	`independent-position-and-mass-tables`	The alternative loses a forced count, type or retained distinction.
`enumeration`	`complete-product`	`selected-subset`	A subset cannot establish uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9ec46ff5c95b2fd48a31bc690e9bf84766a3b1ffdf1341ee4dee16431b5b479a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:c1dfc1914fc49f2342ab4c3ef9c0275934af2f03335bbd7983d791f51a3aed09`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:ae8180a6ae41198e1486c567c366381774c279ac9099ec8943d4dc1978381f32`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:abf06429178c30c9d1499e9968d6db8547a9181d561da41d39885f03fa8cf4b4`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:003e0a517d302c9b660ea8a3dcc7083bc8eaeb53d3b9480c6eaf1bcb61e975b0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4`. Independent certificate: `sha256:b4ec2c038542041df9d42be1edf37a296c1686a090288661014740f84bee158c`. Engine external-validation hash: `sha256:731e0a163bfec04a79482e3ccdc3c93914404c41b83735890fc18e3ba6ea4824`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:cc112ad075219f223756ab606e34c4979a6fff9bbd4733570352cb652916ae03; engine receipt sha256:bd01069f06d421e37672d6a58657824c529f0bf33ce2286dc723a40d1d7855e4 at receipts/engine/model_admitted/SFT-PHYS-MATTER-MIRROR-MASS-CLOSURE-003-bd01069f06d421e3.json; empirical-validation hash None; measurement receipt None.

175. Mixing-row coupling return law

Claim identity: SFT-PHYS-MATTER-INTER-ENTRY-COUPLING-003

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The first row of each exact alignment matrix returns its own generating lock after one complete One is held out: the quark row returns two-thirds and the lepton row returns half-One.

> *The quark row sums to 5/3 and returns 2/3; the lepton row sums to 3/2 and returns 1/2.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-MIXING-CORRESPONDENCE-003
- SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002

Generated grammar and closure boundary. Generate the complete eight-axis product for mixing-row coupling return law from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__first-row-residue-returns-generating-lock__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	first-row-residue-returns-generating-lock	independent-row-normalizations	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:534a23bfab502320e3aec41886fcc96ea1acbb81bce1303e015beb7551aecbf4. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt

sha256:53ee82af1f684449f3abdbea5fb6b136bd55d010c973619074fd546b78f7cc25.

- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:ac116620d6b74cfb550484b4131b1a9fa0fba58e35916d6a8d81e9f4275dd119.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:83f7e568165c611deb04ab0918666b221ed83002a3ad2c8353d25272e14e97e3.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:1193336759f76dc52770c3acb8a4a80fc07c783c87fb25d19049083f4b6fdc08.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4. Independent certificate: sha256:7159d522ec42e11a1789670be8544263e7a917ab561037dff7da0076b2980c08. Engine external-validation hash: sha256:6cf97db02aa7a8085d86430429ecf1b6b628fce538a784427eb354983117a76b.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S017 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:f5d787dfeaa767ad67d5d3b4389c64742ffe4dfcf50619513936a8e372f9ef6d; engine receipt sha256:c1ef281840dac57df5ac309768874c793c7880fcbbfe5299d1b4deff3f2531ea at receipts/engine/model_admitted/SFT-PHYS-MATTER-INTER-ENTRY-COUPLING-003-c1ef281840dac57d.json; empirical-validation hash None; measurement receipt None.

176. Common generation return depth

Claim identity: SFT-PHYS-MATTER-GENERATION-DEPTH-003

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. All three generated lepton sites traverse one generator-three action to the shared half-One waypoint and one Fold action to the One, forcing equal two-operation depth despite distinct positions.

> *Sites 1/6, 1/2 and 5/6 all reach half-One after tripling and the One after one Fold; their common generator-operation depth is two.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-MIXING-CORRESPONDENCE-003

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001

Generated grammar and closure boundary. Generate the complete eight-axis product for common generation return depth from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__one-tripling-plus-one-Fold-common-return__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`imported-parameter-table`	A table has no Fold provenance.
`dependency`	`admitted-root-trace`	`prior-answer-premise`	A prior result cannot select V3.
`relation`	`one-tripling-plus-one-Fold-common-return`	`generation-specific-coupling-depths`	The alternative loses a forced count, type or retained distinction.
`enumeration`	`complete-product`	`selected-subset`	A subset cannot establish uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:c34f9858c4d40073f21cc0b619c683df8e417496b6d2a5958de06080ab2de1b5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:d070d1a38a9f793bdd976358859df7f00dfa35d239432ba46d07788fe33f8b89`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:70b95a2d2cd3e50f23d8a7154f674325af2057d6c6438c0a699983301409c5a4`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8f30e25a78d4faca49dcfce018fe0ea6ea0624d82dacefcb37083e78adf35754`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:5dadf6ad9b24c7e8e3dacea28aca0a939fb72cfc51dcb835109a18d1c2d52661`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4`. Independent certificate: `sha256:1e31a785a752e048f5a2c7d57f7b952bd5a1004a5e5761dae18c924c06eefc5a`. Engine external-validation hash: `sha256:a0e248347df405deb85786191034fda3b58477c2d13e646489df5f6b05006bf9`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:173a98a17650553e593e97fa9882cc6f5ba74b783a0ede98df20e631e81cf5d8`; engine receipt `sha256:a120ad261a4308fc3008ccaab35d38939f6152f82c31f4f00c773582e3f2d726` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-GENERATION-DEPTH-003-a120ad261a4308fc.json`; empirical-validation hash `None`; measurement receipt `None`.

177. Lightest-quark confinement lift

Claim identity: `SFT-PHYS-MATTER-CONFINEMENT-LIFT-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The generator-three sharpening acts on the complete colour-tower predecessor at the independently forced down and up depths, while the sole lightest unclosed generation receives the complete two-fibre confinement action once.

> *The sharpened down/up products are 3/1454 and 3/13118, and the lightest-carrier confinement lift is exactly the Fold fibre count two.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-QUARK-CUBICS-003`
- `SFT-PHYS-MATTER-QUARK-INVARIANTS-003`
- `SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for lightest-quark confinement lift from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__colour-channel-sharpening-plus-one-two-fibre-light-lift__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`imported-parameter-table`	A table has no Fold provenance.
`dependency`	`admitted-root-trace`	`prior-answer-premise`	A prior result cannot select V3.
`relation`	`colour-channel-sharpening-plus-one-two-fibre-light-lift`	`measurement-selected-quark-rescaling`	The alternative loses a forced count, type or retained distinction.

Axis	Forced form	Eliminated form	Elimination reason
`enumeration`	`complete-product`	`selected-subset`	A subset cannot establish uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:8239d56c52092676eb989075291d6a8299930949e94df3fb3eed399b5c1f1d1c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:1a448fe27f4aecb62ef3cb8dec4b0eb060555ce3b287baa87957c8f5684268fe`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:e1a7d6501000853476073c8b6663ed8593169d41136066d9580091ff11a6dd83`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:83a2544c740cb324dd0ad054cec80e316c0837608cadb041659e6d114a3a3912`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:00730b4fbf6ba7db829ad6ceac234db8610661ec657d2905518fe8270d76b617`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4`. Independent certificate: `sha256:5f6bd05f4e59b83fcf89d425a3c8527017529859834 7f164027a77a8b588d3a0`. Engine external-validation hash: `sha256:525e79a933241afd45233f11cbe657ff63e801318f98d9b28975b38974ba9491`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

Shared claim-record clause `PHYS-S004` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:022b73b16a6366287fb7a90cbd6c836e9eea19e3255568c74500f32e5ed79102`; engine receipt `sha256:53f2509cb96b422753fb8211e2d7fdb93b351d64fb85eb1a90a7dacc717e646a` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-CONFINEMENT-LIFT-003-53f2509cb96b4227.json`; empirical-validation hash `None`; measurement receipt `None`.

178. Terminal physical CKM transport

Claim identity: `SFT-PHYS-MATTER-CKM-TERMINAL-004`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The admitted physical CKM slope named terminal quark dressing as a dependency but its leading function transported only the two bare root slopes. Completing that declared carrier forces one terminal electromagnetic share across three colour channels, retained through the already forced depth-seven upper-quark factor and appended to the positive slope separation.

> *The terminal CKM contribution is exactly alpha times [A/(A+7)] divided by three, equivalently 1/[3(A+7)], where A is the admitted terminal inverse-alpha; it appends to the leading positive s23 slope gap, while the existing s12 and six-channel s13 relations are preserved.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-CKM-PHYSICAL-003`
- `SFT-PHYS-MATTER-QUARK-DRESSING-003`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for terminal physical ckm transport from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__leading-slope-gap-plus-colour-shared-terminal-alpha-under-up-retention__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`imported-parameter-table`	A table has no Fold provenance.
`dependency`	`admitted-root-trace`	`prior-answer-premise`	A prior result cannot select V3.
`relation`	`leading-slope-gap-plus-colour-shared-terminal-alpha-under-up-retention`	`leave-declared-terminal-dependency-unused-or-target-select-a-mixing-shift`	The alternative loses a forced count, type or retained distinction.
`enumeration`	`complete-product`	`selected-subset`	A subset cannot establish uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:0e62b057b1a7307e8c234d4dfb9107b033bc3356a3e1bc69d610c353ac77977d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:fe3eec0be13f9f525a03306c6a999497d0c9d2ff2f7d19173b14744317dd4f25`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:c4352e2e942341ffdda9795e3022e1b5890d2380334047acd6ff14740b1342e9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0417c11fe4f2247781f74107ca13f86aa39fb1d49dfb7f52dc430422fc153837`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:e511a716293d36318f800889ed8ff7ff3f38074d0680c24c953cfee5ad1a9501`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ce3435925d6dd0b23b93b50097de951cb6daee8b0dc62b00589ede5117dcc9ae`. Independent certificate: `sha256:fbd90e63b1844637cba91aa60a1bad8c85899c974a05f279d805b6140f598077`. Engine external-validation hash: `sha256:ad40d2bd3fedd0497e7ad5a1728b0cbb4d68707021cfd171426fbdabd90a1ed0`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026

Observed comparison records:

- PDG-CKM-S12-COMPLETE: predicted terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed; observed terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed; exact match True
- PDG-CKM-S23-COMPLETE: predicted terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed; observed terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed; exact match True
- PDG-CKM-S13-COMPLETE: predicted terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed; observed terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed; exact match True
- PDG-CKM-J-COMPLETE: predicted terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed; observed terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Any sealed CKM interval misses its complete registered source interval, a row is omitted, the target enters the relation, or observational provenance is hidden.

Measurement receipt: `sha256:68760b3a29a6320d698a6b7854cf00f4b59d739971c6217985cbc2bb44be72d3`. Isolation certificate:

`sha256:1edd78773190d6b1ed21a62183a75ef26c586e1765cbc29a2b2672bcd80ab32a`. Custody certificate: `sha256:8ff87b19f8434aa20865a75447abc6999ba88742e5525deab228e75b47a77b5c`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-bbang-cosmology.pdf (sha256:652b186e29c835c9cf8fdfd3393fe19c6a81917666989331c5ca060dc789ae9e)

Registered target identities:

- PDG-CKM-S12-COMPLETE from MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026
- PDG-CKM-S23-COMPLETE from MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026
- PDG-CKM-S13-COMPLETE from MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026
- PDG-CKM-J-COMPLETE from MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal
- no claim that this post-adverse completion was a blind forward discovery

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:5da1e4df4b4f9b56eec1f9dea499e480507a6af74c4731b9316b35d12ec92ada`; engine receipt `sha256:b97a60bba913a679c25137427491abe5c13dc24c641d00125059ce6cb3db90fb` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-CKM-TERMINAL-004-b97a60bba913a679.json`; empirical-validation hash `sha256:dd7d3fbbbfc083f60f3fcd094cabbe1bc146628c29a05faedd004bd639238a08`; measurement receipt `sha256:68760b3a29a6320d698a6b7854cf00f4b59d739971c6217985cbc2bb44be72d3`.

179. Terminal baryon-to-photon transport

Claim identity: `SFT-PHYS-MATTER-BARYON-PHOTON-TERMINAL-004`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The already admitted baryon-residue law consumes the complete terminal CKM graph rather than the leading graph; its unique Fold imbalance share remains half-One, so no abundance scale or efficiency is introduced.

> *The terminal baryon-to-photon prediction is eta = J_terminal^2/2, with J_terminal^2 formed from the terminal CKM squared-support graph and no cosmological value in the relation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-CKM-TERMINAL-004`
- `SFT-PHYS-MATTER-BARYON-PHOTON-003`
- `SFT-PHYS-COSMO-COMPLETE-BUDGET-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for terminal baryon-to-photon transport from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__terminal-Jarlskog-square-through-half-One-imbalance__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-carrier`	`imported-parameter-table`	A table has no Fold provenance.
`dependency`	`admitted-root-trace`	`prior-answer-premise`	A prior result cannot select V3.
`relation`	`terminal-Jarlskog-square-through-half-One-imbalance`	`retain-superseded-leading-mixing-or-import-baryon-abundance`	The alternative loses a forced count, type or retained distinction.
`enumeration`	`complete-product`	`selected-subset`	A subset cannot establish uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-shortcut`	A shortcut may hide a free choice.
`measurement`	`seal-before-comparison`	`target-visible-before-seal`	That reverses empirical direction.
`record`	`full-polynomial-trace-census-controls`	`answer-only`	An answer cannot reproduce the result.
`extension`	`no-extra-rule`	`free-extra-rule`	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9d3afbb1a889bd0528310f9b755d753bb0928a4cdb9203f2423e5be768105d3c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:4fc7e50c9e4299e64e95d3adae7f384d9e89d29279995e653d71944758b95642`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:c0f168f6e37b6e74d121a47297bbd2f979d86045cfc9f58c17b3e289b15e4939`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ceaf7ee28f7412d9d5c4c5aee94c1a103ab8559bf141aebb801bcaaf8aaf87ee`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:f27cc9c43a309ae202de93ac8ae420824339ecf0ed864a0b717dea8bac3b79a0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ce3435925d6dd0b23b93b50097de951cb6daee8b0dc62b00589ede5117dcc9ae`. Independent certificate: `sha256:a0ac7c3918ab6a5f35f5cd67b7017b1efe2ddf3c4f0cc65e8c31d6c1906d558d`. Engine external-validation hash: `sha256:0cd155c0799377a073850c428bfb1c416035594b6f020b1e7540300a8aed5b37`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026

Observed comparison records:

- PDG-BBN-ETA-COMPLETE: predicted terminal-baryon-to-photon-prediction-lies-inside-independent-PDG-BBN-and-Planck-derived-intervals__observational-derivation-disclosed; observed terminal-baryon-to-photon-prediction-lies-inside-independent-PDG-BBN-and-Planck-derived-intervals__observational-derivation-disclosed; exact match True
- PDG-PLANCK-ETA-COMPLETE: predicted terminal-baryon-to-photon-prediction-lies-inside-independent-PDG-BBN-and-Planck-derived-intervals__observational-derivation-disclosed; observed terminal-baryon-to-photon-prediction-lies-inside-independent-PDG-BBN-and-Planck-derived-intervals__observational-derivation-disclosed; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The sealed eta prediction leaves either complete registered interval, a route is omitted, or a measured abundance enters the law.

Measurement receipt: `sha256:21c5eaf6456549ce5c2f0d6bdb40ec676ec6349ae70a188d39ac76426563d1c3`. Isolation certificate: `sha256:88462b4c13e117cc75385fcf40c3d76519df74eda3824e3a42929bc6a9d91f60`. Custody certificate: `sha256:1ede510c7cf85ca24d8c654812437699e360ca5e064b117934bfea9eaf1cec0a`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-bbang-cosmology.pdf (sha256:652b186e29c835c9cf8fdfd3393fe19c6a81917666989331c5ca060dc789ae9e)

Registered target identities:

- PDG-BBN-ETA-COMPLETE from MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026
- PDG-PLANCK-ETA-COMPLETE from MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal
- no claim that this post-adverse completion was a blind forward discovery

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:7706a6e587ec3420b86f9b3b256a48cdaea60653056684fb2d8b6591e0d75612`; engine receipt `sha256:e5bd6eede834a53c25e7b33a41c4bf83a8eeb439e5a89de8fe87f8fd0997680a` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-BARYON-PHOTON-TERMINAL-004-e5bd6eede834a53c.json`; empirical-validation hash `sha256:8f80b514031b4a572840a540a369242d71719e5359dfdf48d447fdb5d22b1328`; measurement receipt `sha256:21c5eaf6456549ce5c2f0d6bdb40ec676ec6349ae70a188d39ac76426563d1c3`.

180. Terminal composite proton/electron precision law

Claim identity: `SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The admitted proton/electron algebraic graph receives the complete electromagnetic self-coupling of its three-depth colour composite. Binary-by-colour-cubed support has heavy complement 53; the complete colour bulk and its three carried boundary channels supply support 27+3=30; both charged Fold ends force alpha squared; and the sole terminal return transports the two Fold labels through down-depth five and colour volume 27 once. The retained ratio is therefore the admitted algebraic ratio times One held by 53 alpha^2/30 times (One plus 2 alpha/135).

> *Let R3 be the admitted exact algebraic proton/electron ratio and alpha the reciprocal of the admitted terminal inverse-alpha. The unique retained successor is R4 = R3 * [1 - delta] in conventional notation, represented only as positive_take(One,delta), where delta = (53/30) alpha^2 [1 + (2/135) alpha].*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-PROTON-ELECTRON-003`
- `SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003`
- `SFT-PHYS-MATTER-CONFINEMENT-LIFT-003`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete eleven-axis product of base graph, composite depth, complement, coupling order, bulk/boundary support, terminal return, transport, orientation, custody, provenance and extension forms. The declared exact boundary is: All first terminal electromagnetic dressings of the admitted proton/electron algebraic graph that use each already admitted composite-depth, colour-volume, boundary-channel, Fold-label, cover-depth and terminal-alpha carrier exactly once in its typed role. The generator produced `2048` named candidates and the decision support contains `2048` one-for-one decisions. Exactly one candidate survived: `retain-admitted-proton-graph__generator-three-composite-depth__complete-depth-three-heavy-complement__one-alpha-per-charged-Fold-end__complete-colour-bulk-plus-carried-boundary__one-terminal-alpha-successor__both-labels-through-down-depth-and-volume__hold-dressing-from-existing-ratio__measurement-inaccessible-to-executable-relation__registered-observational-prediction-protocol__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`base`	`retain-admitted-proton-graph`	`replace-or-refit-base-ratio`	Replacing the admitted algebraic graph erases its receipt.
`depth`	`generator-three-composite-depth`	`selected-composite-depth`	A selected depth can tune the correction.
`complement`	`complete-depth-three-heavy-complement`	`free-multiplicity`	A free multiplicity is a parameter.
`order`	`one-alpha-per-charged-Fold-end`	`selected-alpha-order`	A selected order can read the mismatch.
`support`	`complete-colour-bulk-plus-carried-boundary`	`partial-bulk-or-boundary-support`	Omitting either carrier loses part of the composite.
`terminal`	`one-terminal-alpha-successor`	`leading-order-only-or-extra-series`	Leading-only omits the declared terminal carrier; an open series has no closure.
`transport`	`both-labels-through-down-depth-and-volume`	`free-return-weight`	A free return weight is a fitted coefficient.
`orientation`	`hold-dressing-from-existing-ratio`	`append-unbound-mass`	Appending a new mass carrier breaks the normalised composite ledger.
`target`	`measurement-inaccessible-to-executable-relation`	`measurement-readable-relation`	That would fit the known mismatch.
`provenance`	`registered-observational-prediction-protocol`	`unregistered-target-readable-fit`	Target-readable execution cannot issue an independent prediction seal.
`extension`	`no-extra-rule`	`additional-fit-term`	A further term is not supplied by the exhausted first-terminal grammar.

Base and successor. The closure proof is sealed by derivation hash `sha256:7698a7c0c7992df10057596d93069ec68a8a1210898cb3ee083b225c45508450`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:6da8830d7839592209d46483244091a89ada278cce758eff2c7649e1befccb46`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:318343b5344c2afffea78d447779ff1015f5a3f9e225721fa927d0eccfc730e2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e44a69a0e2643fe6c150a20a38555ecfe2a7dc20ecb80d43f25f58d492107161`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:5ad270ae68f84ba1ed0883d3935585922e73076242ad3e3205aed8fe99461965`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0fa8728b5e9260f04e84a1d640d54dadffcddca7181dec03acfb9a8b1abc2d55`. Independent certificate: `sha256:9502cb05b282e7b7f7c7a40405577cf03d6363dbe77e5bd6a43e203d0eab275d`. Engine external-validation hash: `sha256:7234d679a2622ec660398d5c16201419535c40de7c51db941fda3524f20f6eb4`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- MATTER-FLAVOUR-TERMINAL-PROTON-AUTHORITATIVE-2022-2026

Observed comparison records:

- NIST-CODATA-PROTON-ELECTRON-COMPLETE: predicted terminal-target-free-proton-electron-prediction-contained-inside-complete-CODATA-standard-uncertainty__observational-prediction-protocol-passed; observed terminal-target-free-proton-electron-prediction-contained-inside-complete-CODATA-standard-uncertainty__observational-prediction-protocol-passed; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The sealed prediction enclosure leaves the complete registered CODATA interval, any uncertainty or adverse receipt is omitted, the target enters the relation, or observational provenance is hidden.

Measurement receipt: `sha256:f30e3cb373c7008f1841159bb00a206c15e92639942ad2aa83cacefcd5a46fbf`. Isolation certificate: `sha256:08fec209361ef1eeb2d05f2c6e1afd8ea1e8d7e15125fe563d2899395bd0345d`. Custody certificate: `sha256:f92cff29d38a0e839bec6372924028626399bf64d216e772b51a7e509e9482d0`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:a2fd002521902242e3db10eac896389d2190f88169b827c3575640e9a31f2790)

Registered target identities:

- NIST-CODATA-PROTON-ELECTRON-COMPLETE from MATTER-FLAVOUR-TERMINAL-PROTON-AUTHORITATIVE-2022-2026

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured proton/electron value or uncertainty in the executable relation
- no fitted coefficient, selected correction neighborhood or added scale
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no conflation of observationally informed derivation with target-readable execution

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:fb25648f29e5ae0816de8805a75e7f20eb65abebfd67e6ef3180d0537e6fc8d5`; engine receipt `sha256:eb6a7b8b32dd79a9971ce3a64e1b7b29ebfca54bbb6264fe9be3984dd60fc5bf` at `receipts/engine/model_admitted/SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004-eb6a7b8b32dd79a9.json`; empirical-validation hash `sha256:8054c74de41fa8dc06907bd4147c9c50600ebb99737ece784c3396ce32434d66`; measurement receipt `sha256:f30e3cb373c7008f1841159bb00a206c15e92639942ad2aa83cacefcd5a46fbf`.

181. Terminal Fold scattering, Rutherford angular law and Compton transfer

Claim identity: `SFT-PHYS-SCATTERING-RUTHERFORD-COMPTON-TERMINAL-006`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A scattering amplitude carrier is the exact phase-compatible overlap support of one transition leg; an observed differential channel pairs the preparation-to-transfer and transfer-to-record legs, so its weight is their exact self-composition without an imported complex magnitude. For a conserved Coulomb source the rank-two transfer boundary forces differential support proportional to the inverse square of the exact direction-transfer part, with charge-product-squared and incident-energy-inverse-squared scale. At the comparison boundary this is the Rutherford inverse fourth-power half-angle law. For closed photon-electron transfer, wavelength increases by two direction-transfer parts of the action-over-inertia-speed carrier; outgoing energy is incident energy divided by One plus twice the incident/rest-energy ratio times that transfer part. Right-angle and backscatter high-energy ceilings are respectively one and half-One electron rest-energy support. Forward nondeflection remains the structural empty transfer class.

> *Paired phase-compatible Fold overlap legs force exact Coulomb differential support proportional to the inverse square of the positive direction-transfer part, with charge-product-squared and incident-energy-inverse-squared scale. Closed photon-electron transfer forces wavelength increase by two transfer parts of the action-over-inertia-speed carrier, exact outgoing-energy loss, and right-angle/backscatter high-energy ceilings of One and half-One electron rest support; forward nondeflection is structural absence.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-SCATTERING-001`
- `SFT-PHYS-QUANTUM-PHYSICAL-STATE-001`
- `SFT-PHYS-QUANTUM-WEIGHT-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INVERSE-SQUARE-001`
- `SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-SPACETIME-LIMIT-SPEED-001`
- `SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001`
- `SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete product of every exact overlap-leg account, every cross-section measure, every Coulomb angular power, every Coulomb charge/energy scale, every photon wavelength-transfer law, every photon energy-transfer law, both target-custody states and both extension states. The declared exact boundary is: Every exact positive rational nonforward direction-transfer part not exceeding the One; the structural empty forward class; phase-compatible incoming/transfer/outgoing legs; source-normalized differential boundary support; conserved charged-source and incident-energy carriers; one photon, one initially held inertial word and closed energy-direction transfer; sealed or exposed targets; and empty or free extension. The induction certificates are independent of a finite Fold depth. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: paired-phase-compatible-overlap-legs__paired-weight-per-incident-boundary-support__inverse-transfer-part-squared__charge-product-squared-energy-inverse-squared__two-transfer-parts-times-action-over-inertia-speed__rest-over-rest-plus-two-energy-transfer-parts__sealed-before-release__empty-extension. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
amplitude	paired-phase-compatible-overlap-legs	single-unpaired-channel-count	Rejected by computed Fold predicates: amplitude_carrier.
cross_section	paired-weight-per-incident-boundary-support	unnormalized-outgoing-count	Rejected by computed Fold predicates: cross_section_measure.
coulomb_angle	inverse-transfer-part-squared	inverse-transfer-part-first-power	Rejected by computed Fold predicates: coulomb_angular_law.
coulomb_scale	charge-product-squared-energy-inverse-squared	charge-product-linear-energy-inverse	Rejected by computed Fold predicates: coulomb_scale_law.
photon_shift	two-transfer-parts-times-action-over-inertia-speed	one-transfer-part-times-action-over-inertia-speed	Rejected by computed Fold predicates: photon_shift_law.
photon_energy	rest-over-rest-plus-two-energy-transfer-parts	incident-energy-preserved	Rejected by computed Fold predicates: photon_energy_law.
target	sealed-before-release	readable-before-seal	Rejected by computed Fold predicates: target_boundary.
extension	empty-extension	free-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash sha256:dfe407aa03c635803c44c57974d102983d3873a525baa241ca3cc4c31e913180. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected Reject an angle-independent Coulomb response and elastic photon energy.; observed Exact transfer-part and conservation predicates reject the false angular law.; receipt sha256:611a3831f0a7d21d03f350bb88a985e400da613244a25f9c57546a545c5b7962.
- tampered_source: passed; expected Reject any changed claimant source identity.; observed The changed identity differs from the registered source manifest.; receipt sha256:0de6112a4fa89e23a8625b623b57f00f6f65f6e8de7f00f0265c4d1e451b09ae.
- tampered_artifact: passed; expected Reject a duplicated or incomplete candidate census.; observed The deliberate duplicate fails complete candidate identity.; receipt

`sha256:85f7f2fccaaba82e12423de0043e0cde43f60a6c59bbf25b4c3079af293bc64a`.

- `boundary`: passed; expected Reject a numerical forward null, pre-seal target access and free correction.; observed Forward transfer is structural absence; sealed custody and empty extension are required.; receipt `sha256:7f83757007f5533bab0a7d8d12ed5c9237eeba52b2c9c1626a8cbf5ebd8d3ff0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0d35b75c4957b1b77081f92b1cfcf6cfda6596302a2ae3b324a7242b1a03accb`. Independent certificate: `sha256:633aaa0ff4b45f924cfe3a7ef076b2ddac8083b464eada0999874de472b80e99`. Engine external-validation hash: `sha256:2dfad50881002854fd881204b6420b5c76d8e524fa066ab8b1c028b92b14eda5`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAEA-STI-PUB-1196-RADIATION-PHYSICS
- NIST-CODATA-2022

Observed comparison records:

- IAEA complete Rutherford record retained: inverse-square Coulomb source, charge/energy scale and inverse-fourth-power half-angle differential law
- IAEA historical Geiger-Marsden record retained: approximately one in 10^4 alpha particles above 90 degrees versus the obsolete diffuse-model order 10^-3500
- IAEA complete Compton record retained: free stationary electron boundary, closed energy-momentum transfer, delta-lambda relation and h/(m_e*c) carrier
- NIST CODATA complete seven-row scattering vector retained with every reported uncertainty
- exact h/(m_e*c) interval m: (Fraction(6310543, 2600880509804446332), Fraction(18931629, 7802641524616659668))
- NIST Compton wavelength interval m: (Fraction(121315511731, 50000000000000000000000), Fraction(121315511807, 50000000000000000000000))
- NIST right-angle high-energy ceiling interval MeV: (Fraction(51099895053, 100000000000), Fraction(10219979017, 20000000000))
- NIST backscatter high-energy ceiling interval MeV: (Fraction(51099895053, 200000000000), Fraction(10219979017, 40000000000))
- first-power, angle-independent, elastic-energy, numerical-forward, incomplete-row and changed-source controls rejected

Falsification condition: Reject if the complete IAEA record does not retain inverse-square Coulomb force, the inverse-fourth-power half-angle Rutherford relation, nonempty large-angle Geiger-Marsden support, closed Compton energy-momentum transfer or the wavelength-shift relation; if exact h/(m_e*c) does not overlap the NIST Compton-wavelength interval; if the right-angle and backscatter energy ceilings disagree with the NIST electron rest-energy vector and IAEA limits; if any registered row, uncertainty, scope condition or adverse control is omitted; or if target content enters before seal.

Measurement receipt: `sha256:d6c7057f63d6f29411d7fbddef78e3a8cd96aeeadcf8f8ef1348b0c395c17cd0`. Isolation certificate: `sha256:e90f382c158c571f9386300054242fa1cdff5aaf259ee1b5183c1fadee9e181f`. Custody certificate: `sha256:6345f82fc72048f621af014d5e3ce39930ffe9c02d03cbf94d60ccfb50e0f53a`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, certificate, answer table or survivor identity in the derivation runtime
- no Rutherford formula, Compton formula, measured angle, cross section, wavelength, energy or source locator in formal candidate decisions
- no claimant-supplied expected survivor and no submission-controlled admission flag
- no numerical-zero proof value, negative, irrational, imaginary, floating or imported complex proof magnitude
- no path integral, fitted coupling, target-selected angular power, selected correction or stochastic oracle
- no evaluation of the forward empty class by a divergent numerical expression
- no target access before prediction sealing and no free extension after closure

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:e5d2d0eef6a42f2e3af50b8fa33d8312b17c518a603b44ee9c1bf05046f9edc5`; engine receipt `sha256:2c2e6212e5bb27a2b7647e8eca1583bc75775c886f2185885a7925d83e195b91` at `receipts/engine/model_admitted/SFT-PHYS-SCATTERING-RUTHERFORD-COMPTON-TERMINAL-006-2c2e6212e5bb27a2.json`; empirical-validation hash `sha256:4bd3b98fb21c2f4758f64c4b2bba3df272ab97ea863c67477988b192679cf56e`; measurement receipt `sha256:d6c7057f63d6f29411d7fbddef78e3a8cd96aeeadcf8f8ef1348b0c395c17cd0`.

182. Terminal Fold decay widths, exact branching partitions and lifetimes

Claim identity: `SFT-PHYS-DECAY-WIDTH-BRANCHING-LIFETIME-TERMINAL-006`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Every distinguishable open decay channel has exact positive partial width equal to its paired transition weight times available generated output support per action carrier. Closed channels are structural empty forms. The total width is the complete ordered positive sum of the open partial widths; each branching part is its partial width over that total, and the mutually exclusive complete branching family sums exactly to the One. Duration is the action carrier over total width, so increasing transition support increases width and strictly shortens lifetime. The base and channel-successor laws hold for every nonempty finite generated family.

> *Every distinguishable open decay channel has exact positive partial width equal to paired transition support times available generated output support per action carrier; closed channels are structural emptiness; total width is the complete positive sum; branch parts are partial width over total and partition the One; lifetime is action over total width and strictly decreases with width. The complete PDG 2026 W-width and decay-mode vectors and the NIST reduced-action transport pass the registered post-seal comparison without subset double counting or a fitted correction.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-DECAY-001`
- `SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005`
- `SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005`
- `SFT-PHYS-QUANTUM-WEIGHT-001`
- `SFT-PHYS-MECH-DURATION-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001`
- `SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete product of every paired or unpaired transition carrier, every open/closed channel domain, every total-width aggregator, every branching normalization, every completeness partition, every width-lifetime relation, every induced ordering, both target-custody states and both extension states. The declared exact boundary is: Every nonempty finite generated family of mutually distinguishable open decay channels; exact positive rational overlap, available-output, action, partial-width and duration carriers; structurally empty closed channels; complete channel composition; sealed or exposed targets; and empty or free extension. The base and successor certificates are independent of a finite Fold depth and introduce no numerical null state. The generator produced `8748` named candidates and the decision support contains `8748` one-for-one decisions. Exactly one candidate survived: `paired-transition-weight-times-output-support-per-action__positive-open-channels-closed-as-empty__ordered-positive-sum-of-partial-widths__partial-width-over-total-width__complete-exclusive-partition-of-one__action-over-total-width__greater-width-shorter-lifetime__sealed-before-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`partial`	`paired-transition-weight-times-output-support-per-action`	`unpaired-transition-leg-times-output-support`	Rejected by computed Fold predicates: partial_width_carrier.
`domain`	`positive-open-channels-closed-as-empty`	`numeric-null-closed-channels`	Rejected by computed Fold predicates: channel_domain.
`total`	`ordered-positive-sum-of-partial-widths`	`largest-partial-width-only`	Rejected by computed Fold predicates: total_width_law.
`branch`	`partial-width-over-total-width`	`total-width-over-partial-width`	Rejected by computed Fold predicates: branching_law.
`partition`	`complete-exclusive-partition-of-one`	`incomplete-open-channel-subset`	Rejected by computed Fold predicates: partition_law.
`lifetime`	`action-over-total-width`	`total-width-over-action`	Rejected by computed Fold predicates: lifetime_law.
`ordering`	`greater-width-shorter-lifetime`	`greater-width-longer-lifetime`	Rejected by computed Fold predicates: ordering_law.
`target`	`sealed-before-release`	`readable-before-seal`	Rejected by computed Fold predicates: target_boundary.
`extension`	`empty-extension`	`free-correction`	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:42d6264dd3d0731a3920f4c904f688ccf53e028f356a0be3ff910c6041dc0f6d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject incomplete branching and a width-proportional lifetime.; observed Complete normalization and exact inverse duration reject both false laws.; receipt `sha256:1f11a1f6c70596215f0cd6e40f1fc35fac6a7542f354c765400f4b1a3d562c69`.
- `tampered_source`: passed; expected Reject any changed claimant source identity.; observed The changed identity differs from the registered source manifest.; receipt `sha256:0de6112a4fa89e23a8625b623b57f00f6f65f6e8de7f00f0265c4d1e451b09ae`.
- `tampered_artifact`: passed; expected Reject a duplicated or incomplete candidate census.; observed The deliberate duplicate fails complete candidate identity.; receipt `sha256:85f7f2fccaaba82e12423de0043e0cde43f60a6c59bbf25b4c3079af293bc64a`.
- `boundary`: passed; expected Reject a numerical null channel, pre-seal target access and free correction.; observed Closed channels are structural absence; sealed custody and empty extension are required.; receipt

`sha256:89137a6f44d4b9f7604a1d09999b21276005254103306653d01e9d7fc79e424a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b0291ebf59a77a2abc9bb3ba2774893f5738d22b048abf4b280be93fd0788e16`. Independent certificate: `sha256:29c5bc317aac67c022d87e417828f668fd1d0748855e6d0eab968f7047c00910`. Engine external-validation hash: `sha256:72baee80b3adaeac88de833ce60ce9ab2f0c8885919a6d53aadc763fd5a44068`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2026-W-BOSON-LISTING
- NIST-CODATA-2022

Observed comparison records:

- PDG 2026 complete W-width table retained: sixteen rows, every uncertainty, use-status and scale-factor note
- PDG 2026 complete thirteen-row W decay-mode table retained with aggregate, exclusive, inclusive-subset and upper-limit classes kept distinct
- PDG exclusive individual vector retained: e, mu, tau and hadronic branch intervals contain the One jointly
- PDG universality fit retained: B(hadrons)=One take 3*B(ell nu) exactly; the forced complement lies inside the reported hadronic interval
- PDG total width interval GeV: (Fraction(209, 100), Fraction(219, 100))
- PDG exact individual partial-width enclosure GeV: (Fraction(1038103, 500000), Fraction(552537, 250000))
- NIST reduced-action ellipsis retained as an adjacent rational enclosure, never as an exact rational proof value
- derived W lifetime interval s from NIST action and PDG width: (Fraction(6582119569, 21900000000000000000000000000000000), Fraction(658211957, 2090000000000000000000000000000000))
- omitted branch, double-counted inclusive subset, inverse branch, direct width-lifetime and incomplete-source controls rejected

Falsification condition: Reject if the complete registered PDG W-width table, thirteen-row decay-mode table or either exclusive branching organisation is incomplete; if PDG does not identify each branch as Gamma_i/Gamma or does not close the hadronic branch as One take three leptonic branches; if the complete measured exclusive intervals do not contain the One, if the exact complements forced from the printed leptonic values miss the printed hadronic interval, or if inclusive subsets are double counted; if the NIST action-prefix enclosure and PDG width interval do not yield a finite positive inverse-width lifetime interval and ordering; if any registered row, uncertainty, source identity or unfavourable control is omitted; or if target content enters before seal.

Measurement receipt: `sha256:d48b51e08f77bfd7850e90dcc4f0bea4b17a68f8588730baa1d0627dd9376f36`. Isolation certificate: `sha256:6cefc1a860ab53a4ffd00d2c2dace7b786342d3fb7cd0d33829d4b63766e7083`. Custody certificate: `sha256:95a979a4ed2600d2d93d1a25e76b28f6e9b5fa9a9afdd8a2e19e5a14d980ef65`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, certificate, answer table or survivor identity in the derivation runtime
- no particle identity, measured width, branching fraction, lifetime, source locator or PDF text in formal candidate decisions

- no claimant-supplied expected survivor and no submission-controlled admission flag
- no numerical-zero proof value, negative, irrational, imaginary, transcendental or floating proof magnitude
- no continuum exponential, Breit-Wigner line shape, imported perturbative amplitude, fitted coupling, selected correction or measurement-selected candidate
- no double counting of aggregate, inclusive-subset, exclusive-subset or limit rows as independent primary branches
- no promotion of the NIST ellipsis to an exact rational proof value
- no target access before the prediction seal and no free extension after closure

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d2750e9b099ed37535180507e4ef1eca7b678d6c4ce14a16b21c9cd8ec5ee688`; engine receipt `sha256:a4e14f20a7c1f6920115f0e5ad0b911b5462b690aad35ee79aac39b691d12d3f` at `receipts/engine/model_admitted/SFT-PHYS-DECAY-WIDTH-BRANCHING-LIFETIME-TERMINAL-006-a4e14f20a7c1f692.json`; empirical-validation hash `sha256:5f55eaf606e42e1746e073343e52e184ebe55bd62288aa5c0339e4834511eef7`; measurement receipt `sha256:d48b51e08f77bfd7850e90dcc4f0bea4b17a68f8588730baa1d0627dd9376f36`.

183. Terminal Fold running functions and common-support convergence

Claim identity: `SFT-PHYS-COUPLING-RUNNING-CONVERGENCE-TERMINAL-006`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete forced prime sectors p in two, three, five and seven share one generated scale carrier with One base and binary successor R_(n+1)=2R_n. Each sector's exact finite running share is g_p(R)=(p+R-1)/(p+R), with positive shortfall 1/(p+R). Every binary successor raises g_p by R/[(p+R)(p+2R)]. For p<q the exact gap is (q-p)/[(p+R)(q+R)], it strictly shrinks at every successor, and for every finite positive tolerance a generated binary support gives a smaller gap. Thus all four sector functions converge constructively on the common support axis without a completed infinity, imported beta function, fitted parameter or measurement-selected correction. The sealed self-source-range and screening-exposure laws determine the opposite physical energy orientations used in post-seal comparison.

> *The complete prime sectors two, three, five and seven share One-base binary support R. Each exact finite running share is (p+R-1)/(p+R), its positive shortfall is 1/(p+R), its binary successor take is R/[(p+R)(p+2R)], and every ordered pair gap is (q-p)/[(p+R)(q+R)]. Every share rises, every gap strictly shrinks, and every finite positive tolerance has a generated binary-support witness. Current authoritative PDG comparison retains the complete QCD and electromagnetic vectors: alpha_s decreases with increasing transfer scale while electromagnetic alpha increases as screening is removed, exactly as the separately sealed carrier-specific coordinate orientations require. The common-support four-sector convergence is an exact standing prediction, not a fitted value or a falsely claimed direct measurement.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002`
- `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003`
- `SFT-PHYS-STRONG-RUNNING-DIRECTION-002`
- `SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003`
- `SFT-PHYS-MEAS-BOUNDARY-GROWTH-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-MATH-LIMIT-CONTINUUM-002`
- `SFT-MATH-SELF-SIMILAR-CONVERGENCE-002`

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete product of every closed or selected sector domain, every binary, linear or target-assigned scale carrier, every holding-share, fixed or target-assigned coupling form, every successor direction, every exact, constant or target-assigned pair-gap law, every constructive, completed-infinity or visual convergence form, every carrier-specific, universal-sign or measurement-selected physical translation, both target-custody states and both extension states. The declared exact boundary is: The complete independently forced prime-sector ladder two, three, five and seven; every generated finite positive depth level with One base and binary successor; exact positive rational holding shares, shortfalls, successor takes and ordered pair gaps; every finite positive tolerance denominator; the already forced self-source-range and screening-exposure translation classes; sealed or exposed targets; and empty or free extension. No completed infinity, continuum scale or numerical null state is admitted. The generator produced `8748` named candidates and the decision support contains `8748` one-for-one decisions. Exactly one candidate survived: `complete-prime-sectors-through-seven__One-base-binary-depth-successor__holding-share-of-sector-plus-support__binary-support-successor-raises-holding-share__exact-generator-gap-over-paired-sources__finite-positive-epsilon-witness-for-every-pair__carrier-specific-self-source-range-and-screening-exposure__sealed-before-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
sectors	complete-prime-sectors-through-seven	selected-familiar-sectors-only	Rejected by computed Fold predicates: sector_domain.
scale	One-base-binary-depth-successor	imported-linear-scale-grid	Rejected by computed Fold predicates: scale_generation.
coupling	holding-share-of-sector-plus-support	fixed-bare-sector-share	Rejected by computed Fold predicates: coupling_form.
successor	binary-support-successor-raises-holding-share	binary-support-successor-lowers-holding-share	Rejected by computed Fold predicates: successor_law.
gap	exact-generator-gap-over-paired-sources	constant-sector-gap	Rejected by computed Fold predicates: pair_gap_law.
convergence	finite-positive-epsilon-witness-for-every-pair	completed-infinity-as-proof-value	Rejected by computed Fold predicates: convergence_law.
translation	carrier-specific-self-source-range-and-screening-exposure	one-imported-energy-sign-for-every-carrier	Rejected by computed Fold predicates: physical_translation.
target	sealed-before-release	readable-before-seal	Rejected by computed Fold predicates: target_boundary.
extension	empty-extension	free-running-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:ba74e810bf7fdb5eaba6092188c295e809742ad65a33f8e4bc192f3a01326964`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject reversed running and a completed infinity as a proof value.; observed Exact positive successor takes and finite epsilon witnesses reject both false laws.; receipt `sha256:615b161de849d45bd7c2991db2cae052a01ca04b8213f4229e19d60b8dae8c55`.
- `tampered_source`: passed; expected Reject any changed claimant source identity.; observed The changed identity differs from the registered source manifest.; receipt `sha256:0de6112a4fa89e23a8625b623b57f00f6f65f6e8de7f00f0265c4d1e451b09ae`.
- `tampered_artifact`: passed; expected Reject a duplicated or incomplete candidate census.; observed The deliberate duplicate fails complete candidate identity.; receipt `sha256:85f7f2fccaaba82e12423de0043e0cde43f60a6c59bbf25b4c3079af293bc64a`.

- `boundary`: passed; expected Reject pre-seal target access, free correction and a numerical null base.; observed The scale begins at the One; sealed custody and empty extension are required.; receipt `sha256:1282d4e9761ba33212fc7dbf3768f3ad708c64cb4bcc97ef5eae8e4168396988`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d6afafe211ebca3297f1e684cae091e39ca0cc80931b70c3cf11924360fdb9a1`. Independent certificate: `sha256:dfa5fe2d5d8b63a7964d01f099c74abfb48ba490275147c4286b07f1b4645280`. Engine external-validation hash: `sha256:305e1659813dc637a2e779b3578008bc1144c3318872ad7f8d9a6c0ee6efe283`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2026-QCD-REVIEW
- PDG-2026-ELECTROWEAK-REVIEW
- PDG-2025-ELECTROMAGNETIC-RUNNING-ENDPOINT

Observed comparison records:

- PDG 2026 QCD complete Table 9.1 retained: seven rows, every reported averaging column and uncertainty
- PDG 2026 QCD Figure 9.4 retained by immutable source hash and visual audit: nine measurement classes across the displayed multi-scale range
- PDG strong low-scale interval at Q=1.37 GeV: (Fraction(331, 1000), Fraction(559, 1000))
- PDG strong m_Z-scale interval: (Fraction(1171, 10000), Fraction(1189, 10000))
- The separated exact intervals confirm that alpha_s decreases as transfer energy rises; the full figure and page-43 conclusion retain the wider multi-scale evidence
- PDG 2026 electromagnetic scope retained: five alpha(0) determinations, eleven hadronic Delta-alpha evaluations, the published running relation and direct L3/OPAL observations
- PDG explicit prior Thomson inverse-alpha interval: (Fraction(137035999063, 1000000000), Fraction(27407199821, 200000000))
- PDG explicit prior Z-scale inverse-alpha interval: (Fraction(63961, 500), Fraction(63969, 500))
- The inverse-alpha intervals decrease, hence electromagnetic alpha increases toward the Z scale
- Opposite physical energy directions pass only under the sealed carrier-specific self-source-range and screening-exposure translations; a universal imported sign is rejected
- The exact common-support convergence of the four Fold sector functions remains a standing falsifiable prediction and is not mislabelled as a direct measurement

Falsification condition: Reject if any of the seven PDG QCD Table 9.1 rows, nine Figure 9.4 measurement classes, exact low- and m_Z-scale strong-coupling rows, five current alpha(0) rows, eleven current hadronic Delta-alpha rows, published electromagnetic running relation, direct LEP observation or explicit prior Z-scale endpoint is omitted or altered; if the separated strong-coupling intervals do not decrease from 1.37 GeV to m_Z; if the explicit inverse electromagnetic endpoint intervals do not decrease from the Thomson limit to the Z scale; if a universal physical energy sign is imposed despite the sealed carrier-specific coordinates; if the exact Fold support successor, running-share increment, pair-gap shrinkage or finite tolerance witness fails; if the formal convergence prediction is falsely labelled a direct measurement; or if any target content enters before the prediction seal.

Measurement receipt: `sha256:f7b8eb2dd10c0c689ee5dedb5b1add4a6b69ff8466d70e2727964d94289993bc`. Isolation certificate: `sha256:11efa3cdf27c3e412e3a48235b1c6bfb258890a6b4778cadc35c3bbc6e6eb794`. Custody certificate: `sha256:4252ba538115e1ee971b038a463ef6f1d24b97cebda59e8a96094076c93aabcb`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, certificate, answer table or survivor identity in the derivation runtime
- no measured coupling, energy, beta function, renormalization-group curve, unification group, source locator or PDF text in formal candidate decisions
- no claimant-supplied expected survivor and no submission-controlled admission flag
- no numerical-zero proof value, negative magnitude, irrational, imaginary, transcendental or floating proof magnitude
- no completed infinity or continuous scale used as a proof object
- no fitted coefficient, selected scale, running correction or measurement-selected orientation
- no universal physical energy sign imposed on physically distinct self-source and screening-exposure coordinates
- no plot digitization inventing coordinates that the authoritative source does not print
- no promotion of exact common-support convergence to a direct experimental observation
- no target access before the prediction seal and no free extension after closure

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:8e5b0d59dd3626f39dcb398cba57819bac31eca897583d65bb5b99084d67f3f9`; engine receipt `sha256:b9263f585f01a2401261f465ad250f405a9914da5c53e57ba113f076aeb4f309` at `receipts/engine/model_admitted/SFT-PHYS-COUPLING-RUNNING-CONVERGENCE-TERMINAL-006-b9263f585f01a240.json`; empirical-validation hash `sha256:2f1893b335cbcd3be69e125cc935f6b69ab16c56d65aedcc028cfe9c3e85d3cb`; measurement receipt `sha256:f7b8eb2dd10c0c689ee5dedb5b1add4a6b69ff8466d70e2727964d94289993bc`.

184. Simultaneous massless, One-speed and confined strong-carrier law

Claim identity: `SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The generated colour-three sector has eight non-singlet carrier states and coupling two-thirds. Its carrier has an empty mass and rest-capture record, so its local phase advances exactly one support cell per tick. The same carrier bears the colour it mediates and therefore self-sources a fixed half-One transverse tube: every positive separation successor adds the same two-thirds work carrier, exceeding every positive exact finite bound at a generated finite witness. Local massless One-speed propagation and asymptotic confinement are thus simultaneous non-equivalent records, not contradictory alternatives.

> *Three colour labels force eight non-singlet strong carriers. The carrier has empty mass and rest-capture records and advances one support cell per tick with retained phase. Because it carries the colour it mediates, it self-sources a fixed half-One tube whose work rises by exact two-thirds at every separation successor and exceeds every positive exact bound at a finite witness. Local massless One-speed propagation and asymptotic confinement therefore coexist without a fitted parameter or free correction.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003`
- `SFT-PHYS-STRONG-RUNNING-DIRECTION-002`
- `SFT-PHYS-SPACETIME-EXACT-INTERVAL-003`
- `SFT-PHYS-WAVE-EXACT-OPERATIONS-003`

- `SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005`
- `SFT-PHYS-MATTER-COMPOSITE-HADRONS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

Generated grammar and closure boundary. Generate the complete product of every generated, free or target-assigned colour sector; every empty, numeric/fitted or target-assigned mass law; every One-speed, free-speed or target-assigned causal law; every colour self-source, chargeless or target-assigned carrier law; every fixed-tube, spreading/bounded or target-assigned confinement law; every simultaneous, exclusive or target-assigned composition; both target custody states; and both extension states. The declared exact boundary is: The generated colour-three carrier at every positive finite causal tick and separation depth. Masslessness is the empty mass/rest-capture record and One-cell-per-tick local phase propagation; confinement is fixed-width self-sourced colour transport whose positive separation work exceeds every nominated positive exact bound at a finite witness. No completed infinity or claim about an observable isolated colour carrier enters. The generator produced `2916` named candidates and the decision support contains `2916` one-for-one decisions. Exactly one candidate survived: `generated-colour-three-sector-with-eight-nonsinglet-carriers__empty-mass-label-with-no-rest-capture__One-support-cell-per-tick-with-retained-phase__colour-carrying-mediator-resources-its-own-channel__fixed-half-One-tube-and-two-thirds-work-successor__local-massless-One-speed-and-asymptotic-confined-records-coexist__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`sector_law`	`generated-colour-three-sector-with-eight-nonsinglet-carriers`	`free-sector-or-mediator-count`	Rejected by computed Fold predicates: sector.
`mass_law`	`empty-mass-label-with-no-rest-capture`	`numerical-zero-or-positive-fitted-mass`	Rejected by computed Fold predicates: mass.
`causal_law`	`One-support-cell-per-tick-with-retained-phase`	`free-sub-or-super-One-speed`	Rejected by computed Fold predicates: causal.
`self_source_law`	`colour-carrying-mediator-resources-its-own-channel`	`chargeless-linear-carrier`	Rejected by computed Fold predicates: self_source.
`confinement_law`	`fixed-half-One-tube-and-two-thirds-work-successor`	`spreading-flux-or-bounded-separation-work`	Rejected by computed Fold predicates: confinement.
`composition_law`	`local-massless-One-speed-and-asymptotic-confined-records-coexist`	`masslessness-and-confinement-treated-as-exclusive`	Rejected by computed Fold predicates: composition.
`target_boundary`	`sealed-before-observation-release`	`observation-readable-before-seal`	Rejected by computed Fold predicates: target.
`extension`	`empty-extension`	`free-carrier-correction`	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:40144181c27ae46335d82275c66df9cba4743451c49ed8a8700e0fb4e3eceebe`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject treating empty mass and empty isolated-colour records as the same claim or as numerical zero.; observed Local propagation survives while isolated asymptotic colour remains empty.; receipt `sha256:de98ae67df337f751884db61a7bff6821a5b74354eeaae2f79cdd62fa9a451c7`.

- `tampered_source`: passed; expected Reject changed source identity.; observed Identity differs.; receipt `sha256:7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83`.
- `tampered_artifact`: passed; expected Reject duplicate candidates.; observed All identities are unique.; receipt `sha256:0b5a37b200f6544185da03decea0e8f27bba78ad0ff859e5920232c0a780f049`.
- `boundary`: passed; expected Reject numerical/fitted mass, false exclusivity, pre-seal target access and free correction.; observed Only empty mass, simultaneous records, sealed custody and empty extension survive.; receipt `sha256:e74908f15eb59a896903336653189ab9920bb7e4dd97ca7a41b9cbf680c116cf`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9fb24ffba02632292d1b725b216db5ebcd2511a9b53aecefd27b54ea9647efdc`. Independent certificate: `sha256:a9b8d1ca2b0cf318f9c1107d3e76f790f8c924d4751876c7d8cb4d638a829a90`. Engine external-validation hash: `sha256:5cc5486ed41238fd730f87a266c7456cc8c07e4d6c6063c694b49ea4fb690dd4`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- D10F-STRONG-CARRIER-MASSLESS-CONFINED-COMPARISON

Observed comparison records:

- The complete V1 D10f row, ten admitted prerequisite certificates and three complete current PDG sources are hash-locked and released only after the V3 formal commit and in-run prediction seal.
- V3 independently forces three colour labels, eight non-singlet carriers, empty mass and rest-capture records, retained phase advancing one support cell per tick, and no fitted propagation value.
- The same colour-bearing carrier self-sources a fixed half-One tube whose exact separation work increases by two-thirds at every successor and exceeds every positive exact bound at a finite witness.
- The PDG gluon listing records an SU(3) colour octet and m=0, matching the frozen structural classification after release.
- The PDG explicitly classifies m=0 as a theoretical value and says a mass as large as a few MeV may not be precluded; this qualification is retained and no direct free-gluon time-of-flight measurement is claimed.
- The PDG QCD review records Nc=3, Nc-squared-minus-One=8, the adjoint gluon representation, nonlinear field term, three-gluon vertex and four-gluon vertex.
- The same review states that neither quarks nor gluons are observed as free particles and that hadrons are colour-singlet combinations, matching the frozen asymptotic observation boundary.
- Collider colour-factor measurements CA=2.89+/-0.03(stat)+/-0.21(syst) and CF=1.30+/-0.01(stat)+/-0.09(syst) contain the published SU(3) expectations 3 and 4/3 within exact conservative summed uncertainty boundaries.
- The PDG also records clear experimental evidence that gluon jets are softer and broader than light-quark jets; the quark-model review describes physical states as colour singlets and colour as believed permanently confined.
- External measurements and classifications test only the already sealed result: they do not select its survivor, enter its proof arithmetic, fit a parameter or add a carrier correction.

Falsification condition: Reject if the V1 D10f row, any admitted prerequisite or any complete PDG source changes or is omitted; if three colour labels, eight non-singlet carriers, empty mass/rest-capture, One-cell-per-tick propagation, self-source, fixed half-One tube, positive two-thirds work successor or empty isolated-colour record fails; if the PDG theoretical m=0 listing is misreported as a direct measurement or its few-MeV caveat is omitted; if measured colour-factor intervals cease to contain their published SU(3) expectations; if a free coloured state is observed; or if any external value selects the survivor, becomes a formal proof scalar, is fitted, or supplies a free correction.

Measurement receipt: `sha256:798dc0308ac250c03b0c1a8685e9192e95bb1f950344edc37b3c1fa1defd4447`. Isolation certificate: `sha256:1d8ad63be3229d00c3c6fcd4137e4271b221fd4045c96c7aed217c209e3865dd`. Custody certificate: `sha256:be921ad0fb8b3b28c013100c797ddb677217e66cda7ae90ec4f7de764a0ac8c8`.

Registered source descriptions:

- PDG-2026-GLUON-LISTING: https://pdg.lbl.gov/encoder_listings/g021.pdf (sha256:fcc152780d2efb6cb66d2f9351abd034d30cab0e914fd87a0daee46a0344353d)
- PDG-2026-QCD-REVIEW: https://pdg.lbl.gov/2026/reviews/rpp2026-rev-qcd.pdf (sha256:c04c628d76b18610c5fa2a919c6081918a25b55fb971b6af5829f4ca2baa386f)
- PDG-2026-QUARK-MODEL-REVIEW: https://pdg.lbl.gov/2026/reviews/rpp2026-rev-quark-model.pdf (sha256:6aa98fa53857122f27b638c59081af1a2857d787c4205370d8fa38fcb6b70ff0)

Registered target identities:

- D10F-WITHHELD-HISTORICAL-PREREQUISITE-AND-PDG-RECORD from D10F-STRONG-CARRIER-MASSLESS-CONFINED-COMPARISON

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- imported QCD Lagrangian
- imported gauge-group axiom
- numerical zero as the empty mass record
- free carrier mass
- free causal speed
- free confinement potential
- completed infinity
- direct observation of an isolated colour carrier
- irrational or imaginary proof value
- measurement-selected survivor
- measured fit
- free carrier correction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:c523e40c2f6b2f01c6c825148b83d9dfe2d0c20fffe768152a77196764da1742`; engine receipt `sha256:bae91c8dc0b88c6130260c7e10ea22e9433ca322bd5ad3a8f9077653315f57e1` at `receipts/engine/model_admitted/SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013-bae91c8dc0b88c61.json`; empirical-validation hash `sha256:76113e290c7e36ae64b9325312733981bc88d5dd06a3620245b74140a7a16b05`; measurement receipt `sha256:798dc0308ac250c03b0c1a8685e9192e95bb1f950344edc37b3c1fa1defd4447`.

185. Strong self-source iteration and finite fixed-point boundary

Claim identity: `SFT-PHYS-STRONG-FIELD-NONLINEAR-FIXED-POINT-TERMINAL-014`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A chargeless carrier has an empty self-source correction and holds its matter source. The admitted gravitational scalar self-source has one admissible quarter-One fixed point with shrinking corrections. A colour-bearing carrier instead remains inside its own source update: every successor retains the prior source and appends the complete binary carrier, so the correction is exact two and never shrinks. For every positive exact finite F, F plus two is strictly above F, leaving the strong finite fixed-point record empty; the same successor law exceeds every positive exact finite bound at a generated finite witness. This is the depth-independent confinement boundary.

> *The exact three-way field-update census admits only the chargeless held-source, gravitational quarter-One contraction and colour self-source F-plus-two persistence discriminator. The strong correction is exactly two at every positive finite*

> *successor, so no positive exact finite fixed point exists and every positive exact finite bound has a generated finite witness above it. Post-seal PDG and lattice-QCD comparison supports non-abelian self-interaction and the linear unbroken string branch while explicitly retaining full-QCD string breaking, ground-state saturation, uncertainties and single-ensemble scope.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STRONG-RUNNING-DIRECTION-002`
- `SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009`
- `SFT-PHYS-FIELD-SOURCE-RESPONSE-001`
- `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003`
- `SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

Generated grammar and closure boundary. Generate the complete product of every colour-self-sourced, external-only or target-assigned source; every binary-successor, free or target-assigned update; every persistent, shrinking/free or target-assigned correction; every empty, selected or target-assigned finite fixed-point law; every exact three-way, conflated or target-assigned comparator; every arbitrary-bound, finite-prefix/completed-infinity or target-assigned confinement law; both target custody states; and both extension states. The declared exact boundary is: Every positive finite iteration depth of the admitted colour self-source successor, every positive exact finite fixed-point candidate and work/source bound, the admitted gravitational scalar self-source channel, and the chargeless held-source control. The result distinguishes structural source recursion from polynomial degree and does not claim a conventional renormalized Yang-Mills solution. The generator produced `2916` named candidates and the decision support contains `2916` one-for-one decisions. Exactly one candidate survived: `colour-carrier-retained-inside-its-own-source-update__each-successor-retains-prior-source-and-appends-complete-binary-carrier__persistent-exact-binary-correction-at-every-successor__empty-positive-finite-fixed-point-record-by-F-plus-two-order__neutral-hold-gravity-contraction-strong-persistence-three-way-discriminator__finite-witness-above-every-positive-exact-bound__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`source_composition`	`colour-carrier-retained-inside-its-own-source-update`	`external-matter-source-only`	Rejected by computed Fold predicates: source.
`update_law`	`each-successor-retains-prior-source-and-appends-complete-binary-carrier`	`free-linear-contracted-or-oscillatory-update`	Rejected by computed Fold predicates: update.
`correction_law`	`persistent-exact-binary-correction-at-every-successor`	`shrinking-or-free-correction`	Rejected by computed Fold predicates: correction.
`fixed_point_law`	`empty-positive-finite-fixed-point-record-by-F-plus-two-order`	`selected-finite-strong-fixed-point`	Rejected by computed Fold predicates: fixed.
`comparator_law`	`neutral-hold-gravity-contraction-strong-persistence-three-way-discriminator`	`conflate-all-self-source-iterations`	Rejected by computed Fold predicates: comparator.
`confinement_law`	`finite-witness-above-every-positive-exact-bound`	`bounded-prefix-or-completed-infinity`	Rejected by computed Fold predicates: confinement.

Axis	Forced form	Eliminated form	Elimination reason
`target_boundary`	`sealed-before-observation-release`	`observation-readable-before-seal`	Rejected by computed Fold predicates: target.
`extension`	`empty-extension`	`free-strong-field-correction`	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:9764f4e0b2309d8a96191eff92927bb5cd80838266b70108e8f94ea5bbf7713a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject conflating empty, shrinking and persistent correction records.; observed Neutral, gravity and strong updates retain distinct exact correction classes.; receipt `sha256:e4a7af5577b7ed98d709925f9db75a53724b7348abaf318dffdb4857ab64d3b4`.
- `tampered_source`: passed; expected Reject changed source identity.; observed Identity differs.; receipt `sha256:7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83`.
- `tampered_artifact`: passed; expected Reject duplicate candidates.; observed All identities are unique.; receipt `sha256:0b5a37b200f6544185da03decea0e8f27bba78ad0ff859e5920232c0a780f049`.
- `boundary`: passed; expected Reject selected fixed point, finite-prefix/completed-infinity inference, target access and free correction.; observed Only universal positive-order exclusion, finite witnesses, sealed custody and empty extension survive.; receipt `sha256:01fbecf349784fbb255dfecf26b57add49eece9e3bc3fee506da4ac196466b6e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a27960b8343ae01d0fafd4fd7327d118319bd93b780b87b43142e8a304d20759`. Independent certificate: `sha256:4009d2e7479a082ceda554a357c72b83f4c0b9056676c4fa7539c21ed758b4a1`. Engine external-validation hash: `sha256:35920b4ede6319913138c6ef2d31e39e51545b7e4412f6a86af0fbb90dab9d07`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- D10E-STRONG-FIELD-NONLINEAR-FIXED-POINT-COMPARISON

Observed comparison records:

- The complete V1 D10e row, nine admitted prerequisites, the complete current PDG QCD review and complete primary lattice-QCD paper are hash-locked and released only after the frozen V3 commit and in-run prediction seal.
- V3 independently forces an empty chargeless correction with held One source, the admitted gravity quarter-One fixed point with shrinking corrections, and a colour self-source successor F plus two with the exact binary correction retained at every successor.
- For every positive exact finite F, F plus two is strictly above F, so the strong positive finite fixed-point record is empty; for every positive exact finite bound, the same law constructs a finite source witness above it without a completed infinity.
- The PDG QCD review independently retains non-abelian colour, colour-bearing gluons, three- and four-gluon self-interactions and the absence of observed free quarks.
- The lattice study independently reports that the gluons-only static energy grows linearly at asymptotically large separation with string tension sigma and models the unbroken string state as V-hat(r)=V-hat-0+sigma*r in the breaking region.
- The same study observes the required adverse boundary: dynamical light-quark pair creation mixes the string with broken-string states, and the full-QCD ground state tends to two noninteracting static-light mesons rather than growing without bound.
- The reported light and strange breaking distances, 1.224(15) fm and 1.293(16) fm, are retained with lattice spacing 0.06426(76) fm and the 280 MeV pion and 460 MeV kaon masses.

- Those distances come from one ensemble with nonphysical quark masses and are not promoted to universal physical constants; they neither select the Fold survivor nor enter its formal arithmetic.
- The exact correspondence is therefore branchwise: persistent Fold colour self-source matches the unbroken string branch, while pair-created state mixing supplies a distinct full-QCD saturation channel that the validation retains explicitly.
- No external value fits a parameter, supplies a free correction or changes the frozen generated grammar.

Falsification condition: Reject if the complete V1 D10e row, an admitted prerequisite, the current PDG review or the complete primary lattice study changes or is omitted; if the exact chargeless hold, quarter-One gravitational contraction, persistent binary strong correction, empty positive finite strong fixed-point record or finite witness above every positive exact bound fails; if QCD ceases to carry non-abelian gluon self-interaction; if the unbroken string branch ceases its linear growth; if pair-created broken-string mixing and full-QCD ground-state saturation are omitted; if the single-ensemble, nonphysical-mass or non-universal-threshold limits are hidden; or if an external target selects the survivor, enters formal arithmetic, fits a parameter or adds a correction.

Measurement receipt: `sha256:ef22864dfb7236904932ef92516fdedfd7b22bd5d2b471a9a1c3ad1d6e0fbf95`. Isolation certificate: `sha256:fbf7d48bf0d7cc2239a427718dbca7e59534210dc2face57fa133e5c8bb36793`. Custody certificate: `sha256:c0eadbbe6154dbe534b7602036251380ac7f2caaf9b121e1725283ecd6ceb5ea`.

Registered source descriptions:

- PDG-2026-QCD-REVIEW: https://pdg.lbl.gov/2026/reviews/rpp2026-rev-qcd.pdf (sha256:c04c628d76b18610c5fa2a919c6081918a25b55fb971b6af5829f4ca2baa386f)
- BULAVA-ET-AL-2019-STRING-BREAKING: https://arxiv.org/pdf/1902.04006 (sha256:6c6de4ebdfc86d7976fb343a35f490ff108ab0d56dd1450a1ee27539e2641bb5)

Registered target identities:

- D10E-WITHHELD-HISTORICAL-PREREQUISITE-PDG-AND-LATTICE-RECORD from D10E-STRONG-FIELD-NONLINEAR-FIXED-POINT-COMPARISON

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- imported Yang-Mills field equation
- imported gauge-group axiom
- numerical zero as the empty fixed-point record
- negative, irrational or imaginary proof value
- completed infinity
- finite-prefix inference presented as an unbounded law
- measurement-selected survivor
- measured fit
- free strong-field correction
- unbounded full-QCD ground-state potential

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:4af939c69d947be443ff772e7030b49ca3104b4b090febee3ec396240c01ff61`; engine receipt `sha256:641cdf7a00a56d3889b1a31731dff04c70dbe068465924f005d000d3fc100cdc` at `receipts/engine/model_admitted/SFT-PHYS-STRONG-FIELD-NONLINEAR-FIXED-POINT-TERMINAL-014-641cdf7a00a56d38.json`; empirical-validation hash `sha256:6a20eb16725a453c3f37f22c5b0b63324e1a9cc9cc3912738b6584eb73650d4f`; measurement receipt `sha256:ef22864dfb7236904932ef92516fdedfd7b22bd5d2b471a9a1c3ad1d6e0fbf95`.

186. Finite accumulated Fold-coupling separation

Claim identity: `SFT-PHYS-COUPLING-ACCUMULATED-SEPARATION-TERMINAL-015`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For the admitted binary and generator-three running sectors, every positive finite partial accumulation of their exact common-support gaps lies in [1/12, 11/60); every successor from the second gap onward is below half its predecessor, and every positive rational tolerance has a generated finite tail witness.

> *Every positive finite partial sum of the binary/generator-three Fold coupling gaps lies in the exact bracket [1/12, 11/60); after the second level every successor gap is below half its predecessor, and every positive rational tolerance has a generated finite tail witness.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-COUPLING-RUNNING-CONVERGENCE-TERMINAL-006`
- `SFT-MATH-SELF-SIMILAR-CONVERGENCE-002`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-FOUNDATION-COUNT-001`

Generated grammar and closure boundary. Generate the complete product of predecessor, gap term, first-step, contraction, envelope, finite-boundary, tolerance and extension forms. The declared exact boundary is: Every positive finite common-support level generated from the One by binary succession; the exact binary/generator-three pair gap; every positive finite partial accumulation and every positive rational tolerance represented by a positive denominator. The generator produced `512` named candidates and the decision support contains `512` one-for-one decisions. Exactly one candidate survived: `admitted-common-support-gap-law__exact-adjacent-sector-gap__retain-one-twelfth-and-one-twentieth__exact-strict-half-contraction-after-level-two__eleven-sixtieths__every-positive-finite-partial-support__generated-finite-tail-witness__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`admitted-common-support-gap-law`	`imported-or-selected-coupling-table`	A conventional or selected table is not an admitted predecessor.
`term`	`exact-adjacent-sector-gap`	`unsigned-or-fitted-difference`	A fitted difference can encode a target.
`first`	`retain-one-twelfth-and-one-twentieth`	`assert-half-contraction-from-first-gap`	The transition 1/12 to 1/20 shrinks but is not below half.
`contraction`	`exact-strict-half-contraction-after-level-two`	`visual-or-imported-convergence`	Visual approach and an imported series theorem are not Fold proof objects.
`envelope`	`eleven-sixtieths`	`one-sixth`	Later exact partial sums exceed this candidate.
`boundary`	`every-positive-finite-partial-support`	`completed-infinite-sum`	Completed infinity is outside the SFT domain.
`tolerance`	`generated-finite-tail-witness`	`limit-symbol-without-witness`	A limit label alone supplies no finite certificate.
`extension`	`no-extra-rule`	`free-tail-correction`	A free correction is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:5ab1dace9fee93e8153b8079d416bcb0d8a7b6f77af74ad8c27931ddf9ce1beb`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt

sha256:e6b9d33fb638063e5ba0234bcd82d4893ce84bd7de2e3cd9bdc1a4800c024ad4.

- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:59fc2a7bc6963288e74ddf7ecec05a7d87ab2912550cda0b8100f98a348f930b.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:9895bbbd82df65ced4fbba2385eced8ba8abee2c31bdf0ae5725ff30b43ea464.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:a563d412cbe3b2652716fb85b51c254c26c5a2dec563675a60639c7eaa467bdc.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:2cc5ffedbd8c84347cf51184f9d36cfc05a3e969dacad6592956690b77d3f34b. Independent certificate: sha256:180c8807b7d767624480aae2662b1589e08ec8c7bbfff5d132d609f47a84ce47. Engine external-validation hash: sha256:525894b57a38373dfc48ecad834d5b588803d3b69779fc2101a093beac584626.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S017 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, stored partial-sum table or prior survivor in execution
- no imported coupling equation, infinite-series formula, logarithm or continuum limit
- no numerical-zero, negative, irrational, imaginary or floating proof value
- no measured coupling, target value, fitted coefficient or selected stopping depth
- no completed infinity and no claim of an attained terminal sum

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:b12b82925d6cfe4cc7d55b920957f5a3afceb5362164da3818782126d0d3e85b; engine receipt sha256:723ff785cf91a40c287d2f2b38394a63fd8bfaa7a0be7a34be2e9e973202f4bf at receipts/engine/model_admitted/SFT-PHYS-COUPLING-ACCUMULATED-SEPARATION-TERMINAL-015-723ff785cf91a40c.json; empirical-validation hash None; measurement receipt None.

187. Exact two-fibre scattering partition and reciprocal mean free path

Claim identity: SFT-PHYS-SCATTERING-PARTITION-PATH-TERMINAL-017

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The registered one-target Fold support partitions into scatter half-One and pass half-One; cross section is successful/incident support; and mean free path is the reciprocal of positive density times cross section, with larger section forcing shorter path.

> *The registered one-target Fold support partitions exactly into scatter half-One and pass half-One; cross section is successful/incident support; and mean free path is the exact reciprocal of positive density times cross section, with larger section forcing shorter path.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CLASSICAL-PROBABILISTIC-001
- SFT-PHYS-MATTER-SCATTERING-001
- SFT-PHYS-SCATTERING-RUTHERFORD-COMPTON-TERMINAL-006
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of target geometry, outcome support, exclusivity, outcome measure, cross-section measure, density, path, causal interpretation and extension forms. The declared exact boundary is: The one-target geometry with exactly two equipotent Fold outcome cells; all positive finite incident/successful support counts; and all exact positive target densities and cross sections. The generator produced `512` named candidates and the decision support contains `512` one-for-one decisions. Exactly one candidate survived: `registered-one-target-Fold-geometry__complete-scatter-pass-pair__mutually-exclusive-held-labels__equipotent-two-cell-support__successful-over-incident-support__positive-density-times-section__reciprocal-encounter-support__deterministic-held-outcomes__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
geometry	registered-one-target-Fold-geometry	arbitrary-scatterer	Arbitrary geometry does not force equal outcome cells.
outcomes	complete-scatter-pass-pair	sampled-or-omitted-channel	An omitted channel cannot establish certainty.
exclusivity	mutually-exclusive-held-labels	overlapping-outcome-cells	Overlap double-counts one event.
measure	equipotent-two-cell-support	assumed-equal-probability	An assumed probability imports the answer.
section	successful-over-incident-support	fitted-dimensional-number	A fitted number is not the structural measure.
density	positive-density-times-section	density-omitted	Omitting density makes reciprocal path dimensionally incomplete.
path	reciprocal-encounter-support	selected-path-length	A selected length has no encounter trace.
cause	deterministic-held-outcomes	stochastic-transition-oracle	A random cause is not supplied by deterministic Fold structure.
extension	no-extra-rule	free-geometry-or-correction	A free correction defeats zero-parameter closure.

Base and successor. The closure proof is sealed by derivation hash `sha256:e429e239bc740c04992863e402d178f69f6a39f6e0b4a4f98c0df4cce80e7b4a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:f12cced3188c8fd3c53734121199b399596f46210cd1717cc718adc4c6ee6b4a`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:4caec307ab95c9decfce61920cbc4a6d0a92d5ad49da1793b3dc81f27f72b6c4`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f3e060037ddaf61b1b2fe110639e9d5b88d3f9f254bb77ec77a195940bf4f9b8`.

- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:ba7b87b2808814ec99f967b0dfdd6c22be77466992cc5052594752fb451071f8`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:965070318a614d6cb29833c963c07a4aefa2d05f692222ac1a1f74855deeb597`. Independent certificate: `sha256:3bfa663478d1cbc7a15fe9223a42e5d8b023a36cc1ab063d2f9ed05fc39927aa`. Engine external-validation hash: `sha256:6534bb6de6dd67d5b84d7d490b5a06d74692e33d1585e2820dd60c143238dfeb`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no assertion that arbitrary scattering has equal scatter/pass measures
- no stochastic causal selector or nondeterministic transition premise
- no numerical-zero, negative, irrational, imaginary or floating proof value
- no fitted cross section, measured path or target-selected correction
- no omission of target density from the general reciprocal law

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:b741febb3bdb061e559fa6ba2ff10235d96a2d0ddf4952acfb5954ef0cb4a316`; engine receipt `sha256:b4b42d9e9e0a52a703a059c265cc94325c4f56907da3422c4cb83b0facd9efdf` at `receipts/engine/model_admitted/SFT-PHYS-SCATTERING-PARTITION-PATH-TERMINAL-017-b4b42d9e9e0a52a7.json`; empirical-validation hash `None`; measurement receipt `None`.

188. Preserved/broken Fold channels and inverse mediator range

Claim identity: `SFT-PHYS-MEDIATOR-RANGE-CHANNEL-TERMINAL-020`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Complete-One channels have empty mass record; proper channels have positive shortfall, exact conserved forward/rest transfer, finite native trace and reciprocal mass range; larger mass means shorter range while massless inverse-square response remains positive at every finite radius.

> *Complete-One channels have an empty mass record and no finite transport endpoint; proper channels have positive shortfall, exact conserved forward/rest transfer, a finite native trace and reciprocal mass range; larger mass means shorter range, while massless inverse-square response remains positive at every finite radius.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-DYNAMICS-001`
- `SFT-FOUNDATION-PART-EQUIVALENCE-001`
- `SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002`
- `SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003`
- `SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003`

- SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-SPACETIME-LIMIT-SPEED-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of carrier, channel completion, mass record, forward/rest transfer, conservation, range scale, mass ordering, massless transport, post-seal comparison and extension forms. The declared exact boundary is: Every exact positive proper Fold channel and every complete channel sum; every positive finite mass share, its complete positive subtraction trace and reciprocal range; every exact positive finite radius; and the already sealed W/Z, electroweak-share and mediator-range comparison records. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: held-forward-and-rest-carriers__combination-reassembling-the-One__positive-shortfall-or-empty-record__exact-positive-forward-to-rest-transfer__forward-plus-rest-reassembles-One__positive-trace-and-reciprocal-mass-scale__larger-mass-shorter-range__held-forward-and-positive-finite-radius__inherit-sealed-electroweak-and-range-records__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	held-forward-and-rest-carriers	untracked-field-value	An untracked value cannot conserve transferred support.
channel	combination-reassembling-the-One	named-unbroken-channel	A name does not prove completion.
mass	positive-shortfall-or-empty-record	assigned-mass-label	An assigned label imports the classification.
transfer	exact-positive-forward-to-rest-transfer	sink-or-imported-exponential	A sink loses support and an exponential is not generated.
conservation	forward-plus-rest-reassembles-One	forward-loss-without-ledger	That violates closed transfer.
range	positive-trace-and-reciprocal-mass-scale	selected-cutoff-distance	A selected cutoff is a free parameter.
ordering	larger-mass-shorter-range	independent-range-order	Independent ordering can contradict the carrier.
massless	held-forward-and-positive-finite-radius	finite-hard-stop	No generated positive mass exists to supply a stop.
comparison	inherit-sealed-electroweak-and-range-records	W-Z-photon-selected-before-law	Named targets cannot select the discriminator.
extension	no-extra-rule	free-Yukawa-or-cutoff-rule	A free profile or cutoff adds a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:99da829e577bcbc3cc137f224acb0f9d8e17b1bfb14a3ae8896dc950fb8cb0a4. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:69e754afee5837b3eda670ba0f811559a1aa35d07851f1adbac32f6519d9a306.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:c03d0aeca1d46b9f7df23ddee15a2125eadf000da02f0240e71dd48d2404c1fd.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:6db4abf41d52e7f0e129e09fd4099afe11c4a198267f17d86fb05bc06a405de1.

- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:f49676a98e78127dfcfb2b2f1a2ecc870035f253200fd40bbcb708e7ac3b451a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8ab6340e93e19ee8a863181f592c6834e42deb555ca7472fa81c525ff2ea7a32`. Independent certificate: `sha256:ecbb5b36bc6221e4a891fbca2720ed86be8747bddd32314ce1cf3836ea5a0ee3`. Engine external-validation hash: `sha256:573b48a888abd5f135aa27b44db4f4d9b8be0eb3521a2443ff71710c185cd9e9`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, answer table, selected particle name or stored survivor
- no imported exponential, fitted range, hard physical numerical-zero cutoff or sink
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no claim that the finite native subtraction trace is a universal physical field profile
- no target access before the predecessor electroweak and mediator-range seals

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:e2c3444e331c2693253b4e161eed9d6566d0edd036f774a26749476b033dd86a`; engine receipt `sha256:3ad2016a90d7adc2c1c5c5afdfdaf5510f73ade146e2ea2e0a097d9158621f54` at `receipts/engine/model_admitted/SFT-PHYS-MEDIATOR-RANGE-CHANNEL-TERMINAL-020-3ad2016a90d7adc2.json`; empirical-validation hash `None`; measurement receipt `None`.

189. Fold baryogenesis dependency and retained-residue law

Claim identity: `SFT-PHYS-BARYOGENESIS-DEPENDENCY-TERMINAL-021`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. From exact paired particle/antiparticle support, a positive retained baryon residue exists exactly when baryon tally change, CP-distinguished conjugate paths and a nonequilibrium reverse-completion hold are all present; the other seven complete combinations return the empty-One residue record.

> *From exact paired particle/antiparticle support, a positive retained baryon residue exists exactly when baryon tally change, CP-distinguished conjugate paths and a nonequilibrium reverse-completion hold are all present; the other seven complete combinations return the empty-One residue record.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-DYNAMICS-001`
- `SFT-FOUNDATION-PART-EQUIVALENCE-001`
- `SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`
- `SFT-PHYS-MATTER-CKM-TERMINAL-004`

- SFT-PHYS-MATTER-BARYON-PHOTON-TERMINAL-004
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of initial carrier, baryon-tally action, conjugate-path relation, reverse-completion relation, residue ledger, dependency composition, enumeration, physical correspondence, provenance disclosure and extension forms. The declared exact boundary is: Every finite minimal process beginning from one positive paired particle/antiparticle carrier, with each of baryon-tally change, conjugate-path distinction and reverse-completion hold either present or absent, plus the already admitted terminal CP and baryon-abundance comparison records. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `exact-paired-particle-antiparticle-carrier__one-explicit-tally-changing-transition__CP-carrier-distinguishes-conjugate-paths__nonequilibrium-reverse-completion-held__positive-oriented-residue-or-empty-One__single-three-condition-process-composition__all-eight-presence-absence-combinations__inherit-sealed-CKM-and-baryon-photon-records__observational-reconstruction-explicit__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`exact-paired-particle-antiparticle-carrier`	`preloaded-unpaired-residue`	A preloaded residue assumes the result.
`tally`	`one-explicit-tally-changing-transition`	`every-transition-preserves-baryon-tally`	A preserving transition cannot create a tally difference from paired support.
`conjugacy`	`CP-carrier-distinguishes-conjugate-paths`	`exactly-paired-conjugate-paths`	Equal conjugate appends preserve the paired tally.
`reverse`	`nonequilibrium-reverse-completion-held`	`complete-reverse-recurrence`	Complete reversal closes every newly appended carrier.
`ledger`	`positive-oriented-residue-or-empty-One`	`signed-or-unrecorded-net-number`	A signed or missing ledger violates positive exact accounting.
`composition`	`single-three-condition-process-composition`	`named-independent-conditions`	Names alone do not prove their joint necessity.
`enumeration`	`all-eight-presence-absence-combinations`	`selected-successful-combination`	Selecting the familiar triple is not uniqueness.
`comparison`	`inherit-sealed-CKM-and-baryon-photon-records`	`abundance-selects-dependency-law`	A measured abundance cannot choose the formal conditions.
`provenance`	`observational-reconstruction-explicit`	`mislabel-as-blind-discovery`	The dependency triple was known before this V3 reconstruction.
`extension`	`no-extra-rule`	`free-efficiency-or-extra-condition`	An inserted efficiency or condition is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:97be399ff1fc4749dff9e890d336b1888f8ce4735e8a5fd642f065ef5acbfffa`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:e3bae526e83cc28f339b9d5a9119cad60a18dc63bdb69e86fceefa06ba8ceb5c`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:0e3b154bfe87d76d7f8129c5874028dc59692ca2a8aa7d3b2635f6b3cdb7a4c0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:29ad32c527daca782aa19ff642ad10b565934de369d47aac2730a7ff659fbf1b`.

- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:0f15328abb659835ff35fdb4efba0235867f24814e58f0230bd90c896b51d2e8`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c8f6b747b46a795c765a24c8f311aa2d4d7f8e4e4082ed396e7290bf6a168091`. Independent certificate: `sha256:098fcbe06ef474f48600931558aa9ca3ba817711e07989ed650609ec262d57df`. Engine external-validation hash: `sha256:9acd0a3e91324547d90bd78059493ffe57c7a37963659598ad08e2afd459738f`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported Sakharov theorem or conventional kinetic equation as a premise
- no V1/V2 executable, result table, measurement or stored survivor
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no measured baryon abundance, fitted efficiency or selected mechanism in the derivation
- no claim that the V3 reconstruction was a blind historical discovery
- no cosmic-history claim beyond the universal physical dependency law

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:b51006c613c918a6983ff9d857f3e515f4c955917ce1aeca6552c7a1316f25bb`; engine receipt `sha256:738dd11360b3cb43280ce6b595023c6f6408166ab6ef96ad4f8e9a6d4592c7b4` at `receipts/engine/model_admitted/SFT-PHYS-BARYOGENESIS-DEPENDENCY-TERMINAL-021-738dd11360b3cb43.json`; empirical-validation hash `None`; measurement receipt `None`.

190. Prime-sector standing-mode, interaction and period unification

Claim identity: `SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The forced sectors (2,3,5,7) share one exact m-indexed interaction table; half-One uniquely unifies the odd-sector standing modes; all finite common-support sector gaps are positive and shrink without finite coincidence; the gravity/EM/strong Fold periods are 1/2/3 with joint recurrence 6; and the inconsistent flat-EM slope bundle is rejected in favour of the sealed terminal carrier-specific running laws.

> *The forced sectors (2,3,5,7) share one exact m-indexed interaction table; half-One uniquely unifies the odd-sector standing modes; all finite common-support sector gaps are positive and shrink without finite coincidence; the gravity/EM/strong Fold periods are 1/2/3 with joint recurrence 6; and the inconsistent flat-EM slope bundle is rejected in favour of the sealed terminal carrier-specific running laws.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-DYNAMICS-001`
- `SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002`
- `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003`

- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-STRONG-RUNNING-DIRECTION-002
- SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003
- SFT-PHYS-COUPLING-RUNNING-CONVERGENCE-TERMINAL-006
- SFT-PHYS-VALIDATION-FORCE-SECTOR-ANCHORS-003
- SFT-PHYS-VALIDATION-VACUUM-POLARIZATION-003
- SFT-MATH-ORBIT-NUMBER-THEORY-002
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of sector domain, standing-mode carrier, coupling table, mass relation, common support, sector order, coincidence boundary, period dictionary, empirical succession and extension forms. The declared exact boundary is: Every forced prime sector through seven, every interior standing mode, every positive binary common-support successor, every ordered sector pair, the exact Fold periods of One, one-third and one-seventh, and the already sealed complete force/running comparison records. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `complete-prime-ladder-through-seven__unique-self-antipodal-half-One-mode__single-m-indexed-exact-table__positive-shortfall-from-unison__common-positive-binary-support__exact-positive-m-difference-gap__finite-triple-coincidence-forbidden__complete-one-two-three-period-dictionary__terminal-carrier-specific-running-and-anchors__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
sector	complete-prime-ladder-through-seven	selected-familiar-sectors	Selection omits generated prime sectors or the exclusion boundary.
standing	unique-self-antipodal-half-One-mode	named-unification-point	A name does not prove membership or uniqueness.
table	single-m-indexed-exact-table	independent-fitted-couplings	Independent values violate the one-generator law.
mass	positive-shortfall-from-unison	inserted-mass-parameter	An inserted mass is not forced by the holding relation.
support	common-positive-binary-support	independent-scale-axes	Independent axes cannot define exact separation or convergence.
order	exact-positive-m-difference-gap	target-named-physical-order	A physical label cannot choose an inequality.
coincidence	finite-triple-coincidence-forbidden	asserted-unification-crossing	A crossing contradicts the positive pair-gap identity.
period	complete-one-two-three-period-dictionary	selected-force-periods	Selected labels provide no Fold traces.
comparison	terminal-carrier-specific-running-and-anchors	flat-EM-and-inconsistent-slope-bundle	It conflicts internally and with admitted electromagnetic running.
extension	no-extra-rule	free-group-or-crossing-scale	An added group or fitted crossing scale is a free premise.

Base and successor. The closure proof is sealed by derivation hash `sha256:286814bb194316d6ca1b4ba21051191f7ae874f2b884b018593d5a189349f367`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:11291e339eeb79bc8fd240fa7f224091ae8d90181454b0e5f7dc54036db11e0f`.

- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:3f3679f7090946e7e35c8c24b3404ea6d728db8441fbcee8d15c9340a77c896c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4bc46d67f83fe4f2fd275512e8d0fe0b5fbf7b4d79bd1c8c8a2af05dff6ad616`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:f15b3d34a2b7c0ee7df313404be18a2a1be08f98b252e38425e7f7c638322e0c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:912c5c89d9d541e061c7403105de6f544a261de230b53815c44e381fc3a03606`. Independent certificate: `sha256:761d3decd7f1740758885b49c4b939885091bc84bd9b04ba311257a8cecd52ee`. Engine external-validation hash: `sha256:b3654b21b34c6f939fa95016665043430242b00e09da2b02541fd3988504c9a8`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported gauge-unification group, beta function, coupling crossing or mass model
- no V1/V2 executable, answer table, measured target or stored survivor
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no physical-sector ordering substituted for the exact common-support m ordering
- no retention of the internally inconsistent slope=m-1 and slope(2)=0 bundle
- no target access before the inherited force-sector and running seals

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:33d4c53c23e99c7b5b4b44662cc05b6531060e42157a5153505898281139c14a`; engine receipt `sha256:464f20ad41d716a52836a9d5952a62991e76b488a2ea70539ea6f4c0ee8c8f0e` at `receipts/engine/model_admitted/SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025-464f20ad41d716a5.json`; empirical-validation hash `None`; measurement receipt `None`.

191. Finite Fold colour-singlet positive excitation gap

Claim identity: `SFT-PHYS-YANG-MILLS-SINGLET-GAP-TERMINAL-026`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Within the complete finite Fold colour grammar, the least observable colour singlet has normalised gap one-third above the empty excitation record at every positive finite depth; confinement work advances by two-thirds; the two shares form a period-two partition of the One; and the locally massless confined carrier is retained without being mistaken for a free gapless physical singlet.

> *Within the complete finite Fold colour grammar, the least observable colour singlet has normalized gap one-third above the empty excitation record at every positive finite depth; confinement work advances by two-thirds; the two shares form a period-two partition of the One; and the locally massless confined carrier is retained without being mistaken for a free gapless physical singlet.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013
- SFT-PHYS-STRONG-FIELD-NONLINEAR-FIXED-POINT-TERMINAL-014
- SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of colour sector, observable support, vacuum boundary, gap carrier, confinement work, local-carrier distinction, Fold orbit, depth closure, prior-claim correction and extension forms. The declared exact boundary is: The generated colour-three Fold sector, its complete antipodal and three-member singlet supports, every positive finite separation/depth successor, the locally massless confined carrier and the normalized One partition. No continuum field action or completed infinite-volume limit is asserted. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `generated-colour-three-sector__complete-colour-singlet-support__empty-excitation-record__exact-positive-one-third-complement__fixed-two-thirds-positive-work-successor__retain-local-massless-physical-gapped-distinction__complete-period-two-Fold-partition__positive-finite-successor-induction__correct-free-singlet-spectrum-boundary__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
sector	generated-colour-three-sector	named-Yang-Mills-sector	A conventional name does not generate colour structure.
support	complete-colour-singlet-support	isolated-colour-carrier	The admitted confinement law leaves the isolated-colour observation record empty.
vacuum	empty-excitation-record	numerical-zero-vacuum	Numerical zero is outside the Fold proof domain.
gap	exact-positive-one-third-complement	massless-or-fitted-gap	A fitted value or an empty mass label conflates local carrier and physical singlet.
confinement	fixed-two-thirds-positive-work-successor	bounded-or-free-separation-work	Bounded or free work permits an unrecorded free colour state.
carrier	retain-local-massless-physical-gapped-distinction	conflate-local-carrier-with-physical-singlet	The local carrier is not an observable isolated colour state.
orbit	complete-period-two-Fold-partition	selected-gap-coupling-pair	A selected pair does not prove closure or recurrence.
depth	positive-finite-successor-induction	bounded-depth-census-or-completed-infinity	A bounded sample or completed infinity cannot certify every finite successor.
prior	correct-free-singlet-spectrum-boundary	repeat-no-massless-strong-excitation	That wording contradicts the admitted locally massless strong carrier.
extension	no-extra-rule	free-scale-action-or-continuum-limit	An added scale or action is a new premise.

Base and successor. The closure proof is sealed by derivation hash `sha256:17d2c3576cd428a73269198c334acb0f1cf802b25f6e497ed6bacfca9b0190a4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:c99ca2dfdaa551521dbaf64ffa4fe813fb7ebf7aefd9fe92a43cc9acc3490815`.

- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:7dfce3e24867adf20b37bd68a2974b3187df30d0a3fd70d622a21cb0ad1425ef`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f0a61704c36c3b494f221da8100787eed72ce7f23f01f0bf4ccc2768b06fa77c`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:7cb2b037e883a22a7ffc40dc3dfc3ce55a2407fe78599ed0360d879f2713a4e6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9036da93a060fe7853f080b732a716bc5660a07b3dff95439a56848c43e86f33`. Independent certificate: `sha256:7e8adb8391a99dcbd0a57c2801f6a9929eedd49ef4ebe7aeb5419dda306c6226`. Engine external-validation hash: `sha256:fcf8862fd656cdff3479ca59a90b686dd0ad9ed1d8d1b5844e42682bfd549305`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported Yang-Mills action, gauge-group axiom, Hamiltonian or continuum mass-gap proof
- no V1/V2 executable, answer table, lattice mass or stored survivor
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no fitted dimensionful glueball scale or claim that one-third equals a mass in MeV
- no completed infinity or inference from a finite simulation to a conventional continuum theorem
- no conflation of the empty local carrier mass label with the least physical singlet gap

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:98fe4e9d3c65ed14b2416232187dd76ef9816f3774b984860d87138ad775fb82`; engine receipt `sha256:899e904600c1018041286e947de6acefe6c1afb50f1f8da202cfda587fa9568a` at `receipts/engine/model_admitted/SFT-PHYS-YANG-MILLS-SINGLET-GAP-TERMINAL-026-899e904600c10180.json`; empirical-validation hash `None`; measurement receipt `None`.

192. Post-seal lattice-spectrum boundary for the colour-singlet gap

Claim identity: `SFT-PHYS-YANG-MILLS-SINGLET-GAP-EMPIRICAL-027`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The sealed Fold prediction of a positive physical colour-singlet gap matches every retained PDG pure-gauge lattice interval: 0++ at 1653+/-26 MeV, 2++ at 2376+/-32 MeV and 0-+ at 2561+/-40 MeV are each strictly positive and their intervals are ordered and disjoint. These are external lattice records, not fitted Fold values. Quenching, state mixing, full-QCD requirements and the lack of an unambiguous experimental glueball remain explicit boundaries.

> *The sealed Fold prediction of a positive physical colour-singlet gap matches every retained PDG pure-gauge lattice interval: 0++ at 1653+/-26 MeV, 2++ at 2376+/-32 MeV and 0-+ at 2561+/-40 MeV are each strictly positive and their intervals are ordered and disjoint. These are external lattice records, not fitted Fold values. Quenching, state mixing, full-QCD requirements and the lack of an unambiguous experimental glueball remain explicit boundaries.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-YANG-MILLS-SINGLET-GAP-TERMINAL-026`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of physical carrier, positive-gap relation, provenance, prediction isolation, measurement separation, complete spectrum-row retention, finite successor closure and extension. The declared exact boundary is: The admitted finite Fold colour-singlet gap and every registered PDG 2026 pure-gauge glueball mass/uncertainty row, ground/excitation ordering row, quenched, mixing, full-QCD and experimental-identification limitation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__positive-colour-singlet-gap-with-complete-spectrum-scope__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`positive-colour-singlet-gap-with-complete-spectrum-scope`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b91011e3dc72496d920bcc5b4aed304377a9c66b2f4de7e181c4ecb7a3175197`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:67966cb00d6a0c9fd1864193d149e7e29bc3e4a4678bb14be657f7527f7419db`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f303362e4cc90b1b2c95b5098b87e69832a55e960e23b88ca6df495dd0df0dae`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8feb5e24377ab231eb8515c33743564d296e3e046311f170e02121a0cef32202`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

`sha256:f2b80835649f5c2ba9eb104f5c7dbe64338493657e15e5b076ce067128d5a8d8`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4f4480d8febbdb8825c6729c2869fd8f455314664312e2be313925fafec7b9ad`. Independent certificate: `sha256:b6b4b4ec429281b86ec1e5485d020d152dcb0bb939f415377433ec4648bbde77`. Engine external-validation hash: `sha256:241e721b4221701046a50bdd437bb7c219336e8e53cf0d6ab0f93767d5b1f364`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2026-QUARK-MODEL-GLUEBALL-BOUNDARY

Observed comparison records:

- The PDG 2026 source record and complete PDF are hash-locked and released only after the Fold prediction seal.
- All three registered pure-gauge lattice mass and uncertainty rows are retained and evaluated with exact rational inscriptions.
- The 0++ interval 1627-1679 MeV, 2++ interval 2344-2408 MeV and 0-+ interval 2521-2601 MeV are each strictly positive and mutually ordered without overlap.
- The source identifies 0++ as the lattice ground state and 2++ as the first excitation; these quantum-number labels are retained external records, not separately forced by this Fold claim.
- The quenched approximation, neglected quark loops, glue/quark-state mixing, full-QCD requirements and high-statistics limitation are retained.
- The PDG nonidentification boundary is retained: no scalar below 2 GeV is established as predominantly glueball, and no unambiguous experimental glueball is claimed.
- No dimensionful lattice value selects, fits or numerically identifies the exact normalised Fold gap one-third.
- A deliberately altered lowest-mass row breaks interval ordering and is rejected.

Falsification condition: Reject if either source snapshot changes; if any registered spectrum or limitation row is omitted; if the least retained pure-gauge mass interval reaches the empty boundary; if registered intervals lose their stated ordering; if an external mass is imported into the Fold proof; if a lattice prediction is relabelled as direct detection; or if the target is available to the capability-closed predictor before sealing.

Measurement receipt: `sha256:3835a4b4efc277cbd7a7a69ec3afb546254ca82d841b405b6f5abb3bb9100b0f`. Isolation certificate: `sha256:f0903f9280b06201a1a69b2e3297c8741e7ad4ef85af1ae70c9956984e919fa8`. Custody certificate: `sha256:6e92af1f8314d310047e62c9a970652cb5ce748cdf6a635c001b9934a40c35b0`.

Registered source descriptions:

- Particle Data Group: https://pdg.lbl.gov/2026/reviews/rpp2026-rev-quark-model.pdf (sha256:6aa98fa53857122f27b638c59081af1a2857d787c4205370d8fa38fcb6b70ff0)

Registered target identities:

- PDG-2026-WITHHELD-COMPLETE-GLUEBALL-BOUNDARY from PDG-2026-QUARK-MODEL-GLUEBALL-BOUNDARY

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no PDG, lattice mass, uncertainty or quantum-number ordering in the formal predecessor derivation
- no claim that normalised one-third equals 1653 MeV or any dimensionful mass
- no omission of quenched, mixing, full-QCD or unresolved-identification limitations
- no claim of direct experimental glueball detection or conventional continuum Yang-Mills proof

- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude
- no target access before the prediction seal and no fitted correction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:24e898b09f1ff360265f53be7b7fb3c7ef2555cdc64056bc0118c2c0fa00f61b`; engine receipt `sha256:d85984ba94789e15d9fdb97588e63d318b014b3e84315ace56cb50c16cbeb2fe` at `receipts/engine/model_admitted/SFT-PHYS-YANG-MILLS-SINGLET-GAP-EMPIRICAL-027-d85984ba94789e15.json`; empirical-validation hash `sha256:b96e76915046971454a96f7d502c2a1f81c6b84d810d1c21cab293c1e99a1e6c`; measurement receipt `sha256:3835a4b4efc277cbd7a7a69ec3afb546254ca82d841b405b6f5abb3bb9100b0f`.

193. Depth-independent composite confining-sector orbit law

Claim identity: `SFT-PHYS-COMPOSITE-CONFINING-SECTOR-TERMINAL-031`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete composite-sector census is: sector 8, denominator 7, orbits (1,2,4)/(3,6,5), three antipodal pairs and coupling 7/8; sector 12, denominator 11, one ten-mode orbit, five pairs and 11/12; sector 18, denominator 17, two eight-mode orbits, eight pairs and 17/18; sector 24, denominator 23, two eleven-mode orbits, eleven pairs and 23/24; sector 30, denominator 29, one twenty-eight-mode orbit, fourteen pairs and 29/30. The general odd-support inverse proves denominator confinement and complete recurrence at every positive finite Fold depth.

> *The complete composite-sector census is: sector 8, denominator 7, orbits (1,2,4)/(3,6,5), three antipodal pairs and coupling 7/8; sector 12, denominator 11, one ten-mode orbit, five pairs and 11/12; sector 18, denominator 17, two eight-mode orbits, eight pairs and 17/18; sector 24, denominator 23, two eleven-mode orbits, eleven pairs and 23/24; sector 30, denominator 29, one twenty-eight-mode orbit, fourteen pairs and 29/30. The general odd-support inverse proves denominator confinement and complete recurrence at every positive finite Fold depth.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-COUNT-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-FOUNDATION-FOLD-DYNAMICS-001`
- `SFT-FOUNDATION-PART-EQUIVALENCE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-ORBIT-NUMBER-THEORY-002`
- `SFT-PHYS-FOLD-UNIVERSE-TRANSPORT-TERMINAL-024`
- `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003`
- `SFT-PHYS-VALIDATION-FORCE-SECTOR-ANCHORS-003`
- `SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025`

Generated grammar and closure boundary. Generate the complete product of sector scope, held support, standing-mode census, Fold action, confinement, orbit partition, antipodal closure, pair-count law, holding coupling, external correspondence boundary and extension form. The declared exact boundary is: Every positive finite even sector beyond two; its odd positive predecessor denominator; every nonempty numerator label; every exact Fold successor and first return; every complement pair; the five registered composite-sector instances; and the already sealed force-sector anchor boundary. The generator produced `2048` named candidates and the decision support contains `2048` one-for-one decisions. Exactly one candidate survived: `every-generated-even-sector__positive-predecessor-denominator__all-nonempty-predecessor-labels__double-and-cast-denominator-wholes__odd-support-exact-inverse__complete-disjoint-first-return-partition__unique-positive-One-complements__predecessor-of-denominato` `r-over-two__all-but-one-predecessor-over-sector__sealed-eight-anchor-and-explicit-standing-cases__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`scope`	`every-generated-even-sector`	`five-selected-case-table`	A selected table does not prove the next even sector.
`support`	`positive-predecessor-denominator`	`borrowed-or-measured-denominator`	A named or measured denominator can select the orbit.
`modes`	`all-nonempty-predecessor-labels`	`selected-mode-subset`	A subset cannot establish exhaustive confinement.
`action`	`double-and-cast-denominator-wholes`	`selected-orbit-permutation`	A listed permutation imports its answer.
`confinement`	`odd-support-exact-inverse`	`finite-table-assertion`	A table alone does not close arbitrary successor depth.
`orbits`	`complete-disjoint-first-return-partition`	`named-or-overlapping-orbits`	Named or overlapping rows omit a complete partition proof.
`antipodes`	`unique-positive-One-complements`	`borrowed-pairing-rule`	A borrowed pair rule does not prove exhaustiveness.
`pair_count`	`predecessor-of-denominator-over-two`	`ambiguous-sector-minus-one-over-two`	That notation is not a whole count for even s.
`coupling`	`all-but-one-predecessor-over-sector`	`measured-or-fitted-coupling`	Measurement cannot select a holding law.
`comparison`	`sealed-eight-anchor-and-explicit-standing-cases`	`unobserved-cases-counted-as-confirmed`	A standing prediction is not a measurement.
`extension`	`no-extra-rule`	`free-orbit-or-coupling-correction`	An inserted row or correction is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:f2a635c4e730cc95000af2f718ae9c0ea451904511d39264b7aa4669be821395`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:d64de949d0073feb311cb6b076043542586f4bb0f6d164c1c5ae451b8344c2f8`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:de67bce63f980db5ef3b82782a6980729c913a1abb9e175ed3ce896e696d48ce`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8720f86993e5ca432ed6f38fd4bc5f64f288b42723ba14aa0f57729cc611249a`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:11ce4408c46af0dca4b297e5207ff15539732f5323461ae6ca66338a157e79e4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d0a35a4a0cf61ded5a2a3438c5811e48274991518e20380c4132a18b441f638c`. Independent certificate: `sha256:d3cb445ee39882793a1669e1a0519589f5c1d7353d1f4f8f2c1d1822de9d0cac`. Engine external-validation hash: `sha256:a4b6ff145b49bdf9e7752f6b380faa2559e021103010f6efd6932681e2fe9e7e`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, stored orbit table or prior survivor in the derivation runtime
- no imported gauge group, particle taxonomy, measured coupling or target-selected sector
- no semantic numerical zero, negative, irrational, imaginary, floating or completed-infinity proof value
- no claim that every arithmetic composite sector is already an observed independent physical force
- no relabelling of the standing sectors twelve, eighteen, twenty-four or thirty as measured confirmations
- no ambiguous half-whole pair count: the exact whole formula is (s-2)/2

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:fdfab054033e97c405ab0a5d9237425c3b5ee4acaf1e3ccc154e47160c5454ab`; engine receipt `sha256:eb8b53e1662db23eff80d371b5b6db094e3c7d4665b5ea0a6882ba3072e29ec8` at `receipts/engine/model_admitted/SFT-PHYS-COMPOSITE-CONFINING-SECTOR-TERMINAL-031-eb8b53e1662db23e.json`; empirical-validation hash `None`; measurement receipt `None`.

194. Terminal general Fold-fibre, particle-mode and generation-transport law

Claim identity: `SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every positive finite m>=2, the m-Fold has exactly m positive preimages and m-2 interior fixed modes. The four generation coordinate records are complete ordered three-label systems and are not masses. The exact 27-cell state volume forces binary cover depth five and colour cover depth three. Every finite recurrent particle class injects into internal Fold modes without adding a spatial dimension. Generation order transports to the admitted terminal lepton, quark and neutrino polynomial carriers; superseded site-as-mass readings are rejected. The exact dual seeds are 1/95, 1/383 and 1/485; at every positive depth the heavy/light ratio equals the subtraction reach N-1; and the quarter-mode 1:1:4 record is a structural squared-separation multiplicity, not a universal measured mixing-rate ratio.

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Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-ONE-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-FOLD-DYNAMICS-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`

- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003
- SFT-PHYS-MATTER-QUARK-INVARIANTS-003
- SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001
- SFT-PHYS-NEUTRINO-POSITIVE-MASS-003
- SFT-PHYS-MATTER-GENERATION-DEPTH-003
- SFT-PHYS-MATTER-CKM-TERMINAL-004

Generated grammar and closure boundary. Generate the complete product of fibre, fixed-mode, generation-coordinate, cover-depth, particle-placement, mass-transport, dual, reach, transition and extra-rule forms. The declared exact boundary is: Every positive finite m-Fold by depth-independent index induction; all four complete three-label coordinate systems; the exact 27-cell cover; every finite recurrent particle-trace census; all admitted terminal charged-lepton, quark, neutrino and CKM carriers; and every superseded site-as-mass and separation-as-observed-rate alternative. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `complete-m-offset-fibre__complete-fixed-index-range__three-order-isomorphic-coordinate-systems__least-cover-of-generator-volume__internal-recurrent-trace-mode__order-transport-to-terminal-polynomials__complete-colour-binary-dual__exact-structural-reach-identity__structural-separation-multiplicity__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`fibre`	`complete-m-offset-fibre`	`selected-binary-or-colour-count`	A selected special case is not a general law.
`mode`	`complete-fixed-index-range`	`named-three-mode-pattern`	Naming three modes does not force the general count.
`generation`	`three-order-isomorphic-coordinate-systems`	`one-coordinate-system-selected-as-mass`	A coordinate value is not automatically a physical mass.
`cover`	`least-cover-of-generator-volume`	`chosen-depth-five-or-seven`	A selected depth is a free choice.
`placement`	`internal-recurrent-trace-mode`	`extra-spatial-dimension`	An extra spatial coordinate is not generated.
`mass`	`order-transport-to-terminal-polynomials`	`site-fraction-is-mass`	This superseded identification conflicts with the terminal spectra.
`dual`	`complete-colour-binary-dual`	`lookup-depth-fractions`	Looked-up fractions are not forced.
`reach`	`exact-structural-reach-identity`	`lifetime-equals-mass-ratio-postulate`	A dimensional lifetime equality is not forced by the count.
`transition`	`structural-separation-multiplicity`	`universal-observed-rate-ratio`	The old 1:1:4 physical-rate claim omits interaction carriers.
`extension`	`no-extra-rule`	`extra-dimension-or-fit`	Either addition is an unforced parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:c475546316355bcfdc77aac50982f522bb2e5d2837eec8b3175f6691668e0508`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:89d95f25da0998aeee0e2c12ef972877ed435ba65c15ba347f4ad8cc2464ef27`.

- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:49d418baf34f44b4527e9317cc194779bc21f294b0dfd0d94461154a0bb2fd63`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:847552ca2ada1c1c2f5999d26c3c4e4278c5ba52eae77a5d4ff309df95948836`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:4c316148e986b887cbd73d1b0bbce7b8c7dd31124b99dca5d77a7040247ff17f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d7e58bd3b815e0c502770da64b136f2235d69191a2118750f48392f9020b1738`. Independent certificate: `sha256:e3a787096b77a1d8c83910dd1d3de2908658797d5c75947f31aa4abc1c01b97e`. Engine external-validation hash: `sha256:c7a4c26b64bed04fe7a5cb92b586a03223af71bd5f707908a514762f11d6b4e8`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 result artifact, string-theory dimension rule, particle table or measured mass as a formal premise
- no identification of 1/6, 1/2, 5/6, 1/3, 2/3, One, 1/4 or 3/4 with terminal measured masses
- no claim that 1:1:4 is a universal observed mixing or decay-rate spectrum
- no extra spatial dimension, compactification choice, landscape count or target-selected mode
- no fitted mass coefficient, lifetime coefficient, mixing shift or omitted terminal sector
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:4e6de203282d2e77866724630a7ca291d01785551605bda4768e703600efb464`; engine receipt `sha256:18e4aac0792a01c613d31f4d980951b0e2da69107155c4e50967ed80c9eaa700` at `receipts/engine/model_admitted/SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051-18e4aac0792a01c6.json`; empirical-validation hash `None`; measurement receipt `None`.

195. Terminal dark relic, Smithion spectrum and flavour-transition law

Claim identity: `SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Four exact penta/hepta cubics at depths (7,10,9,12) each have three disjoint positive roots, yielding twelve exact rationally enclosed Smithion mass-ratio predictions. Their coloured product family cross-locks the two quark values 3/1454 and 3/13118. Complete neutral p-member singlets and unique lightest roots force the stable relic class within the closed grammar. Dark/baryon is 27/5 and matter/baryon 32/5. Flavour weights are exactly 1/32, 5/96 and 5/24, reducing to 3:5:20 with tau-to-e over tau-to-mu equal to 4. Sector beta slopes are p-1.

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grammar. Dark/baryon is 27/5 and matter/baryon 32/5. Flavour weights are exactly 1/32, 5/96 and 5/24, reducing to 3:5:20 with tau-to-e over tau-to-mu equal to 4. Sector beta slopes are p-1.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003`
- `SFT-PHYS-MATTER-CONFINEMENT-LIFT-003`
- `SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051`
- `SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001`
- `SFT-PHYS-FIELD-INVERSE-SQUARE-001`
- `SFT-PHYS-COMPOSITE-CONFINING-SECTOR-TERMINAL-031`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`

Generated grammar and closure boundary. Generate the complete product of sector, kind, depth, invariant, root-census, mass-ratio, neutral-singlet, stability, abundance, flavour-weight, measurement-boundary and extension forms. The declared exact boundary is: Prime sectors five and seven; down/up cover powers three/four; the complete exact cubic grid; all three root intervals; complete neutral sector fibres; all three lepton transitions; exact positive whole/fraction carriers; and no physical target before seal. The generator produced `2048` named candidates and the decision support contains `2048` one-for-one decisions. Exactly one candidate survived: `both-forced-prime-sectors__complete-down-up-dual__least-binary-cover-of-sector-power__same-coloured-product-family__complete-grid-and-rational-halving__squared-rational-enclosures-with-light-lift__least-complete-neutral-singlet__generation-volume-over-cover-depth__squared-separation-times-parent-part__all-targets-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`sector`	`both-forced-prime-sectors`	`selected-particle-sector`	A selected sector is not a complete census.
`kind`	`complete-down-up-dual`	`one-selected-kind`	One kind omits half the generated spectrum.
`depth`	`least-binary-cover-of-sector-power`	`chosen-depth`	A chosen exponent fits a hierarchy.
`invariant`	`same-coloured-product-family`	`independent-mass-parameter`	A mass parameter forks the law.
`roots`	`complete-grid-and-rational-halving`	`chosen-decimal-roots`	Chosen decimals import an irrational model.
`ratios`	`squared-rational-enclosures-with-light-lift`	`floating-central-estimates`	A central decimal loses exact root custody.
`relic`	`least-complete-neutral-singlet`	`named-dark-particle`	A name does not force stability or neutrality.
`abundance`	`generation-volume-over-cover-depth`	`freezeout-fit-or-density-input`	A cross-section or density would be a parameter.
`flavour`	`squared-separation-times-parent-part`	`fitted-branching-fractions`	Fitted rates cannot force a ratio.
`measurement`	`all-targets-inaccessible-until-seal`	`search-result-readable-before-seal`	That would tune the theory to current limits.
`extension`	`no-extra-rule`	`extra-scale-sector-or-correction`	Any added choice violates zero parameters.

Base and successor. The closure proof is sealed by derivation hash `sha256:3351ece9cf5b191e15416884e44dce897be9cb7cfaa2359d088604779ae3602e`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:3084e9fee79537fb19204951d2bc8e82d13335ddf0ad5d68a2c9054f0c7d738d`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:3187efb26cdc55782a7d746a3e39775a4088b3aa65b92c31e188588d54803f10`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1a08b1cc107e3eb8504a130861084b840018b107d069e3991364b95b3eda5f90`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:5d51b9f8d04a4587584e88486b2a63c0fc0677fb27663430bf131394d110606a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4eb56b948d11e8219bfddf29dac76fae3eb2a7d8e6cc994888e2fec4432e29c4`. Independent certificate: `sha256:f53233046528aef9158e3a637b87e7a874856c29aa86d1808a8568bc89eb5699`. Engine external-validation hash: `sha256:0d0180e4437cf2371bdd914f9b6b5519bbeaf33ba5d192dc4761bc66689728b7`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 code, answer table, survivor identifier or measurement in formal execution
- no fitted mass, coupling, cross-section, freeze-out efficiency, density, branching fraction or search limit
- no negative, irrational, imaginary, floating, NaN, infinite or continuum proof scalar
- no claim that an unobserved Smithion mass has already been experimentally measured
- no claim that non-detection alone identifies the relic

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:dfbaa171ffbee41651dcf56b1a0e0097d2cd088376d003686d56f094f7c8113a`; engine receipt `sha256:71a131119ffd152690ec772019177fb06599474cd5e89b60a62c1b9b69ec5762` at `receipts/engine/model_admitted/SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061-71a131119ffd1526.json`; empirical-validation hash `None`; measurement receipt `None`.

196. Terminal strong-CP alignment and proton/baryon stability law

Claim identity: `SFT-PHYS-STRONG-CP-BARYON-STABILITY-TERMINAL-063`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Strong-sector CP is the aligned One; the strong-violation, electric-dipole and added-compensator records are empty-One. The weak one-hand phase remains the self-antipodal half-One. All 83 generated mediators are fibre-preserving; all

202 ordered cross-fibre label pairs and all six colour-to-binary pairs lack a generated carrier. The proton carries baryon One = 3*(1/3), all 27 generated colour-three images retain it, and every finite composition remains in the colour fibre. Proton decay is absent within the complete declared grammar.

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Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-WEAK-PARITY-FIBRE-002`
- `SFT-PHYS-NEUTRINO-CP-PHASE-002`
- `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003`
- `SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-MATTER-COMPOSITE-HADRONS-001`
- `SFT-PHYS-NUCLEON-BINDING-TERMINAL-005`
- `SFT-PHYS-BARYOGENESIS-DEPENDENCY-TERMINAL-021`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`

Generated grammar and closure boundary. Generate the complete product of immutable predecessor, CP carrier, handed support, strong-sector product, compensator, sector-action typing, baryon tally, finite transition, baryogenesis boundary, measurement custody and extension forms. The declared exact boundary is: All nonempty subsets of the exact two-hand fibre; complete colour-three times hand support; every mediator cell in prime sectors two, three, five and seven; every ordered cross-fibre label pair; every colour-three word; and every finite composition of the generated typed actions. The generator produced `2048` named candidates and the decision support contains `2048` one-for-one decisions. Exactly one candidate survived: `compose-immutable-predecessors__aligned-One-or-self-antipodal-half-One__complete-two-hand-support__paired-parity-returns-aligned-One__empty-One-no-extra-compensator__all-generated-actions-are-fibre-preserving__three-one-third-parts-close-to-One__finite-composition-preserves-the-fibre__retain-explicit-cosmological-boundary__targets-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`compose-immutable-predecessors`	`rewrite-or-bypass-predecessors`	A successor cannot change an admitted receipt.
`cp`	`aligned-One-or-self-antipodal-half-One`	`continuum-angle-or-fitted-small-value`	A continuum angle or small fitted value is inadmissible.
`hand`	`complete-two-hand-support`	`selected-single-hand`	Selecting one hand omits half the colour-times-hand product.
`strong`	`paired-parity-returns-aligned-One`	`weak-antipode-reused-for-strong`	A one-hand weak carrier contradicts complete strong support.
`compensator`	`empty-One-no-extra-compensator`	`add-axion-or-free-compensator`	An extra carrier has no discrepancy to act on and forks the closed inventory.
`sector`	`all-generated-actions-are-fibre-preserving`	`invent-cross-fibre-mediator`	A two-sector signature is absent from every complete sector action table.
`baryon`	`three-one-third-parts-close-to-One`	`untyped-particle-name`	A name alone does not establish conservation.

Axis	Forced form	Eliminated form	Elimination reason
`transition`	`finite-composition-preserves-the-fibre`	`assume-an-unregistered-decay-step`	An unregistered fibre crossing inserts the desired decay.
`baryogenesis`	`retain-explicit-cosmological-boundary`	`conflate-distinct-process-grammars`	A cosmological tally-changing premise is not a low-energy mediator.
`measurement`	`targets-inaccessible-until-seal`	`target-readable-before-seal`	That would tune alignment or stability to observation.
`extension`	`no-extra-rule`	`free-angle-carrier-or-operation`	Any such addition is an unforced parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:d9f8979ae65fb3c20f78a9ab79262c68e94e265ee6c4ca439f4dfe72eae49a84`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:290a02b4e4f473a3ce4642758d4802afc041ce55fa1bdd67b6bfa67362291492`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:52f4575af714aac932f9d9b34d2ffa0e5c9f931c5e2344837a94e396c9dad408`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:552ae613a5e02f35cf12b163fba2b2bab42f8d26c5f5fa288400676c03132929`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:35261f06a81f2466962ec820e1b6b4ec55dd52c60bdf13960ff56c23e2f0db76`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:158f6d0db3af5495b5377fbaa556c17f4010b3ef24b7274d6c60f7e0c4608e67`. Independent certificate: `sha256:3fe766ec11ea0dba5bf00a0d2acd889ca3523c97ef518c6f35d6379d0c32fcab`. Engine external-validation hash: `sha256:fc2a863920358c2c6138acdbecd364cce3472795ef6a85c45b2f9695b2e4e9fe`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table, survivor identifier or measurement as a premise
- no conventional theta continuum, fitted small angle, negative, irrational, imaginary, floating, NaN or infinite proof scalar
- no neutron electric-dipole bound or proton-lifetime limit in formal execution
- no imported Peccei-Quinn, axion, grand-unification or Standard Model mechanism as a proof premise
- no contradiction of the separately admitted cosmological baryogenesis dependency law

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:751d04511116f683c0771afbc6752af128a9360cedbb8b98ad9ba08a2cbc6818`; engine receipt `sha256:6bd69d2eb78150c11fefc82f512bdfd775eaa2f3a2667127407d7a34bd695657` at `receipts/engine/model_admitted/SFT-PHYS-STRONG-CP-BARYON-STABILITY-TERMINAL-063-6bd69d2eb78150c1.json`; empirical-validation hash `None`; measurement receipt `None`.

197. Terminal displaced-ground and Higgs mass/self-coupling law

Claim identity: `SFT-PHYS-HIGGS-SYMMETRY-TERMINAL-065`

Shared claim-record clause `PHYS-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The displaced ground is uniquely half-One; the retained leading Higgs rungs are 1/2, 1/4 and 1/8. Complete six-cell scalar support, its unique held return and the independently equal cover depth five force terminal m_H/v = 1/2 + (6/5) alpha = 2563352914777/5038463954690. The exact native self-coupling is lambda = (m_H/v)^2/2 = 6570778165695741824959729/50772238045420788745992200. Both routes cross-lock in the squared domain; no irrational root is formed.

> *The displaced ground is uniquely half-One; the retained leading Higgs rungs are 1/2, 1/4 and 1/8. Complete six-cell scalar support, its unique held return and the independently equal cover depth five force terminal m_H/v = 1/2 + (6/5) alpha = 2563352914777/5038463954690. The exact native self-coupling is lambda = (m_H/v)^2/2 = 6570778165695741824959729/50772238045420788745992200. Both routes cross-lock in the squared domain; no irrational root is formed.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-ONE-001`
- `SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002`
- `SFT-PHYS-MEDIATOR-RANGE-CHANNEL-TERMINAL-020`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`

Generated grammar and closure boundary. Generate the complete product of ground carrier, displacement, leading rungs, scalar directional support, held return, cover cross-lock, terminal transport, coupling composition, route equality, measurement custody and extension forms. The declared exact boundary is: Structural empty-One, half-One and One ground forms; every binary successor of the displaced ground through the leading coupling rung; the complete binary-hand times generator-direction product; its unique held return; the least cover of generator volume; exactly one terminal alpha transport; exact positive rational operations; and no physical target before seal. The generator produced `2048` named candidates and the decision support contains `2048` one-for-one decisions. Exactly one candidate survived: `positive-proper-ground-carrier__unique-self-antipodal-half-One__retain-half-quarter-eighth-controls__complete-two-by-three-product__hold-unique-return-leaving-five__least-cover-of-generation-volume__one-six-over-five-alpha-return__squared-excitation-shared-over-two-fibres__exact-two-route-cross-lock__all-targets-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`ground`	`positive-proper-ground-carrier`	`absence-as-numerical-ground`	Structural absence is not a field value.

Axis	Forced form	Eliminated form	Elimination reason
displacement	unique-self-antipodal-half-One	unison-or-selected-part	Unison is not displaced and a selected part is a parameter.
rungs	retain-half-quarter-eighth-controls	discard-or-rewrite-leading-rungs	A terminal successor cannot erase its lower-order receipt.
directions	complete-two-by-three-product	selected-direction-subset	A subset omits a generated hand or generator direction.
return	hold-unique-return-leaving-five	retain-all-six-as-active	The invariant return cannot also be an active displaced direction.
cover	least-cover-of-generation-volume	free-denominator	A denominator chosen for agreement is a parameter.
transport	one-six-over-five-alpha-return	fitted-offset-or-repeated-series	An offset or truncation is a fit.
coupling	squared-excitation-shared-over-two-fibres	import-potential-or-free-lambda	An imported potential or free lambda would add an axiom.
route	exact-two-route-cross-lock	independent-mass-and-coupling-routes	Independent routes introduce a dial.
measurement	all-targets-inaccessible-until-seal	target-readable-before-seal	That would fit the coefficient to the measured mass.
extension	no-extra-rule	extra-mass-potential-or-correction	Any added term is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:b032f36759d5585227839385874ee6a097037af0b1142ebf0738a290b480bff5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:92eb8365d3cc30c6263094989f3acef1ac062d7e661401ab949da0b36db56255`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:c1821578d2ccf8237255f446bbde7e3117de5edcd59b00b37071f3e485c6a946`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7f5b272fccfbef1f1ce7eabace8825d5f3c8073c754e1e4b9ab230a2b07b530d`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:ea6d7a5ec8dbbde9369629b7a0919e4233a1aec722a5894371da443dd9818951`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:89f6565f52c17b5f9eb1f217a3dabd38f0142336cd4e23900204d370aa4e78bc`. Independent certificate: `sha256:348c2daebd8443b15a0943ea97c6336d0767529492929a71a15fa788a4a30d53`. Engine external-validation hash: `sha256:3482618408366fc7c558674ede8b2c5ee86d7b0d6e80b42a2373df5bc0ca70db`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, target value, candidate table or stored survivor as a premise
- no measured Higgs mass, VEV, Fermi constant or self-coupling in formal execution
- no imported Higgs potential, Standard Model mass relation, fitted coefficient, uncertainty or correction series
- no semantic numerical zero, negative, irrational, imaginary, floating, NaN, continuum or infinite proof scalar
- no claim that the leading 1/2 mass ratio or 1/8 coupling is the terminal result

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:2c07deac3b179ef81471403a48877e0da72f003cae44166043302a1f0f0a973b`; engine receipt `sha256:3f84e4ede48a489bc78a8cc47eac935bbff3566d1e561c26ef0059d7bbcbfb1e` at `receipts/engine/model_admitted/SFT-PHYS-HIGGS-SYMMETRY-TERMINAL-065-3f84e4ede48a489b.json`; empirical-validation hash `None`; measurement receipt `None`.

198. Complete penta sector phenotype

Claim identity: `SFT-PHYS-PENTA-COMPLETE-PHENOTYPE-088`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Sector 5 has exact shortfall 1/5, coupling 4/5, 5 charge kinds, 24 mediators and 2 complete antipodal confinement pairs. Its carrier has an empty-One mass/rest record, advances one exact support cell per tick, and self-confines in the half-One tube that closes to the One. Every antipodal pair and the complete 5-constituent fibre are neutral at the One.

> *Sector 5 has exact shortfall 1/5, coupling 4/5, 5 charge kinds, 24 mediators and 2 complete antipodal confinement pairs. Its carrier has an empty-One mass/rest record, advances one exact support cell per tick, and self-confines in the half-One tube that closes to the One. Every antipodal pair and the complete 5-constituent fibre are neutral at the One.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003`
- `SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013`
- `SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for the sector-5 predecessor, boundary, exact phenotype, carrier, confinement, inventory, measurement direction and extension form. The declared exact boundary is: The admitted prime-sector ladder at sector 5; all 4 nonidentity charge parts; their complete antipodal pairing; the p-squared-less-One carrier census; mass/rest, propagation and tube records; and all 256 structural forms. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `retain-admitted-prime-sector-predecessors__generated-penta-hepta-boundary__sector-5-complete-forced-phenotype__empty-mass-One-speed-carrier__half-One-tube-and-complete-antipodes__category-clean-finite-inventory__formal-seal-before-observation__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`retain-admitted-prime-sector-predecessors`	`rewrite-or-bypass-predecessors`	Bypassing predecessors severs the root-directed derivation.
`sector`	`generated-penta-hepta-boundary`	`open-or-target-selected-sector`	An open or target-selected sector is a free choice.
`relation`	`sector-5-complete-forced-phenotype`	`free-phenotype-or-slope`	A free phenotype or slope is a parameter.
`carrier`	`empty-mass-One-speed-carrier`	`massive-or-speed-fitted-carrier`	A fitted mass or speed imports an unforced magnitude.

Axis	Forced form	Eliminated form	Elimination reason
confinement	half-One-tube-and-complete-antipodes	unbounded-width-or-unpaired-charge	Unbounded width or an unpaired charge does not close the fibre.
inventory	category-clean-finite-inventory	mixed-category-or-open-inventory	Mixed counting or an open list cannot certify completeness.
measurement	formal-seal-before-observation	target-selected-survivor	Target selection is fitting.
extension	no-extra-rule	free-extra-rule	An exception is an unforced parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9290edfc390d9f420b6bac817d7e52b4d7cb4c2200360cc93d2d71d81682face`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:cadc4d05ba430a6fca7212a36e9a7fd9cd5693f3c20849958a098e7cd1b16edc`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:e0538f1682a782cc2bd335ca27aabad15bb8572f61442c414121f1c0ffe1e4b5`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4e6ce5084099d383bbc6b9071b62330a5891e2d38e3b6bf022e25579691f3598`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:e74a5fd2440d54bafa52f5e96e5752c11aadbeb7938705285148939b36423574`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b771bec2c075a5bba9e57101e0c4ff26f6b6b5a1b69a5a98d3fff1d62c50449f`. Independent certificate: `sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60`. Engine external-validation hash: `sha256:34e47ff260f4e763bf6c9e2baa79c483be31a1b39657947b8665b74885222c80`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, desired phenotype, experimental target or conventional force model as a premise
- no free coupling, fitted slope, invented cross-fibre mediator or open-ended sector
- no patent, collider anomaly, null search or reputation selecting a formal survivor
- no confirmed discovery asserted for a penta/hepta carrier or Smithion
- no numerical-zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:ebbc6d9bffbe29c333bb3d6571d6b342b118d22a52fd00b84cda9bc4c7ac72ce`; engine receipt sha256:11362854f921cda3962cfed7578380b32cca2663c85de8528d4aa17895fe20c4 at receipts/engine/model_admitted/SFT-PHYS-PENTA-COMPLETE-PHENOTYPE-088-11362854f921cda3.json; empirical-validation hash None; measurement receipt None.

199. Complete hepta sector phenotype

Claim identity: SFT-PHYS-HEPTA-COMPLETE-PHENOTYPE-089

Shared claim-record clause PHYS-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Sector 7 has exact shortfall 1/7, coupling 6/7, 7 charge kinds, 48 mediators and 3 complete antipodal confinement pairs. Its carrier has an empty-One mass/rest record, advances one exact support cell per tick, and self-confines in the half-One tube that closes to the One. Every antipodal pair and the complete 7-constituent fibre are neutral at the One.

> *Sector 7 has exact shortfall 1/7, coupling 6/7, 7 charge kinds, 48 mediators and 3 complete antipodal confinement pairs. Its carrier has an empty-One mass/rest record, advances one exact support cell per tick, and self-confines in the half-One tube that closes to the One. Every antipodal pair and the complete 7-constituent fibre are neutral at the One.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-PENTA-COMPLETE-PHENOTYPE-088
- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product for the sector-7 predecessor, boundary, exact phenotype, carrier, confinement, inventory, measurement direction and extension form. The declared exact boundary is: The admitted prime-sector ladder at sector 7; all 6 nonidentity charge parts; their complete antipodal pairing; the p-squared-less-One carrier census; mass/rest, propagation and tube records; and all 256 structural forms. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: retain-admitted-prime-sector-predecessors__generated-penta-hepta-boundary__sector-7-complete-forced-phenotype__empty-mass-One-speed-carrier__half-One-tube-and-complete-antipodes__category-clean-finite-inventory__formal-seal-before-observation__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	retain-admitted-prime-sector-predecessors	rewrite-or-bypass-predecessors	Bypassing predecessors severs the root-directed derivation.
sector	generated-penta-hepta-boundary	open-or-target-selected-sector	An open or target-selected sector is a free choice.
relation	sector-7-complete-forced-phenotype	free-phenotype-or-slope	A free phenotype or slope is a parameter.
carrier	empty-mass-One-speed-carrier	massive-or-speed-fitted-carrier	A fitted mass or speed imports an unforced magnitude.
confinement	half-One-tube-and-complete-antipodes	unbounded-width-or-unpaired-charge	Unbounded width or an unpaired charge does not close the fibre.
inventory	category-clean-finite-inventory	mixed-category-or-open-inventory	Mixed counting or an open list cannot certify completeness.
measurement	formal-seal-before-observation	target-selected-survivor	Target selection is fitting.
extension	no-extra-rule	free-extra-rule	An exception is an unforced parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:97e7a62c1009e36e891034570068d64b5e85e2ec517fcfc81b42a9d77db071a0. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:1d9f0770fb9bfa3a6786ff210a6c6d0179b407fbbf5da3c00ab48bb273d1e0f9`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:d66fb60c2b54d272470ba7e79540e5989aa9b6350dcfcdb1ebc2071aea3720e7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:aad3dadc3eb9d130e8cd20ea1c5410b9a5c6bff82dabcc95d07355364d6bcebf`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:0a24b302ea63b65961c11d7f2dc11f3d3c0164f7ddc1f32e5ffd108595c3ab56`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b771bec2c075a5bba9e57101e0c4ff26f6b6b5a1b69a5a98d3fff1d62c50449f`. Independent certificate: `sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60`. Engine external-validation hash: `sha256:237a501fa276f11b2eefb73a0ce140605393e7453a747bff31e3153d779f9384`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, desired phenotype, experimental target or conventional force model as a premise
- no free coupling, fitted slope, invented cross-fibre mediator or open-ended sector
- no patent, collider anomaly, null search or reputation selecting a formal survivor
- no confirmed discovery asserted for a penta/hepta carrier or Smithion
- no numerical-zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:f258bfa0660ab5e93f09f4bd325e4865ff987c86b498260688322ea51949c4b5`; engine receipt `sha256:57d1a7b380211d15d035e2853e74fa81dc7cd150f4acadef9a55aa1d8efb7a1a` at `receipts/engine/model_admitted/SFT-PHYS-HEPTA-COMPLETE-PHENOTYPE-089-57d1a7b380211d15.json`; empirical-validation hash `None`; measurement receipt `None`.

200. Independent sector-5 beta-slope law

Claim identity: `SFT-PHYS-PENTA-BETA-SLOPE-090`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For sector 5, the exact coupling (4)/5 divided by the exact shortfall 1/5 is the positive whole 4. Thus the sector-5 beta-slope carrier is independently 4. A measured sector-5 slope unequal to 4, after a sector-5 carrier is identified, is its distinct falsifier and does not retire the other sector's law.

For sector 5, the exact coupling (4)/5 divided by the exact shortfall 1/5 is the positive whole 4. Thus the sector-5 beta-slope carrier is independently 4. A measured sector-5 slope unequal to 4, after a sector-5 carrier is identified, is its distinct

> *falsifier and does not retire the other sector's law.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-PENTA-COMPLETE-PHENOTYPE-088`
- `SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for the independently unbundled sector-5 beta slope, carrier, confinement, inventory, measurement and extension forms. The declared exact boundary is: Exact sector 5, its coupling and shortfall, every positive exact ratio reconstruction, its distinct slope falsifier and all 256 structural alternatives. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `retain-admitted-prime-sector-predecessors__generated-penta-hepta-boundary__sector-5-coupling-divided-by-shortfall-equals-4__empty-mass-One-speed-carrier__half-One-tube-and-complete-antipodes__category-clean-finite-inventory__formal-seal-before-observation__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`retain-admitted-prime-sector-predecessors`	`rewrite-or-bypass-predecessors`	Bypassing predecessors severs the root-directed derivation.
`sector`	`generated-penta-hepta-boundary`	`open-or-target-selected-sector`	An open or target-selected sector is a free choice.
`relation`	`sector-5-coupling-divided-by-shortfall-equals-4`	`free-phenotype-or-slope`	A free phenotype or slope is a parameter.
`carrier`	`empty-mass-One-speed-carrier`	`massive-or-speed-fitted-carrier`	A fitted mass or speed imports an unforced magnitude.
`confinement`	`half-One-tube-and-complete-antipodes`	`unbounded-width-or-unpaired-charge`	Unbounded width or an unpaired charge does not close the fibre.
`inventory`	`category-clean-finite-inventory`	`mixed-category-or-open-inventory`	Mixed counting or an open list cannot certify completeness.
`measurement`	`formal-seal-before-observation`	`target-selected-survivor`	Target selection is fitting.
`extension`	`no-extra-rule`	`free-extra-rule`	An exception is an unforced parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:6484ac72f2922ba56544ec63fdd3b1be388d23eb623fac1e262d22933b805b67`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:89ca4cc433c4294e07640e102168569b20d3c850930cd8619f4c23e9aa9f71e3`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:59131819e39083116d3843c2765660c82379661bc6ddc2650543cb181763327d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:71bda1f62e8c960774a90182216d99365e37f1ee511453eccfd43ef4ca956820`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:fa464e52939f5e4ad525c1c567458ee60d0a653b7d40a8550eeaae3c519bc734`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b771bec2c075a5bba9e57101e0c4ff26f6b6b5a1b69a5a98d3fff1d62c50449f`. Independent certificate: `sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60`. Engine external-validation hash: `sha256:66a72ce615c96316e2b4b116ce02e700603433b2642d24aa3ce2d6f3da9d0f1a`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, desired phenotype, experimental target or conventional force model as a premise
- no free coupling, fitted slope, invented cross-fibre mediator or open-ended sector
- no patent, collider anomaly, null search or reputation selecting a formal survivor
- no confirmed discovery asserted for a penta/hepta carrier or Smithion
- no numerical-zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:3cb36e78037a8eeca5f9f42a399151ae6d0f6d0849f6609d2a098eefde194f34`; engine receipt `sha256:df6eeb2137767de6ae27e4dde9a6f4655629411248f30136c5cbdf82c3df9198` at `receipts/engine/model_admitted/SFT-PHYS-PENTA-BETA-SLOPE-090-df6eeb2137767de6.json`; empirical-validation hash `None`; measurement receipt `None`.

201. Independent sector-7 beta-slope law

Claim identity: `SFT-PHYS-HEPTA-BETA-SLOPE-091`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For sector 7, the exact coupling (6)/7 divided by the exact shortfall 1/7 is the positive whole 6. Thus the sector-7 beta-slope carrier is independently 6. A measured sector-7 slope unequal to 6, after a sector-7 carrier is identified, is its distinct falsifier and does not retire the other sector's law.

> *For sector 7, the exact coupling (6)/7 divided by the exact shortfall 1/7 is the positive whole 6. Thus the sector-7 beta-slope carrier is independently 6. A measured sector-7 slope unequal to 6, after a sector-7 carrier is identified, is its distinct falsifier and does not retire the other sector's law.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-HEPTA-COMPLETE-PHENOTYPE-089`
- `SFT-PHYS-PENTA-BETA-SLOPE-090`
- `SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for the independently unbundled sector-7 beta slope, carrier, confinement, inventory, measurement and extension forms. The declared exact boundary is: Exact sector 7, its coupling and shortfall, every positive exact ratio reconstruction, its distinct slope falsifier and all 256 structural alternatives. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `retain-admitted-prime-sector-predecessors__generated-penta-hept` a-boundary__sector-7-coupling-divided-by-shortfall-equals-6__empty-mass-One-speed-carrier__half-One-tube-and-complete-antipodes__category-clean-finite-inventory__formal-seal-before-observation__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	retain-admitted-prime-sector-predecessors	rewrite-or-bypass-predecessors	Bypassing predecessors severs the root-directed derivation.
sector	generated-penta-hepta-boundary	open-or-target-selected-sector	An open or target-selected sector is a free choice.
relation	sector-7-coupling-divided-by-shortfall-equals-6	free-phenotype-or-slope	A free phenotype or slope is a parameter.
carrier	empty-mass-One-speed-carrier	massive-or-speed-fitted-carrier	A fitted mass or speed imports an unforced magnitude.
confinement	half-One-tube-and-complete-antipodes	unbounded-width-or-unpaired-charge	Unbounded width or an unpaired charge does not close the fibre.
inventory	category-clean-finite-inventory	mixed-category-or-open-inventory	Mixed counting or an open list cannot certify completeness.
measurement	formal-seal-before-observation	target-selected-survivor	Target selection is fitting.
extension	no-extra-rule	free-extra-rule	An exception is an unforced parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:8cd99d8af546bf79596df7041f25f570b2704d069ef2130537111ab8f58b319f. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:59668e1a8836b68942e73812945d05e8ccb1c88fdd69f9d7cd6090318eb1283b.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:cefc9e4ab171f4d58ca559432dc3a4275f1867dc53c43589840b5c0df8b53ccf.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:d8ff54f2acceebd518bfc93895e3ff68db2ce8359c8b5a8412f7bec3c1f8083b.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:802664a10d97cabbfd570646d032224daf5bf21b9c3bc798f6160cac1dfde56b.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:b771bec2c075a5bba9e57101e0c4ff26f6b6b5a1b69a5a98d3fff1d62c50449f. Independent certificate: sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60. Engine external-validation hash: sha256:1b45956fc4a6c05bbddb9859269d1eab081b1e3f63d406884faf936bcaf7a466.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, desired phenotype, experimental target or conventional force model as a premise
- no free coupling, fitted slope, invented cross-fibre mediator or open-ended sector
- no patent, collider anomaly, null search or reputation selecting a formal survivor
- no confirmed discovery asserted for a penta/hepta carrier or Smithion
- no numerical-zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:f44f6908696035fe7ffc327f03664d0b7706f6dbdb5e5158093f36c77f653c09`; engine receipt `sha256:bcdbca84f9778bcf437fc0fe253f13f2ab4672db1414d5e1cccf78c7cc38607e` at `receipts/engine/model_admitted/SFT-PHYS-HEPTA-BETA-SLOPE-091-bcdbca84f9778bcf.json`; empirical-validation hash `None`; measurement receipt `None`.

202. Category-clean complete particle census

Claim identity: `SFT-PHYS-CATEGORY-CLEAN-PARTICLE-CENSUS-092`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The category-clean inventory contains 83 sector gauge carriers (3, 8, 24 and 48), three named nonsector bosons (photon, graviton and Higgs), twelve known fermion kinds (three charged leptons, three neutrinos and six quarks), and twelve Smithion kinds. These disjoint categories total 110 named fundamental kinds under the declared counting convention; 72 of the gauge carriers and all twelve Smithions are standing predictions, not measured discoveries.

> *The category-clean inventory contains 83 sector gauge carriers (3, 8, 24 and 48), three named nonsector bosons (photon, graviton and Higgs), twelve known fermion kinds (three charged leptons, three neutrinos and six quarks), and twelve Smithion kinds. These disjoint categories total 110 named fundamental kinds under the declared counting convention; 72 of the gauge carriers and all twelve Smithions are standing predictions, not measured discoveries.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-HEPTA-BETA-SLOPE-091`
- `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003`
- `SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051`
- `SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061`

Generated grammar and closure boundary. Generate the complete eight-axis product for retained predecessors, finite sector boundary, category-clean counts, carrier, confinement, inventory, measurement and extension. The declared exact boundary is: All four admitted prime-sector carrier counts, photon/graviton/Higgs, the twelve category-distinct known fermion kinds, the twelve generated Smithion kinds, every category assignment and all 256 structural alternatives. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `retain-admitted-prime-sector-predecessors__generated-penta-hepta-boundary__category-clean-total-110__empty-mass-One-speed-carrier__half-One-tube-and-complete-antipodes__category-clean-finite-inventory__formal-seal-before-observation__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`retain-admitted-prime-sector-predecessors`	`rewrite-or-bypass-predecessors`	Bypassing predecessors severs the root-directed derivation.
`sector`	`generated-penta-hepta-boundary`	`open-or-target-selected-sector`	An open or target-selected sector is a free choice.

Axis	Forced form	Eliminated form	Elimination reason
relation	category-clean-total-110	free-phenotype-or-slope	A free phenotype or slope is a parameter.
carrier	empty-mass-One-speed-carrier	massive-or-speed-fitted-carrier	A fitted mass or speed imports an unforced magnitude.
confinement	half-One-tube-and-complete-antipodes	unbounded-width-or-unpaired-charge	Unbounded width or an unpaired charge does not close the fibre.
inventory	category-clean-finite-inventory	mixed-category-or-open-inventory	Mixed counting or an open list cannot certify completeness.
measurement	formal-seal-before-observation	target-selected-survivor	Target selection is fitting.
extension	no-extra-rule	free-extra-rule	An exception is an unforced parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:50439459fb73fc405eac43886e7124c719c913d20922273b52660329f186505d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:35848ff6d6a680135d1c5e2466d5a9796d73e746a87c1fd88f9d13af059d2b38`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:15244d6d0284b216a3c13da8dd3f6f30ac87e2d2330509029357a30af65b77fe`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:37a7d996a5c8a72f1d51358975fb48d80b2a6b0444ceecfd6f706bdf7b066a25`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:c214e140a607910d9f7cda0c1541d4230e8f4b7e51d85f7ab85017a442398eb4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b771bec2c075a5bba9e57101e0c4ff26f6b6b5a1b69a5a98d3fff1d62c50449f`. Independent certificate: `sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60`. Engine external-validation hash: `sha256:02f612f058efaa1a8be71f6e6ea71ccd663818ce6977e4619469499e0eb09717`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, desired phenotype, experimental target or conventional force model as a premise
- no free coupling, fitted slope, invented cross-fibre mediator or open-ended sector
- no patent, collider anomaly, null search or reputation selecting a formal survivor
- no confirmed discovery asserted for a penta/hepta carrier or Smithion
- no numerical-zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:3bb655f533511a3347dd60ca0a093a7e0d990b54905074a4adb62c7477272e7b; engine receipt sha256:3d96ec5a6eb16daa8bc8971bfdb34d876fa76fb082da1737583c7eb209ec1e06 at receipts/engine/model_admitted/SFT-PHYS-CATEGORY-CLEAN-PARTICLE-CENSUS-092-3d96ec5a6eb16daa.json; empirical-validation hash None; measurement receipt None.

203. No-extra-sector and outside-particle falsification boundary

Claim identity: SFT-PHYS-NO-EXTRA-SECTOR-PARTICLE-BOUNDARY-093

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The admitted confining-sector ladder ends at prime seven and the first excluded prime is eleven. The 110-kind category-clean inventory is therefore extension-open only through a new lawful derivation that changes this boundary; under the present law, a confirmed fundamental axion, an additional sterile/fourth neutrino kind, a fundamental supersymmetric partner, a confining prime sector beyond seven, or any other confirmed fundamental kind outside the list falsifies this census. A null search is not such a discovery and does not close a standing prediction.

> *The admitted confining-sector ladder ends at prime seven and the first excluded prime is eleven. The 110-kind category-clean inventory is therefore extension-open only through a new lawful derivation that changes this boundary; under the present law, a confirmed fundamental axion, an additional sterile/fourth neutrino kind, a fundamental supersymmetric partner, a confining prime sector beyond seven, or any other confirmed fundamental kind outside the list falsifies this census. A null search is not such a discovery and does not close a standing prediction.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CATEGORY-CLEAN-PARTICLE-CENSUS-092
- SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002
- SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051
- SFT-PHYS-STRONG-CP-BARYON-STABILITY-TERMINAL-063

Generated grammar and closure boundary. Generate the complete eight-axis product for the finite sector ceiling, first excluded prime, category-clean list, explicit outside-list falsifiers, measurement direction and extension. The declared exact boundary is: Prime sectors through seven, the next prime eleven, the complete category-clean total, all five named outside-list falsifier classes and all 256 structural alternatives. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: retain-admitted-prime-sector-predecessors__generated-penta-hepta-boundary__first-excluded-prime-eleven-and-explicit-outside-list-falsifiers__empty-mass-One-speed-carrier__half-One-tube-and-complete-antipodes__category-clean-finite-inventory__formal-seal-before-observation__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	retain-admitted-prime-sector-predecessors	rewrite-or-bypass-predecessors	Bypassing predecessors severs the root-directed derivation.
sector	generated-penta-hepta-boundary	open-or-target-selected-sector	An open or target-selected sector is a free choice.
relation	first-excluded-prime-eleven-and-explicit-outside-list-falsifiers	free-phenotype-or-slope	A free phenotype or slope is a parameter.
carrier	empty-mass-One-speed-carrier	massive-or-speed-fitted-carrier	A fitted mass or speed imports an unforced magnitude.
confinement	half-One-tube-and-complete-antipodes	unbounded-width-or-unpaired-charge	Unbounded width or an unpaired charge does not close the fibre.
inventory	category-clean-finite-inventory	mixed-category-or-open-inventory	Mixed counting or an open list cannot certify completeness.

Axis	Forced form	Eliminated form	Elimination reason
`measurement`	`formal-seal-before-observation`	`target-selected-survivor`	Target selection is fitting.
`extension`	`no-extra-rule`	`free-extra-rule`	An exception is an unforced parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:eccd7560ea58c86022bafb593c418053ccc720c93acc3c3c618896bd70bf1501`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:bf406d79b8ec30e929930bac30a14aaac82db161ed1dd4b49c33f07e21f03315`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:15eff8736c3223226b171c3113c5b68a8e25e4b92b37cc6a8ebea75fd78f47ad`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f4e3bd28abf00543f8717763a9290dd03740b0d9a26dd384fdf6bc024744c22b`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:ad92600518e4dd1319791397fd984c8e4ca4353df80dd027efa9c91fb260eb6a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b771bec2c075a5bba9e57101e0c4ff26f6b6b5a1b69a5a98d3fff1d62c50449f`. Independent certificate: `sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60`. Engine external-validation hash: `sha256:e09b123e994c614d72b80bfc2c74b4479978ad84b3785f16bf3a6dbfd0e8994e`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, desired phenotype, experimental target or conventional force model as a premise
- no free coupling, fitted slope, invented cross-fibre mediator or open-ended sector
- no patent, collider anomaly, null search or reputation selecting a formal survivor
- no confirmed discovery asserted for a penta/hepta carrier or Smithion
- no numerical-zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude

Shared claim-record clause `PHYS-S004` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:03e43449e90b9110fd01c531e1386121e22eca9374c14c7366503f45efe900b6`; engine receipt `sha256:aee65ca87af0ec74235a3e62c475d6254301ef623dbbebef5c1582aab0bb9cd9` at `receipts/engine/model_admitted/SFT-PHYS-NO-EXTRA-SECTOR-PARTICLE-BOUNDARY-093-aee65ca87af0ec74.json`; empirical-validation hash `None`; measurement receipt `None`.

204. Exact Smithion interaction and search-signature law

Claim identity: `SFT-PHYS-SMITHION-INTERACTION-SEARCH-SIGNATURES-094`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For both penta and hepta Smithions, the complete new-sector singlet has an empty-One electromagnetic-charge record and no generated cross-fibre carrier for an ordinary nuclear-recoil interaction. Universal gravitational response remains at the One. If produced in a sufficiently energetic interaction, sector-five and sector-seven carriers form distinct confining classes with 24 and 48 mediator states, neutral 5- and 7-constituent bound states and a lightest neutral missing-carrier record. These are exact search classes; no cross section, threshold, rate or discovery is claimed without an apparatus successor.

> *For both penta and hepta Smithions, the complete new-sector singlet has an empty-One electromagnetic-charge record and no generated cross-fibre carrier for an ordinary nuclear-recoil interaction. Universal gravitational response remains at the One. If produced in a sufficiently energetic interaction, sector-five and sector-seven carriers form distinct confining classes with 24 and 48 mediator states, neutral 5- and 7-constituent bound states and a lightest neutral missing-carrier record. These are exact search classes; no cross section, threshold, rate or discovery is claimed without an apparatus successor.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-NO-EXTRA-SECTOR-PARTICLE-BOUNDARY-093`
- `SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061`
- `SFT-PHYS-STRONG-CP-BARYON-STABILITY-TERMINAL-063`
- `SFT-PHYS-GRAVITY-EQUIVALENCE-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for each new sector's fibre, interaction carrier, confinement, category-clean signature inventory, measurement direction and extension. The declared exact boundary is: Both new sectors, their complete neutral singlets, all admitted cross-fibre carrier records, universal gravitational response, 24/48 carrier jet classes, the lightest neutral relic record and all 256 structural alternatives. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `retain-admitted-prime-sector-predecessors__generated-penta-hepta-boundary__electromagnetically-dark-no-generated-nuclear-recoil-gravity-and-confining-jet-signatures__empty-mass-One-speed-carrier__half-One-tube-and-complete-antipodes__category-clean-finite-inventory__formal-seal-before-observation__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`retain-admitted-prime-sector-predecessors`	`rewrite-or-bypass-predecessors`	Bypassing predecessors severs the root-directed derivation.
`sector`	`generated-penta-hepta-boundary`	`open-or-target-selected-sector`	An open or target-selected sector is a free choice.
`relation`	`electromagnetically-dark-no-generated-nuclear-recoil-gravity-and-confining-jet-signatures`	`free-phenotype-or-slope`	A free phenotype or slope is a parameter.
`carrier`	`empty-mass-One-speed-carrier`	`massive-or-speed-fitted-carrier`	A fitted mass or speed imports an unforced magnitude.
`confinement`	`half-One-tube-and-complete-antipodes`	`unbounded-width-or-unpaired-charge`	Unbounded width or an unpaired charge does not close the fibre.
`inventory`	`category-clean-finite-inventory`	`mixed-category-or-open-inventory`	Mixed counting or an open list cannot certify completeness.
`measurement`	`formal-seal-before-observation`	`target-selected-survivor`	Target selection is fitting.
`extension`	`no-extra-rule`	`free-extra-rule`	An exception is an unforced parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:1d1ff4a75559905705ea958f090d2d5bc5ca690fd06b174757854377455745ee`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:47df96fe68496731a7b39a77d08e2b2ce1f36a4beae5d1e008f01a4496cb5a79`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:d980575bcdd9c2abf5534f44a01145ee606c2ed2e7dbf6670c61deb584616861`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5fad426b19ec6174072a8504e49b362603508850144l393db52dca6f0018c9d6`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:82824c16b6e75cdca8e9c23fceed31412ffd933bcf4e4400079e92ce380ccaf1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b771bec2c075a5bba9e57101e0c4ff26f6b6b5a1b69a5a98d3fff1d62c50449f`. Independent certificate: `sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60`. Engine external-validation hash: `sha256:efe2cc8d8f5809e1d622447daf39a17ee0e6ef5f3481bb0b4e043ecd3b47d42f`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S017` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, desired phenotype, experimental target or conventional force model as a premise
- no free coupling, fitted slope, invented cross-fibre mediator or open-ended sector
- no patent, collider anomaly, null search or reputation selecting a formal survivor
- no confirmed discovery asserted for a penta/hepta carrier or Smithion
- no numerical-zero, negative, irrational, imaginary, floating or completed-infinite proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:9364baaf2a3765bc3cbf4390575aafb17df21dc3b149184afbfb6ab4ab34ff7f`; engine receipt `sha256:5e1ffc9a06c962f4d8301714cf57d5a6e306884cf9c4c4fcfed4b43365c101b7` at `receipts/engine/model_admitted/SFT-PHYS-SMITHION-INTERACTION-SEARCH-SIGNATURES-094-5e1ffc9a06c962f4.json`; empirical-validation hash `None`; measurement receipt `None`.

19. Atomic Molecular

Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry.

205. Atomic orbit-cell capacity from three-space boundary orientation

Claim identity: `SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. At positive orbit rank r, three-space supplies one central orientation and the rank-two boundary supplies one held pair for every predecessor step. Two exclusion-distinct Fold labels occupy each orientation, forcing capacity 2(1+2(r-1)) and the sequence 2, 6, 10, 14, 18 without observation.

> *For every positive orbit rank r, capacity is exactly 2(1+2(r-1)); ranks one through five have capacities 2, 6, 10, 14 and 18.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete product of carrier, central class, successor extension, boundary multiplicity, Fold label, exclusion, capacity, generality, target custody and extension forms. The declared exact boundary is: All orbit-cell capacity laws generated from one central orientation, positive orbit succession, the complete rank-two boundary, two held Fold labels and exclusion. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `held-orientation-cell__one-central-orientation__one-boundary-orientation-pair__forced-rank-two-pair__both-Fold-labels__one-of-each-label-per-cell__labels-times-complete-orientations__constant-boundary-pair-successor__widths-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`held-orientation-cell`	`unlabelled-cell-count`	An unlabelled count loses orientation identity.
`central`	`one-central-orientation`	`empty-numerical-origin`	Numerical zero is not an SFT carrier.
`successor`	`one-boundary-orientation-pair`	`free-rank-increment`	A free increment is a parameter.
`boundary`	`forced-rank-two-pair`	`single-or-three-direction-addition`	One omits a boundary side and three re-adds the held normal.
`label`	`both-Fold-labels`	`label-erased-or-selected`	Erasure or selection violates complete Fold support.
`exclusion`	`one-of-each-label-per-cell`	`duplicate-cell-label`	Duplication destroys distinguishability.
`capacity`	`labels-times-complete-orientations`	`linear-or-doubling-without-boundary`	Those former survivors ignore the newly admitted boundary discriminator.
`generality`	`constant-boundary-pair-successor`	`finite-width-list`	A list has no next-rank proof.
`target`	`widths-inaccessible-until-seal`	`width-list-visible`	That reverses derivation and test.
`extension`	`no-extra-rule`	`extra-degeneracy-rule`	An added degeneracy is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:d97b49ecab2334cd7ab5c9089e53add64d6b9b53f6894a207434c195e870ca2d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:1ed87436e21f59d043967ca432026eaebd71e5e1f8be1907692eab24d946c6d5`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:625315de8974967651ba198f65bcbd903e36bfac7a0cc349d449aed6298445b7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4bf7aa1fcc6d37bda661b6b898af0d9b3a48ba75d0f12afa04f8db6b363f4d61`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:1c45ec73b2de2f2c722af6fcddecc09f844cab496f9bb77bdbdda1f0c85f80ab`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:86c0b909d7fb32a35993571e1c3646d129604a21264f1da55aabad2b205c428e`. Independent certificate: `sha256:dd12a5fc73eacfd47d2eeaaf16cc2925d756b0b88926041d958ecc2867474ec0`. Engine external-validation hash: `sha256:ab724acf6f42444e7b65bc6eeac91a59f5ac455d2dc8b55f81a01cfc413ad772`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S019` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:c2552746629a5f793d7ba62cc2b81a5f7e9292868e3e451c54ebd7d8b105364f`; engine receipt `sha256:1f3ba9a0a82d5ea4f7b36b71f2bc3b135be8f92d2f5f1c3e77182f7b8979e02c` at `receipts/engine/model_admitted/SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001-1f3ba9a0a82d5ea4.json`; empirical-validation hash `None`; measurement receipt `None`.

206. Exact atomic existence boundary

Claim identity: `SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A neutral atomic coordinate remains structurally admissible only while its positive whole charge does not exceed the exact inverse fine-structure Fold ratio. The unique greatest whole below that sealed ratio is 137, with 137 below and its successor 138 above the boundary.

> *The unique greatest positive whole atomic coordinate under the exact Fold binding ceiling is 137; 137 <= 503846395469/3676744786 < 138.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-ATOMIC-NUMBER-001`
- `SFT-CHEM-ELEM-PERIODIC-ORDER-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`

Generated grammar and closure boundary. Generate the complete product of charge carrier, coupling, ceiling, endpoint construction, lower witness, upper witness, generality, target custody and extension forms. The declared exact boundary is: All greatest-positive-whole atomic-coordinate conclusions under the One binding ceiling and the independently sealed exact inverse fine-structure ratio. The generator produced `512` named candidates and the decision support contains `512` one-for-one decisions. Exactly one candidate survived: `positive-held-charge-count__charge-over-sealed-inverse-ratio__complete-One-ceiling__greatest-whole-division-certificate__exact-lower-inequality__successor-above-plus-order__depth-independent-greatest-whole-law__observations-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`charge`	`positive-held-charge-count`	`unbound-number`	An unbound number has no element identity.
`coupling`	`charge-over-sealed-inverse-ratio`	`fitted-binding-equation`	A fitted equation imports the endpoint.
`ceiling`	`complete-One-ceiling`	`external-critical-charge`	An external critical charge is target input.
`endpoint`	`greatest-whole-division-certificate`	`bounded-neighborhood-walk`	A chosen search interval does not close all successors.
`lower`	`exact-lower-inequality`	`endpoint-label-only`	A label has no inequality witness.
`upper`	`successor-above-plus-order`	`single-next-value-check-only`	One check without order closure does not exclude later values.
`generality`	`depth-independent-greatest-whole-law`	`finite-coordinate-census`	A finite census leaves later coordinates open.
`target`	`observations-inaccessible-until-seal`	`observed-elements-visible`	That would turn an observation boundary into a law.
`extension`	`no-extra-rule`	`extra-size-parameter`	A size correction is a different, parameterized model.

Base and successor. The closure proof is sealed by derivation hash `sha256:550cd3fa2c1b453057cc26ac7b17de19a6b7b45308ae4682e86b214660450b9d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:4160fd712b17226fbf3a8cc047e5446f5375e7bd3fa14fd1c64641c517a5e38e`.

- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:eba05c35e0e176c852c05a3f09315b28f7bd0cbe56b2511e8b5f44780bd2472a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fe957e7c888b14fab4f760796244349be17a05c08bb8a440bb09417b28d8a928`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:a37c7574414b2ddd6b071742183623ac6758ef060b905ba5bbf38a6617730b62`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:55d5c09b7d7696f1d73843966a64fb278a666537289fd9472b7c3e557a83e55c`. Independent certificate: `sha256:d4b358d1dc307a24df8800cec04b9db1f575e9e52f920356fabfbf7739b7e2d2`. Engine external-validation hash: `sha256:381f7b512bae94ad41b60401e178bb3751701c3eac70833b176436db373f2e7b`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S019` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:1ed349863e322e77f6c9ddfe683227ea0520c64b2331b364be1d06496e7791f9`; engine receipt `sha256:7ef00e2c6474de6640916036a369c6bb49c988e45332eb04529113f58b45e2b7` at `receipts/engine/model_admitted/SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001-7ef00e2c6474de66.json`; empirical-validation hash `None`; measurement receipt `None`.

207. Cubic atomic support and exact neighbour balance

Claim identity: `SFT-PHYS-ATOMIC-CUBIC-SUPPORT-004`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Three-space supplies three axes and the binary Fold supplies the two held directions on each axis, forcing six nearest neighbours. Dividing the half-One balance equally over those complete neighbours forces weight one-twelfth per neighbour and their exact sum is the half-One.

> *The unique cubic support has 2*3=6 nearest neighbours, exact weight 1/12 per neighbour and complete balance 6/12=1/2.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-FOUNDATION-HALF-ONE-001`

- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis cubic-support product. The declared exact boundary is: All nearest-neighbour organizations using every held direction on every forced spatial axis and an exact equal partition of the half-One balance. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__complete-typed-support__exact-positive-Fold-operation__held-label-orientation__forced-counted-depth__target-inaccessible-formal-relation__complete-root-directed-trace__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	untyped-observed-label	A label alone is not a generated law.
support	complete-typed-support	selected-partial-support	A selected subset can tune the result.
operation	exact-positive-Fold-operation	imported-continuum-operation	An imported continuum operation violates the exact domain.
orientation	held-label-orientation	signed-or-erased-direction	Signed proof values or erased direction violate the domain.
depth	forced-counted-depth	selected-depth	A chosen depth is a parameter.
provenance	target-inaccessible-formal-relation	measurement-readable-relation	A measured target cannot construct or select the survivor.
trace	complete-root-directed-trace	result-without-dependency-trace	An untraced result cannot be admitted.
extension	no-extra-rule	free-extra-rule	An extra rule introduces a choice.

Base and successor. The closure proof is sealed by derivation hash sha256:98ba9e4afaabf76679c1d6f1011d96fca9d2bad2cf9728b0bf3bc3cdda30eef8. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:3962c8446b0f2c5cc4da2bb6f72bca60722b395df7888a482d679d9dcfe83efe.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:19b698c707c9a455606734e8059b4e78d4f8095991c62ad7d34ff516beb53611.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:b0cf6ad66e665745c70a806342def3ec57c78117c33486c7415675a8a3fce98f.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:47d6285a90253f07f50b5b152957fb5d8f652f63123263ad7b805d1e763c2bab.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:b32467e7d48c61fa7f961f38a35ead41a4c7b2cd871d1d51aabedb13aa09b1e6. Independent certificate: sha256:2e36a22788368f121f778a56291dc5c0f9254233375e4fcaee7bad546218fd98. Engine external-validation hash: sha256:ca1b84299c80439ee2c7550a2094b18db30925cc5f6674754054af51357a960d.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S019 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported cubic lattice
- no measured coordination number
- no floating or signed proof value
- no selected neighbour subset

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:afbd1739770387458474abfbc467386a6746ceb7ab03ecf7a3ff9630a08b0da6; engine receipt sha256:8fa8b8bb81ea31d715fae3c5f3ceb869b5e4d8afad6c959ff86f12ab2ea7471a at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-CUBIC-SUPPORT-004-8fa8b8bb81ea31d7.json; empirical-validation hash None; measurement receipt None.

208. Depth-independent exact hydrogen spectral ladder

Claim identity: SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every supplied positive finite principal count n, the binary Fold forces the exact bound support one over n squared. An emitted line is the guarded positive difference between the lower and upper supports, forcing Lyman-alpha three-quarters and Balmer-alpha five-thirty-sixths without a continuum solver.

> *For every positive finite n, H(n)=1/n^2; H(1)-H(2)=3/4 and H(2)-H(3)=5/36, with every emitted line an exact guarded positive difference.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003
- SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis hydrogen-support product and the positive finite principal-count successor. The declared exact boundary is: All exact positive finite Coulomb-bound principal supports generated by the binary exponent and all ordered emitted transition gaps between them. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__complete-typed-support__exact-positive-Fold-operation__held-label-orientation__forced-counted-depth__target-inaccessible-formal-relation__complete-root-directed-trace__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	untyped-observed-label	A label alone is not a generated law.
support	complete-typed-support	selected-partial-support	A selected subset can tune the result.
operation	exact-positive-Fold-operation	imported-continuum-operation	An imported continuum operation violates the exact domain.
orientation	held-label-orientation	signed-or-erased-direction	Signed proof values or erased direction violate the domain.

Axis	Forced form	Eliminated form	Elimination reason
depth	forced-counted-depth	selected-depth	A chosen depth is a parameter.
provenance	target-inaccessible-formal-relation	measurement-readable-relation	A measured target cannot construct or select the survivor.
trace	complete-root-directed-trace	result-without-dependency-trace	An untraced result cannot be admitted.
extension	no-extra-rule	free-extra-rule	An extra rule introduces a choice.

Base and successor. The closure proof is sealed by derivation hash `sha256:2324379b0dded9987f1c5298650eca06fe03487674562cd942e8c3aa1c7a6170`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:9ad1dcf544500f7bfee7f57d704b81e8ba36b249fe9ec348be5bcd7e91042fa8`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:a503e335734dce29a3e66cd48009361f4756281238383caef59dc7d2e9b1ff2c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:10c26242805a62fe6560b570a7d524ba4713f416ee3fcfa13d43c478064220a1`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:23d2dd786a0239ae2cd67d3cf3a0805e5115eca5a39de1aaa1d0d6c96baa857c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b32467e7d48c61fa7f961f38a35ead41a4c7b2cd871d1d51aabedb13aa09b1e6`. Independent certificate: `sha256:2e36a22788368f121f778a56291dc5c0f9254233375e4fcaee7bad546218fd98`. Engine external-validation hash: `sha256:10f3e86c883efea5fe0bb93c9b0353434dac24e8becaa30091736bfd1450a9ae`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S019` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured wavelength or Rydberg value
- no continuum Schroedinger solution
- no completed infinity
- no numerical zero or negative energy

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:a0d511dc4a594dc2724d449a85150580a244d8b41e08614e5f5009855fa5c5bb`; engine receipt `sha256:ea9b3b9f34311e1f1f41962a5405ffad6250fcd74d75cf107a3b92d35dd2d464` at `receipts/eng` ine/model_admitted/SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004-ea9b3b9f34311e1f.json; empirical-validation hash None; measurement receipt None.

209. Exact atomic gross, fine and first bound-return hierarchy

Claim identity: SFT-PHYS-ATOMIC-CORRECTION-HIERARCHY-004

Shared claim-record clause PHYS-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The forced inverse fine-structure carrier supplies alpha exactly. Two held Fold directions force the relative fine scale alpha squared below gross; one complete bound-state self-return applies the same fine carrier once more, placing the first Lamb-family support at alpha to the fourth below gross.

> *Gross is the One scale, fine/gross is alpha^2, and the first complete bound-return support is alpha^4 of gross; the gross Rydberg carrier in electron-rest units is alpha^2/2.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003
- SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003

Generated grammar and closure boundary. Generate the complete eight-axis atomic-correction hierarchy product. The declared exact boundary is: The first two complete binary-direction correction layers on the admitted gross atomic support, with alpha transported once per direction and the first closed bound-state self-return applied once. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__complete-typed-support__exact-positive-Fold-operation__held-label-orientation__forced-counted-depth__target-inaccessible-formal-relation__complete-root-directed-trace__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	untyped-observed-label	A label alone is not a generated law.
support	complete-typed-support	selected-partial-support	A selected subset can tune the result.
operation	exact-positive-Fold-operation	imported-continuum-operation	An imported continuum operation violates the exact domain.
orientation	held-label-orientation	signed-or-erased-direction	Signed proof values or erased direction violate the domain.
depth	forced-counted-depth	selected-depth	A chosen depth is a parameter.
provenance	target-inaccessible-formal-relation	measurement-readable-relation	A measured target cannot construct or select the survivor.
trace	complete-root-directed-trace	result-without-dependency-trace	An untraced result cannot be admitted.
extension	no-extra-rule	free-extra-rule	An extra rule introduces a choice.

Base and successor. The closure proof is sealed by derivation hash sha256:4164bd671351dd706a0ec7b5d8cb71bd90ecd2dec356d7127759caa3a9e6246a. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt

sha256:ef9efe1ac0aa2d75e506bdefba5a10692f86c3d7e5077a18a3c075d0c2f44264.

- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:ecdb546f6df8f215021e8573c3a3a547043042fb47bc7dc4855956e3bb3b495a.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:1f9d9869ec2d0fd20479b04fb2d82a6bb36cf19cd71ae03a4731ca230df41595.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:05c8fc8f10c190261575325bf3dd9f0350b0ccf20a9f8c8cbdb04cf10f93d3a1.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:b32467e7d48c61fa7f961f38a35ead41a4c7b2cd871d1d51aabedb13aa09b1e6. Independent certificate: sha256:2e36a22788368f121f778a56291dc5c0f9254233375e4fcaee7bad546218fd98. Engine external-validation hash: sha256:e61d7937202716a6815f4ae472749d2bb25cb04df677e0bffba48b3da280581e.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S019 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported QED coefficient series
- no measured splitting
- no irrational or floating proof value
- no selected correction order

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:1e410425b3bb9024c271c6dde057a448c61cdbfaf058d73fdfbfa724bb7fe382; engine receipt sha256:a6239e61a184cf72e869fabe9c7d2dcc433366e30ef4bfdff68ebd0ff2a0fd9d at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-CORRECTION-HIERARCHY-004-a6239e61a184cf72.json; empirical-validation hash None; measurement receipt None.

210. Atomic transition, selection and field-splitting law

Claim identity: SFT-PHYS-ATOMIC-TRANSITION-SELECTION-004

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. One atomic Fold act transfers one generated orbital unit while preserving the conserved carrier trace. The binary fibre supplies the two magnetic orientations and their field-dependent separation; the complete three-axis held-direction support supplies six spatial orientations for electric displacement.

> *An elementary admitted transition changes one orbital support unit, retains its carrier, has two magnetic orientations and six complete cubic electric orientations.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-ATOMIC-CUBIC-SUPPORT-004

- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001

Generated grammar and closure boundary. Generate the complete eight-axis atomic-transition and field-orientation product. The declared exact boundary is: All one-act transitions between adjacent generated orbital supports with conserved carrier records, binary magnetic orientation and complete cubic electric orientation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__complete-typed-support__exact-positive-Fold-operation__held-label-orientation__forced-counted-depth__target-inaccessible-formal-relation__complete-root-directed-trace__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	untyped-observed-label	A label alone is not a generated law.
support	complete-typed-support	selected-partial-support	A selected subset can tune the result.
operation	exact-positive-Fold-operation	imported-continuum-operation	An imported continuum operation violates the exact domain.
orientation	held-label-orientation	signed-or-erased-direction	Signed proof values or erased direction violate the domain.
depth	forced-counted-depth	selected-depth	A chosen depth is a parameter.
provenance	target-inaccessible-formal-relation	measurement-readable-relation	A measured target cannot construct or select the survivor.
trace	complete-root-directed-trace	result-without-dependency-trace	An untraced result cannot be admitted.
extension	no-extra-rule	free-extra-rule	An extra rule introduces a choice.

Base and successor. The closure proof is sealed by derivation hash sha256:8da56a20c4d39d81de9d3bf34574686fbe413b10f9f291412be2e86ee24f2b27. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:b70e71d093e3aa59669d78ef48d7f2c73928b3cbb44e1a5acdd9dfc12e3b89bc.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:c9dfb5349a68bac1833ae88ce2b8865d8a7c0588e3e799f8067e27a31b0a1762.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:e5325454908fb5d544376777ffedb775560e959164a880f3339a2fee087fa26f.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:ceb37351349b876c94eb5c093777a9aafc486f8686426bf96a59d0e6d2359527.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:b32467e7d48c61fa7f961f38a35ead41a4c7b2cd871d1d51aabedb13aa09b1e6. Independent certificate: sha256:2e36a22788368f121f778a56291dc5c0f9254233375e4fcaee7bad546218fd98. Engine external-validation hash: sha256:50724e86eeedf70a3394ffb6309ab0abb8aaa08c0d273ba23661b1f39012ea29.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S019` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured line intensity
- no imported angular-momentum algebra
- no unrecorded multi-unit elementary jump
- no signed field proof value

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:8cffe700c5ddd7de94f9813647f587fe3653a6c650a2145240f2e3d96ae7785d`; engine receipt `sha256:34ea2b8ac5ee23d6cfe9e2ca37a190682c73830e53fb3b0c32ece6506e7f8de2` at `receipts/engine/model_admitted/SFT-PHYS-ATOMIC-TRANSITION-SELECTION-004-34ea2b8ac5ee23d6.json`; empirical-validation hash `None`; measurement receipt `None`.

211. Exact molecular rotational-vibrational spectral hierarchy

Claim identity: `SFT-PHYS-MOLECULAR-SPECTRUM-HIERARCHY-004`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The atomic electronic carrier occupies the first binary half-support. Molecular rotation and vibration are the next complete composed Fold level at the quarter-One; two such molecular quanta exactly recompose the electronic half-support, forcing the spectral ordering without choosing a measured frequency.

> *Electronic support is 1/2, molecular rotational-vibrational support is 1/4, and two molecular quanta exactly recompose the electronic support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004`
- `SFT-PHYS-WAVE-RESONANCE-001`
- `SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003`
- `SFT-CHEM-BOND-COVALENT-001`

Generated grammar and closure boundary. Generate the complete eight-axis electronic/molecular spectral-level product. The declared exact boundary is: The first atomic electronic half-support and its next complete binary-composed rotational/vibrational support, each used exactly once in the cross-level relation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__complete-typed-support__exact-positive-Fold-operation__held-label-orientation__forced-counted-depth__target-inaccessible-formal-relation__complete-root-directed-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`untyped-observed-label`	A label alone is not a generated law.
`support`	`complete-typed-support`	`selected-partial-support`	A selected subset can tune the result.
`operation`	`exact-positive-Fold-operation`	`imported-continuum-operation`	An imported continuum operation violates the exact domain.

Axis	Forced form	Eliminated form	Elimination reason
orientation	held-label-orientation	signed-or-erased-direction	Signed proof values or erased direction violate the domain.
depth	forced-counted-depth	selected-depth	A chosen depth is a parameter.
provenance	target-inaccessible-formal-relation	measurement-readable-relation	A measured target cannot construct or select the survivor.
trace	complete-root-directed-trace	result-without-dependency-trace	An untraced result cannot be admitted.
extension	no-extra-rule	free-extra-rule	An extra rule introduces a choice.

Base and successor. The closure proof is sealed by derivation hash `sha256:2b5bfc6bdb7efe2e8b64e023a8a1e49071caa364e8e50a134c26315d5bc880a4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:c56488a117fb913327e10bcc2de8a462ec19479f7c7cab13a6d86b4c43107d1c`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:a77c0590f6f02e1afb77f47d70e44b15d08f53466c73b017c9157559a9789a1b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e42410e762f0869baeb3c2205938d8ae6a31ba8faf12002ce55d9a1c6406ac9d`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:59475a84e7590e322b655736716d01b1a23b17b089f5298c52fe78a64580fb09`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b32467e7d48c61fa7f961f38a35ead41a4c7b2cd871d1d51aabedb13aa09b1e6`. Independent certificate: `sha256:2e36a22788368f121f778a56291dc5c0f9254233375e4fcaee7bad546218fd98`. Engine external-validation hash: `sha256:719e52706bce11716cbc1a7d0661304d9e97d3ff837d59b31ae1f3cdc39999c7`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S019` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no selected molecular frequency
- no imported rigid-rotor or harmonic-oscillator parameter
- no fitted isotope factor
- no floating proof value

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:087830fbf4a75d48ba2a44a86ee67841a65ba408f686797d8350a7eb0ccf1d4e`; engine receipt `sha256:234231a79ab030c84486fbee91cd79a5498ebb4f88c6467c366f38815ff681d0` at `receipts/engine/model_admitted/SFT-PHYS-MOLECULAR-SPECTRUM-HIERARCHY-004-234231a79ab030c8.json`; empirical-validation hash `None`; measurement receipt `None`.

212. Terminal exact hydrogen Lamb-shift successor

Claim identity: `SFT-PHYS-ATOMIC-LAMB-SHIFT-TERMINAL-005`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The immutable leading live-vacuum direction and alpha-four hierarchy are retained. The terminal n=2 Lamb frequency is the admitted Rydberg carrier multiplied by alpha cubed and the positive held support 53/64, after the down-depth/binary return 10 alpha/109, the complete colour-heavy-support return alpha squared/(3*53*64), and the terminal alpha-cubed return/(53*64) have each acted once.

> *The unique target-inaccessible ratio is alpha^3[53/64-(10/109)alpha-alpha^2/(3*53*64)-alpha^3/(53*64)].*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ATOMIC-CORRECTION-HIERARCHY-004`
- `SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003`
- `SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004`
- `SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete ten-axis terminal Lamb successor product and exhaust the typed heavy/support/bridge/return carriers. The declared exact boundary is: All first terminal alpha-cubed completions of the admitted live-vacuum n=2 Lamb direction using depth-three heavy complement, predecessor-up binary support, down-depth transport, generator support and the sole terminal return exactly once. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `retain-and-version-predecessor__complete-typed-terminal-support__ordered-positive-held-parts__finite-carrier-exhaustion-order__role-preserving-exact-composition__no-generated-carrier-remains__target-inaccessible-until-seal__disclosed-observational-prediction-protocol__complete-root-directed-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`retain-and-version-predecessor`	`replace-or-erase-predecessor`	An immutable admitted receipt cannot be rewritten.
`support`	`complete-typed-terminal-support`	`selected-partial-support`	A selected subset can tune a precision value.
`operation`	`ordered-positive-held-parts`	`signed-floating-or-complex-series`	Signed, floating, irrational or complex proof values violate the Fold domain.
`order`	`finite-carrier-exhaustion-order`	`selected-or-open-order`	A selected truncation or open series is a free choice.
`composition`	`role-preserving-exact-composition`	`untyped-coefficient-list`	An untyped list does not force a relation.
`closure`	`no-generated-carrier-remains`	`unbounded-extra-correction`	An unbounded correction family is not closed.
`target`	`target-inaccessible-until-seal`	`measurement-readable-execution`	A target-readable execution cannot issue a prediction seal.
`provenance`	`disclosed-observational-prediction-protocol`	`hidden-fit-or-forward-claim`	Hiding prior target knowledge or calling it blind forward forcing is invalid.
`trace`	`complete-root-directed-trace`	`result-without-root-trace`	An untraced value is inadmissible.

Axis	Forced form	Eliminated form	Elimination reason
extension	no-extra-rule	free-extra-term	An extra term is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:176b961bc07508f9ab48d66136a630768b05210eccb3f06fee03376b6509cfb4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:79c96a63e821917aed95d01bbcca4c2c12f7f7e45bddffd8f52e7e7069e1d02c`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:0d685e259ae1b1a3a4fa44c477dd2f1143eedd3d0896eb384612c75a13e91d7d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:49be95fbf390f5d43c86405883086dc5f15d624e66a1f7f5364f9c97faf85e1e`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:1a8d7bd33be27dd3a475d9a6e28e457c56ff23279c6ae9ada098392b3a6fde0b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:583121aedb9600ea571fbe58823868c8b69d465b23aec19b1b4895c95f80b266`. Independent certificate: `sha256:68da17e92bbcc2ed6199a3310a84710801221d25ccb73bf55d7ddb13ac84f9b0`. Engine external-validation hash: `sha256:fde079e56f76bb45693979a6faa00daeb5549c68f86628f1877d9177418031d4`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026

Observed comparison records:

- NIST-ATOMIC-LAMB-COMPLETE: predicted terminal-Lamb-prediction-contained-in-complete-NIST-Bezginov-interval__observational-prediction-protocol-passed; observed terminal-Lamb-prediction-contained-in-complete-NIST-Bezginov-interval__observational-prediction-protocol-passed; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The sealed lamb prediction fails its complete registered interval decision, a source or uncertainty changes, a target becomes readable before sealing, a measured value selects the formal survivor, observational provenance is hidden, or a predecessor receipt is erased.

Measurement receipt: `sha256:85484cf7048f1b161ce841e7180eebc08ef9505be94a24d5fd4a1199545f6c26`. Isolation certificate: `sha256:a160bd23548fc91b860e9e05b1bf1bad84ff5faf094d484cf0301f49cebc42a9`. Custody certificate: `sha256:100d3d60044988751714bc908804396cd6a2c0cf4002066a796d67b474da8bf9`.

Registered source descriptions:

- National Institute of Standards and Technology, CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

- National Institute of Standards and Technology, 2022 CODATA least-squares adjustment: https://doi.org/10.1063/5.0279860 (sha256:4d7e7f34b98ab2fc4df68b38247f818f6fc8bdf7f25f91abcdfbc329e22d2f32)
- National Institute of Standards and Technology, Atomic Spectroscopy Compendium: https://physics.nist.gov/Pubs/AtSpec/node03.html (sha256:33d048a7f7a8d83e5c2b0b96e442b495cf23b888ad1386d7205b17930af9f152)

Registered target identities:

- NIST-ATOMIC-LAMB-COMPLETE from ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026

Shared claim-record clause `PHYS-S019` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured Lamb frequency or Rydberg value in execution
- no imported QED coefficient series
- no fitted coefficient or selected truncation
- no negative, irrational, imaginary or floating proof value
- no rewriting of the leading alpha-four receipt

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:7217873a2f9f893fa1cdc2c7e2455ca2552bccc70d1ba0c7fb8e20bb75027ab8`; engine receipt `sha256:40acfad3f08aa1358a859a5ab95388d74d055c7d708a9521702b27a2c172ab75` at `receipts/engine/model_admitted/SFT-PHYS-ATOMIC-LAMB-SHIFT-TERMINAL-005-40acfad3f08aa135.json`; empirical-validation hash `sha256:7c520df62c053f0b10ed52ba24237d4c82d70364902eef8df02a43ad27cbba76`; measurement receipt `sha256:85484cf7048f1b161ce841e7180eebc08ef9505be94a24d5fd4a1199545f6c26`.

213. Terminal exact hydrogen fine-structure successor

Claim identity: `SFT-PHYS-ATOMIC-FINE-STRUCTURE-TERMINAL-005`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The n=2 fine splitting begins at alpha squared distributed over the complete sixteen-cell terminal support. The binary-squared first return adds alpha/4; the terminal-turn denominator over down support times the generator boundary holds 113 alpha squared/288; the down-support return holds alpha cubed/32; and binary-through-down transport over the atomic endpoint holds 10 alpha fourth/137.

> *The unique target-inaccessible ratio is alpha^2/16[1+alpha/4-(113/288)alpha^2-alpha^3/32-(10/137)alpha^4].*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ATOMIC-CORRECTION-HIERARCHY-004`
- `SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003`
- `SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004`
- `SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete ten-axis terminal n=2 fine-structure product and exhaust the turn, down-support, boundary, endpoint and terminal-support carriers. The declared exact boundary is: All first terminal completions of the admitted alpha-squared fine/gross scale on the n=2 sixteen-cell support, using the exact turn, down cover, generator boundary and atomic endpoint once in their typed roles. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `retain-and-version-predecessor__complete-typed-terminal-support__ordered-positive-held-parts__finite-carrier-ex` haustion-order__role-preserving-exact-composition__no-generated-carrier-remains__target-inaccessible-until-seal__disclosed-observational-prediction-protocol__complete-root-directed-trace__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	retain-and-version-predecessor	replace-or-erase-predecessor	An immutable admitted receipt cannot be rewritten.
support	complete-typed-terminal-support	selected-partial-support	A selected subset can tune a precision value.
operation	ordered-positive-held-parts	signed-floating-or-complex-series	Signed, floating, irrational or complex proof values violate the Fold domain.
order	finite-carrier-exhaustion-order	selected-or-open-order	A selected truncation or open series is a free choice.
composition	role-preserving-exact-composition	untyped-coefficient-list	An untyped list does not force a relation.
closure	no-generated-carrier-remains	unbounded-extra-correction	An unbounded correction family is not closed.
target	target-inaccessible-until-seal	measurement-readable-execution	A target-readable execution cannot issue a prediction seal.
provenance	disclosed-observational-prediction-protocol	hidden-fit-or-forward-claim	Hiding prior target knowledge or calling it blind forward forcing is invalid.
trace	complete-root-directed-trace	result-without-root-trace	An untraced value is inadmissible.
extension	no-extra-rule	free-extra-term	An extra term is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:023322cf80b418873b48af1b4cd99bcd855c1e61c4632b9c7a2c758457804c33. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:52114f33baac8e07ce48b362d3450a166fa9b9f2661d8d382d1317f030307cd0.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:1f4c4918b3bd51f788556dc51a642b4161904c00ff654b372eb08d865370b1e2.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:d410f54f1e07cbaf9ee72d9f2aa92867b594ae7220072d907ac1fcbc80dc09c0.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:cc6c5bd8fad1123f2f152f19a6666bc86a4f7e762d5e5d25efa2d7de76a04158.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:583121aedb9600ea571fbe58823868c8b69d465b23aec19b1b4895c95f80b266. Independent certificate: sha256:68da17e92bbcc2ed6199a3310a84710801221d25ccb73bf55d7ddb13ac84f9b0. Engine external-validation hash: sha256:f4a2dda6b13d03cb4307feba9a94ce28dbb2988f3bdab339587f6497025b3b5f.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026

Observed comparison records:

- NIST-ATOMIC-FINE-COMPLETE: predicted terminal-fine-prediction-contained-in-conservative-direct-NIST-experimental-interval__observational-prediction-protocol-passed; observed terminal-fine-prediction-contained-in-conservative-direct-NIST-experimental-interval__observational-prediction-protocol-passed; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The sealed fine prediction fails its complete registered interval decision, a source or uncertainty changes, a target becomes readable before sealing, a measured value selects the formal survivor, observational provenance is hidden, or a predecessor receipt is erased.

Measurement receipt:
`sha256:6f54600e250a22faf1db02a1edd03e29b67ba617a2eda072b715244efc64a320`. Isolation certificate:
`sha256:65e7b9b1c81cb1b858422ae931aebde56d698448ab3306dd3de60bdef8e85d44`. Custody certificate:
`sha256:1dcb4dd7ffded5abb21d4873804d0813dc9451a4835f6127e566dac4e838bea4`.

Registered source descriptions:

- National Institute of Standards and Technology, CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)
- National Institute of Standards and Technology, 2022 CODATA least-squares adjustment: https://doi.org/10.1063/5.0279860 (sha256:4d7e7f34b98ab2fc4df68b38247f818f6fc8bdf7f25f91abcdfbc329e22d2f32)
- National Institute of Standards and Technology, Atomic Spectroscopy Compendium: https://physics.nist.gov/Pubs/AtSpec/node03.html (sha256:33d048a7f7a8d83e5c2b0b96e442b495cf23b888ad1386d7205b17930af9f152)

Registered target identities:

- NIST-ATOMIC-FINE-COMPLETE from ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026

Shared claim-record clause `PHYS-S019` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured fine splitting or Rydberg value in execution
- no consensus Ritz value as a premise
- no fitted coefficient or selected correction
- no negative, irrational, imaginary or floating proof value
- no rewriting of the leading fine/gross receipt

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:4d650fc1ca21915b8f46bb26f19801dd4377821d950bce96123244e1f83fbe92`; engine receipt
`sha256:12eb3bdc8a9340c15fa621a3d37bee91c32fd1336491fa29ddd70f03c8bb1773` at `receipts/engine/model_admitted/SFT-PHYS-ATOMIC-FINE-STRUCTURE-TERMINAL-005-12eb3bdc8a9340c1.json`; empirical-validation hash
`sha256:a02e8b92becfa9f084ac41acfc04b1c3fc7aed9456388dde153c20d961ea929a`; measurement receipt `sha256:6f54600e250a22faf1db02a1edd03e29b67ba617a2eda072b715244efc64a320`.

214. Terminal exact hydrogen hyperfine and twenty-one-centimetre successor

Claim identity: `SFT-PHYS-ATOMIC-HYPERFINE-TERMINAL-005`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The terminal proton/electron algebraic enclosure supplies the inverse mass carrier and exact cubic reduced-mass retention. Both spin ends over generator three force 16/3 alpha squared. The proton magnetic projection is the finite turn held by one generator share and both alpha hands, returned by 5 alpha/16, held by 16 alpha squared/53 and 7 alpha cubed/80, and closed by alpha fourth/135.

> *For each exact terminal proton/electron ratio rho, nu_hfs/(R-infinity*c)=(16/3)alpha^2 rho^-1[rho/(rho+1)]^3 M, where M=T-1/3-2alpha+(5/16)alpha-(16/53)alpha^2-(7/80)alpha^3+alpha^4/135; exact algebraic rho isolation gives a strict rational prediction enclosure.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004`
- `SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004`
- `SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete ten-axis terminal ground-state hyperfine product and exhaust the mass, reduced-mass, spin, finite-turn, down/up, support and terminal-volume carriers. The declared exact boundary is: All first terminal hyperfine completions using the admitted algebraic proton/electron enclosure, its exact reduced-mass transport, both spin hands, generator distribution, terminal finite turn, depth-three complement and terminal support exactly once. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `retain-and-version-predecessor__complete-typed-terminal-support__ordered-positive-held-parts__finite-carrier-exhaustion-order__role-preserving-exact-composition__no-generated-carrier-remains__target-inaccessible-until-seal__disclosed-observational-prediction-protocol__complete-root-directed-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`retain-and-version-predecessor`	`replace-or-erase-predecessor`	An immutable admitted receipt cannot be rewritten.
`support`	`complete-typed-terminal-support`	`selected-partial-support`	A selected subset can tune a precision value.
`operation`	`ordered-positive-held-parts`	`signed-floating-or-complex-series`	Signed, floating, irrational or complex proof values violate the Fold domain.
`order`	`finite-carrier-exhaustion-order`	`selected-or-open-order`	A selected truncation or open series is a free choice.
`composition`	`role-preserving-exact-composition`	`untyped-coefficient-list`	An untyped list does not force a relation.
`closure`	`no-generated-carrier-remains`	`unbounded-extra-correction`	An unbounded correction family is not closed.
`target`	`target-inaccessible-until-seal`	`measurement-readable-execution`	A target-readable execution cannot issue a prediction seal.
`provenance`	`disclosed-observational-prediction-protocol`	`hidden-fit-or-forward-claim`	Hiding prior target knowledge or calling it blind forward forcing is invalid.
`trace`	`complete-root-directed-trace`	`result-without-root-trace`	An untraced value is inadmissible.
`extension`	`no-extra-rule`	`free-extra-term`	An extra term is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:f257126024ff51d6ab58036d63ea59295705aaffb0e665875fc6fbb43d47e378`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:202da18f8c49f533508fc653fbd7cad2bd05424edaa99ff240adbe31ce40f668`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:a6d64ac4f3238ccffd8cbe848a37e2aafcb54971e6474f6b5abe197e33be3f3e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3a81cc4622e0b65e6bfd33dfb8f0e48079cca52a501b97613c36f5543db23665`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:917b749628833a2b10ab79c97c7ef007b7dd9a6c57697a248c4e079d9abed7f9`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:583121aedb9600ea571fbe58823868c8b69d465b23aec19b1b4895c95f80b266`. Independent certificate: `sha256:68da17e92bbcc2ed6199a3310a84710801221d25ccb73bf55d7ddb13ac84f9b0`. Engine external-validation hash: `sha256:56c598f2b1e5c01459da3f3847c15b13facb5121356693ba1bde89be5ebe6035`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026

Observed comparison records:

- NIST-ATOMIC-HYPERFINE-COMPLETE: predicted terminal-hyperfine-central-prediction-inside-NIST-21cm-interval-and-complete-carrier-interval-overlaps__observational-prediction-protocol-passed; observed terminal-hyperfine-central-prediction-inside-NIST-21cm-interval-and-complete-carrier-interval-overlaps__observational-prediction-protocol-passed; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The sealed hyperfine prediction fails its complete registered interval decision, a source or uncertainty changes, a target becomes readable before sealing, a measured value selects the formal survivor, observational provenance is hidden, or a predecessor receipt is erased.

Measurement receipt: `sha256:efe690404acb3a76b5b16e7786f08939d14c395c654928d314707b97b7f296cb`. Isolation certificate: `sha256:00b7c45797d40d68910892ef0225b0dc4269caf67e46d89768771c50ae8ab3fc`. Custody certificate: `sha256:54f57ef885645b222619ef97b19e2c13e7034b0cd39862576ece7e50d8947a03`.

Registered source descriptions:

- National Institute of Standards and Technology, CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)
- National Institute of Standards and Technology, 2022 CODATA least-squares adjustment: https://doi.org/10.1063/5.0279860 (sha256:4d7e7f34b98ab2fc4df68b38247f818f6fc8bdf7f25f91abcdfbc329e22d2f32)
- National Institute of Standards and Technology, Atomic Spectroscopy Compendium: https://physics.nist.gov/Pubs/AtSpec/node03.html (sha256:33d048a7f7a8d83e5c2b0b96e442b495cf23b888ad1386d7205b17930af9f152)

Registered target identities:

- NIST-ATOMIC-HYPERFINE-COMPLETE from ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026

Shared claim-record clause `PHYS-S019` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured 21-cm frequency or Rydberg value in execution
- no imported proton magnetic moment
- no fitted magnetic coefficient or hidden target knowledge
- no negative, irrational, imaginary or floating proof value
- no collapse of the algebraic rho enclosure to a decimal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:68ded2bc46504ed344008c4c5c6b0d1da6412ead07b9e5262e468b8341183452`; engine receipt `sha256:0e39c9521ebfe7ad1e66f06b203534155ee1762c660d907d118757cdd9dc346a` at `receipts/engine/model_admitted/SFT-PHYS-ATOMIC-HYPERFINE-TERMINAL-005-0e39c9521ebfe7ad.json`; empirical-validation hash `sha256:048fa0301b1f4e7de1f44e04573b6024051d7fd87f7b4f30af042a174ad41ed3`; measurement receipt `sha256:efe690404acb3a76b5b16e7786f08939d14c395c654928d314707b97b7f296cb`.

215. Terminal exact molecular rotational, vibrational and isotope spectroscopy

Claim identity: `SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The preserved Fold hierarchy separates molecular rotation, harmonic vibration and anharmonic return. Positive rotational ordinals force E_J/B=J(J+1) and adjacent gap 2J; positive vibrational ordinals force the odd half-step oscillator ladder and ordered anharmonic narrowing. The exact alpha carrier then closes distinct hydrogen rotational/vibrational and anharmonic/vibrational ratios and first-isotope rotational and squared-vibrational transports without importing a molecular measurement.

> *E_J/B=J(J+1), Delta_J/B=2J, and G_n=omega(2n-1)/2 Take omega_x(2n-1)^2/4. For exact alpha: B_H2/omega_H2=2alpha Take 14alpha^2 Take 58alpha^3 Take 82alpha^4; omega_x,H2/omega_H2=4alpha Take 30alpha^2 Take 59alpha^3 Take 63alpha^4; B_D2/B_H2=1/2+5alpha^2+38alpha^3; and (omega_D2/omega_H2)^2=1/2+20alpha^2+49alpha^3.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MOLECULAR-SPECTRUM-HIERARCHY-004`
- `SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003`
- `SFT-PHYS-WAVE-RESONANCE-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete ten-axis terminal molecular-spectroscopy product and exhaust the distinct ladder, alpha-return, isotope and provenance carriers. The declared exact boundary is: All first terminal molecular successors preserving the admitted hierarchy and adverse receipt while using the complete J, odd-oscillator, down/up, generator, heavy, terminal and isotope carriers once in their typed roles. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `retain-and-version-predecessor__distinct-generated-rotational-and-vibrational-carriers__counted-JJplusOne-and-odd-oscillator-ladders__ordered-positive-held-Takes__count-refined-linear-and-squared-transport__finite-typed-carrier-exhaustion__target-inaccessible-until-seal__disclosed-observational-prediction-protocol__complete-root-directed-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	retain-and-version-predecessor	replace-or-erase-predecessor	A versioned successor cannot rewrite an admitted receipt or delete an adverse row.
carrier	distinct-generated-rotational-and-vibrational-carriers	one-universal-quarter-carrier	One numerical carrier repeats the already rejected equality.
ladder	counted-JJplusOne-and-odd-oscillator-ladders	imported-continuum-spectrum	A continuum spectrum is outside the exact finite Fold grammar.
operation	ordered-positive-held-Takes	signed-floating-series	Signed or floating proof scalars violate the permitted domain.
isotope	count-refined-linear-and-squared-transport	free-isotope-factor	A chosen scale factor is a parameter.
order	finite-typed-carrier-exhaustion	selected-or-open-correction-order	A selected truncation or open correction list is not closed.
target	target-inaccessible-until-seal	measurement-readable-execution	A target-readable execution cannot issue a sealed empirical prediction.
provenance	disclosed-observational-prediction-protocol	concealed-development-provenance	Concealing prior observation would misstate the empirical protocol.
trace	complete-root-directed-trace	result-without-root-trace	An untraced numerical relation is inadmissible.
extension	no-extra-rule	free-extra-term	An ungenerated extra term is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:0ea91c00e707f57346b080151ab99ec054a7619504a122cbbd171ba6b3828c03. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:f01d0ff1ae2a565038b11a1fbb8d6f8dd676ccc0b317f24beb33633680006376.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:7357d35b1e9d73a17a643f7ddd74659f8520e5e38d72843eeca43b1a73a2d66b.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:80ad2ac3a4b100486bfa95508dace8af040368b7f66f7e96d1bd0078a39dbfbf.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:45a74a3e4905cc6e2ef65119261f21696ab3bb97a16c357ecf8cdd50468168eb.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:8fd64c9b1d709bc057502777c2459cd4e9da0fb1e6e318728db625dcb75fd0db. Independent certificate: sha256:103ad110876cbd3f6d10ff61c00b8e980ef8c3baf9eb4ecc01d12ab708ed6a98. Engine external-validation hash: sha256:cebb62be6a7b8a8621f3e6a50a96ced47ffa6901d6caf46b12f748f7f0c5dbd2.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-MOLECULAR-SPECTROSCOPY-H2-D2-SUCCESSOR

Observed comparison records:

- NIST-MOLECULAR-H2-D2-COMPLETE-VECTOR: predicted terminal-molecular-JJplusOne-anharmonic-and-isotope-relations-contained-in-complete-NIST-H2-D2-displayed-resolution-vector__observational-prediction-protocol-passed; observed terminal-molecular-JJplusOne-anharmonic-and-isotope-relations-contained-in-complete-NIST-H2-D2-displayed-resolution-vector__observational-prediction-protocol-passed; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The claim fails if any sealed ratio leaves its complete registered exact displayed-resolution interval, the J(J+1), 2J or positive anharmonic ladder identity fails, either isotope hierarchy or source hash changes, a target becomes readable before sealing, development provenance is hidden, or an immutable leading or adverse receipt is erased.

Measurement receipt: `sha256:1dc9c5c049ef7d2dd0081440f03490ab68af16f4fa6a689d30e0adcf7344364c`. Isolation certificate: `sha256:2857ffe2b8f7c6baa494ee27dce0f953c5e65b9a2d8d7a379319b834584c402c`. Custody certificate: `sha256:322f837623047dc59924a095e5a8a7b2ae53824d76486028de4b227a3f0718b5`.

Registered source descriptions:

- National Institute of Standards and Technology, Chemistry WebBook SRD 69: https://webbook.nist.gov/cgi/cbook.cgi?ID=C1333740&Mask=1828 (sha256:18036a188088f880122249544ceb6b384fabfba93b300b4f9f0fa01aa0ed9b24)
- National Institute of Standards and Technology, Chemistry WebBook SRD 69: https://webbook.nist.gov/cgi/cbook.cgi?ID=C7782390&Mask=1000 (sha256:790276cd493dbede03fc9e83db95bc0808a36838eb31473aaf923493b4749936)

Registered target identities:

- NIST-MOLECULAR-H2-D2-COMPLETE-VECTOR from NIST-MOLECULAR-SPECTROSCOPY-H2-D2-SUCCESSOR

Shared claim-record clause `PHYS-S019` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no H2 or D2 measurement, wavenumber, source record or displayed digit in execution
- no imported rigid-rotor parameter, oscillator frequency or isotope multiplier
- no numerical-zero state, negative proof scalar, irrational root, imaginary value or floating proof value
- no concealed observational development and no erasure of the adverse predecessor receipt
- no additional correction after the typed grammar is exhausted

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:7b12b8e6f6d03040c6845733b99a0a6a17d4d989f326442c497176b12dd3eaef`; engine receipt `sha256:17e6e8ac57a97e1c23334300ebf9bcfddad8a5c9f0dca822119fef004c00762c` at `receipts/engine/model_admitted/SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005-17e6e8ac57a97e1c.json`; empirical-validation hash `sha256:e0b1b828ca16c16aed6e3d08f5e75cfb3a3a3f70e50ab92634d5d2d00be89ca4`; measurement receipt `sha256:1dc9c5c049ef7d2dd0081440f03490ab68af16f4fa6a689d30e0adcf7344364c`.

216. Terminal exact hydrogen reduced-mass and Rydberg-scale completion

Claim identity: `SFT-PHYS-ATOMIC-HYDROGEN-RYDBERG-TERMINAL-005`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The terminal proton/electron enclosure forces the finite-nuclear-mass retention rho/(rho+1). The exact Fold carriers then add alpha squared over the down support and alpha cubed over the binary support, hold the complete binary-colour alpha-four return, and restore the up-square alpha-five terminal cell. Multiplication by alpha squared over two gives the hydrogen ionization/electron-rest carrier, while the immutable three-quarter and five-thirty-sixth gaps give Lyman-alpha and Balmer-alpha.

For exact terminal rho, H/R_infinity=rho/(rho+1)+alpha^2/5+alpha^3/2 Take 6alpha^4+49alpha^5; H/(m_e c^2)=alpha^2/2 times that carrier; Lyman-alpha/R_infinity is three-quarters and Balmer-alpha/R_infinity is five-thirty-sixths of the same carrier.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004`
- `SFT-PHYS-ATOMIC-CORRECTION-HIERARCHY-004`
- `SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004`
- `SFT-PHYS-ATOMIC-HYPERFINE-TERMINAL-005`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete ten-axis terminal hydrogen Rydberg-scale product and exhaust the reduced-mass, down, binary, colour, direction, up-square and line-gap carriers. The declared exact boundary is: All first terminal finite-proton-mass completions of the admitted alpha-squared-over-two hydrogen scale, using the terminal rho enclosure and each generated alpha-return carrier exactly once before the immutable line gaps act. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `retain-and-compose-predecessors__exact-reduced-mass-from-terminal-rho__role-typed-alpha-return-chain__ordered-positive-held-Take__finite-carrier-exhaustion-order__alpha-square-times-terminal-scale__target-inaccessible-until-seal__disclosed-observational-prediction-protocol__complete-root-directed-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`retain-and-compose-predecessors`	`replace-or-erase-predecessor`	A successor cannot rewrite either immutable derivation.
`mass`	`exact-reduced-mass-from-terminal-rho`	`imported-measured-mass-factor`	A measured multiplier would become a proof parameter.
`correction`	`role-typed-alpha-return-chain`	`untyped-coefficient-list`	An untyped list cannot force placement or orientation.
`operation`	`ordered-positive-held-Take`	`signed-floating-series`	Signed or floating proof scalars violate the exact Fold domain.
`order`	`finite-carrier-exhaustion-order`	`selected-or-open-order`	A chosen truncation or open series is a free choice.
`composition`	`alpha-square-times-terminal-scale`	`answer-only-decimal`	An answer-only decimal erases alpha, mass and gap provenance.
`target`	`target-inaccessible-until-seal`	`measurement-readable-execution`	A target-readable execution cannot issue a sealed prediction.
`provenance`	`disclosed-observational-prediction-protocol`	`concealed-development-provenance`	Hidden prior observation would misstate the protocol.
`trace`	`complete-root-directed-trace`	`result-without-root-trace`	An untraced physical ratio is inadmissible.
`extension`	`no-extra-rule`	`free-extra-term`	An ungenerated term is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:c71981e09cb1fef4c6c62401c24677acb20127e5f5a34b3823b238766af2ff96`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt

`sha256:f82b7235a6e12a498a923c82aef74d997febeca07e40cda7b8fb571c4022ffce`.

- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:da567131b75798adf2a0584bde41872f1a396fa7dacd0a00cc55b933bb3aae0f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e579dfe41bad3f3bd0deb223cfd8a538364698cb6e4576a8fb7c985eba144400`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:b391a351cdf19e1f414e6e7543040bdfbf869b6a32e578dcfc0f999187892c27`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:108dd4fd13bd52da50840c9a1d4f3b879fbcb05e2f2acf86ed1ee6cb3ba45d1c`. Independent certificate: `sha256:3d4260ccd137743c362ea95b6744808114dd7e8b769a135cc6fdc87d5d638a60`. Engine external-validation hash: `sha256:a306e48e142800169816ae9a9f0df7aa37f74340eaf73eb82c70e15d2beefd9b`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-HYDROGEN-RYDBERG-SUCCESSOR-2022-2026

Observed comparison records:

- NIST-HYDROGEN-RYDBERG-IONIZATION-LINE-COMPLETE-VECTOR: predicted terminal-hydrogen-reduced-mass-and-radiative-scale-contained-in-NIST-ionization-Rydberg-ratio__absolute-ionization-Lyman-Balmer-vector-passed; observed terminal-hydrogen-reduced-mass-and-radiative-scale-contained-in-NIST-ionization-Rydberg-ratio__absolute-ionization-Lyman-Balmer-vector-passed; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The claim fails if the sealed scale leaves the complete ionization/Rydberg interval, its translated ionization leaves the displayed-resolution enclosure, either line misses its full interval, any source hash or uncertainty changes, target access precedes sealing, development provenance is hidden, or a predecessor receipt is rewritten.

Measurement receipt: `sha256:1087060f8b6669d00ba27c6f5d3b154505adfa0d21bbf94bfe1a00ccdfd34b52`. Isolation certificate: `sha256:265d8b0ffa1ba0dd6f7e797837298dbbc6712f0aebf43f487a2cfb474bd75663`. Custody certificate: `sha256:8b635f1be853b0c4790fc20a790755e0f6bd9bd587ba101dc4eb478fb5f72618`.

Registered source descriptions:

- National Institute of Standards and Technology, CODATA 2022: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)
- National Institute of Standards and Technology, Atomic Spectra Database: https://physics.nist.gov/PhysRefData/Handbook/Tables/hydrogentable1.htm (sha256:cc54c774518d62cbb7e95f17b3b7fcd9d1faf0aa90043a607b91dff0a43e2087)
- National Institute of Standards and Technology, Atomic Physics Division: https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=842564 (sha256:86ccaeb875fcdb4445e727618509621ceee656e5dc02295140ecdb6e1d2dd443)

Registered target identities:

- NIST-HYDROGEN-RYDBERG-IONIZATION-LINE-COMPLETE-VECTOR from NIST-HYDROGEN-RYDBERG-SUCCESSOR-2022-2026

Shared claim-record clause `PHYS-S019` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured Rydberg, ionization, electron-rest energy or line value in execution

- no imported reduced-mass factor or atomic correction series
- no numerical-zero state, negative proof scalar, irrational, imaginary or floating proof value
- no concealed observational development or rewritten predecessor receipt
- no correction beyond the exhausted typed carrier grammar

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:b85d9184a979cd5f3aa2b64ce106297f4ba59aecca0e785cd085e1e9523c94db`; engine receipt `sha256:5d7e0c49ec4455a01a886ceedfaaa5522323d691247fb9455dd41afd9c158d90` at `receipts/engine/model_admitted/SFT-PHYS-ATOMIC-HYDROGEN-RYDBERG-TERMINAL-005-5d7e0c49ec4455a0.json`; empirical-validation hash `sha256:721ec82bae74b9db8ed1a512ba4b30b04fc5010b1d1574cbf97b82f2e0b38115`; measurement receipt `sha256:1087060f8b6669d00ba27c6f5d3b154505adfa0d21bbf94bfe1a00ccdfd34b52`.

217. Terminal exact atomic shell, filling and periodicity completion

Claim identity: `SFT-PHYS-ATOMIC-SHELL-PERIODICITY-TERMINAL-005`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Every positive principal shell n contains all orbit ranks One through n, so the admitted capacities sum exactly to 2n squared. Increasing joint cover with retained principal-rank tie order generates the complete filling recurrence. The first filled cell and each later filled rank-two outer boundary force closure coordinates 2, 10, 18, 36, 54, 86 and 118 and period widths 2, 8, 8, 18, 18, 32 and 32. The physical ionization signature is a rising period envelope followed by a closure-to-successor reset, not a false claim of strict monotonicity inside every period.

> *For every positive n, sum from orbit rank One through n of 2(1+2(r-1)) equals 2n^2. The generated closures are 2,10,18,36,54,86,118 and their successive widths are 2,8,8,18,18,32,32; ionization comparison tests endpoint-above-start and endpoint-to-next-start reset while retaining internal dips.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-ORDER-LATTICE-001`

Generated grammar and closure boundary. Generate the complete ten-axis atomic shell, cover-order, closure, recurrence, ionization-envelope, custody, trace and extension product. The declared exact boundary is: All positive principal shells, every valid principal/orbit cell in increasing joint-cover order, the admitted atomic endpoint, and the first complete empirical period/reset vector. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `compose-immutable-orbit-capacity__all-positive-orbit-ranks-through-n__exact-complete-capacity-sum__increasing-joint-cover-principal-tie__first-cell-then-complete-rank-two-boundary__successor-after-held-closure__rising-envelope-and-boundary-reset__target-inaccessible-until-seal__complete-root-directed-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`compose-immutable-orbit-capacity`	`replace-or-relabel-capacity`	A successor may not rewrite an immutable receipt.

Axis	Forced form	Eliminated form	Elimination reason
shell	all-positive-orbit-ranks-through-n	selected-width-list	A width list is an imported answer.
sum	exact-complete-capacity-sum	asserted-two-n-square	Asserting the result supplies no derivation.
order	increasing-joint-cover-principal-tie	memorized-filling-table	A memorized table would select the result.
closure	first-cell-then-complete-rank-two-boundary	chosen-period-endpoints	Chosen endpoints are measured inputs.
recurrence	successor-after-held-closure	free-reset-rule	A reset rule is a parameter.
ionization	rising-envelope-and-boundary-reset	strict-stepwise-monotonicity	Strict monotonicity erases genuine within-period distinctions.
target	target-inaccessible-until-seal	external-data-readable	Target-readable execution cannot seal a prediction.
trace	complete-root-directed-trace	answer-without-dependencies	An untraced answer is inadmissible.
extension	no-extra-rule	free-exception-or-width	An ungenerated exception is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9bef6a459bb5c21c3ccba869d36eec0782c722d262a55e43899698f1322221a7`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:4e78c1737aa8aa63b9dc904637766a7b4256e9f491c968754ad1769305088c1a`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:102a3bd8225297b93ad36f4a78620d42dbd77c34852dbf94fed73d248a85069f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bc0ef6e00fbca12d8562d2e45e194fe3aa092474ffbe38e74c18a9e26812ad9b`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:92918c4a35853f3faa2d01f66ac0712babe628aa62d6de100b01dbace0b2e522`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9277b76711c49b11d9c2e7fa5f8ba4dfc8a3f8e462d4f444c1ec1bfcdf2c24a4`. Independent certificate: `sha256:c961db81ab6ffefa476252c89efe3e8a0afaa9bd2c211e2e4cd5d7d6d4e188b5`. Engine external-validation hash: `sha256:4363ca220035df9a50f988bac512fad6fdad9d8adb756681ed618f1fcd6ef117`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IUPAC-NIST-ATOMIC-SHELL-PERIODICITY-2022-2026

Observed comparison records:

- IUPAC-NIST-ATOMIC-PERIODICITY-COMPLETE-VECTOR: predicted exact-shell-sum-and-generated-period-widths-match-IUPAC__NIST-configurations-and-ionization-envelope-reset-vector-passed-with-local-dips-retained; observed exact-shell-sum-and-generated-period-widths-match-IUPAC__NIST-configurations-and-ionization-envelope-reset-vector-passed-with-local-dips-retained; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The claim fails if any generated closure or width differs, a registered NIST configuration disagrees, a period envelope or reset fails, a retained local dip is erased, any source identity changes, target access precedes sealing, or observational development is concealed.

Measurement receipt: `sha256:7202b1fdc7cb7790b8d3fa83a25cccffcc6a98bd2a115548894c95ad2153b827`. Isolation certificate: `sha256:7fc84c9223bc7dc7e8a10dc720d8def4d85ffebe7621c01db69aaa8d9366f97e`. Custody certificate: `sha256:c80e2a3dc99a5b30230981facf04df61804069b9361f9136682994cba33219b8`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://iupac.org/wp-content/uploads/2022/05/IUPAC_Periodic_Table-04May22.pdf (sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)
- National Institute of Standards and Technology: https://physics.nist.gov/cgi-bin/ASD/ie.pl (sha256:3132a0088e0346af7d0ae0b88937765b4c6dfad951d79bb9677103aa6affbdbb)

Registered target identities:

- IUPAC-NIST-ATOMIC-PERIODICITY-COMPLETE-VECTOR from IUPAC-NIST-ATOMIC-SHELL-PERIODICITY-2022-2026

Shared claim-record clause `PHYS-S019` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported periodic table, aufbau list, electron configuration or ionization value in execution
- no conflation of shell capacities 2,8,18,32 with actual period widths 2,8,8,18,18,32,32
- no claim of strict monotonic ionization inside a period
- no numerical-zero state, negative proof scalar, irrational, imaginary or floating proof value
- no hidden observational development, free exception or rewritten predecessor receipt

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:fc6874f7a94acfae2f63dfce5a3d103070fea429ac8f02eda9e2c931e6df81c0`; engine receipt `sha256:d7f0cce03863186e4f97aa36ec68a00797edd4103268c44dc18faceb05c40cf7` at `receipts/engine/model_admitted/SFT-PHYS-ATOMIC-SHELL-PERIODICITY-TERMINAL-005-d7f0cce03863186e.json`; empirical-validation hash `sha256:ee740d5729032a5997144bfeb96f34ff1ce8ea1feebc839090c89ac89bbbc3f0`; measurement receipt `sha256:7202b1fdc7cb7790b8d3fa83a25cccffcc6a98bd2a115548894c95ad2153b827`.

218. Terminal exact atomic transition-rate, lifetime and multipole completion

Claim identity: `SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The admitted one-unit elementary transition retains an exact positive level gap. Transport through all three forced spatial directions makes the E1 rate proportional to gap cubed times held line strength divided by complete upper-state weight. Complete positive decay channels force lifetime to be the reciprocal of their rate sum. Electric multipole rank L has exponent 2L+1, so each successor is conditionally suppressed by exact gap squared at equal strength and weight. Magnetic and other typed channels retain distinct strengths; the label forbidden alone is not promoted to a false universal ordering.

> *Normalized E1 rate is gap^3*S/g; lifetime is 1/(sum_i A_i); normalized electric rank-L rate is gap^(2L+1)*S/g and the equal-strength/equal-weight successor ratio is gap^2. The elementary orbital step remains One, while non-electric typed channels require their own held strengths.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ATOMIC-TRANSITION-SELECTION-004`

- SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete ten-axis transition-act, gap, spatial-volume, strength, weight, lifetime, multipole, typed-channel, custody and extension product. The declared exact boundary is: All exact positive normalized electric-multipole rates generated from an admitted level gap, held strength, positive upper-state weight and every finite positive multipole rank, plus complete finite positive decay-channel sums. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `retain-one-unit-elementary-act__exact-positive-level-gap__complete-generator-three-volume__held-exact-line-strength__divide-by-complete-positive-weight__reciprocal-complete-rate-sum__append-held-boundary-pair__typed-channel-conditional-suppression__target-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`retain-one-unit-elementary-act`	`replace-selection-law`	A successor cannot rewrite the immutable one-unit receipt.
`gap`	`exact-positive-level-gap`	`selected-decimal-frequency`	A selected frequency would be a target-bearing parameter.
`space`	`complete-generator-three-volume`	`free-rate-exponent`	A chosen exponent is a parameter.
`strength`	`held-exact-line-strength`	`erase-or-fit-strength`	Erasure or fitting cannot distinguish lines.
`weight`	`divide-by-complete-positive-weight`	`unweighted-or-fitted-factor`	A hidden multiplicity changes rates freely.
`lifetime`	`reciprocal-complete-rate-sum`	`independent-lifetime-parameter`	A separately chosen lifetime is not forced.
`multipole`	`append-held-boundary-pair`	`chosen-power-list`	A list has no successor theorem.
`forbidden`	`typed-channel-conditional-suppression`	`universal-slower-label`	M1 shares the cubic exponent and distinct strengths can reverse an unqualified ordering.
`target`	`target-inaccessible-until-seal`	`external-target-readable`	Target-readable execution cannot seal a prediction.
`extension`	`no-extra-rule`	`free-coefficient-or-exception`	An ungenerated coefficient is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:aa0277fa92edcafc18899a086e2fd274ef5946f5b97e63f0b089384a583398f9`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:226dec5ab4ab69dfcadebf543257b117191a1957bc41ea3b8b1cb0fc168834ba`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:8399e7e82be3c0197772f68a0dde646adf6b3f7ed8366a2eb96028584992ffda`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:8a207dc13bd9188a6a113f07e03bc0d6d92b5013ac03193bd037350cd0f05399.

- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:35918069bd6ff102c9048d4c82c03488f0dba11fbaa7763cef032e25bb99bf2a.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:fce8dcbb269092987146a5e52ddbf37c4934cb2219b0c6d34bf5185db9ec6f56. Independent certificate: sha256:767d9605573e4eccaed519642e3c083f8436af9f8e824c4c7df691a4b32c7aea. Engine external-validation hash: sha256:d3ba0cfe14588b1ca303f397f1c0895e515517a55ac6c08b94c4a7423bd5b9d1.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ATOMIC-TRANSITION-RATE-LIFETIME-2026

Observed comparison records:

- NIST-ATSPEC-RATE-LIFETIME-EQUATIONS: predicted sealed-E1-gap-cube-E2-gap-five-and-reciprocal-lifetime-match-NIST-equations__hydrogenic-rate-four-cross-locks__complete-metastable-lifetime-vector-passed; observed sealed-E1-gap-cube-E2-gap-five-and-reciprocal-lifetime-match-NIST-equations__hydrogenic-rate-four-cross-locks__complete-metastable-lifetime-vector-passed; exact match True
- NIST-AR9-M1-METASTABLE-LIFETIME: predicted sealed-E1-gap-cube-E2-gap-five-and-reciprocal-lifetime-match-NIST-equations__hydrogenic-rate-four-cross-locks__complete-metastable-lifetime-vector-passed; observed sealed-E1-gap-cube-E2-gap-five-and-reciprocal-lifetime-match-NIST-equations__hydrogenic-rate-four-cross-locks__complete-metastable-lifetime-vector-passed; exact match True
- NIST-AL27-CLOCK-METASTABLE-LIFETIME: predicted sealed-E1-gap-cube-E2-gap-five-and-reciprocal-lifetime-match-NIST-equations__hydrogenic-rate-four-cross-locks__complete-metastable-lifetime-vector-passed; observed sealed-E1-gap-cube-E2-gap-five-and-reciprocal-lifetime-match-NIST-equations__hydrogenic-rate-four-cross-locks__complete-metastable-lifetime-vector-passed; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The claim fails if a sealed exponent or lifetime relation differs, hydrogenic exponent four does not cross-lock, either complete metastable interval fails, M1's adverse boundary is erased, a source hash or uncertainty changes, target access precedes sealing, or observational provenance is concealed.

Measurement receipt: sha256:f32cc32b701515a000f43cacfc95b84b5cde8f18b3b11a7b1782ba0ebd1cfda1. Isolation certificate: sha256:b518b45b8ff4e9d72141479c02bb09854e948587471177552095827b44f4991d. Custody certificate: sha256:d17509c170dfd3cfeba831b2cf4b0c610f35ef6d3ef25f71b8a47d38cfbb4e77.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/Pubs/AtSpec/node17.html (sha256:35f4c76e7ed640958c6ef184cb731fc91c540608bda87bab7ab63130abb7f7a2)
- National Institute of Standards and Technology: https://www.nist.gov/publications/lifetime-metastable-2p12-state-f-ar9-isolated-compact-penning-trap (sha256:77bb5da60432c1dd9d5979e952254ace888073af666605ecce7c8f5fda4eb5a4)
- National Institute of Standards and Technology: https://www.nist.gov/publications/observation-1s0-3p0-clock-transition-27al (sha256:77bb5da60432c1dd9d5979e952254ace888073af666605ecce7c8f5fda4eb5a4)

Registered target identities:

- NIST-ATSPEC-RATE-LIFETIME-EQUATIONS from NIST-ATOMIC-TRANSITION-RATE-LIFETIME-2026
- NIST-AR9-M1-METASTABLE-LIFETIME from NIST-ATOMIC-TRANSITION-RATE-LIFETIME-2026

- NIST-AL27-CLOCK-METASTABLE-LIFETIME from NIST-ATOMIC-TRANSITION-RATE-LIFETIME-2026

Shared claim-record clause `PHYS-S019` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured wavelength, rate, line strength, lifetime or source access in execution
- no imported continuum radiation equation or floating proof value
- no numerical-zero state, negative rate, irrational or imaginary proof scalar
- no universal claim that every forbidden-channel rate is slower independent of strength
- no fitted exponent, coefficient, lifetime or multipole exception

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:6c3e15385eb400dc4f936be254d0fbfd7a829ab9a55e53c5b03f041f4335b8bb`; engine receipt `sha256:41ac193f7b4d9e904e3557ad5e01d9fc529acc0d54d8fd0e141b43a99b666b25` at `receipts/engine/model_admitted/SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005-41ac193f7b4d9e90.json`; empirical-validation hash `sha256:35122805bb7add54e8fa16280f7bfb168266af44afb2e7a0241e542f75b752ba`; measurement receipt `sha256:f32cc32b701515a000f43cacfc95b84b5cde8f18b3b11a7b1782ba0ebd1cfda1`.

219. Terminal exact atomic Zeeman and Stark field-splitting completion

Claim identity: `SFT-PHYS-ATOMIC-FIELD-SPLITTING-TERMINAL-005`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Complete held angular support forces 2J+1 magnetic sublevels. Opposing directions remain labels and every displacement magnitude is positive; one weak-field act therefore forces linear Zeeman shifts g*m*B with equal adjacent spacing g*B. In electric response a retained degenerate partner preserves the first dipole act and forces a linear Stark shift. Without that partner the opposed first acts close to the empty form and the first retained response uses two field acts, forcing a quadratic shift.

> *A J level has 2J+1 magnetic classes; Zeeman magnitude is g*m*B with adjacent spacing g*B; retained degeneracy gives dipole*E while nondegenerate closure gives polarizability*E^2/2.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ATOMIC-TRANSITION-SELECTION-004`
- `SFT-PHYS-ATOMIC-SHELL-PERIODICITY-TERMINAL-005`
- `SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`

Generated grammar and closure boundary. Generate the complete ten-axis field-predecessor, multiplicity, held-direction, magnetic response, spacing, degenerate/nondegenerate electric response, order, custody and extension product. The declared exact boundary is: Every finite positive angular support, exact positive field magnitude, held side/orientation label and the two complete electric support classes distinguished by whether a degenerate partner is retained. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `compose-immutable-field-predecessors__complete-held-orientation-successor__held-side-label-and-positive-distance__one-field-act-per-held-orientation__one-common-positive-field-step__held-first-order-dipole-channel__paired-first-act-closes-then-square__degeneracy-typed-response-order__target-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	compose-immutable-field-predecessors	replace-field-predecessors	A successor cannot rewrite admitted receipts.
multiplicity	complete-held-orientation-successor	selected-line-count	A chosen multiplet width imports the answer.
direction	held-side-label-and-positive-distance	signed-energy-scalar	Negative proof scalars violate the domain.
magnetic	one-field-act-per-held-orientation	chosen-field-polynomial	A chosen response polynomial is a parameter.
spacing	one-common-positive-field-step	unequal-free-spacing	Free spacings destroy the generated orientation order.
degenerate	held-first-order-dipole-channel	erase-first-displacement	Erasure loses the held cross-state channel.
nondegenerate	paired-first-act-closes-then-square	free-first-order-shift	A first-order scalar without a retained partner is ungenerated.
order	degeneracy-typed-response-order	universal-single-order	One order erases the degeneracy discriminator.
target	target-inaccessible-until-seal	external-target-readable	Target-readable execution cannot seal a prediction.
extension	no-extra-rule	free-coefficient-or-rule	An ungenerated response is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:076b6ce41326063fe4919a4e7761d8be1d0e628a712c0abceb8f0af1b61aa5a7. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:a5c77c740fa2357d0e8e4fb7d8c0454e1a125f10d4ade55e7719ba3cf31a37ba.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:4a585aad8d67b2153cd331d6bfdaf88d4d062aaa0cb4ac2430b1481133958eab.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:0ab10ce95a892f62966c9a951b6530881d217dd8e7654a185db62d410d33a6e5.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:8a0996c55d448f033a7b0ead29c8cda092982015d0839183f3b0e9e32a001b6b.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:01a7cf130a10dc80276c9a07f338f4a493eba8be6c0c02fa14368c07bf367963. Independent certificate: sha256:7beefcdf3c0679968c67d8ab49fe6087dc1641ef92910b06edfd98690bef1737. Engine external-validation hash: sha256:f20af1bc490241ef3cf48db5ed4e51cd2862d4aaac7419756e3e5cdcff8a0da5.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-NBS-ATOMIC-FIELD-SPLITTING-2026

Observed comparison records:

- NIST-WEAK-FIELD-ZEEMAN: predicted sealed-two-J-plus-one-and-linear-Zeeman-match-NIST__degenerate-hydrogen-linear-Stark-matches-NBS__nondegenerate-cesium-quadratic-Stark-matches-NIST; observed sealed-two-J-plus-one-and-li near-Zeeman-match-NIST__degenerate-hydrogen-linear-Stark-matches-NBS__nondegenerate-cesium-quadratic-Stark-matches-NIST; exact match True

- NBS-HYDROGEN-LINEAR-STARK: predicted sealed-two-J-plus-one-and-linear-Zeeman-match-NIST__degenerate-hydrogen-linear-Stark-matches-NBS__nondegenerate-cesium-quadratic-Stark-matches-NIST; observed sealed-two-J-plus-one-and-linear-Zeeman-match-NIST__degenerate-hydrogen-linear-Stark-matches-NBS__nondegenerate-cesium-quadratic-Stark-matches-NIST; exact match True
- NIST-CESIUM-QUADRATIC-STARK: predicted sealed-two-J-plus-one-and-linear-Zeeman-match-NIST__degenerate-hydrogen-linear-Stark-matches-NBS__nondegenerate-cesium-quadratic-Stark-matches-NIST; observed sealed-two-J-plus-one-and-linear-Zeeman-match-NIST__degenerate-hydrogen-linear-Stark-matches-NBS__nondegenerate-cesium-quadratic-Stark-matches-NIST; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The claim fails if multiplicity or a field order differs, the degeneracy discriminator fails, the exact voltage/spacing translation fails, a displayed value or uncertainty status changes, target access precedes sealing, or observational provenance is concealed.

Measurement receipt: `sha256:cf170c1d95baa5030c8cfca839145219a52bab2be16e2e7a2a1d62fdeb1b1449`. Isolation certificate: `sha256:4717b7bb4b2d8fa0675a75566fc0c1dea71899f66663d54fef1a1ee55a1ddb7a`. Custody certificate: `sha256:a233fd690200cf6b6479c41cfae6179b44220ecfef5eb1a9ff07592386dcda91`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/Pubs/AtSpec/node12.html (sha256:5093c824cec5438a98f82fd5e75460e737a001a11471d12dbd7cd67b930661df)
- National Bureau of Standards: https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nbsspecialpublication617.pdf (sha256:5093c824cec5438a98f82fd5e75460e737a001a11471d12dbd7cd67b930661df)
- National Institute of Standards and Technology: https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=933361 (sha256:5093c824cec5438a98f82fd5e75460e737a001a11471d12dbd7cd67b930661df)

Registered target identities:

- NIST-WEAK-FIELD-ZEEMAN from NIST-NBS-ATOMIC-FIELD-SPLITTING-2026
- NBS-HYDROGEN-LINEAR-STARK from NIST-NBS-ATOMIC-FIELD-SPLITTING-2026
- NIST-CESIUM-QUADRATIC-STARK from NIST-NBS-ATOMIC-FIELD-SPLITTING-2026

Shared claim-record clause `PHYS-S019` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured Zeeman coefficient, Stark coefficient, splitting or source access in execution
- no numerical-zero central state or negative signed proof scalar
- no irrational, imaginary or floating proof value
- no universal Stark order that erases the degeneracy discriminator
- no fitted g factor, dipole, polarizability or response coefficient

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:5c0102ba97c50ee7f4074dfd64e096c867f1ff60dc87f8c8f327ce5fe7ed21ab`; engine receipt `sha256:4c1b970f750e939d90fa57d37316a7fbaeb9883643fb582613aea733e38b9c4b` at `receipts/engine/model_admitted/SFT-PHYS-ATOMIC-FIELD-SPLITTING-TERMINAL-005-4c1b970f750e939d.json`; empirical-validation hash `sha256:38f2d39e9d8abe07d3fd177f2f6c417315d51b36060f1f151db4e5c1a317225f`; measurement receipt `sha256:cf170c1d95baa5030c8cfca839145219a52bab2be16e2e7a2a1d62fdeb1b1449`.

20. Nuclear Hadronic

Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries.

220. Nuclear binding and mass relation

Claim identity: `SFT-PHYS-NUCLEAR-BINDING-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Nuclear binding is the closed joint-word recurrence whose separated constituent inertial support exceeds the bound-word inertial support by exactly the energy released into named carriers during composition.

> *Nuclear binding pairs a positive released energy carrier with the exact reduction of separated constituent inertial support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-MATTER-MIXING-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__composition-energy-inertial-support-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`composition-energy-inertial-support-pairing`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:41ce5dd6049ce6d8c29a6259adca52d9a1daa2d1ce060c73c9e587d2693c57eb`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:fd81a904c6c352b69b65f42e7480ce913ebcfba32e8ecf6697542de0587ecfe5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7319c9d7e19e03ea80eda886ea4656be56455b7390a319a4ce06e0f54790b57c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:195fca22a5ce2ebd1b964d027360a7ce9191601530fb05ea2c1b697f7ff04d37`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f357cf87333e729a5c24cb2024a070e081048c02ff807adc26935fa8e3e998df`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:13230a01de73f6c8dc99c99cc990cfe58483ed3d9c24013610fc2d941f3b834a`. Independent certificate: `sha256:ab40eb1ab93ae74c3d7c9804d11fdf239d29147c5926a59521295a02f0545202`. Engine external-validation hash: `sha256:b5fb7e3788cf07d165ec72915679ccb28ad9c6ec0b374103b2d3d8a69e4ea54f`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAEA-ENSDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-binding-1: predicted nuclear-binding-mass-relation; observed nuclear-binding-mass-relation; exact match True
- nuclear-binding-2: predicted nuclear-binding-mass-relation; observed nuclear-binding-mass-relation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:c742e2f226709803fe09dcf898572c36f2b8b7b58ae51eb689aa2d5ffb9549ff`. Isolation certificate: `sha256:240fa9bac3af622b43a3cc0cd73cf383451974aea26c9f87e72f84701a151281`. Custody certificate: `sha256:39ebe251b139549e5ca3e0a7c3222d813829154e68502b62ca41ea061b21537b`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://nds.iaea.org/ensdf/ (sha256:c17d03b30d117a48b776fd143a2e4224bd4f53bc774d394c1343645329a0ce93)

Registered target identities:

- nuclear-binding-1 from IAEA-ENSDF-CURRENT-2026-07-23
- nuclear-binding-2 from IAEA-ENSDF-CURRENT-2026-07-23

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:a3d6a1bfb80fe1c0e401a07d5b799f77e6fc3aa0f5693b81b46c55a07a751cc3`; engine receipt `sha256:b7d9018630e2f7bc82b25b4f0814dcf216773f2ae4b07242ce80b099ac5546d2` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-BINDING-001-b7d9018630e2f7bc.json`; empirical-validation hash `sha256:360c3e56f3cf45fad87e444b2aa732ed7669956c5efbf15cb376adcb01d6195a`; measurement receipt `sha256:c742e2f226709803fe09dcf898572c36f2b8b7b58ae51eb689aa2d5ffb9549ff`.

221. Nuclear levels and transitions

Claim identity: `SFT-PHYS-NUCLEAR-LEVELS-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Nuclear levels are the exact boundary-closed recurrence classes of a composite nuclear word; transitions are conserved carrier differences between named classes.

> *Nuclear levels are discrete recurrence classes of bound composite Fold support with source-bound transitions.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-NUCLEAR-BINDING-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__composite-boundary-recurrence-classes__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`composite-boundary-recurrence-classes`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:1d1e276a544e6571c434d85768955d353cc7063fd0953e86607331335aaa55df`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:327a71cc8de73d2a1c78a174dc6970825c5522d443715a42c4395d6b1d8af7e7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:653c74bcd47a95a1311142e94053c696505967b25fd90664363116fbecdc26e4`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:300df7b3eb317cfdfeb054c30f8bf07707fb63158e6be614b57d9e13805e0c2e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ebfadd76b8db50a577d6da9f56e084e6260e6f00893d29aa7afe61517b56a497`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:57005aa5b4ea43540c6c5c2b6c90a9d03f7ccabdf65f58e8b37041d3847fa204`. Independent certificate: `sha256:bba43a62fb1fccfdb86973be39bb8b5fca71241a3ca036d137ac74bf796514b2`. Engine external-validation hash: `sha256:c3b7043246d5c7310ba96c8e84645e136bdee576495d1c7425fa27cb671a78c7`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAEA-ENSDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-levels-1: predicted nuclear-level-recurrence; observed nuclear-level-recurrence; exact match True
- nuclear-levels-2: predicted nuclear-level-recurrence; observed nuclear-level-recurrence; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:64bf8280d66199f2fac6c626b55217f1206c34133bb52ca9f935198f20da63e3`. Isolation certificate: `sha256:e7c1be300263a29e3437b2c9b6a550e31bbea10dbbd22fed1be691cbf3d2d77e`. Custody certificate: `sha256:aefd92d8c0df9c01cea86b3bf978766065a29830b239c3ed36a5b22bf8c31135`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://nds.iaea.org/ensdf/ (sha256:c17d03b30d117a48b776fd143a2e4224bd4f53bc774d394c1343645329a0ce93)

Registered target identities:

- nuclear-levels-1 from IAEA-ENSDF-CURRENT-2026-07-23
- nuclear-levels-2 from IAEA-ENSDF-CURRENT-2026-07-23

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species

- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:b5ced32a6a1a278dcf9fb9d371d425cca12dc49fdbdf7ea1ba7e3b1bb8b33313`; engine receipt `sha256:c44425e09076186d367c47513afb11ccf0f5685fd00df98d7a8ed7beb14b2916` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-LEVELS-001-c44425e09076186d.json`; empirical-validation hash `sha256:eef6d8a7df3f8614f712c843ad7736358eed5472cc0417dba0ec473127303e16`; measurement receipt `sha256:64bf8280d66199f2fac6c626b55217f1206c34133bb52ca9f935198f20da63e3`.

222. Radioactive transition support

Claim identity: `SFT-PHYS-NUCLEAR-RADIOACTIVITY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Radioactivity is a nuclear recurrence with lawful output channels but no stable self-return on the observed support; exact event weights arise from complete deterministic hidden-path census.

> *Radioactivity is deterministic complete-path transition from a nuclear word lacking stable observed recurrence.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-NUCLEAR-LEVELS-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__nonreturning-nuclear-transition-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`nonreturning-nuclear-transition-support`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:6b98f20b4d4443f9491a266859ca1de0ad7d7c5acb77ba39c79c5746ad317386`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7e9ccd542cf7e94950e6e4aa1e519f1337c9368433868c2ca3703d920a9562e4`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:83f0826fdd6d0e6db144c50f30bdbcc9be7c25e0aa72c829226abc11ca3e6314`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:404a845a88efd25dbe794b73e078814ad4636aeff85114ecb1a00b642a80963f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:eaa65ad66b7acc1ad1ded6aa28a4ca7d3f4bda1c0265fe92bc6cadcfc21bfad7`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8bc09f9171ba803b1633192af125e6695a777ea38bf18ad5d9da6015d11f0924`. Independent certificate: `sha256:c8c4cf90ccf96270f8e98d74e35afb66cdb835387dfa783b2f8778240884f741`. Engine external-validation hash: `sha256:8c6f11383d1a9c9ae7b4f7ad33db8d21c9455878503f4e0a324958b776a99b53`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAEA-ENSDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-radioactivity-1: predicted radioactive-transition-support; observed radioactive-transition-support; exact match True
- nuclear-radioactivity-2: predicted radioactive-transition-support; observed radioactive-transition-support; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:fc8ad9bf968d6f5096490a2cf2e3562a235904f18606074ba638448815df9cae`. Isolation certificate: `sha256:7b8f166d9a7f0f44ceeb8a082c1087d4bde650f16be695d99e3666bd52298da6`. Custody certificate: `sha256:aad0327ad01a5cb173fe2e5f0fbb5bcaaae11cbb5ba6774734bac6cc8067e639`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://nds.iaea.org/ensdf/ (sha256:c17d03b30d117a48b776fd143a2e4224bd4f53bc774d394c1343645329a0ce93)

Registered target identities:

- nuclear-radioactivity-1 from IAEA-ENSDF-CURRENT-2026-07-23
- nuclear-radioactivity-2 from IAEA-ENSDF-CURRENT-2026-07-23

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set

- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:00aed5f8cc17893380baf4b6f07d19df98b984e64948cac5415473f12182756e`; engine receipt `sha256:4e90565b4e6d54f5247bec21bd83c2deb7f3748fe72c100d3c4a5473811266d7` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-RADIOACTIVITY-001-4e90565b4e6d54f5.json`; empirical-validation hash `sha256:99f8a32f3520b41b9c8a9c0b34024e6a24cfe0a48e548006a079e43c38a12782`; measurement receipt `sha256:fc8ad9bf968d6f5096490a2cf2e3562a235904f18606074ba638448815df9cae`.

223. Nuclear reaction and channel accounting

Claim identity: `SFT-PHYS-NUCLEAR-REACTIONS-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A nuclear reaction is a complete map between incident and outgoing composite words in which every material, energy, momentum and held conserved label is paired across the boundary.

> *Nuclear reactions are exact closed maps across complete generated incident and product channels.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-NUCLEAR-RADIOACTIVITY-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__complete-nuclear-channel-accounting__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`complete-nuclear-channel-accounting`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:cb8f8ae45b91cf6bdf5ca5b9ddbafea5d7ac82998c177a7a744ea6e264da3165`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:65887188ea7c159786c06b3c69f0295b3bb31780ee1b2a068f1b91429f2cd077`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:51f7f0eac3a58bed76c33fa2c48b1790fd7134b7f80ccdb63877b6c421e655bd`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3201fac045e03496efadcc814789850c6a1dec7f2e47278b0ebfaef3ed7decb1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6bf41deed8cdec0fd0fde987f16271b96d3490911393a5366d923a4f6e52828a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5dae5b17cd7967171984f76fe3a5fc7931906d98c4d43ef699ca0bdc2213f099`. Independent certificate: `sha256:0222736e5d94d5fd9354aac097ea8eb3337b66bf92a6858f908a9e7918252573`. Engine external-validation hash: `sha256:dca0bdaec4dd150587f5a94bf5703ff74eed35d68e99521d77c9a766a47501dc`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAEA-ENDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-reactions-1: predicted nuclear-reaction-channel-closure; observed nuclear-reaction-channel-closure; exact match True
- nuclear-reactions-2: predicted nuclear-reaction-channel-closure; observed nuclear-reaction-channel-closure; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:332c4b633980e86388ef3e1565a8b61b06bbe99615bdc4a4c956cd84039e0830`. Isolation certificate: `sha256:3e5cea9680c46732f1f596e1b2df8bb5794820a6d6e0c5946fd07475afc15cd9`. Custody certificate: `sha256:3004d40f657b161e964b45e44312aced9283d96edb7250519316ab5008be75ef`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://www-nds.iaea.org/endf/ (sha256:16b8187ef7ef9024914aa7b54b6af07928a2f1d3bb97a6999f7563fa676ccbd9)

Registered target identities:

- nuclear-reactions-1 from IAEA-ENDF-CURRENT-2026-07-23

- nuclear-reactions-2 from IAEA-ENDF-CURRENT-2026-07-23

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:bb9f2c5a0caf772761ef940797a2036515ed92a51ca06a6eae6390653d9fed7b`; engine receipt `sha256:a66b3922831670f991c59f828fefce5d5bf36e157f2f16af7ac56886ad38be24` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-REACTIONS-001-a66b3922831670f9.json`; empirical-validation hash `sha256:d734737d2d8fc901f8e8adf05457ebb4bf23bd07d52dd864147e7116249b4ef7`; measurement receipt `sha256:332c4b633980e86388ef3e1565a8b61b06bbe99615bdc4a4c956cd84039e0830`.

224. Fission decomposition

Claim identity: `SFT-PHYS-NUCLEAR-FISSION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Fission is a nuclear transition whose output support contains multiple separately recurrent composite words and named released carriers, with all labels conserved across the decomposition.

> *Fission is conserved decomposition of one bound nuclear Fold word into multiple recurrent product words and released carriers.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-NUCLEAR-REACTIONS-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__bound-word-multiple-product-decomposition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`bound-word-multiple-product-decomposition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d598fce6425a71b2da54939f8f447c830c271a490a7d19c4c0963485b8b7b091`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8945ba3b665b979784ca31d3ea3de9e420d5ec2ca8c234a5719fbb9bf9e64de5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b69d6af60d92b523ec3246adfb4b1039b12979d2e4b73f1128080af26d5d2008`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:446277cc78578caf8243313372fa8da0542b513e37fca900760b935dde0d3165`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1a1d0d29304c8b346c5eea4f464658c45618a20cb1cc5135b55cecdd87bb657c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ac12c4c5196542894a2ab1002ece360d2c5ad971332c46e66f77d4be44587307`. Independent certificate: `sha256:c378af85ed65ad3e447c5497f76e416380c3d70bfcb24eeb663135f589afa458`. Engine external-validation hash: `sha256:eebb21e27726095e8c0e43cb0e7f0251a0535e70b5f256b1cec163143f011548`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAEA-ENDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-fission-1: predicted nuclear-fission-decomposition; observed nuclear-fission-decomposition; exact match True
- nuclear-fission-2: predicted nuclear-fission-decomposition; observed nuclear-fission-decomposition; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:2cae604cb73ef7900ffb8152792b672f4d22fe9f749a6dc4716a01a85963200d`. Isolation certificate: `sha256:a4d714c48a8ed2a075069d36caa72dee7bd6e493a61ea410c8543896eec17ee0`. Custody certificate: `sha256:78266d1894b13816bbf800b2edc7492be321feaf7743ef442979f36836564128`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://www-nds.iaea.org/endf/ (sha256:16b8187ef7ef9024914aa7b54b6af07928a2f1d3bb97a6999f7563fa676ccbd9)

Registered target identities:

- nuclear-fission-1 from IAEA-ENDF-CURRENT-2026-07-23
- nuclear-fission-2 from IAEA-ENDF-CURRENT-2026-07-23

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:6a4ac47c2e1a2554864bdac2e30702204e9c0025a76e78be1560aaf66dc19382`; engine receipt `sha256:3686041dfb9a9bfd34d127884e1642a257f331162f92295faefa8bea1584dc60` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-FISSION-001-3686041dfb9a9bfd.json`; empirical-validation hash `sha256:66b6b36ea9cfc97db0e5f7360e29e4f98b06b5e5733031ced625c97bca796f8e`; measurement receipt `sha256:2cae604cb73ef7900ffb8152792b672f4d22fe9f749a6dc4716a01a85963200d`.

225. Fusion composition

Claim identity: `SFT-PHYS-NUCLEAR-FUSION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Fusion is a transition from multiple incident nuclear words to one more tightly boundary-closed recurrent product word plus the exact positive carriers released by the mass-energy relation.

> *Fusion is conserved composition of multiple nuclear Fold words into a bound recurrent product and exact released carriers.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-NUCLEAR-FISSION-001`

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__multiple-word-bound-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	multiple-word-bound-composition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a1360d08edb3cc8a0f28e642c027176f528968e07fcd667d7813406acc61eb6f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:dcf009d6b2b76f5c957ff2982be5c3854b417e1d327c14c98dfb98c4b66ee982`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1a789c350ea0afb3aee5c9f53654677f9a627b2f0c989b3a34eee08369dca1d0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:de889777d2509a339723f1f13e24af5fb765b008fb431795f7f6e06dc10b9161`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8f1439a35ff8a600cf9fde87e3351fedad3eacc9488c343d570f0fc1d2654c16`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:290be909c8c1da1aeb994b4ed9541ada13c006702379e79ed7c326f7032ce4dc`. Independent certificate: `sha256:aaa7e043eec1826489229193189f1c25aad22db59b2635e678317e3c9a58f54e`. Engine external-validation hash: `sha256:14dc0df02354d966eef61665c897028a4a8fd8e65621162aea2429c1e69fac39`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAEA-ENDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-fusion-1: predicted nuclear-fusion-composition; observed nuclear-fusion-composition; exact match True
- nuclear-fusion-2: predicted nuclear-fusion-composition; observed nuclear-fusion-composition; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S018` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:2935b5dc01eaa21d4ddb0a79710e7a454b918eb1285487a64df1892ddce383b8`. Isolation certificate:

sha256:fa83c46945988e05f9f15d06ec4a29066cd790271d6218d1b9c2c64f50bb1443. Custody certificate: sha256:091ee8ae005a05c244143bf409b70d7bd46616e35b88b72c159681dac5267774.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://www-nds.iaea.org/endf/ (sha256:16b8187ef7ef9024914aa7b54b6af07928a2f1d3bb97a6999f7563fa676ccbd9)

Registered target identities:

- nuclear-fusion-1 from IAEA-ENDF-CURRENT-2026-07-23
- nuclear-fusion-2 from IAEA-ENDF-CURRENT-2026-07-23

Shared claim-record clause PHYS-S020 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:ce6089de05553400fc6c994fe49e6cb9b3c3faab2279d3f31710a9d0e20ca9d2; engine receipt sha256:30416752fa5273c24001c4536bab90b3e2eeb804086a18b3c3858485492c91d7 at receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-FUSION-001-30416752fa5273c2.json; empirical-validation hash sha256:efa9bfbae8a9f6453ac7a7f4775f3ebd10d338f0d3e6f572bf044ffe8093f695; measurement receipt sha256:2935b5dc01eaa21d4ddb0a79710e7a454b918eb1285487a64df1892ddce383b8.

226. Exact colour-sector Fold coupling

Claim identity: SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete two-label Fold distinction transported over the generator-three carrier forces the exact positive coupling ratio two-thirds, with no measured or fitted interaction strength.

> *The exact generator-three colour-sector Fold coupling is 2/3.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate every product of carrier, numerator, denominator, orientation, normalization, target custody and extension forms. The declared exact boundary is: All exact normalized ratios formed by transporting the complete Fold-label support across one generator-three interaction carrier. The generator produced 128 named candidates and the decision support contains 128 one-for-one decisions. Exactly one candidate survived: generator-three-carrier__both-Fold-labels__complete-generator-support__labels-held-through-carrier__two-over-three__closures-inaccessible-until-seal__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generator-three-carrier	free-colour-count	A free carrier count is a parameter.
numerator	both-Fold-labels	selected-single-label	One label omits half the Fold distinction.
denominator	complete-generator-support	external-normalization	External normalization is fitted.
orientation	labels-held-through-carrier	labels-collapsed	Collapse loses the interaction distinction.
normalization	two-over-three	reciprocal-or-unscaled-count	Those forms reverse roles or omit normalization.
target	closures-inaccessible-until-seal	magic-list-visible	A closure list would fit the coupling.
extension	no-extra-rule	extra-coupling-rule	An added strength is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:01a06556273c98111733f4d5f9e30ca0a0b83b04562e1400683508b426ae0600`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:e4e7687a9a1d832e556174d015e0aa64f11f64f92e06734a6051d3b987b676db`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:e2edad0e6d45799bbf3bb96bb8c4e89f8be5a715fc0d921e462d57b1d7d2e8f0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:12e6496b82b65e70e5458cdc2c1f49daecc9625414c9a58da9cda056fea5e5c6`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:5947e11fa098dddaf9eb6ce47ae72cd6b0492d99a75eeb6452299129b69e0996`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:babedcbfa7852074ccccf0885740154b359515c4b90ca34e584e8544bc59f31b`. Independent certificate: `sha256:8a8b86ffa63d4555d21aaab69918f4d60af798ad0b31d7e9c2014eb38779646e`. Engine external-validation hash: `sha256:a15be402310b73709b09d09f921b1f528900085c679701da9369955eb0604a8b`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:429ccbb5d38a6e36918c858af4e9cec9fddcb1e31eb23d9531ed71f7db93f44d`; engine receipt `sha256:5ac1c6ce8b4094ba57fd03f96cb30be011ea724f3856903a9ed0b0ab5ca3ddfd` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001-5ac1c6ce8b4094ba.json`; empirical-validation hash `None`; measurement receipt `None`.

227. Complete Fold nuclear-closure sequence

Claim identity: `SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Complete weak compositions across three spatial directions with two exclusion labels force the base filled-shell counts. The independently sealed two-thirds colour coupling reaches its first whole spin-orbit gap at angular count three and forces the reordered closure sequence 2, 8, 20, 28, 50, 82, 126, 184 with a depth-independent successor law.

> *The depth-independent nuclear closure law gives the exact prefix 2, 8, 20, 28, 50, 82, 126, 184; its next proton closure after 82 is 126 and the next closure after 126 is 184.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001`
- `SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-NUCLEAR-BINDING-001`
- `SFT-PHYS-NUCLEAR-LEVELS-001`
- `SFT-PHYS-NUCLEAR-REACTIONS-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete product of spatial composition, label multiplicity, base closure, coupling source, shift, threshold, reordering, recurrence, target custody and extension forms. The declared exact boundary is: All filled-shell closure schedules built from complete weak compositions over forced three-space, both exclusion labels and the first whole-gap action of the independently forced colour coupling. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `complete-three-direction-weak-compositions__two-exclusion-distinct-labels__positive-rank-composition-form__sealed-two-thirds-colour-coupling__top-angular-half-label-pair__least-positive-whole-gap__complete-top-orbit-capacity__piecewise-positive-successor-law__sequence-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`composition`	`complete-three-direction-weak-compositions`	`selected-shell-degeneracy`	A selected degeneracy imports a model.
`labels`	`two-exclusion-distinct-labels`	`single-or-duplicated-label`	Omission or duplication violates Fold support and exclusion.
`base`	`positive-rank-composition-form`	`closure-list`	A list has no generative proof.
`coupling`	`sealed-two-thirds-colour-coupling`	`fitted-spin-orbit-strength`	A fitted strength reads the closures.

Axis	Forced form	Eliminated form	Elimination reason
shift	top-angular-half-label-pair	free-shell-shift	A free shift is a parameter.
threshold	least-positive-whole-gap	selected-rank-threshold	A selected threshold fits the sequence.
reorder	complete-top-orbit-capacity	arbitrary-intruder-count	An arbitrary intruder count is fitted.
recurrence	piecewise-positive-successor-law	finite-eight-value-table	A table does not close later ranks.
target	sequence-inaccessible-until-seal	magic-numbers-visible	That is target leakage.
extension	no-extra-rule	extra-nuclear-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:67d186b6229c40775300a18be245b6a176a8afc16e8dd5336933222b707a81d4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:5c771cf5592a53cc672843d7cdadcfbe7e340db992858decb1ab49eb0d695e8b`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:6e2dc9948609f6a9bcffed8b9e06971fa9e84b851c60221af6c0726a7a4bf601`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8069e2016e09a7983c9c51f36313fbec85fe529516e27d9f31877bb68d776e78`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:9df3facdacb8b6591dcebf69efd2b75b8b300056734782cf444b7e08cdb38fc6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0c39e11b369cf6d388ac93d2942d49a037aa20f3cb2550601df04805eee0dcaa`. Independent certificate: `sha256:988e6fd342d2c09cabea9a075e383ac3d673f1e74279b820e4d79c2bbb3a37ea`. Engine external-validation hash: `sha256:a5900eda24e4ab5c9787189865afe525369245cb57accdca58373e45b1cbcee2`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:a6f2ce3b05bdd9695225e199028e32fc75ad9403517bd1c9de3e7490f27522ae`; engine receipt `sha256:eb48cd3f1935f2539cc128ad8259b18deb47ba9338585ceaa9f2ec6b6a4d7295` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001-eb48cd3f1935f253.json`; empirical-validation hash `None`; measurement receipt `None`.

228. Terminal nucleon composition, binding dominance and mass ordering

Claim identity: `SFT-PHYS-NUCLEON-BINDING-TERMINAL-005`

Shared claim-record clause `PHYS-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The generated period-three colour cycle closes as 1/7+2/7+4/7=One. Complete positive u/d count enumeration then uniquely gives the unit-charged word uud and empty-neutral word udd. At the admitted upper-quark depth seven, one of 128 cells is the explicit valence share and the positive predecessor 127/128 is the held-cycle share. Exact light-quark root enclosures place down above up by more than the admitted terminal proton electromagnetic dressing, so replacing one up by one down forces the neutron above the proton.

> *The proton word is uud and the neutron word is udd; the exact structural mass ledger is bare 1/128 and held-cycle 127/128, hence bare below 1/100 and held above 99/100; the exact down-minus-up enclosure exceeds the proton electromagnetic dressing and forces neutron mass above proton mass.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-MATTER-COMPOSITE-HADRONS-001`
- `SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001`
- `SFT-PHYS-MATTER-QUARK-CUBICS-003`
- `SFT-PHYS-MATTER-CONFINEMENT-LIFT-003`
- `SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete twelve-axis predecessor, colour, composition, depth, ledger, dominance, flavour, electromagnetic, ordering, custody, provenance and extension product. The declared exact boundary is: All minimal positive u/d count splits over the complete generated colour-three cycle, complete depth-seven binary support, the admitted light-quark algebraic enclosures and the admitted terminal proton electromagnetic dressing. The generator produced `4096` named candidates and the decision support contains `4096` one-for-one decisions. Exactly one candidate survived: `compose-immutable-predecessors__complete-period-three-One-cycle__enumerate-complete-three-flavour-charge-words__admitted-upper-quark-depth-seven__one-cell-and-complete-positive-predecessor__bare-below-one-percent-held-above-ninety-nine__exact-light-root-enclosure-order__compare-admitted-terminal-proton-dressing__one-down-replacement-forces-neutron-heavier__target-inaccessible-until-seal__registered-observational-prediction-protocol__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`compose-immutable-predecessors`	`rewrite-predecessor-laws`	A successor cannot alter admitted receipts.
`colour`	`complete-period-three-One-cycle`	`listed-three-colour-names`	Names do not prove a closed orbit.
`composition`	`enumerate-complete-three-flavour-charge-words`	`import-proton-neutron-labels`	Imported particle labels would select the result.

Axis	Forced form	Eliminated form	Elimination reason
`depth`	`admitted-upper-quark-depth-seven`	`selected-binding-depth`	A chosen depth can tune a mass share.
`ledger`	`one-cell-and-complete-positive-predecessor`	`free-binding-percentage`	A percentage read from observation is a parameter.
`binding`	`bare-below-one-percent-held-above-ninety-nine`	`qualitative-most-without-bound`	An unbounded slogan is not falsifiable.
`flavour`	`exact-light-root-enclosure-order`	`borrowed-quark-mass-order`	Borrowing a measured ordering reverses the derivation.
`electromagnetic`	`compare-admitted-terminal-proton-dressing`	`ignore-or-fit-electromagnetic-effect`	Ignoring or fitting the competing carrier cannot close the sign.
`ordering`	`one-down-replacement-forces-neutron-heavier`	`assert-measured-mass-order`	An asserted ordering has no Fold certificate.
`target`	`target-inaccessible-until-seal`	`external-target-readable`	Target access cannot seal a prediction.
`provenance`	`registered-observational-prediction-protocol`	`conceal-observational-development`	Concealment misstates the derivation history.
`extension`	`no-extra-rule`	`free-mass-term-or-percentage`	An ungenerated term is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:51d9fefec823ee956714b8e7eaebb9c38298d8aacb78e006d422d24f435b7cbd`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:54063610f9d68deac3d5bedebe77094f529e4c1797a39007600ee7ff74366834`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:f726e25682e5a2c5d4e6d66241b58d9ebe7d9a1fc97f541ad69fb788dc30f0ce`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:62b1a594364f0ae5025d07323672a17bf9e216d471708087ad46956473d0e28a`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:8afc24456299ab469c4baf3fa778a81b6eb71b94762f7a2872b38455570697f2`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a2fd2e24c2d9788f10dfab09cc2acc2b26cdb1612a23f11a278a74a27d1444c6`. Independent certificate: `sha256:b85a840ca3397b64ecb53beac739fbb11ab7cab78b1801ef32447aed79ae1ee0`. Engine external-validation hash: `sha256:8e103f8fd33e1d728c7e58ea37adace2dc9b567394dda63f41b0a9c32a338da7`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-NIST-NUCLEON-BINDING-2022-2026

Observed comparison records:

- PDG-THREE-COLOUR-QQQ-SINGLET: predicted sealed-three-colour-uud-udd-singlet-correspondence-passes-PDG__structural-one-over-128-and-PDG-uud-current-share-both-below-one-percent__nonidentity-of-those-two-objects-retained__held-residuals-above-99-percent__down-up-and-neutron-proton-ordering-pass-complete-PDG-NIST-intervals; observed sealed-three-colour-uud-udd-singlet-correspondence-passes-PDG__structural-one-over-128-and-PDG-uud-current-share-bot h-below-one-percent__nonidentity-of-those-two-objects-retained__held-residuals-above-99-percent__down-up-and-neutron-proton-ordering-pass-complete-PDG-NIST-intervals; exact match True

- PDG-LIGHT-QUARK-MASS-DOMINANCE: predicted sealed-three-colour-uud-udd-singlet-correspondence-passes-PDG__structural-one-over-128-and-PDG-uud-current-share-both-below-one-percent__nonidentity-of-those-two-objects-retained__held-residuals-above-99-percent__down-up-and-neutron-proton-ordering-pass-complete-PDG-NIST-intervals; observed sealed-three-colour-uud-udd-singlet-correspondence-passes-PDG__structural-one-over-128-and-PDG-uud-current-share-both-below-one-percent__nonidentity-of-those-two-objects-retained__held-residuals-above-99-percent__down-up-and-neutron-proton-ordering-pass-complete-PDG-NIST-intervals; exact match True
- NIST-NUCLEON-MASS-ORDERING: predicted sealed-three-colour-uud-udd-singlet-correspondence-passes-PDG__structural-one-over-128-and-PDG-uud-current-share-both-below-one-percent__nonidentity-of-those-two-objects-retained__held-residuals-above-99-percent__down-up-and-neutron-proton-ordering-pass-complete-PDG-NIST-intervals; observed sealed-three-colour-uud-udd-singlet-correspondence-passes-PDG__structural-one-over-128-and-PDG-uud-current-share-both-below-one-percent__nonidentity-of-those-two-objects-retained__held-residuals-above-99-percent__down-up-and-neutron-proton-ordering-pass-complete-PDG-NIST-intervals; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The claim fails if qqq colour closure differs, either complete dominance inequality fails, structural and scheme-dependent quantities are conflated, d/u or neutron/proton intervals cease to order, the direct mass difference leaves propagated intervals, a source/uncertainty changes, or target access precedes sealing.

Measurement receipt: `sha256:13acf48c26f08b73be838cf251df8c9e4faa3668fd6d431f806dd07f0149501c`. Isolation certificate: `sha256:98536a5223bbe845bc6842539332efd7df4af50b41301170dfd7f2cc4c9fc6c1`. Custody certificate: `sha256:6a4e8ba45dffcc4cf444dd3b6fe741a3db05cded0d37d76abf54da921be8d3ba`.

Registered source descriptions:

- Particle Data Group: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-model.pdf (sha256:2f75c5124ce570fad3becd1ff5901f6062c6dea05b5a41a1892819006fa6290e)
- Particle Data Group: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf (sha256:d544d099aa15739ec83d87711bd4c5b1e0a1032d6f70aaa4847ec78601f7aeae)
- NIST / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- PDG-THREE-COLOUR-QQQ-SINGLET from PDG-NIST-NUCLEON-BINDING-2022-2026
- PDG-LIGHT-QUARK-MASS-DOMINANCE from PDG-NIST-NUCLEON-BINDING-2022-2026
- NIST-NUCLEON-MASS-ORDERING from PDG-NIST-NUCLEON-BINDING-2022-2026

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer value or measured mass as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted binding percentage, constituent mass, electromagnetic correction or nucleon scale
- no claim that a scheme-dependent PDG current-quark mass ratio is identical to the structural 1/128 cell
- no target access before the derivation and prediction seals

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:8b0b929301167e783277929312f485a521a307eba5e8ebdafa2e92426965cfed`; engine receipt `sha256:f61d912f53959182fd1b5b3544dd750bd0beb095b4c72adc11323f375df90762` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEON-BINDING-TERMINAL-005-f61d912f53959182.json`;

empirical-validation hash `sha256:f57e81e2e81bdf41eee71bb3108abe10ba45bd0d3c5148e372b5e79f7229d6c4`; measurement receipt `sha256:13acf48c26f08b73be838cf251df8c9e4faa3668fd6d431f806dd07f0149501c`.

229. Terminal light-hadron multiplets and depth-independent Regge support

Claim identity: `SFT-PHYS-HADRON-REGGE-TERMINAL-005`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Generator-three flavour support forces nine ordered flavour/antiflavour cells, partitioned as one invariant cell class and its eight-class positive predecessor. Complete three-place flavour support forces twenty-seven cells; direct exchange enumeration partitions them as ten symmetric, two eight-cell mixed hands and one antisymmetric class. Independently, one retained fixed-tube act per positive spin successor forces exact affine squared support. The exact theorem governs normalised fixed-carrier support; whether physical resonance masses realize literal equal spacing is a separate post-seal measurement.

> *Light mesons have exact flavour support 9=8+1; light three-place baryons have exact support 27=10+8+8+1; normalized fixed-tube squared support is Q(J)=J and every positive successor difference is exactly One.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-MATTER-COMPOSITE-HADRONS-001`
- `SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001`
- `SFT-PHYS-MATTER-CONFINEMENT-LIFT-003`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete ten-axis predecessor, flavour, meson, partition, baryon, exchange, tube, affine-support, custody and extension product. The declared exact boundary is: Every ordered word over the complete generated three-member light-flavour support at widths two and three, every permutation-exchange class, and every positive rank produced by repeated application of one held fixed-tube successor act. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `compose-immutable-hadron-predecessors__complete-generator-three-flavour-support__enumerate-three-by-three-ordered-pairs__one-invariant-and-positive-predecessor__enumerate-three-place-flavour-words__enumerate-exchange-symmetry-classes__one-retained-step-per-spin-successor__depth-independent-affine-successor__target-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`compose-immutable-hadron-predecessors`	`rewrite-hadron-predecessors`	A successor cannot alter admitted receipts.
`flavour`	`complete-generator-three-flavour-support`	`import-named-flavour-table`	Names do not generate a support.
`meson`	`enumerate-three-by-three-ordered-pairs`	`assert-nonet`	A conventional name does not prove its size.
`partition`	`one-invariant-and-positive-predecessor`	`import-octet-singlet-representation`	An imported representation would select the answer.
`baryon`	`enumerate-three-place-flavour-words`	`assert-octet-and-decuplet`	Named multiplets are not a derivation.
`exchange`	`enumerate-exchange-symmetry-classes`	`import-SU-three-decomposition`	A conventional group decomposition cannot be a premise.
`tube`	`one-retained-step-per-spin-successor`	`chosen-spin-dependent-increment`	A different free increment at each rank destroys structural closure.
`Regge`	`depth-independent-affine-successor`	`fitted-mass-polynomial`	A fitted polynomial is not forced.

Axis	Forced form	Eliminated form	Elimination reason
`target`	`target-inaccessible-until-seal`	`external-target-readable`	Target access cannot seal a prediction.
`extension`	`no-extra-rule`	`free-residual-or-tension-fit`	An ungenerated correction would be a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:467a7cbe4dacc0e1f6fde3a90214820d1ca8d2cc9aedfc23ac9983bc6d6de760`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:a51239286297ee3bbf8b2985502ac279a2d0028270b705005f1328f1837f935d`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:792f08f5e1624f6c8026ac2fb851d08fe34493dac3f85501c655ddfabdcf537e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ce1e1df74076e998d8540199220566bda0b71667a0f23a7b0648f9a952ab3bbd`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:29a2e04e67f50893cc95aaf395d84a178a79f4bd01e8616165309005cc110d5e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:99065993d853f987480ba1f1dd34c0811d501a9607b9002c7e8a03496acb261f`. Independent certificate: `sha256:7e65311b23611ed4c357b35b21a1abeed3e0d6895139fe5ec355147f22c73113`. Engine external-validation hash: `sha256:2d286df3ce8433fea887ae36d82a9ec2bcd937e5e74f808466c37c35fbb0777e`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-HADRON-MULTIPLETS-REGGE-2025

Observed comparison records:

- PDG-LIGHT-MESON-MULTIPLET: predicted sealed-nine-equals-eight-plus-one-and-twenty-seven-equals-ten-plus-eight-plus-eight-plus-one-match-PDG__complete-five-state-natural-parity-mass-vector-is-strictly-ordered__literal-common-squared-mass-step-is-rejected-by-complete-reported-intervals__exact-affine-Fold-law-retained-at-normalised-fixed-carrier-boundary__rho5-summary-omission-status-retained; observed sealed-nine-equals-eight-plus-one-and-twenty-seven-equals-ten-plus-eight-plus-eight-plus-one-match-PDG__complete-five-state-natural-parity-mass-vector-is-strictly-ordered__literal-common-squared-mass-step-is-rejected-by-complete-reported-intervals__exact-affine-Fold-law-retained-at-normalised-fixed-carrier-boundary__rho5-summary-omission-status-retained; exact match True
- PDG-LIGHT-BARYON-MULTIPLETS: predicted sealed-nine-equals-eight-plus-one-and-twenty-seven-equals-ten-plus-eight-plus-eight-plus-one-match-PDG__complete-five-state-natural-parity-mass-vector-is-strictly-ordered__literal-common-squared-mass-step-is-rejected-by-complete-reported-intervals__exact-affine-Fold-law-retained-at-normalised-fixed-carrier-boundary__rho5-summary-omission-status-retained; observed sealed-nine-equals-eight-plus-one-and-twenty-seven-equals-ten-plus-eight-plus-eight-plus-one-match-PDG__complete-five-state-natural-parity-mass-vector-is-strictly-ordered__literal-common-squared-mass-step-is-rejected-by-complete-reported-intervals__exact-affine-Fold-law-retained-at-normalised-fixed-carrier-boundary__rho5-summary-omission-status-retained; exact match True
- PDG-NATURAL-PARITY-TRAJECTORY: predicted sealed-nine-equals-eight-plus-one-and-twenty-seven-equals-ten-plus-eight-plus-eight-plus-one-match-PDG__complete-five-state-natural-parity-mass-vector-is-strictly-ordered__literal-common-squared-mass-step-is-rejected-by-complete-reported-intervals__exact-affine-Fold-law-retained-at-normalised-fixed-carrier-boundary__rho5-summary-omission-status-retained; observed sealed-nine-equals-eight-plus-one-and-twenty-seven-equ

als-ten-plus-eight-plus-eight-plus-one-match-PDG__complete-five-state-natural-parity-mass-vector-is-strictly-ordered__literal-common-squared-mass-step-is-rejected-by-complete-reported-intervals__exact-affine-Fold-law-retained-at-normalised-fixed-carrier-boundary__rho5-summary-omission-status-retained; exact match True

- deliberately tampered unfavourable control rejected

Falsification condition: The claim fails if either sealed partition differs, the normalised affine successor ceases to be exact, any registered mass/uncertainty/status row is lost, strict ordering changes, the exact common-step result is misstated, a fitted correction is introduced, or target access precedes sealing.

Measurement receipt: `sha256:f43ce49f3e2ad612efbbd2f3036bbbceb1383e70920b098f8904946f25e709e5`. Isolation certificate: `sha256:94f12450697b4db9c3262c0ba17e4d9a8e52c8903f38f9890b0e460699c45662`. Custody certificate: `sha256:5da6bbe00d8154e94767fcff8e6627db0ddced2d3716e7e5e8aefa1e061f7e2d`.

Registered source descriptions:

- Particle Data Group: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-model.pdf (sha256:c6e671212984fa6b6956a3d748c12c81879686b9198ca5a0818670fe63301358)
- Particle Data Group: https://pdg.lbl.gov/2025/tables/rpp2025-sum-mesons.pdf (sha256:c6e671212984fa6b6956a3d748c12c81879686b9198ca5a0818670fe63301358)
- Particle Data Group: https://pdg.lbl.gov/2025/listings/rpp2025-list-rho5-2350.pdf (sha256:c6e671212984fa6b6956a3d748c12c81879686b9198ca5a0818670fe63301358)

Registered target identities:

- PDG-LIGHT-MESON-MULTIPLET from PDG-HADRON-MULTIPLETS-REGGE-2025
- PDG-LIGHT-BARYON-MULTIPLETS from PDG-HADRON-MULTIPLETS-REGGE-2025
- PDG-NATURAL-PARITY-TRAJECTORY from PDG-HADRON-MULTIPLETS-REGGE-2025

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer value or conventional SU(n) representation as a premise
- no measured resonance mass, uncertainty, trajectory slope or source access in execution
- no numerical-zero state, negative, irrational, imaginary or floating proof value
- no fitted tube tension, intercept, residual, correction term or selected trajectory subset
- no claim that non-minimal colour-neutral composites are forbidden
- no target access before derivation and prediction seals

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:2b95046c74ce7250d50ab9ca6fe980f6907fe73c953e469fa665f3525ba87cd7`; engine receipt `sha256:66bc011c6612641467f7deb8554279abbc55311f86f34efab51ac3fa2f718b29` at `receipts/engine/model_admitted/SFT-PHYS-HADRON-REGGE-TERMINAL-005-66bc011c66126414.json`; empirical-validation hash `sha256:41b2181924a6893266bd4dff6210830833f7d4d5c2b5a8b90a85ae3ee0b18821`; measurement receipt `sha256:f43ce49f3e2ad612efbbd2f3036bbbceb1383e70920b098f8904946f25e709e5`.

230. Terminal residual nuclear interaction and inverse-mediator range

Claim identity: `SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A complete colour-three composite closes every leading colour label inside its boundary, so the external leading act is the empty form. The first nonempty exchange must retain one boundary act from each of two neutral composites; the forced half-One therefore enters twice and fixes the second-order structural support at one-quarter. This is an exact Fold order, not a universal dimensional cross section. A present mediator mass is a positive carrier and forces the finite reciprocal range One/m; an absent mass label has no finite subtraction endpoint. Thus lighter massive mediators reach farther, while finite boundary cells and finite range force locality and saturation without an imported exponential or hard numerical-zero cutoff.

> *Leading colour exchange outside a singlet is the empty form; the first retained two-boundary order is (1/2)^2=1/4 and returns 1/4 to 1/2 to One in two Folds; every positive mediator has finite normalized range 1/m and m1<m2 forces 1/m1>1/m2.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-HALF-ONE-001`
- `SFT-FOUNDATION-FOLD-DYNAMICS-001`
- `SFT-PHYS-MATTER-COMPOSITE-HADRONS-001`
- `SFT-PHYS-MATTER-CONFINEMENT-LIFT-003`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-NUCLEON-BINDING-TERMINAL-005`
- `SFT-PHYS-HADRON-REGGE-TERMINAL-005`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete ten-axis predecessor, neutral closure, residual order, strength scope, mass class, reciprocal range, ordering, saturation, custody and extension product. The declared exact boundary is: Every exchange between two complete colour-three neutral composite boundaries, the forced half-One primary carrier, absent or exact positive mediator-mass labels, and every positive mass ordering under reciprocal range. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `compose-immutable-neutral-composite-predecessors__leading-colour-act-closes-to-empty-form__paired-half-One-boundary-act__structural-order-not-universal-dimensional-strength__empty-mass-label-or-positive-mass-carrier__exact-reciprocal-mass-scale__lighter-positive-mass-has-greater-reciprocal__finite-boundary-cells-with-finite-range__target-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`compose-immutable-neutral-composite-predecessors`	`rewrite-nucleon-predecessors`	A successor cannot alter admitted receipts.
`neutral`	`leading-colour-act-closes-to-empty-form`	`export-raw-colour-label`	A raw colour label would violate the admitted singlet boundary.
`residual`	`paired-half-One-boundary-act`	`reuse-primary-half-One`	A first-order carrier ignores both neutral closures.
`strength`	`structural-order-not-universal-dimensional-strength`	`identify-quarter-with-all-cross-sections`	Measured channels need not share one dimensional strength.
`mass`	`empty-mass-label-or-positive-mass-carrier`	`numerical-zero-mass`	Numerical zero is outside the proof domain.
`range`	`exact-reciprocal-mass-scale`	`chosen-decay-length`	A selected length is a parameter.
`ordering`	`lighter-positive-mass-has-greater-reciprocal`	`import-particle-range-ranking`	A named particle table cannot prove the order.
`saturation`	`finite-boundary-cells-with-finite-range`	`all-to-all-independent-links`	Unlimited independent links ignore finite boundary support and finite range.
`target`	`target-inaccessible-until-seal`	`external-target-readable`	Target access cannot seal a prediction.
`extension`	`no-extra-rule`	`free-decay-profile-or-coupling`	An ungenerated profile or coefficient is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:86de94911553fc9e84d7c586f2c45c0d46aa4cf3c3be1d2bae96e4d8aaa75f37`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:34099156fc06f273a51f09f7da7504d4bd59a4eebe2319e5b955a7f71fd57238`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:c536cc85949216dab482a7c2500ddd85ba2311b6a3c958047544ed7ba5ffd6de`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:14fd7b1b50d0e825f3c03fc2beddd9d0826b31ddc790aa5b78197be3b1b0b322`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:f3bfca3b8afbef80625227c4b545a628b8d00e68e0d6b07b958364f0d1e2828a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4555d9828b920fb140c8fcaf9160cb9766d156d878865c635d4243704ffbe94a`. Independent certificate: `sha256:a5e981f99ef37dcc7a01729bee70b109b49ec1e697bdd8050f145aea02f578d2`. Engine external-validation hash: `sha256:29b710e08f664d2a8b09c2c6e406c29e2b4f7d71747fe058ad362a9ba3c6e777`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-NIST-NUCLEAR-RESIDUAL-RANGE-2025-2026

Observed comparison records:

- PDG-MEDIATOR-MASS-RANGE: predicted sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; observed sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; exact match True
- NIST-HBAR-C-RANGE-TRANSLATION: predicted sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; observed sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; exact match True
- NIST-NEUTRON-SCATTERING: predicted sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; observed sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PD

G-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; exact match True

- NIST-PION-LONG-RANGE-CONTEXT: predicted sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; observed sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The claim fails if neutral leading closure or quarter order changes, reciprocal ordering fails, the complete pion and vector ranges overlap, either scattering interval ceases to be positive, channel dependence is erased, one-quarter is relabelled as a measured strength, or target access precedes sealing.

Measurement receipt: `sha256:8f5016866c9a285b44d40db24b0910e2c9904da4d66e263bb58fbdb1a609a6b1`. Isolation certificate: `sha256:3c04fbeab4cc1174b730b02f647067c8802c00ca0dba7a84d771b0d4055c25a7`. Custody certificate: `sha256:29ec184f1e0a7db74d47f095ac9b9a87ff8fdd30847ea872e5b064f122bb1477`.

Registered source descriptions:

- Particle Data Group: https://pdg.lbl.gov/2025/tables/rpp2025-sum-mesons.pdf (sha256:898f2c17eaadb3419aa5ee9be4519b21eeeaf5523045111bb0a04017d18a11d1)
- National Institute of Standards and Technology / CODATA: https://physics.nist.gov/cuu/pdf/all.pdf (sha256:898f2c17eaadb3419aa5ee9be4519b21eeeaf5523045111bb0a04017d18a11d1)
- NIST Center for Neutron Research: https://www.ncnr.nist.gov/resources/activation/scattering_table.html (sha256:898f2c17eaadb3419aa5ee9be4519b21eeeaf5523045111bb0a04017d18a11d1)
- National Institute of Standards and Technology: https://nvlpubs.nist.gov/nistpubs/jres/110/3/j110-3pag.pdf (sha256:898f2c17eaadb3419aa5ee9be4519b21eeeaf5523045111bb0a04017d18a11d1)

Registered target identities:

- PDG-MEDIATOR-MASS-RANGE from PDG-NIST-NUCLEAR-RESIDUAL-RANGE-2025-2026
- NIST-HBAR-C-RANGE-TRANSLATION from PDG-NIST-NUCLEAR-RESIDUAL-RANGE-2025-2026
- NIST-NEUTRON-SCATTERING from PDG-NIST-NUCLEAR-RESIDUAL-RANGE-2025-2026
- NIST-PION-LONG-RANGE-CONTEXT from PDG-NIST-NUCLEAR-RESIDUAL-RANGE-2025-2026

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer value, Yukawa potential or consensus interaction model as a premise
- no particle name, measured mass, cross section, length, uncertainty or source access in execution
- no numerical-zero state, negative, irrational, imaginary or floating proof value
- no identification of structural one-quarter with a universal dimensional nuclear-force strength
- no fitted coupling, exponential decay profile, hard cutoff or selected mediator scale
- no target access before derivation and prediction seals

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:73c5f4b87b1827e6b3f5290aca04c417613f9e7e6517c382dd69c7b8e24b7df1`; engine receipt `sha256:918406dee504d2f2b2ce5f178a98d88a3fb7cc3dd52fea8e948a675da034201d` at `receipts/eng` `ine/model_admitted/SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005-918406dee504d2f2.json` ; empirical-validation hash `sha256:095bb2fc70fcb27e010e4c48a68d4bca8af701602766ac9aff509295d6486283`; measurement receipt `sha256:8f5016866c9a285b44d40db24b0910e2c9904da4d66e263bb58fbdb1a609a6b1`.

231. Terminal zero-parameter nuclear binding curve and stability maximum

Claim identity: `SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Short-range quarter-order residual recurrence and exact electromagnetic opposition force one positive bulk/surface/Coulomb/asymmetry/pairing ledger with no fitted coefficient. The surface share is exactly One over the binary-plus-generator interface carrier five; Coulomb loss is the admitted exact alpha over every ordered distinct proton path; asymmetry and pairing carry the admitted quarter-order. Exact rational radius enclosures, complete concave integer/parity enumeration below the forced mass 2^(5+7), and a two-case tail induction uniquely maximize normalised binding at mass count 62, charge count 28 and neutron count 34. No nuclide name or measured binding value enters execution.

> *The unique zero-parameter normalized binding maximum over every positive proton/neutron composition is A=62, Z=28, N=34. The exact ledger is bulk One; radial surface loss (1/5)/A^(1/3) represented only by rational enclosures; Coulomb loss alpha*Z*(Z-1)/A^(4/3); quarter-order asymmetry loss (1/4)*(N-Z)^2/A^2; and quarter-order parity gain/loss (1/4)/A^(4/3).*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-NUCLEAR-BINDING-001`
- `SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005`
- `SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete ten-axis predecessor, bulk, surface, Coulomb, asymmetry, pairing, radius, maximum, target-custody and extension product. The declared exact boundary is: Every positive proton/neutron composite, exact saturated bulk, forced 1/5 interface share, exact alpha ordered charge path, quarter-order imbalance/pairing class, rational rank-two/rank-three enclosure and every mass beyond the forced 4096 tail boundary. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `compose-immutable-predecessors__one-saturating-bulk-support__one-over-binary-plus-generator-interface__exact-alpha-times-ordered-charge-paths__quarter-order-unmatched-label-square__quarter-order-paired-gain-unpaired-loss__rational-enclosures-until-order-separates__complete-concave-census-plus-forced-tail-induction__target-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`compose-immutable-predecessors`	`rewrite-predecessor-laws`	A successor cannot alter admitted receipts.
`bulk`	`one-saturating-bulk-support`	`unbounded-pairwise-bulk-growth`	Short-range saturation forbids growth with every distant constituent.
`surface`	`one-over-binary-plus-generator-interface`	`fitted-surface-coefficient`	A fitted coefficient is a parameter.

Axis	Forced form	Eliminated form	Elimination reason
coulomb	exact-alpha-times-ordered-charge-paths	import-semi-empirical-coulomb-term	An imported mass formula cannot select the law.
asymmetry	quarter-order-unmatched-label-square	chosen-asymmetry-penalty	A chosen coefficient is not forced.
pairing	quarter-order-paired-gain-unpaired-loss	fitted-pairing-term	A fitted parity correction is a parameter.
radius	rational-enclosures-until-order-separates	evaluate-irrational-radius	Irrational values are outside the exact proof domain.
maximum	complete-concave-census-plus-forced-tail-induction	selected-isotope-window	A chosen neighborhood cannot prove a global maximum.
target	target-inaccessible-until-seal	external-target-readable	Target access would fit the isotope coordinate.
extension	no-extra-rule	free-coefficient-or-residual	An ungenerated correction destroys zero-parameter closure.

Base and successor. The closure proof is sealed by derivation hash `sha256:52b045b235c55325dd947ece9092ee10b90faaeda53a9bd6f01cbbb60023b1bf`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:aa5bac9837a7818ed70ba9ca95fa92717770e5029da870ed653a7c8f483cf173`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:1d88222acb46af18d14b0cb9c13022ee6d500a6454fd8b6639c38e0590b8eb90`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:cdf01f744269da954e96e5160deadcb7d7a46e6b4f481fbe58a74825b0e37a62`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:a02ce3ddca54d64008d3aa110b8826d6ea8f659fc72c768e70774eb375ea5646`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e39db94459ddc98b0c3e5814f690c31c7fe6f1b712bf53bd925067c384a0f74d`. Independent certificate: `sha256:15f531e8c5394402b340056decd1679d7d6b65be5e837b6766b4b29daf33b97d`. Engine external-validation hash: `sha256:43e840827408a7c49a848af7f82095afd54f2abaac48e57c0afd86352bfaacb6`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- AMDC-AME2020-MASS-1-BINDING-2021

Observed comparison records:

- AME2020-COMPLETE-BINDING-CENSUS: predicted sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; observed sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; exact match True

- AME2020-GLOBAL-MAXIMUM: predicted sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; observed sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; exact match True
- AME2020-LIGHT-TO-IRON-NICKEL-RISE: predicted sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; observed sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; exact match True
- AME2020-HEAVY-FALL: predicted sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; observed sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; exact match True
- AME2020-IRON-ONLY-ADVERSE: predicted sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; observed sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The claim fails if the complete measured table has another maximum, the predicted coordinate differs, uncertainty intervals erase the separation, either curve direction fails, any numeric rival is omitted, or target access precedes the formal seal.

Measurement receipt: `sha256:bc92b527d85825a9450fdb69e80dd9577ef5b3600fc9283bbdc0ef6fda36e1f8`. Isolation certificate: `sha256:158a1f7451bfb1eac45f914b1e2867187b67a3334e2f3f70ce13d391677830db`. Custody certificate: `sha256:5315c0ba0d5c124ff56134799321c8d0f270c4ca2b6e29c0cef7be6b61781e58`.

Registered source descriptions:

- Atomic Mass Data Center, Institute of Modern Physics, Chinese Academy of Sciences: https://amdc.impcas.ac.cn/masstables/Ame2020/mass_1.mas20 (sha256:d10a9474253b2d29fac71ee21352039659f2e5f4b7d75395416d361878a18ddc)

Registered target identities:

- AME2020-COMPLETE-BINDING-CENSUS from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-GLOBAL-MAXIMUM from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-LIGHT-TO-IRON-NICKEL-RISE from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-HEAVY-FALL from AMDC-AME2020-MASS-1-BINDING-2021

- AME2020-IRON-ONLY-ADVERSE from AMDC-AME2020-MASS-1-BINDING-2021

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, semi-empirical mass formula or fitted coefficient as a premise
- no nuclide name, measured mass, binding energy, uncertainty, table row or source access in execution
- no numerical-zero state, negative, irrational, imaginary or floating proof value
- no evaluated cube root; exact rational enclosures refine only until order separates
- no selected isotope neighborhood, finite scan standing in for the unbounded tail or hidden parity class
- no target access before derivation and prediction seals
- no free bulk, surface, Coulomb, asymmetry, pairing or residual correction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:1b1b51bff98360abcccd08859caf83fb533ff45de725ed644614448cc4f74d8c`; engine receipt `sha256:08d480f86053d84de80bb82879ba57fc11d0dbcba02557045244e7998cfa65a5` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005-08d480f86053d84d.json`; empirical-validation hash `sha256:d55a83b0ab2af76a602619323114e6a70dd1ade29b858bed31d34222bd18b906`; measurement receipt `sha256:bc92b527d85825a9450fdb69e80dd9577ef5b3600fc9283bbdc0ef6fda36e1f8`.

232. Terminal radioactive transition topology and exact half-life survival

Claim identity: `SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Complete nuclear transition structure forces exactly three primitive nonidentity topologies: boundary release/decomposition, held charge-label conversion at retained constituent support, and internal level de-excitation at retained support and charge. Alpha, beta and gamma are the canonical representatives, not an exclusion of proton, neutron, cluster, capture, conversion, fission or delayed composite channels. Complete deterministic predecessor paths split into the two Fold fibres at each half-life carrier, forcing the exact positive survival part 1/2^k for every positive finite k without an ontic random choice, continuum exponential, fitted decay constant or numerical-zero endpoint.

> *Exactly three primitive radioactive transition topologies exist. Alpha realizes mass-four/charge-two boundary release, beta realizes retained-support held-label conversion with lepton records, and gamma realizes internal level lowering with a radiative record. Every other named decay is one primitive or an ordered composition. Survival after positive half-life count k is exactly 1/2^k and remains positive at every finite k.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-NUCLEAR-RADIOACTIVITY-001`
- `SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001`
- `SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005`
- `SFT-PHYS-QUANTUM-TUNNELLING-001`
- `SFT-PHYS-WEAK-PARITY-FIBRE-002`
- `SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-INFO-ENTROPY-UNCERTAINTY-001`

Generated grammar and closure boundary. Generate the complete ten-axis predecessor, topology, alpha, beta, gamma, composition, survival, determinism, target-custody and extension product. The declared exact boundary is: Every finite nuclear nonidentity transition distinguished by constituent-support release, held charge-label conversion and recurrence lowering; every ordered composition of those primitives; every positive finite half-life count and positive transported time carrier. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `compose-immutable-predecessors__three-structural-transition-topologies__binary-square-cluster-boundary-release__held-label-conversion-with-lepton-records__internal-level-lowering-with-radiative-record__compose-the-three-primitive-traces__positive-rational-binary-geometric__deterministic-hidden-path-partition__target-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`compose-immutable-predecessors`	`rewrite-predecessor-laws`	A successor cannot alter admitted receipts.
`topology`	`three-structural-transition-topologies`	`three-named-particles-only`	Nuclear data contain additional emissions, capture, fission and composite delayed channels.
`alpha`	`binary-square-cluster-boundary-release`	`borrowed-alpha-name`	A name alone has no constituent trace.
`beta`	`held-label-conversion-with-lepton-records`	`stochastic-particle-appearance`	Unrecorded random appearance violates complete conservation.
`gamma`	`internal-level-lowering-with-radiative-record`	`continuum-energy-leak`	An unrecorded energy leak violates recurrence and conservation.
`composition`	`compose-the-three-primitive-traces`	`declare-each-a-new-primitive`	Adding a primitive for every code destroys exhaustive topology.
`survival`	`positive-rational-binary-geometric`	`import-continuum-exponential`	A continuum exponential and transcendental value are outside the proof domain.
`determinism`	`deterministic-hidden-path-partition`	`unrecorded-random-choice`	An unrecorded choice violates the superdeterministic complete-path census.
`target`	`target-inaccessible-until-seal`	`external-target-readable`	Target access could select a topology or rate.
`extension`	`no-extra-rule`	`free-rate-or-extra-primitive`	A selected rate or topology is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:91e20a1a06c5090effcee4cb0efe8e2d85b69ac897a88f47bb5ea588e4f68c61`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:d7c6bc41c703c74f311ce61e4dd3b0eef07613062a28000af97833ef858fa3f2`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:02519f10ba73e8b4bf83b02f50856c0df91988192d7518e04978894e91d3a173`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9f2d5c82fceb4b3b77d93a6350ab1d6cdd8c1f8720a00007a56cff68408b31d2`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:ff691f49cf2dfb372f2b19144882b53ab03086c3097daceb6d9a20d1b146d627`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:521ea0c5560600d89eb379569a950a4c97aca02867b01edbd08ba98480317419`. Independent certificate: `sha256:363f509d49d688652a8a6eed058c1b81f11486302efe43b0e64bc87fee6fbcfd`. Engine external-validation hash: `sha256:31c8caa9ffe5620c3940cdf8579a32a8cc4dca5544e54a2daf0bffe567f7de08`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- AMDC-NUBASE2020-DECAY-HALFLIFE-2021

Observed comparison records:

- NUBASE2020-COMPLETE-DECAY-CENSUS: predicted sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; observed sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; exact match True
- NUBASE2020-COMPLETE-HALFLIFE-CENSUS: predicted sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; observed sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; exact match True
- NUBASE2020-RELEASE-REPRESENTATIVES: predicted sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; observed sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; exact match True
- NUBASE2020-CONVERSION-REPRESENTATIVES: predicted sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; observed sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; exact match True
- NUBASE2020-DEEXCITATION-REPRESENTATIVE: predicted sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; observed sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; exact match True
- NUBASE2020-ONLY-THREE-ADVERSE: predicted sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; observed

sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; exact match True

- deliberately tampered unfavourable control rejected

Falsification condition: The claim fails if any NUBASE2020 decay code cannot be expressed by the three primitive topologies or their composition, a positive half-life cannot transport exact finite survival, a row or uncertainty is omitted, the literal only-three-code claim is retained, or target access precedes sealing.

Measurement receipt: `sha256:aaa3a2c9ebffb428bcf48ae68fc5469b7ef29194c72418fbfddc12bfed40108f`. Isolation certificate: `sha256:58caa0ad72c5f9ce87bfcc5d75e1e8ad596aff567b73db2d6c6208374898c49a`. Custody certificate: `sha256:9ab177b5a045ca8d912ba52dfef4ae098dfac5baee6587588c53ad50501cb859`.

Registered source descriptions:

- Atomic Mass Data Center, Institute of Modern Physics, Chinese Academy of Sciences: https://amdc.impcas.ac.cn/masstables/Ame2020/nubase_4.mas20 (sha256:9a2275f1c4f82f0bdf3d41b25a1c8af535fefb7451ea28bad4f6cb2beec32391)

Registered target identities:

- NUBASE2020-COMPLETE-DECAY-CENSUS from AMDC-NUBASE2020-DECAY-HALFLIFE-2021
- NUBASE2020-COMPLETE-HALFLIFE-CENSUS from AMDC-NUBASE2020-DECAY-HALFLIFE-2021
- NUBASE2020-RELEASE-REPRESENTATIVES from AMDC-NUBASE2020-DECAY-HALFLIFE-2021
- NUBASE2020-CONVERSION-REPRESENTATIVES from AMDC-NUBASE2020-DECAY-HALFLIFE-2021
- NUBASE2020-DEEXCITATION-REPRESENTATIVE from AMDC-NUBASE2020-DECAY-HALFLIFE-2021
- NUBASE2020-ONLY-THREE-ADVERSE from AMDC-NUBASE2020-DECAY-HALFLIFE-2021

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, decay table, isotope name, measured lifetime or intended mode code as a premise
- no numerical-zero state, negative, irrational, imaginary, transcendental or floating proof value
- no continuum exponential, ontic random choice or fitted decay constant
- no literal claim that alpha, beta and gamma are the only named nuclear decay codes
- no unrecorded particle, energy, lepton, fragment, capture or delayed channel
- no target access before derivation and prediction seals

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:50490ef7f3fd9e133b25fe68d5e97c7003e3da3e069c2e50ef341a68fca80ccc`; engine receipt `sha256:6c9903242fbe62385d16d103f5d7b27a214f4e6d1e2adbeec47f845daf62363f` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005-6c9903242fbe6238.json`; empirical-validation hash `sha256:5c82afb9c9c769013e4ca908559618a46160a317c5672ee235a43c9058f6148c`; measurement receipt `sha256:aaa3a2c9ebffb428bcf48ae68fc5469b7ef29194c72418fbfddc12bfed40108f`.

233. Terminal fusion/fission direction, barrier and release law

Claim identity: `SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The least nonidentity nuclear junction and decomposition are binary. Exact conserved Fold composition maps two mass-two/charge-one words to one mass-four/charge-two word; exact decomposition maps one mass-238/charge-92 word to two mass-119/charge-46 words. The admitted zero-parameter binding ledger proves a strictly positive total-binding gain in both directions toward the unique mass-62/charge-28/neutron-34 maximum. Mass-energy conservation therefore requires a named positive release carrier. The normalised reaction barrier is the forced Half-One. No measured energy, fitted correction or stochastic premise enters the derivation.

> *Binary fusion and fission conserve exact nuclear counts, increase total binding toward the unique unbounded 62/28/34 maximum, require a held positive mass-energy release record, and carry the normalized Half-One barrier; AME2020 confirms both exact directions after reported uncertainties across the complete 2,548-row positive-composite census.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-HALF-ONE-001`
- `SFT-PHYS-NUCLEAR-FUSION-001`
- `SFT-PHYS-NUCLEAR-FISSION-001`
- `SFT-PHYS-QUANTUM-TUNNELLING-001`
- `SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete product of three possible fusion maps, three possible fission maps, both binding directions, every first-two-Fold barrier part, both release-record states, both global-peak certificates, both target-custody states and both extension states. The declared exact boundary is: Every identity, least nonidentity binary junction and least nonidentity binary decomposition over the registered two-fibre nuclear word; both exact binding-order orientations; all quarter-grid parts generated by two Fold refinements; complete or missing release records; finite-only or depth-independent peak closure; pre/post-seal target access; and empty or free extensions. The generator produced `1152` named candidates and the decision support contains `1152` one-for-one decisions. Exactly one candidate survived: `binary-junction__binary-decomposition__toward-higher-binding__half-One__complete-held-release__unique-unbounded-peak__sealed-before-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`fusion`	`binary-junction`	`identity`	Rejected by computed Fold predicates: fusion_map.
`fission`	`binary-decomposition`	`identity`	Rejected by computed Fold predicates: fission_map.
`binding`	`toward-higher-binding`	`toward-lower-binding`	Rejected by computed Fold predicates: binding_direction.
`barrier`	`half-One`	`quarter-One`	Rejected by computed Fold predicates: barrier.
`release`	`complete-held-release`	`unrecorded-release`	Rejected by computed Fold predicates: energy_accounting.
`peak`	`unique-unbounded-peak`	`selected-finite-peak`	Rejected by computed Fold predicates: peak_closure.
`target`	`sealed-before-release`	`readable-before-seal`	Rejected by computed Fold predicates: target_boundary.
`extension`	`empty-extension`	`free-correction`	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:3b5ba8c1b1c7cef52e338ec2656e5043717dbb8f9d23040986d9325c1b25302c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject a binding-loss direction for both exact representative reactions.; observed Exact interval ordering gives a positive binding gain for both representatives.; receipt `sha256:bae4807a0cb3647819ceeb0d3ebc9fe67b58f7053aaf0f5f2fd589b31ed537bf`.
- `tampered_source`: passed; expected Reject any changed claimant source identity.; observed The changed source identity differs from the registered manifest.; receipt `sha256:a9afcf0815909f62fdf159bee3d9b5e65c53dbf73156c3f19f7e0e9f89badbe4`.
- `tampered_artifact`: passed; expected Reject a duplicated candidate as an incomplete or nonunique census.; observed The deliberately duplicated candidate fails exact census uniqueness.; receipt `sha256:147ddd65e940c7b98a128bd62ebeecbf1e8c6a2ff7dad6aedc88f85b2824138a`.
- `boundary`: passed; expected Reject a non-Half-One normalized barrier and any target-readable derivation.; observed Quarter-One and pre-seal target exposure each fail a computed boundary predicate.; receipt `sha256:b12f93f19a2e1bbe0b9c76dacfd4ef176d9888ff2b934af4b61678ca2ec719c1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:76a335d018ddd2437b61f284d82383b85803971dbbe0e500feb639ca6a970ce3`. Independent certificate: `sha256:5ca4f105af45becb2d736b592b744de0f59405642563449fa97a03216771cefa`. Engine external-validation hash: `sha256:7e0c9230c8e5f9d819dea83227fcae05ee1e46f621ade23abe44782c31745316`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- AMDC-AME2020-MASS-1-BINDING-2021

Observed comparison records:

- complete AME2020 positive-composite census retained: 2548 rows
- fusion anchor exact intervals keV/nucleon: D [11122829/10000, 11122833/10000], He-4 [35369577/5000, 35369579/5000]; strict rise True
- fission anchor exact intervals keV/nucleon: U-238 [75701199/10000, 3028053/400], Pd-119 [83688901/10000, 83690287/10000]; strict rise True
- complete-census maximum: A=62, Z=28, N=34; separated from every rival True
- both external singleton 0.0 inscriptions retained only as empty-binding boundaries
- reversed fusion and fission directions, incomplete census and displaced peak controls rejected

Falsification condition: Reject if either sealed exact binding direction is absent from AME2020 after reported uncertainties, if mass/charge coordinates differ, if the sealed global maximum is not the complete-census maximum, if any positive-composite row or singleton boundary inscription is omitted, if a reversed-direction control is accepted, if a source identity changes, or if target content enters prediction before its seal.

Measurement receipt: `sha256:722810443b2187a7ddf03f88a97bd8fd7b1123f536427cd976857f14552d35ee`. Isolation certificate: `sha256:262fe874dacac1094654742a8ac05cd6be9e2ff3d5f9934bdbdef781f523d50d`. Custody certificate: `sha256:b89abecbb1d4af6705d6b925ca4de257608d015c8de9ba2f7246f954fc1baef3`.

Registered source descriptions:

- Atomic Mass Data Center, Institute of Modern Physics, Chinese Academy of Sciences: https://amdc.impcas.ac.cn/masstables/Ame2020/mass_1.mas20 (sha256:d10a9474253b2d29fac71ee21352039659f2e5f4b7d75395416d361878a18ddc)

Registered target identities:

- AME2020-COMPLETE-POSITIVE-COMPOSITE-CENSUS from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-SINGLETON-EMPTY-BINDING-BOUNDARIES from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-DEUTERIUM-HELIUM4-FUSION-ANCHORS from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-URANIUM238-PALLADIUM119-FISSION-ANCHORS from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-GLOBAL-BINDING-MAXIMUM from AMDC-AME2020-MASS-1-BINDING-2021

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no AME row, element name, measured binding energy or dimensional reaction energy in formal execution
- no imported semi-empirical coefficient, fitted correction or selected residual
- no numerical-zero, negative, irrational, imaginary, floating or stochastic proof value
- no finite isotope window standing in for the inherited unbounded peak certificate
- no unrecorded released carrier and no target access before prediction seal
- no claim that normalised Half-One is a universal dimensional activation energy

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:2a000fca551d443077a846fad52175406aa63aebc37ea397ac555ecb3a3c61c2`; engine receipt `sha256:fbbdba722e6fb27d738ad3cd1df1f62e4caa979038b212147120ca814f0e25fa` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005-fbbdba722e6fb27d.json`; empirical-validation hash `sha256:8938824e23d81c3a0af6f06a90f74d4517541d3ef935743d19ee7ce1a0992d97`; measurement receipt `sha256:722810443b2187a7ddf03f88a97bd8fd7b1123f536427cd976857f14552d35ee`.

234. Fusion/fission yield ordering and distinct threshold carriers

Claim identity: `SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For the least light binary fusion representative and the registered symmetric heavy fission representative, exact zero-parameter binding-score enclosures force the fusion gain per conserved nucleon strictly above the fission gain per conserved nucleon, while the heavy reaction's larger nucleon count forces its total release strictly above the light reaction's total. Fusion requires two charged incident boundaries to approach through Coulomb opposition before short-range residual closure; fission has one parent and therefore proceeds through a neutral-capture or internal-surface carrier. Half-One and quarter-order structures remain normalised relations, not one universal dimensional threshold. Dimensional reaction energies and thresholds are opened only after sealing.

> *Exact zero-parameter binding enclosures force the least light fusion representative above the registered heavy fission representative per conserved nucleon, while the heavy representative is greater in total release. Fusion carries a charged two-boundary approach barrier; fission carries neutral capture or an internal surface barrier. Dimensional reaction energies and thresholds remain reaction-specific post-seal measurement records.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005`
- `SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003`
- `SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005`
- `SFT-PHYS-QUANTUM-TUNNELLING-001`
- `SFT-PHYS-THERMO-TEMPERATURE-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of all three exact per-nucleon orders, all three total-release orders, all generated fusion and fission threshold-carrier classes, all threshold scopes, all access-carrier classes, both metric-retention states, both target-custody states and both extension states. The declared exact boundary is: The least light binary junction D+D to He-4 and the registered symmetric heavy decomposition U-238 to two Pd-119 words; every strict or equal ordering of their exact binding-score enclosures; every incident charged, neutral or internal boundary class; normalized versus dimensional threshold scope; thermal or directed access support; separate versus conflated release metrics; sealed or exposed target custody; and empty or free extension. The certificate is not bounded by a Fold search depth. The generator produced `5832` named candidates and the decision support contains `5832` one-for-one decisions. Exactly one candidate survived: `fusion-greater-per-nucleon__fission-greater-total__charged-boundary-approach__neutral-capture-or-internal-surface__normalized-structure-with-reaction-specific-dimensions__thermal-or-directed-energy-support__retain-per-nucleon-and-total-separately__sealed-before-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
per_nucleon	fusion-greater-per-nucleon	equal-per-nucleon	Rejected by computed Fold predicates: per_nucleon_relation.
total	fission-greater-total	fusion-greater-total	Rejected by computed Fold predicates: total_release_relation.
fusion_threshold	charged-boundary-approach	neutral-capture-approach	Rejected by computed Fold predicates: fusion_threshold_carrier.
fission_threshold	neutral-capture-or-internal-surface	charged-boundary-approach	Rejected by computed Fold predicates: fission_threshold_carrier.
scope	normalized-structure-with-reaction-specific-dimensions	one-universal-dimensional-threshold	Rejected by computed Fold predicates: threshold_scope.
access	thermal-or-directed-energy-support	one-universal-temperature-value	Rejected by computed Fold predicates: access_carrier.
metrics	retain-per-nucleon-and-total-separately	conflate-per-nucleon-with-total	Rejected by computed Fold predicates: metric_retention.
target	sealed-before-release	readable-before-seal	Rejected by computed Fold predicates: target_boundary.
extension	empty-extension	free-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:9206289cf47782eef20fdfe8c8f6602338e0746b0a29f668edf950c8b5b90b4e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject a reversed fusion/fission gain-per-nucleon order.; observed Exact lower-versus-upper separation rejects the reversed order.; receipt `sha256:45fa9ae8e1cfa4571b1b2811e710b45e6e8f2cb2b14c522e5a74df1f250af94c`.
- `tampered_source`: passed; expected Reject any changed claimant source identity.; observed The changed identity differs from the registered source manifest.; receipt `sha256:0de6112a4fa89e23a8625b623b57f00f6f65f6e8de7f00f0265c4d1e451b09ae`.
- `tampered_artifact`: passed; expected Reject any duplicated or incomplete candidate census.; observed The deliberate duplicate fails complete census uniqueness.; receipt

`sha256:9d47a2bed75a232a7ccef5e496679b0c00950d7bc18a8c16404a7dc8ae280226`.

- `boundary`: passed; expected Reject one universal dimensional threshold and any pre-seal target access.; observed Distinct incident carriers and sealed custody reject both boundary violations.; receipt `sha256:955d7ff449ed8aa55a801fafc2ce3d2341ff5741519175cece8712c78da7bea7`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8750255e3be661050ac71b20b5bbb248cd929265755728l8455919da3c7eefa1`. Independent certificate: `sha256:0f3a80971c1639dd5665b1c7bb5faefcd77c90fa44fac7832a5da17f47a4cf2c`. Engine external-validation hash: `sha256:d4fe6a92b4eee8e74b4106e2acc2c35333e42cd247fe5e0728a91d23b0db7fc0`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- AMDC-AME2020-MASS-1-BINDING-2021
- IAEA-WORLD-FUSION-OUTLOOK-2023
- IAEA-TECDOC-2024-NUCLEAR-COGENERATION
- IAEA-ENDF-102-DATA-FORMATS-PROCEDURES
- IAEA-NDS-LEXFOR-FISSION-2006

Observed comparison records:

- complete AME2020 positive-composite census retained: 2548 rows
- AME2020 exact gain intervals keV/nucleon: fusion (Fraction(59616321, 10000), Fraction(59616329, 10000)); fission (Fraction(998447, 1250), Fraction(499318, 625)); fusion strictly greater
- AME2020 exact total-gain intervals keV: fusion (Fraction(59616321, 2500), Fraction(59616329, 2500)); fission (Fraction(118815193, 625), Fraction(118837684, 625)); fission strictly greater
- IAEA D-T release and per-incident-nucleon values MeV: 88/5 and 88/25
- IAEA U-235 fission average/released totals MeV: 200 and 207; average per parent nucleon 40/47
- all twelve fusion-energy rows and all seven fission-energy component rows retained
- charged low-or-absent true threshold versus effective Coulomb threshold, neutral neutron evaluation class, spontaneous internal-barrier class and captured-particle class all retained
- rounded illustrative fission masses were not substituted for the exact AME calculation; reversed orders, conflated metrics and incomplete-row controls were rejected

Falsification condition: Reject if the sealed fusion-greater-per-nucleon relation or fission-greater-total relation fails the exact AME2020 uncertainty intervals or the registered IAEA D-T/U-235 energy comparison; if the IAEA charged effective-barrier, neutron-trigger, spontaneous/internal-barrier or captured-particle class is absent; if per-nucleon and total metrics are conflated; if any AME, fusion-table, fission-ledger, threshold or limitation row is omitted; if a source identity changes; or if target content enters prediction before its seal.

Measurement receipt: `sha256:550034e2999fda2c58cd3c1d8dadca9e5833baee6d8445271718653240a5de8d`. Isolation certificate: `sha256:a1d48ce8acda5a3973e9e1fad3621e5911a1bf6a6115a4009da3f418cf87a577`. Custody certificate: `sha256:0a66cc5340cd48c51515b89932c504dbfdaed8b3d1b244d715e644b64a82fa9f`.

Registered source descriptions:

- Atomic Mass Data Center, Institute of Modern Physics, Chinese Academy of Sciences: https://amdc.impcas.ac.cn/masstables/Ame2020/mass_1.mas20 ()
- International Atomic Energy Agency: https://www-pub.iaea.org/MTCD/Publications/PDF/FusionOutlook2023_web.pdf ()
- International Atomic Energy Agency: https://www-pub.iaea.org/MTCD/Publications/PDF/TE-2024web.pdf ()

- International Atomic Energy Agency: https://nds.iaea.org/public/endf/endf-manual.pdf ()
- International Atomic Energy Agency Nuclear Data Section: https://nds.iaea.org/nrdc/nrdc_doc/iaea-nds-0208-200601.pdf ()

Registered target identities:

- AME2020-COMPLETE-POSITIVE-COMPOSITE-CENSUS from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-FUSION-FISSION-YIELD-ANCHORS from AMDC-AME2020-MASS-1-BINDING-2021
- IAEA-COMPLETE-FUSION-REACTION-ENERGY-TABLE from IAEA-WORLD-FUSION-OUTLOOK-2023
- IAEA-FISSION-ENERGY-LEDGER from IAEA-TECDOC-2024-NUCLEAR-COGENERATION
- IAEA-FUSION-FISSION-THRESHOLD-CLASSES from IAEA-ENDF-102-DATA-FORMATS-PROCEDURES+IAEA-NDS-LEXFOR-FISSION-2006

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, certificate, answer table or survivor identity in the derivation runtime
- no AME or IAEA target, nuclide name, dimensional reaction energy, temperature, cross section or threshold in formal candidate decisions
- no universal dimensional threshold inferred from normalised Half-One or quarter-order structure
- no numerical-zero proof value, negative, irrational, imaginary or floating proof magnitude
- no imported nuclear reaction equation, stochastic premise, fitted coefficient or selected correction
- no omission or conflation of per-nucleon and total release metrics
- no target access before the prediction seal and no free extension after closure

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:7a0b18b87f3a1df38a9679fef0ed10a6e72fa9d06851a8a38f38626db55c1db8`; engine receipt `sha256:1b13169930d9f4c22d75008ef7debe615a8c1dfdedee7a5f495b85c467444e50` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006-1b13169930d9f4c2.json`; empirical-validation hash `sha256:83a6f01dfa45d115b8e784d5904e99a605913f66e348ffdbae5b7daf54eaa085`; measurement receipt `sha256:550034e2999fda2c58cd3c1d8dadca9e5833baee6d8445271718653240a5de8d`.

235. Terminal deuteron binding, spin and dinucleon exclusion

Claim identity: `SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete two-label spin word has three exchange-preserving readings with support three-quarters and one exchange-alternating reading with support one-quarter. The admitted two-boundary residual act is one-quarter: preserving support therefore retains exactly half-One as a bound recurrence, while alternating support is exhausted and closes to the empty form. In the least spatial ground recurrence, the alternating proton-neutron charge hand permits the preserving spin channel; identical proton-proton and neutron-neutron charge hands require the alternating spin channel to preserve total fermionic exchange. Hence the proton-neutron pair alone is bound, with complete spin One and three readings; both identical two-nucleon pairs are unbound. Proton Coulomb opposition is an additional pp boundary and cannot explain or alter the nn result. Dimensional binding energy and scattering records open only after the structural seal.

> *Complete two-label exchange support partitions as preserving three-quarters and alternating one-quarter; the admitted quarter-One residual act leaves half-One only in the preserving channel. The complete spatial, spin and charge exchange ledger therefore binds proton-neutron with total spin One and three readings, while proton-proton and neutron-neutron close to empty binding support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-HALF-ONE-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-NUCLEON-BINDING-TERMINAL-005
- SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005
- SFT-PHYS-NUCLEAR-BINDING-001
- SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete product of all exchange partitions of the four two-label spin states, every ground spatial/spin/charge exchange allocation, every quarter-residual binding discriminator, every three-pair binding table, every composite-spin reading class, both charged-pair roles, both target-custody states and both extension states. The declared exact boundary is: Every ordered word over the two forced held spin labels; preserving and alternating exchange hands; the least exchange-preserving spatial recurrence; preserving identical or alternating proton/neutron charge support; the admitted quarter-One residual boundary act; all proton-neutron, proton-proton and neutron-neutron ground channels; sealed or exposed target custody; and empty or free extension. The certificate is independent of a finite Fold search depth. The generator produced `2592` named candidates and the decision support contains `2592` one-for-one decisions. Exactly one candidate survived: `three-preserving-one-alternating__pn-preserving-spin-ppnn-alternating-spin__preserving-leaves-half-alternating-empty__pn-only-bound__complete-One-with-three-readings__pp-positive-opposition-secondary-to-singlet-exclusion__sealed-before-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`exchange`	`three-preserving-one-alternating`	`two-preserving-two-alternating`	Rejected by computed Fold predicates: exchange_partition.
`ground`	`pn-preserving-spin-ppnn-alternating-spin`	`pn-alternating-spin-ppnn-preserving-spin`	Rejected by computed Fold predicates: ground_exchange_ledger.
`residual`	`preserving-leaves-half-alternating-empty`	`quarter-act-binds-both-spin-hands`	Rejected by computed Fold predicates: residual_discriminator.
`binding`	`pn-only-bound`	`all-pairs-bound`	Rejected by computed Fold predicates: binding_table.
`spin`	`complete-One-with-three-readings`	`empty-spin-with-one-reading`	Rejected by computed Fold predicates: composite_spin.
`charged_pair`	`pp-positive-opposition-secondary-to-singlet-exclusion`	`charged-path-selects-both-identical-outcomes`	Rejected by computed Fold predicates: charged_pair_role.
`target`	`sealed-before-release`	`readable-before-seal`	Rejected by computed Fold predicates: target_boundary.
`extension`	`empty-extension`	`free-correction`	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:a618c2d97aa7b0e98f2b95f55e582ae9432238180faa3b17c53e0450b5c6ab94`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject binding of the identical pairs with exclusion of proton-neutron.; observed The computed exchange and quarter-support ledger rejects the reversed outcome.; receipt `sha256:2ba04fbaa83807074b2a3d7350a0dd5e74a5b5aac0caf51f1bdd5ff2368cbd95`.
- `tampered_source`: passed; expected Reject any changed claimant source identity.; observed The changed identity differs from the registered source manifest.; receipt `sha256:0de6112a4fa89e23a8625b623b57f00f6f65f6e8de7f00f0265c4d1e451b09ae`.
- `tampered_artifact`: passed; expected Reject a duplicated or incomplete candidate census.; observed The deliberate duplicate fails complete candidate identity.; receipt `sha256:85f7f2fccaaba82e12423de0043e0cde43f60a6c59bbf25b4c3079af293bc64a`.
- `boundary`: passed; expected Reject pre-seal target access and every ungenerated correction.; observed Sealed custody and the empty extension are both required by computed predicates.; receipt `sha256:0dfa0edf0904171cdd0da22079b20109bee460b9c155ae4353b37466b4f164ef`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c570667345abb2e4acb914a92e88a08afdb8050dbf9696f508cb0fae4e0f1f77`. Independent certificate: `sha256:09b05f9132863ba86a8807390fef592e706e53b45b72e9406eab6a570eb1e714`. Engine external-validation hash: `sha256:cceefee5c41149be61fff0a214f71e987fb4cc83dea73c842847275d6ec2f9bf`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- AMDC-AME2020-MASS-1-2021
- AMDC-NUBASE2020-2021
- NIST-CODATA-2022
- IAEA-NDS-JENDL1-INDC-JAP-45
- IAEA-THEORY-NUCLEAR-STRUCTURE-STI-PUB-249

Observed comparison records:

- complete AME2020 numeric coordinate census retained: 3,558 rows; exactly one A=2 coordinate (2H)
- complete NUBASE2020 state census retained: 5,843 rows; exactly one A=2 state, stable 2H with directly measured Jpi=1+
- NIST CODATA proton+neutron taking deuteron binding interval MeV: (Fraction(111228251, 50000000), Fraction(55614193, 25000000)); central 222456637/100000000
- AME2020 deuteron total binding interval MeV: (Fraction(11122829, 5000000), Fraction(11122833, 5000000)); central 11122831/5000 keV
- CODATA mass-difference and AME2020 binding intervals overlap exactly after all reported uncertainties
- IAEA JENDL complete two-set spin-channel table retained; singlet scattering lengths are externally signed opposite to both positive triplet entries, with every effective-range uncertainty
- IAEA records the bound deuteron, parallel neutron/proton spins, triplet association and spin-dependent cross-section separation
- reversed pp/nn binding, Coulomb-only nn explanation, incomplete-census and changed-source controls rejected

Falsification condition: Reject if the complete AME2020 or NUBASE2020 census contains a different A=2 bound-state inventory; if the directly measured deuteron spin/parity is not 1+; if the exact CODATA proton-plus-neutron taking deuteron interval does not overlap the complete AME2020 deuteron binding interval; if IAEA does not retain distinct singlet/triplet support, a bound parallel-spin deuteron or the reported spin-channel separation; if any source, uncertainty, row or unfavourable control is omitted; or if target content enters before seal.

Measurement receipt:
`sha256:517fdc91b68661db3cf84eb534f213a8acfdb139ab490279a5aed8744ae9241d`. Isolation certificate: `sha256:1bc4132cc9b3f3454a85ac694fded29649157bc1d5892117f5d2a99fd3cdf36b`. Custody certificate: `sha256:dbc76018471956c03fcd443307afeeca2fbbf4d56cf5731eba650eec50b23bc3`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, certificate, answer table or survivor identity in the derivation runtime
- no nuclide identity, measured spin, mass, binding energy, scattering length, cross section or source locator in formal candidate decisions
- no claimant-supplied expected survivor and no submission-controlled admission flag
- no numerical-zero proof value, negative, irrational, imaginary or floating proof magnitude
- no imported nuclear potential, stochastic premise, fitted coefficient, selected correction or measurement-selected candidate
- no Coulomb-only explanation for neutron-neutron exclusion
- no target access before the prediction seal and no free extension after closure

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:cf08b6132dc2c644a2a6316b6fc508ae22948dbb880c4ffd460115e699b0b605`; engine receipt `sha256:c2a2ca11504bbe66760011c57dff2e8459f171c836b754d24b2ae3c200fa4cbc` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006-c2a2ca11504bbe66.json`; empirical-validation hash
`sha256:05f87ae0d5eee06b72dbf7925eb6aa00b14b45a59a12a9bd2df572552b92be32`; measurement receipt `sha256:517fdc91b68661db3cf84eb534f213a8acfdb139ab490279a5aed8744ae9241d`.

236. Terminal dimensional light-hadron Regge carrier

Claim identity: `SFT-PHYS-HADRON-REGGE-DIMENSIONAL-TERMINAL-059`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Every positive spin rank has the exact squared resonance-support carrier Q(J)=(6J-3)/5. The base is 3/5; every successor step is 6/5; the first five carriers are 3/5, 9/5, 3, 21/5 and 27/5.

> *Every positive spin rank has the exact squared resonance-support carrier Q(J)=(6J-3)/5. The base is 3/5; every successor step is 6/5; the first five carriers are 3/5, 9/5, 3, 21/5 and 27/5.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-HADRON-REGGE-TERMINAL-005`
- `SFT-PHYS-RELATIVITY-TWO-HAND-DIRAC-SQUARE-003`
- `SFT-PHYS-MATTER-CONFINEMENT-LIFT-003`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`

Generated grammar and closure boundary. Generate the complete product of predecessor custody, motion share, tube hands, base rank, successor act, closed form, unit boundary, target custody and extension forms. The declared exact boundary is: The admitted exact 3/5 motion share, exactly two Fold tube hands, every positive spin successor, exact positive fractions, a post-seal common unit label and no measurement in survivor selection. The generator produced `512` named candidates and the decision support contains `512` one-for-one decisions. Exactly one candidate survived: `immutable-receipt-custody__admitted-three-fifths-motion-share__both-forced-Fold-hands__two-times-three-fifths__first-positive-rank-three-fifths__depth-independent-induction__postseal-common-positive-unit__target-inaccessible-until-formal-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`predecessor`	`immutable-receipt-custody`	`rewrite-or-ignore-admitted-laws`	A successor may not replace an admitted dependency.
`motion`	`admitted-three-fifths-motion-share`	`selected-or-fitted-intercept`	A selected intercept reads the spectrum.
`hands`	`both-forced-Fold-hands`	`chosen-trajectory-multiplicity`	A chosen multiplicity changes the slope.
`successor`	`two-times-three-fifths`	`fitted-common-slope`	A measured slope is not a derivation.
`base`	`first-positive-rank-three-fifths`	`numerical-zero-or-free-offset`	Neither numerical nothingness nor an offset is generated.
`closure`	`depth-independent-induction`	`finite-five-row-pattern`	Five inspected rows do not establish a general law.
`units`	`postseal-common-positive-unit`	`unit-name-selects-coefficients`	A unit label cannot choose a ratio.
`target`	`target-inaccessible-until-formal-seal`	`target-readable-before-seal`	That would fit the known spectrum.
`extension`	`no-extra-rule`	`free-spin-correction`	An ungenerated correction is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9f7d24de80fb141e1123b67d6c9295b208fb454e4fb95d34e54fc5d21689043b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:f8d54892b5d1ed3de6ba27737dd5b9c621492a76f12a7ac16c91ae9adbe70762`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:da01e8d6d5522b0b8e723c54fdd47faa177bba7822980c9fd58585ff2ed16c23`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ab20aee40cd9249b6bf57a500d8ba2e70e83717f2eb6edbfe97b6b0982951749`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:bc80d96fa5308bfe83ed5785f06f0a5a4cb2407237f8386e2aadad901aa56e2d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d8eb0ca9a875d53c79c84e15781c7b8432d9e5e0e36c184f5953831babba8d0b`. Independent certificate: `sha256:33518183cdd429114c0ead42400024b705b56090b4981d0ffcf41b0c0e059bc7`. Engine external-validation hash: `sha256:fdbe35e195df34916562fc5484c6c7c5f813b09c2de7688d49580c3e08bdd9d4`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S020` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no fitted or measured Regge slope, intercept, mass, width, residual or selected trajectory row
- no imported string-tension parameter or continuum string equation
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof scalar
- no pre-seal physical-unit calibration or target access

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:6ffe106a351c8ba56384198d304d4fa489454b66823ee3ec41e608c59fd69daf`; engine receipt `sha256:f82510d61a2e1472c8d090f2fd1f63b53491bab9f875ce4ae27b0e0b6b5fc54e` at `receipts/engine/model_admitted/SFT-PHYS-HADRON-REGGE-DIMENSIONAL-TERMINAL-059-f82510d61a2e1472.json`; empirical-validation hash `None`; measurement receipt `None`.

21. Spacetime Gravitation

Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise.

237. Relational spacetime event

Claim identity: `SFT-PHYS-SPACETIME-EVENT-RELATION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A spacetime event is a physical Fold transition located only by exact relations to generated recurrence clocks, spatial support and causal predecessor events.

> *A physical spacetime event is a source-bound transition with exact clock, spatial and causal relations.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__event-as-clock-space-causal-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	event-as-clock-space-causal-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:cb4c53dede16c698a2a594bf0d66f53e42db1161b858df897910b3ee2a209316`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b7773e6b2ff8d06c5214ded33789524114c64d4484a5670e1a4f605059e96789`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8477eec669255c0e170818818a86a100ecda40bae844ba4019c1752c9e65db54`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b18bfc28a43fc2854e21066cbc6d1247cb14cecc546dc46ce09de064318cf37f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:fecb336074e0c41f4d6ec1eec8923db6075bf2c4a69c0a19ed362d29e889d150`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:7fb975e8ff2ba01379b95f1223d62e94f244d303e7486926461154f0e3e40e29`. Independent certificate: `sha256:ceb24ff9b7ff53a8e2f1279b4659c17bfce039260279ca8494f93b65139f003b`. Engine external-validation hash: `sha256:69bb77e2786480544de17c082db5ac3c339d677690806bb1a764ae5185a8bf5c`.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GWOSC-V2-CATALOGUES-2026-07-23

Observed comparison records:

- spacetime-event-relation-1: predicted relational-spacetime-event; observed relational-spacetime-event; exact match True
- spacetime-event-relation-2: predicted relational-spacetime-event; observed relational-spacetime-event; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause PHYS-S021 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:5370bd3f8497444bcdb5a2e46aa19044005c077bea80e4a0155d183f97ac6d63`. Isolation certificate:
`sha256:b99a380ca10c10ea2f24b934b071842e1a16a12e627fc736f3b8dae84d3b1080`. Custody certificate:
`sha256:8844e921b77af969a585aaa6a526428ca7e5b6593b29b35f4d946485e27b52a1`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- spacetime-event-relation-1 from GWOSC-V2-CATALOGUES-2026-07-23
- spacetime-event-relation-2 from GWOSC-V2-CATALOGUES-2026-07-23

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:4d3830a18644af7dc4f6c0647a81a52c666ad2d7806d151eb85dec652f4b807a`; engine receipt
`sha256:c4ac9143c7a478a7432bd5dd1ffca1ff505b170bef1e91f51457b02bdf887002` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-EVENT-RELATION-001-c4ac9143c7a478a7.json`;
empirical-validation hash
`sha256:f9818523102782e5712ec08d9de7e916288ed85d51fdfc6d4f66ddc18e6847c8`; measurement
receipt `sha256:5370bd3f8497444bcdb5a2e46aa19044005c077bea80e4a0155d183f97ac6d63`.

238. Invariant event interval

Claim identity: `SFT-PHYS-SPACETIME-INTERVAL-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The interval between events is the canonical equivalence class of their exact clock recurrence and oriented propagation-support separation under reference relabeling.

> *Spacetime interval is the reference-invariant Fold class of exact temporal and spatial event separation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-SPACETIME-EVENT-RELATION-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__reference-invariant-event-separation-class__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	reference-invariant-event-separation-class	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:969d464ff5704d7df784733a4abb5aa8ddc23a72141274a01931f80b811dbba7. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:ce6b28bb82bc59bdd033240abe690be1d0975058ebd0eeea5a81cfef20c143a7.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:743da5168c1ece2e75de4e5c4d4e98f8aa2a750be492e54da39ffffb955995af.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:c8a03de0d32ff02986b3511d82d721222393e34eaa6192d6b70ee1c7bd2f5307.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:c5068957099301afd9005fc768ab8997fcdb26b8b587fd1ece663c56d13b325d.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:1c2f6139e6822a4f6143d4e727b590f756a442630bfb5b8fa0c6694f8072d72a. Independent certificate: sha256:50c3ed5aa9bee665adf665dabbef51f40cebd1394415c1e8d4a1843c91d6308d. Engine external-validation hash: sha256:07ef591e6701f315a2ddcd45391ee501b0806fd2a4601fdbe08552bfdee9c9b7.

Shared claim-record clause PHYS-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- spacetime-interval-1: predicted invariant-spacetime-interval; observed invariant-spacetime-interval; exact match True

- spacetime-interval-2: predicted invariant-spacetime-interval; observed invariant-spacetime-interval; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S021` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:607e308a418eadfb03677c6525930befa24d63ee87297670bb8a29d595f69bc0`. Isolation certificate: `sha256:ed9361321aa818ffe0e5a22a89574355527e9a765c07d2954af7d50bdd3a6e3c`. Custody certificate: `sha256:42cd9cc37145ad571de5ddf034ca28bc8f5a31a9a4b4b4d50a7abc493a93ef86`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- spacetime-interval-1 from NIST-CODATA-2022-ALL-CONSTANTS
- spacetime-interval-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d20d124068af15cb5713e9d712c085c732bd34b29533df6659d45501cd69fdca`; engine receipt `sha256:ff799557cb3f67eb01e19774155c51c2be6cd39efd8ef827d6764790e6ffd35b` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-INTERVAL-001-ff799557cb3f67eb.json`; empirical-validation hash `sha256:44a0d8cef90df1220adb2c587e30151d80d4db6eb9d103c010c898438d896975`; measurement receipt `sha256:607e308a418eadfb03677c6525930befa24d63ee87297670bb8a29d595f69bc0`.

239. Causal order and accessibility cone

Claim identity: `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Event A can precede event B exactly when a generated adjacent propagation path from A reaches B within the limiting recurrence support; the complete reachable set is the causal cone.

> *Causal order is exact generated path reachability bounded by limiting propagation support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`

- SFT-PHYS-SPACETIME-INTERVAL-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__limiting-path-reachability-order__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	limiting-path-reachability-order	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:c5a61427ff534fba36b65cd490eefe64ae034bfb4ae38c6dfab02f6014f9e6e4. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:e94442b5bb53ef2b5de6881020d27b9d07013f476e63ff55922e7c574c7b0285.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:8be3895f058cdc5dbb68c13f8f5b74cf024aac7b5f19f6ca8c896d79208e9b5f.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:0ecf97d49c8e894ca1a702750e0dcf902e1287d8aa6a7907f2456279e0cc0fa1.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:d59f9e9634e3091169f87061ee34a4ebe45752d4dc067ddda43e8e0b0b483615.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:d997d62bd528d094409a6febb7838577ed38e68e6594d69616cf0c8eebdf2768. Independent certificate: sha256:118cf1eb0c90c71598c96f9d3e8a2f8bfd9885ea5133bd149f50dad48a4ead8c. Engine external-validation hash: sha256:b5fd573a0c464aa8af221827df16915f468f0bb645be88d8e6a873891b48bba6.

Shared claim-record clause PHYS-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GWOSC-V2-CATALOGUES-2026-07-23

Observed comparison records:

- spacetime-causal-order-1: predicted spacetime-causal-order; observed spacetime-causal-order; exact match True
- spacetime-causal-order-2: predicted spacetime-causal-order; observed spacetime-causal-order; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S021` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:7d83cc797a4ec25bf64882b6bf29f44d072147cb8e2d19fe3526ed2cc5e00f1e`. Isolation certificate: `sha256:cd634809928a113606fd0562ab70d3c31b40b2098bb4255564a2cbbdc42b44cd`. Custody certificate: `sha256:8d92c4b8aa7326eab0bc27096f5cb6e8aca48086625e61231b01d5592c58105f`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- spacetime-causal-order-1 from GWOSC-V2-CATALOGUES-2026-07-23
- spacetime-causal-order-2 from GWOSC-V2-CATALOGUES-2026-07-23

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:fae8fd18cd615688bad2ec48fa6f16e3298e42fd0738aff59c707b219d70db5b`; engine receipt `sha256:b4989c3a9812b65a2902f0c81fe585c555c7f3de0e8060c9b63fc5402fa172f5` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-CAUSAL-ORDER-001-b4989c3a9812b65a.json`; empirical-validation hash `sha256:c2bd535fb4b33c9b8100218207be3160aa357f1084b0cac7ab40d4c1f66a6453`; measurement receipt `sha256:7d83cc797a4ec25bf64882b6bf29f44d072147cb8e2d19fe3526ed2cc5e00f1e`.

240. Inertial frame correspondence

Claim identity: `SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. An inertial-frame transformation is the unique bijective relabeling between uniform recurrence/translation records that preserves event coincidence, interval class and limiting causal paths.

> *Inertial frames correspond by exact interval- and causality-preserving Fold bijections.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`

- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__interval-preserving-frame-bijection__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	interval-preserving-frame-bijection	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:832454f065ef6a75d6bf444f7c0c47e6ef4d9add5cadd01c49a6f9154a34de44`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f9297fdf7205dde4aa959fb5ff0c226f839fb3bf06aa40b29fac6e3a2a23cddd`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:18d929aa636d7afda9393834957574c0fbc60fc64c1c0ea4c0a122f011c08b56`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c44d24816d8e519c021582ddc7b5e6dc80a1d0118982f7005e1f0d6c0e5c317d`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:879f262781f49174bae996767595daa69cf72f1d253b7cfa8f546f530ab1de8d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:6e481b4aa80234a618d5d8a306575f66818fc68bffd9e0a8f9ecb7a409db3e4b`. Independent certificate: `sha256:52baf3a0668d6cd44be3f6bf6380c3ef2ff6a548bb046038bd2bad31ffc7809e`. Engine

external-validation hash:
`sha256:63344f7e4ada50c5b98f9f607d2d9821317565c5385cbefe179b38353b8c48fc`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- spacetime-inertial-transformation-1: predicted inertial-frame-correspondence; observed inertial-frame-correspondence; exact match True
- spacetime-inertial-transformation-2: predicted inertial-frame-correspondence; observed inertial-frame-correspondence; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S021` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:f9e804c3c4487001e7c3113e2936f1dd99ab81f7a270b43362bb1b48567e447e`. Isolation certificate: `sha256:4e635150b4db929b6b6d2a98f5ea483b2c9de5cc4c44e301b2e64d11c370dfe1`. Custody certificate: `sha256:e377f6c14024f55b07bbd121f89cef9d38cc05bab003b05cf2641cd10614fb1e`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- spacetime-inertial-transformation-1 from NIST-CODATA-2022-ALL-CONSTANTS
- spacetime-inertial-transformation-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:4363f723ab86a9dc002553820ce3a960a0ef952511c25bd47e73a2dad7b20a3f`; engine receipt `sha256:00540e0aabda81441fef87902d4ee27007641146a290d4f50729b05eec1a4c64` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001-00540e0aabda8144.json`; empirical-validation hash
`sha256:ddbffcd3b15628bfb0fd22807624ce57f0141230c1677b6d36736c521120eb28`; measurement receipt `sha256:f9e804c3c4487001e7c3113e2936f1dd99ab81f7a270b43362bb1b48567e447e`.

241. Invariant limiting propagation speed

Claim identity: `SFT-PHYS-SPACETIME-LIMIT-SPEED-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The limiting speed is the exact spatial support advanced by one elementary causal recurrence; every inertial correspondence must preserve this boundary ratio.

Limiting propagation speed is the invariant exact ratio of elementary causal support to recurrence duration.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__elementary-causal-support-per-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	elementary-causal-support-per-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:da5a613550dae702d0357821532b6fd246f3855a1c34af1ac8318bf02aaf9749. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:3f24907ad869c851e94358c6d2e1589485d8dfc57a3e47429d9e67fec3b1b807.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:f0065fbc2140dc09cb06db2d8f0fed9292f7e037bc3543987d690031ab0fc8f9.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:d403016dd04c1b0dbe76390cc38dcb92301e6b787b9c7d905ee0c6e4a5172e64.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:70e1003982b55a829dedd23bbd2d3c5a8b811ed7f9bc72eacc9d8f3309a6fc1b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f9f2b248a9aa95dd1cd4f693c9ec4d5bce43afe14d0559a70bed45bcd9308135`. Independent certificate: `sha256:9229fc9c2465f5e749ccf72df5c46a67dfe460021a0128ede2badd6dde62c28a`. Engine external-validation hash: `sha256:79bed6bb49c3c715d0b3bceb1f9526084a260fa161fd11951241b0a06e06272b`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- speed of light in vacuum: exact predicted interval [1616237000000000/5391307, 1616273000000000/5391187]; external interval [299792458/1, 299792458/1]; overlap True
- deliberately displaced unfavourable interval rejected

Falsification condition: The sealed exact Planck-support/recurrence interval does not contain the released CODATA limiting speed or the displaced control is accepted.

Measurement receipt: `sha256:6bff73fc80277a04a78c6462f80577c64feeb66567f7dd444e66813adea818c2`. Isolation certificate: `sha256:c6e3b55e6c377701525c0aa22a61210e7b1c9c3fd8d71a31326a053586f09d8e`. Custody certificate: `sha256:c696612f5b2cfdbd0d21475bc22e91b5e5d70dfc671b08c85719c66905e1e1ed`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- spacetime-limit-speed-1 from NIST-CODATA-2022-ALL-CONSTANTS
- spacetime-limit-speed-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d36cd54cb90b3cbf7b73cdb4bf158be85e088b812f05dd3132069f394469256a`; engine receipt `sha256:9dd1cbeef40d76297b1a96f762ad084770998e4744f4d437e83d4bbb3e620ec7` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-LIMIT-SPEED-001-9dd1cbeef40d7629.json`; empirical-validation hash `sha256:fcb635a39ccaf1a716ef720448b0eb40c18eb15e1d7f0f83a671e90095bff610`; measurement receipt `sha256:6bff73fc80277a04a78c6462f80577c64feeb66567f7dd444e66813adea818c2`.

242. Relative clock-rate relation

Claim identity: `SFT-PHYS-SPACETIME-CLOCK-RATE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A clock rate is the exact recurrence count along one causal path; between inertial paths it transforms by the unique positive ratio preserving the common event interval and limiting boundary.

> *Relative clock rate is exact recurrence-count correspondence between causal paths sharing invariant event separation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-SPACETIME-LIMIT-SPEED-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__path-recurrence-ratio-at-invariant-interval__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`path-recurrence-ratio-at-invariant-interval`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:cfe0b3e569b9a6c09eb5dc3466a844059cc00f84e5f208f0c1adcaec443a1649`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:943cf51afadc3eea33e1917e2654588cd5ffb852a9b66d055f31edb06a920b7d`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:cb4c45ed3892212961529a7859a527ca986a6ffb4e44f83ce60fc5016d7fac6b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:575773c48e443966b9c33c32c94c1ea43757f7839befd628e08b309b52f8278b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:49b8effffa4e055079087bdc448a43dd780c1cb38eae3e71224b8d58cee7a105`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:22904c14e9462cef8a843160336eb4f8b6cc423c4f567aa2155f81febeed99a5`. Independent certificate: `sha256:9eec8c27eeab2fd5cb74ae22d8d68cde4653e993b1ee0f1cb459951c561224db`. Engine external-validation hash: `sha256:1077be71b4394675c18c81bb0314828de84323da51b2f611de79fc3ddc5b3cc8`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GWOSC-V2-CATALOGUES-2026-07-23

Observed comparison records:

- spacetime-clock-rate-1: predicted relative-clock-rate; observed relative-clock-rate; exact match True
- spacetime-clock-rate-2: predicted relative-clock-rate; observed relative-clock-rate; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S021` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:3ffddc542866319fddee0e1a5fe45e817ab3907715056d246fbd6c225519a502`. Isolation certificate: `sha256:36b8318927a83827db7d73366087bf79ac106df6eceb9e612a5a5d290ec01bd6`. Custody certificate: `sha256:0d009cedc1457d0b6dc9f80891aa4f660a4be3b00c9b549932707012239dad0f`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- spacetime-clock-rate-1 from GWOSC-V2-CATALOGUES-2026-07-23
- spacetime-clock-rate-2 from GWOSC-V2-CATALOGUES-2026-07-23

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:2d2220a337fcbc2f9de55ec8b45f43c70d2e454940bb8295f48b7fb627aef385`; engine receipt `sha256:3fb742859b688cf0a9fcaf65f92b15d654f33f46c2bb193d577117ba6d0ad2d8` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-CLOCK-RATE-001-3fb742859b688cf0.json`; empirical-validation hash `sha256:92fdecb55207fa29fab6821c2287d7560ee94e0e3c1d425b73a8445361a47bb7`; measurement receipt `sha256:3ffddc542866319fddee0e1a5fe45e817ab3907715056d246fbd6c225519a502`.

243. Relative spatial-extent relation

Claim identity: `SFT-PHYS-SPACETIME-LENGTH-RELATION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Spatial extent is the simultaneous support count under one clock partition; changing inertial partition forces the exact extent ratio paired with its clock-rate ratio so the event interval is preserved.

> *Relative length is exact support-count correspondence under interval-preserving change of clock partition.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__clock-partitioned-support-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`clock-partitioned-support-ratio`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:263aae32f3e663919fd2d25fad6e89781024fd93bbbb5263dba5db01dde789c2`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:01fbbc74d963f944aec1a95d254af9a06b05424fef643181c1e652078e61cffa`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9eebb4ebf15a0efc40bd52f8a03163195485f5c32e1d62e802380539f86a8619`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5c03ab3952ed6a4bc31c5fe759f384a92af1015b79645066ebfbe551a7f61e21`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ee13c0e3ae4a4b7073813fa16fd78284ce2aa8b1c28a2a48addbe4a1832475a1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a4a71b3973fbd61b62fc84b2bb2e009fc86edeada7c2bbebe2255f0ce7eb9817`. Independent certificate: `sha256:d035014e134e4cb6cecc72653f05c6bc1ca0d36bba21361fb670e65c303eded6`. Engine external-validation hash: `sha256:776a74272dae5e602fa41e0c67d0bd06bae3d9dbd00a0bf7514d3a2981cde958`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- spacetime-length-relation-1: predicted relative-spatial-extent; observed relative-spatial-extent; exact match True
- spacetime-length-relation-2: predicted relative-spatial-extent; observed relative-spatial-extent; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S021` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:6b1c09d5587b6e21bcd71549bb98d6579df70df238035a93aeae2bdebb1e711a`. Isolation certificate: `sha256:07b5aa8c478b12f04051312c16b07241a53a0449c5348a4cbc12952975f08bda`. Custody certificate: `sha256:87d3ad1e766e525b0d06f797480fcbded3e8353e4ab65942b284c916162f2fde`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- spacetime-length-relation-1 from NIST-CODATA-2022-ALL-CONSTANTS
- spacetime-length-relation-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity

- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:c3a4afdfdf240c367600ba77d1ac710555cce7430c55124b0608924071bfa7ea`; engine receipt `sha256:7c3c508f453b0e55987abd30fa084844ea3904397ca514024ffd34b660a5e05e` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-LENGTH-RELATION-001-7c3c508f453b0e55.json`; empirical-validation hash `sha256:45c13ce916bf4fc6e6dbb7eaa656328ef9fbcceeac316c9f9cf01e25e226b01b`; measurement receipt `sha256:6b1c09d5587b6e21bcd71549bb98d6579df70df238035a93aeae2bdebb1e711a`.

244. Inertial-gravitational equivalence

Claim identity: `SFT-PHYS-GRAVITY-EQUIVALENCE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Local free response to a gravitational source is observationally equivalent to uniform change of the relational reference support when every constituent follows the same causal path class.

> *Inertial and gravitational response are locally equivalent complete Fold traces under universal free propagation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-SPACETIME-LENGTH-RELATION-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__universal-free-response-reference-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`universal-free-response-reference-equivalence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:12ddd5afa113ddfe5d5545bab9097492461756b320bf20df495da7652900deb6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:73aa11a193c0b20cc2a035d9f42bc3740a8a3ce235bc3da7f1badbffe27147a5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:92df6588bcf888aaad40ca734ec326adc9765e2fc6bd382c5e746b6cf2368d0d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8862ade98a9553248fe679256e7b4d33b52f0dfaa585520ff00e310bd9e1adc9`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f041cf78472d82e37e466383945ec3bc74481b6b52ad05634a1a27f122ac85e3`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:3c06fca37a2b1be088f1a2c5652ccc01ab5951b29fd61569ec0e2ce49eea9b99`. Independent certificate: `sha256:bd958e9791dc9ae04d5aa573f9d94372281a1ae9b2f163f93e43d158bdab4d82`. Engine external-validation hash: `sha256:cafbb4a19f07dc05ac1dd96994ed6f28eaea582b40c89be357a0eaf90c82b4af`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GWOSC-V2-CATALOGUES-2026-07-23

Observed comparison records:

- gravity-equivalence-1: predicted gravity-inertia-equivalence; observed gravity-inertia-equivalence; exact match True
- gravity-equivalence-2: predicted gravity-inertia-equivalence; observed gravity-inertia-equivalence; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S021` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:97569d15494bb74bd27b4929f2470330e3d1fc53072aaf5226315e921a1ce058`. Isolation certificate: `sha256:a686a35cd8c0b462e97e2b88268750f9cc474d6b3b34315b6410ddf70c306c1f`. Custody certificate: `sha256:87d925f49cb47af391aae82daf64996fa6840b3dbf2776c75eccf7a05bd121f9`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-equivalence-1 from GWOSC-V2-CATALOGUES-2026-07-23

- gravity-equivalence-2 from GWOSC-V2-CATALOGUES-2026-07-23

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:4f54f68bd104a19856c1103d25575a589fedd64d8487f98bf2a9f2715b103a2d`; engine receipt `sha256:c49d9f4c5550a7448d5a882431585823d85c51b2829b2becd99d5cc2af0cab52` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-EQUIVALENCE-001-c49d9f4c5550a744.json`; empirical-validation hash `sha256:8ee730f771d3f95a4d08db9b8da3a864a1f672a394f5d0acedf7d025def92244`; measurement receipt `sha256:97569d15494bb74bd27b4929f2470330e3d1fc53072aaf5226315e921a1ce058`.

245. Source-linked relational curvature

Claim identity: `SFT-PHYS-GRAVITY-CURVATURE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Curvature is the failure of source-linked adjacent event relations to close to the same transported orientation around distinct generated paths.

> *Gravitational curvature is exact source-bound path dependence of transported Fold event relations.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-GRAVITY-EQUIVALENCE-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__path-dependent-orientation-closure__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`path-dependent-orientation-closure`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:590ebc5532f8fd426e4fcfd82dc741c937dae79ad438fbb0a6fc23273752018a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:cc296bc892c14918ca793090c889169d6a420f86cda52974fc4454296a920104`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:296028aa42392a91f94031efd1eda98594633cd11e145432a956394e715d8d2f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:267e1318436939856318206e4807adee62447f2541e0602f82d97a15cfd24771`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:5427c13bc35996d5dc625b40524d58f651b52064201989dca085bb5c20eff4e0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:60dc41684edf0c9909df0e2dfa1dd501118544d2b9d3e08bb2ee5f7aebb88b8f`. Independent certificate: `sha256:4338bb649a7ef62c7b507d93622192e8c473ba74270c4380cc37ff1a332ca050`. Engine external-validation hash: `sha256:28f14bb2537fe01e810d086705c1c5181ec87546c1ac41744ffe6e87af569ec8`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GWOSC-V2-CATALOGUES-2026-07-23

Observed comparison records:

- gravity-curvature-1: predicted source-linked-relational-curvature; observed source-linked-relational-curvature; exact match True
- gravity-curvature-2: predicted source-linked-relational-curvature; observed source-linked-relational-curvature; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S021` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:b8e46a0d25cecd420ee787a65aff11afb68b9fda780c2d7791c14fad33bda9b1`. Isolation certificate:

`sha256:aa09bb191c1b2e7cb82002db7a77c652c141701697ed7e2ab7a851266ddc0745`. Custody certificate: `sha256:d673b079df6d3ce1934330668ad60d00259c02dd762795b02525b174b1b5e049`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-curvature-1 from GWOSC-V2-CATALOGUES-2026-07-23
- gravity-curvature-2 from GWOSC-V2-CATALOGUES-2026-07-23

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:e08b55584d2841656e68a6c52772cef2b4b563eadf226a2a6095111e9b0eb7fa`; engine receipt `sha256:20c01653cd3fea0fca9ac38af60f26ea532e08d7b2797a12004fc79c2f87c200` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-CURVATURE-001-20c01653cd3fea0f.json`; empirical-validation hash `sha256:e7d7c0e0cc9c1295c5430f8a6ecaae51622d464a2248bcf4cead40b8d651596f`; measurement receipt `sha256:b8e46a0d25cecd420ee787a65aff11afb68b9fda780c2d7791c14fad33bda9b1`.

246. Free propagation on curved relation

Claim identity: `SFT-PHYS-GRAVITY-GEODESIC-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A geodesic is the locally unforced adjacent path that preserves the propagating word's inertial recurrence while following the complete source-curved event relation.

> *Free gravitational motion is locally inertial Fold propagation along the minimal complete curved-support path.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__locally-inertial-curved-support-path__source-bound-proof-trace__capability-c` losed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	locally-inertial-curved-support-path	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:e0be350416fc1dc1ba9bef26da882a573d31ee010c52f89a5afb5451a3df9db8. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:c0dd995225e3068b28acc56c8e6f66418abd154c843e41d3148593bbea25d211.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:9982d9cbdfb704442072dc38365d63f2a65efed5391a5ff4f9e2f9a2ef03d976.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:6b9d7defb3d6154f7fffcf998c8750eca69274a2760a27dab5dbe6f299e88d36.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:d779b05ac5f9cc3a8bfd895b3e03292d2761344d1f2d09b3cd6968798e61ae45.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:a34d39b8e9a0e93721e04115fe1c670f9194546f60bd479e2aafe9a19f652181. Independent certificate: sha256:1aaae7eae0755b0584d825c67752d6059a5e66c1abbecc7bbffba773b19be9a6. Engine external-validation hash: sha256:32b1fdb7e7d4898d29494a1dc8151e12262bd8217ce1ca020d1290e3febfc73e.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GWOSC-V2-CATALOGUES-2026-07-23

Observed comparison records:

- gravity-geodesic-1: predicted gravity-geodesic-propagation; observed gravity-geodesic-propagation; exact match True
- gravity-geodesic-2: predicted gravity-geodesic-propagation; observed gravity-geodesic-propagation; exact match True

- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S021` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:e7489ebf5ed47b17d81215fe14474c9a8f156dcdb48462f962e5edfc51e5446c`. Isolation certificate: `sha256:265c1d67f82c3f22dbde609b9685a9d51ef6f584e236d1cee47f1ff48aa1e658`. Custody certificate: `sha256:0f7a7a505add623a2cd655c077d22c310f688748042cf05e8c1cac5929a4fed0`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-geodesic-1 from GWOSC-V2-CATALOGUES-2026-07-23
- gravity-geodesic-2 from GWOSC-V2-CATALOGUES-2026-07-23

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:2782d0f159a7416e8a61322b247d804599708430e598ec3fbb1467e5e4847c31`; engine receipt `sha256:1817eff144b27f906eaa4b4add71f5a101bbd594cfbc57a9f82eda3ffa7be7d9` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-GEODESIC-001-1817eff144b27f90.json`; empirical-validation hash
`sha256:1ed888263f62a975c449a8301d34fd434c131dcb94131ce9615f413cd6d736c4`; measurement receipt `sha256:e7489ebf5ed47b17d81215fe14474c9a8f156dcdb48462f962e5edfc51e5446c`.

247. Gravitational field-source closure

Claim identity: `SFT-PHYS-GRAVITY-FIELD-SOURCE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Every gravitational relational change is paired with an exact inertial-energy source carrier and its propagation trace; closed boundaries pair interior change with outward field transfer.

> *Gravitational field closure pairs inertial-energy source, relational curvature and boundary propagation on complete support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`
- `SFT-PHYS-GRAVITY-GEODESIC-001`

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__inertial-energy-source-curvature-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`inertial-energy-source-curvature-pairing`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:388dae914685978c7baa09d40128292f0863e1bee86ff14eb9be6e66127d88a0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6de7b89644fb09a81e548b3a6503b3ae4903f29a50146da6a9213c0cb3701be8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ec1b9c32b20f0b63851cbaf3c076294a09139217c74e0b40d6211ceda7ad63c9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:31847463ec37ed5d0dcfc0131a741bf1cc77146584900bdb0b9ae432fec15b57`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:3f224f13c7965a287af13df8a33132cbe7f70ce8aca7796cc86109d47a7e3d31`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f0ed74962fa452e895824dceff5799f47a305d51bd3625919fb3d7dccd6ba72e`. Independent certificate: `sha256:8cb3c50c0d9bed56fa6c08931db1d4ad43098c2fa827d7973af9c6907313aa9e`. Engine external-validation hash: `sha256:df8974c0fec7b12e3b4aefda114966c4fb984d78b4071a2038fc8679c9fdc13c`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GWOSC-V2-CATALOGUES-2026-07-23

Observed comparison records:

- gravity-field-source-1: predicted gravity-field-source-closure; observed gravity-field-source-closure; exact match True
- gravity-field-source-2: predicted gravity-field-source-closure; observed gravity-field-source-closure; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S021` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:4830be11fafa38d4e78c03514d86398ddd480524221c468c01d0c803d118829d`. Isolation certificate: `sha256:5563c253006886e12e79d7add4e90c0bdc90d4017eae579f942e5421753a1bfb`. Custody certificate: `sha256:b660649c07acc11456f933befc98f66716af6046dec8b6964cf723e15c6414ac`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-field-source-1 from GWOSC-V2-CATALOGUES-2026-07-23
- gravity-field-source-2 from GWOSC-V2-CATALOGUES-2026-07-23

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:ea7542aca5e0865825218851f66686c7d0b9e49fd80488baad04293af4857676`; engine receipt `sha256:ca8327c8264470eda6d345068c3565835c80243f2ea069429a672977bec22c79` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-FIELD-SOURCE-001-ca8327c8264470ed.json`; empirical-validation hash `sha256:e16604ad24ae73b3d120005eb3bb7dc6b4d6127731a96c5048c29e5c9c732070`; measurement receipt `sha256:4830be11fafa38d4e78c03514d86398ddd480524221c468c01d0c803d118829d`.

248. Gravitational-wave propagation

Claim identity: `SFT-PHYS-GRAVITY-WAVE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A changing source-curvature relation that closes into transverse recurrence propagates at the limiting causal speed and transports energy-momentum with held polarization.

> *Gravitational waves are limiting-speed source-curvature recurrences carrying energy, momentum and polarization.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MECH-EVENT-CHANGE-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`

- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-GRAVITY-FIELD-SOURCE-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__self-propagating-curvature-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`self-propagating-curvature-recurrence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:79ad72e859ba9745a15d0f358b41b168bac29774f0beb1ed0506255f465273dc`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a4668b977a7dc243e8d7cffb5281dd522f7e141c9d57a414c7fa4b9d7ff5c07b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:99c266006d03d674de429c1914a35d332b4e2fb65b4d77de084eb84f01ec84bc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5d26782234031d446a0c0401268a54c749a24761281beb8753a6d453d8e5b91f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8b6850e209743b529c53b98b2167da4edab5bf543e2b22f1b57f3feb3e79deec`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:62dcb4de31ba440c9b80227a2b922bdb5b626eafabaa6041c4b368edeefb57fb`. Independent certificate: `sha256:de8127768a7fdeb63d0f4d9e9da3ab09f73ba114d18a96b60074a66d8d08e4be`. Engine

external-validation hash:
`sha256:4a6ea9e931e0b5122fb7526e6c8f8eb3d68f24c2e2c102e0f65cee2173126734`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GWOSC-V2-CATALOGUES-2026-07-23

Observed comparison records:

- gravity-wave-1: predicted gravitational-wave-propagation; observed gravitational-wave-propagation; exact match True
- gravity-wave-2: predicted gravitational-wave-propagation; observed gravitational-wave-propagation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S021` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:4a923055756f49afde559a4abd24e8ffab800bf895fd2126dfbe6023d9d9871b`. Isolation certificate:
`sha256:2b19ab43fce244485e8c407bf78d6bee007e39067dfd70ad7f4a2e066c4e1235`. Custody certificate:
`sha256:aa49c02c765843f70d9bfc1d18a215dad4397d72db3232168da33e4838d9bf56`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-wave-1 from GWOSC-V2-CATALOGUES-2026-07-23
- gravity-wave-2 from GWOSC-V2-CATALOGUES-2026-07-23

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:275f479048f80a1c7d19818586d630d946c8569a9b749f5566048fb7c50f96e4`; engine receipt
`sha256:0749b38eecb66d4117113353706603a5344e93996220b1a359f78d667369218d` at
`receipts/engine/model_admitted/SFT-PHYS-GRAVITY-WAVE-001-0749b38eecb66d41.json`;
empirical-validation hash
`sha256:8b193f6b95319c5174f8b2e49025a75978a28ede73ad4b42bc78f7147d556daa`; measurement
receipt `sha256:4a923055756f49afde559a4abd24e8ffab800bf895fd2126dfbe6023d9d9871b`.

249. Causal horizon boundary

Claim identity: `SFT-PHYS-GRAVITY-HORIZON-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A horizon is the generated boundary separating events with a limiting-speed outward path to the observer from events whose every outward path remains source-confined.

> *A gravitational horizon is the exact Fold boundary of outward causal accessibility on source-curved support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-GRAVITY-WAVE-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__outward-causal-path-accessibility-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	outward-causal-path-accessibility-boundary	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:def43752e829b2bbf25eae2e3e4df1497ad1c5e343b13964456585ab5d633158`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:90c176ed08a43d27a61c034b0213577c6468523dd8cd8a190252fb874fdb3686`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:158073e681f1b8a87a714e63e93fef55959987dcadfc116ceee7a281f1c6eb0a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:aefc1806213b3b9c4642d09fc10e8ea8a2663ef634218e22bfcac424218d4d22`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1a3295b020e67bd329f48fcaedc43302734345fba5abfe1ead76a577d5b8e4e0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:04d5416a76e6cae6eed78368afc175b43ea8899624372c23524cd3ff369b6219`. Independent certificate: `sha256:3782e0a1753794145e8d5cc5e1b84d7e8e490a067d75dd3203db16660946f9ad`. Engine external-validation hash: `sha256:6f453111ec6cc349180d4e844817a20fd86c054c27a295ff2d8419e1a4c24b8f`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GWOSC-V2-CATALOGUES-2026-07-23

Observed comparison records:

- gravity-horizon-1: predicted gravitational-causal-horizon; observed gravitational-causal-horizon; exact match True
- gravity-horizon-2: predicted gravitational-causal-horizon; observed gravitational-causal-horizon; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S021` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:7886166e1883b3ee28f30e03150d01264e0fd2ee3efff776f66e8e0f5987f228`. Isolation certificate: `sha256:e144f7e3cc0eb608ed677716599d54f1d446c3e657f47b0b9d685da01a5117ff`. Custody certificate: `sha256:f2283e4c5a80d4f234a2cc0acca4ffd559e611e743e3e53fbdef9991d6503347`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/api/v2/catalogs (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-horizon-1 from GWOSC-V2-CATALOGUES-2026-07-23
- gravity-horizon-2 from GWOSC-V2-CATALOGUES-2026-07-23

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:b4d8649bf625d85555719a297b2752f5752c822561d3559799f6e2798b896a66`; engine receipt `sha256:3b1a314d1646e9db9d8213fc14565e9b9dac7cf9cd024251f89cb1cbf6420266` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-HORIZON-001-3b1a314d1646e9db.json`; empirical-validation hash `sha256:ad5522d2cf676c784fcbc933b8df5ea99e9f9e5617d964ef2ea517fbb705c4c1`; measurement receipt `sha256:7886166e1883b3ee28f30e03150d01264e0fd2ee3efff776f66e8e0f5987f228`.

250. Weak gravitational potential and Gauss closure

Claim identity: `SFT-PHYS-GRAVITY-WEAK-FIELD-FLUX-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A positive inertial-energy source distributed over the forced rank-two boundary has inverse-square response, and field times boundary growth reconstructs the same source at every generated radius.

For source 1/8, radii 1/4 and 1/2 both return flux 1/8; fields are 2 and 1/2 and clock potentials are 1/2 and 3/4.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-FIELD-INVERSE-SQUARE-001`
- `SFT-PHYS-GRAVITY-FIELD-SOURCE-001`
- `SFT-PHYS-MECH-CONSERVATION-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for weak gravitational potential and gauss closure from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__positive-source-over-rank-two-boundary__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-spacetime-object`	The object has no generated Fold provenance.
`dependency`	`admitted-V3-root-trace`	`asserted-prior-answer`	A prior answer is not a V3 premise.
`relation`	`positive-source-over-rank-two-boundary`	`imported-Newton-profile-or-free-exponent`	The alternative loses conservation, causality or a required record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	An omitted carrier may silently choose the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access reverses the empirical direction.
`record`	`complete-causal-source-control-record`	`answer-only`	An answer cannot reproduce its causal/source ledger.
`extension`	`no-extra-rule`	`free-extra-rule`	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:536252bb17f9ca456c07c08b13be87fc626e4f0daa38dc7c63cfb95c744dc2ab`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:cf6b59676d1adc6d639b0032d305cc79c0bebfb082ad3b124027655716448475`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:67e2899f9784ba430c1d9a152f4bd10f8ee35f00b78ec95a5a4d0978b7b52109`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:139890ccf0b6c0c76335901373433e22d031be49f72159f2576a386808e77aa2`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:53d773b9122b5823346f7c3b3eb874aa41d779c8e386c53e6caf4b7bf06a7f58`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash:

sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5. Independent certificate: sha256:6700a3ce895d0265f7e8bd0fc61aa9075b0b62d853ac9184eac8fd75075055c9. Engine external-validation hash: sha256:70b8ab007d5e20ae0e6428e0c9b0b3e04065af604fffb6bd66683d55a7ef2510.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S022 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:7c0505b506dfe931f6cc18545096842c688511616b5a689011fec0919398570a; engine receipt sha256:35145c1a594d6707add50a3f2e72dd3a2162092390fd9b3db03eed38b79ee147 at receipts/engine/model_admitted/SFT-PHYS-GRAVITY-WEAK-FIELD-FLUX-003-35145c1a594d6707.json; empirical-validation hash None; measurement receipt None.

251. Exact causal interval

Claim identity: SFT-PHYS-SPACETIME-EXACT-INTERVAL-003

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The spatial square is positively taken from the temporal square; the generated 3-4-5 partition gives massive interval square 16/25 and proper carrier 4/5, while equality is an empty massless-boundary record.

> *Temporal One and spatial 3/5 retain interval square 16/25 and proper time 4/5; equal temporal/spatial support yields the empty massless boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-WEAK-FIELD-FLUX-003
- SFT-PHYS-SPACETIME-INTERVAL-001
- SFT-PHYS-RELATIVITY-TWO-HAND-DIRAC-SQUARE-003

Generated grammar and closure boundary. Generate the complete eight-axis product for exact causal interval from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-root-trace__positive-temporal-square-retains-spatial-take__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	positive-temporal-square-retains-spatial-take	signed-imported-Minkowski-form	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:2c7bd4d76a5bc514dae6ad6a306edfc3d537e572bbc4d77cea94f2fa9842dffe`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:e67e32a67e79785404f6cd1167b3b4d8e9a9945b223ad59e6ea44677436fd288`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:e8086dd9e72526956535496de02b3ec653f209d9d1803d32241d2d3cb032993e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fcf2b65da1cda8b4ba64962e0e41bf8f139277c8888315ab05fa51b2ebe27427`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:d1d0beea696c30473418bdd6f27a5ba78355af46a27098b55432500446442968`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:c7646e73d8e016cd482c23b904b0a33cc42d4a3c147be42aca4346a788d5f801`. Engine external-validation hash: `sha256:241816ae38caab731db6c17066cee42e7c27c0dd2028e508fcb36ef236776a52`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S022` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor

- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:388e83da2499c5e29b9f4c9a54b10d6728cb9fc9900ae60fb561b65608b9ee27`; engine receipt `sha256:c5560ae617b8b077c8d945e3f9b40475f2ca9c4b64028b4a7ea339753d936c4f` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-EXACT-INTERVAL-003-c5560ae617b8b077.json`; empirical-validation hash `None`; measurement receipt `None`.

252. Static Fold-covariant clock

Claim identity: `SFT-PHYS-GRAVITY-STATIC-CLOCK-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The static clock square is the positive remainder of well depth from the One and commutes with Fold exposure; at well 7/16 the generated clock rate is exactly 3/4.

> *At well 1/8, Fold(7/8)=3/4=One-Fold(1/8); at well 7/16, metric square 9/16 has clock rate 3/4; the horizon yields an empty clock record.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-EXACT-INTERVAL-003`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003`

Generated grammar and closure boundary. Generate the complete eight-axis product for static fold-covariant clock from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__One-take-well-depth-with-Fold-covariance__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-spacetime-object`	The object has no generated Fold provenance.
`dependency`	`admitted-V3-root-trace`	`asserted-prior-answer`	A prior answer is not a V3 premise.
`relation`	`One-take-well-depth-with-Fold-covariance`	`imported-clock-dilation-function`	The alternative loses conservation, causality or a required record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	An omitted carrier may silently choose the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access reverses the empirical direction.
`record`	`complete-causal-source-control-record`	`answer-only`	An answer cannot reproduce its causal/source ledger.
`extension`	`no-extra-rule`	`free-extra-rule`	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9ac8ee4d8bd9978d82f2d9b052a5de9dd727dda7f1fe27068b5b1ae374f977fd`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:1036fb1fbd40f1bffdfca1f05f7308fb1a1373acce57b8e5244b024ba514f126`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:7225468d3a83f38d08d71615cf5cd054548be6adeb48ae8e78ca3ec242700682`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:525f6761831cfb14c1c2364067135af6decbd4c7886144652822c1b1d991f37b`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:7de2d029797fa95cc5f26c00cac79f244e002f075a91896e88c2f0bcec978149`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:a4c0900c2684ce125957ab918d0edfec0ac8d0b553f6b8ec1bacfea83643b311`. Engine external-validation hash: `sha256:8223650a7e3328ecf0fbc18cee075e301774f0bd683690217dda0bdb1b9d01cf`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:bf88511ad49c19ee24bae6de64e2f8e2977ed8e31e5c0f473640c27d71332bee`; engine receipt `sha256:8c03d3fd2812c980509616560286fd28ee1b29400c611cde02ef129d15fa8cb4` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-STATIC-CLOCK-003-8c03d3fd2812c980.json`; empirical-validation hash `None`; measurement receipt `None`.

253. Gravitational-redshift and acceleration equivalence

Claim identity: `SFT-PHYS-GRAVITY-REDSHIFT-EQUIVALENCE-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The same exact acceleration-height carrier appears as weak gravitational redshift and acquired-speed Doppler shift, leaving no local observation label that distinguishes the complete traces.

At acceleration 1/4 and height One, gravitational and accelerated shifts are both 1/4.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-GRAVITY-STATIC-CLOCK-003`
- `SFT-PHYS-GRAVITY-EQUIVALENCE-001`
- `SFT-PHYS-MECH-ACCELERATION-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for gravitational-redshift and acceleration equivalence from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__gravity-redshift-equals-accelerated-Doppler__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-spacetime-object`	The object has no generated Fold provenance.
`dependency`	`admitted-V3-root-trace`	`asserted-prior-answer`	A prior answer is not a V3 premise.
`relation`	`gravity-redshift-equals-accelerated-Doppler`	`independent-fitted-redshift-laws`	The alternative loses conservation, causality or a required record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	An omitted carrier may silently choose the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access reverses the empirical direction.
`record`	`complete-causal-source-control-record`	`answer-only`	An answer cannot reproduce its causal/source ledger.
`extension`	`no-extra-rule`	`free-extra-rule`	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:26be961265bea114935fd9ef430e954092c0b698a37e61fd23cf7c20f952aee7`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:34445064638bbce6286f2f2d83a969d6a045a927ab41c4320d4f2479d1be6a52`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:c7a539be9215ee867ac689964bfd2c3356d64e9ca6b2a619687e88c27a71fdb1`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6801215f1eba777f94256999621282c0d46a0312b5fce2f452b579df9e3aafcd`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:bf76521c08fb69ef09c4ef1db6d1a3680bfd9460cf891261958852825f732267`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:e60c9095875c1c45f456f06745990d777374a1844757509a1a2acee82cb27cd5`. Engine external-validation hash: `sha256:9dadae88048f1d3fe5eadb1ab6fa440d48341070a60e5677b1545aabf0ebb129`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:9ac9a202b994c37ead3a9befcaac43fa469fc5b49ca0eb2a46241e1dfe9262a3`; engine receipt `sha256:6726f8bb1dd6a005b03f9058957774e3e97b2d14f7850901331494aefef7185c` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-REDSHIFT-EQUIVALENCE-003-6726f8bb1dd6a005.json`; empirical-validation hash `None`; measurement receipt `None`.

254. Exact lattice curvature family

Claim identity: `SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The exact second difference of the square carrier is binary curvature at every positive spacing; summing complete axes yields Laplacians 2, 4 and 6 in one-, two- and three-dimensional support.

> *At spacings 1/8 and 1/16, per-axis curvature is 2; complete dimension counts give Laplacians 2,4,6.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-GRAVITY-REDSHIFT-EQUIVALENCE-003`
- `SFT-PHYS-GRAVITY-CURVATURE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for exact lattice curvature family from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__binary-curvature-per-generated-axis__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	binary-curvature-per-generated-axis	continuum-derivative-or-fitted-curvature	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:4b1c60e6cc08f4a63afcdd01fd0282aedf0f8ed95e7b50b2e2793386f7873fa8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:4162c08533363fc40c8a047f13d104ef86273ffc0122a78417b7669e8952ad22`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:1188ac93e7ea3b9d8aadb79be92bdaa90e52d5ddf385dd5571be04fc171a198e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f16881aa678b81018a623faf9f769b7563e09e03d10bac2d7c227123c6c2cd75`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:a732758c32ac5963478fa788dfd5d60c16eeeee14676fee382f2f7da885c1856`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:cc9083a1d111b532d331004348ac65dafb2a21273f68113279a6641c4cd264f8`. Engine external-validation hash: `sha256:86dcc72575bccbfb2b13af05f2722738c5da2b81d0a589f719d234134e610010`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor

- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:7e746116ea5adbf38e0424307cef139670ef7b261813a4e196ba3d94f78cb787`; engine receipt `sha256:f10d0261e22c0cdfbce4d91bd75653851064c132325530b1b57e2da5e06ee457` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003-f10d0261e22c0cdf.json`; empirical-validation hash `None`; measurement receipt `None`.

255. Gravity self-sources through field energy

Claim identity: `SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The positive gravitational field carrier has an exact square energy carrier that couples through the same half-One relation and therefore appends a separately retained second-order source correction.

> *Source 1/3 at coupling 1/2 gives linear field 1/6, field energy 1/36 and self-source correction 1/72.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003`
- `SFT-PHYS-GRAVITY-FIELD-SOURCE-001`
- `SFT-PHYS-MATTER-MASS-ENERGY-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for gravity self-sources through field energy from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__field-square-reenters-same-source-channel__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-spacetime-object`	The object has no generated Fold provenance.
`dependency`	`admitted-V3-root-trace`	`asserted-prior-answer`	A prior answer is not a V3 premise.
`relation`	`field-square-reenters-same-source-channel`	`linear-only-gravity-or-free-nonlinearity`	The alternative loses conservation, causality or a required record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	An omitted carrier may silently choose the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access reverses the empirical direction.
`record`	`complete-causal-source-control-record`	`answer-only`	An answer cannot reproduce its causal/source ledger.
`extension`	`no-extra-rule`	`free-extra-rule`	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9c6cf17dcb875e2fd58df7c4dbfdc7c208d3b7311b9e1c724a90c4305ad848bc`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:d54c6721eaa4476367500ed60d0dbda0486e12e04a8d0df8a129fd477795ad49`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:23476aa444ae59176da16d0f4ecf82202065d9897a56154e5e02ee45d9a4ade0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c2efcd39ebc636c66c75c91d913a37acf11ee43befab73c914db68656d9ec364`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:8e75fa19591b5c7c5cfc055211ae29e06686c7bdc6c820b28bbd6a74d6423280`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:89dd8e216bad3895aa17475cc1f2b8132c9d21a6c9279876cc2ab46edaa2ff61`. Engine external-validation hash: `sha256:96c1f150187c3ce4f7544129801d3f1a5387906da62bd3fb5553c23eecfcccb1`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:8ac8e53f3269e2eb63aae1ac45af265f8731882d171ae9d5c120f2b640d93670`; engine receipt `sha256:ce5bb88d7247a7c82c3aa1050b1e4a5261acb4e4cbc301489b504668a87c5fbe` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003-ce5bb88d7247a7c8.json`; empirical-validation hash `None`; measurement receipt `None`.

256. Graviton polarization count

Claim identity: `SFT-PHYS-GRAVITY-GRAVITON-POLARIZATION-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A symmetric four-dimensional metric has ten component slots; eight coordinate/gauge slots leave exactly two physical polarization carriers, while planar spacetime leaves an empty propagating record.

> *For D=4, D(D+1)/2=10 and 2D=8 leave two; for D=3, six and six leave the empty structural record.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-GRAVITY-WAVE-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for graviton polarization count from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__symmetric-metric-count-less-coordinate-pairs__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-spacetime-object`	The object has no generated Fold provenance.
`dependency`	`admitted-V3-root-trace`	`asserted-prior-answer`	A prior answer is not a V3 premise.
`relation`	`symmetric-metric-count-less-coordinate-pairs`	`asserted-two-polarizations`	The alternative loses conservation, causality or a required record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	An omitted carrier may silently choose the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access reverses the empirical direction.
`record`	`complete-causal-source-control-record`	`answer-only`	An answer cannot reproduce its causal/source ledger.
`extension`	`no-extra-rule`	`free-extra-rule`	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:011a120c156ebbf6960704219c027b415ceadb1b37c3c6c6cb8018e66d970105`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:bbd6f292164cb3936a7dea5be06de8c91b87dc172cfe9eba3b4211b1565ba80e`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:ba6d3b400ce067415bd4e8529d76d343a22cd74e3593a3b32be3fa3c75457054`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8f4e2caefee982d3a49a0378ae738862a87c1ee6cc367ad2e04df09d668e67d3`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:7c0e82c4ff69db1fb17a9b18ef5ff4ec7c4dca12f0fce32a9c0bb74e01dac5b8`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash:

sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5. Independent certificate: sha256:221f6da122f88601e6e92c4fbc6bcb156440ba10013c6cd79ff095db63502557. Engine external-validation hash: sha256:fc8100f1407297b1ffae018a9de7a20a15bdbb7b23e6f165873b6bfcc8925728.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S022 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:4a4e93a0ec30fe32dc6426ba37a59997f1ba955a54ac323f1f16a72e9b8145b8; engine receipt sha256:708fffd5a309a9cee077c2db46aef2bf8374435ec2c7bdf4685ef1fb719d0d73 at receipts/engine/model_admitted/SFT-PHYS-GRAVITY-GRAVITON-POLARIZATION-003-708fffd5a309a9ce.json; empirical-validation hash None; measurement receipt None.

257. One-speed quadrupole gravitational propagation

Claim identity: SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A detached gravitational recurrence advances at the same One causal speed as the electromagnetic carrier; conserved source and momentum freeze monopole and dipole records, leaving changing second differences as the first radiative moment.

> *Eight ticks retain exact One-speed phase; uniform linear motion has equal first differences, while the generated cubic drive has changing second differences 12 and 18.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-GRAVITON-POLARIZATION-003
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003

Generated grammar and closure boundary. Generate the complete eight-axis product for one-speed quadrupole gravitational propagation from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-root-trace__One-speed-recurrence-with-first-unfrozen-quadrupole__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	One-speed-recurrence-with-first-unfrozen-quadrupole	separate-fitted-gravity-wave-speed-or-dipole-radiation	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:6149a0c9454a4a7602a469e8ca6eda0ebc4a283731080167308a9697be0aa08e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:3dcc8a4686ae1db4f931da19670c6d4f5e35360b4c8b4e409099e36dbcaeeba0`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:54e4f803ef2e864ecf1da37bbc319b56642fa5368fcaef466ba178900d8cb3fc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e0414f864b09343c80f7cf6924fbfcccfcde6807d17409ca4dd3ca0818dfbc74`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:25f9f7cc3d8ee870e68b69f14f1f97871a3a0be0ab9b120c1a79738ae8e8bad6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:4c196aeafd9b0d9849ccaa9ff6a014fe3315e99840227633bf0ea9ecbec0e09f`. Engine external-validation hash: `sha256:fa445e1b3181849903d1611aa36d0ad3405bcd6c40252c859ae71266eb1eb983`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor

- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:213090faa82308f6a578c468bf789350ae23d79bc26ca90da85020678f8a4415`; engine receipt `sha256:e0769c5998c353edbc1b4d0a6df76c0c409591e722297dff3803cabe99e73ab1` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003-e0769c5998c353ed.json`; empirical-validation hash `None`; measurement receipt `None`.

258. Strong-field horizon and finite floor

Claim identity: `SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The horizon radius is the Fold of the quarter-One mass, exactly twice the mass; two binary halvings force the quarter area coefficient, and every finite depth retains a positive distance floor so no singular empty distance enters the physical grammar.

> *Mass 1/4 has radius 1/2; entropy coefficient is 1/4; area 32 has eight boundary records; normalized thermal carrier is positive 1/4; every finite depth k has distance floor 1/2^k.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003`
- `SFT-PHYS-GRAVITY-HORIZON-001`
- `SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003`

Generated grammar and closure boundary. Generate the complete eight-axis product for strong-field horizon and finite floor from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__Fold-mass-radius-quarter-area-and-positive-finite-floor__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-spacetime-object`	The object has no generated Fold provenance.
`dependency`	`admitted-V3-root-trace`	`asserted-prior-answer`	A prior answer is not a V3 premise.
`relation`	`Fold-mass-radius-quarter-area-and-positive-finite-floor`	`singular-point-or-imported-horizon-equation`	The alternative loses conservation, causality or a required record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	An omitted carrier may silently choose the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access reverses the empirical direction.
`record`	`complete-causal-source-control-record`	`answer-only`	An answer cannot reproduce its causal/source ledger.
`extension`	`no-extra-rule`	`free-extra-rule`	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:cb9a8b69637f9ab33a00c5824cbbf3801b7a410e9f7ca918604f964be2f83302`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:7a4c685139dd6cf0d9e7f814bbf57bf2e588bca2480aea881ce0679dda59959a`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:4d75656d316adcfde311a5e5a025cf860ef12933bb37fb5b8d3da2c8cf12f669`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8e3af091ba2dc1a355f7bd49af657b9a29bd65eb0dbb02d6e420bc5ee9079606`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:22ef5f6a7305f72d218d9da6e0152437777b7edafac6bb9a9a3b8bd2e7865614`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:8dfa5a51a288e241dd824d7f2f85c3aa2863c69b14b8deaafc418cc69b286d68`. Engine external-validation hash: `sha256:14c9288cae27098ff04e1269768f66df859d74bdbc6e2324ce25480bd1bc3bd9`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger
- no claim that normalised quarter-One alone fixes a measured black-hole temperature in kelvin

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:b722ecbf789c1bbc44be3a51f17133c93acc9f519572e5e27c93156ba959c6db`; engine receipt `sha256:3c02d56ccea9efaed7a9fc9f065e27145f2b6597504a447e3b4ac623e61cc6af` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003-3c02d56ccea9efae.json`; empirical-validation hash `None`; measurement receipt `None`.

259. Horizon information closure and reconstruction

Claim identity: `SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003`

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Local horizon observation closes access to the predecessor as an empty record, while the complete boundary-cell and inverse-process records retain the distinctions required for global reconstruction; loss and conservation are therefore observation-relative parts of one ledger.

> *The 32-cell horizon at coefficient 1/4 retains eight boundary records; local predecessor access is empty, and reconstruction is lawful exactly when the complete inverse record is held.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-QUANTUM-MEASUREMENT-001

Generated grammar and closure boundary. Generate the complete eight-axis product for horizon information closure and reconstruction from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-root-trace__local-closure-with-complete-boundary-inverse-record__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	local-closure-with-complete-boundary-inverse-record	absolute-destruction-or-unrecorded-reconstruction	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:8145ca63db017e3b2ed21f59e936062e32be7b81cd6c9da2648593352aee1f53. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:ebc1967b3bc60b951b615326a4f86ffe446b397698fad8f4b208f81866d08b95.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:26aa4548c05926d4a8bb0ffa996396f36dc3301fa5c5bda6076932217bce61bf.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7672c01641308dd654c1ebee7de58faeb02a2534d0b22983804f0023133f8ac5`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:a79b2b8cb938b2dfedf3e8a03358244b2e3bf2200802338557605e4207cf2177`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:104954c35efa854449d14b8828bdca7e2471cc95ba46e53d2e860a119ab65a30`. Engine external-validation hash: `sha256:b44cecbfb46d816b6c808929529cea33e78d5ddf9c72b848dea9926fd7e04812`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d578e7a1a921310db0c147efd4ec00f3b615bcbd2778ead49ce42d1b7ccef823`; engine receipt `sha256:db27fe2ffddbd49475498a04db13944c1443b625794229c3b222676e143520a2` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003-db27fe2ffddbd494.json`; empirical-validation hash `None`; measurement receipt `None`.

260. Wormhole admissibility boundary

Claim identity: `SFT-PHYS-SPACETIME-WORMHOLE-ADMISSIBILITY-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A shortcut relation is physically admissible only when both mouths and the link are generated adjacent support with a positive source and complete causal trace; a named remote identification without that construction is rejected.

> *The current grammar forces the admissibility conditions but does not force an observed or constructible wormhole; positive source plus generated link plus complete trace is accepted, and every omission is rejected.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for wormhole admissibility boundary from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__positive-source-generated-link-complete-causal-trace__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-spacetime-object`	The object has no generated Fold provenance.
`dependency`	`admitted-V3-root-trace`	`asserted-prior-answer`	A prior answer is not a V3 premise.
`relation`	`positive-source-generated-link-complete-causal-trace`	`declared-remote-mouth-equivalence`	The alternative loses conservation, causality or a required record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	An omitted carrier may silently choose the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access reverses the empirical direction.
`record`	`complete-causal-source-control-record`	`answer-only`	An answer cannot reproduce its causal/source ledger.
`extension`	`no-extra-rule`	`free-extra-rule`	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:44ee49488de167b953bbe36a675d7fd32eabc14820a9aa77dc6dd2bd51e5fca1`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:442b7fc900478a517577c7bc8d1a77fe4bdabd594e71fbc6f26034054815bd33`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:f8e7957d5b934b9fc99cbb7bff00adac033933d50e28028af7840271e9d60272`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:693c48f320606f49532921cea4a4f4367e1e6957b6c6989d66edf72cb5af04f1`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:83e769e2435711ad1f4afb3e408cb2e662f36e0910562fdc2131bb2fed287c33`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:0b323d1864572b705051ef560561f938544549d47d614e4d33a00dce12976bff`. Engine external-validation hash: `sha256:e245ab0be905c2faf0a57d1a75b87fdfbcb49dfb49098f0fc2d65ce7ed0e1f41`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger
- no claim that an astrophysical or engineered traversable wormhole has been observed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:67c1666443b8ed04ce9307ab397b0928be0efcceed8444bdafec9817c69d7cc7`; engine receipt `sha256:ad75329a73a9f31385c1e0e991770bed8415bafa318be84afb8f278ca322d96d` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-WORMHOLE-ADMISSIBILITY-003-ad75329a73a9f313.json`; empirical-validation hash `None`; measurement receipt `None`.

261. Warp-support admissibility boundary

Claim identity: `SFT-PHYS-SPACETIME-WARP-ADMISSIBILITY-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A propagation-support redistribution is admissible only as a positive conserved transfer whose destination carrier equals the source carrier and whose apparatus/source return ledger is complete; negative energy and unrecorded net work are not generated.

> *The current grammar forces the admissibility ledger but no achieved warp device: equal positive source/destination with complete return passes; changed support or incomplete return fails.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-WORMHOLE-ADMISSIBILITY-003`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-VACUUM-COMPLETE-CYCLE-LEDGER-003`

Generated grammar and closure boundary. Generate the complete eight-axis product for warp-support admissibility boundary from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__positive-conserved-support-redistribution-with-return-ledger__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-spacetime-object`	The object has no generated Fold provenance.

Axis	Forced form	Eliminated form	Elimination reason
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	positive-conserved-support-redistribution-with-return-ledger	negative-energy-bubble-or-open-work-ledger	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:3cd2419fa83378d59d112007eab55513bdb45dc4dbef0332c8aa1a20660cb5aa`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:a6668bb1deb89cb401deada96196fe3f0b7e3a0dca9858d7046dfbb006fc4196`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:97c5713c84fffb19408b8d6f30e624c16a4ba2d4de2613edb6e5832df09a095b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:2d26588bd043dc2b7f89dff1cd9f95976fe18ab0bfd0de82ccf4a6bfe85925d6`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:b3eb8a8389ff47a26d4620a363c33a77cebea423f7895c966768a33015d800c3`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:89e785b01dc67bccb19cb3b90a4c67d513bd1558f170b6e0eae9843e0cfaed87`. Engine external-validation hash: `sha256:754bd2a1b36e60d3ce3d296ded8a55b2143b98c7e9fc6de7850c39a1116863b3`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization

- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger
- no claim of observed superluminal transport or engineered spacetime distortion

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d190525037fcae71f32029e1d2d030ea68ffcd51ca42aa7cc7fdf2f0b69cf098`; engine receipt `sha256:b864f95be8c08840d7e5e99ab664e84404d6121fa17385ed3fe751d5a53e5ab9` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-WARP-ADMISSIBILITY-003-b864f95be8c08840.json`; empirical-validation hash `None`; measurement receipt `None`.

262. Closed-timelike recurrence admissibility boundary

Claim identity: `SFT-PHYS-SPACETIME-CLOSED-TIMELIKE-ADMISSIBILITY-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A closed causal recurrence is consistent only when it returns the complete physical and proof record exactly; any self-ancestor cycle that changes its own retained premise has no fixed record and is rejected.

> *Exact recurrence of the complete state is admissible as a cycle; changed self-ancestor, missing trace or answer-altering loop is not. No physical closed timelike curve is forced by this boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-WARP-ADMISSIBILITY-003`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-COMP-CBL-HALTING-001`

Generated grammar and closure boundary. Generate the complete eight-axis product for closed-timelike recurrence admissibility boundary from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__exact-first-return-of-complete-state-and-proof-record__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`generated-exact-Fold-carrier`	`imported-spacetime-object`	The object has no generated Fold provenance.
`dependency`	`admitted-V3-root-trace`	`asserted-prior-answer`	A prior answer is not a V3 premise.
`relation`	`exact-first-return-of-complete-state-and-proof-record`	`changed-self-ancestor-paradox`	The alternative loses conservation, causality or a required record.
`enumeration`	`complete-registered-product`	`selected-neighbourhood`	A selected neighbourhood cannot prove uniqueness.
`minimality`	`every-omission-rejected`	`uncontrolled-omission`	An omitted carrier may silently choose the answer.
`measurement`	`formal-result-sealed-first`	`target-visible-before-seal`	Target access reverses the empirical direction.
`record`	`complete-causal-source-control-record`	`answer-only`	An answer cannot reproduce its causal/source ledger.
`extension`	`no-extra-rule`	`free-extra-rule`	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:16319bd99f53812a381b0aae9cdaf0e89813d4d36d651870ea00e232a89af879`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:9957180f4f1faba948d8c4074204dcdcc8a92a4bc2917eaa35c8f390b44eb0bf`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:fd47b425c66c6b1ee1c7039b67cfd9a5623660eb8660367860cc8a4b9b8739e1`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9c30db967de953e4f00529d924afccc03b9a91a1fc08cc48391f64a254ec1337`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:5812ff52b9b2bc581092d182f10152473f49eec88a92f6bf3574b4414b8424bf`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:1b6adb99315a575631f149094dde7dc5f46a2549c73c970874eb9ff3433d254b`. Engine external-validation hash: `sha256:f24538578ffb99077f9087b2b24b885ece06b008703e45300c0de79dce8f6980`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger
- no claim that a physical closed timelike curve has been observed or constructed

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:9bdb9f8c120d617b2cab104453a64776034f03b12b936fb070f853cbd129420b`; engine receipt `sha256:101b8251369135bcf188c580c15f045d3f54cc502ad46813ded8b478b60b19cb` at `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-CLOSED-TIMELIKE-ADMISSIBILITY-003-101b8251369135bc.json`; empirical-validation hash `None`; measurement receipt `None`.

263. Exact gravitational self-source fixed point and correction contraction

Claim identity: `SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Binary support and generator-three colour force matter source seven-sixteenths; half-One coupling and positive quadratic field feedback force the exact map f-next=half-One times (seven-sixteenths plus f squared), whose sole admissible Fold-part fixed point is quarter-One and whose positive errors and added corrections contract at every successor.

> *The exact Fold scalar self-source map f-next=half-One times (seven-sixteenths plus f squared) has one admissible fixed point, quarter-One; the larger seven-quarters root lies outside the Fold-part domain, and every positive error and added correction contracts at every successor by an exact factor below quarter-One.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-FOUNDATION-HALF-ONE-001`
- `SFT-MATH-SELF-SIMILAR-CONVERGENCE-002`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

Generated grammar and closure boundary. Generate the complete product of every structural, free or target-assigned source; every forced, free or measured coupling; every quadratic, linear or target-assigned action; every admissible, larger-root or measured fixed point; every exact, finite-only or assumed convergence law; every Fold-scalar, universal-GR or historical-only scope; both target custody states; and both extension states. The declared exact boundary is: The exact self-source scalar channel generated by binary support, generator-three colour, half-One coupling, positive Fold parts, quadratic field-energy feedback and every positive iteration depth. The theorem does not claim numerical convergence of every post-Newtonian expansion for every physical source. The generator produced `2916` named candidates and the decision support contains `2916` one-for-one decisions. Exactly one candidate survived: `binary-colour-source-seven-sixteenths__forced-half-One__matter-plus-field-square__unique-admissible-quarter-One__exact-error-and-correction-contraction__exact-Fold-scalar-self-source-channel__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`source_construction`	`binary-colour-source-seven-sixteenths`	`free-matter-source`	Rejected by computed Fold predicates: source.
`coupling`	`forced-half-One`	`free-coupling`	Rejected by computed Fold predicates: coupling.
`nonlinear_action`	`matter-plus-field-square`	`linear-no-self-source`	Rejected by computed Fold predicates: action.
`fixed_point`	`unique-admissible-quarter-One`	`select-larger-root`	Rejected by computed Fold predicates: fixed.
`convergence_law`	`exact-error-and-correction-contraction`	`finite-prefix-only`	Rejected by computed Fold predicates: convergence.
`scope`	`exact-Fold-scalar-self-source-channel`	`universal-general-relativity-convergence`	Rejected by computed Fold predicates: scope.
`target_boundary`	`sealed-before-observation-release`	`observation-readable-before-seal`	Rejected by computed Fold predicates: target.
`extension`	`empty-extension`	`free-convergence-correction`	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:b443f34a942066bb6f369cf46db12084b53e1e6a11c34fec8885781148f21e6c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject larger algebraic root as a Fold part.; observed Seven-quarters exceeds the One.; receipt `sha256:9d321bdfe55b3a09a66ffa3205e3e797a328bdcce871aba5a6ea04bdb48f5b80`.
- `tampered_source`: passed; expected Reject changed source identity.; observed Identity differs.; receipt `sha256:7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83`.
- `tampered_artifact`: passed; expected Reject duplicate candidates.; observed All identities unique.; receipt `sha256:5f9c0ad545be40ca825517099110bae199281727e06768826c0280cdfae9b73b`.
- `boundary`: passed; expected Reject universalization, target access and correction.; observed Only scalar scope, sealed target and empty extension survive.; receipt `sha256:66585422a19fd2f9635ec272fe744036151e2d1526d840a8a0bc556374ef3340`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:69137be32b8e94461a24e7d56ced4562972f567b11309e82700fc850779156d7`. Independent certificate: `sha256:962c960e94d9be73927d46a2bea89e53280cad82bcffa4a8669cd87698c6d97f`. Engine external-validation hash: `sha256:56c9d7f6b1d51e58026b50449cf798adcc8516c3b9f902df8c30382ab37575a3`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- D9M-POST-NEWTONIAN-FIXED-POINT-COMPARISON

Observed comparison records:

- The V1, V2 and primary PN review identities are hash-locked and opened only after the V3 prediction seal.
- V3 independently forces source seven-sixteenths, coupling half-One, admissible fixed point quarter-One and excludes the seven-quarters algebraic root.
- Ten exact successor steps retain positive increasing fields below quarter-One while every error and every added correction shrinks by a factor below quarter-One.
- The complete V1 correction vector seven-fiftieths, seven-one-hundred-fiftieths, seven-thirteen-hundred-fiftieths and seven-one-hundred-nine-thousand-three-hundred-fiftieths is retained and strictly decreasing.
- The V2 self-source map, matter source, coupling and fixed point are reproduced exactly.
- The external review retains nonlinear iteration, the linear-plus-quadratic 1PN scalar structure, weak-field slow-motion near-zone scope, exterior matching and formal any-order hierarchy.
- External formal iteration is not inflated into a claim that every physical PN series converges numerically; the depth-independent convergence proof belongs to the exact registered Fold scalar map.

Falsification condition: Reject if the V1, V2, PN or source-record identity changes; if any historical correction, map, source, coupling, fixed-point, external scheme, nonlinearity, scalar structure, domain, order or scope-limiting row is omitted; if the independently reconstructed map has another admissible fixed point; if any error or correction fails to shrink; if external formal iteration is misreported as universal numerical convergence; if a fitted parameter or free correction enters; or if targets enter before sealing.

Measurement receipt: `sha256:527fd031517257181bd22ccd2b9152f5a09ce8690dd0917818c00961f784f0e6`. Isolation certificate: `sha256:6bc2a7ca92ef2d92b9dc7155422d08af6813762a233c210736ec0d7e98504deb`. Custody certificate: `sha256:0a13b5c88cd7098d69630b3a31f007426e333b7cff0430648736e47321fb6ff9`.

Registered source descriptions:

- BLANCHET-POST-NEWTONIAN-THEORY: https://arxiv.org/abs/1310.1528 (sha256:c3b1b7c454e15b65bade4245a8e6d61b6e5c0546baaee436a17b99ef6983b4a8)

Registered target identities:

- D9M-WITHHELD-HISTORICAL-AND-PN-RECORD from D9M-POST-NEWTONIAN-FIXED-POINT-COMPARISON

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- historical correction selected law
- external target selected law
- free matter source
- free coupling
- larger non-Fold root
- assumed universal series convergence
- fitted correction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:f84ebef827868c003789aad44f9088adf3c06844f66bbaecb17d26ee53ca2c48`; engine receipt `sha256:b0e30a2cb8fa9d9f2232df8f8790ab82d8fc2f8e6555326543e15fce2fb48bba` at `receipts/engine/model_admitted/SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009-b0e30a2cb8fa9d9f.json`; empirical-validation hash `sha256:ca10b7b14f51a35455ae8ad25496eb963f4e4f9c9da7fe69085349663f8f364d`; measurement receipt `sha256:527fd031517257181bd22ccd2b9152f5a09ce8690dd0917818c00961f784f0e6`.

264. Four-coordinate symmetric curvature-source conservation law

Claim identity: `SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Four generated coordinate labels force ten symmetric rank-two slots paired one-to-one with one energy, three momentum and six stress source carriers; generated coordinate shifts force four exact carried/opposed divergence-ledger identities at every finite depth, and deletion of one flow term breaks local conservation.

> *Four generated coordinate labels force ten symmetric curvature-source slots partitioned into one energy, three momentum and six stress carriers; each of four generated commuting-difference divergence directions has exactly matching 48-term carried and opposed ledgers at every finite depth, while removal of one flow term breaks conservation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-GRAVITY-GRAVITON-POLARIZATION-003`
- `SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003`
- `SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009`
- `SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003`
- `SFT-PHYS-SPACETIME-EXACT-INTERVAL-003`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

Generated grammar and closure boundary. Generate the complete product of every complete, scalar-only or target-assigned component organization; every componentwise, aggregate or target-assigned field pairing; every generated-shift, imported-continuum or target-assigned derivative algebra; every exact, asserted or target-assigned conservation identity; every complete, unlabelled or target-assigned source ledger; every detecting, ignored or target-assigned leak test; both target custody states; and both extension states. The declared exact boundary is: Every exact symmetric rank-two carrier over four generated coordinate labels, its complete one-energy/three-momentum/six-stress source ledger, and every finite product-lattice depth whose held coordinate shifts commute. This theorem is the generated commuting-difference curvature channel, not an assertion that every nonlinear lattice discretization has the same operator form. The generator produced `2916` named candidates and the decision support contains `2916` one-for-one decisions. Exactly one candidate survived: `complete-ten-symmetric-source-slots__one-curvature-slot-to-one-source-slot__commuting-generated-shift-differences__four-exact-opposed-ledger-identities__energy-one-momentum-three-stress-six__missing-flow-breaks-exact-balance__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`component_organization`	`complete-ten-symmetric-source-slots`	`scalar-source-only`	Rejected by computed Fold predicates: components.
`field_pairing`	`one-curvature-slot-to-one-source-slot`	`aggregate-source-equation`	Rejected by computed Fold predicates: pairing.
`derivative_algebra`	`commuting-generated-shift-differences`	`imported-continuum-derivatives`	Rejected by computed Fold predicates: derivative.
`conservation_identity`	`four-exact-opposed-ledger-identities`	`asserted-conservation-only`	Rejected by computed Fold predicates: conservation.
`source_ledger`	`energy-one-momentum-three-stress-six`	`unlabelled-ten-slots`	Rejected by computed Fold predicates: source.
`leak_test`	`missing-flow-breaks-exact-balance`	`leak-ignored`	Rejected by computed Fold predicates: leak.
`target_boundary`	`sealed-before-observation-release`	`observation-readable-before-seal`	Rejected by computed Fold predicates: target.
`extension`	`empty-extension`	`free-conservation-correction`	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:ab511510c585b97e523ac81b1b37b516dfe1451ee07af973d78e0c3f348f0b19`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject a source ledger missing one opposed flow term.; observed The exact multisets no longer match.; receipt `sha256:49286d6288e8c1870761485f13ecfe4cd14de2785e6d46532d0acecb723741f4`.
- `tampered_source`: passed; expected Reject changed source identity.; observed Identity differs.; receipt `sha256:7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83`.
- `tampered_artifact`: passed; expected Reject duplicate candidates.; observed All identities are unique.; receipt `sha256:0b5a37b200f6544185da03decea0e8f27bba78ad0ff859e5920232c0a780f049`.
- `boundary`: passed; expected Reject imported operators, pre-seal observation and free correction.; observed Only generated shifts, sealed targets and empty extension survive.; receipt `sha256:c29f33fe86181bd8646bae70e39a16e747ed8c9f6692072c73003c3ae9d1b8db`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d1eef07aacb5859461d3e13ba7ed0524f5385d5ab68850051665a349b5fe85df`. Independent certificate: `sha256:b2f4684dc85a52b77caffa778654275b3d240a9974dda285c7133bdd09cf738a`. Engine external-validation hash: `sha256:943f4901d70ba986a43a84cd7ebdb491d98abe58deaaddbe7c809bac04e50606`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- D9N-SYMMETRIC-SOURCE-CONSERVATION-COMPARISON

Observed comparison records:

- The V1 D9n observation, admitted polarization certificate and all three complete primary documents are hash-locked and opened only after the V3 prediction seal.
- V3 independently forces ten symmetric slots, one energy plus three momentum plus six stress source carriers and a ten-pair componentwise field/source ledger.
- Each of four held source directions expands to 48 carried and 48 opposed terms with 31 distinct canonical derivative terms; every exact multiset matches.
- All 64 three-shift words commute under canonical held-coordinate ordering, and the successor proof extends the identity to every finite product-lattice depth.
- Removing one opposed flow term breaks balance, reproducing the adverse leaking-source boundary.
- Carroll's complete continuum rows retain the symmetric source and geometric tensors, contracted identity, componentwise field equation and source-conservation consequence.
- Hamber-Kagel's exact discrete identity, four contracted identities, ten equations, four gauge conditions, two physical degrees and arbitrary-curvature validity are retained together with the crucial fact that its general curvature relation is nonlinear and not identified with V3's operator.
- GW170817 retains six generic metric modes, two GR tensor modes, the sky-constrained log-Bayes comparisons 20.81+/-0.08 and 23.09+/-0.08, and the limitation to pure-mode hypotheses; no mixed-mode result is claimed.
- No historical, conventional or measured value is fitted into the formal V3 law.

Falsification condition: Reject if any historical, prerequisite, primary-theory, discrete-gravity or GW source identity changes; if any of the thirty-two registered favourable or limiting rows is omitted; if the formal component count, one-three-six source partition, componentwise pairing, four conservation ledgers or leak rejection fails; if the exact Regge identity is conflated with the V3 commuting-difference operator; if the pure-mode GW comparison is inflated into a mixed-mode result; if a measured or fitted value enters the V3 proof; or if targets enter before sealing.

Measurement receipt: `sha256:34d28a05fe1a7fcfb4fd61bb56d7b4a0977ed55509006d5f77e994f3e78793f0`. Isolation certificate: `sha256:33e0147f88792f18ad853089aa8a29b889d7bb71bdd011ee5e0adef16cd16dfa`. Custody certificate: `sha256:44007b0d51691a1ea8a63709eb6b5bc5b30164bf871ef5fcb0d69fab15592d9f`.

Registered source descriptions:

- CARROLL-GENERAL-RELATIVITY-NOTES: https://arxiv.org/abs/gr-qc/9712019 (sha256:d4cbfeed3a18370cb7e1a256da6fa7de722982ad7f6a146b4fc0fe14a6bebf6c)
- HAMBER-KAGEL-EXACT-REGGE-BIANCHI: https://arxiv.org/abs/gr-qc/0107031 (sha256:98499dc16e8963c56be885911f4a0476036f97607f01dfb221ec1c9bc85f6d6c)
- LIGO-VIRGO-GW170817-GR-TESTS: https://dcc.ligo.org/LIGO-P1800059/public (sha256:d86fc8bc15abdac47b75960218aefd247f7adeddc00237e9c329b492962584b7)

Registered target identities:

- D9N-WITHHELD-HISTORICAL-THEORY-AND-GW-RECORD from D9N-SYMMETRIC-SOURCE-CONSERVATION-COMPARISON

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- continuum equation selected operator
- historical target selected components

- aggregate scalar source
- asserted conservation
- ignored leaking source
- signed numerical proof scalar
- fitted measurement
- free correction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d431ad929bc7f92e5dbefee79f3035e70ea388ffcb3a410c4200932ef6096424`; engine receipt `sha256:79b12fc32760832dc52e87fcf8d743575ebe2f944c18e7cb24cbb41a6d8a5b22` at `receipts/engine/model_admitted/SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010-79b12fc32760832d.json`; empirical-validation hash `sha256:ec7c1dffc9a180ff3f16ed4ab83a2919d2642fb06239e9288d65aff94393cd45`; measurement receipt `sha256:34d28a05fe1a7fcfb4fd61bb56d7b4a0977ed55509006d5f77e994f3e78793f0`.

265. Exact static spherical exterior coefficient, flux and clock law

Claim identity: `SFT-PHYS-STATIC-EXTERIOR-CLOCK-TERMINAL-011`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The generated boundary is twice the positive mass carrier; rank-two conservation forces field equal to boundary over radius squared and well share equal to boundary over radius; its positive One-complement is the exact exterior clock-square coefficient, with conserved flux, finite outward successor closure, empty boundary record and farther exact-square clocks running faster.

> *The doubled positive mass carrier fixes the exterior boundary; rank-two conservation forces field equal to boundary over radius squared and well share equal to boundary over radius; its positive complement from the One is the exact static clock-square coefficient, with empty boundary record, conserved flux at every finite radius and farther exact-square clocks running faster.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-GRAVITY-WEAK-FIELD-FLUX-003`
- `SFT-PHYS-GRAVITY-STATIC-CLOCK-003`
- `SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003`
- `SFT-PHYS-GRAVITY-REDSHIFT-EQUIVALENCE-003`
- `SFT-PHYS-VALIDATION-GRAVITY-CLOCK-EQUIVALENCE-003`
- `SFT-PHYS-FIELD-INVERSE-SQUARE-001`
- `SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001`
- `SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

Generated grammar and closure boundary. Generate the complete product of every generated, free or target-assigned horizon normalization; every conserved inverse-square, free-power or target-assigned field; every radius-derived, independent or target-assigned well; every positive complement, signed import or target-assigned coefficient; every exact-square, unrestricted-root or target-assigned clock; every finite-successor, completed/inward or target-assigned boundary; both target custody states; and both extension states. The declared exact boundary is: Every exact positive mass whose generated doubled boundary does not exceed the One, every exact radius at or outside that boundary, and every finite outward binary successor.

The coefficient theorem covers every exact exterior radius; numerical clock rates are evaluated only on generated exact-square carriers, and the boundary is an empty clock record rather than a numerical zero. The generator produced `2916` named candidates and the decision support contains `2916` one-for-one decisions. Exactly one candidate survived: `twice-mass-generated-horizon__inverse-square-with-conserved-horizon-flux__radius-times-field-gives-horizon-over-radius-share__positive-One-complement-of-well__exact-square-carriers-and-empty-horizon-record__finite-successor-flat-boundary-and-vacuum__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`horizon_normalization`	`twice-mass-generated-horizon`	`free-radius-scale`	Rejected by computed Fold predicates: horizon.
`field_law`	`inverse-square-with-conserved-horizon-flux`	`free-field-power`	Rejected by computed Fold predicates: field.
`well_law`	`radius-times-field-gives-horizon-over-radius-share`	`independent-potential-profile`	Rejected by computed Fold predicates: well.
`coefficient_law`	`positive-One-complement-of-well`	`signed-subtraction-import`	Rejected by computed Fold predicates: coefficient.
`clock_law`	`exact-square-carriers-and-empty-horizon-record`	`unrestricted-irrational-root`	Rejected by computed Fold predicates: clock.
`boundary_law`	`finite-successor-flat-boundary-and-vacuum`	`completed-infinity-or-interior-extension`	Rejected by computed Fold predicates: boundary.
`target_boundary`	`sealed-before-observation-release`	`observation-readable-before-seal`	Rejected by computed Fold predicates: target.
`extension`	`empty-extension`	`free-exterior-correction`	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:1439d8057d90d99b24bb7e9544bf9818d939eb2902a0d94382f30ad91f23f918`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject a numerical boundary coefficient or exterior clock record at complete well depth.; observed The boundary is the empty structural record.; receipt `sha256:41b03ba80ef3029577651ebe4fa916e917f1e0176ab802661359e2becc45ca73`.
- `tampered_source`: passed; expected Reject changed source identity.; observed Identity differs.; receipt `sha256:7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83`.
- `tampered_artifact`: passed; expected Reject duplicate candidates.; observed All identities are unique.; receipt `sha256:0b5a37b200f6544185da03decea0e8f27bba78ad0ff859e5920232c0a780f049`.
- `boundary`: passed; expected Reject signed import, pre-seal observation and free correction.; observed Only positive complement, sealed target and empty extension survive.; receipt `sha256:d1ecae9cbfce14712e209bda40343a65dc3218f4e5fd75074cefd7d08432b29b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:811cdfabd0655a1eb9a41beab5e44a584ccfc67eec401c7b3868bd7e862419a9`. Independent certificate: `sha256:75f710b586b5bd2577261040cb9cd21ee902d18e4926b7b1c56888eb8285bbf4`. Engine external-validation hash: `sha256:14f497c68b2096e266e353ccb8bf66a89d1dbcf0da7a80cb313d08f26b10ef92`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- D9O-STATIC-EXTERIOR-CLOCK-COMPARISON

Observed comparison records:

- The V1 D9o row, five admitted V3 certificates, complete primary exterior-theory paper, complete NIST/JILA paper and earlier gravity vector are hash-locked and open only after the V3 prediction seal.
- V3 independently forces normalised horizon half-One from mass quarter-One, coefficient equal to the positive complement of horizon-over-radius, and empty clock record at the boundary.
- Across twelve exact outward successors, well share halves, field quarters, coefficient increases and field times radius squared remains the identical horizon carrier.
- Exact-square near and far coefficients nine-sixteenths and sixteen-twenty-fifths force clock rates three-quarters and four-fifths and far-over-near rate sixteen-fifteenths without an irrational proof value.
- The primary exterior treatment retains spherical-vacuum uniqueness, staticity, source-free equation, the One-take-two-GM-over-radius coefficient, reciprocal radial coefficient, mass/horizon unit restoration, asymptotic flatness and absent spherical radiation.
- The NIST/JILA paper retains predicted gradient -1.09e-19 per millimetre, corrected full-cloud result -9.8(2.3)e-20, corrected split-cloud result -1.28(27)e-19 and fractional uncertainty 7.6e-21.
- NIST/JILA observes the higher region ticking faster and reports agreement with gravitational redshift after its full systematic treatment.
- The NIST result does not measure universal Fold fractions, and the external coordinate solution does not select the V3 law.
- No completed infinity, numerical boundary zero, negative or irrational proof scalar, measured fit or free correction enters V3.

Falsification condition: Reject if any historical, admitted-prerequisite, primary-theory, NIST measurement or earlier gravity-vector identity changes; if any of the thirty-four favourable or limiting rows is omitted; if the formal doubled-mass boundary, inverse-square flux, well share, positive complement, empty boundary record, finite successor or exact clock direction fails; if completed infinity, an irrational clock value, a signed/forbidden proof scalar, measured fit or free correction enters; if the external coordinate solution or NIST result is overgeneralized; or if targets enter before sealing.

Measurement receipt: `sha256:15a204fcfe5f0dff58a1f20ab334170998568827dde8b70e681adee9e7537d3b`. Isolation certificate: `sha256:7ed192061df91d2056009f454223742bfbf166f0683d21c7d9f4ac3f379be309`. Custody certificate: `sha256:80ca4da24fb7b4073ba16cad4f86f6d9c1f2476fe1c7fa431fb673dd7b73abd1`.

Registered source descriptions:

- HEINICKE-HEHL-SCHWARZSCHILD-KERR-INTRODUCTION: https://arxiv.org/abs/1503.02172 (sha256:13288f2e0ada228df6e567506c6ac1541e2559a423a740d1e2844cc9a4fe7085)
- NIST-JILA-MILLIMETRE-REDSHIFT-2022: https://www.nist.gov/publications/resolving-gravitational-redshift-across-millimetre-scale-atomic-sample (sha256:7d4376361233f17082814d99cc46cd3263c9febfdbf273bb9bfbe9699723e2b3)

Registered target identities:

- D9O-WITHHELD-HISTORICAL-THEORY-AND-CLOCK-RECORD from D9O-STATIC-EXTERIOR-CLOCK-COMPARISON

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- imported exterior equation
- free horizon scale
- free field exponent
- signed coefficient

- numerical boundary zero
- completed infinity
- irrational clock rate
- measured fit
- free correction

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:e5f2822b8dd9d6c891579c00b272e1f47f6f682a772df43caa5e5965f7863e74; engine receipt sha256:fca4c68a06026daea19121fd5b71d52bc98e2410be1f83eaef13ff62600d880c at receipts/engine/model_admitted/SFT-PHYS-STATIC-EXTERIOR-CLOCK-TERMINAL-011-fca4c68a06026dae.json; empirical-validation hash sha256:eca6a4597b64466858c074f25b32e63047298bd700645bf2fc6939183255f393; measurement receipt sha256:15a204fcfe5f0dff58a1f20ab334170998568827dde8b70e681adee9e7537d3b.

266. Exact quadrupole third-difference radiated-power law

Claim identity: SFT-PHYS-QUADRUPOLE-RADIATED-POWER-TERMINAL-012

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Held source and momentum records close monopole and dipole radiation, leaving the quadrupole as the first radiative moment. One more generated difference gives its radiative rate. The admitted field-energy square and binary half-One coupling force power equal to half-One paired with that third-rate carrier twice. A static quadrupole closes to an empty radiation record, amplitude scaling is exactly squared, and rank-two shell dilution preserves total power at every finite radius.

> *Held monopole and dipole records leave the quadrupole first; its third generated difference is the radiative rate; half-One times that positive rate paired with itself is total radiated power; static support closes to an empty record, amplitude scaling is exactly squared, and rank-two outward dilution preserves total power at every finite shell.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003
- SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010
- SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001

Generated grammar and closure boundary. Generate the complete product of every quadrupole-first, lower-moment or target-assigned moment law; every third-generated, free-order or target-assigned rate; every squared, linear/free or target-assigned energy carrier; every admitted, free or target-assigned coupling; every rank-two conserved, free or target-assigned shell law; every structural-empty, numerical/linear or target-assigned adverse control; both target custody states; and both extension states. The declared exact boundary is: Every finite exact positive quadrupole-magnitude trace generated by a positive cubic tick carrier and positive amplitude, every positive tick index under the all-positive cubic successor identity, and every finite outward binary shell. Static silence is structural emptiness. The theorem is the leading positive-magnitude quadrupole channel; tensor orientation and physical-unit comparison are retained post-seal. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: held-monopole-and-dipole-leave-quadrupole-first__third-generated-quadrupole-difference__positive-square-of-third-rate__admitted-binary-half-One-coupling__rank-two-density-dilutio `n-with-conserved-total-power__static-empty-and-amplitude-square-scaling__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`moment_law`	`held-monopole-and-dipole-leave-quadrupole-first`	`lower-moment-radiation`	Rejected by computed Fold predicates: moment.
`rate_law`	`third-generated-quadrupole-difference`	`free-difference-order`	Rejected by computed Fold predicates: rate.
`energy_law`	`positive-square-of-third-rate`	`linear-or-unsigned-rate-energy`	Rejected by computed Fold predicates: energy.
`coupling_law`	`admitted-binary-half-One-coupling`	`free-radiation-coupling`	Rejected by computed Fold predicates: coupling.
`shell_law`	`rank-two-density-dilution-with-conserved-total-power`	`unconserved-or-free-shell-power`	Rejected by computed Fold predicates: shell.
`adverse_control`	`static-empty-and-amplitude-square-scaling`	`numerical-zero-or-linear-scaling`	Rejected by computed Fold predicates: control.
`target_boundary`	`sealed-before-observation-release`	`observation-readable-before-seal`	Rejected by computed Fold predicates: target.
`extension`	`empty-extension`	`free-radiation-correction`	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:c8e4934943b15932c31efcff4d375375735fcda0691dde4e2010c45c441b0135`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject numerical-zero static power or linear amplitude scaling.; observed Static radiation is empty and doubled third rate gives fourfold power.; receipt `sha256:e5d34752bf8c897a64aa0754ccdaa54c0d3200eb23175fa89cf22fb2632490cc`.
- `tampered_source`: passed; expected Reject changed source identity.; observed Identity differs.; receipt `sha256:7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83`.
- `tampered_artifact`: passed; expected Reject duplicate candidates.; observed All identities are unique.; receipt `sha256:0b5a37b200f6544185da03decea0e8f27bba78ad0ff859e5920232c0a780f049`.
- `boundary`: passed; expected Reject free derivative order, pre-seal target access and free correction.; observed Only generated third rate, sealed target custody and empty extension survive.; receipt `sha256:b4b67644c9dcf9c22ff5b0902cd3465f324534fcbd35e159a14cc378341037ed`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0894076545fc7d5a702ce4cb4bd542f03f3fd4188508dda1d1c2feb15c1b0192`. Independent certificate: `sha256:85ed385605982d16e63203b60b40ba5ae092b000287283048bfe1d05002b440c`. Engine external-validation hash: `sha256:980d6f821d33052300a2f84690d44a58e35ba9547927280e702a8199b5de4854`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- D9Q-QUADRUPOLE-RADIATED-POWER-COMPARISON

Observed comparison records:

- The complete V1 D9q row, eight admitted prerequisite certificates and complete 56-page primary Double Pulsar paper are hash-locked and released only after the V3 formal commit and in-run prediction seal.

- V3 independently forces held monopole and dipole records, third generated quadrupole difference, half-One times its positive square, structural static silence and exact amplitude-square scaling.
- The all-positive cubic opposition identity has retained third-rate six at every successor; the exact witness has power eighteen and doubled amplitude has power seventy-two.
- Rank-two outward transport quarters density at every binary radius successor while density paired with radius twice reconstructs the same total power.
- The 16-year PSR J0737-3039A/B record covers a 2.45-hour mildly eccentric orbit and approximately sixty thousand tracked orbits.
- The primary paper retains Shklovskii, Galactic and spin-down mass-loss corrections before reporting measured GW-driven orbital decay -1.247782(79)e-12.
- The primary quadrupole prediction -1.247810(+6/-7)e-12 and retained 3.5PN correction -1.75e-17 give total prediction -1.247827(+6/-7)e-12.
- Measured over predicted decay is 0.999963(63): its exact rational uncertainty interval contains the formal normalised One, and the published 95-percent agreement boundary is 1.3e-4.
- The extreme equation-of-state rerun 0.999958(64), timing-error dominance, future kinematic limit, material-binary regime and compact-object comparison caveats are all retained.
- The external negative decay direction and decimal measurements remain observational records; no measured value, fitted parameter, target-selected survivor or free radiation correction enters V3.

Falsification condition: Reject if the V1 row, any admitted prerequisite, the complete primary timing paper or any of its forty-eight favourable, correction, uncertainty or scope rows changes or is omitted; if the formal quadrupole-first, third-difference, half-One square, static-empty, amplitude-square or rank-two shell identity fails; if the published normalised decay no longer contains the formal One within its registered uncertainty; if the mildly relativistic material-binary result is universalized to every compact-object regime; if any external sign or decimal becomes a formal proof scalar; or if target access, fitting or a free correction precedes closure.

Measurement receipt: `sha256:86923e0713e593e62ee8cade4399e2321732f802fceb921aaf303a58707d554b`. Isolation certificate: `sha256:61154b4a9ff8917ddf69691cb9ade45de69128efeae713376842971d7470f44d`. Custody certificate: `sha256:67821312675e15942c10a54436b21309444ef307b8fba8f9e99af0657519e038`.

Registered source descriptions:

- KRAMER-ET-AL-DOUBLE-PULSAR-2021: https://arxiv.org/abs/2112.06795 (sha256:9d1b5b4dc304b1d6b1b11059a26d96ca97d1c0d43857523058f973c12bddc6a4)

Registered target identities:

- D9Q-WITHHELD-HISTORICAL-PREREQUISITE-AND-DOUBLE-PULSAR-RECORD from D9Q-QUADRUPOLE-RADIATED-POWER-COMPARISON

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- imported quadrupole power equation
- lower-moment gravitational radiation
- free difference order
- free radiation coupling
- numerical static zero
- negative external decay sign as proof scalar
- irrational or imaginary proof value
- measurement-selected survivor

- measured fit
- free radiation correction

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:d66fcf0085cdaf7086a73cd43170aba452784f91acf319dce4ce84281e43833d; engine receipt sha256:e43448d651ade64487e4cbd44620c4812ffafcb62d20bd71c9c3bde51cdb4db6 at receipts/engine/model_admitted/SFT-PHYS-QUADRUPOLE-RADIATED-POWER-TERMINAL-012-e43448d651ade644.json; empirical-validation hash sha256:fdd1e90d58113fe690623d58a08292113881bfcb55f9bb2d3407edcd63b7aedd; measurement receipt sha256:86923e0713e593e62ee8cade4399e2321732f802fceb921aaf303a58707d554b.

267. Single finite Fold quantum-gravity composition

Claim identity: SFT-PHYS-FINITE-QUANTUM-GRAVITY-TERMINAL-023

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. At every positive finite Fold depth, one shared lattice jointly retains complete quantum support, three spatial directions, a symmetric rank-two gravity carrier with two physical polarizations, empty mass/rest record, One-cell causal advance, a positive distance floor, a finite rational loop sum and a quarter-area horizon ledger; no extra spatial direction or completed infinity is generated.

> *At every positive finite Fold depth, one shared lattice jointly retains complete quantum support, three spatial directions, a symmetric rank-two gravity carrier with two physical polarizations, empty mass/rest record, One-cell causal advance, a positive distance floor, a finite rational loop sum and a quarter-area horizon ledger; no extra spatial direction or completed infinity is generated.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-QUANTUM-PHYSICAL-STATE-001
- SFT-PHYS-QUANTUM-EVOLUTION-001
- SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003
- SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010
- SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003
- SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003
- SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003
- SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003
- SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003
- SFT-PHYS-VALIDATION-FINITE-LOOPS-003
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of shared carrier, dimensional support, rank-two source, polarization reduction, mass/causal carrier, quantum support, finite-loop closure, horizon ledger, comparison boundary and extension forms. The declared exact boundary is: Every positive finite Fold depth on the admitted three-space lattice, its complete binary quantum support, four-coordinate symmetric rank-two source slots, finite exact loop prefix, positive distance floor and complete quarter-area horizon record. Continuum completion is outside the boundary. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: single-finite-Fold-lattice__three-space-plus-one-process-direction__complete-symmetric-rank-two-source__ten-take-four-take-four-leaves-two__empty-mass-record-and-One-cell-per-tick__complete-finite-binary-word-support__complete-finite-rational-loop-prefix__positiv e-floor-and-quarter-area-record__inherit-sealed-wave-loop-and-horizon-comparisons__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	single-finite-Fold-lattice	separate-quantum-and-gravity-substrates	Separate substrates do not establish a composition law.
dimension	three-space-plus-one-process-direction	chosen-or-extra-dimensional-space	A chosen or added dimension is not generated by the stability theorem.
rank	complete-symmetric-rank-two-source	scalar-or-selected-tensor	A scalar or selected tensor omits the admitted complete source slots.
polarization	ten-take-four-take-four-leaves-two	assigned-spin-or-mode-count	An assigned label does not derive physical degrees.
propagation	empty-mass-record-and-One-cell-per-tick	fitted-mass-or-speed	A fitted mass or speed imports the target.
quantum	complete-finite-binary-word-support	continuum-amplitude-space	A continuum adds ungenerated support.
loops	complete-finite-rational-loop-prefix	completed-infinity-or-counterterm	A completed infinity and correction are outside the Fold grammar.
horizon	positive-floor-and-quarter-area-record	unbounded-or-unrecorded-boundary	An unbounded or unrecorded boundary loses the finite ledger.
comparison	inherit-sealed-wave-loop-and-horizon-comparisons	target-selects-composition	External programmes cannot choose the joint model.
extension	no-extra-rule	extra-dimension-or-free-regulator	Either addition changes the generated model.

Base and successor. The closure proof is sealed by derivation hash sha256:f096b575fff7c59ef6ae9d4a7f92f9e78b8ca7f901d67a6329218d8003dc4db1. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:bc21475047b4e738c5bdd65495fdad246c91a59c4b3f1db1db7944785e6715af.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:cd38d0775100d18bfc3cf208b6993ae2b2b1443141df70112c6ef0232c1c1522.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:a6791b69b5fb4b37c86d8b6c5d774414fb1d896e586f28e383a38937a06801b9.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:610f2dae6849e9463367f5c6e25fea21fd1120b805ea63b4af0ddd004cffabb4.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:50fe71d3ae3dc9f2a84f9956a1fe508b355fef7d5f6e5d71b824dc9753908084. Independent certificate: sha256:9c22d51a095f633f83a21d93863618612c1d42c3816c08a4d29efe4dd73f5110. Engine external-validation hash: sha256:a3c69b5deb330c8717e1f165a1469a1fbfb5ceeecf6016c9a0345ff18f31c4d7.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported continuum quantum field, Einstein quantization or conventional quantum-gravity programme as a premise
- no V1/V2 executable, answer table, measured value or stored survivor
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no completed infinity, fitted counterterm, free regulator or extra dimension
- no claim of direct graviton detection or proof of a continuum Yang-Mills/Einstein completion
- no external comparison before the inherited wave, loop and horizon seals

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:3a1a03aa2db41ddfec0d52aa4cef5f80c5bd2448cfeeb8c3c862209bc4c63672`; engine receipt `sha256:a627c3db6aff944a007b53df0a06975454cec52c113e672da62a440ebfd191e7` at `receipts/engine/model_admitted/SFT-PHYS-FINITE-QUANTUM-GRAVITY-TERMINAL-023-a627c3db6aff944a.json`; empirical-validation hash `None`; measurement receipt `None`.

268. Exact positive Fold velocity-composition law

Claim identity: `SFT-PHYS-SPACETIME-VELOCITY-COMPOSITION-TERMINAL-033`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The unique exact same-direction Fold composition is (u+v)/(One+uv). Typed rest is a two-sided identity; the limiting One is absorbing; positive sublimit parts remain strictly sublimit; pair multiplication makes the law associative and reconstructible; and the complete identity-compatible bilinear census contains four candidates with exactly one survivor.

> *The unique exact same-direction Fold composition is (u+v)/(One+uv). Typed rest is a two-sided identity; the limiting One is absorbing; positive sublimit parts remain strictly sublimit; pair multiplication makes the law associative and reconstructible; and the complete identity-compatible bilinear census contains four candidates with exactly one survivor.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-ONE-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-EXACT-OPERATIONS-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-PHYS-MECH-SPEED-VELOCITY-001`
- `SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001`
- `SFT-PHYS-SPACETIME-LIMIT-SPEED-001`

Generated grammar and closure boundary. Generate the complete product of motion domain, typed rest, forward/held representation, identity-compatible numerator cross term, identity-compatible denominator cross term, limiting-speed behavior, closure, associativity, held-record reversibility, empirical custody and extension form. The declared exact boundary is: Every exact positive rational speed part through the limiting One; typed rest; typed empty held part; both present/absent choices for the bilinear cross term after two-sided identity forces the left, right and denominator-One terms; every exact triple composition; and no signed or continuum proof scalar. The generator produced `2048` named candidates and the decision support contains `2048` one-for-one decisions. Exactly one candidate survived: `exact-positive-parts-and-held-direction__typed-r` est-state__forward-held-pair__cross-term-absent__cross-term-present__limiting-One-is-absorbing__strictly-inside-One__exact-associative-product__retain-forward-and-held-parts__postseal-only-comparison__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
domain	exact-positive-parts-and-held-direction	imported-real-line-velocity	A continuum imports the target model.
rest	typed-rest-state	numerical-zero	Numerical nothing is not an admitted physical part.
representation	forward-held-pair	single-unrecorded-scalar	One scalar discards the reverse distinction.
numerator	cross-term-absent	cross-term-present	It prevents the limiting One from remaining fixed.
denominator	cross-term-present	cross-term-absent	Then composing the limiting One with motion exceeds the One.
limit	limiting-One-is-absorbing	limit-moves-or-is-exceeded	That violates the admitted causal boundary.
closure	strictly-inside-One	unbounded-sum	Ordinary addition can leave the domain.
associativity	exact-associative-product	order-dependent-rule	Frame grouping cannot change one physical relation.
record	retain-forward-and-held-parts	discard-held-part	Discarding the held part makes reversal ambiguous.
measurement	postseal-only-comparison	Fizeau-readable-before-seal	A measured fringe slope could select the survivor.
extension	no-extra-rule	free-coefficient-or-correction	A coefficient would be a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:e223da6d28c5b7d688dad577ff84b1390f312761a376a600f0ed1e14502193dd. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:1a79d62f8ac2f0a5fe2ffe1d3f2dca509e9f18574bc167ecdb1dab811e6204de.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:5813a6c7d53a6d4e3115124335f2123031a0e24752459a9b03375bb0c7ec5938.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:e38d5fdee7058292ad1eadfc03d800a6406911159ddd0706be07124cb04bf09e.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:1260794679b17e38744d0f6ff5ab19124bbd9d70cf8176ba23e0e66d5f86bb13.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:aae0a1d576d4398f4ec1cc2d1bae82141125702742896d55b4effd574b61e078. Independent certificate: sha256:99a423ca52573da9a5770fac70fb2414a1e0949e4acb826714bf285e58b5458f. Engine external-validation hash: sha256:c9cf1eef018a668be095cd8d378c4ceda7d99d0ccd70b166758188ed9b250d85.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S022 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, stored velocity-addition formula or measured optical result in the forcing runtime
- no numerical zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof value
- no imported Lorentz transform, rapidity, hyperbolic function or consensus kinematic equation as a premise
- no target-selected coefficient, fitted denominator or hidden premarked survivor
- no claim that the same-direction scalar law alone supplies the full non-collinear rotation law

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:1ae6884e89287d60530400baa6c2765a3339fc5329619cc83d6270e09516fada; engine receipt sha256:337fccb78afd2207d9976e8034d89d1db75ba9fed2f63c44a3b36ae0e90935a0 at receipts/engine/model_admitted/SFT-PHYS-SPACETIME-VELOCITY-COMPOSITION-TERMINAL-033-337fccb78afd2207.json; empirical-validation hash None; measurement receipt None.

269. Terminal compact-object and horizon-thermodynamics law

Claim identity: SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every positive whole q, complete exclusion packing has q^3 occupants, forced momentum depth q, relativistic support q^4 and gravitational paired support q^6; q^6 is strictly greater after the exact base. The loaded exclusion threshold is 3/4 and its Fold balance is 1/2. Exactly two pre-horizon support families exist. For every positive exact mass m, horizon radius is 2m, area support is (2m)^2, entropy support is one quarter of that area, and temperature is 1/(16m), preserving exact product mT=1/16. At m=1/4 the admitted radius and temperature are respectively 1/2 and 1/4. Every finite halving emission trace retains positive mass, temperature, area and information support; no numerical zero or completed infinity enters.

> *For every positive whole q, complete exclusion packing has q^3 occupants, forced momentum depth q, relativistic support q^4 and gravitational paired support q^6; q^6 is strictly greater after the exact base. The loaded exclusion threshold is 3/4 and its Fold balance is 1/2. Exactly two pre-horizon support families exist. For every positive exact mass m, horizon radius is 2m, area support is (2m)^2, entropy support is one quarter of that area, and temperature is 1/(16m), preserving exact product mT=1/16. At m=1/4 the admitted radius and temperature are respectively 1/2 and 1/4. Every finite halving emission trace retains positive mass, temperature, area and information support; no numerical zero or completed infinity enters.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-VACUUM-ODD-RECURRENCE-003
- SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003
- SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003
- SFT-PHYS-FINITE-QUANTUM-GRAVITY-TERMINAL-023

- SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069

Generated grammar and closure boundary. Generate the complete twelve-axis product of occupation, momentum, support scaling, gravity scaling, threshold, endpoint-family, horizon-radius, area, entropy, thermal, finite-floor and extension forms. The declared exact boundary is: Every positive finite perfect-cube exclusion support and successor; the complete binary endpoint fibre; the admitted quarter-One horizon witness; every positive exact mass carrier; every finite binary emission depth; and the complete twelve-axis alternative product, with dimensional observations inaccessible. The generator produced `4096` named candidates and the decision support contains `4096` one-for-one decisions. Exactly one candidate survived: `one-fermion-per-generated-cell__cube-side-momentum-depth__occupants-times-forced-depth-q4__paired-source-q6__three-quarter-preimage-to-half-One__exactly-two-binary-fibre-families__one-Fold-mass-doubling__rank-two-radius-pair__quarter-area-boundary-record__fixed-thermal-mass-product__positive-finite-floor-at-every-reached-depth__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`occupation`	`one-fermion-per-generated-cell`	`multiply-occupied-cell`	That violates the admitted exclusion law.
`momentum`	`cube-side-momentum-depth`	`selected-momentum-scale`	A selected scale is a free parameter.
`support`	`occupants-times-forced-depth-q4`	`imported-pressure-equation`	An imported equation cannot select the Fold law.
`gravity`	`paired-source-q6`	`linear-unpaired-source`	Gravity couples the complete source support to itself.
`threshold`	`three-quarter-preimage-to-half-One`	`chosen-dimensional-mass`	A measured mass cannot choose a structural threshold.
`endpoint`	`exactly-two-binary-fibre-families`	`open-ended-remnant-list`	An open list is not generated by the binary fibre.
`radius`	`one-Fold-mass-doubling`	`imported-metric-radius`	A conventional metric is not a premise.
`area`	`rank-two-radius-pair`	`volume-support`	A volume count contradicts boundary rank two.
`entropy`	`quarter-area-boundary-record`	`selected-entropy-coefficient`	A selected coefficient is an extra rule.
`thermal`	`fixed-thermal-mass-product`	`constant-or-fitted-temperature`	A constant or fitted temperature loses the scale recurrence.
`floor`	`positive-finite-floor-at-every-reached-depth`	`numerical-zero-or-completed-infinity`	Neither is an admissible Fold proof form.
`extension`	`no-extra-rule`	`free-family-or-correction`	An ungenerated addition destroys uniqueness.

Base and successor. The closure proof is sealed by derivation hash `sha256:56d27e2bf2befa6f84307356e39c47f3e504c62df9abf7b2308422a1b55e2b02`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:0878afc8bf2c2c8f737679a1c61c32e38401807d20596f05b5f47dc537e086a4`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:50471e1307e0c04b3cf08a18994d9af7ecaa4ef648402663b012ab188253ab85`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6910b46decbdf4d41ac76a288239eef5b0ca41b5d6e8e3a7b9fa96066e7c3f5a`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt

sha256:b6638c8870605306c719cd98a4b5ccdd7c66c10d1dafd55134832de03f8d1977.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:dec2163b532cbc88444fee5bb33e4db12b914fe99dbc46cad6b9edbce6e3a676. Independent certificate: sha256:ee8bdb300a7efc9e7e3f2bed94d58781b80605853c27c1f8e06fb0f523da8bce. Engine external-validation hash: sha256:d1a6ef1df492186113ff9b23ce189642cc691394402e6275109743177960b4d1.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S022 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, conventional compact-object equation, catalogue mass or Hawking target in formal execution
- no fitted mass limit, equation-of-state knob, temperature calibration, area normalization or evaporation correction
- no numerical zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinite proof scalar
- no claim that a normalised Fold temperature alone is a direct kelvin observation
- no third pre-horizon family, hidden support source, volume entropy or information deletion

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:244b4f8381f20d52b62cb6b81bb90e001fcdb6aecf409046330a78c7d7b4c6fd; engine receipt sha256:69152da01b20c6b1a7ab6ecfab4e44cf836d588cba4ea6f3b02d5827cb5a0b3f at receipts/engine/model_admitted/SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071-69152da01b20c6b1.json; empirical-validation hash None; measurement receipt None.

270. Terminal gravitational-wave chirp, merger and ringdown law

Claim identity: SFT-PHYS-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-TERMINAL-073

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every positive finite binary shrink depth, separation and period squared strictly fall, exact squared orbital and gravitational-wave frequencies strictly rise, and every successor retains a positive radiated take. The quadrupole wave frequency is exactly twice the orbital frequency. At the first exact horizon contact, two compact source supports join one remnant with the complete radiation and component ledger held. The remnant mass/turn key fixes one tone class and its positive amplitude halves at every finite return. The unique causal waveform order is inspiral with rising chirp, merger, then damped ringdown; no negative, irrational, imaginary, continuum, numerical-zero or completed-infinite proof scalar enters.

For every positive finite binary shrink depth, separation and period squared strictly fall, exact squared orbital and gravitational-wave frequencies strictly rise, and every successor retains a positive radiated take. The quadrupole wave frequency is exactly twice the orbital frequency. At the first exact horizon contact, two compact source supports join one remnant with the complete radiation and component ledger held. The remnant mass/turn key fixes one tone class and its positive amplitude halves at every finite return. The unique causal waveform order is inspiral with rising chirp, merger, then damped ringdown; no negative, irrational, imaginary, continuum, numerical-zero or completed-infinite proof scalar enters.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MECH-ANGULAR-MOTION-001
- SFT-PHYS-WAVE-PERIOD-FREQUENCY-001
- SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009
- SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003
- SFT-PHYS-QUADRUPOLE-RADIATED-POWER-TERMINAL-012
- SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003
- SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071

Generated grammar and closure boundary. Generate the complete twelve-axis product of source, radiative moment, transfer ledger, separation, orbital balance, wave recurrence, contact, merger, remnant tone, damping, finite-floor and sequence forms. The declared exact boundary is: Every finite positive binary separation successor ending at the admitted horizon-contact support; every exact squared period/frequency and positive radiation-take record; the complete two-to-one source join; every finite binary remnant return; and all 4096 registered alternative combinations. The generator produced `4096` named candidates and the decision support contains `4096` one-for-one decisions. Exactly one candidate survived: `held-two-source-orbit__admitted-quadrupole-rate__positive-retained-take__strictly-shrinking-positive-separation__inverse-square-period-square-equals-radius-cube__two-wave-cycles-per-orbit__first-horizon-boundary-contact__one-remnant-with-held-ledger__held-remnant-mass-turn-class__binary-half-One-contraction__positive-at-every-finite-depth__inspiral-chirp-merger-ringdown`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`source`	`held-two-source-orbit`	`unheld-single-source`	A single unheld source has no generated binary orbit.
`moment`	`admitted-quadrupole-rate`	`monopole-or-dipole-radiation`	Conserved source and momentum close those records.
`transfer`	`positive-retained-take`	`negative-or-erased-energy`	A negative scalar or erased distinction violates the Fold ledger.
`separation`	`strictly-shrinking-positive-separation`	`fixed-or-expanding-orbit`	That does not conserve the positive binding and radiation ledger.
`balance`	`inverse-square-period-square-equals-radius-cube`	`imported-or-selected-frequency-law`	An imported or selected law is not forced.
`wave`	`two-wave-cycles-per-orbit`	`one-or-free-wave-cycle-per-orbit`	A free multiplier loses the two-lobed source record.
`contact`	`first-horizon-boundary-contact`	`chosen-merger-time`	A chosen time is a parameter.
`merger`	`one-remnant-with-held-ledger`	`erased-or-two-source-remnant`	It either deletes provenance or fails to join.
`tone`	`held-remnant-mass-turn-class`	`fitted-tone`	A fitted tone is measurement-selected.
`damping`	`binary-half-One-contraction`	`negative-exponential-or-free-decay`	That imports a forbidden scalar or parameter.
`floor`	`positive-at-every-finite-depth`	`numerical-zero-or-completed-infinity`	Neither is an admissible Fold proof endpoint.
`sequence`	`inspiral-chirp-merger-ringdown`	`permuted-or-omitted-stage`	It violates contact and source causality.

Base and successor. The closure proof is sealed by derivation hash `sha256:32121943d933e156c025ac6e64a1ea453122f13d421c0cc1a6b7c1fe56d9aa5b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:5cfebdc73d3ed910e8d7e650bd05623f61c3784e9893a4315e79132d2e2c3ba8`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:a2f756d006fe4276cceb5b5f90db01e157663e808cbd4c37e4a49756582e2388`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:54cb462cc3670586005c76324d6abbc96ca1d0949db6ed365b1a0f97703851b2`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:0230f911c50b7e075bd6308cd0d401310ea086b772d57fb59c1388ae88f5fb8b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0775b4e9ea9962f5407b6321178bd5571184eef331f569e088bd1326c6409981`. Independent certificate: `sha256:331dde20de8d262dd8f1086f919f30c3ddaa36eca49d78de9d8438dc3ec48069`. Engine external-validation hash: `sha256:c2a8a1d0a98c5fed6e75d2cf5634af2244239b42e188f9c82a3d1f9250cf4809`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S022` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, LIGO waveform, dimensional frequency, measured mass or conventional inspiral equation in formal execution
- no fitted chirp mass, merger time, waveform template, damping time, tone, amplitude or correction
- no numerical zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinite proof scalar
- no erased radiation, source, component, contact or remnant record
- no claim that normalised Fold frequency is a measured hertz value

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:fb5a228f7f149ff8eb6b404af09b762c1c49fb051219104e070f9d80ff528571`; engine receipt `sha256:7add0b09e795aee79758286268637f442b40155f4af5c83a4a123f90e2150e19` at `receipts/engine/model_admitted/SFT-PHYS-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-TERMINAL-073-7add0b09e795aee7.json`; empirical-validation hash `None`; measurement receipt `None`.

22. Continua Collective Matter

Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance.

271. Finite coarse-grained continuum correspondence

Claim identity: `SFT-PHYS-CONTINUUM-COARSE-GRAIN-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A continuum description is the observation quotient of a completely generated finite cell network when adjacent cell records lie in the same retained macro-class.

> *Physical continua are scale-declared observation quotients of complete finite Fold cell networks.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-cell-observation-quotient__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`finite-cell-observation-quotient`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:1ce27cef7d1ed74840ac979a45f08172d39465f5a5e9f6f7f732de5749e11a03`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d6bce9c5b6034211c3297165bb209983ecfcda45e5061635706fa854895e9b27`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ce09b9ee426c6690cbbd588473c9e32e0298a7911dfbda3b4453219f8598860c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7d15bff6005e58bb09c1a112387b838e5228cf03409007959e7e994af98e04a9`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:be709c0e451aa29ed4ee66afac372db7eb3ee5c6f1eaf735d720c38950c4b44d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:1a51fc9553d35360f449393d395f3ea0d072eb3095fa63397d888131fb34cfcf`. Independent certificate: `sha256:1efded5d8390c6f7d639702f9f91759b6aacb5442b8ccee2e9fead7d5280239c`. Engine external-validation hash: `sha256:9d487e7713b481ba4836058311df1bbb33224a376dc637e86486c2c0de10fb2c`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- continuum-coarse-grain-1: predicted finite-continuum-correspondence; observed finite-continuum-correspondence; exact match True
- continuum-coarse-grain-2: predicted finite-continuum-correspondence; observed finite-continuum-correspondence; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S023` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:a21a75859c14fbc5797c2c73b94238d653a7824cf3b73191eb4365f13c654059`. Isolation certificate: `sha256:64fde06b0865ba6435b1d7810c41bbe8e849070ef753885227a940cedb51b95b`. Custody certificate: `sha256:88b1d4c9eddf8020acbd951f8062e68db181e796f3bc73b462f38ffdf4588eea`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- continuum-coarse-grain-1 from NIST-SRD-INDEX-2026-07-23
- continuum-coarse-grain-2 from NIST-SRD-INDEX-2026-07-23

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:aad2e1c1d2a3af02f0219b3f6c8347130d5f593f0a280a61823e18c26e1f1dc5`; engine receipt sha256:50129d20eaabd2b42eb571667be8ccb5bbfa01aec3e7e39282dd28d74fcadefd at receipts/engine/model_admitted/SFT-PHYS-CONTINUUM-COARSE-GRAIN-001-50129d20eaabd2b4.json; empirical-validation hash sha256:312274385dfd1d0b5c5770b6d8165dc12b53dbc54570a67a31f5a9882faf451c; measurement receipt sha256:a21a75859c14fbc5797c2c73b94238d653a7824cf3b73191eb4365f13c654059.

272. Density as exact content-to-support relation

Claim identity: SFT-PHYS-FLUID-DENSITY-001

Shared claim-record clause PHYS-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Density is the exact positive material-content count or inertial carrier divided by the generated spatial support count of the same observation region.

> *Fluid density is an exact content-to-support ratio on a complete generated region.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-CONTINUUM-COARSE-GRAIN-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__content-over-generated-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	content-over-generated-support	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:e1135eb1e37026a68a66535735e48c35acf1719d85114cf9d6bbb7a97bc8e05a. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d6b53ecc44e62942f0af864b87f43a598c49def06da854548f868257c5bb21ee`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:878e57a9dda01e2d5c20bbdeb5eea018d2b2d3ae390cfd4cab2ca48fc9368008`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4acd658a8909be1728fc3579dc413b6181aeaf168f7519d34b98036850c37a43`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:3f265911c3f4a001248376002fe6c3ff2d0a07b17a94b7009487f9f7824043f9`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:39d3d3ca8f88165fea6399c5b3889e5a0708134a39f25135c8e21b1df4d138aa`. Independent certificate: `sha256:fc4ec7e77d06740835e907c86bdb712e8f4a5cc2d38f7af667a5a06fc0c66a05`. Engine external-validation hash: `sha256:ed25d13dc67b64c09887a35755599593e3c3318cce8e1494a6f3c5cf6af26aaa`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- molar mass constant: exact predicted interval [625000000461504249/625000000000000000000, 48828125066636203/48828125000000000000]; external interval [50000000037/50000000000000, 12500000017/12500000000000]; overlap True
- deliberately displaced unfavourable interval rejected

Falsification condition: The sealed exact mass/count interval does not overlap the released CODATA molar-mass interval or the displaced control is accepted.

Measurement receipt: `sha256:e498d17d351fe8a07e3093c658a7e309084838dc5441aeb35b5ac0bf322ccc49`. Isolation certificate: `sha256:a5890fcb16d593590da7172cea6e562bc1ec53cb3159c01982750153156631c9`. Custody certificate: `sha256:d39628defe147d0af0aec320453442d4514f0c905b3f0a9e480e43c58e3e9bce`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-density-1 from IAPWS-RELEASES-2026-07-23
- fluid-density-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude

- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:00e525b6865cb06e8731ff2a594ea2e25a3f5f048592c33740f479af1653f4e7`; engine receipt `sha256:5de8d82f4c7c74a083bf5b10cb8f369dc2edb4106e56887704b0fdd3c6d9b86e` at `receipts/engine/model_admitted/SFT-PHYS-FLUID-DENSITY-001-5de8d82f4c7c74a0.json`; empirical-validation hash `sha256:f241ec59f21507d70b70567251198827f768fc4fa7eb470d33f9e6eb4492a599`; measurement receipt `sha256:e498d17d351fe8a07e3093c658a7e309084838dc5441aeb35b5ac0bf322ccc49`.

273. Pressure and stress transport

Claim identity: `SFT-PHYS-FLUID-PRESSURE-STRESS-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Stress is oriented momentum transfer per generated boundary support and recurrence; pressure is its orientation-closed normal observation class.

> *Stress is exact oriented momentum flux and pressure is its normal orientation-closed Fold observation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-FLUID-DENSITY-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__momentum-transfer-over-boundary-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`momentum-transfer-over-boundary-support`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.

Axis	Forced form	Eliminated form	Elimination reason
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:4b8e2807810b62bc672cbb947cf06ea428f85a58e6adc504766a946b7ba38ec7. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:fb7fa3ccfb18ebce02d1140a68e3555e7593365a1e5ca5c14ca402e044e7d6da.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:92447125cc883fb979c3cde345b03c998c687a9bfd04232867dae08b530ba12e.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:429ba780f58f3ff79a79b892f1c98dfb54cbcf84380a1b291c26ab05f649ea56.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:4be9f236a07b5d7b7e6a9996faf428ba56d911ade74f2569ab2a032a4b52896a.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:dbd7d23c7160c57567f66f104c05c598d4c86b8e7d2096e1957bd5138b6f6204. Independent certificate: sha256:30d13302ce554ba52ecf0fe8e5cd7044a5f3a1850cf73ab3189034001eafa953. Engine external-validation hash: sha256:55d54aa39906f121a850637fdea6ef30a9bc7a6d80dd49620b29281d70914ca2.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- fluid-pressure-stress-1: predicted fluid-pressure-stress; observed fluid-pressure-stress; exact match True
- fluid-pressure-stress-2: predicted fluid-pressure-stress; observed fluid-pressure-stress; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause PHYS-S023 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: sha256:8047d6f7637bbf91809f57d6d1380eaf1e8c296e04e967d4340fc8f7d4ac10cc. Isolation certificate: sha256:3521573258c3addd909c56705c1323d4a2ddab64858f070d12770a5e6d6905c9. Custody certificate: sha256:74f1693c3d291005c4aef90b67a8365041cc346bebbce3a1d57942cc6826639c.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-pressure-stress-1 from IAPWS-RELEASES-2026-07-23
- fluid-pressure-stress-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause PHYS-S024 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:a6a010b81025e255d529429be1c93c2cdd1b9a01f1618587aa102d38992aab09; engine receipt sha256:86c5ce303dc984aca71e28aeec89beadb8d727c08a50f18f06a369aadc7860d4 at receipts/engine/model_admitted/SFT-PHYS-FLUID-PRESSURE-STRESS-001-86c5ce303dc984ac.json; empirical-validation hash sha256:6dcc35497697c045a106614945c1453f7b55659b91225048525876ae87e832f4; measurement receipt sha256:8047d6f7637bbf91809f57d6d1380eaf1e8c296e04e967d4340fc8f7d4ac10cc.

274. Flow conservation

Claim identity: SFT-PHYS-FLUID-CONSERVATION-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For every complete cell region, material, momentum and energy changes pair exactly with named transfers across its generated boundary.

> *Fluid conservation is exact pairing of interior carrier change with oriented boundary flow.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__interior-change-boundary-flux-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-success-or-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	interior-change-boundary-flux-pairing	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9dec7a845c22850fb65978dd5223afc80510e1bf41dd923823109fca7955eb38`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6c2960986f4d55e887ca5bcba3b6ae11b3c5dbc032efec8e183b0a7ca2146e21`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8905302ed286a1762db5fbb766b12964a3b6646cd3c4e09e7a9e46f466742f0b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:02ec9fd11b8633c435c5a45764bc7e6dd4b02e881b1fdf86755bd78e360aa3f2`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ac8b616e32bac9ad28c3cbe03c4ca63bde1df101ba74d479f5f6d06f957620b6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4436963e2bcb6dc07b5a59fde089e6de3c5263fde0fa90e6a6c19c3d8acd969b`. Independent certificate: `sha256:185dc33929ce6029c79c250a4e28bc2c1eb622501a7838fd233b6d446f906830`. Engine external-validation hash: `sha256:694e29af4501215cf4daa382a7003f22ad034848aa57dc7b662a98d8de42191e`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- fluid-conservation-1: predicted fluid-flow-conservation; observed fluid-flow-conservation; exact match True
- fluid-conservation-2: predicted fluid-flow-conservation; observed fluid-flow-conservation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S023` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:023f7f6fd78674225a402c4160b3a74445890b5475c311c2684c0d422fbebf7b`. Isolation certificate: `sha256:e24d63e7417b95779dd2c5fe2c3853ce9a49f63317e114b2bd710a1a24998692`. Custody certificate: `sha256:e1d0ca42df1c4cb9985d43e30f9e5aa04371b89b85f3ba08bddcd1833d015a0f`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-conservation-1 from IAPWS-RELEASES-2026-07-23
- fluid-conservation-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:15c277ee9a81d10b5f8e276e20bde33a07790b3fd3659a5768cf19cf1ffe4f21`; engine receipt `sha256:712b58981a69e077c8fb50ddafea70ddb5b390d3a2df0dd03b9ca71fa09794b1` at `receipts/engine/model_admitted/SFT-PHYS-FLUID-CONSERVATION-001-712b58981a69e077.json`; empirical-validation hash `sha256:3cfdb55e275962048965cf26322c9870aa48fbf02ad05d00bebb8559f5845d63`; measurement receipt `sha256:023f7f6fd78674225a402c4160b3a74445890b5475c311c2684c0d422fbebf7b`.

275. Inviscid flow correspondence

Claim identity: `SFT-PHYS-FLUID-INVISCID-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Inviscid flow is the boundary where adjacent cells exchange normal pressure carriers but retain their tangential motion labels without intercellular transfer.

> *Inviscid flow is the exact fluid observation with pressure transfer and no resolved tangential momentum exchange.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-FLUID-CONSERVATION-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__no-tangential-momentum-transfer-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	no-tangential-momentum-transfer-boundary	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9fea5ff3856d9c6927e50148d8f561a31de0e3c4cde8f33c9728dc7f0ba5ecb5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2dfd81c580bb59efca01f992f8cc1573ced5ceeb378ec1cb2fc54c598d76c618`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:518face2fa90408cb1537676e5ae2dc65a40dafcd9dddcbfcf09675c18b61ed9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a996c280c1b2b3383a30b0b528487bb4789d5b1a04fb9c144bf8194547ec8fca`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a1b5cfa22dfeededb22eb4d43fddd0b01e467092976b426d90f7ee9ba26ef751`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4be4eec85be994c725ed2e8c41fe1f8b05b5fccfda1c37693de5235861561d7b`. Independent certificate: `sha256:274aeb210d2fc6e06167b3c816daeac66339cd3d4d3137099997026ca200d934`. Engine external-validation hash: `sha256:b8a906ea1673dba85b6fb6f29aa616bafe62bb93a6d073e3ea21787cafe900ff`.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- fluid-inviscid-1: predicted inviscid-flow-boundary; observed inviscid-flow-boundary; exact match True
- fluid-inviscid-2: predicted inviscid-flow-boundary; observed inviscid-flow-boundary; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause PHYS-S023 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:162fb589ec4ff445db5e16c382011008c3d0b9e97907b96a1ef20245f15866a1`. Isolation certificate:
`sha256:b1b99dab0ff9b031d069bcfd8be2c4b8a8215e5a9a7b34dee048b70023a20df1`. Custody certificate:
`sha256:9aef35775d2d45c7b4c55871dcc4161bf3f3281f24126dbc5ae375c8f9ecef24`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-inviscid-1 from IAPWS-RELEASES-2026-07-23
- fluid-inviscid-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:31d8a07f00c94e480c22a85dcbf57a5d9fae7a0e04cfe68795db0a05c9bb28a0`; engine receipt
`sha256:952a9eb1da2d46708380c440328bb5f77838ecc74808a5d510911f370e2addbb` at
`receipts/engine/model_admitted/SFT-PHYS-FLUID-INVISCID-001-952a9eb1da2d4670.json`;
empirical-validation hash
`sha256:86bc3d2785d29eb4931be3614a0e821a74914fc03f6ab4a724f907dc89cab243`; measurement
receipt `sha256:162fb589ec4ff445db5e16c382011008c3d0b9e97907b96a1ef20245f15866a1`.

276. Viscous momentum transport

Claim identity: `SFT-PHYS-FLUID-VISCOSITY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Viscosity is the exact tangential momentum transferred between adjacent velocity-labelled cell classes per interface support and recurrence difference.

> *Viscosity records exact adjacent tangential momentum transport conditioned on velocity recurrence difference.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-FLUID-INVISCID-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__tangential-adjacent-momentum-transfer__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`tangential-adjacent-momentum-transfer`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:5db91bbb39d36ae50e5638db692c61aa0ee7a7b5b3d711b7c1370b05ebd19d16`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:243b176f433d91fd498e68242b70e2bb66b2c1db852a059a126084da67db8fef`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:dfb3417d8cdf99d0bd693593953db666c9a0a1b443675f187fffc9a50a4360ab`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b0bda1d0ed11869cb1296026168adba986e70db0635133673b73975fc618acad`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e65a0f8e56f9657a81b05bd6cfbc99dcafaad256167c24be0dd6d3def944c411`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:dd2ea2c2404d6fb46987bb7c28819da623e8b20a855d6fe34ab68fc4294d8570`. Independent certificate: `sha256:88c8c29b5b3ad197093011b38958cbdba4cc0e1bad0af766083087ebd9723dcd`. Engine external-validation hash: `sha256:57a16aedaf6176aed0e9fd89fe104205452fcef9b3fb0ecbd38156e8a2e1a7b6`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- fluid-viscosity-1: predicted viscous-momentum-transport; observed viscous-momentum-transport; exact match True

- fluid-viscosity-2: predicted viscous-momentum-transport; observed viscous-momentum-transport; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S023` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:614327c713b6095ad408d501c790141e159b2a9b8bb1c97194bcc293ac8019f9`. Isolation certificate: `sha256:91cba8c275ec055107e6a1efb4496d7c21a65861f15a8181c9227e1847315829`. Custody certificate: `sha256:b48a1dce4417e81db3be18b5035579e31c4bbadcf87b4133cb8b3fc52bf92f2a`.

Registered source descriptions:

- International Association for the Properties of Water and Steam: https://www.iapws.org/release.html (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-viscosity-1 from IAPWS-RELEASES-2026-07-23
- fluid-viscosity-2 from IAPWS-RELEASES-2026-07-23

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:34866177da341a81ccbbcc33f79eccdb22c8861f4bd7aba83f18434e43ef9653`; engine receipt `sha256:070b3ca4b4303c214c11ef666aaa9cb44b57bdb09cd81a752e519867872a814d` at `receipts/engine/model_admitted/SFT-PHYS-FLUID-VISCOSITY-001-070b3ca4b4303c21.json`; empirical-validation hash `sha256:6698af438d9dabbb70a12c50baacff3738eeeea376ea99c1a260be9fa6fcf438`; measurement receipt `sha256:614327c713b6095ad408d501c790141e159b2a9b8bb1c97194bcc293ac8019f9`.

277. Finite turbulence and cascade boundary

Claim identity: `SFT-PHYS-FLUID-TURBULENCE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Finite turbulence is nonperiodic multi-scale recurrence on a complete flow network, with conserved carriers transferred among generated support refinements; cascade claims close only at the enumerated scale boundary.

> *Turbulence is deterministic multiscale Fold flow with exact carrier transfer and an explicit finite census boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-FLUID-VISCOSITY-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__multiscale-flow-recurrence-transfer__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`multiscale-flow-recurrence-transfer`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:52d5283a1936f0b0c4d874b86e5d08eb3d5d6a7926a863ab320cd6e065cfc010`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a73887a8fc98a863be96b9f31736681c2e9fcb1954acc76cd4f49a6e03a526b2`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f9f3711bbcac605f36d171f20e96d6353e8dd48c3b973fb7a8cd91b9d1e43a6e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:533dac907b77fc5c47d01f99e297d5f36749f922d27a4ab684160748ba86528a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:33b889a2994f57a1e38443abc4b3013b70f1d44f495c9345f75daf08293cc931`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:15da7684842244e814ad50b42670a8cf90fe83b67039dde3ebbaf3f65b5060bc`. Independent certificate: `sha256:97481c4173ee2b5fa96182afa033951fdcf78787204793058f1343605430479a`. Engine external-validation hash: `sha256:d4d0aef86a4a08a59d55ba7c8d70f256aa652347eec778b66d5500ac60f7cc8f`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- fluid-turbulence-1: predicted finite-turbulence-cascade; observed finite-turbulence-cascade; exact match True
- fluid-turbulence-2: predicted finite-turbulence-cascade; observed finite-turbulence-cascade; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S023` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:044ddbbffd81f43da1501fb97ec4e7e288822395a3e21442fb16f9dcfa25b61c`. Isolation certificate: `sha256:73aebe86db9913ac8bcdbddaa1ffb2d9ac059389a972e6b0228c4983f4cc6044`. Custody certificate: `sha256:6512b5673d5e2bee1e04715b829d98e2fcc4741604163959d8302ba94f9c9acc`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- fluid-turbulence-1 from NIST-SRD-INDEX-2026-07-23
- fluid-turbulence-2 from NIST-SRD-INDEX-2026-07-23

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:0fcac0621f95107970bd0bd87eeaa74befbde2fb61fa7e9828de98b20d2e9423`; engine receipt `sha256:fabb7a737b98b1cd23f182021914f63de08b58d715aeff9ff3798f9fee2d9eff` at `receipts/engine/model_admitted/SFT-PHYS-FLUID-TURBULENCE-001-fabb7a737b98b1cd.json`; empirical-validation hash `sha256:b4bdb5d3556e92eccfbcaf7bd6e0201d01ef5aedbc05e3ea92d345ea8250688c`; measurement receipt `sha256:044ddbbffd81f43da1501fb97ec4e7e288822395a3e21442fb16f9dcfa25b61c`.

278. Plasma collective response

Claim identity: `SFT-PHYS-PLASMA-COLLECTIVE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A plasma is a mobile two-orientation charge-carrier network whose long-range source-response support couples local constituent motion into collective field recurrence.

> *Plasma collective behavior is source-bound electromagnetic coupling across a mobile held-charge Fold network.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`

- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-FLUID-TURBULENCE-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__mobile-charge-network-field-coupling__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	mobile-charge-network-field-coupling	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:f56ca3205b7b83fb31200095de50ab61ff71ecb6657809dc4585d22fe5f02035`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a89f9ffa8ee5ae6d2a673c5e62fd60a6c313076e86184c2417c1ca9b3216d22e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d9ab01dd2e6633d7a172025c5f01c04d1247d6177636d4d81f3085c9c84871f8`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c9f04994806a3462214b469cce636a358c665007a36ab75140155e59e7cfbbbf`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:321294b1ba2e87e4b07d90a4815ddffc2bb23729469f21499c9e8d9fe5a7971d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2926da1c4cc5aa86a14edf2ae9353aa3c87a2e7b05fde8ea1fd505bb63162e1d`. Independent certificate: `sha256:ee358c5d98fe74f5a041dfc0a4cb8794420c2d0880172a452adf402ed78c9365`. Engine

external-validation hash:
`sha256:168a3c19a9e382e49363a192ee91c7eb72a0640e71fdcdffae94bd99d0ab3a3c`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- Faraday constant: exact predicted interval [120606665154137523/1250000000000, 120606665154137523/1250000000000]; external interval [2412133303/25000, 9648533213/100000]; overlap True
- deliberately displaced unfavourable interval rejected

Falsification condition: The sealed exact charge/count interval does not overlap the released CODATA Faraday interval or the displaced control is accepted.

Measurement receipt:
`sha256:80aa74c895d0f99f10829ec3d83a407b5681f81d89155d449a4fddec0fe45f4c`. Isolation certificate: `sha256:5adefbde9005ccf1b32e59dd324b84d658b3f20d8688169e2c675ac6ee0735d8`. Custody certificate: `sha256:859ac36aad6b034cd66615ff95dbd0706aa16c7d41e85a919f1ff343f72d7449`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- plasma-collective-1 from NIST-ASD-5.12
- plasma-collective-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:fba059bd1650b9469b4a633bbaa8bc3d4222d03f1bd286358ff4b21947a985cf`; engine receipt
`sha256:c10a601659f4bc15de4150c9e599c2f50d4a8bb30118724f44aa56bbd190dd8e` at
`receipts/engine/model_admitted/SFT-PHYS-PLASMA-COLLECTIVE-001-c10a601659f4bc15.json`;
empirical-validation hash
`sha256:5c48bf2ec334804827385dc14338e7fe4e78221a47ac2188f3a9891e7baff3dc`; measurement receipt `sha256:80aa74c895d0f99f10829ec3d83a407b5681f81d89155d449a4fddec0fe45f4c`.

279. Plasma oscillation and screening

Claim identity: `SFT-PHYS-PLASMA-OSCILLATION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Displacing held charge orientations creates a restoring field recurrence; phase-mixed mobile responses redistribute until distant observation classes close the source distinction, producing screening.

> *Plasma oscillation is charge-separation recurrence and screening is closure of its source distinction by complete mobile response.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-PLASMA-COLLECTIVE-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__charge-separation-restoring-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	charge-separation-restoring-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:5947a67cf435cb34ccad72f64843e48171248253247d559a875c2fdb131b3a12`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:426274129891558468084830d17dbe35712d20f0c3a86ff0a1710366f2664387`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:711474c269625d4f16c0a6dc355a0879c88aaf11cd9469298fa8234a20bc6fed`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:19a9f5fe7f3a6a1f91a35c7b3a4f4dc0c6c9abc2eca2ded59c1ff60bcd415aa4`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c2581546a9b49c3c450d381bfdbc49f43a1c3919f89e39649c06e1a7e4f632f2`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2acf9e656c728c6d6ba48f454d426f61e8b538b29873bf9b356571ce2b19416d`. Independent certificate: `sha256:24c27df396a58d082fb49a4efb2043962553983784ee0eb71dc0b21f6918c958`. Engine external-validation hash: `sha256:b7107968b68abb4037587277366cdef6422f8467bed90c66b14bc0c772d2e80a`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- plasma-oscillation-1: predicted plasma-oscillation-screening; observed plasma-oscillation-screening; exact match True
- plasma-oscillation-2: predicted plasma-oscillation-screening; observed plasma-oscillation-screening; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S023` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:254a5d09d418db604a6f7d12dec85f980012729689ff5cbeaab4aeef3f83aa88`. Isolation certificate: `sha256:b7964b572f53b949694c372a133b1237f32bc9ac0f4d4bbe2df09fc37bb8a227`. Custody certificate: `sha256:42c01830908f4dc895bb1b4fdbb413169836a10a6765a8e6a018eb8c81ac223d`.

Registered source descriptions:

- National Institute of Standards and Technology: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- plasma-oscillation-1 from NIST-ASD-5.12
- plasma-oscillation-2 from NIST-ASD-5.12

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:bc4354921d5b3db4d9086a705509b43e323132265c3b120330588c6948029dcc`; engine receipt `sha256:2888b7efaca20d0c7f08be59011d936c2345b2bb8c459ef7f59cd04ec24f6498` at `receipts/engine/model_admitted/SFT-PHYS-PLASMA-OSCILLATION-001-2888b7efaca20d0c.json`; empirical-validation hash `sha256:94732ad537ab4174743ca5149d76dd78c3f0728153634f05312eddcda1d2e565`; measurement receipt `sha256:254a5d09d418db604a6f7d12dec85f980012729689ff5cbeaab4aeef3f83aa88`.

280. Magnetofluid composition

Claim identity: `SFT-PHYS-PLASMA-MHD-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Magnetofluid behaviour is the compositional closure of conserved fluid flow with oriented charge transport and magnetic source-response on the same generated cell network.

> *Magnetohydrodynamics is the exact joint Fold process of fluid conservation and electromagnetic charge flow.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-PLASMA-OSCILLATION-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__fluid-flow-electromagnetic-joint-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`fluid-flow-electromagnetic-joint-composition`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:e4273ce7f67d8334803acda8ec38f1e47a016ab747e076252df8a477058ae4de`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7ac127a6edcf37c11b9d4123d9d0dd9158cab0aa3a8fd744eed289494f624b03`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:731f7b6746cb28d3129ee5047a211addaaa884961de1792249a10515532126b2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5c6de5e550bd5e51ca3b2d461671df02ec347b65c2a831034beeda1eb2f967b1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dfeb70b8f6190c73fe7c021d94bda529f597d4f32a1d2f699ac928cf3097c1a1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9b9af7f6ff1f479de604d080ede292c2367aca22e4a244c29bf4f0ec70872582`. Independent certificate: `sha256:048987901a7d00652dd28d6b307ed8caac174301fdd75a358afc547be3376929`. Engine external-validation hash: `sha256:21e2d92b98c4c54d3b4a02510f9b06a1c5ccd4044d6b2e18acc8e36782be5620`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- plasma-mhd-1: predicted magnetofluid-composition; observed magnetofluid-composition; exact match True
- plasma-mhd-2: predicted magnetofluid-composition; observed magnetofluid-composition; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S023` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:bfe77b0dec41d0abe4adc60f64ca544a33adf42c2309915c8e901fbe81115bc9`. Isolation certificate: `sha256:601e31ba05fc4b750c628028a27ce62bc764aa3c6ccfa6c5caaa87b1e86d9c67`. Custody certificate: `sha256:0dc20cbaac4ebac49785e61773b5d10f0ac86a6ce320780618952b7108b168f9`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- plasma-mhd-1 from NIST-SRD-INDEX-2026-07-23
- plasma-mhd-2 from NIST-SRD-INDEX-2026-07-23

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:cc8a3f076a4dd89c6aa8df8976670ece2f18a8a3edb84c5d4fa0173aa13d7857`; engine receipt `sha256:6cca90cf9ab0c0a753f0aac31257648c0164daf92d57347119bf09406431f404` at `receipts/engine/model_admitted/SFT-PHYS-PLASMA-MHD-001-6cca90cf9ab0c0a7.json`;

empirical-validation hash
`sha256:6eb58003f3fb4625c04affd580c15750d52f94a7d1e4cb697fcd1f88f41861b0`; measurement receipt `sha256:bfe77b0dec41d0abe4adc60f64ca544a33adf42c2309915c8e901fbe81115bc9`.

281. Lattice order and excitation

Claim identity: `SFT-PHYS-CONDENSED-LATTICE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A lattice is a finite adjacency network whose local constituent word and neighbor relation recur under generated translations; excitation is a source-bound departure propagated through that recurrence.

> *Condensed lattice order is translation recurrence of a finite Fold adjacency network and excitation is its propagated labelled deviation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-PLASMA-MHD-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__translation-recurrent-adjacency-network__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`translation-recurrent-adjacency-network`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash
`sha256:2d9aaf394c91a79b77497e14a28e755cc078b9d8e9dd0d01c7fcbfeddc38f2f6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b1cc8064129d3557b22b833afa9a83a704ab693dea418f8b40835f3df2f3d8ba`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8ed9ef2406b9bd6616e7f7cdd01a97adc21e8c5ed9bf668b5b67fa9f4f4e4017`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:07bab574e37e3d080d2079c1f26a7ab081f926d4f752ade2a505022ad9528967`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4013bffdbbb665a8dabe7e065bb971764b0c102b235cc83180695b542cc35eae`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e8d201f82909c9dc391dcc47256dc3fda9b22754bf0d69c9f55e8cf5242d3eed`. Independent certificate: `sha256:90a98406eb4fbb2f21d6615ea1ace1c4bcc5c80290eabba33b2cc4b92dda55b4`. Engine external-validation hash: `sha256:ec9de4ad31f948480c6337b7ba52147485be2a56493af771528d9e1daf3ed7b2`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- condensed-lattice-1: predicted condensed-lattice-excitation; observed condensed-lattice-excitation; exact match True
- condensed-lattice-2: predicted condensed-lattice-excitation; observed condensed-lattice-excitation; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S023` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:423a77816fae19c8507e439b5be3e6a45efc5a1cc093cb60c11c864f42c542e4`. Isolation certificate: `sha256:bc5f9c9155f76aa5713e85ea6961a6d1ab82240343eef4d7e339f9507bb2e7a7`. Custody certificate: `sha256:0a5f8632e1d90523b31acc9ea81ce4867ad92d882b0edbef2a06be16862dbe0a`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- condensed-lattice-1 from NIST-SRD-INDEX-2026-07-23
- condensed-lattice-2 from NIST-SRD-INDEX-2026-07-23

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase

- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:6929b114e85948dda3e3d23a5723cd60492959f498235b8555b8199731be3158; engine receipt sha256:9e62f34c6fca48262068bcb2b937689e1059d336414ba6df6a4a946dd206251c at receipts/engine/model_admitted/SFT-PHYS-CONDENSED-LATTICE-001-9e62f34c6fca4826.json; empirical-validation hash sha256:bd9683ad99dd28899fb738038fdfa496c4ed2e37c570e729890deebf1006158b; measurement receipt sha256:423a77816fae19c8507e439b5be3e6a45efc5a1cc093cb60c11c864f42c542e4.

282. Band support and transport

Claim identity: SFT-PHYS-CONDENSED-BAND-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Band support is the complete set of phase-compatible constituent recurrences on a translation-recurrent network; gaps are observation classes with no compatible generated path.

> *Bands are phase-compatible Fold recurrence classes of a lattice and gaps are complete-support absences at the declared boundary.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-CONDENSED-LATTICE-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__lattice-phase-compatible-recurrence-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	lattice-phase-compatible-recurrence-support	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.

Axis	Forced form	Eliminated form	Elimination reason
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4caef9434e4cd64492f82ad0406a7789badeda6b21055b501e7047278a980015`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:85b0b149042eefbc0921b56113028b1ec5a08a7cbf25076eaf8a71115bfa4c1e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:038dd4e7c0ee189026496752a144279c36795917b9005b100802ed42d08e9b27`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1e383fb722a2a19061bf3948397e286dbfbfb8768aa42018acd8c5eb07a310bb`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:13b3e4761ad16cd3e803b3254843b144ed07e75271da670f911db4bd18f516de`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:6fe2d12b549ad0d66bc826b8346857be50e5047b7e8696df922a4049cdcfc31d`. Independent certificate: `sha256:e52c462dbedfce2764ab700ba086e59a0cf92943d3d9b817f85ebf71ba8c2367`. Engine external-validation hash: `sha256:91c580d9e27ef011a7d4853bbffcd8b00e43bc5f2b89b141896ac4a40bebfb7a`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- condensed-band-1: predicted condensed-band-support; observed condensed-band-support; exact match True
- condensed-band-2: predicted condensed-band-support; observed condensed-band-support; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S023` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:45e359946f7ea9f8e6d15a5d68276b12c22f6b9e21fd68aa2106ca85b4fd39c5`. Isolation certificate: `sha256:966542bfb5bf562594c3eb3ccab5c52ddb1baaa1c0a9d736db7fa5aa25fe63e8`. Custody certificate: `sha256:59d884ef5f3e7260ea42bc1aa282a440c7d6428282fa4c08f740bd9fd0444027`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- condensed-band-1 from NIST-SRD-INDEX-2026-07-23
- condensed-band-2 from NIST-SRD-INDEX-2026-07-23

Shared claim-record clause PHYS-S024 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:12b92b81da03f9a83749c90244816d4dd51ceb5a6a7d2eb8ab8534017bce1bf1`; engine receipt `sha256:2c38da1478ea6ad1a9edfcad1bb1727d7511f937fdea85cd18ab7a7ed6e38c76` at `receipts/engine/model_admitted/SFT-PHYS-CONDENSED-BAND-001-2c38da1478ea6ad1.json`; empirical-validation hash `sha256:0598855fc1c16067d5379ef5767792af2e6ae4ea9933b9103b610280b480db03`; measurement receipt `sha256:45e359946f7ea9f8e6d15a5d68276b12c22f6b9e21fd68aa2106ca85b4fd39c5`.

283. Collective order and phase transition

Claim identity: `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Collective order is a shared recurrence label retained across a connected support fraction; a phase transition changes the stable macro-observation fibre and its adjacency symmetry class.

> *Condensed phase order is shared Fold recurrence over connected support; transition is provenance-preserving change of macro-fibre and symmetry.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-CONDENSED-BAND-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__connected-shared-recurrence-order__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`connected-shared-recurrence-order`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b6f31fff778821a4dc6e8959c0a9ad5e481c0b0943200e877f1ac526ebaf9ef6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e1921214fd5a796742099527107d08bc253142690a0a512c1fc37c6d2ab883cb`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:151cbb565925f8890c997ccad95fcc1527c320948f80cf34a1b7701021a1433d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b562b5f1cf958f42a347f9408cc549b2ee44d2b8324c40590f7166d452f8af7f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ffe3538d4e78cb676aeee96d370bd15b9cd0814d8e838e92bfa11f0083bab4ba`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:125652ffac712d8fcd5648b50691367d6bda8c6041d39b69c72f68dc4e2d7393`. Independent certificate: `sha256:43d1226773c7d47f0390a4fbe3152c93d5fc630c6e8a0c3b44bdb3c168e4a3d6`. Engine external-validation hash: `sha256:cccf0ccbef7c3212547b2885fd5d02c61ec212c02d9edbed7a0ab48b598f452c`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- condensed-phase-order-1: predicted collective-phase-order; observed collective-phase-order; exact match True
- condensed-phase-order-2: predicted collective-phase-order; observed collective-phase-order; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S023` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:7be026f4d7be8a4c9e53c6300b76558cf5348208bddca293337e70041606c2c7`. Isolation certificate: `sha256:db2d92525f690d9976ff58e2e0f6722259b63bfbf07c60e120e897f20d8459d5`. Custody certificate: `sha256:246ade689294a7e27fc61f425a22d121807695770fc78c396bde5e9cc7f37331`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- condensed-phase-order-1 from NIST-SRD-INDEX-2026-07-23
- condensed-phase-order-2 from NIST-SRD-INDEX-2026-07-23

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:de862e7658c028c9b671017a1aff5ef7135a58a5f3fd7b6ef23b410553e24693`; engine receipt `sha256:3b4572a30962211d40ab1e494710304df352743d3478bab102f73c92a43e4663` at `receipts/engine/model_admitted/SFT-PHYS-CONDENSED-PHASE-ORDER-001-3b4572a30962211d.json`; empirical-validation hash `sha256:511de23eb34b66be0865bd5d68080e0872237e1ad2eac65394b9b73c651878e8`; measurement receipt `sha256:7be026f4d7be8a4c9e53c6300b76558cf5348208bddca293337e70041606c2c7`.

284. Superconducting coherent transport

Claim identity: `SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Superconducting transport is a phase-locked joint carrier recurrence whose accessible support contains no momentum-randomizing transition compatible with the retained collective label.

> *Superconductivity is phase-coherent collective Fold transport with structurally closed dissipative transition support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__phase-locked-scattering-closed-transport__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	phase-locked-scattering-closed-transport	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:89ac0d447fc62d5262c5b7826e36fa17cb7e4207757655f3c03ae9dea01dbc26`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d223d8cd28aabe2b9f36107ddde2e6e8c811a34a78fb47914256490babf5e5dc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:11c0231899eca9549f63ab1f75820adc479df5559982abe9bee90b00249037e9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fe78fcc493a1e048e85a7cbfa55e5bd3e71f74518243bc3fdca8f318fe01cc4f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:fcd2a75b76e9935e1034b9e0a2477ce7849f7ff90320cdada985cdd1aa306dea`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d8554ea60284f22785ca2689d890bae4e78117970a0f1d45ebf2876bf94f4a14`. Independent certificate: `sha256:c18546fc09850ef3db9f7ba1f96d7c0996ab108991fb44df787f7306f9d6ec71`. Engine external-validation hash: `sha256:6bd35287c0f9d2e9cee27134c7c49787dc32ee8d00aae2eab6efae727abc6a49`.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- condensed-superconductivity-1: predicted superconducting-coherent-transport; observed superconducting-coherent-transport; exact match True
- condensed-superconductivity-2: predicted superconducting-coherent-transport; observed superconducting-coherent-transport; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S023` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:724bc1d5e5bcd04099f7d3f23658fe3310b05074b73276901e94c32753779f42`. Isolation certificate:
`sha256:74c26678d693b20d1635fda0f87dcefffffb710871d46798f13e76c2177b2387`. Custody certificate:
`sha256:3b80bbe76d9c5ed422941d63b5a228ec76d02d8593b239164b7ea879d75acc76`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- condensed-superconductivity-1 from NIST-CODATA-2022-ALL-CONSTANTS
- condensed-superconductivity-2 from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:cf708d2202191d94e5d569179420cc85fc4d83324f7cd00c135315e4482e052b`; engine receipt
`sha256:28d4be47dd3e3e3ea7e064dccfa81b4c165deb39fdc07ce034b507d8b39cf632` at `receipts/engine/model_admitted/SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001-28d4be47dd3e3e3e.json`;
empirical-validation hash
`sha256:639401cf70eec279b25bbe3ec78a1f79c261da321dd3fa73872de0f0614fa56b`; measurement
receipt `sha256:724bc1d5e5bcd04099f7d3f23658fe3310b05074b73276901e94c32753779f42`.

285. Topological collective-state correspondence

Claim identity: `SFT-PHYS-CONDENSED-TOPOLOGICAL-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A collective state is topological when its complete global path/adjacency class is invariant under every generated local deformation that preserves connectivity and boundary records.

> *Topological collective order is invariance of the global Fold path class under all connectivity-preserving local transformations.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-FIELD-CONSERVED-SOURCE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001`

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__global-path-class-local-deformation-invariance__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`global-path-class-local-deformation-invariance`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0359889fea55090c908d33edb444f2cad98a2dcdd7ddb431a380a48d77280de6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bb8a4e31e8197f486fdb8e593f2109bc41d28c1f3614edf3ff948fabcb3c07c8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:380060ab86ed9165d21af7087a574b93e6ba01d9c9e469c87f9afc77f9ee9f06`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5493ae6329eac437a63678ee19a7a4ca8371a4a133f85ecdfec1e2069c5bffad`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:70337bca18cd7f4f94b5b20c543b1cd7de6747f07a0534b809af74e2d10f6db6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:07694800478ee3eaef088816a80ceccd07cd8fc888af6a1fa111a4229664372e`. Independent certificate: `sha256:ed10d3766585d02702ad385d9fd6cd5a9c2bcd7251ede206f9c03b9fe5453657`. Engine external-validation hash: `sha256:fcbacf8b09002ad913e042b971eebe9e97384f2524acab6400dbee64219aff06`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- condensed-topological-1: predicted topological-collective-state; observed topological-collective-state; exact match True
- condensed-topological-2: predicted topological-collective-state; observed topological-collective-state; exact match True
- deliberately tampered unfavourable control rejected

Shared claim-record clause `PHYS-S023` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:cd62859b44d6dd8ab70248d1a40d8f1e767d3d86e965092adca8a55ea8d639c0`. Isolation certificate:
`sha256:e270cb8a0ed8c98f04c187dd013f4a9c16c530df4ca190070b9a47ea3c2dfbcf`. Custody certificate:
`sha256:988db4e9953f335d644a6c34492abd1d047e91e6b9410c3f0f109d9c024e3d2a`.

Registered source descriptions:

- National Institute of Standards and Technology: https://www.nist.gov/srd (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- condensed-topological-1 from NIST-SRD-INDEX-2026-07-23
- condensed-topological-2 from NIST-SRD-INDEX-2026-07-23

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:508242d2b7f517d0ab3542ff4d8363f1489eec57e52cebd2c4b5337fdccf4372`; engine receipt
`sha256:5d769fc8e72acb923996a2284b2c552f0a25dbc4a175576a9cb27f2e2bfb121f` at `receipts/engine/model_admitted/SFT-PHYS-CONDENSED-TOPOLOGICAL-001-5d769fc8e72acb92.json`;
empirical-validation hash
`sha256:4c0784fb7a6dbed000be9965875510c93342d1814171400626b0571999a088af`; measurement receipt `sha256:cd62859b44d6dd8ab70248d1a40d8f1e767d3d86e965092adca8a55ea8d639c0`.

286. Exact Fold lattice operators, causal cones and phase modes

Claim identity: `SFT-PHYS-LATTICE-OPERATOR-TERMINAL-022`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The exact Fold nearest-neighbour family retains half-One at the centre and shares half-One over 2d neighbours; point-source peak and opposition-ring magnitudes are 2d; causal balls at ticks 1,2,3 are (3,5,7), (5,13,25) and (7,25,63) for dimensions one, two and three; and every N-site cycle has N exact phase-mode carriers with one stationary identity mode.

> *The exact Fold nearest-neighbour family retains half-One at the centre and shares half-One over 2d neighbours; point-source peak and opposition-ring magnitudes are 2d; causal balls at ticks 1,2,3 are (3,5,7), (5,13,25) and (7,25,63) for dimensions one, two and three; and every N-site cycle has N exact phase-mode carriers with one stationary identity mode.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-DYNAMICS-001`

- SFT-FOUNDATION-PART-EQUIVALENCE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003
- SFT-PHYS-FIELD-MAXWELL-PLANAR-CLOSURE-003
- SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003
- SFT-PHYS-WAVE-DISPERSION-001
- SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003
- SFT-PHYS-VALIDATION-ATOMIC-CUBIC-SUPPORT-004
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of site carrier, neighbour family, conserved stencil, positive operator, point-source ring, causal support, cycle modes, dimension induction, comparison boundary and extension forms. The declared exact boundary is: Every positive finite one-, two- and three-direction nearest-neighbour Fold lattice; every positive tick and oriented causal cell; every positive finite cyclic site count of at least three; and the already sealed dispersion, Maxwell and cubic-support comparison records. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `positive-held-site-presence__complete-two-per-generated-axis-ring__half-One-held-half-One-equally-distributed__positive-peak-with-held-opposition__two-dimension-peak-and-complete-ring__complete-oriented-taxicab-ball__complete-exact-phase-mode-carriers__axis-and-oriented-step-successor__inherit-sealed-dispersion-and-cubic-records__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`positive-held-site-presence`	`signed-field-number`	A signed scalar imports cancellation and an unheld absence value.
`neighbours`	`complete-two-per-generated-axis-ring`	`selected-or-remote-neighbours`	Selection or remote adjacency breaks locality and completeness.
`stencil`	`half-One-held-half-One-equally-distributed`	`free-dimension-specific-weights`	Independent weights introduce parameters.
`operator`	`positive-peak-with-held-opposition`	`signed-Laplacian-value`	A signed result is not a positive Fold record.
`source`	`two-dimension-peak-and-complete-ring`	`chosen-peak-or-partial-ring`	A chosen peak or incomplete ring does not conserve the neighbour ledger.
`causality`	`complete-oriented-taxicab-ball`	`continuum-or-selected-front`	A continuum or selected sample omits generated cells.
`modes`	`complete-exact-phase-mode-carriers`	`evaluated-cosine-or-mode-table`	An imported cosine or table introduces nonnative and potentially irrational proof values.
`induction`	`axis-and-oriented-step-successor`	`bounded-count-list`	A short list does not close the next generated axis or tick.
`comparison`	`inherit-sealed-dispersion-and-cubic-records`	`measurement-selects-operator`	External targets cannot choose the stencil or cone.
`extension`	`no-extra-rule`	`free-coefficient-or-extra-neighbour`	That changes the generated conservative family.

Base and successor. The closure proof is sealed by derivation hash `sha256:31f2aed551b31a4e2dbdbbbdfe60aa10d749cd91e0bca328cdc7da8409bd5362`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt

sha256:62664c6125390b5cecdcdb762b22312db24644f6901cbd965944a7de259418e2.

- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:c8f5dac5af8783ca95453b0fab2a430910ce74fae14f2fd406a8a34d2fd0b14a.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:fb404db2158783d426d842850836a63421288fb3383c8598b89671690c645cdd.
- boundary: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:fbcc3aa0b64e314009868b01a0345a4cecda24e0ad4121391e5647ec8b4bd955.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:507f7e32537e49871acf519e2834df4c5c6318e561bb2b49069218004bf10d30. Independent certificate: sha256:3ee7bd89b082732dac0bc7118a197cf982b2f12db163add3de4d79a8d05913be. Engine external-validation hash: sha256:4e4449c05d228fc505dc6fce3b16d66fc7e557de66befaf5de6af0ef92c7e90c.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S024 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported continuum, signed Laplacian, Fourier cosine or monatomic-chain equation as a premise
- no V1/V2 executable, answer table, measured spectrum or stored survivor
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no target-selected neighbour, coefficient, mode or causal-front count
- no claim that the exact phase carrier evaluates an irrational conventional eigenvalue
- no external comparison before the inherited dispersion and cubic-support seals

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:355f6b2232eae04527ae5f7d859449a4649f32cd1eaeb7f4f0b0ca4db254fc2b; engine receipt sha256:2d54a071b1a994f26b3a05083d0d57ec33659b9d5430b17467f5dfd03d525fb8 at receipts/engine/model_admitted/SFT-PHYS-LATTICE-OPERATOR-TERMINAL-022-2d54a071b1a994f2.json; empirical-validation hash None; measurement receipt None.

287. Terminal finite radiation, acoustic, laser, plasma and Alfvén response law

Claim identity: SFT-PHYS-COLLECTIVE-RADIATION-RESPONSE-TERMINAL-041

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Every finite cavity occupation is an exact rational mean over the complete microcanonical word census; all modes above total support are empty and common scale changes preserve the complete word set. Wien covariance is exact at the relation level. In stable three-space P(gT)/P(T)=g^4, so doubling gives 16. Acoustic modes are f_n=n f_1. Laser inversion is greater than 1/2, threshold is gain=loss, above-threshold requires gain>loss, and linewidth times coherence ticks is One. Plasma omega_p^2=nq^2/(m epsilon), lambda_D^2=epsilon T/(nq^2), and Alfvén v_A^2=B^2/(mu rho), all as exact positive rational carriers.

> *Every finite cavity occupation is an exact rational mean over the complete microcanonical word census; all modes above total support are empty and common scale changes preserve the complete word set. Wien covariance is exact at the relation level. In stable three-space P(gT)/P(T)=g^4, so doubling gives 16. Acoustic modes are f_n=n f_1. Laser inversion is greater than 1/2, threshold is gain=loss, above-threshold requires gain>loss, and linewidth times coherence ticks is One. Plasma omega_p^2=nq^2/(m epsilon), lambda_D^2=epsilon T/(nq^2), and Alfvén v_A^2=B^2/(mu rho), all as exact positive rational carriers.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-ONE-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001`
- `SFT-PHYS-THERMO-TEMPERATURE-001`
- `SFT-PHYS-MATTER-FERMION-BOSON-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`
- `SFT-PHYS-FIELD-RADIATION-001`
- `SFT-PHYS-WAVE-RESONANCE-001`
- `SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005`
- `SFT-PHYS-PLASMA-COLLECTIVE-001`
- `SFT-PHYS-PLASMA-OSCILLATION-001`
- `SFT-PHYS-FLUID-DENSITY-001`
- `SFT-PHYS-FIELD-MAGNETIC-001`

Generated grammar and closure boundary. Generate the complete product of finite occupation, radiative dimension, acoustic boundary, stimulated mode, gain threshold, linewidth, plasma balance and magnetofluid response forms. The declared exact boundary is: Every finite positive-whole mode-cost list and positive total-quanta support; every common positive-whole scale; every exact positive temperature ratio; every positive acoustic mode count; every positive laser population, gain, loss and coherence count; and every exact positive charge, mass, permittivity, temperature, field, permeability and density carrier. Empty occupation is a structural empty record, not a physical zero magnitude. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-finite-boson-word-census__three-mode-powers-plus-one-energy-power__positive-whole-harmonic-ladder__duplicated-identical-held-mode__gain-loss-equality-with-strict-half-One-inversion__positive-reciprocal-coherence-carrier__charge-stiffness-over-inertia-and-thermal-balance__magnetic-tension-over-fluid-inertia`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`occupation`	`complete-finite-boson-word-census`	`continuum-exponential-premise`	An imported exponential violates finite exact generation.
`radiation`	`three-mode-powers-plus-one-energy-power`	`fitted-Stefan-exponent`	A fitted exponent is target-selected.
`acoustic`	`positive-whole-harmonic-ladder`	`selected-frequency-list`	A named spectrum cannot select the law.

Axis	Forced form	Eliminated form	Elimination reason
`stimulated`	`duplicated-identical-held-mode`	`unrelated-spontaneous-output`	An unrelated output does not amplify a held mode.
`threshold`	`gain-loss-equality-with-strict-half-One-inversion`	`free-gain-setting`	A free setting adds a parameter.
`linewidth`	`positive-reciprocal-coherence-carrier`	`zero-width-monochromatic-idealization`	A numerical-zero width erases the finite record.
`plasma`	`charge-stiffness-over-inertia-and-thermal-balance`	`named-empirical-fit`	A named fit cannot select the carrier.
`magnetofluid`	`magnetic-tension-over-fluid-inertia`	`selected-wave-speed`	A measured speed is not a derivation.

Base and successor. The closure proof is sealed by derivation hash `sha256:b55f7422998c6bf29035fca2e4d0c8636e26da1d9c58a4541e17630ef5ebf43c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:d2f95bbcb1a590cd99cadb1377bebfb4976fe85d32db6c8b4dd2afb248109883`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:8bab28793b529c5c4c48d5a1397fa044d6ec84f1a70a5852f1ab0741eb3fadc7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6848a259aebfde6e574cca1866c660f3f962a9bdb705035700064ff9c179efee`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:bd27fdedb838b7a2218f203a53ca54b64809e258d407881ab8f0b3a44615bd99`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c0623faae3b4b8928150de44e49170e249d77076c1310271af05579a2811fba1`. Independent certificate: `sha256:7b21c482e17e56c9fac34cee25c1a428562ac59c0bbf82a3bb51318854fe8dba`. Engine external-validation hash: `sha256:c2471891deaf2b417e317d017a3b53a8c59e638145c98c27539c71eb8de8815d`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no Planck exponential, Bose-Einstein continuum distribution, fitted Wien constant or Stefan coefficient as premise
- no measured spectrum, laser setting, plasma frequency, screening length or Alfvén speed available to selection
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof magnitude
- no square-root proof value; squared frequency, length and speed carriers are the exact native forms
- no free gain, loss, linewidth, plasma, acoustic or magnetofluid coefficient

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:b500ac7de380339137eca53bb1d67451c29a9e5d3b479b9d43d7ebf0077c32ab`; engine receipt `sha256:50974ab152438d8b84fe8220281e0d98737df0880f9a6384d50bac6060987b07` at `receipts/engine/model_admitted/SFT-PHYS-COLLECTIVE-RADIATION-RESPONSE-TERMINAL-041-50974ab152438d8b.json`; empirical-validation hash `None`; measurement receipt `None`.

288. Terminal exact phase-criticality, universality and turbulence-scaling law

Claim identity: `SFT-PHYS-CRITICALITY-UNIVERSALITY-TURBULENCE-TERMINAL-047`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. For the binary self-antipodal local-order class, the threshold is 1/2 and the exact exponent carriers are beta=1/2, nu=1/2, gamma=1 and delta=3, with alpha and eta typed as empty One records. Widom gamma=beta(delta-1), Rushbrooke 2beta+gamma=2 and Fisher gamma=2nu all close exactly. For the distinct generator-three conserved cascade, the second-order structure exponent is 2/3 and the energy-spectrum exponent has falling orientation with positive magnitude 5/3. At every positive whole scale q and depth d, structure^3=length^2 and spectrum-divisor^3=wavenumber^5, providing depth-independent exact witnesses without irrational evaluation.

> *For the binary self-antipodal local-order class, the threshold is 1/2 and the exact exponent carriers are beta=1/2, nu=1/2, gamma=1 and delta=3, with alpha and eta typed as empty One records. Widom gamma=beta(delta-1), Rushbrooke 2beta+gamma=2 and Fisher gamma=2nu all close exactly. For the distinct generator-three conserved cascade, the second-order structure exponent is 2/3 and the energy-spectrum exponent has falling orientation with positive magnitude 5/3. At every positive whole scale q and depth d, structure^3=length^2 and spectrum-divisor^3=wavenumber^5, providing depth-independent exact witnesses without irrational evaluation.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-ONE-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-HALF-ONE-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`
- `SFT-PHYS-FLUID-TURBULENCE-001`
- `SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008`

Generated grammar and closure boundary. Generate the complete product of critical threshold, order, correlation, response, critical-field, scaling-identity, cascade and spectrum forms. The declared exact boundary is: The binary self-antipodal local-order class and the generator-three conserved-cascade class; every positive whole perfect-power scale base and depth; exact rational exponent carriers; and typed empty exponent records. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `binary-self-antipodal-half-One__square-excess-to-linear-order-carrier__square-excess-to-linear-inverse-correlation__reciprocal-linear-excess-response__generator-three-order-cube__typed-empty-alpha-eta-with-exact-identities__three-branch-cube-to-square-transfer__falling-held-orientation-with-five-thirds-magnitude`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
threshold	binary-self-antipodal-half-One	measured-critical-temperature	A measured temperature would select the law.
order	square-excess-to-linear-order-carrier	fitted-decimal-exponent	A fitted exponent is not forced.
correlation	square-excess-to-linear-inverse-correlation	continuum-correlation-length-ansatz	A continuum ansatz is outside the Fold grammar.
response	reciprocal-linear-excess-response	selected-response-power	A selected power adds a parameter.
field	generator-three-order-cube	imported-field-polynomial	An imported polynomial is an independent model.
identities	typed-empty-alpha-eta-with-exact-identities	numerical-zero-residuals	Conventional zero is not a proof scalar.
cascade	three-branch-cube-to-square-transfer	dimensional-analysis-assumption	Named dimensional analysis does not force the branch count.
spectrum	falling-held-orientation-with-five-thirds-magnitude	fitted-negative-five-thirds	A fitted signed exponent violates the grammar.

Base and successor. The closure proof is sealed by derivation hash `sha256:b05e501fa57215bf093b880f7960323996494347ac2c58befc10f8200cfe0a58`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:ff266871a7c66c6e020794234f45ed42e85407e9de5ac1c729d78a7ee5f38f5b`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:d811c2f0fbf0ad2a0508722d88072c64994579e55ac5c2a190d9a414c5a3e50c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0c0ff1a6ce8765f94f653f179c2b8eb457f45d130612ef1a2cb2637d5e324fa5`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:424effe32a4dedfaa352c97e4e845e6436d547e40bdbcdb6e092474b09ac8b2b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:bf784f179216305a8029a2fe648735d072b7bfe6b34f81662d523f8b3ed02859`. Independent certificate: `sha256:8c5ffb7c980bc8fc76da3665d7fffeaadaa514ecf820d5ee98c56430ab402a64`. Engine external-validation hash: `sha256:db340f6a2e87fa45d45b46f167e02cf12947996b22ebd1a310ab29db4a84231d`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S024` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no renormalization-group, continuum field theory, Landau functional or Kolmogorov law imported as a premise
- no measured critical exponent, transition temperature, structure function or spectrum slope available to candidate selection
- no claim that every physical transition belongs to one class; equality requires the complete generated class key
- no fitted exponent, intermittency correction or Reynolds-number calibration
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof magnitude; spectral decrease and empty exponents are typed records

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:90edf0a55d6c31e17fd8092433e55b4bc14c422006998aa12e67f29ebefd170e`; engine receipt `sha256:0601d19640943c4b99eb8cccf061e3115c520773eac5e762bf7c5b7440339b25` at `receipts/engine/model_admitted/SFT-PHYS-CRITICALITY-UNIVERSALITY-TURBULENCE-TERMINAL-047-0601d19640943c4b.json`; empirical-validation hash `None`; measurement receipt `None`.

23. Physical Cosmology Boundary

The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology.

289. Propagation redshift relation

Claim identity: `SFT-PHYS-COSMO-REDSHIFT-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Redshift is the exact ratio change between emitted and observed recurrence carriers after source, propagation and observer clock relations are retained.

> *Cosmological redshift is an exact source-to-observer Fold recurrence ratio with complete path provenance.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`
- `SFT-PHYS-GRAVITY-WAVE-001`
- `SFT-PHYS-GRAVITY-HORIZON-001`

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__emission-observation-recurrence-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`emission-observation-recurrence-ratio`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:fbff204b5860100876cebcf0bbb84ba8b599dc6d469a11fb959d2a2676207c78`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9597ef38113a9a2375194761d7288a9920dbafa12a79e4212ba7efeecea59c44`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:bd4d7ace5d03153879e4d4f67b95c4ac791180d31b65e2b9c58fc1e56a13d9d5`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:56f45be90500b8eee597a38e7b2eb11e7cd8e61296df3da3fafe7be046d3d90c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d09e058c6a6886e403b4fd29935aaa5cf253af5b5c402dcacc06049310a41d1f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d09a1391267a174beda0f5d62f8adb0b7ebda82318099ff975f233d451899d7e`. Independent certificate: `sha256:0dbb95ae8ca617e091662cc252d9aeb4b9c6efdbf49b3e104cdf12a0f8408369`. Engine external-validation hash: `sha256:93f7c57ab6352786c1e8a1631c45af1f9c2d20ed49d6baadc36e499729155f06`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GWOSC-GW231123-135430-V3-2026-07-23

Observed comparison records:

- redshift-1: predicted cosmological-redshift-relation; observed cosmological-redshift-relation; exact match True
- redshift-2: predicted cosmological-redshift-relation; observed cosmological-redshift-relation; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:92717ed4f3ba2bcf6e783bdc5f5f3e6da7b372011ee4555cda20c169623ff51e`. Isolation certificate: `sha256:5ee949022428358b56bb7a6e1be1a546f926f502b6dbe43903942361ed71e325`. Custody certificate: `sha256:68bf98e8157386a12fedcbb99e0326e40da12a88350d0b5458c3ae67e43557d5`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/eventapi/json/GWTC-4.1/GW231123_135430/v3/ (sha256:67266a5406cbee1dc98a9853beb42d771595fa474f5ddd33ab4b368d5b71c45d)

Registered target identities:

- redshift-1 from GWOSC-GW231123-135430-V3-2026-07-23
- redshift-2 from GWOSC-GW231123-135430-V3-2026-07-23

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favourable observation or unrecorded source/path

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:c8ecbd91bdea42c8b56aeb5faeac6c598dc628f7a78fa878c0530e32b51f5db8`; engine receipt `sha256:011c8f49c26e4ff32c5202fcf398b1e5e7934b9b20b9a6a8de3eb31baaadfd6e` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-REDSHIFT-001-011c8f49c26e4ff3.json`; empirical-validation hash `sha256:8417063792a39c45d8f0b5cb542c337e1f92e4b42cff9fe787970a435d2724cc`; measurement receipt `sha256:92717ed4f3ba2bcf6e783bdc5f5f3e6da7b372011ee4555cda20c169623ff51e`.

290. Relational expansion observation

Claim identity: `SFT-PHYS-COSMO-EXPANSION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Relational expansion is coherent change of source-separation support relative to observation-clock recurrence across multiple source-bound paths; it is not motion into external space.

> *Expansion is coherent Fold change of relational source support per observation recurrence over a declared finite census.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`
- `SFT-PHYS-GRAVITY-WAVE-001`
- `SFT-PHYS-GRAVITY-HORIZON-001`
- `SFT-PHYS-COSMO-REDSHIFT-001`

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__coherent-source-separation-clock-change__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	coherent-source-separation-clock-change	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d33f908ea6611093fbd1f136a764bdc137c92dbf0ae1b2a5b4765443d98bc1cf`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a2dbb0b406192e90d75444c33a1a0cc7bb0f88d0e13b4e41145f977438ba145f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b5ae118031c6058ace28ac8f9026ffb3493dfe178c532404c893156cac44ebce`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ee430e29142cb3594511f207700c87ea69687f6058b67f30d75c8df741a03357`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ff266063a5b0223e3d2a5cd0c274b948cd09fc341c0e0cbd98ddfc094ae379e7`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e22e8687206a89a280e1b93bac0efe44993a1487ad0ac1de544a4f4779bc672d`. Independent certificate: `sha256:d1d7a1544b2aebaa6b78c3af0af668553ff3b955fff404639b1e8b73e7f0aba1`. Engine external-validation hash: `sha256:2cce4e0ba3d4db4b1c93f96153f8267774ff3aa4bbdb2971f9871a0da58652d1`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NASA-LAMBDA-2026-07-23

Observed comparison records:

- expansion-1: predicted relational-expansion-observation; observed relational-expansion-observation; exact match True
- expansion-2: predicted relational-expansion-observation; observed relational-expansion-observation; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt:
`sha256:180c34948af83fce3115ad4cef3de0524f89e3cf1c2f694cd1522e27922bd4b9`. Isolation certificate:
`sha256:55cd6a3bd7c1f966b7edb76f70c4fe28cd92a7e6a35c8b6612fe94371aba9956`. Custody certificate:
`sha256:e9e17560c8b0d3726b6cdcbd428636b897c23965fc6230e3e4c1d4462943ff77`.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center: https://lambda.gsfc.nasa.gov/ (sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddedffe94e50b2d4275915)

Registered target identities:

- expansion-1 from NASA-LAMBDA-2026-07-23
- expansion-2 from NASA-LAMBDA-2026-07-23

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favourable observation or unrecorded source/path

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:a269499bba0d5b326890c33a3b04021b4899a5d96cd11425dc24f54e2652d77f`; engine receipt
`sha256:f3294516adbba51c7a38681bd1e4df12fd2b9818b85b73472ea9e2fec76ed5c6` at
`receipts/engine/model_admitted/SFT-PHYS-COSMO-EXPANSION-001-f3294516adbba51c.json`;
empirical-validation hash
`sha256:b441e22e3470023097ee73b739446b382033a47da46aef7ec92a96fe52c78d37`; measurement
receipt `sha256:180c34948af83fce3115ad4cef3de0524f89e3cf1c2f694cd1522e27922bd4b9`.

291. Background-radiation field relation

Claim identity: `SFT-PHYS-COSMO-BACKGROUND-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Background radiation is a near-omnidirectional recurrence field whose thermodynamic observation class is invariant across registered frequency support after local foreground records are separated.

> *Cosmic background radiation is a source-history-bound, frequency-consistent thermal Fold recurrence field.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`
- `SFT-PHYS-GRAVITY-WAVE-001`
- `SFT-PHYS-GRAVITY-HORIZON-001`
- `SFT-PHYS-COSMO-EXPANSION-001`

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__om` nidirectional-frequency-invariant-thermal-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	omnidirectional-frequency-invariant-thermal-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:dfbb61b81831741f1cc7a25696488fc6d1ecd405919f62a68c2dfd6de36853be. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:8cc7e810ff2e58c660c84249ac346725d8b2e9d7a99b29cebf699bd8266b3ea8.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:932c0513c1bb93eb7e383ca9c0efd027c242eebe1796f73e3f3ab4a35518d326.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:d58a5c5c3ad1b7c2116b9a85bdcf7458049dc99b65136d6901944d3eeacd9669.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:651bcce56f70730f691d13534b72f1ad7dedb56692f9948c5401e78a6396ac4d.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:64dd57d72ca9cf7c24e56b6c7e3aaf3caaf3ad3c0b77317ac71a66a3fb93b717. Independent certificate: sha256:cac3fc2eef8135972ca5755eac53ecebfab08ef65bbd56ca971fb89948f389a4. Engine external-validation hash: sha256:00991c0c822b140c708152e3732053349ee786c8eaf73e34f90fbd5bb85b3bd1.

Shared claim-record clause PHYS-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NASA-LAMBDA-COBE-FIRAS-2026-07-23
- NASA-LAMBDA-COBE-FIRAS-MONOPOLE-2026-07-23

Observed comparison records:

- FIRAS background interval predicted (Fraction(2723, 1000), Fraction(2727, 1000)); independently released monopole interval (Fraction(681, 250), Fraction(1363, 500)); overlap True
- deliberately displaced unfavourable interval rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:9210aff0c6e319bf03a32716e6fb0d3b0db28c9034a90a9dea81ce726d815f0a`. Isolation certificate: `sha256:32ee283818c2235434698ef39a62215ae6bddd9e7524180968249d3f110be6bb`. Custody certificate: `sha256:ea4c0ce215615ff7c43e58d7087e4f5aa901110280c92723a7c09af7dcfa0a8b`.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center / COBE FIRAS: https://lambda.gsfc.nasa.gov/product/cobe/about_firas.html (sha256:5eae175f5ea06f7a5cb863866eeee9d87a38d870937d9a93ebf28629af0d6ef9)
- National Aeronautics and Space Administration, Goddard Space Flight Center / COBE FIRAS: https://lambda.gsfc.nasa.gov/product/cobe/firas_monopole_spect.html (sha256:e42bcf7e3e2eccac3ac8c210d4287e2477c394ad03d76ac7f719515d29753f2a)

Registered target identities:

- firas_monopole_temperature from NASA-LAMBDA-COBE-FIRAS-MONOPOLE-2026-07-23

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favourable observation or unrecorded source/path

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:49220e097dfb2d3a61ca0cebb54bcc489dd4e02d4de267d3ab4798ded1742126`; engine receipt `sha256:65950d7051f8a1cc5261cb6a1051b1888b4e4b9dff9d52960798ac40e7df7c02` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-BACKGROUND-001-65950d7051f8a1cc.json`; empirical-validation hash `sha256:2ff56a1f3f0ca301762e25a62e41028fa3cd2cbf16ac51af7396654bafeaa928`; measurement receipt `sha256:9210aff0c6e319bf03a32716e6fb0d3b0db28c9034a90a9dea81ce726d815f0a`.

292. Gravitational lensing relation

Claim identity: `SFT-PHYS-COSMO-LENSING-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Lensing occurs when source-curved event support generates multiple causal propagation paths between one emitter and observer, changing direction, arrival recurrence and observed support while preserving source labels.

> *Gravitational lensing is multiple source-bound causal Fold paths through relational curvature to one observation support.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`

- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-GRAVITY-HORIZON-001
- SFT-PHYS-COSMO-BACKGROUND-001

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__multiple-curved-source-observer-paths__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	multiple-curved-source-observer-paths	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:c11cae52e3a2ec2a6d8efe86ccaeac241ae589b1b0176c4738ec3f6c2e1d7a31`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3cc2a9e0f5dfd8bf2a27996c4f376e3a512b0c3c176108f1b10a868461af918a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f4e6b91bd814d859f13b3adafad665231ba76827cb01a8b25abfa05843dc5606`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1293463a7a30cb51089c59d24177fdd106c31613e039b12ed6871cc7c9e2f89c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e74c60c8dd88ff2f5a79298eeafec74420d3d4a42b0c9589b9354bba36906f0c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f0125e16e59c6103aa719aff85e8c32e19455717c544a0cab29cf1a92362c292`. Independent certificate: `sha256:0f2974648713cade0dea1dc5e85c4fe4130a680fe87a1319f1a72a7ec58ba150`. Engine external-validation hash: `sha256:a753b7187e6c335a85131e859dc66413e9216cefadbcac1a074486ce9f0bc47b`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NASA-LAMBDA-2026-07-23

Observed comparison records:

- lensing-1: predicted gravitational-lensing-relation; observed gravitational-lensing-relation; exact match True
- lensing-2: predicted gravitational-lensing-relation; observed gravitational-lensing-relation; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:57f6faa3f4ac43599bfb4cecf25f40c23edb939db795aefaa53950a0189c68ac`. Isolation certificate: `sha256:b6b51c68452d3661a9c4444aca676208680fa839040a42eb2cd53014b802bdff`. Custody certificate: `sha256:99be4345446763a718ac16c5c0c56d70097245ee03f176a1fc604fcdee8cab42`.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center: https://lambda.gsfc.nasa.gov/ (sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddedffe94e50b2d4275915)

Registered target identities:

- lensing-1 from NASA-LAMBDA-2026-07-23
- lensing-2 from NASA-LAMBDA-2026-07-23

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favourable observation or unrecorded source/path

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:e1cc3ef991bf0adf07dda89b3ae1326296593f35b634305e01a7a59304e141de`; engine receipt `sha256:c7e4807f21bea4889c3f78cd9c3e02b73375658afd99cc46da93556c3d4e0189` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-LENSING-001-c7e4807f21bea488.json`; empirical-validation hash `sha256:e786550028a5b83cd92754ba4078f1c137e14cec4529d87a64d007a1f674d1cd`; measurement receipt `sha256:57f6faa3f4ac43599bfb4cecf25f40c23edb939db795aefaa53950a0189c68ac`.

293. Causal distance and observation-time relation

Claim identity: `SFT-PHYS-COSMO-DISTANCE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Cosmological distance is the exact equivalence class of source-to-observer causal path support conditioned on emission and observation recurrence records; different distance notions are distinct retained projections.

> *Cosmological distance is a declared Fold projection of exact causal support and observation-time relations.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`

- SFT-PHYS-SPACETIME-CLOCK-RATE-001
- SFT-PHYS-GRAVITY-CURVATURE-001
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-GRAVITY-HORIZON-001
- SFT-PHYS-COSMO-LENSING-001

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__causal-path-support-conditioned-on-clock-records__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	causal-path-support-conditioned-on-clock-records	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:648e1b6b768af2d7dc3fc55088055f4becd92ce00d93cc99304b854779192641`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:aa383062c59596ea69991db0272859b5ac291a76ee550f5438a54416dcbf5522`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f03495752b1f8a3690542261a0db6ca12a1130d48a1a1ca86b52ece949c40932`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ead8c15e0123454492674e2991a83303cd7b9c50188c91d33ccf00d539c543b5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:72c6f70adb91d4a3030d8086e4c3b58234b48e54d4c05613afedeafb023fd6e2`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d92bab1acf1fc9757e600c3d188a5747dcb24385dc817a1cd7d948d5ecd3392d`. Independent certificate: `sha256:05c01482d13acea0e21a1464be758598c2f44d3d5dd3ce03654bbfeaac5a1981`. Engine external-validation hash: `sha256:be3475999e02f417dbbfcb6fdf089202bcfb2a9cb1f169c1573c0ab9c78b3cc5`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GWOSC-GW231123-135430-V3-2026-07-23

Observed comparison records:

- distance-1: predicted cosmological-causal-distance; observed cosmological-causal-distance; exact match True
- distance-2: predicted cosmological-causal-distance; observed cosmological-causal-distance; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:48cd4970fa0a4ba3f0d2adcf8e11aca3c702ec5ce21d07dff512d28009647667`. Isolation certificate: `sha256:08593b541e0482822e3be9b12c8b34ff49e192746b19f5d0181a7667e27a8674`. Custody certificate: `sha256:10669e9353e935060fff6f4e3bd6bee23315b3b8f9dc9adc4412a655620cf709`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: https://gwosc.org/eventapi/json/GWTC-4.1/GW231123_135430/v3/ (sha256:67266a5406cbee1dc98a9853beb42d771595fa474f5ddd33ab4b368d5b71c45d)

Registered target identities:

- distance-1 from GWOSC-GW231123-135430-V3-2026-07-23
- distance-2 from GWOSC-GW231123-135430-V3-2026-07-23

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favourable observation or unrecorded source/path

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:3e8f5b7766a0f5516179dd9d013964a99a976388ee4a20d6d2c80c8c5e5e5ee1`; engine receipt `sha256:8c54ba35f01cd4082993baabb8e8725cb47d9c42ccc8a2c1ebfb1b33a6e13cc9` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-DISTANCE-001-8c54ba35f01cd408.json`; empirical-validation hash `sha256:176841cb106bd80e489a81e90ec622f1ac4d0f3674cf35c4060be5083d7aea22`; measurement receipt `sha256:48cd4970fa0a4ba3f0d2adcf8e11aca3c702ec5ce21d07dff512d28009647667`.

294. Finite structure-growth relation

Claim identity: `SFT-PHYS-COSMO-STRUCTURE-GROWTH-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Structure growth is deterministic evolution of a finite source-density distinction network under locally forced interaction and expansion relations; weights describe unresolved initial/history fibres rather than ontic randomness.

> *Cosmic structure growth is finite deterministic Fold network evolution under registered gravitational and expansion relations.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`
- `SFT-PHYS-GRAVITY-WAVE-001`
- `SFT-PHYS-GRAVITY-HORIZON-001`
- `SFT-PHYS-COSMO-DISTANCE-001`

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-density-network-dynamical-growth__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`finite-density-network-dynamical-growth`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3e6861001d51e501bb3809f75fbec4a7cd36e767a7fa7eef73bd68c1812e3f3f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:4e6a66e9753cfc7df280705717c0fcf90fabcc3c968ab2d95191d72dba295567`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6294b6a7219d87acbd7ac9cd3fd06eeef95f57487d3408af2cafcad072092a45`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6533e6345df8c7b31418838b2fd1c63bfe822ddd7f37fb0307e841748f779a62`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7e741f8ba12c27a8e4a76258e9fb1f384f4c067b450ea022c48bad8b8d21bdeb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:eb896632cf27d04f3c63a7581367588cebcbf9d67de990119b9522392de880c2`. Independent certificate: `sha256:2e2581a18ed8db4826faf50042bf61a441363d25c9ba0c02ebf744d886b53ff4`. Engine external-validation hash: `sha256:2d9a545f449e77107216c437c9f42d56848318c86fb5e28f4530e0c6c0739438`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NASA-LAMBDA-2026-07-23

Observed comparison records:

- structure-growth-1: predicted finite-cosmic-structure-growth; observed finite-cosmic-structure-growth; exact match True
- structure-growth-2: predicted finite-cosmic-structure-growth; observed finite-cosmic-structure-growth; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:ae106ffd3d3065ef4946f880466fc7e43aa96a04209cce61b5e76eef473dc838`. Isolation certificate: `sha256:6543bf6b7a08b99dcce963996fc5a826d4bb470c2a7d78160946f24d5c9c7b8b`. Custody certificate: `sha256:e70c21b47356aa63855256eca5f65de81ddc81a8a322a3778cf6564c306d67d1`.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center: https://lambda.gsfc.nasa.gov/ (sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddedffe94e50b2d4275915)

Registered target identities:

- structure-growth-1 from NASA-LAMBDA-2026-07-23
- structure-growth-2 from NASA-LAMBDA-2026-07-23

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favourable observation or unrecorded source/path

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:63196aa699533d04e2f690d8b649069c7b9578830d0800e14a77c209d0e9d5be`; engine receipt `sha256:4b216f643356092cdc0906d603b2a3f726e42227e89d2783bd4504aa5d8763dd` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-STRUCTURE-GROWTH-001-4b216f643356092c.json`; empirical-validation hash `sha256:6e6b90f90753116ebfd720894398cfb893ab026c3b6f76e9a953a0a11a103145`; measurement receipt `sha256:ae106ffd3d3065ef4946f880466fc7e43aa96a04209cce61b5e76eef473dc838`.

295. Physics-to-astronomy/cosmology handoff boundary

Claim identity: `SFT-PHYS-COSMO-BRANCH-BOUNDARY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Physics closes the universal source, propagation, observation and interaction relations; object census, historical initial state and inferred cosmic chronology pass as explicit measured inputs to the later Astronomy/Cosmology branch.

> *The branch handoff preserves closed Physics laws and exports only source-bound finite observations for later astronomical reconstruction.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`
- `SFT-PHYS-SPACETIME-CLOCK-RATE-001`
- `SFT-PHYS-GRAVITY-CURVATURE-001`
- `SFT-PHYS-GRAVITY-WAVE-001`
- `SFT-PHYS-GRAVITY-HORIZON-001`
- `SFT-PHYS-COSMO-STRUCTURE-GROWTH-001`

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__law-versus-historical-state-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`law-versus-historical-state-boundary`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:1b4ee906fd8e39af5a7396cca3c92d365af4fa7b928dc17695694ba42fee0b0c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:43993065c8994e8e6caf618318ed6ae948b22df1bbd82a4678c2865d80ba89b6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9579c3b9ecba7097e4caae2712a7b0aeea983ae70b5d83fb8bb1b1a4290856a1`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9b6a0c84cbb54a1ac283a2806ecbf11c23551aa2ca7639fecb9fed1bdf4d349b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:fff8bff53425a262dc269b5a84aba203bc7d05d2b4d16d384c5a3e830d104e52`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e34d327c6aef4d6ee556576812a387fe8696bd13e948e144e11c271e682b73c3`. Independent certificate: `sha256:d3c90ff6e0e964eff72260f53e9e2ff541241215ecfb3f945e51bb9286b525e0`. Engine external-validation hash: `sha256:17b524c7708914115f8efccfd6b1e89a85d8c1008bdd1dc0b26c8ab125a36060`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NASA-LAMBDA-2026-07-23

Observed comparison records:

- branch-boundary-1: predicted physics-cosmology-handoff; observed physics-cosmology-handoff; exact match True
- branch-boundary-2: predicted physics-cosmology-handoff; observed physics-cosmology-handoff; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:a987f58380a86fc56df8b4d560562005584547b0b7b32692ccc9cdaad0834775`. Isolation certificate: `sha256:863acf77aed121a9467ff8fb7d8979db31de6d037f678d8fa1f8540e3a5279f3`. Custody certificate: `sha256:ad997a85533fa5984ab3e5d1b1dd10c601cf8db4ca4fc954381ee3f65dc38c88`.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center: https://lambda.gsfc.nasa.gov/ (sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddedffe94e50b2d4275915)

Registered target identities:

- branch-boundary-1 from NASA-LAMBDA-2026-07-23
- branch-boundary-2 from NASA-LAMBDA-2026-07-23

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favourable observation or unrecorded source/path

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:1a0342a4382ae7963c70d4d9781e4de7e26061fcf3f2410e02e4ba3223f6ecc6`; engine receipt `sha256:23d1f1fc5dfb84012ce63e1c1a63da2c73e3cea268aab818089d15eb42070b4c` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-BRANCH-BOUNDARY-001-23d1f1fc5dfb8401.json`; empirical-validation hash `sha256:66f42826b6d159c8060ef3679cd685a4fe93a226e5538827f032a86226013537`; measurement receipt `sha256:a987f58380a86fc56df8b4d560562005584547b0b7b32692ccc9cdaad0834775`.

296. Generation-cover dark-to-baryon fraction

Claim identity: `SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Generator three over the forced three-space supplies volume 27. Its least complete binary cover has depth five and support 32, forcing leading baryon/dark shares 5/32 and 27/32 and ratio 27/5. The unique period-five orbit floor 31 deepens the same depth once, forcing 279/52.

> *The leading dark-to-baryon ratio is 27/5 with shares 27/32 and 5/32; the unique native period-five deepening is exactly 279/52.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-FOUNDATION-COUNT-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`

Generated grammar and closure boundary. Generate the complete product of generation carrier, spatial volume, cover, partition, leading ratio, recurrence floor, deepening, measurement boundary, provenance and extension forms. The declared exact boundary is: All exact first-recursive dark/baryon partitions generated from the admitted generator-three spatial volume, its least binary cover and the first-return orbit native to that cover depth. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `one-complete-matter-cover__generator-over-forced-space__least-complete-binary-cover__depth-and-volume-over-cover__volume-over-depth__own-depth-first-return-floor__one-positive-orbit-part-appended__densities-inaccessible-until-seal__prior-question-machine-independent-reconstruction__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`one-complete-matter-cover`	`named-dark-and-baryon-scalars`	Names do not construct a common conserved support.
`volume`	`generator-over-forced-space`	`selected-cosmological-volume`	A selected volume reads the target.
`cover`	`least-complete-binary-cover`	`free-cover-depth`	A free depth is a fitted parameter.
`partition`	`depth-and-volume-over-cover`	`independent-normalizations`	Separate denominators need not conserve one matter whole.
`leading`	`volume-over-depth`	`measured-ratio-insertion`	A measured ratio is not a derivation.
`orbit`	`own-depth-first-return-floor`	`tower-or-up-depth-floor`	The tower is pre-periodic and the up-depth belongs to another carrier.

Axis	Forced form	Eliminated form	Elimination reason
deepening	one-positive-orbit-part-appended	selected-sign-or-bare-depth	A chosen orientation or bare repeat adds no forced recurrence record.
target	densities-inaccessible-until-seal	density-readable-construction	That would fit the ratio.
provenance	prior-question-machine-independent-reconstruction	old-answer-runtime-input	Earlier answer artifacts cannot select the V3 law.
extension	no-extra-rule	extra-density-or-fit-term	An added term is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:00937cfc05f8ce49be9b1ddfec0e666241d0cf9d8a082719dfb115745bc13399`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:13b87ee58afe7e2f83654afb81b6fcb56e86bd8cf9ef0bb0c5100b83cbfdb528`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:4655befa0bb2e2327eb6a52c79ee6bfaa11e74f1e3e90651a0efe99374e58591`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ea4e84735302478c251e5608af61e891a7ed5edf79829dada9455096541c7c1a`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:01bacc06ab1cdf3734a9b0d49205526e6878f557b380dee5fedccf17345520d1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:675d474c832285d80cb109d73394f0a5568cc55361437841ff048913967a99d1`. Independent certificate: `sha256:ed22d598f85cc461219ca381a91a8f098ee968271527e934a9d49eb066ecfbc0`. Engine external-validation hash: `sha256:93ab300854cbc1d02290b6b015586c989618ee5f3a8dda27b229b1d4de08cfe3`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PLANCK-2018-VI-ABSTRACT-DENSITIES

Observed comparison records:

- PLANCK-2018-BARYON-DENSITY: predicted sealed-leading-and-native-deepened-ratios-inside-complete-planck-density-interval; observed sealed-leading-and-native-deepened-ratios-inside-complete-planck-density-interval; exact match True
- PLANCK-2018-COLD-DARK-DENSITY: predicted sealed-leading-and-native-deepened-ratios-inside-complete-planck-density-interval; observed sealed-leading-and-native-deepened-ratios-inside-complete-planck-density-interval; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Either sealed ratio lies outside the complete density interval, either density row or uncertainty is omitted, source custody changes, or a tampered comparison is accepted.

Measurement receipt: `sha256:a0fd402bb0effe64765ad44c943854268bcb09537abc0e91d4acc3fe4c13aa97`. Isolation certificate:

`sha256:9b58bdce54fe4cc0f9949f5aaeacbc518bb1d7ea8d83b0127227d35fb5a6b75a`. Custody certificate: `sha256:b848576337389f7bb3a59c13d42730413bb3bebd8bf9ab12fccb2881088d0037`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 answer artifact or cosmological density in the executable derivation
- no numerical zero, negative, irrational, imaginary or floating proof value
- no fitted density, tolerance, selected floor or additional cosmological parameter
- no assertion that an external cosmological model is an SFT premise

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:9071bab4809a9bd764583e4da1dd2b644d5094018910663d51dc25f9635f9f1c`; engine receipt `sha256:d8a5088d10154314b6c018e2daf4ab64bf2c4c94e6f4ee77a814bbf56048f3f3` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001-d8a5088d10154314.json`; empirical-validation hash `sha256:bd2ae0ccaefb149e9437818bb2cd7b6fc2d1b1add265354ccda820c273261661`; measurement receipt `sha256:a0fd402bb0effe64765ad44c943854268bcb09537abc0e91d4acc3fe4c13aa97`.

297. Fold calibration ratio between early and late expansion routes

Claim identity: `SFT-PHYS-COSMO-HUBBLE-CALIBRATION-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. One retained formed class and the complete two-fibre open complement partition the generator-three cosmic carrier into exact matter and vacuum shares 1/3 and 2/3. Distributing the open share over the depth-three Fold support eight forces correction 1/12 and late/early calibration ratio 13/12. The independently forced depth-seven first-return floor 127 supplies the unique registered first-recursive deepening, giving exact ratio 3305/3048.

> *The exact matter/vacuum shares are 1/3 and 2/3; the leading correction and late/early ratio are 1/12 and 13/12; the unique registered depth-seven recurrence refinement is 3305/3048.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-FOUNDATION-ONE-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-COUNT-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`

Generated grammar and closure boundary. Generate the complete product of cosmic carrier, occupancy distinction, partition, calibration support, leading correction, deep cover, recurrence floor, recursive correction, target custody, provenance and extension forms. The declared exact boundary is: All exact leading and one-native-recurrence calibration ratios formed from the complete One, the Fold fibres, generator three, depth-three support and the independently forced depth-seven orbit. The generator produced `2048` named candidates and the decision support contains `2048` one-for-one decisions. Exactly one candidate survived: `one-complete-cosmic-carrier__one-formed-class-and-two-fibre-open-complement__one-third-formed-two-thirds-open__complete-depth-three-Fold-support__open-share-over-complete-support__independently-forced-up-cover-depth__depth-seven-first-return-floor__one-positive-orbit-part-inside-open-share__both-routes-inaccessible-until-seal__observational-reconstruction-with-independent-runtime__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	one-complete-cosmic-carrier	independent-density-scalars	Independent scalars do not close one conserved whole.
occupancy	one-formed-class-and-two-fibre-open-complement	named-matter-and-vacuum	Names do not force multiplicity or orientation.
partition	one-third-formed-two-thirds-open	selected-or-reversed-shares	Selection or reversal loses the formed/open orientation.
support	complete-depth-three-Fold-support	free-calibration-scale	A free scale is a fitted parameter.
leading	open-share-over-complete-support	measured-ratio-correction	A measured correction would select the answer.
depth	independently-forced-up-cover-depth	selected-cosmological-depth	A selected depth is a parameter.
orbit	depth-seven-first-return-floor	arbitrary-denominator	An arbitrary denominator fits a correction.
deepening	one-positive-orbit-part-inside-open-share	selected-sign-or-extra-term	A sign choice or external term is not forced.
target	both-routes-inaccessible-until-seal	hubble-values-readable-before-seal	That is fitting.
provenance	observational-reconstruction-with-independent-runtime	mislabel-as-unobserved-discovery	The prior result was already observed.
extension	no-extra-rule	extra-calibration-rule	An added rule is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:a8187c340c09417dac359fd9a0e1eca209db8f599683128d5558d1417078b02c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:e6a7e7f78a3e8655d01c36d33b35c823f5215d24eb4073dbf881de3c29eeba4c`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:793381e00887efeac6ded724dc0422e9ba3039650cf85e5f288d49c70cbf0480`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3d16267cb16626a1efecb424ffc8cdc455c3733937fcaa52e48dd53ea0dd9f70`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:6a6c3eef48a63aab3af0d8af496a5247c390b8c5dbaccee665af301eaf944782`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash:

`sha256:990b8774815230213ce4e33207c0362f06a6685321085ae27872cecc79e0708c`. Independent certificate: `sha256:b9a45aaa7f5eab1950f7a8df2fce83966bd9a39efa88de12a7439730d8f63e89`. Engine external-validation hash: `sha256:2826092c2e1792ea2152cb55f5b80d11367178584b2df1fcb3720cda1a9fa0f5`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PLANCK-2018-VI-HUBBLE
- SHOES-2022-BASELINE-HUBBLE

Observed comparison records:

- PLANCK-2018-HUBBLE-INTERVAL: predicted sealed-leading-and-depth-seven-ratios-inside-complete-planck-shoes-ratio-interval; observed sealed-leading-and-depth-seven-ratios-inside-complete-planck-shoes-ratio-interval; exact match True
- SHOES-2022-HUBBLE-INTERVAL: predicted sealed-leading-and-depth-seven-ratios-inside-complete-planck-shoes-ratio-interval; observed sealed-leading-and-depth-seven-ratios-inside-complete-planck-shoes-ratio-interval; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Either sealed exact ratio lies outside the complete outward-propagated late/early interval; either route, central value, uncertainty, unit or qualification is omitted; source custody changes; or a tampered comparison is accepted.

Measurement receipt: `sha256:fd88f265f3e8b7ebe77e7556a972cbc4c9b417883a7eac490edd3f4b5cf13f72`. Isolation certificate: `sha256:b659b92d75865983c13a078318f6fe42bc8c0d555eb940ae09314cdc6f9b00fa`. Custody certificate: `sha256:64e85a6e009e889b38edd07d1108c5c2ee6155d917ffc5a60d60990811802a68`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 answer artifact or Hubble measurement in executable forcing
- no numerical zero, negative, irrational, imaginary or floating proof value
- no fitted density, selected tolerance, calibration scale, depth, sign or extra term
- no claim that either external measurement route is itself an SFT premise
- no claim that the observational reconstruction was an unobserved blind discovery

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:b0b7c10ed6f813c7310824e9db25cfa199a1a313dab0f5cea56ef74b1da70d37`; engine receipt `sha256:b58587667e05026f07be77365d7b81e121a46114aedb0cfde51b566d1ebb365b` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-HUBBLE-CALIBRATION-001-b58587667e05026f.json`; empirical-validation hash `sha256:5f1990ed5244edf1dea48d96b49f4cb5d18328cba5dee219fd9c1acd585cb839`; measurement receipt `sha256:fd88f265f3e8b7ebe77e7556a972cbc4c9b417883a7eac490edd3f4b5cf13f72`.

298. Complete-partition spatial flatness and absent curvature remainder

Claim identity: `SFT-PHYS-COSMO-SPATIAL-FLATNESS-001`

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Every generated cosmic component is an exact positive part of one common total carrier. A complete component family exhausts that One, so a separate curvature remainder is the empty One form rather than a numerical zero. Every positive repartition preserves the same closure at arbitrary finite depth.

> *Every finite complete exact positive cosmic-component partition closes to the One; the curvature remainder is the empty One form, and every positive parent-preserving refinement retains that result.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-PART-EQUIVALENCE-001
- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-PHYS-COSMO-BRANCH-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete product of total carrier, component identity, normalization, completeness, remainder role, absence representation, repartition, temporal scope, target custody, provenance and extension forms. The declared exact boundary is: All finite exact positive cosmic-component partitions of one common total carrier and every finite positive refinement that preserves the parent part. The generator produced 2048 named candidates and the decision support contains 2048 one-for-one decisions. Exactly one candidate survived: one-common-total-carrier__exact-positive-held-part__common-One-normalization__every-generated-part-exhausts-One__unassigned-part-remainder__empty-One-form__positive-parent-preserving-split__all-finite-repartitions__curvature-record-inaccessible-until-seal__observational-reconstruction-with-independent-runtime__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	one-common-total-carrier	independent-component-totals	Independent totals need not form one conserved whole.
component	exact-positive-held-part	signed-or-unbound-density-number	An unbound number is not a held part.
normalization	common-One-normalization	separate-normalizations	Separate normalizations cannot close jointly.
completeness	every-generated-part-exhausts-One	selected-visible-components	A selected list can leave an unregistered remainder.
remainder	unassigned-part-remainder	extra-curvature-scalar	An extra scalar after exhaustion double-counts support.
absence	empty-One-form	numerical-zero-curvature	Numerical zero is not an SFT proof carrier.
refinement	positive-parent-preserving-split	epoch-specific-fraction-table	A table has no general preservation law.
scope	all-finite-repartitions	single-observed-epoch	One epoch cannot force persistence.
target	curvature-record-inaccessible-until-seal	curvature-record-readable-before-seal	That would import the desired result.
provenance	observational-reconstruction-with-independent-runtime	mislabel-as-unobserved-discovery	The prior flatness result was already observed.
extension	no-extra-rule	extra-cosmological-closure-rule	An added rule is a free premise.

Base and successor. The closure proof is sealed by derivation hash sha256:8adf40bc2691d39a9bf8429f87620624c105ead29fe71164476801c53274d739. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:75dad868142a0a34a404a85fbbc39d70a82b419d49390835a6254659f37d5131`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:60473ebc3e3cbb47602a0b5657066aaa39dcd3f4c0307d694ae87c0f8693aee9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e9c4b946bca30d903b8e277776feb03fdf005094151fae375d33420348370c40`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:9d589341c63a50dcfca228e1b99a4e320a5d1e2f33205b66b28254303d70cca0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:40818b5b12e926755cf41400949fcd6aa049e5438a291041e90fe079461f1366`. Independent certificate: `sha256:a18c1dd8c93e9f17769729de3035bef04e20cabf573ebc81956327fea57b5b22`. Engine external-validation hash: `sha256:73b1a66288cf6ab9d93889af64689263e89ba2ee95b4ba79c1bed3ede32c9e3b`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PLANCK-2018-VI-CURVATURE-47B

Observed comparison records:

- PLANCK-2018-BAO-CURVATURE-47B: predicted sealed-empty-curvature-remainder-inside-complete-planck-bao-signed-interval; observed sealed-empty-curvature-remainder-inside-complete-planck-bao-signed-interval; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The complete registered curvature interval excludes structural absence; its orientation, central magnitude, uncertainty, confidence region or data combination is omitted; source custody changes; or a tampered comparison is accepted.

Measurement receipt: `sha256:5a71aa469576894ab2ef534823d10539274b39608b1bf2db7ec91bec96790208`. Isolation certificate: `sha256:b148715a75aa6ab39e3d1ce1352a109a07fcbfbb9804f04ffd290f0d5dc20ac9`. Custody certificate: `sha256:36509a6514639fa3f2476ddcebcbddbe35f88a28e72545fa839c49e8e546d764`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured Omega_K value or standard cosmological field equation in executable forcing
- no numerical zero, negative, irrational, imaginary or floating proof value
- no curvature remainder treated as a signed proof scalar

- no selected epoch, incomplete component list or extra closure rule
- no claim that Planck parameter inference is an SFT premise

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:a470e426ab1c63abab074b7acad016b608612c5b0085eb3902c7bee39e2f1e37`; engine receipt `sha256:f0302179aa36d55df5b54595b6ece20566ec9c5e7a3f00f9e302ef705f5f504d` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-SPATIAL-FLATNESS-001-f0302179aa36d55d.json`; empirical-validation hash `sha256:dfc659ca193243af87f989371a4f446fb5a8c8c6c4bb68821184726dfec8f028`; measurement receipt `sha256:5a71aa469576894ab2ef534823d10539274b39608b1bf2db7ec91bec96790208`.

299. Depth-five Fold cosmic energy budget

Claim identity: `SFT-PHYS-COSMO-COMPLETE-BUDGET-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The least binary support covering the generator-three spatial volume has depth five and 32 states. The complete Fold boundary pair pins two states at each depth, forcing ten pinned matter states and twenty-two free vacuum states: matter 5/16 and vacuum 11/16. The independently reconstructed five-to-twenty-seven matter partition then forces baryon 25/512 and cold dark 135/512.

> *The refined present budget is vacuum 11/16, total matter 5/16, baryon 25/512 and cold dark 135/512; baryon plus cold dark equals matter and vacuum plus matter equals the One.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001`
- `SFT-PHYS-COSMO-SPATIAL-FLATNESS-001`
- `SFT-FOUNDATION-COUNT-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar and closure boundary. Generate the complete product of support, boundary incidence, pinning, free complement, global partition, internal matter partition, composition, closure, target custody, provenance and extension forms. The declared exact boundary is: All exact present-budget partitions formed from the least depth-five binary cover, one complete boundary pair per depth and the independently admitted five-to-twenty-seven matter partition. The generator produced `2048` named candidates and the decision support contains `2048` one-for-one decisions. Exactly one candidate survived: `least-generator-volume-binary-cover__complete-two-state-boundary-pair__all-depth-boundary-pairs__complete-unpinned-complement__five-sixteenths-and-eleven-sixteenths__admitted-five-to-twenty-seven-partition__global-matter-times-internal-shares__exact-four-part-One-closure__all-planck-rows-inaccessible-until-seal__disclosed-successor-observational-reconstruction__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
support	least-generator-volume-binary-cover	selected-density-denominator	A selected denominator fits observations.
boundary	complete-two-state-boundary-pair	free-pinned-count	A free count is a parameter.
pinning	all-depth-boundary-pairs	named-matter-share	A name does not construct a share.
free	complete-unpinned-complement	selected-vacuum-share	A selected share reads the target.
global	five-sixteenths-and-eleven-sixteenths	leading-one-third-two-thirds-only	The earlier leading partition ignores the admitted depth-five boundary discriminator.
internal	admitted-five-to-twenty-seven-partition	new-baryon-dark-parameter	A new ratio is free.
composition	global-matter-times-internal-shares	independent-component-normalizations	Independent denominators need not close.
closure	exact-four-part-One-closure	separate-rounded-total	Rounded values cannot prove closure.
target	all-planck-rows-inaccessible-until-seal	planck-budget-readable-before-seal	That would fit four targets.
provenance	disclosed-successor-observational-reconstruction	erase-leading-law-or-mislabel-discovery	The leading law and later observed refinement must both remain visible.
extension	no-extra-rule	extra-budget-rule	An added rule is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:fb0ca90bec0ed1065a6ff037d391fcc1b91045e609991ffefa3da0129d59f919`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:e81d1523b010a9f8a9afdc38fa6d75d1b6a71a2f668fd5e458e75158a2885e56`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:4c9cf1328e2b99e37fd5530c3176c7bf5a2c22cd5ba908aa7335770f36029259`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:485b256c1b7845b037ee4d3664b2f5f8887919e64ff95e89fd02de409e9f8ff9`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:c3dc60a5559ebdbb06247db197bb11549765fa5140920078750c6578ae3701fb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a88c40db509140589a9cd7606cef716d5bcbda246a38e268f57b06cd4c6ed419`. Independent certificate: `sha256:146f3c550bedff74d628ff8e3feafee3a5dd7c1264e491ca91efceda1f31a1bd`. Engine external-validation hash: `sha256:5304204042761661cce089b78e4a719d2533a0ea8d613419a6b64f3a8ac9ccbf`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PLANCK-2018-BAO-BASELINE-BUDGET

Observed comparison records:

- PLANCK-BAO-VACUUM-FRACTION: predicted sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; observed sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; exact match True
- PLANCK-BAO-MATTER-FRACTION: predicted sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; observed sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; exact match True
- PLANCK-BAO-BARYON-FRACTION: predicted sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; observed sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; exact match True
- PLANCK-BAO-COLD-DARK-FRACTION: predicted sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; observed sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Any refined component lies outside its complete interval; any earlier leading component is falsely labelled as passing; any required source row or uncertainty is omitted; custody changes; or a tampered comparison is accepted.

Measurement receipt: `sha256:5d9c0ed1e48fcc6c1a82abb4e5a25830968e7a2cee49bc7cbd359b7d80023ef7`. Isolation certificate: `sha256:c23bae73b804aba9135d992b6498aabed2960d114751a6a81f5dfb225f8c5f5c`. Custody certificate: `sha256:479b0be1093254e86ed4605ff7e3d741bbd4124e7cd156f177ba062cbf2ee38a`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no Planck value or V1/V2 answer artifact in executable forcing
- no numerical zero, negative, irrational, imaginary or floating proof value
- no fitted density, selected tolerance, free pinned count or extra component rule
- no erasure of the earlier 1/3 and 2/3 leading approximation
- no claim that the external base-Lambda-CDM parameter analysis is an SFT premise

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d9039bd7e94afa2586b76d24fd94e5b77c581f2751b9b066722899b15ea2248d`; engine receipt `sha256:616ac87b6340810e8dee1c13237b4de2569b7ee827ec9863e982b03ccbef8507` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-COMPLETE-BUDGET-001-616ac87b6340810e.json`; empirical-validation hash `sha256:753b3e09330fdcb1559b455879eb2ff587548e2a1a617c986d9de5e64d165bc7`; measurement receipt `sha256:5d9c0ed1e48fcc6c1a82abb4e5a25830968e7a2cee49bc7cbd359b7d80023ef7`.

300. Composite Fold-orbit topology and inter-universe transport boundary

Claim identity: `SFT-PHYS-FOLD-UNIVERSE-TRANSPORT-TERMINAL-024`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Coprime odd-denominator Fold components have an exact bijective composite map that commutes with Fold and has least-common-multiple recurrence, forcing joint lockstep correlation; native Fold dynamics preserves each denominator, one component cannot steer another, and the literal physical inter-component signalling/travel record is empty.

Coprime odd-denominator Fold components have an exact bijective composite map that commutes with Fold and has least-common-multiple recurrence, forcing joint lockstep correlation; native Fold dynamics preserves each denominator, one component cannot steer another, and the literal physical inter-component signalling/travel record is empty.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-DYNAMICS-001`
- `SFT-FOUNDATION-PART-EQUIVALENCE-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-ALGEBRA-001`
- `SFT-MATH-ORBIT-NUMBER-THEORY-002`
- `SFT-PHYS-QUANTUM-ENTANGLEMENT-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`
- `SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001`
- `SFT-PHYS-SPACETIME-CAUSAL-ORDER-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete product of orbit identity, denominator dynamics, composite topology, component map, period law, correlation record, signal test, travel test, physical interpretation and extension forms. The declared exact boundary is: Every positive reduced odd-denominator Fold orbit; every coprime pair and its complete composite unit-residue product; every positive finite Fold depth; and the admitted locality, causal-order, entanglement and no-signalling boundaries. The generator produced `1024` named candidates and the decision support contains `1024` one-for-one decisions. Exactly one candidate survived: `exact-odd-denominator-orbit-component__denominator-preserved-at-every-step__complete-coprime-composite-product__bijective-commuting-component-map__least-common-multiple-period__joint-lockstep-support-correlation__target-trajectory-independent-of-source__empty-cross-component-transition-record__arithmetic-correspondence-with-causal-boundary__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`orbit`	`exact-odd-denominator-orbit-component`	`unbounded-cosmological-world`	A cosmological world is not generated by denominator arithmetic.
`dynamics`	`denominator-preserved-at-every-step`	`selected-denominator-change`	No Fold transition generates a new denominator.
`topology`	`complete-coprime-composite-product`	`asserted-network-edge`	An asserted edge has no exact state map.
`mapping`	`bijective-commuting-component-map`	`lossy-or-selected-projection`	Selection or loss breaks exact component recovery.
`period`	`least-common-multiple-period`	`chosen-or-multiplied-period`	A chosen/product period ignores shared recurrence factors.
`correlation`	`joint-lockstep-support-correlation`	`independent-unrelated-trajectories`	That discards the shared composite origin.
`signal`	`target-trajectory-independent-of-source`	`correlation-relabeled-as-signal`	Correlation alone supplies no sender-selected change.
`travel`	`empty-cross-component-transition-record`	`description-map-relabeled-as-travel`	A preparation/projection map is not a native transition path.
`physical`	`arithmetic-correspondence-with-causal-boundary`	`literal-multiverse-transport-asserted`	Arithmetic components do not by themselves establish multiple physical cosmologies or traversal.
`extension`	`no-extra-rule`	`free-portal-or-time-channel`	A portal or temporal channel is an unforced extra law.

Base and successor. The closure proof is sealed by derivation hash `sha256:1b13d077a862deef71159fe0510fe85a55df8692127f59bae10afa4c58ec1612`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:42d217e1ec9558cef2ed9e8eae23b48fca3b93c20eb6f6105f3358d7453499da`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:84b93776a59aed706e545a356e4b0a22671d1c01b0a7f233e0d7cdf836981623`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:21843056311f55d0f352361d3b45405dfbcedd28f78885292121975ac0942a3d`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:bcdd2b1f4aa185c175a837aa171b4065c0c5c805f59c6b78f31904f8b349283d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f9152ab79a3d9b2dbf20612699ffa53bf0dd91a02cf8f94dacf517bb1c2df172`. Independent certificate: `sha256:c7aaba8e82e50a15947f40eaab615a661c035b5635fe1c62922855f44a103591`. Engine external-validation hash: `sha256:acbbb2768ee213ad6f25d2b834c6d534a16f448de4cb8dcc24b0a509b4cbbf42`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported cosmological multiverse, wormhole, closed-timelike-curve or portal premise
- no V1/V2 executable, answer table, physical target or stored survivor
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no relabelling of arithmetic composition/projection as literal physical travel
- no relabelling of deterministic correlation as directed communication
- no physical universe plurality without a separately admitted observational discriminator

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:11495f5fc8cfb40b8b30ef1b5ad2d6566c2aedf6acef48066fb6d1ca39703b92`; engine receipt `sha256:b03563b484c95269254dd2bb409ae42efbd2a71989eba3ac1b3dd8171c7e467d` at `receipts/engine/model_admitted/SFT-PHYS-FOLD-UNIVERSE-TRANSPORT-TERMINAL-024-b03563b484c95269.json`; empirical-validation hash `None`; measurement receipt `None`.

301. Terminal cosmic component transport and expansion law

Claim identity: `SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Matter, radiation and vacuum have respectively third-power, fourth-power and Fold-invariant transport. The terminal late-time law is E2(r)=(11+5r^3)/16; Omega_m(r)=5r^3/(11+5r^3); Omega_v(r)=11/(11+5r^3); matter-vacuum equality is r^3=11/5; acceleration onset is r^3=22/5; and the present accelerating magnitude is 17/32. The static-vacuum correspondence is tension of One magnitude. All quantities are exact positive whole/fractional Fold forms or typed empty/orientation structures.

> *Matter, radiation and vacuum have respectively third-power, fourth-power and Fold-invariant transport. The terminal late-time law is E2(r)=(11+5r^3)/16; Omega_m(r)=5r^3/(11+5r^3); Omega_v(r)=11/(11+5r^3); matter-vacuum equality is r^3=11/5; acceleration onset is r^3=22/5; and the present accelerating magnitude is 17/32. The static-vacuum correspondence is tension of One magnitude. All quantities are exact positive whole/fractional Fold forms or typed empty/orientation structures.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-COUNT-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-FOUNDATION-FOLD-DYNAMICS-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001`
- `SFT-PHYS-COSMO-REDSHIFT-001`
- `SFT-PHYS-COSMO-EXPANSION-001`
- `SFT-PHYS-COSMO-SPATIAL-FLATNESS-001`
- `SFT-PHYS-COSMO-COMPLETE-BUDGET-001`
- `SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`

Generated grammar and closure boundary. Generate the complete product of spatial carrier, matter transport, radiation transport, vacuum invariant, terminal expansion, density fractions, equality threshold, acceleration threshold, orientation typing, physical scale transport, target custody, prior-correction and extension forms. The declared exact boundary is: Every positive exact rational stretch; the three matter/radiation/vacuum transport classes; the admitted terminal 5/16 and 11/16 late-time shares; every exact derived fraction and cube threshold; all thirty-two registered cosmic-chronometer rows; the registered acceleration, equation-of-state, Planck-ratio and DESI adverse rows; and no imported continuum or signed proof scalar. The generator produced `4096` named candidates and the decision support contains `4096` one-for-one decisions. Exactly one candidate survived: `admitted-generator-three-volume__inverse-volume-third-power__volume-plus-one-recurrence-power__Fold-invariant-One__terminal-eleven-five-curve__component-over-complete-rate__exact-cube-eleven-fifths__exact-seventeen-thirty-seconds-and-twenty-two-fifths__positive-magnitude-with-held-orientation__post-seal-held-reference-transport__all-targets-open-only-after-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`space`	`admitted-generator-three-volume`	`borrowed-three-dimensional-volume`	Borrowing the exponent imports a premise.

Axis	Forced form	Eliminated form	Elimination reason
`matter`	`inverse-volume-third-power`	`selected-matter-power`	A selected exponent is a free law.
`radiation`	`volume-plus-one-recurrence-power`	`selected-radiation-power`	A selected fourth power is not derived.
`vacuum`	`Fold-invariant-One`	`selected-static-density`	A named cosmological constant cannot select invariance.
`expansion`	`terminal-eleven-five-curve`	`superseded-two-thirds-curve`	The leading budget is rejected by the admitted terminal discriminator.
`fractions`	`component-over-complete-rate`	`fitted-epoch-density`	An epoch fit is a parameter.
`equality`	`exact-cube-eleven-fifths`	`decimal-root-selected`	A decimal root may be target-selected and irrational.
`acceleration`	`exact-seventeen-thirty-seconds-and-twenty-two-fifths`	`old-half-magnitude-and-four-cube`	The old result inherits a superseded budget.
`orientation`	`positive-magnitude-with-held-orientation`	`negative-proof-scalars`	Negative scalars violate the Fold proof language.
`scale`	`post-seal-held-reference-transport`	`fitted-Hubble-normalization`	Fitting H0 to the chronometer rows would select the result.
`target`	`all-targets-open-only-after-seal`	`measurements-readable-before-seal`	Target access can manufacture the law.
`extension`	`no-extra-rule`	`free-component-correction`	An inserted coefficient is a fit.

Base and successor. The closure proof is sealed by derivation hash `sha256:8efa797d1e574e5c87696c84bbdd034a73bc59b71b68d3f60850c8c00172f2e0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:946f81f183439e0e95006a602f8bd8deac1eb8098846018d89fce73b82b02833`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:ce283a1afae5385be857dd9d15f628774982e125943bff0717b59b0b164d2b3a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:78728a9bbebf9ad4da51d496b0edaac4e14e3f739ad4c4d316fb5c9961d41648`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:af2dc89f3d4177fba3f98917394e50ada6e45233730681b1c90e7830eaef38b0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0368583227e567a96df47d2c3a39f3914cf6d3222bf26981b35db371739a33a9`. Independent certificate: `sha256:1c27cf58211873ad12e183709250ba9d3a9b35179287eabe4817ffc07624630c`. Engine external-validation hash: `sha256:1912e7d1290e48f17710a1214bb1b33ff3979ce747d8b85f9b9194589ec3db8f`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GOMEZ-VALENT-2023-CCH-32
- PLANCK-2018-BAO-BASELINE-BUDGET
- GOMEZ-VALENT-2019-ACCELERATION
- ESCAMILLA-2024-DARK-ENERGY-STATE

- DESI-DR2-2025-COSMOLOGY

Observed comparison records:

- CCH-H2-01: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-02: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-03: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-04: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-05: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-06: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-07: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-08: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-09: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-10: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-11: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-12: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-13: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-14: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-15: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-16: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-17: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-18: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-19: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-20: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-21: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True

- CCH-H2-22: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-23: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-24: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-25: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-26: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-27: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-28: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-29: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-30: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-31: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- CCH-H2-32: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- PLANCK-EQUALITY-CUBE: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- Q0-TYPED-MAGNITUDE: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- ACCELERATION-TRANSITION-CUBE: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- VACUUM-TENSION-MAGNITUDE: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- DESI-DYNAMIC-ADVERSE-ROW: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Any of the 32 CCH intervals fails the registered squared two-uncertainty comparison; either threshold cube lies outside its complete external interval; the q or tension magnitude lies outside its interval; any source hash, row, custody record or adverse DESI statement changes; or a tampered target is accepted.

Measurement receipt: `sha256:a33e0831e949329261b8cf0d72892053904c03dc38d4e21f086127b16196697a`. Isolation certificate: `sha256:47678f36ed211ed8e386676c03ea0525a0a85131a1e903322e8167783559f414`. Custody certificate: `sha256:f7088cb50ea54bd246562b80fced3192aa0c79f7ff1cea970b3650fc404efc15`.

Registered source descriptions:

- Cosmic chronometers to calibrate the ladders and measure the curvature of the Universe. A model-independent study: https://academic.oup.com/mnras/article/523/3/3406/7188316 (sha256:cbd38055c3c9b27b2cbe1fb7456c60dfd7df3eeb366182932b7899d25207b128)
- Planck 2018 plus BAO baseline parameter table: https://wiki.cosmos.esa.int/planck-legacy-archive/images/4/43/Baseline_params_table_2018_68pc_v2.pdf (sha256:a8525585688e7ba818f8650fc0f8b73a449823d5e199a8be36e37b8e73a0e612)
- Quantifying the evidence for the current speed-up of the Universe with low and intermediate-redshift data: https://arxiv.org/abs/1810.02278 (sha256:55100ed889f8aad5ada957f33759d954466b944d21b461b4c63c62f0b0fb45d0)
- The state of the dark energy equation of state circa 2023: https://arxiv.org/abs/2307.14802 (sha256:67c8d75e79fd85d727012e72e8838651de6720a2e8898e225e251601814c20a5)
- DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints: https://arxiv.org/abs/2503.14738 (sha256:1e82f26e4cc3901b16168cd147f252bfa804f9c3caad3f4f7e3532640d237841)

Registered target identities:

- CCH-H2-01 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-02 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-03 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-04 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-05 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-06 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-07 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-08 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-09 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-10 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-11 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-12 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-13 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-14 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-15 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-16 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-17 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-18 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-19 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-20 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-21 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-22 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-23 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-24 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-25 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-26 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-27 from GOMEZ-VALENT-2023-CCH-32

- CCH-H2-28 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-29 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-30 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-31 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-32 from GOMEZ-VALENT-2023-CCH-32
- PLANCK-EQUALITY-CUBE from PLANCK-2018-BAO-BASELINE-BUDGET
- Q0-TYPED-MAGNITUDE from GOMEZ-VALENT-2019-ACCELERATION
- ACCELERATION-TRANSITION-CUBE from GOMEZ-VALENT-2019-ACCELERATION
- VACUUM-TENSION-MAGNITUDE from ESCAMILLA-2024-DARK-ENERGY-STATE
- DESI-DYNAMIC-ADVERSE-ROW from DESI-DR2-2025-COSMOLOGY

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, old two-thirds/one-third survivor or external equation in forcing
- no numerical-nothing zero, negative, irrational, imaginary, floating, NaN or completed-infinity proof scalar
- no continuum scale factor, differential field equation, fitted Hubble value, fitted density or selected redshift
- no use of a measured q, w, transition, H row or DESI interpretation before the derivation seal
- no erasure of the superseded one-half/four-cube results or the current DESI dynamic-dark-energy tension
- no claim that the terminal late matter-vacuum slice supplies a measured present radiation normalization

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:e1004c307761e6e384dfebeb1cacc43157ac49b91d746bfe9cef867d59fadc1d`; engine receipt `sha256:504257d441fad45a8f173fad45015c7d723caa7636c517e71d27fddc43a2c42d` at `receipts/engine/model_admitted/SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032-504257d441fad45a.json`; empirical-validation hash `sha256:2356d7946091e0bf3fc556d6750ecf94f76435c1dedb3ef864edc81d59b87a9e`; measurement receipt `sha256:a33e0831e949329261b8cf0d72892053904c03dc38d4e21f086127b16196697a`.

302. Terminal inflation, primordial-support and structure-growth law

Claim identity: `SFT-PHYS-INFLATION-GROWTH-TERMINAL-039`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The unique generator-volume inflation cover is five Fold doublings with support 32. The complete primordial partition is scalar 31/32 and least tensor-pair 1/32. The structural exit is the first complete cover of volume 27. The least resolved perturbation grows exactly 1/4 to 1/2 to One in two Fold steps. For every exact positive later scale growth g, matter retention is 1/g^3, radiation retention is 1/g^4 and their relative growth is g. A conventional natural-log e-fold count is a comparison label, not an SFT proof scalar.

> *The unique generator-volume inflation cover is five Fold doublings with support 32. The complete primordial partition is scalar 31/32 and least tensor-pair 1/32. The structural exit is the first complete cover of volume 27. The least resolved perturbation grows exactly 1/4 to 1/2 to One in two Fold steps. For every exact positive later scale growth g, matter retention is 1/g^3, radiation retention is 1/g^4 and their relative growth is g. A conventional natural-log e-fold count is a comparison label, not an SFT proof scalar.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-ONE-001`
- `SFT-FOUNDATION-FOLD-001`

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-FOUNDATION-PART-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-GRAVITY-GRAVITON-POLARIZATION-003
- SFT-PHYS-COSMO-EXPANSION-001
- SFT-PHYS-COSMO-SPATIAL-FLATNESS-001
- SFT-PHYS-COSMO-STRUCTURE-GROWTH-001
- SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032

Generated grammar and closure boundary. Generate the complete product of spatial volume, binary cover, duration carrier, scalar support, tensor support, exit boundary, perturbation growth and component-transfer form. The declared exact boundary is: Generator-three volume in stable three-space; every positive binary support until the unique least cover; every exact complete doubling ratio; the complete thirty-two-record primordial partition; the quarter/half/One Fold growth trace; and every exact positive rational later scale-growth carrier under admitted third/fourth transport. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `generator-three-space-volume__least-binary-depth-five-cover__five-exact-doubling-transitions__one-boundary-record-short-of-complete__one-tensor-pair-record-in-thirty-two__first-complete-generator-volume-cover__quarter-half-One-two-step-growth__third-versus-fourth-power-relative-growth`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
volume	generator-three-space-volume	named-observable-universe-size	A measured horizon cannot select the support.
cover	least-binary-depth-five-cover	selected-sixty-efold-story	A conventional model-dependent duration is not a generated Fold count.
duration	five-exact-doubling-transitions	irrational-logarithmic-efolds	A continuum logarithm is outside exact Fold arithmetic.
scalar	one-boundary-record-short-of-complete	fitted-spectral-index	A decimal index selected from CMB data is a fit.
tensor	one-tensor-pair-record-in-thirty-two	free-tensor-amplitude	A tunable tensor amplitude adds a parameter.
exit	first-complete-generator-volume-cover	chosen-inflaton-potential-exit	A chosen potential imports an independent model.
growth	quarter-half-One-two-step-growth	stochastic-or-decaying-seed	Ontic noise or decay is not forced by the Fold.
transfer	third-versus-fourth-power-relative-growth	fitted-transfer-function	A fitted spectrum is not a law.

Base and successor. The closure proof is sealed by derivation hash `sha256:038cb81edb90bea3adf8f8d01c98503c453223fc79676f8eac63aff1b3c9ee5f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:716a5ab4fb9d1de1475efcf5399bcd85b405d32abf2fc73e383c859b06ee8809`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:648551c6e3e7563a2f5261e04e0076c4c28d64b7a019bd14c1c847ac8854a32d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d454a7c0e4d9470179577a2c03aa3a0484980f8c4a5db65b88be3c391e1b7fc4`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:cb7f3f69be4ae22f13eb9fc0311effa52d977dbe2d6e25e9a504045d15533a4d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:75b56683b979a9f537c5a4675e5e83e6fe2a0caf4247b9e8799b8779fa082470`. Independent certificate: `sha256:b6fe904029ba98e5a266135a0550b50b142ec04df91beb8de888382c67442541`. Engine external-validation hash: `sha256:48aadd0587f83014412a792ae95266b11fd4c61d4cca3ae27464033b082f31ff`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no inflaton potential, slow-roll equation, conventional natural logarithm or selected sixty-e-fold premise
- no CMB spectral index, tensor bound, matter spectrum or measured horizon available to candidate selection
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof scalar
- no ontic stochastic seed, fitted transfer function, amplitude or free exit scale
- no claim that five Fold doublings equal sixty conventional e-folds

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:abf31c25a28d3d16ce81601347077dccdff7ec229f558a98ad6170602bdb32d5`; engine receipt `sha256:00aa52b47f8ebfe57fb5a90530a54614ba175d87f6708c69e454fcf9fd888aa7` at `receipts/engine/model_admitted/SFT-PHYS-INFLATION-GROWTH-TERMINAL-039-00aa52b47f8ebfe5.json`; empirical-validation hash `None`; measurement receipt `None`.

303. Terminal stellar hydrostatics, galactic response and tidal-lock law

Claim identity: `SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Hydrostatic balance is exactly (1/2,1/2). Radial pressure and gravity responses are 5/3 and 4/3, certified only through q^5 and q^4 perfect powers: every compression is outward restoring and every expansion inward restoring. The two terminal homologous luminosity exponents are three and four, forcing lifetime-fall magnitudes two and three. Circular balance is v^2=M(r)/r; flat v forces M(r) proportional to r and excludes a finite visible-only asymptote. The baryonic Tully-Fisher exponent is exactly four. Every finite rational isolated circular tidal mismatch exhausts to 1:1; eccentric or externally forced resonances remain an explicit separate boundary rather than a counterexample hidden from the census.

Hydrostatic balance is exactly (1/2,1/2). Radial pressure and gravity responses are 5/3 and 4/3, certified only through q^5 and q^4 perfect powers: every compression is outward restoring and every expansion inward restoring. The two terminal homologous luminosity exponents are three and four, forcing lifetime-fall magnitudes two and three. Circular balance is v^2=M(r)/r; flat v forces M(r) proportional to r and excludes a finite visible-only asymptote. The baryonic Tully-Fisher exponent is exactly four. Every finite rational isolated circular tidal mismatch exhausts to 1:1; eccentric or externally forced resonances remain an explicit separate boundary rather than a counterexample hidden from the census.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-ONE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-FIELD-INVERSE-SQUARE-001`
- `SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009`
- `SFT-PHYS-FLUID-PRESSURE-STRESS-001`
- `SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043`
- `SFT-PHYS-WAVE-RESONANCE-001`
- `SFT-PHYS-MECH-ANGULAR-MOTION-001`
- `SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061`
- `SFT-PHYS-COSMO-STRUCTURE-GROWTH-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`

Generated grammar and closure boundary. Generate the complete twelve-axis product of hydrostatic balance, radial response, stellar scaling, lifetime, circular balance, flat-curve support, dark carrier, Tully-Fisher exponent, resonance, tidal dissipation, measurement custody and extension forms. The declared exact boundary is: Exact positive normalized shares; three-space and two-fibre counts; exact q above the One; perfect-power response comparisons; homologous volume and radiative-recurrence stellar endpoints; circular inverse-square balance; finite enclosed-support and velocity ratios; finite rational spin/orbit cycle counts; inherited neutral relic and resonance laws; and no measurement before seal. The generator produced `4096` named candidates and the decision support contains `4096` one-for-one decisions. Exactly one candidate survived: `complete-self-antipodal-half-One__five-thirds-versus-four-thirds-perfect-power-test__three-space-and-volume-plus-recurrence__one-fuel-carrier-over-complete-luminosity__inverse-square-balance-v-squared-equals-M-over-r__enclosed-support-grows-linearly-with-radius__admitted-neutral-stable-relic-under-fixed-gravity__three-space-plus-one-orbital-recurrence__finite-low-denominator-common-refinement__finite-mismatch-exhaustion-to-one-to-one__all-targets-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`hydrostatic`	`complete-self-antipodal-half-One`	`selected-pressure-gravity-ratio`	A selected ratio is a parameter.
`radial`	`five-thirds-versus-four-thirds-perfect-power-test`	`asserted-self-correction`	An assertion does not compare response carriers.
`stellar`	`three-space-and-volume-plus-recurrence`	`fitted-single-power`	A fitted exponent is selected by stellar data.
`lifetime`	`one-fuel-carrier-over-complete-luminosity`	`independent-lifetime-fit`	An independent lifetime exponent adds a dial.

Axis	Forced form	Eliminated form	Elimination reason
circular	inverse-square-balance-v-squared-equals-M-over-r	imported-Kepler-answer	An inherited named answer is not a Fold derivation.
flat	enclosed-support-grows-linearly-with-radius	finite-visible-asymptote	A fixed mass makes M/r fall outward.
dark	admitted-neutral-stable-relic-under-fixed-gravity	modified-law-or-named-particle	A named alternative is not structurally selected.
tully	three-space-plus-one-orbital-recurrence	measured-or-fitted-slope	A slope selected from galaxies is a parameter.
resonance	finite-low-denominator-common-refinement	irrational-continuum-period	An irrational continuum period violates the generated finite recurrence grammar.
tidal	finite-mismatch-exhaustion-to-one-to-one	unbounded-or-selected-terminal-ratio	An unbounded path or selected ratio does not close.
measurement	all-targets-inaccessible-until-seal	target-readable-before-seal	That would fit the result.
extension	no-extra-rule	free-response-scale-or-correction	An added response forks the law.

Base and successor. The closure proof is sealed by derivation hash `sha256:8e8cf180731600a6dbe1310cb2d9a83ed0bf793c01f957c0884313ecc37e12b4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:152c1d169c9d2e59069fb8b6598ba79f5c79b2e519d16f6ce902b13da9fcf324`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:9e6f6491ddb20e53761897e5ef7bac5f4d7630c039a7428b3df40be1fa405274`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:111d6aef6b9dd85abd6b0d078cc84264abebb621f9fd85ed8a32dda84ecd76c9`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:22ef586869d3ff3a3e257e57eae785a3e2625dda7f5e808bb7e129b0f01f35f4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e4cc3ed8767fe932277c90468e576b14f06536c91f8aa3f31259c738975fa89d`. Independent certificate: `sha256:e1a5dced711c5b5a9425bd16975c0b9c082e6d74e77812bae1beeb5e795d6527`. Engine external-validation hash: `sha256:d78e63db5478ae7a4f1531ab83b8a78e0b89c20ec0161c8206c913c8474f2949`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, stellar catalogue, galaxy slope, lunar period, survivor identifier or measured target in formal execution
- no fitted exponent, mass-to-light ratio, halo profile, tidal quality factor, age, metallicity or normalization
- no universal claim that every stellar mass domain has one exponent and no concealment of separate microphysical regimes
- no universal 1:1 claim for eccentric or externally maintained spin-orbit resonances
- no negative, irrational, imaginary, floating, NaN, continuum or infinite proof scalar

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:4c30c363ac59c1fb301ea0197223c174dc7c25defaa5d15e5012177f0684cb2a`; engine receipt `sha256:24674dfa019c208012414ab3c587656ee404512f20d00ab44e26ee137be22ae2` at `receipts/engine/model_admitted/SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067-24674dfa019c2080.json`; empirical-validation hash `None`; measurement receipt `None`.

304. Terminal stellar nuclear chain, collapse and heavy-element law

Claim identity: `SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Charged access paths are exactly n(n+1) and strictly increase for every positive stage successor; finite stage supports are strictly nested. Exothermic ordinary fusion terminates at the independently forced global binding maximum (mass 62, charge 28, neutron count 34), beyond which its release is the empty One form. Retained inward gravity then forces the support-loss collapse channel, with compressed nuclear support and horizon closure as the complete endpoints. The distinct linked-fuel recurrence has no interior fixed point and ends only at finite fuel exhaustion, forcing thermonuclear unbinding. Neutral capture has the empty inter-boundary charge path; repeated capture followed by registered radioactive rebalance forces the heavy-element channel beyond the ordinary-fusion terminal. Dimensional ignition temperatures and observed events remain post-seal checks.

> *Charged access paths are exactly n(n+1) and strictly increase for every positive stage successor; finite stage supports are strictly nested. Exothermic ordinary fusion terminates at the independently forced global binding maximum (mass 62, charge 28, neutron count 34), beyond which its release is the empty One form. Retained inward gravity then forces the support-loss collapse channel, with compressed nuclear support and horizon closure as the complete endpoints. The distinct linked-fuel recurrence has no interior fixed point and ends only at finite fuel exhaustion, forcing thermonuclear unbinding. Neutral capture has the empty inter-boundary charge path; repeated capture followed by registered radioactive rebalance forces the heavy-element channel beyond the ordinary-fusion terminal. Dimensional ignition temperatures and observed events remain post-seal checks.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-ONE-001`
- `SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005`
- `SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005`
- `SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006`
- `SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005`
- `SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005`
- `SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067`
- `SFT-PHYS-GRAVITY-HORIZON-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete twelve-axis product of stage composition, charged boundary order, release direction, global binding terminal, nested support, fusion-support loss, collapse endpoint, thermonuclear recurrence, neutral capture, radioactive rebalance, target custody and extension forms. The declared exact boundary is: Every positive finite stage count and successor; complete charged boundary paths; the inherited all-mass binding maximum and tail induction; half-One hydrostatic shares; empty-One absence of release; both compressed support and horizon endpoint classes; every finite fuel recurrence; neutral incident capture followed by registered radioactive rebalance; and no stellar or supernova target before seal. The generator produced `4096` named candidates and the decision support contains `4096` one-for-one decisions. Exactly one candidate survived: `positive-Fold-stage-successor__strictly-growing-charged-boundary-paths__only-toward-higher-binding__unique-all-mass-62-28-binding-maximum__strictly-nested-access-support__empty-One-fusion-support-with-gravity-retained__compressed-nuclear-support-or-horizon__runaway-to-finite-fuel-exhaustion__neutral-capture-with-empty-charge-path__registered-radioactive-label-successor__all-targets-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
stage	positive-Fold-stage-successor	named-stage-list	A named observational list is not a derivation.
barrier	strictly-growing-charged-boundary-paths	one-universal-dimensional-temperature	One selected temperature erases reaction-specific structure.
release	only-toward-higher-binding	fusion-past-peak-by-assertion	Fusion past the maximum cannot release binding support.
terminal	unique-all-mass-62-28-binding-maximum	selected-iron-name	A conventional name cannot select the endpoint.
nesting	strictly-nested-access-support	unordered-mixed-shells	Higher access cannot occupy a larger lower-access support.
loss	empty-One-fusion-support-with-gravity-retained	hidden-residual-fusion-energy	An ungenerated residual is a free source.
collapse	compressed-nuclear-support-or-horizon	single-selected-remnant	Selecting one outcome omits the distinct support boundary.
thermonuclear	runaway-to-finite-fuel-exhaustion	chosen-partial-burn	A partial terminal is not fixed under positive feedback.
capture	neutral-capture-with-empty-charge-path	charged-fusion-beyond-maximum	That requires inward energy and a Coulomb path.
rebalance	registered-radioactive-label-successor	unrecorded-charge-change	An unrecorded change violates label conservation.
measurement	all-targets-inaccessible-until-seal	target-readable-before-seal	That would fit the channel census.
extension	no-extra-rule	free-channel-or-energy	An ungenerated term destroys zero-parameter closure.

Base and successor. The closure proof is sealed by derivation hash `sha256:3689e53b92ca88280d666ece8306d439f340fd0a80d9977b8c3e878ed6fad13d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:8165d365df02494a33ae9d5360d2c4b970d49ca9b5db7927e9308c441a34b9ad`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:928815f87df9944fe2d6a80762ef34a558d7b736f5d40d1c3f02bce9ac2c0ea2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7301e51c84488a2130717a8bdf54f192be47839475a07674e4996c6d431b7aa0`.

- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:a7a6dfc2cd363681c827ff106cc67147e6c7340aca3f0bbbe90a2a7bd78b2c54`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:874f1168f335a9a58e6c94231d156a36322ff6d890a353d74523c06027f90ee8`. Independent certificate: `sha256:852e960611e5afb743d00ddd6278d85ed07d7df951e6b4e3f0353b442169ec87`. Engine external-validation hash: `sha256:19dc8622b1cea19c71e340328c0b4e55c8bc208a3a331d22389f9540ebd206b9`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S025` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, named stellar stage, ignition temperature, supernova event, abundance, isotope target or survivor identifier in formal execution
- no fitted reaction rate, stellar mass, opacity, metallicity, explosion energy, neutrino efficiency or standard-candle calibration
- no universal dimensional threshold substituted for reaction-specific access carriers
- no fusion release asserted beyond the globally sealed binding maximum
- no unrecorded charge change and no omission of either collapse endpoint or either explosion channel
- no negative, irrational, imaginary, floating, NaN, continuum or infinite Fold proof scalar

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:498653ec652f93ab85ca1466d0954e8fd3a0eacd4996a0d4d5eb4179b66b611b`; engine receipt `sha256:a41b28a935f54de5daabdbe77b6c1aba9286f174526b8feaa60829ec848e08af` at `receipts/engine/model_admitted/SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069-a41b28a935f54de5.json`; empirical-validation hash `None`; measurement receipt `None`.

24. Post Seal Empirical Validation

These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims.

305. Exact molar Planck measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-METROLOGY-MOLAR-PLANCK-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The molar Planck carrier is predicted by exact composition of the Planck carrier with the counted molar carrier.

> *Planck constant composed with Avogadro count predicts the CODATA molar Planck interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`

- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MEAS-UNIT-COMPARISON-001

Shared claim-record clause PHYS-S026 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-positive-interval-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:d71dc20cb4b55ac86cf64e36703439a6a4ec9fb52d2c7aa9ec70456e01c45c5e. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:8dba5666f65856a7a2f6d46a59317ff158e90067b1d68739639625fbe1ff4c48.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b50f43ff8ea8b8b28920191600b58f1084d395bbb64a7c281b26769d5fd5e8cb.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:01234e1a45ed97261d6e6aca6fa6a09c82d85cc82a44e3c8668947aff0f6ed5e.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:9bd844cf923a2862db055f9e90a89436b1ceb8493ccc58ea2b1a9c50e1c5b732.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:5c176bbcc9c5e3304bf168c8624612b897b8c0e0d982f5c4d493520690bc404e. Independent certificate: sha256:7e2cc5601355e962f71607ad460a5e96a5cd116059af12db77640845bd7d3877. Engine external-validation hash: sha256:a925918b091110cb58f01a391a42961a3ec8e81ea8f5cb139ee60cee9ae10a26.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- molar Planck constant: exact predicted interval [19951563564467157/500000000000000000000000000, 19951563564467157/500000000000000000000000000]; external interval [498789089/12500000000000000000, 3990312713/100000000000000000000]; overlap True
- deliberately displaced unfavourable interval rejected

Shared claim-record clause `PHYS-S027` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt:
`sha256:57dffc3d8d1d54c8c9b87040af676d5b017c4e70a4d5db643b8df499a492b06e`. Isolation certificate: `sha256:5bf1cb356f73b26a7fb20b1cf7c6f7a81916006d76c062b6a22118634a13cb35`. Custody certificate: `sha256:419425274ecffb691716d54bc8301ce5c3ffa3df3a89aeb222378c6c02130a92`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- metrology-molar-planck-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:879b6e34c81488195ba96103500294107baac03e2942e9e59e7487255e3959a3`; engine receipt `sha256:da7b1c59b205af44cd642a3e6c357ddf45fccec0684e88493a7c6e659c8f3b9f` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-METROLOGY-MOLAR-PLANCK-001-da7b1c59b205af44.json`; empirical-validation hash `sha256:e14c0b22c133517e2470d5c0b6abca223f5fdb7c7948f8c96e700a1db1bdb086`; measurement receipt `sha256:57dffc3d8d1d54c8c9b87040af676d5b017c4e70a4d5db643b8df499a492b06e`.

306. Exact momentum measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-MECHANICS-MOMENTUM-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The atomic momentum carrier is predicted by exact composition of inertial content with propagation speed.

> *Atomic mass composed with atomic velocity predicts the CODATA atomic momentum interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MECH-MOMENTUM-001`

Shared claim-record clause `PHYS-S026` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-positive-interval-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4528dc68c3a1c39c28339e02e15ce0ba10fe8886bb26725ecec2d3905e8b8d73`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:86f5eb2fbf987c6a15c200773f00f2697c12ae8673795cbcfa43408304a1c8e1`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:32ddd7bd51c73ff395027bcbf5673d17534a4a83309101de6f9eb8697a2ee79b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e52819448d4eb62b03f6356b792ad24afbb305bbf741b588bfb21c460591077a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e93bf332fc53a176bfd807a7582b045628da9fbff31feaee0de3950334533b94`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:96232cb6e8a85b63a038a0dbb0c587d0c8a77c3dfc0570cbbfbab64dca1a6d89`. Independent certificate: `sha256:3e5bba783e37e7e0781b1b92e395e66f8d8791ca74db8886c7d0457e22cac2cd`. Engine external-validation hash: `sha256:2ab67402da38b681a40b60e98e5903441cb9486574903a311b071c9e75b49331`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of momentum: exact predicted interval [9964259572669456670101/5000000000000000000000000000000000000000000000, 15942815331027492267/8000000000000000000000000000000000000000000]; external interval [99642595757/50000000000000000000000000000000000, 24910648947/12500000000000000000000000000000000]; overlap True

- deliberately displaced unfavourable interval rejected

Shared claim-record clause `PHYS-S027` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:ad8649e589af8398bcc3d29744f98566448dcddb86a4a6defe396554697ea769`. Isolation certificate: `sha256:40b2f43d9241d81b0986bfddff6175c64e70bdd605641fe6a059cf7389162688`. Custody certificate: `sha256:533536831e0995d4d3119db8c77a38fafa4a2265d6886ee1e4b35aa683bd7c2f`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- mechanics-momentum-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:cf5101dd2227ef92f30cc59f14b0c11ef10a8d381cf0f238d76a1ac04b3e76b2`; engine receipt `sha256:6712e0c2581044040f8eaac911bd382d500c1a2d839f2b5c78b1e409436ba5e4` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-MECHANICS-MOMENTUM-001-6712e0c258104404.json`; empirical-validation hash `sha256:104a8f65f01db73c7611ef7b22f816bb842cb4e301f98f0de3259cd83d6235ff`; measurement receipt `sha256:ad8649e589af8398bcc3d29744f98566448dcddb86a4a6defe396554697ea769`.

307. Exact force measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-MECHANICS-FORCE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The atomic force carrier is predicted by exact work-energy transfer per spatial support interval.

> *Atomic energy divided by atomic length predicts the CODATA atomic-force interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MECH-FORCE-001`

Shared claim-record clause `PHYS-S026` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-positive-interval-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b8e2d190e4b58c1f7ac66ca1ad951d6b5c87ce351b8c95669409e7fbc1415f3f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7d0e248a615bf7fa85e0c4e63bb059d811484b30daa1e6562655d5a2d9e50005`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e05966041153e32d868b967f176746cd8bbf7ade55fadf948076da6c65fcc6a7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:154a326728312835d420ee7a8f72418b24f9eac1b385b3fbe5bcd514ea6f13cd`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dfd96e21b8455542399ab2e45dac987d56016ced10065bbd4a854c2cb45e6164`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:305c4b0bfef2f469c7b94f691536ca5c5bc9ef5e4d2b21edc8afe150c342f609`. Independent certificate: `sha256:6accffe40e2a38bcd1e59f58794b8201a2756f1d626bd4e6bed722d321b4ead2`. Engine external-validation hash: `sha256:5e37bb95e2bce99367fe31657edd83bcbd83da62d9c0230ee6e673de6b1a4911`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of force: exact predicted interval [10899361805503/132294302656500000000, 10899361805527/132294302615500000000]; external interval [3295489401/40000000000000000, 82387235051/1000000000000000000]; overlap True
- deliberately displaced unfavourable interval rejected

Shared claim-record clause `PHYS-S027` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:be6de102f4119b3168cacfb79e0ae2cd5109dec887963aaf7998d90a6b81bfdc`. Isolation certificate:

`sha256:bfeae631de03df3a9121bc6a24c18171d7dffbd40c32fc1a86da4810818377db`. Custody certificate: `sha256:2c6d653e7b677979bea6dbb9b599fc161763bd3106a050d77e71000429f72277`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- mechanics-force-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d665cef20bf52a8696fdf72b9da2968d5352e148b02cddbe96543cdfa9e76c27`; engine receipt `sha256:d1a6cb9ef380156b36f27835ef0c0f23424005c6dbdb19962fc8d0e5bd2deef3` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-MECHANICS-FORCE-001-d1a6cb9ef380156b.json`; empirical-validation hash `sha256:34c64f9407aa0a15e102c779d581b02caa9f7571e64ae802cfcf1634e14ef501`; measurement receipt `sha256:be6de102f4119b3168cacfb79e0ae2cd5109dec887963aaf7998d90a6b81bfdc`.

308. Exact electric-potential measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-FIELD-ELECTRIC-POTENTIAL-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The atomic electric-potential carrier is predicted by exact energy transfer per elementary held-charge carrier.

> *Atomic energy divided by elementary charge predicts the CODATA atomic electric-potential interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`

Shared claim-record clause `PHYS-S026` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-positive-interval-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:17555e58355774f3b823ed825bb0c93afb30e957bc7a07febb7fe07db6411cfd`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c5d5bc3b55e252b97377e395fd24d91276e5901912f446c273d050a5815e3452`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:60be210b04698f43634a547027bf2d8018948689ca5f24ffaf24138e79792bcf`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:465eda6c270bb44dc6c32e821e599098a9618ae570ddfffe2b3101603ed172f7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:86f4aea99ba7612cf9889c79870777cf6878cec798c0bb551416fdf9ed9a07bb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:eb299544fa0a2cb21cc9c01a748f9da3bcaeb47b0ea1b7ee954a9bbb3db44137`. Independent certificate: `sha256:10ed41b8443c1db52bc830ad1f60d83f261b7995f2ceb65587025a1056685460`. Engine external-validation hash: `sha256:c39ae7a1837468b1e4d9ee819542a5163a34c9f2bb437ac5751eea8736adfa1e`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of electric potential: exact predicted interval [10899361805503/400544158500, 10899361805527/400544158500]; external interval [27211386245951/1000000000000, 27211386246011/1000000000000]; overlap True
- deliberately displaced unfavourable interval rejected

Shared claim-record clause `PHYS-S027` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:e16c0a69688e47b96007e948c03992fd0d0191a6187cad13eea0ba53ada97f78`. Isolation certificate: `sha256:0eab9b93b87eaf53f533af6d9d53682136dcbffab839900260bc866d98db9808`. Custody certificate: `sha256:5742beff7dd81c544f73e648d0149030a62c4f5651725651143f07826e2974ac`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- field-electric-potential-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:a42969465a5d726aca2bdcaa29cf8ef62d03f3c7aa37025a94301156bca8c527`; engine receipt `sha256:a4c49655afebcc07b564e29e47e2248d648ef15b7a8868b7fbc96daee7cbb962` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-FIELD-ELECTRIC-POTENTIAL-001-a4c49655afebcc07.json`; empirical-validation hash `sha256:879228ec29d49aaf3c5b270f024724495ee1808adf8e6487ca1e5aa5db810844`; measurement receipt `sha256:e16c0a69688e47b96007e948c03992fd0d0191a6187cad13eea0ba53ada97f78`.

309. Exact electric-field measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-FIELD-ELECTRIC-STRENGTH-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The atomic electric-field response is predicted by exact potential change per atomic spatial support interval.

> *Atomic electric potential divided by atomic length predicts the CODATA atomic electric-field interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`

Shared claim-record clause `PHYS-S026` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-positive-interval-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:094e7a2918c70d99991e8794fb25bab2f99a0944591d7486c99e609b806a47a3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9935fe5faaef70de04a0fb1096d94b7102ff87ca5c74a8c907f9c2b86e268442`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c94feca21e55eed4366a19704f6e7d9d64ade592f25c5c765c4df02d3984a99b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e699961a2ae9dfb96d886fabba5d3137d5cb6ee283a558e308aaafd0dd70e3ca`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:450854a196136d47100d2bfc73504c78f85dde507440fc34cb3b9a950ac9b2bd`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f0f3fa02e9e303bb6a657ca72723370b1476bb9017f6bc0f60ef840ee05d0e8c`. Independent certificate: `sha256:b563e6fd9fedf5ba12fec207079a96b863ec69bc82d5b8f09497c229a7238ee8`. Engine external-validation hash: `sha256:5c1ad7fdddc03f79c17716c6cea3303934d7cbd59b8956d3c63d186b2480a1b5`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of electric field: exact predicted interval [136056931229755000000000/264588605313, 136056931230055000000000/264588605231]; external interval [514220675032/1, 514220675192/1]; overlap True
- deliberately displaced unfavourable interval rejected

Shared claim-record clause `PHYS-S027` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:41c4ea224b0c40d00d19bb79028d83d209e68b808fcacc8207e2765cb637f7f1`. Isolation certificate: `sha256:6567bb4f072be3188f4eaf6cfd40cd8efc71bb542c469d595e6a44fafe08dc0e`. Custody certificate: `sha256:4adb3b39b88c05eb747fdc1d6ae873dc741a13ab483ef4fa9e39c40d25ff74ab`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- field-electric-strength-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:b9e78bb06f283d94c0e1356d5f175b9d4f2b66482cf60853268940229744c0be`; engine receipt `sha256:7cbdd805803e22814f2ac485a7d14615c126ad9f32fd60663e5f0148fc0d3465` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-FIELD-ELECTRIC-STRENGTH-001-7cbdd805803e2281.json`; empirical-validation hash `sha256:769a41dcc7800a601b10bbc7878c3033a1aa3a3397795d46cd3a564f96776300`; measurement receipt `sha256:41c4ea224b0c40d00d19bb79028d83d209e68b808fcacc8207e2765cb637f7f1`.

310. Exact wave frequency measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-WAVE-FREQUENCY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A registered inverse-length recurrence transported at the limiting propagation speed predicts its recurrence frequency.

> *Rydberg inverse length composed with limiting speed predicts the CODATA Rydberg-frequency interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001`

Shared claim-record clause `PHYS-S026` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-positive-interval-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d144b754107dfae0ab4442d6e08f5fe373a915398572a6fcdb7033a4e0189a00`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d1b1b26ccb8e0f088ad709031fb472de420836598102c0ceec7d13bcb415e688`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:657967992b8f1c2e12d67611da1293b2fb5cde33f726e9b486a9eefa47701483`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:213e3480e3dad607392a3f96d7656c451e251f3e53fcb53a77e67ecea56d5db7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4a60e6dd8a7cffe9b795bdb5dd2806f1a7f7c5d5118a1cb55f26bc43e7cb7d29`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f12b815d77906521b2090244dd8dd4e3afd91e97d51cb9ccbdfb1e7962a32aab`. Independent certificate: `sha256:1c07dd896d1c7074c9cb1e7e3aab7471662f3829120d5536b0534ed959b72048`. Engine external-validation hash: `sha256:754899d3692e155649ca8702917591505f58bdbc7767f91517a62e77aa163e8c`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- Rydberg constant times c in Hz: exact predicted interval [328984196024638405041/100000, 1644920980126789534701/500000]; external interval [3289841960246400/1, 3289841960253600/1]; overlap True
- deliberately displaced unfavourable interval rejected

Shared claim-record clause `PHYS-S027` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:5f6551267787aa01f511dde26f378ce75f3bb283480948c3569aeab02bf32278`. Isolation certificate: `sha256:3738716deccd5f5591477379eb3f8427a4b5cdc7df85de49ef0a72868e51b483`. Custody certificate: `sha256:79faf0c1b912ed0090420a1d286d3f0a79f5fe27401b50db68169033d3d5ab35`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- wave-frequency-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values

- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:f6325612616628d8ec05710efa9c67abca0d20fba0abea8da77c941669193fc9`; engine receipt `sha256:647ababd101acf8a9e533bfd8d8acfe6e34fdfe0524bdf55b42cd846fca2ea6a` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-WAVE-FREQUENCY-001-647ababd101acf8a.json`; empirical-validation hash `sha256:5ed082f3321877266a9e9c89fb20ee4a9a407fb89e23ed74957f3ae26df808ab`; measurement receipt `sha256:5f6551267787aa01f511dde26f378ce75f3bb283480948c3569aeab02bf32278`.

311. Exact wave energy measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-WAVE-ENERGY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. A recurrence frequency composed with the Planck transfer carrier predicts its transported energy.

> *Rydberg inverse length, Planck carrier and limiting speed predict the CODATA Rydberg-energy interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-WAVE-ENERGY-MOMENTUM-001`

Shared claim-record clause `PHYS-S026` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-positive-interval-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:cfe6ed20142513582b1e17682b8e1542813c1197307c237822a0ae7ed90afd34`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:80fababc6aacb0f4cc9169e4d261edb055e531385429c8d6d0712e082e2b8c8f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:dcbdff65034657a9d951b337630fda7c01e70642ab9a1746257825b9f4b3fd42`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4d7cb5e64989a5c0d5d54a87e838a6d7717f0352e221a6ad80c6a7a5f9c963c8`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a5d04f8c4cd2f3b0152f661e985a1d9d0839aece8a9dc6f8319999702392e77a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:6dd06d5f612f4003d26dafa3da17e7fd27e58f1dba04cf89bd205dbf7a056e7e`. Independent certificate: `sha256:2623704f4a5b8842046e10ea7adc04f8be5fbd323819cd0bda228158f507b9ab`. Engine external-validation hash: `sha256:de5c74b49370c64679a106957bce2dfce19e57abdba728212023720e066d6399`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- Rydberg constant times hc in J: exact predicted interval [43597447222012104003715592523/20000000000000000000000000000000000000000000000, 217987236110537267024293705503/100000000000000000000000000000000000000000000000]; external interval [10899361805503/5000000000000000000000000000000, 10899361805527/5000000000000000000000000000000]; overlap True
- deliberately displaced unfavourable interval rejected

Shared claim-record clause `PHYS-S027` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:9227c49e00d7850387fbccc6dd6dd2723d3c5bf2cfc3056dc2bdc57d336ce2b0`. Isolation certificate: `sha256:172aa020966af54716e3a528465f8d8e02e90ff8e445b9e672b16a3f64f16d0c`. Custody certificate: `sha256:0c10b810ba323215950dbe0e0cba3c8c8c2ceab46d9a639d1280ed4f3ee7c7ce`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- wave-energy-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:350bd512a1f1eedee6b9a3ab78b7224d110349ff264d74a1f31a1bd1b24e62e0`; engine receipt `sha256:7398c538ac392b7387169fe5c92eedb34897b9342b5769f6f9fb1d7e8898f915` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-WAVE-ENERGY-001-7398c538ac392b73.json`; empirical-validation hash `sha256:46140e56cab446f1101cd300fb4c3dbe45a47b56ca99f71389398bd7b3d9b54d`; measurement receipt `sha256:9227c49e00d7850387fbccc6dd6dd2723d3c5bf2cfc3056dc2bdc57d336ce2b0`.

312. Exact molar thermal measured-value correspondence

Claim identity: `SFT-PHYS-VALIDATION-THERMO-MOLAR-ENERGY-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The molar thermal-energy carrier is predicted by exact composition of the per-constituent thermal carrier with the counted molar carrier.

> *Boltzmann carrier composed with Avogadro count predicts the CODATA molar gas-constant interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-THERMO-TEMPERATURE-001`

Shared claim-record clause `PHYS-S026` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`exact-positive-interval-relation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:899b2a73feda34234c02b7a77c50c68c9eedae010ca7fcb82888996e82220763`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

`sha256:2d76619ba4fffb95f44490faecc85c9503181235a3b97795c0cd00e91dba3d27`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:189f4aefc0e0d84013394e7c93774b570c9d30738b5bf0eca7850d92fa3f48c6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:18fc3e77d08c08e7911ab7bbecea77efb7cee6a6defc6de717db77287244a504`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1dd0e1d54a27ff5a0f5ebf3731e5842ce31f34eb8975dd1f742ee0d3862503da`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:7c74bfeab70b66c1a68ab28d93377be8592b949cde42658538a08e24dac8c6b5`. Independent certificate: `sha256:94c1f683599f41d967651a055f6c72c4d311a58710642443377dfa8144ce603d`. Engine external-validation hash: `sha256:9c65eac5a3392966f8966ce2b7a7458ea8a2b3a326bfdd68e0f5cbc1c1e68064`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- molar gas constant: exact predicted interval [207861565453831/25000000000000, 207861565453831/25000000000000]; external interval [4157231309/500000000, 8314462619/1000000000]; overlap True
- deliberately displaced unfavourable interval rejected

Shared claim-record clause `PHYS-S027` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Measurement receipt: `sha256:4a11e450a6ef060987c585328d944427535b55d05489ba85cfced48a7cc389b8`. Isolation certificate: `sha256:8cba1da130745fa6b1b01d33003212323735749d0578c746fd3e47d6629c95c1`. Custody certificate: `sha256:c73ae00a7e7ac1d018e630f16b0a00857adef092f46ee5dc52445760fda51d40`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- thermo-molar-energy-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:fc23276d5f98fbca64546ee6d71c58719e3217e481e5f2c6dccc9fedebf7e53d`; engine receipt `sha256:78920ce87ea1f42ad25bed4bb60a3421f288929a14bb97aa5f6eb23241c00349` at `receipts/eng` ine/model_admitted/SFT-PHYS-VALIDATION-THERMO-MOLAR-ENERGY-001-78920ce87ea1f42a.json; empirical-validation hash sha256:d2c8c17f9c2b361e6e75df25448ccbfe498b45b72554d1a00776ee6577a2e559; measurement receipt sha256:4a11e450a6ef060987c585328d944427535b55d05489ba85cfced48a7cc389b8.

313. Post-seal experimental agreement of the inverse-square exponent

Claim identity: SFT-PHYS-VALIDATION-INVERSE-SQUARE-001

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. After the Fold structure independently forces and seals positive integer exponent two, the Williams, Faller and Hill Coulomb-law experiment reports q = (2.7 +/- 3.1) x 10^-16 in the tested form 1/r^(2+q). Converting only the reported finite interval to exact positive rational exponent bounds and enumerating every positive integer inside them leaves two as the sole compatible exponent.

> *The complete reported experimental exponent interval contains exactly one positive integer exponent, two, agreeing with the independently sealed inverse-square law.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-MEAS-BOUNDARY-GROWTH-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete product of physical carrier, inverse-square relation, provenance, capability-closed prediction, separate measurement record, complete rows, positive-successor closure and no-extra-rule forms. The declared exact boundary is: All exact post-seal comparisons between the forced positive integer exponent and the complete finite positive exponent interval reported by the registered primary experiment. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__forced-exponent-two-versus-complete-reported-interval__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	forced-exponent-two-versus-complete-reported-interval	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2ba3cad3055c04f18e765fdb982ee818b418a339345d85a4ab6ee592e24e8c70`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:cf789a8189e966181ed9fce634a62aa9673b28a799764651de82d6ada2967d4f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:adbfc13814bcc801f5be5939b4b014fe3f5ac6c653f3a38c51101ced522375b7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ec4568e8deb5d9d95c85707880cc97bf58db129bade753295ec114c5bf92286f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d3667e2420aed299c2645bde52a7be6730ab235e2c2cf7f3ca342a447acdeec7`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8a4bedf7f45425c480eb95fc57a8224e59106a705245cb0442c2bd05fb683412`. Independent certificate: `sha256:937824d169404edcb5cdb810fdf82d65a2ce065a92fab16469eba917e2fdebde`. Engine external-validation hash: `sha256:0cf56dd06325c73eab0125f0a6d323062158c01ffde7d8c9009ab63ac5f3e99a`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- APS-WILLIAMS-FALLER-HILL-1971-COULOMB

Observed comparison records:

- APS-PRL-26-721-REPORTED-EXPONENT-INTERVAL: predicted reported-exponent-interval-has-sole-positive-integer-two; observed reported-exponent-interval-has-sole-positive-integer-two; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The complete registered reported interval does not contain exponent two, contains another positive integer exponent, differs from its source hash, is opened into the formal derivation, or a tampered comparison is accepted.

Measurement receipt: `sha256:ce1b0821186866e407419d06e9e49510b3b5dcc4c4cd39c545ef85410fec45b7`. Isolation certificate: `sha256:466b96f993119075da005de699a5614b1d1214c7f0f1bf93398613ecc898edd9`. Custody certificate: `sha256:5a2cc4390d7ba630b7c42ea2b388cd65c28523a95b96aa4d1466097838963cb3`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- measured result used by the formal inverse-square derivation
- logarithm, regression, floating fit or best-match selection in the checker
- semantic numerical zero, negative, irrational or imaginary proof value
- omission of the reported central value or uncertainty
- claim that interval agreement proves an exact physical value

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:30f8ed73b40f9c296d26cfd7775111ea30b6dbd07a93456a9620c9fa58961f01`; engine receipt `sha256:e70905b63729a3bf5619b26115f7fe8b78eb04aba53eba78f6697405d78a220b` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-INVERSE-SQUARE-001-e70905b63729a3bf.json`; empirical-validation hash `sha256:590cd21dab3dad6d12ef0d8030ae9d5c91bcf4f931afb318b024a7ad9aa5068b`; measurement receipt `sha256:ce1b0821186866e407419d06e9e49510b3b5dcc4c4cd39c545ef85410fec45b7`.

314. Post-seal CODATA check of the terminal inverse fine-structure ratio

Claim identity: `SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. After V3 independently seals the exact terminal inverse fine-structure ratio, the complete NIST CODATA 2022 central value and stated standard uncertainty are converted to exact rational interval endpoints. The sealed ratio is checked for interval membership without fitting or rewriting it.

> *The exact terminal ratio 503846395469/3676744786 lies inside the complete 2022 CODATA interval 137.035999177 +/- 0.000000021.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal relation, provenance, custody, row, exact-interval and no-extra-rule product. The declared exact boundary is: All exact post-seal comparisons between the immutable terminal ratio and the one complete registered CODATA inverse-alpha row including its uncertainty. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-terminal-ratio-versus-complete-codata-interval__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-terminal-ratio-versus-complete-codata-interval	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b7c0c90dc665b8ea2c4851a1b4b03f49a24824dfbf6fdbcd7dbf24b6e9d24d49`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2460ed9e9112ca2db7f5fec2ba9676993debc0b85cb8d1040342b8a51dd34e68`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:20a184a1749db08e208fa09d5609987741171219db6cfdb7e3c1db52cbf51806`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3921cc46ead2ad22ee5ae1bb3ad7f8b9184ac08981590c2f580ab371eda892a3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9d42267b960bf9dfdd82343e855943f7fd5e75c3f418041c4cee03165d0ad02f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:31aaffa05397abd35a390640284e0b7e2c2bd7c1793d599a60aadc3916ff1695`. Independent certificate: `sha256:4b54876e16a0163c485b0fe3c7b79c5c12fd3c53c49dd90bdd7fa11ed7248116`. Engine external-validation hash: `sha256:c7c4ea246e68c5499740d43f9f5a9664281e668ec04fd3a9faaf54dc36486379`.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- NIST-CODATA-2022-INVERSE-ALPHA: predicted sealed-terminal-inverse-alpha-inside-complete-codata-2022-interval; observed sealed-terminal-inverse-alpha-inside-complete-codata-2022-interval; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The sealed ratio lies outside the complete reported interval, the source row/hash changes, the uncertainty is omitted, or a tampered comparison is accepted.

Measurement receipt:
`sha256:174e9648fe675a9e9aaf93e2819fae94768b2cab092661242b6f37eed5ff4115`. Isolation certificate:
`sha256:33953e6f606dfeaadb101604ed560ae3f7271e380ccd50a37cd6244b900c0cdc`. Custody certificate:
`sha256:a5792b967d652068b45a24ab4cf0f36ba9edf668702ad33a906104abe6d6c706`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- NIST-CODATA-2022-INVERSE-ALPHA from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no CODATA value accessible to the formal derivation
- no fitted digit or best-match selection
- no floating-point comparison
- no omitted uncertainty or unfavourable control

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:67d0db973ac152718d55f9496cf2053fccaa52474a82aaf98bdfce3a7c05a7e8`; engine receipt
`sha256:13cf55d69e81157f6a00ec4968703133953f817c337147c7d3f160a86d52210a` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001-13cf55d69e81157f.json`; empirical-validation hash
`sha256:e595221262923b47ead9892b90ba21d0ab8f57cbf5bacea6a2aad1f3054ff528`; measurement receipt `sha256:174e9648fe675a9e9aaf93e2819fae94768b2cab092661242b6f37eed5ff4115`.

315. Post-seal IAEA check of the nuclear-closure sequence

Claim identity: `SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. After V3 independently seals the positive-rank nuclear recurrence, the complete first eight outputs are compared in order with the registered IAEA Nuclear Data Section magic-number record.

> *The sealed V3 closure prefix (2, 8, 20, 28, 50, 82, 126, 184) exactly matches the complete registered IAEA sequence.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal ordered-sequence, provenance, custody, row-retention and no-extra-rule product. The declared exact boundary is: All exact order-preserving post-seal comparisons between the immutable first-eight closure output and the complete registered IAEA positive magic-number sequence. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-closure-prefix-versus-complete-iaea-sequence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-closure-prefix-versus-complete-iaea-sequence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:48803409d8b03f6a27de4c607b0e9e56240b79755487e59e1623b53ca3f2325c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6f99243f846465531b8e8d04eafd231fa57515365c2ce894180820a930c4f6be`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:856dc18ebff37c0d3ec1fe6a114c55e16f6bd261778ca0abd9f5830bf7c82bb8`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:77a9bb7213b8ff916417351bf0dada8cf20622b8fef94e29c8a977e5c2c2eb9e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:745f64af58a7835963b0d86ea1dd07931f1d68b4035bc3d091798fd4b90ac198`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:305b02c334ed9f2efcdc389e5a3601348dabb2f28d1b4b9e9a0c91f36ba7716b`. Independent certificate: `sha256:0c4548de901d5c3e1ab8a7be0eb1ce51e3801104365a796d733b907dd9ca9642`. Engine external-validation hash: `sha256:1b44ec6e3f06a5aa46ddacffe0c1bbc3cccf2d8f6cbf8793fedd0d8574dddad4`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- IAEA-INDC-NDS-0452-MAGIC-NUMBERS

Observed comparison records:

- IAEA-INDC-NDS-0452-MAGIC-SEQUENCE: predicted sealed-eight-closure-prefix-exactly-matches-iaea-reported-sequence; observed sealed-eight-closure-prefix-exactly-matches-iaea-reported-sequence; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Any sealed/source coordinate differs, either ordered sequence is incomplete, the source identity/hash changes, or a tampered comparison is accepted.

Measurement receipt: `sha256:9d6a3776768080863fc58346713e458f0bcce719fb4ab5c4e6e1b0b2e10d28a6`. Isolation certificate: `sha256:58948401bd9ab2d041e405b3a3977f84159c75d029efbdf844da55304230f468`. Custody certificate: `sha256:23a1397345696cf50a730a232db13f6c2821c918b5c02d1cb136cb23590eec79`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://www-nds.iaea.org/publications/indc/indc-nds-0452-part1.pdf (sha256:ce642d62698b97fa509b8101a5661e578d6524f8705c19e38f7426efe7a0f6a6)

Registered target identities:

- IAEA-INDC-NDS-0452-MAGIC-SEQUENCE from IAEA-INDC-NDS-0452-MAGIC-NUMBERS

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no magic-number value accessible to the formal derivation
- no selected favourable subsequence
- no fitted coupling or threshold
- no omitted source or tampered control

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:2f473f9fceddbeae8dca17c53207c4e8662d9698d60095a45aa871a8aba3fb79`; engine receipt `sha256:0b4634bdeef77436218ece89bbabd86819a7828d1dc9566b0fbaa31970a99b30` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001-0b4634bdeef77436.json`; empirical-validation hash `sha256:2c1266880d27f0e464c70b7bfd819f0285f915e5f9a2aff59a7c071b50f83347`; measurement receipt `sha256:9d6a3776768080863fc58346713e458f0bcce719fb4ab5c4e6e1b0b2e10d28a6`.

316. Post-seal exact Koide comparison

Claim identity: `SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-001`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The exact Fold value two-thirds is sealed before both complete CODATA mass-ratio intervals are opened and converted, with exact rational square-root enclosures, into a conservative Koide interval.

> *The sealed exact Koide value 2/3 lies inside the complete conservative CODATA-derived interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`

- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete post-seal invariant, two-source-row, rational-enclosure, custody and no-extra-rule comparison product. The declared exact boundary is: The full Cartesian uncertainty rectangle of both complete CODATA 2022 charged-lepton mass-ratio rows. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-two-thirds-versus-complete-rational-koide-enclosure__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-two-thirds-versus-complete-rational-koide-enclosure	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:5134dd4a4a0dc10103a2d42fd2110a7f005f7dc3bf261d6acfb26620390db7af. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:0db9171872e00ee706733cfebc55c1c2dedcace1957cd49881073358744b6d2f.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:69e20cd5a6dc9fb152ee9161cb3487f80a300317afd5c386a8aad46c60efe56e.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:3966a66e1a9f0719f16bd35dd82cb84626b18afc8e197d080f8bfad3be3ab561.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:4163c02f9ea2f8ced72407d4d905d426254508a47c6886b9f1f93bbd5600ae3d.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e3cd83ee4f6fafde717e8b0e0cf9225f363bf191cffa99cd90546a5acfe486cd. Independent certificate: sha256:c060fa4e8d243d22657a297a4d45013b3dc832315ad6e3e78ade240d1dbbfc26. Engine

external-validation hash:
`sha256:4d077bb255a05ec80331e7bd47eb7cbd0fd442833e54f715894db1d78b4d6d2e`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- NIST-CODATA-2022-MUON-ELECTRON-MASS-RATIO: predicted sealed-two-thirds-inside-complete-codata-derived-koide-interval; observed sealed-two-thirds-inside-complete-codata-derived-koide-interval; exact match True
- NIST-CODATA-2022-MUON-TAU-MASS-RATIO: predicted sealed-two-thirds-inside-complete-codata-derived-koide-interval; observed sealed-two-thirds-inside-complete-codata-derived-koide-interval; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The exact two-thirds lies outside the outward CODATA-derived interval, either row or uncertainty is omitted, source custody changes, or a tampered comparison is accepted.

Measurement receipt:
`sha256:57c2b84cacc9185a7edc9db2d5df68aea1be3282d10900517ff5aef9bc56a81b`. Isolation certificate:
`sha256:efd6b0a4ea3cfcb65abbe1aca981dd684c32a69bc59c3976d8a1f0c3d0e43da7`. Custody certificate:
`sha256:a3c74caa46b99e06dc09e6c508b9f1adfd26040629a3c596988f50e0c32da531`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured mass ratio in the formal two-thirds derivation
- no floating square root
- no central-value-only comparison
- no omitted uncertainty or row

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:c306c099b3911c2ce92a65b2d651a37602bc770a82005a9ee962444eb397d37f`; engine receipt
`sha256:5d3bb335ef3740f3b1fd8660a4ab95b3d7fc3dfef8534dd90e7dd2b9b69deb89` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-001-5d3bb335ef3740f3.json`; empirical-validation hash
`sha256:27f6abcba7b4156c18573b87e7ca34ea81b0438f63cf0467aaa5d6fd96f36a5f`; measurement receipt `sha256:57c2b84cacc9185a7edc9db2d5df68aea1be3282d10900517ff5aef9bc56a81b`.

317. Blind PDG test of the terminal electroweak Fold share

Claim identity: `SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. After the exact terminal weak share is sealed, it is compared with the PDG on-shell weak angle and with squared W/Z mass intervals from both the all-input world average and PDG's separately published compatible-input evaluation. The resulting pass/pass/tension vector is retained whole.

The sealed terminal share must lie inside the PDG on-shell interval and compatible-input W/Z interval while the all-input W/Z tension is retained, not silently discarded.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal relation, source provenance, custody, exact interval, complete-row, successor and no-extra-rule product. The declared exact boundary is: All exact post-seal comparisons of the immutable terminal sin-squared and cos-squared shares with every registered PDG weak-angle and W/Z mass interval, including the unfavorable all-input result. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-terminal-share-versus-complete-pdg-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-terminal-share-versus-complete-pdg-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:23424a21d7759f4768266f38ab052b9a6118919f82d5ff4fc052a665e606a3f2`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:10878412eb173fed35b73c98461bd25be5849bf70b3649f44954c90a7735f43f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1dc44c2a919a389128cc5f0437f114079e17df55855381463f1da809da19cbb6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:81f26488154f0dd23cf48f2b5a732e7036a01d736cc5633423e4515997df4b7a`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e29e2abdf7db536e975be6c8a5c9a92faec02648ff5a0a80afe0c3f22b987a38`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:fddd39ccabb378e3fa86e639e95933b9dc704b4b75ee59d0fb89d65b3db02e09`. Independent certificate: `sha256:71b523de6f4a0b3c139d99186cfae5fb0bffe38f55c658fb1297dcf3a3b85972`. Engine external-validation hash: `sha256:e911e11a732838f9eb0cb6447f2bf34016c4552d324844a271fa0606a433d3c4`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-ELECTROWEAK-PRECISION-2024-2025

Observed comparison records:

- PDG-ELECTROWEAK-COMPLETE-VECTOR: predicted on-shell-pass__compatible-w-mass-pass__all-input-w-mass-tension-retained; observed on-shell-pass__compatible-w-mass-pass__all-input-w-mass-tension-retained; exact match True

- deliberately tampered unfavourable control rejected

Falsification condition: The sealed weak share leaves the on-shell interval, leaves the compatible-input W/Z interval, enters the presently disjoint all-input interval without a source revision, any registered row/hash changes, or an unfavourable row is omitted.

Measurement receipt: `sha256:87eed9a2f6cd1390c95be0178803be0eb29247542d087f3e120ec81284e3edc5`. Isolation certificate: `sha256:12a728777a2c876898ed928998a35aca59166b7ea3210eecdc04755e67ab66a6`. Custody certificate: `sha256:c0c12ec5060160e05c0c85881f7cdabd03f0c12d446820883c3a138b0934bed5`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf and https://pdg.lbl.gov/2024/listings/rpp2024-list-w-boson.pdf (sha256:1eb3111d343f411a1788120d394899c03ea75279bbea68a0c8284a58f6af89bd)

Registered target identities:

- PDG-ELECTROWEAK-COMPLETE-VECTOR from PDG-ELECTROWEAK-PRECISION-2024-2025

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no PDG value accessible before the formal seal

- no fitted digit or uncertainty widening

- no deletion of the unfavourable all-input row

- no floating-point interval decision

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:dd731a716517ca6416a0fde457ac44d5a1eeb623b0ef5bb9a6f02b487c7641d9`; engine receipt `sha256:f4fd9f4991baf12de960ff990065313fa6292e4c23cda53995b2019f0e9acbc6` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003-f4fd9f4991baf12d.json`; empirical-validation hash `sha256:407021ddf0dc740b3a35418a9836ce8fbda16d2a59a599e255dd8e3c472b1bb3`; measurement receipt `sha256:87eed9a2f6cd1390c95be0178803be0eb29247542d087f3e120ec81284e3edc5`.

318. Blind CODATA test of the terminal proton-to-Planck hierarchy

Claim identity: `SFT-PHYS-VALIDATION-PROTON-PLANCK-TERMINAL-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. After the exact terminal squared hierarchy is sealed, the complete CODATA 2022 Planck-mass and proton-mass intervals are propagated outward through an exact positive squared ratio and tested without forming an irrational root.

> *The sealed terminal squared hierarchy lies inside the complete outward-propagated CODATA 2022 Planck/proton mass interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SCALE-PROTON-PLANCK-TERMINAL-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal relation, source provenance, custody, exact interval, complete-row, successor and no-extra-rule product. The declared exact boundary is: All exact post-seal outward interval propagation from the complete registered CODATA Planck-mass and proton-mass rows to the immutable squared hierarchy. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-terminal-hierarchy-versus-complete-codata-mass-interval__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-terminal-hierarchy-versus-complete-codata-mass-interval`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a79d4b769b204ad71561bd949355aa2ad05f900029a0dec8e71fc9b8d1c8a321`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:be07260dbfa4233d3bb0d739bb40fe35fc81fcf348b89b6e3c5a64fd65bb16ae`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f6dd73cab024961b7b1eee64f41622e759af190152ebe18a140eec0a3c72a023`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b205d74a96169e83e09d899ab57901854b936696b5340728d0b647a56f8d9c4d`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9c21f4d3ceece1acc073200621d0f0fb8b533c14bdac560ba6a229f4bfbc326c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ac90f42e63702458fad777d5aa28b5543ea3c51232db593710c398d66bb5794e`. Independent certificate: `sha256:2f462d0692a8219ad630c17db02f5294bcc88d22783f97d5221fd5d9e2ea33e6`. Engine external-validation hash: `sha256:eff6fe46f7de335a85af3e86b452e0e84127725a95cd50da2c6951e18951b76a`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- NIST-CODATA-2022-PLANCK-PROTON-HIERARCHY: predicted terminal-squared-hierarchy-inside-complete-codata-mass-interval; observed terminal-squared-hierarchy-inside-complete-codata-mass-interval; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The sealed squared hierarchy lies outside the complete propagated CODATA interval, either source row or hash changes, an uncertainty is omitted, or a tampered comparison is accepted.

Measurement receipt: `sha256:292dea2593ff526610453a8332d8e65806c8549b7e460bb5894045d3338823b4`. Isolation certificate: `sha256:3a926f40536b92a346cf5977487ffe20cfdfa00e765335f888e4aa44360d532d`. Custody certificate: `sha256:af15bd7b30c54ddbafcaf56b47164dcc4f1fc383ac09b722c8023bba90bd6685`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- NIST-CODATA-2022-PLANCK-PROTON-HIERARCHY from NIST-CODATA-2022-ALL-CONSTANTS

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no CODATA value accessible before the formal seal
- no fitted hierarchy exponent or correction
- no floating-point comparison
- no omitted source uncertainty
- no irrational square-root proof value

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:c33b6d7ed6ea86aa21807f45992bc9da563d35394cd2e050d1831bc1ec5bf870`; engine receipt `sha256:6483314f23ac951735fa6b201281236b523f2ecbfc33e73813dd92d3501264b8` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-PROTON-PLANCK-TERMINAL-003-6483314f23ac9517.json`; empirical-validation hash `sha256:e18f009abc4c7b7f4b11eca7cc67fd85428423cd1e8c1ebab915673a92324ddd`; measurement receipt `sha256:292dea2593ff526610453a8332d8e65806c8549b7e460bb5894045d3338823b4`.

319. Blind PDG anchor test of the complete force-sector formula

Claim identity: `SFT-PHYS-VALIDATION-FORCE-SECTOR-ANCHORS-003`

Shared claim-record clause `PHYS-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. After the four-sector inventory seals, the same p-fibre and p-squared-less-One formulas are checked against PDG's sector-two weak-mediator count and sector-three colour/gluon counts. Sector-five and sector-seven outputs remain explicitly unmeasured predictions.

> *The sealed common formula reproduces weak mediator count three, colour count three and gluon count eight; sector-five count twenty-four and sector-seven count forty-eight remain standing unmeasured predictions.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed sector carrier, common count relation, provenance, custody, separate record, complete anchors, successor closure and no-extra-rule forms. The declared exact boundary is: Every registered known PDG anchor of the common sector formula, plus a mandatory boundary record preserving every presently unmeasured penta/hepta consequence. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-sector-formula-versus-all-pdg-anchor-counts__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-sector-formula-versus-all-pdg-anchor-counts`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a1e35e13a982c482c7a75ece5032e60e77b5b51789266c24317170a6fbec7033`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2d202d9be6a198b5d95cab2d43a8c243f9f7b209d36ff93b3b428ace957ded84`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9090ffcbf7ce4d101b7103ec1dbd481817bc04276f41e7bd55dc2c598a1a7c08`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0abe7f7af69ef66c3040ae6ca5e510d04d22f4d5e2e979878d94d538d3a06db3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:5de8b3690077050c16a394b4e792a2a6bfd428f94ee5a62b4d1a7a734bf6fabf`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b9ea919cc172d6d9382effbcee8ebc2bc69b8337bc660107411cca3bab6104ee`. Independent certificate: `sha256:ccf55bacab31ba908af95597b9b719fe68f5715a388e391fdad284c70e8346cf`. Engine external-validation hash: `sha256:5fb7b6d50fdc96d718352a8b4a9e87a3a8c387ce4acf5cb3846aff70fab0b766`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-FORCE-SECTOR-ANCHORS-2025

Observed comparison records:

- PDG-FORCE-SECTOR-COMPLETE-ANCHOR-RECORD: predicted sealed-common-formula-matches-three-eight-anchors__penta-hepta-retained-as-unmeasured-predictions; observed sealed-common-formula-matches-three-eight-anchors__penta-hepta-retained-as-unmeasured-predictions; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Any known count differs, a source identity changes, an unmeasured penta/hepta result is omitted or mislabelled as observed, or a tampered row is accepted.

Measurement receipt: `sha256:a1ff4f9334fa84f57a08397926db701829abd2ba8df26889da20ac395fa4f675`. Isolation certificate: `sha256:5feb31c58206086fc062253378b6ab8c817a249ca99814a528b85cceaf198faa`. Custody certificate: `sha256:973b3b0ff5c3b58e6ff60e7cebc51e32e0f992998e28da81e69b11b94d204182`.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-qcd.pdf and https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf (sha256:bb37e82e9515ff526cbaad742167c16ac1960b7bfca32d8b39dd4384c73b7341)

Registered target identities:

- PDG-FORCE-SECTOR-COMPLETE-ANCHOR-RECORD from PDG-FORCE-SECTOR-ANCHORS-2025

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no PDG count visible to the formal sector derivation
- no imported gauge group selecting p-squared less One

- no unobserved prediction represented as confirmation
- no omitted unfavourable or tampered row

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:8c94b4353ef5b93c7e364680d664e558ec886107e5ba50b8343199ef9f1b9058; engine receipt sha256:88a3d6273a4332a6eb1dbe565e812e59d6ec768c094c0e37f1f81b3bcc0c00ba at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-FORCE-SECTOR-ANCHORS-003-88a3d6273a4332a6.json; empirical-validation hash sha256:b855fdf8e536d5cf1d50f5f3d2f69cdb1e4e14897cc46cfd9a0ab1c598bdf42e; measurement receipt sha256:a1ff4f9334fa84f57a08397926db701829abd2ba8df26889da20ac395fa4f675.

320. Post-seal NIST test of the nonempty vacuum floor

Claim identity: SFT-PHYS-VALIDATION-VACUUM-FLOOR-003

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. After the half-One floor and oscillator support are sealed, the full NIST ground-state and Casimir record confirms nonempty residual quantum support and a boundary-dependent vacuum interaction. The absolute half-spacing energy offset is not isolated as a separately measurable scalar, and that boundary is retained.

> *NIST confirms nonempty ground fluctuations and boundary-dependent vacuum force; the sealed half-One coefficient remains a formal structural value rather than a separately isolated measured scalar.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal vacuum-floor comparison product. The declared exact boundary is: Every registered NIST ground-state and Casimir row, including the direct-measurement boundary on the absolute half-spacing offset. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-vacuum-floor-versus-complete-NIST-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-vacuum-floor-versus-complete-NIST-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4e04f45b9dd05e8742a4a809e8781285b648af365f0b469be54c5dd3e0fad61d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d62c7a13817f3409c893bcc756605822fe16f184abf406247d0c4798d2385d92`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f8f27cef4429c37b815e8216cd541a481607b59c0fe113e7368e3fd492c243aa`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:30f479f82606c3cde185295a74313fb4b5d503e2f3a6eb8300e493d27916de9a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1937768f91c7a380043fc9bb4e6bb2605df230ac5ed1c5d30ae594d0bc96b323`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c9d74312a5c205da46ebf64d121135da2109a90abead5867cce56d9e152e1fa7`. Independent certificate: `sha256:30d3e8937f20c12b9096f2336a92eeef982cfba58ecaefb8fecb294154438ae2`. Engine external-validation hash: `sha256:10931e4fc572535fbdde75672ee1669994c3886c0eea0f5510e9f23466becc6b`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Observed comparison records:

- VACUUM-LINEAGE-FLOOR: predicted nonempty-ground-confirmed__casimir-boundary-force-confirmed__half-offset-not-directly-isolated; observed nonempty-ground-confirmed__casimir-boundary-force-confirmed__half-offset-not-directly-isolated; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: NIST ceases to report nonempty ground support or boundary-dependent vacuum interaction, a source hash changes, or the half-offset is falsely promoted from indirect evidence to a direct scalar measurement.

Measurement receipt: `sha256:29cb00823855c7540eb49397f232d8050b97ad52c4b1d79a89b79d3d1ba76956`. Isolation certificate: `sha256:03ad7260b7a0a10a255860795ec419d4bb1fa2bd82a0970f20e08ebeba3a2fd8`. Custody certificate: `sha256:d1ae9c2686595830f446407bdde4050446247368c1beb5c3b79868bec0f9705c`.

Registered source descriptions:

- NIST, Particle Data Group and CNES MICROSCOPE collaboration: https://www.nist.gov/news-events/news/2015/06/vacuum-fluctuations-measuring-unreal; https://www.nist.gov/news-events/news/2013/02/calculating-quantum-vacuum-forces-nanostructures; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf; https://cnes.fr/en/projects/microscope (sha256:327696985650aebaa20436e9da1d7f20f9bbff4205f9e7146ec8bd887535f3a9)

Registered target identities:

- VACUUM-LINEAGE-FLOOR from VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no target access before the formal seal
- no claim that Casimir force alone measures the half-One coefficient
- no deletion of the direct-observability boundary

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:5b5abfc825de946478763798c6ab81a26d028927a0929a06fcc1f6c890ccb1f5`; engine receipt `sha256:f14ff2733c67bd9100858be8410ccf0effba1632d77b481630baa6749930834d` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VACUUM-FLOOR-003-f14ff2733c67bd91.json`; empirical-validation hash `sha256:63e6405e0cd14f185a02b713cdfaae9f60e605ca0adc118a4d046e7bce4199c7`; measurement receipt `sha256:29cb00823855c7540eb49397f232d8050b97ad52c4b1d79a89b79d3d1ba76956`.

321. Post-seal PDG test of the vacuum-polarization running direction

Claim identity: `SFT-PHYS-VALIDATION-VACUUM-POLARIZATION-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The sealed structural direction is compared with both complete PDG inverse-coupling intervals. The low-energy inverse interval lies wholly above the Z-scale interval, so the coupling itself increases with closer/higher-energy probing; PDG also retains direct LEP observations of running.

> *The complete PDG intervals confirm the sealed direction: inverse alpha decreases from 137.035999084(21) to 127.930(8), hence effective alpha increases with scale.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal electromagnetic-running comparison product. The declared exact boundary is: Both full PDG inverse-coupling intervals plus the direct-running observation, with exact interval ordering and no beta-coefficient claim. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-running-direction-versus-complete-PDG-intervals__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-running-direction-versus-complete-PDG-intervals`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:885177a42d0708abac8f7fd97cd6efd3293e2feab3e2933eefd0b5cd63543d87`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3e5c7c6bc91e160a9a5e28c781647e346f7ddcf508b4da31b655b3ac062a706c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3291393287d2b144f83fbd7009da2720d140a65f329f07b967c3b8e004db528f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d7ab76ff08ea358d1ef265b8a4d92babe072de15688644270c28158b976f091e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2a381c3f2497067e41476a7e0a141b243d5861e15be04a42f97559fd456dd71e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ddeb4a98e4fbe345139ee83b5aee74348f08feee0e405ecc042716390c7b4ffa`. Independent certificate: `sha256:24fc04e99b6bc6ef8b5b337fc4839d6ab90467004180cdde94d1cdee9bf378aa`. Engine external-validation hash: `sha256:2dfa9249400a4902546a86fd34fe3074b77782a2b51752a04418b82b545d8990`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Observed comparison records:

- VACUUM-LINEAGE-POLARIZATION: predicted inverse-coupling-decreases-with-scale__effective-coupling-running-direction-confirmed; observed inverse-coupling-decreases-with-scale__effective-coupling-running-direction-confirmed; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The authoritative inverse-coupling intervals reverse or overlap contrary to the registered direction, direct running is withdrawn, a row/hash changes, or a fitted beta law is substituted for the sealed structural claim.

Measurement receipt: `sha256:1db5d59ff856de727e2077baa8139c453df44c1f8227b2b38f8a09373195fa03`. Isolation certificate: `sha256:cd7e278f5be46855489776b58d055eeeb4cfeda1ed8322f489a63ba8e2e031ea`. Custody certificate: `sha256:2490c6423812450302e85e74f791ccd717e93d44d462531317c1adb5837f3e08`.

Registered source descriptions:

- NIST, Particle Data Group and CNES MICROSCOPE collaboration: https://www.nist.gov/news-events/news/2015/06/vacuum-fluctuations-measuring-unreal; https://www.nist.gov/news-events/news/2013/02/calculating-quantum-vacuum-forces-nanostructures;

https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf; https://cnes.fr/en/projects/microscope (sha256:327696985650aebaa20436e9da1d7f20f9bbff4205f9e7146ec8bd887535f3a9)

Registered target identities:

- VACUUM-LINEAGE-POLARIZATION from VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured value accessible before seal
- no fitted running curve
- no omission of either uncertainty
- no claim of an exact beta coefficient from the two endpoints

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:0b47ec757dc884d7cd7d186e585710bffea32a2bfaf9488abac456d2ac38640a`; engine receipt `sha256:2faf46c5d43a2e2026a6ea4dc78cb52557592b262b6801d3ccda2c6a594b1d5e` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VACUUM-POLARIZATION-003-2faf46c5d43a2e20.json`; empirical-validation hash `sha256:037830773e529ab4d45438b054315936b393c1bfaf3170c01eecb5b7de8a707c`; measurement receipt `sha256:1db5d59ff856de727e2077baa8139c453df44c1f8227b2b38f8a09373195fa03`.

322. Post-seal MICROSCOPE anchor test of vacuum-inertia unity

Claim identity: `SFT-PHYS-VALIDATION-VACUUM-INERTIA-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The exact vacuum/inertia unity relation is compared with CNES MICROSCOPE's completed equivalence-principle mission. Universal inertial/gravitational response is supported to the reported 10^-15 mission precision. The stronger claim that a vacuum carrier can be engineered to control inertia was not tested and remains a standing prediction.

> *MICROSCOPE supports composition-independent inertial/gravitational unity response to 10^-15 mission precision; it does not test vacuum-mediated inertia control.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-VACUUM-INERTIA-UNITY-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal vacuum/inertia anchor comparison product. The declared exact boundary is: The complete CNES mission identity, response class and reported precision together with the explicit untested engineering-control boundary. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-unity-relation-versus-MICROSCOPE-anchor__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
`relation`	`sealed-unity-relation-versus-MICROSCOPE-anchor`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9bee87a8141a41e77e749ec45518cd68f30386bd3e08c6d142d3182d8c6d0b23`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:cb29b0894217eb327de698a5f0f4d9762ad5388ecada69f5691e8a1b6dfc4d2e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1a297cfa629c2aa50a1a415db58c4b7d455276d714abdaf84aafc88f7430f435`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c4dead2a3c9d4c128e3e7b2ea42eb34c00f53747c519b61053c8ac7d2e8d09b2`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:bb3f53f22dc0b2f3d35a5f4b68ea3e6e7a3bc453cee3b0f5fd9bd094cc256126`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:072e6dfb7eaee1e1a8eb757fe0e2d06e7a993cd02d00846be83d11a292b01747`. Independent certificate: `sha256:35dca81d6c2ac6fe5294ce6dd18c936099fccf1aca8198f3eac768072e6ece0b`. Engine external-validation hash: `sha256:1ef0280ed56a5f3ee4f930c1453a1bade251563d68acccbad3e0c200806de2a3`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Observed comparison records:

- VACUUM-LINEAGE-INERTIA: predicted equivalence-unity-anchor-confirmed-to-mission-precision__vacuum-control-untested; observed equivalence-unity-anchor-confirmed-to-mission-precision__vacuum-control-untested; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: A verified composition-dependent inertial/gravitational response violates the unity relation, or a claimed direct inertia-control confirmation lacks a separately committed controlled experiment.

Measurement receipt:
`sha256:2b8f1b611d07ad8d12d67459cb28db401d0ceddec45fb9ae2392ebc718d6b9fa`. Isolation certificate:
`sha256:d1155de954cccdc8d994a03fe8b46ce04d2a06cda7d635eaf71c575bfc4309de`. Custody certificate:
`sha256:2a3bc0aa781ca7de7a662a850e64841171bbd0edbe2a458049722702abc148b8`.

Registered source descriptions:

- NIST, Particle Data Group and CNES MICROSCOPE collaboration: https://www.nist.gov/news-events/news/2015/06/vacuum-fluctuations-measuring-unreal; https://www.nist.gov/news-events/news/2013/02/calculating-quantum-vacuum-forces-nanostructures; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf; https://cnes.fr/en/projects/microscope (sha256:327696985650aebaa20436e9da1d7f20f9bbff4205f9e7146ec8bd887535f3a9)

Registered target identities:

- VACUUM-LINEAGE-INERTIA from VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no UAP report or propulsion claim used as measurement
- no inference that equivalence alone demonstrates controllable inertia
- no deleted untested boundary

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:6fb1ea82f3144903c9c33a7bf2142271480b4cc81a91053a94c3e0b21452298a`; engine receipt
`sha256:9dbc21cecc28efb90789178529c54474aa4077fbd1452065ef038cad34c25d2f` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VACUUM-INERTIA-003-9dbc21cecc28efb9.json`;
empirical-validation hash
`sha256:cc553a6e8b5cc6ee4afe5d25ea22ed43a48a4669831ae11c6417d1ecc219d552`; measurement receipt `sha256:2b8f1b611d07ad8d12d67459cb28db401d0ceddec45fb9ae2392ebc718d6b9fa`.

323. Post-seal NIST boundary test of vacuum-work extraction

Claim identity: `SFT-PHYS-VALIDATION-VACUUM-EXTRACTION-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The positive one-sixth formal transfer and complete returned-cycle ledger are tested against the complete NIST ground-state record. NIST reports that no energy can be removed from the ground state and that amplification requires coherent pump energy. Consequently free-standing cyclic vacuum work is not empirically confirmed, while the complete-cycle no-unrecorded-gain ledger is consistent with the measured pump boundary.

> *The formal outward transfer remains sealed; NIST does not observe standalone extractable vacuum work and requires external pump energy for amplification; this supports the complete returned-cycle accounting rather than a source-free gain claim.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-VACUUM-COMPLETE-CYCLE-LEDGER-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-PHYS-VACUUM-ASYMMETRIC-BEAT-EXTRACTION-003`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal vacuum-extraction and returned-cycle comparison product. The declared exact boundary is: Every registered NIST work-removal and pump row, preserving the positive formal transfer, physical translation boundary and complete returned-cycle ledger. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-extraction-ledger-versus-complete-NIST-work-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-extraction-ledger-versus-complete-NIST-work-boundary`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:8a07bba46fc5ae67e22b7200c0c2090f8df7f348efddab8dc409f85970df9dda`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f79f3722d9c99925a830509b85a67a3e498128933a91787104d7028694a932a7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ad367dd6f5d5ba841fb12c20144b3eb2808cf52ee7065b3a60fa1a1e96b7fad8`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:379457c2dec6e112927e7eebc0f3bff6bbadb2ad1749e9d43ba8e9ef04eec879`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6d2bd33127758fa52be0b90cecfc8dba6ab9c800d189faae28b606a5178d3847`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4bed1b1ab95e3e34b9490ba6c787b6d6a8def75d24b9d2b2a5255dc2d75e55f9`. Independent certificate: `sha256:4dc5c2ad5d1020c9cfbe9a28a2fab89247a5c5ab3895719ca17a23ad6950cba5`. Engine external-validation hash: `sha256:66ba8c5ebe75af0cc931f07b4c35251c9f778e413c32e8bae43c2378736b60d8`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Observed comparison records:

- VACUUM-LINEAGE-EXTRACTION: predicted positive-formal-outward-transfer-sealed__standalone-vacuum-work-not-observed__complete-returned-cycle-consistent; observed positive-formal-outward-transfer-sealed__standalone-vacuum-work-not-observed__complete-returned-cycle-consistent; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: A controlled device returns vacuum, apparatus and information records to their initial states while delivering net work without an external source, or any registered NIST source/pump/adverse row is omitted or altered.

Measurement receipt: `sha256:4145ed3f56d68b698bb3542b1e845c1bb50849096248cc290b0e31786ad6e831`. Isolation certificate: `sha256:47c79790b4aba4cd0fb9fca7c11ad85b057c4363c596da8aebe98b99b1931a73`. Custody certificate: `sha256:26b06b29c4d3a02a8b01c96e9d7eae900542eabdd719037226bda12ce9338d38`.

Registered source descriptions:

- NIST, Particle Data Group and CNES MICROSCOPE collaboration: https://www.nist.gov/news-events/news/2015/06/vacuum-fluctuations-measuring-unreal; https://www.nist.gov/news-events/news/2013/02/calculating-quantum-vacuum-forces-nanostructures; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf; https://cnes.fr/en/projects/microscope (sha256:327696985650aebaa20436e9da1d7f20f9bbff4205f9e7146ec8bd887535f3a9)

Registered target identities:

- VACUUM-LINEAGE-EXTRACTION from VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no erasure of the positive formal extraction claim
- no claim of empirical free-energy confirmation
- no omission of external pump or reservoir restoration
- no target access before seal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:8b8c259ee2add8c9d1044ceccb2ed2e712eb699ea918ef1eac6d5f629af2838a`; engine receipt `sha256:ee9efc5016007be792ea5f6a8cce8f5efb6b7b45ef997c464709296d250b5922` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VACUUM-EXTRACTION-003-ee9efc5016007be7.json`; empirical-validation hash `sha256:db45cab4ed11e689adbfe5e7ece07f9158f4b0c13d9168c5ae2d7fec8a61cd8b`; measurement receipt `sha256:4145ed3f56d68b698bb3542b1e845c1bb50849096248cc290b0e31786ad6e831`.

324. Post-seal dynamics and spectra comparison

Claim identity: `SFT-PHYS-VALIDATION-DYNAMICS-SPECTRA-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete NIST ASD and CODATA rows confirm discrete physical energy levels, transitions, wavelengths and energy-frequency carriers. They do not identify the universal dimensionless Fold witness levels as measured atomic energies, and that limitation is retained.

> *Discrete spectra and energy-frequency phase correspondence are externally supported; the particular Fold fractions remain exact structural normalization witnesses, not universal measured atomic energies.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003
- SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal dynamics/spectra comparison product. The declared exact boundary is: All registered NIST discrete-spectrum and CODATA energy-frequency rows together with the exact-normalization unfavorable control. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-Fold-dynamics-versus-complete-NIST-spectrum-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-Fold-dynamics-versus-complete-NIST-spectrum-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d717558526d68d1e7f0d2ed04978d26b8b4a45e6ebed182fbdf48c1e142e5db3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e2c87766879b437a55f2a3b6438e924dbb0dddb65598fb6305410536068aac27`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:efc79bf59416c63d0d2c56a8dd03c72150b389987374bcb6347b48bb3ba1fcf7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:73036f37e4f48171fdf291abb17213ae808e984532a7038d1caa21dd3ad45428`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9fe6b608b8e57e3697202b7fb59ea90b9d5d2f3859395f399c916c233fec7e17`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2cf0988f2e6e454edfe952bb9cdcf19aef533433f9fa684e67aa1a3a9300041b`. Independent certificate: `sha256:107e2ca2c00622c6aac5c93ef03cb1b2000838c4d7d95e9718f2d7b45106eb79`. Engine external-validation hash: `sha256:f7a0d22b83582a3af73f670c42672a1b010fe00e5ebaa86e5e62e31c768f021e`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Observed comparison records:

- RELATIVISTIC-FIELD-DYNAMICS: predicted discrete-spectra-and-energy-phase-correspondence-confirmed__specific-Fold-normalization-structural-not-a-measured-particle-value; observed discrete-spectra-and-energy-phase-correspondence-confirmed__specific-Fold-normalization-structural-not-a-measured-particle-value; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Authoritative spectra cease to exhibit discrete levels/transitions or the structural Fold normalization is promoted to a direct measured value without a committed measurement.

Measurement receipt: `sha256:f532aba0420cbcbadc1cbfad7d19c611b2607233db27aa898b620e120ab01f7f`. Isolation certificate: `sha256:b100e7d64bdfcb1d023b0ac41c5e430d289888696614f6bed521772049562128`. Custody certificate: `sha256:e867bcdaeec43cfa0ba39ad45dad2a92dbbfd47c80c6b4944b7e6a9aaa73ba4b`.

Registered source descriptions:

- NIST, CODATA, Particle Data Group and American Physical Society: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml; https://physics.nist.gov/cuu/Constants/Table/allascii.txt; https://doi.org/10.1103/PhysRevLett.26.721; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf; https://www.nist.gov/news-events/news/2019/12/new-playbook-interference; https://www.nist.gov/publications/nonlinear-networks-arbitrary-optical-synthesis; https://tf.nist.gov/ofm/synthesis/ppln.htm (sha256:97a7631d956cc2eb8142e64603a1f51ece76944195274a28bbe91498edae0308)

Registered target identities:

- RELATIVISTIC-FIELD-DYNAMICS from RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no source access before formal seal
- no claim that NIST measured universal Fold fractions
- no omission of the normalization control

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:a1ee9931e05c4529e60821472005d00eed1d644344c4aa460a5a2363b09261b5`; engine receipt `sha256:b9d360bc08458cdb4507806883d1fd07fdf5bd1ce119192ee95b86ae6bf9c0b7` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-DYNAMICS-SPECTRA-003-b9d360bc08458cdb.json`; empirical-validation hash `sha256:7ea22b062376362f5baabdb6fe2063ca3e407da63fcca8fa9b29d37a2500abc0`; measurement receipt `sha256:f532aba0420cbcbadc1cbfad7d19c611b2607233db27aa898b620e120ab01f7f`.

325. Post-seal relativistic and electromagnetic field comparison

Claim identity: `SFT-PHYS-VALIDATION-RELATIVISTIC-FIELDS-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete PDG, CODATA and APS record supports Dirac/Lorentz physical structure, one invariant vacuum speed, electric/magnetic observable families and the inverse-square Coulomb exponent. It does not measure the Fold 3/5, 4/5 or four-half closure witnesses as universal particle coordinates.

> *Dirac/Lorentz structure, invariant vacuum speed, electric/magnetic observables and inverse-square Coulomb behavior are externally supported; the exact rational Fold closures remain structural witnesses.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003`
- `SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003`
- `SFT-PHYS-FIELD-LORENTZ-TRANSFER-003`
- `SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal relativistic/electromagnetic comparison product. The declared exact boundary is: All registered PDG Dirac/Lorentz, CODATA speed/electromagnetic and APS inverse-square rows plus every rational-witness scope control. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-relativistic-field-laws-versus-complete-authority-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-relativistic-field-laws-versus-complete-authority-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:c3600a83c430f04e8fe3f18831ede721301547b94ca1c6804fe917920d8e72ec`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:0f5d12ebe7a12829793e67cabeeab858be54036267be413409f41f9c937ac6fb.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:1a0bb96564a502d21766adbd7ad508c54e2248ed7c5355a58392d16469b4c52d.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:9452ac6c8fe6d698663a2f665d20c48dc9b60a1e964942cca5e873bf17a7c00f.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:b8f6c200b11fa4f3d5a904e388d13fa4d7c71c1c07bf0b0f2dc226edfc8d91ea.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:0318131d9b374c14facbbbd8dd1de5443bf68ea6bb4488b59488654dd6699f2d. Independent certificate: sha256:59180ae362cc75407c15974fd1714b53cf2724e51cb2a4cd5d8e970edeafd342. Engine external-validation hash: sha256:698e4cffc18a8e7ab27b23045d2b9623fd876fc45a8d6b6de75072d542dbaca3.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Observed comparison records:

- RELATIVISTIC-FIELD-FIELDS: predicted Dirac-Lorentz-invariant-speed-and-Coulomb-inverse-square-correspondence-confirmed__specific-rational-closure-witnesses-structural; observed Dirac-Lorentz-invariant-speed-and-Coulomb-inverse-square-correspondence-confirmed__specific-rational-closure-witnesses-structural; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: A verified non-invariant vacuum propagation speed, non-inverse-square Coulomb exponent outside the registered bound, or absence of Dirac/Lorentz and electric/magnetic physical classes contradicts the correspondence; a changed source or omitted limitation invalidates the validation receipt.

Measurement receipt: sha256:1322935259cab8623e2ba4350d1b1519faf6e2c66ffcbd14a95d439ca1ea01af. Isolation certificate: sha256:965233c0a88fc116e5e5ba5803f2f6f4f28d0c9986bf9940660f9f4318074bf7. Custody certificate: sha256:9d8342ff7e6ceb13616344205ca955cd179ffb0b77e73bd8fe23859c45d3061c.

Registered source descriptions:

- NIST, CODATA, Particle Data Group and American Physical Society: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml; https://physics.nist.gov/cuu/Constants/Table/allascii.txt; https://doi.org/10.1103/PhysRevLett.26.721; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf; https://www.nist.gov/news-events/news/2019/12/new-playbook-interference; https://www.nist.gov/publications/nonlinear-networks-arbitrary-optical-synthesis; https://tf.nist.gov/ofm/synthesis/ppln.htm (sha256:97a7631d956cc2eb8142e64603a1f51ece76944195274a28bbe91498edae0308)

Registered target identities:

- RELATIVISTIC-FIELD-FIELDS from RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Shared claim-record clause PHYS-S028 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no conventional equation selecting the formal law
- no promotion of structural rational witnesses to measured particle values
- no omitted APS uncertainty or CODATA definition boundary

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:e32bfc8fac4701dcf7fe2e138bb2fb77f4a1eb9a39caaf3c591af772f62491c0; engine receipt sha256:be5db432d9abee0d45670c13606e3b5ab4b501ac626c457a458365c530c01c6e at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-RELATIVISTIC-FIELDS-003-be5db432d9abee0d.json; empirical-validation hash sha256:0327b1171045986b81cfa163efc4564eb7393a4dc125fda479ff2ba8238d5884; measurement receipt sha256:1322935259cab8623e2ba4350d1b1519faf6e2c66ffcbd14a95d439ca1ea01af.

326. Post-seal optical-operation comparison

Claim identity: SFT-PHYS-VALIDATION-OPTICAL-OPERATIONS-003

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. NIST experimental records support photon interference, polarization observables and nonlinear sum, difference, second-, third- and four-wave mixing classes. The exact half-One support normalization remains a sealed Fold relation rather than an imported susceptibility or measured universal amplitude.

> *Interference, transverse polarization observables and nonlinear frequency mixing are externally supported; the Fold half-One is not asserted to equal every physical amplitude or susceptibility coefficient.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal optical-operation comparison product. The declared exact boundary is: Every registered NIST interference, polarization and nonlinear-mixing row together with the normalization and material-response controls. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-optical-operations-versus-complete-NIST-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-optical-operations-versus-complete-NIST-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:91fcee665e8f564ad0f8eb596f3657fa72c65f0c679ae9c0e2dd259b1cb88449`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c1d8d482048194f5688ebdcbec17b04d425680acb2a2661c5458588a8b19ff9b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:69070f199be6d34c3f6ae85a96c9a0a3caa25bc0c3d9f21156cdba5be05e886c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f9040489e5a295e0d78a8cddacbe68540ee86dd22b5382f66a9bc93fd5f4e268`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:497b0daf538864ea517c50a5c851c7bcd16f68a8ea3d93b46550946730511aad`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:684223ef4c98beddc30002ffdd503d26235848f5fb341743bb5767940ca6e260`. Independent certificate: `sha256:2d1a16b0a578132bcef06dbe25974644eec2693cc32061469f2a8fffc88a15e1`. Engine external-validation hash: `sha256:5dbdd9291daadc1c591f2e332993abe7546f58c4e46e75dbce80ec115a68fe4c`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Observed comparison records:

- RELATIVISTIC-FIELD-OPTICS: predicted interference-and-nonlinear-frequency-operations-confirmed__held-polarization-supported-by-registered-observables__half-One-normalization-structural; observed interference-and-nonlinear-frequency-operations-confirmed__held-polarization-supported-by-registered-observables__half-One-normalization-structural; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Controlled optical experiments cease to exhibit interference or conservation-governed nonlinear mixing, or the validation silently equates a structural Fold share with an unmeasured universal amplitude.

Measurement receipt: `sha256:6ba2d1a8ef90dd1b12ebb8838bf805f176e351ebbd05dc43e7747ea27fd2d8e0`. Isolation certificate: `sha256:70b91540fc7c3f28162cf49ef9b6e750ceb1b43c81799131b168c15f14fd49e0`. Custody certificate: `sha256:ac1a427cbdfb37962640da3787889f78ae5c1855632f677d7eb9937a9293c484`.

Registered source descriptions:

- NIST, CODATA, Particle Data Group and American Physical Society: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml; https://physics.nist.gov/cuu/Constants/Table/allascii.txt; https://doi.org/10.1103/PhysRevLett.26.721; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf; https://www.nist.gov/news-events/news/2019/12/new-playbook-interference; https://www.nist.gov/publications/nonlinear-networks-arbitrary-optical-synthesis; https://tf.nist.gov/ofm/synthesis/ppln.htm (sha256:97a7631d956cc2eb8142e64603a1f51ece76944195274a28bbe91498edae0308)

Registered target identities:

- RELATIVISTIC-FIELD-OPTICS from RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported optical amplitude
- no fitted nonlinear susceptibility
- no claim that every material realizes every operation at the same strength

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:698500e2a83530185795a180b76af95b35687ca6b5f5bb546304d3a93dcd8410`; engine receipt `sha256:73d24227b39bed29d9648706021b5d098732eb9e054568f70d5144bc29f3881b` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-OPTICAL-OPERATIONS-003-73d24227b39bed29.json`; empirical-validation hash `sha256:7ded85d56ce448f9588e0a8d7ddc568e07f56e85ae3715c73cfe95b3feed23fa`; measurement receipt `sha256:6ba2d1a8ef90dd1b12ebb8838bf805f176e351ebbd05dc43e7747ea27fd2d8e0`.

327. Post-seal finite-loop and renormalization-boundary comparison

Claim identity: `SFT-PHYS-VALIDATION-FINITE-LOOPS-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. PDG retains finite renormalized radiative calculations and schemes, providing correspondence for finite observable loop receipts. These calculations do not directly measure whether the substrate is fundamentally finite, so the V3 law remains a finite generated-support theorem rather than a claimed empirical disproof of every continuum representation.

> *Finite renormalized radiative observable calculations correspond to exact finite loop receipts; the absence of completed continuum support is a formal Fold boundary, not a directly measured ontology claim.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal finite-loop comparison product. The declared exact boundary is: The registered PDG renormalized-calculation row and the explicit distinction between finite generated-support proof and unmeasured substrate ontology. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-finite-loop-law-versus-PDG-radiative-calculation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-finite-loop-law-versus-PDG-radiative-calculation`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:35852c08e6181f510664f37a118a27d5f86feab3a206aff97c02bd57aac8a6f5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b5b8ef99f7d0a4cc654db8a5eb66e0aa622e55639d04e071250335987128259b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a26bebbee3de346e70b9e9276f52f8578a6e1b526f11c327161c72c6075963d6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1a57452db5900cce3c3d0f28b850c3879cdbe6e6b02c2c10c2711d9e99b5cb2f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:38ed69fb363332a574141c377c615d1d1b361d8aa9ed4d819bc02f97062caed0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:cee6eec1e3e0d2658873ad73424245540f059ef095c92c0fcd4dd2af20c98b0a`. Independent certificate: `sha256:3e424a2707ab266aaa10dd700ffd886b615109152be56d64a7c07c9f0ed79684`. Engine external-validation hash: `sha256:49af49f6f4188f70806f08a77e123d1f6fe99ffb0ab2f474028c930a09678744`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Observed comparison records:

- RELATIVISTIC-FIELD-LOOPS: predicted finite-renormalized-radiative-calculations-confirmed__floored-loop-law-is-a-formal-finite-support-boundary-not-a-direct-measurement-of-continuum-nonexistence; observed finite-renormalized-radiative-calculations-confirmed__floored-loop-law-is-a-formal-finite-support-boundary-not-a-direct-measurement-of-continuum-nonexistence; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: A generated finite exact loop census diverges, an unregistered subtraction is required inside the Fold proof, or the validation misstates a conventional calculation as direct measurement of substrate ontology.

Measurement receipt: `sha256:4eaf857db79a6c2b3f7fddd76aa2c89fe5adbaac630bdbd59e0cb5d5cc6b084e`. Isolation certificate: `sha256:21f6f042af802c0d1f75c561a16df55d9de2450d86b7bba3e71ec8912d162fdf`. Custody certificate: `sha256:596ba8104c01280f799ec42a5498248f1c2e9a15aae7f8a2a3bd2a53f221a4e1`.

Registered source descriptions:

- NIST, CODATA, Particle Data Group and American Physical Society: https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml; https://physics.nist.gov/cuu/Constants/Table/allascii.txt; https://doi.org/10.1103/PhysRevLett.26.721; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf; https://www.nist.gov/news-events/news/2019/12/new-playbook-interference; https://www.nist.gov/publications/nonlinear-networks-arbitrary-optical-synthesis; https://tf.nist.gov/ofm/synthesis/ppln.htm (sha256:97a7631d956cc2eb8142e64603a1f51ece76944195274a28bbe91498edae0308)

Registered target identities:

- RELATIVISTIC-FIELD-LOOPS from RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no fitted counterterm in the Fold derivation
- no claim that PDG directly measured substrate finiteness
- no completed infinite support admitted

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:ae266be4c577388a1f5461b4e0ed5c97b608597e76ef6bd23f76ff9350caeac8`; engine receipt `sha256:78f1d28a0a4ca09b87ef08f5ebc95af026e2d90f372a48ddb5d735cdf21140b3` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-FINITE-LOOPS-003-78f1d28a0a4ca09b.json`; empirical-validation hash `sha256:f28d4682eaa02b33f1520fb83f2cde1e56b281e09509835f0eff23cdfc742300`; measurement receipt `sha256:4eaf857db79a6c2b3f7fddd76aa2c89fe5adbaac630bdbd59e0cb5d5cc6b084e`.

328. Post-seal gravity, clock and equivalence comparison

Claim identity: `SFT-PHYS-VALIDATION-GRAVITY-CLOCK-EQUIVALENCE-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. NIST/JILA resolves gravitational clock redshift at millimetre scale and CNES MICROSCOPE tests universal free fall at order 10^-15. These observations support the sealed clock/equivalence classes but do not measure the particular Fold fractions 7/16 or 3/4.

> *Gravitational clock-rate/redshift and inertial-gravitational equivalence phenomena are confirmed; the specific Fold fractions remain structural coordinates rather than measured universal values.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-GRAVITY-STATIC-CLOCK-003`
- `SFT-PHYS-GRAVITY-REDSHIFT-EQUIVALENCE-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal weak-gravity, clock and equivalence comparison product. The declared exact boundary is: Every NIST clock-gradient and CNES equivalence row plus the exact Fold-normalization controls. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-gravity-clock-laws-versus-complete-NIST-CNES-vector__source-bound-proof-trace__capability-closed-prediction__se` parate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-gravity-clock-laws-versus-complete-NIST-CNES-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:a8bbbaa4a43b5a9f899f5a439e6d4f901590f22b341bc07db6e477e3de9736de. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:8869eb7a90f98167a950339bc2b51ea61c87b6af3661a8370a30338246de75a0.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:28558fed02d54eaad97cc6dd65125a9c4127f1444349db0064a997df449c9380.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:41290050eddc2615b2ff0f73eedcfb89d7618691694eec55dee79db31f516ddf.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:0d708207c49bc0c4b4ede688a3cc38145a4030127d647fe54e270e80e17c5282.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:746c57e9ab79b21abdf9eedbe78954bd318e6bbbc3f46d7d32a7718f0531dc0c. Independent certificate: sha256:c430048665501db6da2564a13a9f038aaf173e53863f3fdc12ce25c6e62b5310. Engine external-validation hash: sha256:7fa27d40680cd8a911b8bf698a63336c9f78f82a42bbf5a65b7b127f6704d741.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Observed comparison records:

- GRAVITY-SPACETIME-CLOCK: predicted inverse-square-source-flux-clock-redshift-and-equivalence-phenomena-confirmed__specific-Fold-fractions-structural; observed inverse-square-source-flux-clock-redshift-and-equivalence-phenomena-confirmed__specific-Fold-fractions-structural; exact match True

- deliberately tampered unfavourable control rejected

Falsification condition: Controlled clocks fail to exhibit source-correlated gravitational redshift, free fall becomes composition dependent outside the registered uncertainty, or a scope row/source identity is omitted or altered.

Measurement receipt: `sha256:41515f53cfa1de4ed5421d339f11d336e066c1c8000e13993a461920f7480278`. Isolation certificate: `sha256:c6e797950defd830f7317d909c8d4d985fd2fe17dbb0ef5da9055d7a04c6bb5f`. Custody certificate: `sha256:dbf4b3cfa10209f0cf864b81bcf82e60ff992232b35a84b65ab798aa7dc3c811`.

Registered source descriptions:

- NIST/JILA, CNES MICROSCOPE, LIGO-Virgo-KAGRA/GWOSC and Event Horizon Telescope: https://www.nist.gov/publications/resolving-gravitational-redshift-across-millimetre-scale-atomic-sample; https://cnes.fr/en/projects/microscope; https://dcc.ligo.org/P1700308/public; https://dcc.ligo.org/P2000091/public/; https://eventhorizontelescope.org/press-release-april-10-2019-astronomers-capture-first-image-black-hole; https://gwosc.org/api/v2/catalogs (sha256:b64325355bd993ab32cff6c088023dbe7911ab13278b9439466cd0ae56bedd5e)

Registered target identities:

- GRAVITY-SPACETIME-CLOCK from GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no target access before formal seal
- no claim that NIST measured a universal 3/4 clock factor
- no omission of MICROSCOPE mission precision or scope

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:e5d30d720a64aeddc8384c352d34478ad589f7f981333e46952c0a3f1bedd50f`; engine receipt `sha256:a78e700c5fd4bed0bafe60270e978e25166dcc20ddc31b7d8c62efb6daf15040` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-GRAVITY-CLOCK-EQUIVALENCE-003-a78e700c5fd4bed0.json`; empirical-validation hash `sha256:bbb9e18db57c2672b33ecac7b8a31ecc75fd5d9c18a747a6c88243bac2988149`; measurement receipt `sha256:41515f53cfa1de4ed5421d339f11d336e066c1c8000e13993a461920f7480278`.

329. Post-seal gravitational-wave speed, polarization and quadrupole comparison

Claim identity: `SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVES-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The GW170817/GRB170817A measurement bounds relative gravity/light speed between -3 x 10^-15 and +7 x 10^-16; LIGO-Virgo catalogue tests support tensorial polarization and quadrupolar strong-field modes. Frequency/source coverage remains finite and explicit.

> *Gravity/light common speed is confirmed at the reported multimessenger bound; tensor polarization and quadrupolar strong-field support are confirmed within the registered catalog analyses.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`

- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal gravitational-wave comparison product. The declared exact boundary is: Every registered multimessenger speed interval, polarization, quadrupolar ringdown and finite-coverage control row. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-gravity-wave-law-versus-complete-LIGO-GWOSC-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-gravity-wave-law-versus-complete-LIGO-GWOSC-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:e73a047d1d3894d18fbf4c551d0a686bd0113799a0032774906516465ec006d3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1e0778a3e05b6137c48e1e5f892282830cb4edd8866adf2a2f34990090c923d6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:133903c972c6d02baf629cbd3b5d9099b1c4fda9429a5c8a128f1c8395957b6e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8b2b76b02216f9689ff132c0a33fb4e40bce4910a18392dc6b9b535907d1e7fe`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9a9031b8d09c8940619de30102e51f40f799521eee6d31404529f214d3735d25`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5c372a5e07b98f86aafbeec78736b78a8936cc9ababbb60075fdc68dcd1c2443`. Independent certificate: `sha256:c80726d2bcd9e2bd7673f06100406c75ce596b18244ed4808a34367b445b5e09`. Engine external-validation hash: `sha256:fa9ce852c9dc59c4ee45d70d3f88661c266d3c048fd7827b71153e1bbadae496`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Observed comparison records:

- GRAVITY-SPACETIME-WAVES: predicted gravity-light-speed-equality-confirmed-at-multimessenger-bound__tensor-polarization-and-quadrupolar-strong-field-support-confirmed; observed gravity-light-speed-equality-confirmed-at-multimessenger-bound__tensor-polarization-and-quadrupolar-strong-field-support-confirmed; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: A replicated multimessenger event establishes gravity propagation outside the registered One-speed class, a non-tensor mode replaces the two-tensor class, quadrupole support is absent, or any adverse/scope row is deleted.

Measurement receipt: `sha256:6efde3176b15af0b637ff04ce1694cb6fb068fc48b73fd0dc9f62e0dc2e19996`. Isolation certificate: `sha256:00fea139f60f2689b6b27393a2c9ac11a54a68c2d9e2b5a7f1dac501057a369c`. Custody certificate: `sha256:805122281381faa718e430498f1e2385023535b3fa9188c148ed50d7e26b88b4`.

Registered source descriptions:

- NIST/JILA, CNES MICROSCOPE, LIGO-Virgo-KAGRA/GWOSC and Event Horizon Telescope: https://www.nist.gov/publications/resolving-gravitational-redshift-across-millimetre-scale-atomic-sample; https://cnes.fr/en/projects/microscope; https://dcc.ligo.org/P1700308/public; https://dcc.ligo.org/P2000091/public/; https://eventhorizontelescope.org/press-release-april-10-2019-astronomers-capture-first-image-black-hole; https://gwosc.org/api/v2/catalogs (sha256:b64325355bd993ab32cff6c088023dbe7911ab13278b9439466cd0ae56bedd5e)

Registered target identities:

- GRAVITY-SPACETIME-WAVES from GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no omitted side of the speed interval
- no assertion that all extra modes are excluded at all frequencies
- no target-selected quadrupole law

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:50a6acf7e4e114d7bd6013065229eca023f2b0b4358f61893911118382d2c76d`; engine receipt `sha256:627976ac082693a34548353be3a26eae93d6804108d549a4ab6411c0575f70ea` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVES-003-627976ac082693a3.json`; empirical-validation hash `sha256:903b4e0d071f2cd69f0fbeb0a081ebbdc31a9a6f345ff5c2ebdc756447c05f04`; measurement receipt `sha256:6efde3176b15af0b637ff04ce1694cb6fb068fc48b73fd0dc9f62e0dc2e19996`.

330. Post-seal horizon and information-boundary comparison

Claim identity: `SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. EHT independently reconstructs a persistent horizon-scale shadow and LIGO-Virgo observes compact strong-field ringdown. Neither body directly measures the event horizon, Hawking temperature, singularity avoidance or complete information reconstruction; those distinctions remain explicit.

> *Horizon-scale compact strong-field shadow and ringdown support are confirmed; direct horizon imaging, normalized thermal carrier, Hawking radiation and information reconstruction remain unmeasured predictions or formal boundaries.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003`

- SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal horizon and information comparison product. The declared exact boundary is: All registered EHT shadow, directness, LIGO strong-field and unmeasured Hawking/information rows. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-horizon-laws-versus-complete-EHT-LIGO-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-horizon-laws-versus-complete-EHT-LIGO-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:2cde5bb0238d8ffa9f8e222b4e115e0ec561fe6acbd58d2bdc0a6afd6c1de323. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:29258b931ec34e43228045f1db5c68d63c453b106deb41923301ba152a605ec5.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:bfb9f4f4aae6574ceaaedbe65e5de14f752a4e1e191ba61ea447247ed5078468.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:6c0340298be8d1400af8ca367d0ce0ecc81f553568034cfb076e55bacff931e5.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:852b4c711de58ac6519603d3c242e0d57f9ca68425fd65095d9b87d33f46c9d9.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:9fa4c0074e72a04672cfd6919cde925d4a626a6c6f17eaa8269bf002df996e23. Independent certificate: `sha256:ebf753556858424cab0a30c3de00762beb533baf68ac69815b7c88d9ed3af770`. Engine external-validation hash: `sha256:f3e2f08bd58a00f1e1c19da4096460082ba35507e209e15830321fb506ec669b`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Observed comparison records:

- GRAVITY-SPACETIME-HORIZONS: predicted horizon-scale-shadow-and-compact-strong-field-support-confirmed__event-horizon-Hawking-temperature-and-information-reconstruction-not-directly-measured; observed horizon-scale-shadow-and-compact-strong-field-support-confirmed__event-horizon-Hawking-temperature-and-information-reconstruction-not-directly-measured; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Independent horizon-scale observations cease to support compact shadow/ringdown structure, or an indirect/model-dependent record is mislabeled as direct Hawking, horizon or information-recovery measurement.

Measurement receipt: `sha256:e4efcfa0509c550a7e72e0eb4844bd51b522e3503731cad6538da738a6d50579`. Isolation certificate: `sha256:9f95ac2d4325e277e0097b75d5b7ae7a8d91ec71e9835ba4987c105dfa3628b4`. Custody certificate: `sha256:46468310e28e92780d56342c020a55b8010f6804bb7e2611c50b10fa73f0b755`.

Registered source descriptions:

- NIST/JILA, CNES MICROSCOPE, LIGO-Virgo-KAGRA/GWOSC and Event Horizon Telescope: https://www.nist.gov/publications/resolving-gravitational-redshift-across-millimetre-scale-atomic-sample; https://cnes.fr/en/projects/microscope; https://dcc.ligo.org/P1700308/public; https://dcc.ligo.org/P2000091/public/; https://eventhorizontelescope.org/press-release-april-10-2019-astronomers-capture-first-image-black-hole; https://gwosc.org/api/v2/catalogs (sha256:b64325355bd993ab32cff6c088023dbe7911ab13278b9439466cd0ae56bedd5e)

Registered target identities:

- GRAVITY-SPACETIME-HORIZONS from GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no claim that EHT directly photographed the event horizon
- no empirical Hawking-radiation claim
- no claim that information reconstruction was observed
- no omitted model/reconstruction boundary

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d0aa61b617d79a034c1f8da2407ddb376728774e481bbddaa51681642bd6995d`; engine receipt `sha256:5fab7cea569fcb273a9b39aacdd824a37ed42f3939fd494517dcb62619a95752` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003-5fab7cea569fcb27.json`; empirical-validation hash `sha256:bee619b212faa3ca3d6b4f2c3f2d89c99915f4bb2208db324414e18bdc726f4e`; measurement receipt `sha256:e4efcfa0509c550a7e72e0eb4844bd51b522e3503731cad6538da738a6d50579`.

331. Post-seal nonstandard-spacetime realization boundary

Claim identity: `SFT-PHYS-VALIDATION-NONSTANDARD-SPACETIME-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The sealed wormhole, warp and closed-timelike results are admissibility theorems. The complete registered authority vector contains no confirmed physical realization of any of the three, and current catalogue tests report no nonstandard strong-field signal in their stated analyses.

> *Wormhole, warp and exact closed-timelike admissibility boundaries are formally sealed; no authoritative realization appears in the registered source vector, so all remain unconfirmed physical frontier predictions.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-WORMHOLE-ADMISSIBILITY-003`
- `SFT-PHYS-SPACETIME-WARP-ADMISSIBILITY-003`
- `SFT-PHYS-SPACETIME-CLOSED-TIMELIKE-ADMISSIBILITY-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal nonstandard-spacetime status product. The declared exact boundary is: Every formal admissibility result, registered authority search row, non-detection scope and explicit no-realization statement. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-admissibility-laws-versus-complete-authority-status-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-admissibility-laws-versus-complete-authority-status-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:09480d6700873a056fb513b704c4562df413783fc61c62db6ca6a2a4f90f4cd8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:dd60f9ca7ba68318cd1e3d20ddac505e14dcce896d8c98c561b2c1cc337cd140`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt

`sha256:efcc45c9309466ae7d5dd96f081b24c3641fdfc3cb04a6ce867fe12e915314b8`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:62a8b7f46e6309005d43d3e4e6adcd0c71218b0c14e2f847563b2e0323b044f0`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:12757404cd5714f353e09c4f931bbdae7170e9ec81dc147d5166f531341bc1d3`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c0bbff7c5e0def998f350465d0fb0f4d3153c41fbd1381a4d711bc6e606c97ed`. Independent certificate: `sha256:62d11fef2c37c2f55f0e333b56d2ef6ef4a286d563ab5ec3fded3f436bfaaff3`. Engine external-validation hash: `sha256:f000289f07465eaa9cbe8321efaaffe355f7215c828c50e3a12cff74f62a9bfa`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Observed comparison records:

- GRAVITY-SPACETIME-NONSTANDARD: predicted formal-admissibility-boundaries-sealed__no-authoritative-wormhole-warp-or-closed-timelike-realization-in-registered-source-vector; observed formal-admissibility-boundaries-sealed__no-authoritative-wormhole-warp-or-closed-timelike-realization-in-registered-source-vector; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: A controlled authoritative experiment realizes one of the structures but violates its sealed positive-source, causal or complete-return ledger, or the validation promotes formal admissibility to observed existence without evidence.

Measurement receipt: `sha256:d166b6cfc4d4000c2025621fcdad83e477f8b85b5c6199d1b9e965fad178752e`. Isolation certificate: `sha256:fe85349cb6ad49279d3a21b783f6c053b7759678ce643e64628800a9d1178025`. Custody certificate: `sha256:d73e9908b74067e724e25af60e5d5fbdc91d495f52ddadd5d385f95a3c95dd04`.

Registered source descriptions:

- NIST/JILA, CNES MICROSCOPE, LIGO-Virgo-KAGRA/GWOSC and Event Horizon Telescope: https://www.nist.gov/publications/resolving-gravitational-redshift-across-millimetre-scale-atomic-sample; https://cnes.fr/en/projects/microscope; https://dcc.ligo.org/P1700308/public; https://dcc.ligo.org/P2000091/public/; https://eventhorizontelescope.org/press-release-april-10-2019-astronomers-capture-first-image-black-hole; https://gwosc.org/api/v2/catalogs (sha256:b64325355bd993ab32cff6c088023dbe7911ab13278b9439466cd0ae56bedd5e)

Registered target identities:

- GRAVITY-SPACETIME-NONSTANDARD from GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no inference from formal admissibility to existence
- no UAP or anecdotal evidence
- no unlimited no-go conclusion from a finite source search
- no deletion of the standing prediction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:53d5fa3ce932f20ef4bfec2bd595c9fe03bf1c4c26da4db987db0a940e61d354`; engine receipt `sha256:58e0992e82b5379cb948eeed83b8ea8babc833386599c5c8ab93eafbb4705fb3` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-NONSTANDARD-SPACETIME-003-58e0992e82b5379c.json`; empirical-validation hash `sha256:459b0d490228bf837cd3cd95393fd72bbbd27b51242591f1cbbdda546b2343d2`; measurement receipt `sha256:d166b6cfc4d4000c2025621fcdad83e477f8b85b5c6199d1b9e965fad178752e`.

332. Quark dressing and CKM comparison

Claim identity: `SFT-PHYS-VALIDATION-QUARK-CKM-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Exact polynomial enclosures and terminal dressing are tested against complete PDG quark and CKM intervals; the exact top/charm result remains a closed prediction because no direct scheme-matched measurement is registered, while the CKM s23 comparison is retained.

> *Dressed s/d and b/s overlap their complete PDG intervals; the sealed exact t/c value is a prediction awaiting a direct scheme-matched test; CKM s12, s13 and J overlap three-sigma support while s23 does not.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-QUARK-CUBICS-003`
- `SFT-PHYS-MATTER-QUARK-DRESSING-003`
- `SFT-PHYS-MATTER-CKM-PHYSICAL-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal quark dressing and CKM comparison product. The declared exact boundary is: Every registered source row, uncertainty, scheme/status limitation and unfavourable outcome in this comparison. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-formal-laws-versus-complete-matter-flavour-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-formal-laws-versus-complete-matter-flavour-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:e437cc377ece2d398c187dc2fb3983636757335d67429e3c29f168554da7ae41`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:846dee47e1fba80298a7651c60c0829177b6c77cd558fb2ae358b05406c5eacb`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:da90a8405025cef7a23552e52aab7493542cbdff40d4cbc38f4ac025f83a6dcc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3c7895fc1ab1c63204ecc1b82f0d8d07138378a80a187421533a067e654e0f7a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8ea84eb8cce6629530492c992a373ae73825564f5396d58b54a4b8410a0b32d9`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:26a82e973fc68e78eedccbd187e44f0c6a90513b78e58396ca34acefae9fa805`. Independent certificate: `sha256:0d8d362cd5df041519bf9da5a5bb58227a23719e106e8afc573cf7302f20e579`. Engine external-validation hash: `sha256:73eab5eb22f7b11ac8d9794210efe740e0148636468bb62a395bd4c9a4bad896`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Observed comparison records:

- MATTER-FLAVOUR-QUARK-CKM: predicted dressed-sd-and-bs-inside-complete-PDG-intervals__tc-is-a-sealed-exact-prediction-with-no-registered-direct-scheme-matched-measurement__CKM-hierarchy-and-s12-s13-J-supported-with-s23-three-sigma-tension-retained; observed dressed-sd-and-bs-inside-complete-PDG-intervals__tc-is-a-sealed-exact-prediction-with-no-registered-direct-scheme-matched-measurement__CKM-hierarchy-and-s12-s13-J-supported-with-s23-three-sigma-tension-retained; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Any sealed passing ratio leaves its complete interval, the t/c prediction is weakened because no exact comparator exists, the s23 result is omitted, or source custody changes.

Measurement receipt: `sha256:8ec8d4e93cfe6a4f80844aebe97c082c19d37b99fa5810a3ef1c0169a6d7e79f`. Isolation certificate: `sha256:5e751c6db8c92905c3152239a0bf98d7192bc635442d7b139977f3bb8b9ec974`. Custody certificate: `sha256:b61c13b133d149a066bf82cb0a80a0ed0b5d865f65db7086482455d305685e25`.

Registered source descriptions:

- NIST/CODATA, Particle Data Group, Fermilab Muon g-2 Collaboration and Planck Collaboration: https://physics.nist.gov/cuu/Constants/Table/allascii.txt; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf; https://muon-g-2.fnal.gov/result2025.pdf; https://arxiv.org/abs/1807.06209 (sha256:6e56fcdd94371ebf396b9600a1431fd448075f69073572b9cb0660959329b9fe)

Registered target identities:

- MATTER-FLAVOUR-QUARK-CKM from MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no target access before seal
- no omission of CKM s23 tension
- no relabelling an unmeasured prediction as incomplete
- no cross-scheme top/charm comparison
- no central-value-only comparison

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:bb70efd021393c212e392f33c03bae2b37db63c68c6b6c2c83484acbc5820567`; engine receipt `sha256:42b3d23953c8914d3affd7385c6f52158dca86b01131a4748c45d00600480402` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-QUARK-CKM-003-42b3d23953c8914d.json`; empirical-validation hash `sha256:4e223023e5cc66ccaf59dc8dd70d628e9d7c3e35cca9ff01267404dcbe1a6e5c`; measurement receipt `sha256:8ec8d4e93cfe6a4f80844aebe97c082c19d37b99fa5810a3ef1c0169a6d7e79f`.

333. Proton/electron precision comparison

Claim identity: `SFT-PHYS-VALIDATION-PROTON-ELECTRON-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The sealed algebraic proton/electron interval is compared with the complete CODATA value and uncertainty; the non-overlap is retained as an unfavourable result.

> *The V3 reconstruction is near the 1836 scale but lies outside the complete CODATA uncertainty interval and is not empirically closed to CODATA precision.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-MATTER-PROTON-ELECTRON-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal proton/electron precision-comparison product. The declared exact boundary is: Every registered source row, uncertainty, scheme/status limitation and unfavourable outcome in this comparison. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-formal-laws-versus-complete-matter-flavour-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-formal-laws-versus-complete-matter-flavour-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:45d60e8833350918e2c950fc97f8d37e898a64feee5eee1ad98b6797e1fdd6db`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6d0190e3d472ec90764d25a126a2f1bc87326b6d2e7f142601f3af753b1e838a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:477e008d76e7cda64465017561e232f7c3e749fceb3c58baedbbb219028afd72`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bae3bc9a2b81f87ca9366741449fce8cd3d8cafc7fbdbd630adb71ad71e133ee`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:97a1f46a7a108f7b25ba1c6e20fb1bb996bbc639a7d314f3e8a9785f222d4f69`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b10afaa17622b82df307aa9fc2bd74176fba10e786a4836faefcffebb7595ed1`. Independent certificate: `sha256:bc01921ec3b65a815ef82ca613e604250e945f4add782ecbb1af4b8fba613798`. Engine external-validation hash: `sha256:0cf348da35a65bb3dc7500569344fceebc09fd659503cabfbd67ffad6bcefcf9`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Observed comparison records:

- MATTER-FLAVOUR-PROTON-ELECTRON: predicted sealed-proton-electron-reconstruction-near-scale-but-outside-complete-CODATA-uncertainty__unfavorable-result-retained; observed sealed-proton-electron-reconstruction-near-scale-but-outside-complete-CODATA-uncertainty__unfavorable-result-retained; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: A lawful separately sealed successor overlaps the complete CODATA interval, or this validation falsely labels the current non-overlap as a pass.

Measurement receipt: `sha256:a9e00f3e412a4d1a8ea7b55e949a60e5372cdb8550bf8ee10e23ac6373a78cdc`. Isolation certificate: `sha256:0e8d6a0065d9da0eaae0ac9b8c02b5792a58a2c69b821f337594131fac6dbfe0`. Custody certificate: `sha256:fc7df817e56da899939f319b04caccf936b3e86a5f79004ee96485f2d5fd46d8`.

Registered source descriptions:

- NIST/CODATA, Particle Data Group, Fermilab Muon g-2 Collaboration and Planck Collaboration: https://physics.nist.gov/cuu/Constants/Table/allascii.txt; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf; https://muon-g-2.fnal.gov/result2025.pdf; https://arxiv.org/abs/1807.06209 (sha256:6e56fcdd94371ebf396b9600a1431fd448075f69073572b9cb0660959329b9fe)

Registered target identities:

- MATTER-FLAVOUR-PROTON-ELECTRON from MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no enlarged uncertainty
- no relative-error-only pass label
- no fitted successor
- no omitted unfavourable result

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:63cca8ba830b458087d00df07b520031f1f0119cb9f638153b67f4ae835ccf30`; engine receipt `sha256:c9c81674c4671965244d499e7846cfd6154e3b256db1aaf0e14ef32e58972292` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-PROTON-ELECTRON-003-c9c81674c4671965.json`; empirical-validation hash `sha256:f3788d7eb324b464d9951f4a229b270b543735a722747930f08b0bb096e054f5`; measurement receipt `sha256:a9e00f3e412a4d1a8ea7b55e949a60e5372cdb8550bf8ee10e23ac6373a78cdc`.

334. Neutrino mass, ordering, PMNS and CP comparison

Claim identity: `SFT-PHYS-VALIDATION-NEUTRINO-MASS-MIXING-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The positive mass-square coefficients, splitting law, PMNS triple and CP carrier are translated with one declared solar anchor and tested against complete PDG, KATRIN and Planck rows.

> *All PMNS/CP predictions lie in registered three-sigma support; the positive normal mass sum is about 0.060 eV and remains below current direct and cosmological bounds; normal ordering preference is not yet decisive.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-NEUTRINO-SPLITTING-003`
- `SFT-PHYS-NEUTRINO-POSITIVE-MASS-003`
- `SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal neutrino mass, ordering, PMNS and CP comparison product. The declared exact boundary is: Every registered source row, uncertainty, scheme/status limitation and unfavourable outcome in this comparison. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-formal-laws-versus-complete-matter-flavour-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-formal-laws-versus-complete-matter-flavour-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:956df6479052d1022da663f4f4d200c8b54ca8c4fc36db9c9fcf9a51d6dd3776`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c92895443f7bc85e324ade7e44c9b40903008f7f367c172b150cdc7fcce8256f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ae026dd7454b3249e6388a3646d93ef4f01e128df0434a45b0bee838f77de17f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9b9807b82f27cbcef3074acfd79b2b4b23876f08485cd79466fe5bb25c6463f3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:02f5d3478ccdcda433027aade2e27368808215ea87b1b4e410329e9fc6bc580e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b7a37f21b1d6bacff432eb0e3ad7a4869ad202257c8e3686a50ad140e568a96f`. Independent certificate: `sha256:00ce7b894cbf5c8432010ccc66fec2431b33f18e4d31aec65e9b845217f854a7`. Engine external-validation hash: `sha256:c22c14ffe4d1ee602eaef658b96626d4c0148c1c3ae10b5910004609fa1cda26`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Observed comparison records:

- MATTER-FLAVOUR-NEUTRINO: predicted positive-normal-neutrino-mass-structure-PMNS-and-CP-inside-registered-three-sigma-support__mass-sum-below-direct-and-cosmological-bounds__ordering-preference-not-decisive; observed positive-normal-neutrino-mass-structure-PMNS-and-CP-inside-registered-three-sigma-support__mass-sum-below-direct-and-cosmological-bounds__ordering-preference-not-decisive; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: A PMNS/CP value leaves the registered support, the positive mass sum exceeds a robust bound, inverted ordering becomes decisive, or an uncertainty/scope row is altered.

Measurement receipt: `sha256:d313c3a6c8e4375ce78dd8297f89ab47ec234b1cc79aca2f342ae5ec676ecbb3`. Isolation certificate: `sha256:02dadaa1d44d256e16feb89f5f5dade878890ecf7f935c40515f70687c56c5be`. Custody certificate: `sha256:f4507f3f3e537c7a60298ef96428f4fc529720de0f64556d7197921b0b4d471b`.

Registered source descriptions:

- NIST/CODATA, Particle Data Group, Fermilab Muon g-2 Collaboration and Planck Collaboration: https://physics.nist.gov/cuu/Constants/Table/allascii.txt; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf; https://muon-g-2.fnal.gov/result2025.pdf; https://arxiv.org/abs/1807.06209 (sha256:6e56fcdd94371ebf396b9600a1431fd448075f69073572b9cb0660959329b9fe)

Registered target identities:

- MATTER-FLAVOUR-NEUTRINO from MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no numerical-zero lightest mass
- no claim of decisive measured ordering
- no hidden second scale anchor
- no omitted model dependence of cosmological bound

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:2b749f0f0f535e6456f627c1f3a5caa9b0e7c1245fa9f19f92fef42278e731e3`; engine receipt `sha256:f7ce6f015aa32501a5e90ce65ee001c92b60d9c3543a9b062607c65aa5b84d8f` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-NEUTRINO-MASS-MIXING-003-f7ce6f015aa32501.json`; empirical-validation hash `sha256:3e4f78efa659744b5ea9af78fccb81c4528d118ca50be628c3c90fb69c8738e5`; measurement receipt `sha256:d313c3a6c8e4375ce78dd8297f89ab47ec234b1cc79aca2f342ae5ec676ecbb3`.

335. Majorana and neutrinoless-double-beta status comparison

Claim identity: `SFT-PHYS-VALIDATION-MAJORANA-ZERO-NU-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The formally closed Majorana discriminator and noncancellation prediction are tested against the complete current non-observation and effective-mass limits.

> *Zero-nu-beta-beta is not observed; the sealed positive effective-mass interval remains a closed prediction below current sensitivity and is neither confirmed nor excluded.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-NEUTRINO-MAJORANA-003`
- `SFT-PHYS-NEUTRINO-ZERO-NU-BETA-BETA-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal Majorana and neutrinoless-double-beta status-comparison product. The declared exact boundary is: Every registered source row, uncertainty, scheme/status limitation and unfavourable outcome in this comparison. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-formal-laws-versus-complete-matter-flavour-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-formal-laws-versus-complete-matter-flavour-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ec64c093af8af3b48be0828a6ef06899733c5760e69adcdcf9824f7253f28889`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8d1bb5c25d0ff13d5fe019120e793686249afb2234daa9c816adc0a8d77e9efd`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:fb4766873e090d9a030659dcb6b1896147fc44baf27982def486dde01f768fef`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:2db951c6493b3b153515a51b84f651886421bc3fd102811451c410595a9bdb67`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6ae51fa07fffb6e327f689afeda0f17daa2bd53ded2995570da32d38c77b7859`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e8fe2066d6ded0fe0a91aafbdbbc5dca334aaa786299e75be2a4e3b9c2bcb6f9`. Independent certificate: `sha256:355ba11fd0e20414f29473b4bd2afccbf2d4eb72e8de83561418b98e66af444f`. Engine external-validation hash: `sha256:a81937546412f1f36f76b73b5064254b7dbe7c5d200eb31836483b3bd96a633e`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Observed comparison records:

- MATTER-FLAVOUR-MAJORANA-ZERO-NU: predicted Majorana-and-zero-nu-beta-beta-formally-forced-and-closed-as-predictions__not-yet-observed__predicted-effective-mass-below-current-experimental-upper-bound; observed Majorana-and-zero-nu-beta-beta-formally-forced-and-closed-as-predictions__not-yet-observed__predicted-effective-mass-below-current-experimental-upper-bound; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: A controlled search excludes the complete sealed effective-mass interval under validated matrix elements, a Dirac neutrino is established, or an observation violates the sealed structure.

Measurement receipt: `sha256:d5afa7aa2318b0008d0cca8dfaba1b4bc541d9ffc7a4863660dc20b57486441d`. Isolation certificate: `sha256:95b38574becd4e26b03a645bbbd1266e1f8f6c5cf1222b632e72e91bb5255f07`. Custody certificate: `sha256:f0b0bc231c389c23d83fe3c0c1c1ce488a3ef53bde38dd3e0cb7446c34fb7e00`.

Registered source descriptions:

- NIST/CODATA, Particle Data Group, Fermilab Muon g-2 Collaboration and Planck Collaboration: https://physics.nist.gov/cuu/Constants/Table/allascii.txt; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf; https://muon-g-2.fnal.gov/result2025.pdf; https://arxiv.org/abs/1807.06209 (sha256:6e56fcdd94371ebf396b9600a1431fd448075f69073572b9cb0660959329b9fe)

Registered target identities:

- MATTER-FLAVOUR-MAJORANA-ZERO-NU from MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no claim of existing observation
- no conversion of non-observation to disproof below sensitivity
- no omitted nuclear-matrix uncertainty
- no erased Majorana prediction

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:5dcca01925043b871a6a78a22018331e195848b1dc3e9cc584ebabf26b230516`; engine receipt `sha256:67bc72ab10d4e08f78495417337900a4543ad146d4c03c543e86d5e2ecc6851a` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-MAJORANA-ZERO-NU-003-67bc72ab10d4e08f.json`; empirical-validation hash `sha256:094e6e201e84e0195deb21a2df13bd4adaec42545ff400b1e366c559679eed1b`; measurement receipt `sha256:d5afa7aa2318b0008d0cca8dfaba1b4bc541d9ffc7a4863660dc20b57486441d`.

336. Electron and muon magnetic-anomaly comparison

Claim identity: `SFT-PHYS-VALIDATION-MAGNETIC-ANOMALIES-003`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The bare g, terminal-alpha leading structure and squared mass sensitivity are compared only with the raw NIST electron data and Fermilab final muon measurement; no consensus-model interpretation enters the validation path.

> *The currently sealed leading terminal-alpha/full-turn translation is compared with both raw measured anomalies; the raw measurements remain the sole empirical targets while a separately forced successor is constructed.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal electron and muon magnetic-anomaly comparison product. The declared exact boundary is: Every registered source row, uncertainty, scheme/status limitation and unfavourable outcome in this comparison. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-formal-laws-versus-complete-matter-flavour-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-formal-laws-versus-complete-matter-flavour-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:7be15683d06d05ae3e2c7c5709c1055840314ec58836f0db55712f7cc35e27ec`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:27242c74ceae811d2b7eb09aff8d37f8a58beae1d832c821447a43ba3a2f1081`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:860159bdbbb5be8f18d19c0d3c3a854df65124abccd24a164748ae6ef8b3556d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4e91c5ba807d898c0a7a9c41d0382f2e699418f8ec5debe349aa5bccb829d269`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:053abe2406ecb945558cd762f4f534a607dd1e9c6c9d93388faba2d8a6eb082d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5d03fba565167b06734b4c16c0094ae1dd9597e9883fc7315ea8ef5e14d53fe`. Independent certificate: `sha256:20684d12718d544efba49f0b5581e4f74d5fcb7e4b492a7886e95d72e32dc149`. Engine external-validation hash: `sha256:c38493c980a05122819d02ae03ca2fa14d2ced144d17973bbbdfdbdef2d73eab`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Observed comparison records:

- MATTER-FLAVOUR-MAGNETIC: predicted bare-g-and-terminal-alpha-leading-structure-compared-only-with-raw-electron-and-muon-measurements__consensus-model-interpretation-excluded; observed bare-g-and-terminal-alpha-leading-structure-compared-only-with-raw-electron-and-muon-measurements__consensus-model-interpretation-excluded; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: A complete separately sealed SFT radiative successor fails the measured intervals, or this validation mislabels the current leading-only result as the full anomaly.

Measurement receipt: `sha256:d3d13af47f19842657fd949dcf45fb8416fc1effc442819cd61c8f8a645551af`. Isolation certificate: `sha256:6f9f7dec660940a057eec9fce1a81699e1561fb93f9590739025df92d2892aaa`. Custody certificate: `sha256:0e4bca406095ed5e2285ec1ebf86ba5ea3c1cfc11cd883c6f02b64f616ab73d5`.

Registered source descriptions:

- NIST/CODATA, Particle Data Group, Fermilab Muon g-2 Collaboration and Planck Collaboration: https://physics.nist.gov/cuu/Constants/Table/allascii.txt; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf; https://muon-g-2.fnal.gov/result2025.pdf; https://arxiv.org/abs/1807.06209 (sha256:6e56fcdd94371ebf396b9600a1431fd448075f69073572b9cb0660959329b9fe)

Registered target identities:

- MATTER-FLAVOUR-MAGNETIC from MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported QED series
- no irrational proof value
- no exact-measured-anomaly claim from a leading term
- no consensus-model result as evidence
- no measurement-selected survivor

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:4230e200ef145aba63288bf2a082e5facfd575d9c10909b79e3708594f574a24`; engine receipt `sha256:c203e50408fe3b0bcf653ce46fea4e72b8947ddc0c8103054b1db3f8ab44dfdd` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-MAGNETIC-ANOMALIES-003-c203e50408fe3b0b.json`; empirical-validation hash `sha256:958c2a64e743a1b951c0238cbb3e94f467e7cb2b31d93cc08f72ca4e5ecb61b4`; measurement receipt `sha256:d3d13af47f19842657fd949dcf45fb8416fc1effc442819cd61c8f8a645551af`.

337. Post-seal NIST comparison of the forced six-neighbour cubic support

Claim identity: `SFT-PHYS-VALIDATION-ATOMIC-CUBIC-SUPPORT-004`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. After the exact Fold law independently forces all two held directions on all three generated spatial axes, the sealed coordination count six is compared with the complete registered NIST simple-cubic row.

> *The sealed Fold count 2*3=6 exactly equals the complete NIST simple-cubic nearest-neighbour count six.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ATOMIC-CUBIC-SUPPORT-004`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal exact-count, provenance, custody, row-retention and no-extra-rule product. The declared exact boundary is: All exact comparisons between the immutable six-neighbour Fold coordination and the sole complete registered NIST simple-cubic coordination row. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-six-neighbour-count-versus-complete-nist-row__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-six-neighbour-count-versus-complete-nist-row`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:67eead957f0ae32d6fcf389fd57c9a03581e98a3173095ab817255cbd0e81573`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ff7b9a5b78ee1aaf141fcf80b1d1f1b307914a8907497aa715965e3e9c460a1c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3d80082d15c527741aa56150b21d49ca736a087bd55730012d66ed3f3d624d7c`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f4620a5cb5a3e690961b955ade1e2e4a143f7902c0bf9964160b599364b2b636`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:51a7e2f92614bd6de1faefce57551ac589fe2d21d98f3604b2494c301e380b15`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:997bf380310d08236df0976f30b8115dcd01bbedc9efe4c17c821651f3a650dd`. Independent certificate: `sha256:fd3e9a21c669f14d0e5cc2c5d2ae01c1e70e3664ab42644dac4fd475c2a11fcf`. Engine external-validation hash: `sha256:d8de0b6cd8fd28be251f30eb271c6cfb361eeef6ba6aa4279d0a7125976fe7da`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-NCNR-DCS-SUPERFLUID-HELIUM

Observed comparison records:

- NIST-SIMPLE-CUBIC-NEAREST-NEIGHBOURS: predicted sealed-six-neighbour-cubic-support-exactly-matches-complete-nist-row; observed sealed-six-neighbour-cubic-support-exactly-matches-complete-nist-row; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The registered NIST row is not six, the formal count changes, any source hash changes, target access precedes sealing, or a tampered row is accepted.

Measurement receipt: `sha256:4685784e776563df5bcc95df8e79f4ae3fe640cbfc47520eded62c7b8f8fa06e`. Isolation certificate: `sha256:125acaf4ee56292ac158d2cb8031170df39a6af8a8e0e76fe23ab0ed8393bb55`. Custody certificate: `sha256:9bf4d45e86d752e4be035be15041893908cc8eded8913152edac08a8801b94a2`.

Registered source descriptions:

- National Institute of Standards and Technology, NIST Center for Neutron Research: https://www.nist.gov/document/dcs (sha256:ab723f91d6658bd8b5d19889884fd87e07046d2b0060e23f7df8d7cb6c35fa5b)

Registered target identities:

- NIST-SIMPLE-CUBIC-NEAREST-NEIGHBOURS from NIST-NCNR-DCS-SUPERFLUID-HELIUM

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no NIST value accessible to the formal law
- no imported lattice premise
- no selected neighbour subset
- no fitted count or extra rule

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:cd1893e275d850316ff2a34f1c7b1ccb11c310ec623321e885ddb6d5cf4ae7a4`; engine receipt `sha256:0086a149cdb14e727bfd35a6df637e53220728b2d137a8d442a6009dc4c586fc` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-ATOMIC-CUBIC-SUPPORT-004-0086a149cdb14e72.json`; empirical-validation hash `sha256:bcbbeb1712b7db5e70b25f5839014c1a5a16c1e6c212426348963d6b8c401fc8`; measurement receipt `sha256:4685784e776563df5bcc95df8e79f4ae3fe640cbfc47520eded62c7b8f8fa06e`.

338. Post-seal NIST comparison of the exact hydrogen spectral ladder

Claim identity: `SFT-PHYS-VALIDATION-ATOMIC-HYDROGEN-SPECTRUM-004`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The NIST hydrogen ionization wavenumber is retained solely as the dimensionful input carrier. The already sealed exact Fold gaps three-quarters and five-thirty-sixths are applied without adjustment, then compared with the complete NIST Lyman-alpha and Balmer-alpha measured intervals.

> *The sealed ratios predict Lyman-alpha 82259.078775 cm^-1 and Balmer-alpha 1096787717/72000 cm^-1; these lie respectively inside the complete NIST intervals [82259.02, 82259.30] and [15233.14, 15233.28] cm^-1.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal carrier-times-gap, exact-interval, provenance, custody, complete-row and no-extra-rule product. The declared exact boundary is: All exact post-seal applications of the immutable Fold Lyman-alpha and Balmer-alpha ratios to the one registered NIST hydrogen ionization carrier, compared with both complete reported line intervals. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-hydrogen-gaps-versus-complete-nist-line-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-hydrogen-gaps-versus-complete-nist-line-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:bd694a9355abc5d16e4a4af1cc9c6360e1b33c79b48465919ebfc78d7d341e6c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b558c1ca9f5a44eefe936b86328aa16428f847084bb5e7d8c9828c00e87ab248`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:45512783de9d059b64088e3a3ac4621d0453d407499f05f050a34f9020778bbd`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0f43226e7860b8fd91f5aa133fb59a05b21f227242c7b10cd6630269c7374fa6`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dbfe52d93dd3ac709f8cf3f48918fd95f4423e3940f9fa09334ff99c4fb584e6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:929cb21c17fc7f62f089d25a2cc709c3bf95d82dd73f4287a83a46d5a75f44b0`. Independent certificate: `sha256:b2b10cbd51128d459628d3701b6ad4e44d74fe307130d5f01aaf08870c0a0d0a`. Engine external-validation hash: `sha256:f3a8bf5687c553edb1f8b692676338797914e8c48b9a4af4cce73528d7b71da6`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-KRAMIDA-HYDROGEN-CRITICAL-COMPILATION-2010-2019

Observed comparison records:

- NIST-H-I-LYMAN-ALPHA-INTERVAL: predicted sealed-hydrogen-gross-ratios-lie-inside-both-complete-nist-line-intervals; observed sealed-hydrogen-gross-ratios-lie-inside-both-complete-nist-line-intervals; exact match True
- NIST-H-I-BALMER-ALPHA-INTERVAL: predicted sealed-hydrogen-gross-ratios-lie-inside-both-complete-nist-line-intervals; observed sealed-hydrogen-gross-ratios-lie-inside-both-complete-nist-line-intervals; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: Either exact prediction lies outside its complete NIST interval, a row or uncertainty is omitted, a source hash changes, target access precedes sealing, or a tampered row is accepted.

Measurement receipt: `sha256:5c8410854baba36fb927ab2edd197c037c9ffe849383761dfaacef401c1c38cd`. Isolation certificate: `sha256:e2c08b307cd0a1e21e2963f8df5bbe9caa9b77b61533e1c33607fda1b1e5258f`. Custody certificate: `sha256:20f367dec32c5be68f3ba38654d1e1da7a05fd64531b49933a35187ce406a08e`.

Registered source descriptions:

- National Institute of Standards and Technology, Atomic Spectra Database: https://physics.nist.gov/PhysRefData/Handbook/Tables/hydrogentable1.htm (sha256:cc54c774518d62cbb7e95f17b3b7fcd9d1faf0aa90043a607b91dff0a43e2087)
- National Institute of Standards and Technology, Atomic Physics Division: https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=842564 (sha256:86ccaeb875fcdb4445e727618509621ceee656e5dc02295140ecdb6e1d2dd443)

Registered target identities:

- NIST-H-I-LYMAN-ALPHA-INTERVAL from NIST-KRAMIDA-HYDROGEN-CRITICAL-COMPILATION-2010-2019
- NIST-H-I-BALMER-ALPHA-INTERVAL from NIST-KRAMIDA-HYDROGEN-CRITICAL-COMPILATION-2010-2019

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no line target accessible to the formal law
- no adjusted spectral ratio
- no floating comparison
- no omitted line, uncertainty or adverse row

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:5486e49ffeeebdeef591fd3d721e8e36e34a62fb9b80448d11126cce18faf221`; engine receipt `sha256:bb91ca7db989b8462ce2594703fed2705b738a25036f4c6498ef26b93dadc47c` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-ATOMIC-HYDROGEN-SPECTRUM-004-bb91ca7db989b846.json`; empirical-validation hash `sha256:5f402271ffef3982e8c105ddb4e2ff21e0d3013e494249290c47dfba0b6646d6`; measurement receipt `sha256:5c8410854baba36fb927ab2edd197c037c9ffe849383761dfaacef401c1c38cd`.

339. Post-seal Fizeau discriminator for Fold velocity composition

Claim identity: `SFT-PHYS-VALIDATION-VELOCITY-COMPOSITION-034`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The measured water slope is 274/1000 +/- 3/1000 rad s/m. It lies inside the complete reported relativistic-systematics bracket 248/1000 to 299/1000 and is more than ten times closer to its nearest relativistic row than to ordinary addition at 563/1000. The air result independently rejects ordinary addition and is compatible with relativistic composition. This is a post-seal discriminator, not a fit or precision confirmation; all source-stated systematics remain open.

The measured water slope is 274/1000 +/- 3/1000 rad s/m. It lies inside the complete reported relativistic-systematics bracket 248/1000 to 299/1000 and is more than ten times closer to its nearest relativistic row than to ordinary addition at 563/1000. The air result independently rejects ordinary addition and is compatible with relativistic composition. This is a post-seal discriminator, not a fit or precision confirmation; all source-stated systematics remain open.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPACETIME-VELOCITY-COMPOSITION-TERMINAL-033`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed operation, Fizeau discriminator, provenance, target isolation, measurement separation, complete row/limitation retention, successor closure and extension. The declared exact boundary is: The admitted exact positive velocity operation; the complete registered water measured/uncertainty, raw relativistic, corrected relativistic and ordinary-addition rows; the air result; and every stated systematic limitation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-velocity-composition-with-complete-Fizeau-discriminator__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-velocity-composition-with-complete-Fizeau-discriminator	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2c95368e6cb55afa4f0ff970e6a336df3f2aefbb2eb3d5d84c437e5167934a41`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8009bd07833670e510f10387289b297f705cccb3914d77a5950657cfd4f6b395`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:27812259a34558ed1713aa01e228303994b6a3412740a08d998686ed6e1f7aaa`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:248e0a15eade5b255ce84bb354d5fa31d1189398dcc0efd2df92e545791c45ac`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:83549ee13670779240a1e7473688fc47758bbaa3df875f9bcdcb293b1b96f240`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b9a10021cdd87914bfa0c1b2056460b07cbc35c10e4a923fa9dcd60d6cf6c784`. Independent certificate: `sha256:37966620627fdbe37ededaf13405fbc189d23ffa27d2b5bb93af1c75d1b52efe`. Engine external-validation hash: `sha256:45aff6a43141ce2fe66d8b31ed63704e133c8eb97f065e5513f569319e5297e0`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- ARXIV-1201.0501-FIZEAU-WATER-AIR

Observed comparison records:

- The exact Fold velocity operation was admitted before the Fizeau source was retrieved.
- The water slope is retained as 274/1000 +/- 3/1000 rad s/m.
- The raw and corrected relativistic rows 248/1000 and 299/1000 bracket the measured interval.
- Ordinary addition predicts 563/1000; its distance 289/1000 exceeds the nearest relativistic distance 25/1000 by more than tenfold.

- The air row rejects ordinary addition and is compatible with relativistic composition.
- Flow calibration, turbulent-profile and path-length systematics are retained; no precision overclaim is made.
- A tampered ordinary-addition row equal to the measured central value destroys the discriminator and is rejected.

Falsification condition: Reject if either source identity changes; if a registered measurement, control or limitation is omitted; if the water measured interval leaves the complete reported relativistic-systematics bracket; if ordinary addition is at least as close as both relativistic rows; if the air control no longer disfavors ordinary addition; if any target changes the sealed operation; or if a precision confirmation is claimed beyond the source.

Measurement receipt:
`sha256:9b184c06fd6dd000723fe92c1f69d7a1f7b48afd4b0d9e3d32e88d90e567e0be`. Isolation certificate:
`sha256:8af63e373e75f8c1e898e37db3f6b796cb871e2db1c44176b53fea0e14c6820c`. Custody certificate:
`sha256:901464dd7418ef296ff80c3dee2c1d39c45e51c03c9fe8ea257d20cb2961182a`.

Registered source descriptions:

- Fizeau's aether-drag experiment in the undergraduate laboratory: https://arxiv.org/pdf/1201.0501 (sha256:8c19ceeccda50a3c297aa13c6006a6cf032d41a3c714f128b2be663bd3c5b5b3)

Registered target identities:

- ARXIV-1201.0501-WITHHELD-COMPLETE-FIZEAU-RECORD from ARXIV-1201.0501-FIZEAU-WATER-AIR

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no Fizeau slope, refractive record or external velocity formula in the formal predecessor derivation
- no fit of a Fold coefficient to the water or air record
- no omission of the ordinary-addition control or source-stated systematic limitations
- no precision interval claim where the source reports an eight-percent systematic-level agreement
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude
- no target access before the formal prediction seal

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:70a12111a120963928d953bec47666c73308724f2eb2ec93275f27b25d5e2c8d`; engine receipt
`sha256:54dce92f903c6f102e4d39f3bdb826a89c05e69e80b6d6df66728711dbc03cf4` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VELOCITY-COMPOSITION-034-54dce92f903c6f10.json`; empirical-validation hash
`sha256:932b06b5da3476726d83ce723657ae4e1f0cfb8b57a70670279a7fa99f7d23b9`; measurement receipt `sha256:9b184c06fd6dd000723fe92c1f69d7a1f7b48afd4b0d9e3d32e88d90e567e0be`.

340. Post-seal Planck/CODATA test of the Fold vacuum-density scale law

Claim identity: `SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The exact Fold vacuum share 11/16 is inside Planck's interval [6833/10000,1389/2000]. The normalised Fold magnitude 33/16 is inside [20499/10000,4167/2000]=3 times that interval. Using page-225 H0=67.68 +/- 0.42 km s^-1 Mpc^-1 and exact c=299792.458 km s^-1 transports Lambda to the exact positive interval [(33/16)(67.26/c)^2,(33/16)(68.10/c)^2] Mpc^-2. One/2^20 is below the global interval and is therefore retained only as the separately typed local boundary-energy floor.

> *The exact Fold vacuum share 11/16 is inside Planck's interval [6833/10000,1389/2000]. The normalized Fold magnitude 33/16 is inside [20499/10000,4167/2000]=3 times that interval. Using page-225 H0=67.68 +/- 0.42 km s^-1 Mpc^-1 and exact c=299792.458 km s^-1 transports Lambda to the exact positive interval [(33/16)(67.26/c)^2,(33/16)(68.10/c)^2] Mpc^-2. One/2^20 is below the global interval and is therefore retained only as the separately typed local boundary-energy floor.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-DENSITY-SCALE-TERMINAL-035
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed vacuum relation, Planck/CODATA target, source provenance, capability-closed isolation, proof/measurement separation, complete row and correction retention, successor closure and extension. The declared exact boundary is: The admitted exact local floor, finite mode ledger, 11/16 global share and 33/16 normalized magnitude; Planck table 12.16 page-225 H0, Omega_Lambda and Omega_m rows with complete uncertainties; CODATA exact c; the older H0 transcription correction; and the local/global adverse type control. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-vacuum-density-law-versus-complete-Planck-CODATA-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-vacuum-density-law-versus-complete-Planck-CODATA-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:aac1f5afbedc98835000ee0959231e7e6213eae90e23f1502c48346abd816225`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:80323984b7f3e4556b6a437274b69af93eb908accc7828348411dbf280b6c821`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a0c6725ee5796923141b8a1ed45d77de545e582264fadb7dbe5bf403ef5f1af1`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a89ecd40b7631fe123e2b57bebffdd301066da0cde5dd1abc2e4291838ff98d7`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e0414394833e40ba64335f2decf3b46c98cba1a7dfb295cd38cbf1b095c23f0a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9b4f66424c7fa3b2ffc7943492d8d556da80dcd1b280405db5a9fb1cb0f5520b`. Independent certificate: `sha256:b8101dfd5ee3a439f81be03bd7b4f8a8c88e3c3cc125901e60a32e5bd8113404`. Engine external-validation hash: `sha256:d8ddaf48a77d46203c68ba9687c9b06a2bc92bee78d687688e73cd0fa83b6cab`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PLANCK-PLA-BASE-NRUN-TTTEEE-BAO-LENSING-68PC
- NIST-CODATA-2022-EXACT-C

Observed comparison records:

- The formal vacuum-density law was admitted before the primary Planck PDF was retrieved.
- Historical V1/V2 targets and an older local transcription were known; no historical-blindness claim is made.
- The exact 11/16 vacuum share lies inside the complete Planck 0.6889 +/- 0.0056 interval.
- The exact 33/16 normalised magnitude lies inside the three-direction transported Planck interval.
- Page 225 directly reports H0=67.68 +/- 0.42; the older 67.66 transcription is corrected here without rewriting prior receipts.
- CODATA exact c transports the sealed coefficient to a positive exact inverse-square-megaparsec interval.
- One/2^20 and the finite-ledger mean remain outside the global fraction interval, preserving their distinct types.
- A tampered vacuum central value of 0.6000 rejects the registered comparison.

Falsification condition: Reject if any source identity or registered row changes; if 11/16 leaves the complete Planck vacuum interval; if 33/16 leaves its exact three-direction transported interval; if the Planck central budget ceases to close at the One; if dimensional transport is nonpositive or scale-inconsistent; if One/2^20 is relabeled as the global fraction; if the older H0 transcription is silently substituted; or if any target changes the formal survivor.

Measurement receipt: `sha256:0bb3659038316226f202044b500f6bf1a6867720089a91f659b908e9fe356cf4`. Isolation certificate: `sha256:f285d0debc6ce9b8d4818d3543c820f822146affc4724edbebe6681167e8fada`. Custody certificate: `sha256:e87a3b3313685c954c8c60820553b833343ee076775dec0952b8d5d4b43af12e`.

Registered source descriptions:

- Planck 2018 baseline parameter tables, 68 percent limits, table 12.16: https://wiki.cosmos.esa.int/planck-legacy-archive/images/4/43/Baseline_params_table_2018_68pc_v2.pdf (sha256:03038805021f2f894e09f4b21b0f20418570e352822f095abcc085942919da70)
- 2022 CODATA recommended values - complete ASCII table: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- PLANCK-CODATA-WITHHELD-COMPLETE-VACUUM-SCALE-RECORD from None

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no Planck or CODATA value readable by the formal candidate generator or independent formal validator
- no fit, recalibration or coefficient choice from Omega_Lambda or H0
- no claim of historical blindness: V1/V2 targets and an older local transcription were known

- no silent reuse of the older 67.66 H0 transcription where primary page 225 reports 67.68
- no direct equality between One/2^20, a raw radiative sum, Omega_Lambda and dimensional Lambda
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:f13f4bb53dc4e55f6ffa2503fba63bdc74c8024e6692b57e3f30269ee5c18485`; engine receipt `sha256:6379f0d35a606d217e2351a35c42ae258aea71c52d8d20d2c4abeef2fdd8c202` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036-6379f0d35a606d21.json`; empirical-validation hash `sha256:7695b43af80dd5ba9987cee7e0375f9f91c9dcd81fde97020402fd544559998a`; measurement receipt `sha256:0bb3659038316226f202044b500f6bf1a6867720089a91f659b908e9fe356cf4`.

341. Post-seal thermal-history, nucleosynthesis and recombination test

Claim identity: `SFT-PHYS-VALIDATION-THERMAL-HISTORY-RECOMBINATION-038`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The direct high-redshift CMB temperature exponent interval [49/50,517/500] contains exact One. The external freeze-out sequence is approximately 1/6 then 1/7, matching the separately forced typed transport. Direct helium Yp=0.2458+/-0.0013 gives [489/2000,2471/10000], which excludes 1/4; therefore 1/4 remains the exact analytic helium-family partition and is not relabelled as the exact isotope measurement. Direct primordial deuterium (2.527+/-0.030) times 10^-5 is positive and subordinate. Planck retains z*=1089.92+/-0.25, eighteen finite extrema and seven complete TT peak rows. Those angular peak intervals are not integer multiples of the first, so the internal-whole-mode/observed-projection distinction passes and the old angular-integer reading is rejected.

The direct high-redshift CMB temperature exponent interval [49/50,517/500] contains exact One. The external freeze-out sequence is approximately 1/6 then 1/7, matching the separately forced typed transport. Direct helium Yp=0.2458+/-0.0013 gives [489/2000,2471/10000], which excludes 1/4; therefore 1/4 remains the exact analytic helium-family partition and is not relabelled as the exact isotope measurement. Direct primordial deuterium (2.527+/-0.030) times 10^-5 is positive and subordinate. Planck retains z=1089.92+/-0.25, eighteen finite extrema and seven complete TT peak rows. Those angular peak intervals are not integer multiples of the first, so the internal-whole-mode/observed-projection distinction passes and the old angular-integer reading is rejected.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed thermal relation, complete external target, source provenance, capability-closed isolation, proof/measurement separation, complete row and adverse-result retention, successor closure and extension. The declared exact boundary is: The admitted temperature exponent One, typed freeze-out/capture sequence, analytic quarter family partition, positive minor channels, finite visibility and internal acoustic parity; every registered uncertainty endpoint; all seven Planck TT peak positions and amplitudes; the eighteen-peak mission count; and explicit rejection of exact observed quarter equality and exact observed angular integer multiples. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-thermal-history-law-versus-complete-primary-observation-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-thermal-history-law-versus-complete-primary-observation-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:15bb75cf8556908a2498c9611ae022d5f26c7f8869bf195bfd51e24f992fbd79`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:da3b69464f65ec92f9d718e382f646416b99f6ad2d22ed3400b7422807ee7c40`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9385edfe317a9f525e0d985f2ba5be0917d5ee3c28190101f8ab0bdf91cb537e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:82bdaa2907c33acaf5d625f07b19d93588963bcddf3179924cb10519b0d151d7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a36854cd806c012a590df961b03c57faa3bd44f63ba8f520013c518f424b5091`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:17bd84328167f26310571135007e60049fe99b927637d4416853c2604908960b`. Independent certificate: `sha256:35623ce9812112afaca02c9a38799dca2e331cd1a284756a2bbcdac0dc592516`. Engine external-validation hash: `sha256:437b0ab034ac12219f463c53f2435c820096d84aefea57fa07f05491ef30d126`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NOTERDAEME-ET-AL-2011-CMB-TEMPERATURE
- AVER-ET-AL-2026-PRIMORDIAL-HELIUM
- COOKE-PETTINI-STEIDEL-2018-DEUTERIUM
- PLANCK-2018-OVERVIEW-PEAKS
- PLANCK-2018-COSMOLOGICAL-PARAMETERS
- PDG-2025-BBN-REVIEW

Observed comparison records:

- The formal thermal-history law was admitted before the new primary snapshots were retrieved.
- Direct high-redshift CMB thermometry contains exact exponent One.
- The external freeze-out sequence is approximately One/six then One/seven.
- Current direct helium measurement excludes exact One/four; the old exact measured-equality reading is rejected without fitting.
- Direct primordial deuterium remains a positive minor light-nuclide channel.
- Planck retains finite recombination support, eighteen extrema and all seven TT peak rows.
- Observed angular multipoles reject exact integer multiples of the first peak, preserving the internal-mode/projection distinction.
- A tampered temperature exponent rejects the comparison.

Falsification condition: Reject if any source identity or registered row changes; if exact One leaves the temperature-exponent interval; if the typed One/six to One/seven sequence is absent; if the unfavourable helium exclusion is hidden or used to fit a correction; if deuterium is nonpositive; if recombination support or the acoustic census is erased; if observed peak intervals are falsely relabelled as exact integer multiples; or if any target changes the formal survivor.

Measurement receipt: `sha256:f5710146b77a40129a6f52186fa2622ec68b7bc3ddc765685c6532dadc769f78`. Isolation certificate: `sha256:5823d5b59a85e6b38a07af5cf433d1913c23d7e8ec5c7aa56de869e6f33663d9`. Custody certificate: `sha256:d8fcf914ad1ad8db6fae9575f19e0508a9018033aac51294ff1bec29494dd4e9`.

Registered source descriptions:

- The evolution of the Cosmic Microwave Background Temperature: https://arxiv.org/abs/1012.3164 (sha256:aa79c65170e84d6a8dc71dbd0876538c19619a1c926453ce5a54bd96d3f1efb7)
- The LBT Yp Project IV: A New Value of the Primordial Helium Abundance: https://arxiv.org/abs/2601.22238 (sha256:72f646a345c36b77211e6035d3a78345498f0766cf9fc0a54932ebe0dd670c6a)
- One percent determination of the primordial deuterium abundance: https://arxiv.org/abs/1710.11129 (sha256:1a6ec19d6568b8854a40e9dd587906f99a8ea94ecd2d5c712ad86aa506b93f4a)
- Planck 2018 results. I. Overview and the cosmological legacy of Planck: https://arxiv.org/abs/1807.06205 (sha256:dca932893b7d2724aa2b8d33170fc1b5682c425dd56fcd3c5d7b2098d377db5c)
- Planck 2018 results. VI. Cosmological parameters: https://arxiv.org/abs/1807.06209 (sha256:8e172730faf07c9f4ff3fdcc7043f76ed67df6f76066d47df30d693025b6ce77)
- Big-Bang Nucleosynthesis: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-bbang-nucleosynthesis.pdf (sha256:1843c1d9025aa33c059700708f1041c4462364cdb582325f0c685a6ba1b38484)

Registered target identities:

- THERMAL-HISTORY-WITHHELD-COMPLETE-OBSERVATION-RECORD from None

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no external value readable by the formal candidate generator or formal independent validator
- no fitted cooling exponent, abundance correction, reaction rate, visibility width or angular projection
- no historical-blindness claim because V1/V2 named these targets
- no omission of the current direct helium interval that excludes exact One/four
- no preservation of the old exact observed-quarter or angular-integer claims by weakened comparison
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:f3401b72677923a0ba81abb3f9790bc49b6edf2e2b0a2711b4d6ad51a0d5fe8d`; engine receipt `sha256:7440c78415e834656217aca301aaa8b6f987d6f795544eb5c03b0679970321f4` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-THERMAL-HISTORY-RECOMBINATION-038-7440c78415e83465.json`; empirical-validation hash `sha256:d07b6ae3dcbfb2690cf52cb27769c24b2533d40955a2f338cd8f03a30b622413`; measurement receipt `sha256:f5710146b77a40129a6f52186fa2622ec68b7bc3ddc765685c6532dadc769f78`.

342. Post-seal scalar-support and tensor-support measurement test

Claim identity: `SFT-PHYS-VALIDATION-INFLATION-GROWTH-040`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Planck ns=0.9649+/-0.0042 gives exact interval [9607/10000,9691/10000]; the independently forced 31/32 equals 0.96875 and lies inside that interval. BICEP/Keck plus Planck reports r<0.032=4/125 at 95 percent; the independently forced 1/32=0.03125 lies strictly below the bound by 3/4000=0.00075. The complete exact 31/32+1/32 partition therefore passes both external rows without fitting. Five Fold doublings is not relabelled as a conventional natural-log e-fold count, for which this claim asserts no measured equality.

Planck ns=0.9649+/-0.0042 gives exact interval [9607/10000,9691/10000]; the independently forced 31/32 equals 0.96875 and lies inside that interval. BICEP/Keck plus Planck reports r<0.032=4/125 at 95 percent; the independently forced 1/32=0.03125 lies strictly below the bound by 3/4000=0.00075. The complete exact 31/32+1/32 partition therefore passes both external rows without fitting. Five Fold doublings is not relabelled as a conventional natural-log e-fold count, for which this claim asserts no measured equality.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-INFLATION-GROWTH-TERMINAL-039`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed inflation relation, complete external target, source provenance, capability-closed isolation, proof/measurement separation, complete rows, successor closure and extension. The declared exact boundary is: The admitted scalar support 31/32 and tensor support 1/32; both endpoints of the Planck one-standard scalar interval; the strict BICEP/Planck tensor upper bound; every bound source and custody row; and explicit separation of exact Fold doubling depth from conventional model-dependent e-fold language. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-inflation-growth-law-versus-complete-scalar-index-and-tensor-bound-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-inflation-growth-law-versus-complete-scalar-index-and-tensor-bound-record`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:ac9c38c10ba15714347560effb54dce239f5ef3d293433fecf50442e1cedd5c5. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:f2a877f8ea08632506702369035891 7ac7059f44eb87fa1bc8ae5c211e77e629.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:a639582b262acfd4204bced27e3e4eb9f404e56b99a711698d85e32e9feeb3b8.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:10416829131668b2e9a5f08af885fae861009dc76a59ff081e255215b712fb26.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:2c07eba135d75e0a47a23c88d003a12e3374cc1c163cf8d44a8e065ad5a45657.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:00278934fa11f29736892bf5adbd1c82c2cbd6305f9f47ec5b1dcc352a27b38a. Independent certificate: sha256:3cae59937dfb3791d3ee2c24c32584194b9fa10a1301bd26e6d73fc00e1862e3. Engine external-validation hash: sha256:7a0c0dfaa703412fe6099687c7a2e15459282635cb36047b2e3f79061620f5ee.

Shared claim-record clause PHYS-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PLANCK-2018-INFLATION
- BICEP-PLANCK-2022-TENSOR

Observed comparison records:

- The formal inflation-growth law was admitted before the primary snapshots were bound.
- Exact scalar support 31/32 lies within Planck ns=0.9649+/-0.0042.
- Exact tensor support 1/32 lies strictly below r<0.032 by 3/4000.
- The complete 31/32 plus 1/32 partition is retained without fitting.
- No measured equality is claimed between five Fold doublings and conventional logarithmic e-folds.
- A tampered scalar target rejects the comparison.

Falsification condition: Reject if any source identity or registered row changes; if 31/32 leaves the complete scalar interval; if 1/32 is not strictly below the tensor bound; if the narrow margin is omitted; if conventional e-fold language is asserted as exact equality; or if a target changes the formal survivor.

Measurement receipt: sha256:b642e922c44895919b78a5d1f98b2adc1d42fa97e7d1422b69e23a3fda14fe8b. Isolation certificate:

`sha256:c44e9ae1e05fe66e91a2cd0550c53e62b7d9aec763965e44517219c4246c5a74`. Custody certificate: `sha256:28c44958737414705d1e7638388a7a18a41c26aeee1b568e990d928927ecdf58`.

Registered source descriptions:

- Planck 2018 results. X. Constraints on inflation: https://arxiv.org/pdf/1807.06211 (sha256:fa728dc0198ccfddd7132c0e0a29cca586a1431a045568a33d7a197a0427427d)
- Improved limits on the tensor-to-scalar ratio using BICEP and Planck: https://arxiv.org/pdf/2112.07961 (sha256:5a1add11c98a8e5c4d70fd438eacd5242aa10631ffea0c72358f49e59eda07f2)

Registered target identities:

- INFLATION-GROWTH-WITHHELD-COMPLETE-OBSERVATION-RECORD from None

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no external scalar index or tensor limit readable by the formal generator or formal validator
- no fitted spectral index, tensor amplitude, inflaton potential, transfer function or exit scale
- no historical-blindness claim because V1/V2 named these observables
- no claim that five exact Fold doublings equal a conventional logarithmic e-fold count
- no omission or rounding of the narrow tensor-bound margin
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:6aa90bdf5f84f09a5c3d941b128a685c262a0bbaecc9a665a1ec7e86ccde3066`; engine receipt `sha256:8e06d75a25206d5cce129a9afd2583a54cb1e070a0daf6eeeff92ece99a8d01f` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-INFLATION-GROWTH-040-8e06d75a25206d5c.json`; empirical-validation hash `sha256:272562ca8235316e3bae097e8a3af450c1c4785525266797ca8bf6d012a272e9`; measurement receipt `sha256:b642e922c44895919b78a5d1f98b2adc1d42fa97e7d1422b69e23a3fda14fe8b`.

343. Post-seal blackbody, acoustic, laser, plasma and Alfvén response test

Claim identity: `SFT-PHYS-VALIDATION-COLLECTIVE-RADIATION-RESPONSE-042`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. External blackbody measurement reports exponent 4 exactly, matching the forced rank-three-plus-energy exponent; doubling therefore remains the exact ratio 16. The historical Stefan coefficient is reported within 1/100 of the present value and spectrum shape agreement is retained, but no dimensional coefficient is manufactured from a normalised carrier. Acoustic and microwave cavity resonances were measured to part-per-million precision. NIST reports unmodified laser linewidths of at least 10,000,000 Hz and extended-cavity linewidths no more than 500,000 Hz, so linewidth remains positive and feedback narrows it by at least factor 20. Two NASA flights directly track electron density through plasma frequency, and NASA registers Alfvén waves from 2007 through 2014. All comparisons pass at their declared relation/value type without fitting.

> *External blackbody measurement reports exponent 4 exactly, matching the forced rank-three-plus-energy exponent; doubling therefore remains the exact ratio 16. The historical Stefan coefficient is reported within 1/100 of the present value and spectrum shape agreement is retained, but no dimensional coefficient is manufactured from a normalized carrier. Acoustic and microwave cavity resonances were measured to part-per-million precision. NIST reports unmodified laser linewidths of at least 10,000,000 Hz and extended-cavity linewidths no more than 500,000 Hz, so linewidth remains positive and feedback narrows it by at least factor 20. Two NASA flights directly track electron density through plasma frequency, and NASA registers Alfvén waves from 2007 through 2014. All comparisons pass at their declared relation/value type without fitting.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-COLLECTIVE-RADIATION-RESPONSE-TERMINAL-041
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed collective response, complete targets, bound provenance, capability isolation, proof/measurement separation, complete rows, successor closure and extension. The declared exact boundary is: The sealed finite occupation and exact scaling laws; measured exponent four; coefficient agreement bound one percent; positive measured linewidth ranges; acoustic resonance precision; direct plasma-frequency density dependence; complete Alfvén observation-year boundary; and every source/custody row. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-collective-response-versus-complete-blackbody-acoustic-laser-plasma-Alfven-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-collective-response-versus-complete-blackbody-acoustic-laser-plasma-Alfven-record`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:fdf048a5f1358d5dd699b09cc0df4135a9d45f2f439cd74e13768f4d8fa22d69`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9e95f31ffd65f7e0e7fe88cff91fa56174657df423fe9339a700d16d8af28c8d`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:4a9bf8a263ce6dee9c7b4fecbac1db0490cb2ab1e2dff3a463e0f17f77d7576b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e078e684ee11df95956211e36bb820cf50e1465dc92d49b3bc4b4cfab7f81cb3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:cd21f5572e340e60783f653afb6f6a9081e9a9d32378cb85c23e9dab152bfcb3`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:374078411155f78c3dd1273bad1444aeba59106a0bdc75d2c5b4e3828bbbc021`. Independent certificate: `sha256:d5e36700fbe06d0dbe4cda4afa473cfe7c18d8bc19318a677ccf8a15e776a81a`. Engine external-validation hash: `sha256:cb22b4bcf0d3da68cd1002f5d0d0556664932179bfa9a62796cf3ca65cbf7c1b`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-COBLENTZ-BLACKBODY
- NIST-DIODE-LASER-LINEWIDTH
- NBS-ACOUSTIC-CAVITY-1986
- NASA-PLASMA-FREQUENCY-PROBE-1992
- NASA-STEREO-ALFVEN-LIST

Observed comparison records:

- Measured blackbody exponent four forces doubling ratio sixteen.
- Historical coefficient agreement is within one percent and spectrum shape agreement is retained.
- Acoustic cavity resonances were measured to part-per-million precision.
- Measured laser linewidth remains positive and feedback narrows it by at least factor twenty.
- Two NASA flights track density through plasma frequency.
- NASA registers Alfvén observations across eight calendar years.
- No dimensional coefficient is fitted from normalised Fold structure.
- Tampered exponent three rejects.

Falsification condition: Reject if a source or row changes; if the measured blackbody exponent is not four; if measured linewidth is empty or feedback does not narrow it; if acoustic resonances, direct plasma-frequency density dependence or Alfvén observations are absent; if a dimensional coefficient is fitted from normalised structure; or if a target changes the formal survivor.

Measurement receipt: `sha256:ff7685c1c62d0af80001e60ad5c44ba0199bea46b5a43073dedac0f3332a0dd7`. Isolation certificate: `sha256:888d25acd42241bcf5ca6fd22ade5ec7a4d86826d003f2cbbf0c563e3e38038e`. Custody certificate: `sha256:363a20ce7e39db88f33913b933b8b094dd389cb98329a79f05f46ca81275afcf`.

Registered source descriptions:

- Coblentz and blackbody radiation measurements: https://nvlpubs.nist.gov/nistpubs/sp958-lide/html/007-009.html (sha256:deb600e47fd138dfe82aef3271b19f2030cd7cd3e4fa202ef80f4f1ae82eec71)
- Diode Laser Technology: https://tf.nist.gov/ofm/lasers/diode.htm (sha256:7c0a9a1b4dc19b573cb6603f87453161508784ca59ec26ed698b0354bb7520b6)
- Measurement of the ratio of the speed of sound to the speed of light: https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=925220 (sha256:f4dbf2b01604b8a6f26621bb8bafa856913e25158a8635f7454838ef5360f1b4)
- Measuring ionospheric electron density using the plasma frequency probe: https://ntrs.nasa.gov/citations/19920042037 (sha256:4fff15fe03f66e4009ca34e83f0d0ea02e66b2d40e92ee987ae46f437f39f36c)
- STEREO-A PLASTIC Alfvén Wave List: https://data.nasa.gov/dataset/stereo-a-plastic-alfven-wave-list (sha256:da0eab2e7955e0670d734c905c360758aabe57bb900de277e42da13fa0314385)

Registered target identities:

- COLLECTIVE-RADIATION-WITHHELD-COMPLETE-RECORD from None

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no target readable by the formal generator
- no fitted dimensional coefficient or converted absolute value
- no historical-blindness claim
- no omission of finite linewidth or source limitations
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:e5232310fe7cfa0feeaffd1866117045b7718fee1f1ccc7725a73fc11715683e`; engine receipt `sha256:1d7dd8e6373027c10bd83aa9ff128ebabe3fdc60569b24ad597f9c22abad2455` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-COLLECTIVE-RADIATION-RESPONSE-042-1d7dd8e6373027c1.json`; empirical-validation hash `sha256:83cf47bcba4dc585c59c8f16adf89071a2d100d79b6cf98f197a45ea82a3d8f6`; measurement receipt `sha256:ff7685c1c62d0af80001e60ad5c44ba0199bea46b5a43073dedac0f3332a0dd7`.

344. Post-seal CODATA validation of the terminal charged-lepton invariant

Claim identity: `SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. After the exact terminal charged-lepton invariant is sealed, exact rational root brackets predict both adjacent-root squared mass ratios and compare them with both complete NIST CODATA 2022 intervals.

> *Both sealed terminal-cubic mass-ratio consequences overlap their complete one-standard-uncertainty CODATA 2022 intervals.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`

Generated grammar and closure boundary. Generate the complete eight-axis post-seal terminal-invariant, source, interval, row-retention, custody and no-extra-rule product. The declared exact boundary is: Both exact adjacent-root squared ratios of the sealed terminal cubic against both complete registered CODATA intervals. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-terminal-cubic-ratios-versus-complete-codata-intervals__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-terminal-cubic-ratios-versus-complete-codata-intervals	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4b0f13ecfa647507c2116a80608f8817884eef53fde16d2ca18ab9c9f8edadf0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c44fcfd2be5a25663f279cd0a06772eeccf6e63e389685a64899dcc4a0861008`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b596438bde200a2809c35642cb33eda7a71543904271bf37b296e7518f065e3f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5cb9911daba73b152460a90883b7b8cf5f5b3cfe81f98402dfec24998918f976`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dfee970fb252d4b9a06366c282e3d38063bba95f0b2800b2cae879e18cf322ba`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:024f84cb4b6103c422fa67d656620f53d80ea7dceff6c720aa1f02201de5ae31`. Independent certificate: `sha256:bb85213071aba6fa4ef1f4ce506c214a9fa4bebe181d02881e36c2b705057381`. Engine external-validation hash: `sha256:355eb559ee89354203142ccb3b59c17e0081caa150f150c0778bddc056239ffb`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- NIST-CODATA-2022-MUON-ELECTRON-MASS-RATIO: predicted terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; observed terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; exact match True
- NIST-CODATA-2022-MUON-TAU-MASS-RATIO: predicted terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; observed terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; exact match True

- deliberately tampered unfavourable control rejected

Falsification condition: Either complete terminal prediction interval fails to overlap its source interval, a row/hash changes, an uncertainty is enlarged or omitted, or a tampered comparison is accepted.

Measurement receipt: `sha256:b1bc18c4ca4396fcaf2a88e4a9a8fc34bd1acf30ae4d80949faa6d1c67ec44e7`. Isolation certificate: `sha256:bb11637734c577c68efa087896e41a65b07db3971be5a6c9bfc12b01eab1f9ec`. Custody certificate: `sha256:2f89d8555574629bf6360dd461ca6c73a9c117a6e67cd5b5e8e445f66a9417e9`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no CODATA value accessible to formal forcing
- no fitted terminal coefficient or root
- no floating comparison
- no omitted failed row or enlarged uncertainty

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:73e9e100453602ec568ccfebb17a875ea8e6387b9d1559d50f2ad5d36aca92f3`; engine receipt `sha256:1df007d68f197e5a8a8a58dad28190bbaf10229ab2f8136365a25622eba5d5d8` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002-1df007d68f197e5a.json`; empirical-validation hash `sha256:1502cfabdf3a00b544b7fd2a44179bff7f0dc10ed9141b342c0e4ff684c9af26`; measurement receipt `sha256:b1bc18c4ca4396fcaf2a88e4a9a8fc34bd1acf30ae4d80949faa6d1c67ec44e7`.

345. Versioned post-seal exact Koide measurement validation

Claim identity: `SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-002`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The exact Fold value two-thirds is sealed before both complete CODATA mass-ratio intervals are opened and propagated through exact rational square-root enclosures.

> *The sealed exact Koide value 2/3 lies inside the complete conservative CODATA-derived interval.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-001`
- `SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal two-thirds, two-source-row, rational-enclosure, custody and no-extra-rule comparison product. The declared exact boundary is: The full Cartesian uncertainty rectangle of both complete NIST CODATA 2022 charged-lepton mass-ratio rows. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-two-thirds-versus-complete-rational-koide-enclosure-v2__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-two-thirds-versus-complete-rational-koide-enclosure-v2	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:bca9ace286d82c15f81c66da94c8f5d4e3a41caf7ca3c5d72cceb4c447fdf4ec. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:3532da9d5c6d2e27a021faa14ff22e62256b4adcb27b72a74275eb48c11afb19.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e5c839d5cadeaa5f243adec6dbf44c23ebb0cb9872c6359cf1be7151bc93f17c.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:724d004ca77c3a8a5f941c8168ccdede605016800fdf94dad1d07935882fac4d.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:f138b0a8ca62e6462c8e2caa1cdfc6f57077a46208c91c28053c30ac3e6f8438.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:ca563bc0a2c7d291e111a9f6c0511b689c4b0f45314fb520d97c75ead0f3a808. Independent certificate: sha256:6ddde3960e2364184cfbb6e3c47d13d74ab4c5b1f77f827db1d4ead9c0bb7f6b. Engine external-validation hash: sha256:97998128cccbc5765d78232c461abd4155fc23719c1fd13659e8f73769cf49b5.

Shared claim-record clause PHYS-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- NIST-CODATA-2022-MUON-ELECTRON-MASS-RATIO: predicted sealed-two-thirds-inside-complete-codata-derived-koide-interval; observed sealed-two-thirds-inside-complete-codata-derived-koide-interval; exact match True
- NIST-CODATA-2022-MUON-TAU-MASS-RATIO: predicted sealed-two-thirds-inside-complete-codata-derived-koide-interval; observed sealed-two-thirds-inside-complete-codata-derived-koide-interval; exact match True
- deliberately tampered unfavourable control rejected

Falsification condition: The exact two-thirds lies outside the outward CODATA-derived interval, either row or uncertainty is omitted, source custody changes, or a tampered comparison is accepted.

Measurement receipt: `sha256:e7ccd0d3a9e0f675cd2d1e894bd347e753ad9f4d7c9081bc094e8b850fb8baf7`. Isolation certificate: `sha256:6e92f178a47954d84aa0acb09fdefee178f7cbb36533fb38688a89a6f2d81308`. Custody certificate: `sha256:68ac69675369c6d62faa95647520644b327f0ab77851e51e60b40127a8dde030`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured mass ratio in the formal two-thirds derivation
- no floating square root
- no central-value-only comparison
- no omitted uncertainty or row

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:685f4a86e3a630c03aa11251b684b32319b7b27b9f58520b1f1972aa6c076748`; engine receipt `sha256:4e7a64b6751e5622c827612840751a26b40b61ee657bb011eded7ff7914aa776` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-002-4e7a64b6751e5622.json`; empirical-validation hash `sha256:d399bfa96ceabc1eedd3b61d77618e85f20e488f1ffb2db46f619413beac4a7a`; measurement receipt `sha256:e7ccd0d3a9e0f675cd2d1e894bd347e753ad9f4d7c9081bc094e8b850fb8baf7`.

346. Post-seal acoustic and Johnson-noise thermometry test

Claim identity: `SFT-PHYS-VALIDATION-THERMAL-EQUILIBRIUM-044`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. With all values carried as exact integers over 10^30 J/K, exact SI k_B is 13806490/10^30. The acoustic result is (13806484 +/- 28)/10^30 and therefore gives [13806456,13806512]/10^30; the Johnson-noise result is (13806510 +/- 170)/10^30 and gives [13806340,13806680]/10^30. Both intervals contain the exact carrier. The two measurement routes are physically distinct. The acoustic record identifies thermodynamic temperature with mean kinetic energy, and the electronic record reports Johnson noise power as jointly dependent on temperature and resistance to one part per million in its declared regime. No external row claims the Fold dyadic ladder or 3/4:1/4 response coordinates as universal measured values.

> *With all values carried as exact integers over 10^30 J/K, exact SI k_B is 13806490/10^30. The acoustic result is (13806484 +/- 28)/10^30 and therefore gives [13806456,13806512]/10^30; the Johnson-noise result is (13806510 +/- 170)/10^30 and gives [13806340,13806680]/10^30. Both intervals contain the exact carrier. The two measurement routes are physically distinct. The acoustic record identifies thermodynamic temperature with mean kinetic energy, and the electronic record reports Johnson noise power as jointly dependent on temperature and resistance to one part per million in its declared regime. No external row claims the Fold dyadic ladder or 3/4:1/4 response coordinates as universal measured values.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of the sealed thermal law, exact source-bound intervals, target custody, complete row retention and formal/measurement separation. The declared exact boundary is: The sealed finite-population theorem; exact SI k_B; the complete acoustic and electronic k_B standard-uncertainty intervals; the kinetic-temperature and Johnson-noise relation rows; both explicit nonmeasurement boundaries; and every source/custody row. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-thermal-law-versus-complete-acoustic-and-Johnson-noise-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-thermal-law-versus-complete-acoustic-and-Johnson-noise-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:019db88dcdb566512199c250c6841ce242ba1a225cf35f4495a68b1977c7ed37`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3616e5ddecafab7522954ecae0da77f6eec8e9057249d748373534465565149d`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:400a1cdd726220cd82097342db240c8988e9eaf785459534e546c1d842111bcc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5b4a3858a67e89cb7ed7cdcf2a40e56a17718850916d10aef131f36b472cb81b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9892ff517e4a939131153c66697e30fda07f1b043ee104300ee03ce7cfc29eff`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ca67b8df9587363e46d0c18b5d69a4c5d234e99832846b5dcf5a8e5c8e1b009b`. Independent certificate: `sha256:37d234b90b063945ab5f2d13a1be76fe385c6cda55be766f1eda81f4b0aa1304`. Engine external-validation hash: `sha256:51495049d9401534e4174d42f85d8fca21b8a9e8e89eb7ecba2128eb492a0053`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-CODATA-2022-BOLTZMANN
- NIM-NIST-ACOUSTIC-BOLTZMANN-2017
- NIST-JOHNSON-NOISE-BOLTZMANN-2011

Observed comparison records:

- Exact SI k_B is 13806490/10^30 J/K.
- Acoustic interval [13806456,13806512]/10^30 J/K contains exact SI k_B.
- Johnson-noise interval [13806340,13806680]/10^30 J/K contains exact SI k_B.
- Acoustic and electronic routes are physically distinct and both remain complete.
- Acoustic thermometry retains the mean-kinetic-energy temperature relation.
- Johnson noise retains joint temperature/resistance response at the reported one-part-per-million regime.
- No direct universal measurement is claimed for the dyadic ladder or 3/4:1/4 Fold coordinates.
- Tampered acoustic interval excludes the exact carrier and rejects.

Falsification condition: Reject if any source or row changes; either complete measured interval excludes exact SI k_B; the routes are not physically distinct; the kinetic-temperature or Johnson-noise response row is absent; a limitation row is omitted; an internal Fold coordinate is relabelled as directly measured; or target data alter the formal survivor.

Measurement receipt: `sha256:cffa2dbbd885f19cf9a19f08aec070370ba237b7aa12735d2173fc8d77f0e7f6`. Isolation certificate: `sha256:ee10f5a56f89082a81a2e386f8e5f82d14ff84edf39ea3859ba317b2e04b3eb7`. Custody certificate: `sha256:2e674cf79354cc078e08ef624ddad870c176f981f96c15a124d1c6a36730445f`.

Registered source descriptions:

- NIST CODATA 2022 all constants - Boltzmann constant: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)
- Determination of the Boltzmann constant with cylindrical acoustic gas thermometry: https://doi.org/10.1088/1681-7575/aa7b4a (sha256:1cbb21f0e5817270b5e028105aa79fea22017fd84130c2c4b79d1492fb37e418)
- An electronic measurement of the Boltzmann constant: https://doi.org/10.1088/0026-1394/48/3/008 (sha256:187066b5390d57a3c058e0a34f6c7803a659045ba714ca4b25eed7b84b212bbb)

Registered target identities:

- THERMAL-EQUILIBRIUM-WITHHELD-COMPLETE-RECORD from None

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no target readable by the formal generator
- no fitted Boltzmann carrier, temperature scale or response coefficient
- no conventional continuum distribution imported into the formal proof
- no claim that the dyadic ladder or 3/4:1/4 coordinates were directly measured
- no omitted unfavourable, uncertainty or applicability row
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:9c5025b8f068aebe13e964ab1abf997dd143ddb453308512038c162fe8bed065`; engine receipt `sha256:60366227e00c50a3529d98538a3ca42d9620f4da30a174b9cff96f1122097dbd` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-THERMAL-EQUILIBRIUM-044-60366227e00c50a3.json`; empirical-validation hash `sha256:9d94b8c34f0413e7170a0aa78fc95dec6807bae70ea620e75af6f9f131e56a16`; measurement receipt `sha256:cffa2dbbd885f19cf9a19f08aec070370ba237b7aa12735d2173fc8d77f0e7f6`.

347. Post-seal finite BEC, Pauli-blocking and two-turn spinor test

Claim identity: `SFT-PHYS-VALIDATION-SPIN-STATISTICS-CONDENSATION-046`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The finite Bose record measures ground-state occupation, its increase under cooling, and a sharp transition at (94 +/- 5)/100 of the declared trap scale, giving the exact interval [89,99]/100. The finite two-spin-state Fermi record directly observes Pauli blocking and a factor-two reduction in effective collision cross section. The sealed alternating-state return is two complete turns, corresponding to 720 degrees; the neutron-interferometry result is 704 +/- 38 degrees, giving [666,742], which contains 720. All three physical routes agree with their respective sealed structural distinctions. The BEC scale ratio and collision-response factor are retained as external apparatus-dependent measurements rather than relabelled universal Fold proof scalars.

> *The finite Bose record measures ground-state occupation, its increase under cooling, and a sharp transition at (94 +/- 5)/100 of the declared trap scale, giving the exact interval [89,99]/100. The finite two-spin-state Fermi record directly observes Pauli blocking and a factor-two reduction in effective collision cross section. The sealed alternating-state return is two complete turns, corresponding to 720 degrees; the neutron-interferometry result is 704 +/- 38 degrees, giving [666,742], which contains 720. All three physical routes agree with their respective sealed structural distinctions. The BEC scale ratio and collision-response factor are retained as external apparatus-dependent measurements rather than relabelled universal Fold proof scalars.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SPIN-STATISTICS-CONDENSATION-TERMINAL-045`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of the sealed spin-statistics law, finite BEC record, Pauli-blocking record, two-turn spinor interval, custody, full-row retention and formal/measurement separation. The declared exact boundary is: The admitted 256-form formal survivor; all three bound primary records; the complete BEC transition interval and direction rows; the complete Pauli rows; the complete spinor interval and return rows; and every custody, limitation and adverse-control row. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-spin-statistics-condensation-law-versus-complete-finite-BEC-Pauli-and-spinor-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-spin-statistics-condensation-law-versus-complete-finite-BEC-Pauli-and-spinor-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:00c2bf991301e167212f42e5e1cd5c4271d69a5bf5b209d0f397261877c1e853`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:71667d88b3de82c75428da3d963a842ffecca969031fec3d637d34cd953ab4f9`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6023224eea2ee01eb133f80d8898133abfd6bf5f32965a3af344c0b301999eaa`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:cac70197006d203259866f1502a64a75ecaa45ee6e9a9e230ad7fdb161e17354`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9e81125fdafed5f0286f7510956a2e8ac22ed6c4348943017be4f33b4d75088f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2abf34329c76d85d6e1e413feb2cf8ad0445ddd30070ecde9689518fdedf80ea`. Independent certificate: `sha256:9628100ad1af900942a0252fa75c303188d2a2904aef0e65de5543f6ccd83532`. Engine external-validation hash: `sha256:105fb7e4fb72eb3d20b4fe28c0c832a8cdd38e5601c361a134768b37eaaffbd7`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- APS-JILA-BEC-GROUND-OCCUPATION-1996
- NIST-JILA-PAULI-BLOCKING-2001
- RAUCH-COHERENT-SPINOR-ROTATION-1975

Observed comparison records:

- Finite rubidium-87 Bose gas ground occupation is measured and rises as temperature is lowered.
- BEC transition interval is [89,99]/100 of the declared trap scale; its finite-population correction remains retained.
- Two-spin-state potassium-40 gas directly exhibits Pauli blocking and the reported factor-two collision reduction.
- Neutron spinor interval [666,742] degrees contains the sealed two-turn 720-degree return.
- One turn changes the held spinor orientation and two turns restore the identical state.
- Trap-specific BEC and collision-response magnitudes remain external scale records, not formal-law selectors.
- Tampered spinor and reversed-cooling controls reject.

Falsification condition: Reject if any source or row changes; finite bosons fail to accumulate in the measured ground state under cooling; Pauli blocking is absent in the declared identical-fermion regime; the complete spinor interval excludes 720 degrees; a trap-specific or response-specific value is relabelled as a universal formal scalar; an unfavourable row is omitted; or target data alter the sealed formal survivor.

Measurement receipt: `sha256:171baf50d3b3944b9f2c8e14f7bfddf7da778071c52a03bfc33e6c422ea6ff95`. Isolation certificate: `sha256:3df0e1a9705da1b597362b4786897f30b2125a11228ef0823a6dcd24fbd79b09`. Custody certificate: `sha256:cd36776ecd5e074b3c50f48ec1eeb5875970aa180c17018a3382c3897350bdb3`.

Registered source descriptions:

- Bose-Einstein Condensation in a Dilute Gas: Measurement of Energy and Ground-State Occupation: https://doi.org/10.1103/PhysRevLett.77.4984 (sha256:9086058e1c750191fb36eab5acf7c65da1aa7f0e4cdcf643887e282ad3920c93)
- Pauli Blocking of Collisions in a Quantum Degenerate Atomic Fermi Gas: https://www.nist.gov/publications/pauli-blocking-collisions-quantum-degenerate-atomic-fermi-gas-0 (sha256:82555a4057e339aef96cde7484fe3fbd190a8d2cecbf4c3de73996af473b9af7)
- Verification of Coherent Spinor Rotation of Fermions: https://doi.org/10.1016/0375-9601(75)90798-7 (sha256:11638b8c2aaf8fbdd96aa50c9496bb9f7597b308f19d780ab779cc2dd6752824)

Registered target identities:

- SPIN-STATISTICS-CONDENSATION-WITHHELD-COMPLETE-RECORD from None

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no BEC temperature, Pauli response, particle species or spinor angle readable by the formal generator
- no fitted occupation ceiling, critical temperature, collision factor or rotation interval
- no continuum Fermi-Dirac or Bose-Einstein function imported into the formal proof
- no relabelling of the apparatus-specific BEC ratio or collision reduction as a universal Fold proof scalar
- no omitted unfavourable, uncertainty, finite-population or applicability row
- no conventional numerical-nothingness, negative, irrational, imaginary, floating, NaN or continuum Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:76e98275459015406d1fb9dcc615d65b8bc0c2d55deeaf0fde4e4ab8bde88733`; engine receipt `sha256:6c37132ae33b8f23b06e69b7034c5acdd39144d51b2661d7e65f441b400604eb` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-SPIN-STATISTICS-CONDENSATION-046-6c37132ae33b8` `f23.json`; empirical-validation hash `sha256:b19baca0cbfe8f4673e94c3cb40502fba2a954a0172387e9573a3d5bafd4e0fe`; measurement receipt `sha256:171baf50d3b3944b9f2c8e14f7bfddf7da778071c52a03bfc33e6c422ea6ff95`.

348. Post-seal critical-exponent, universality and turbulence-scaling test

Claim identity: `SFT-PHYS-VALIDATION-CRITICALITY-UNIVERSALITY-TURBULENCE-048`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The erbium intervals [46,50]/100 for beta, [90,102]/100 for gamma and [47,51]/100 for nu contain exactly 1/2, One and 1/2. Of the complete five-row manganite record, La00, La04, La06 and La08 contain the entire beta=1/2, gamma=One, delta=three vector; La02 excludes gamma=One and delta=three and therefore cannot be admitted to that generated universality class. The independent turbulence interval [666,692]/1000 contains exactly 2/3. A Reynolds-count-720 physical flow exhibits five-thirds compensated plateaux over 40--4000 Hz by Fourier analysis and 20--2000 Hz by Hilbert analysis. The nonmatching composition, finite-Reynolds correction boundary, structure-function range limitation, normalised-threshold boundary and empty-exponent typing are all retained.

> *The erbium intervals [46,50]/100 for beta, [90,102]/100 for gamma and [47,51]/100 for nu contain exactly 1/2, One and 1/2. Of the complete five-row manganite record, La00, La04, La06 and La08 contain the entire beta=1/2, gamma=One, delta=three vector; La02 excludes gamma=One and delta=three and therefore cannot be admitted to that generated universality class. The independent turbulence interval [666,692]/1000 contains exactly 2/3. A Reynolds-count-720 physical flow exhibits five-thirds compensated plateaux over 40--4000 Hz by Fourier analysis and 20--2000 Hz by Hilbert analysis. The nonmatching composition, finite-Reynolds correction boundary, structure-function range limitation, normalized-threshold boundary and empty-exponent typing are all retained.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-CRITICALITY-UNIVERSALITY-TURBULENCE-TERMINAL-047`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of the sealed criticality/cascade law, exact external intervals, universality membership, complete-row custody, method agreement, limitation retention and proof/measurement separation. The declared exact boundary is: The admitted 256-form formal survivor; all four bound source records; every reported erbium and five-composition manganite exponent interval; the complete turbulence structure interval; both physical spectrum-analysis routes; every unfavorable row, limitation and custody record. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-criticality-and-cascade-law-versus-complete-postseal-exponent-and-turbulence-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-criticality-and-cascade-law-versus-complete-postseal-exponent-and-turbulence-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:37318550316d6549d405a22e3baa9ec6dab4de1c5646452d2a188f234e930e2e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9b5652ebc3f084bc0e3a2c2987fab6356660f2d418e607754557b3d973e67b34`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8a47b83313e602972ed0b81b0c6df8037546083c6d6aff27ee3a08227163ef2c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3663e27042caebc8ec58272e4bf040604e79a8480d8616419b76612ea7d3d4ec`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:84d564af90805459dd95a0b12bbdf10fb2024d47fdb2d7cefc5fcb63b438541a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:664628e4ab961c2fe2bbc9289ecfcae9b3805d45f0f05fd1f2e86dd98e24c37a`. Independent certificate: `sha256:87f93c3da5f96313e0d799a45fce681b7570c0e08bb36a5d12f9b39c551713ad`. Engine external-validation hash: `sha256:84d787b11748ec58ad11057fee614d97e312e5801219e21d9bc4fb85f4992618`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- SPRINGER-CETIN-MANGANITE-CRITICAL-EXPONENTS-2026
- MCMASTER-LIN-ERBIUM-CRITICAL-SCATTERING-1993
- APS-MCCOMB-TURBULENCE-STRUCTURE-2014
- APS-HUANG-TURBULENCE-SPECTRUM-2010

Observed comparison records:

- Erbium beta/gamma/nu intervals contain exactly 1/2, One and 1/2.
- Four of five complete manganite vectors contain exactly beta=1/2, gamma=One and delta=three.
- La02 is retained and rejected as a complete member of that generated exponent class.
- The independent zeta_2 interval [666,692]/1000 contains exactly 2/3.
- The physical channel-flow record exhibits five-thirds compensated plateaux by Fourier and Hilbert routes.
- Finite-Reynolds, structure-function-range, threshold and empty-exponent boundaries remain retained.
- Tampered interval, relabelled nonmatch, removed-row and reversed-spectrum controls reject.

Falsification condition: Reject if any source or reported row changes; the erbium vector excludes a sealed carrier; fewer than the registered four manganite rows match or La02 is concealed or relabelled as a complete match; the zeta_2 interval excludes 2/3; either registered physical spectrum route lacks its five-thirds plateau; any limitation is omitted; or target data alter the sealed survivor.

Measurement receipt: `sha256:eee857690b4e7a3a0c6c6d16b82055bd33a7ae269d887097921428c9f6ece26d`. Isolation certificate: `sha256:59dd6d2b2fe4fc0d7366b399c5888ae5728bcbdd91d489d31b9ca29e020acc0a`. Custody certificate: `sha256:5710aa5047e21e2e8f0994451a3b0d42b9feb38c36a8f27d963753f47336328f`.

Registered source descriptions:

- Structural, magnetic, and magnetocaloric properties of La and Sm co-doped CaMnO3 manganites: https://link.springer.com/article/10.1007/s10854-026-17400-y (sha256:afd0f783bd40b8ed376238eda9b254387c75c0c26ecc57488ab9b81b33d1d455)
- Critical magnetic scattering in erbium: https://experts.mcmaster.ca/scholarly-works/262851 (sha256:d8b6495b7defb46daa4d1162967d11b4680012ebd4a8e6d60e8892b83b9b3564)
- Spectral analysis of structure functions and their scaling exponents in forced isotropic turbulence: https://doi.org/10.1103/PhysRevE.90.053010 (sha256:0c9f4321b52f78fe64b02684b4139a450955c12092c4faf8487f49bf6ff1d0f5)
- Second-order structure function in fully developed turbulence: https://doi.org/10.1103/PhysRevE.82.026319 (sha256:1bff196a24bc3f852dd384469c5f8e4112fcdf717c2416190ad44b0012ac06cb)

Registered target identities:

- CRITICALITY-UNIVERSALITY-TURBULENCE-WITHHELD-COMPLETE-RECORD from None

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no critical exponent, sample class, Reynolds record, structure exponent or spectrum slope readable by the formal generator
- no fitted exponent, uncertainty interval, intermittency correction or measurement-selected universality class
- no omission of La02 or any other unfavourable measurement row
- no conversion of the normalised half-One threshold into a universal laboratory temperature
- no conversion of typed empty alpha or eta into a numerical-zero measurement
- no claim that every physical transition shares the binary self-antipodal exponent vector
- no conventional numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:79e4dae658e266d5768876896a2ffff9d1a8b2adcede60acedbc65409a3430dc`; engine receipt `sha256:075d304c85f48d6886ba7cb1fd44d5db25815cdf3a5d016502da1435654f1bdc` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CRITICALITY-UNIVERSALITY-TURBULENCE-048-075d304c85f48d68.json`; empirical-validation hash `sha256:875b3c350cd9f407f5608d25ef989b2161ffad838c4c4d9805412d07c43aec58`; measurement receipt `sha256:eee857690b4e7a3a0c6c6d16b82055bd33a7ae269d887097921428c9f6ece26d`.

349. Post-seal superconducting-circuit test of the Fold Bell factorization boundary

Claim identity: `SFT-PHYS-VALIDATION-QUANTUM-SUPPORT-UNCERTAINTY-050`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The exact measured S interval [20714,20780]/10000 lies wholly above the local-factorization bound two; its central excess exceeds twenty-two stated standard uncertainties and the reported P value is below 10^-108 across more than one million trials. The complete timing intervals preserve space-like separation, and the analysis retains its memory control. This physically rejects the sealed local-factorization class under the source's declared conditions while preserving deterministic complete-record admissibility, no signalling, and the explicit measurement-independence assumption. The distinct Walsh support-count product remains a formal/computational result and is not relabelled as a variance measurement.

> *The exact measured S interval [20714,20780]/10000 lies wholly above the local-factorization bound two; its central excess exceeds twenty-two stated standard uncertainties and the reported P value is below 10^-108 across more than one million trials. The complete timing intervals preserve space-like separation, and the analysis retains its memory control. This physically rejects the sealed local-factorization class under the source's declared conditions while preserving deterministic complete-record admissibility, no signalling, and the explicit measurement-independence assumption. The distinct Walsh support-count product remains a formal/computational result and is not relabelled as a variance measurement.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-QUANTUM-SUPPORT-UNCERTAINTY-TERMINAL-049`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-QUANTUM-BELL-001`
- `SFT-PHYS-QUANTUM-NO-SIGNALLING-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed Bell boundary, external interval, provenance, target isolation, proof/measurement separation, complete condition retention, successor closure and extension. The declared exact boundary is: The admitted Claim 049 Bell-response census; the complete primary S interval, trial-count, significance, space-like timing, memory-analysis and measurement-independence records; and every stated interpretive limitation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-local-factorization-law-versus-complete-superconducting-circuit-Bell-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-local-factorization-law-versus-complete-superconducting-circuit-Bell-record`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d9d2ffe6f3de6a674f40a65c0f7ba8c9bdf10a54fe7061141ebc9fa73bff4fdc`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ba21cca4d789ae798ee99d66062a03b65b548a0941483202a985b2908c22f2f9`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:99df30fde6e9e24905d717834894b22176a56ac9a56e12e24e618b3da29fd370`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b1e892060c3134158247569c3f75db2a7b153d8e4193d4df177f7ac37ccac46c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d6cfacd47b3e5fa3c8e02c4c0602fb52aed94102f4778a23ca7614efbab26950`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4e5da504fe3f3a28d7e51131cd3113d4e77eb9aaa32ac15577492c21dbc8bd4f`. Independent certificate: `sha256:6e96d67c42eb8074c876d6595ec8c0748b60daacc5f22cfcfb78ae396ba5f232`. Engine external-validation hash: `sha256:40e0fe145c4f95e92ef5cb0d86081a7ae76b5914d1a02b78018bf1c515bc610a`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NATURE-STORZ-LOOPHOLE-FREE-BELL-2023

Observed comparison records:

- The complete measured S interval [20714,20780]/10000 lies above the local-factorization bound two.
- The central excess is greater than twenty-two stated standard uncertainties and P is reported below 10^-108.
- More than one million superconducting-circuit trials are retained.
- The complete trial-duration interval lies below the complete light-travel-budget interval.
- The memory-robust analysis and setting-choice procedure are retained.
- Measurement independence remains a declared source assumption; no ontic randomness premise is imported.
- The result rejects local factorization under those conditions, not determinism as a whole.
- The Bell record does not relabel the separate formal Walsh support product as measured variance.
- Tampered S and timing controls reject.

Falsification condition: Reject if either source identity changes; the complete S interval touches or falls below two; the trial, significance, timing or memory records fail; measurement independence or any stated limitation is omitted; the result is presented as excluding all determinism; ontic randomness is imported; the Walsh support law is called a measured variance; or any target changes the sealed survivor.

Measurement receipt: `sha256:c97b436ea794ddadea73446b5e06a1a88a0a23119e66279f989cba218a71c2b6`. Isolation certificate: `sha256:8a5f0cf9ed57acca9841b1e5ebeccb9f552f33865e270b5a8eacc1d0046eea83`. Custody certificate: `sha256:bf791660650594a56788cae969a412800b6365ee0b08619ddfad007e6d0e108b`.

Registered source descriptions:

- Loophole-free Bell inequality violation with superconducting circuits: https://doi.org/10.1038/s41586-023-05885-0 (sha256:fabe97e4970990cef6246f0541a8b9e80e47b0df58f10234b8f8155d1b4ade81)

Registered target identities:

- NATURE-STORZ-2023-WITHHELD-COMPLETE-BELL-RECORD from None

Shared claim-record clause PHYS-S028 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no Bell value, trial result, significance record or timing record available to formal candidate selection
- no claim that a Bell violation excludes all deterministic or complete-record models
- no import of random-number generation as ontic nondeterminism
- no omission of the measurement-independence assumption or the authors' stated assumption boundary
- no relabelling of support cardinality as statistical moment variance
- no claim that this Bell experiment separately measures the Walsh support product
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:10b33ae9570072c13faeaaf5093bd33adb08f6cc638bfc9eb8534039a811a012; engine receipt sha256:92401779cd8580dbaf3962207ae25e0b7fc75787f91e27681d1ac290de403085 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-QUANTUM-SUPPORT-UNCERTAINTY-050-92401779cd8580db.json; empirical-validation hash sha256:1c47006310cbe6ee2c744184c71d29789299e7ccf62f3d81b51de8b4fb68a4b7; measurement receipt sha256:c97b436ea794ddadea73446b5e06a1a88a0a23119e66279f989cba218a71c2b6.

350. Terminal electroweak measured-value correction

Claim identity: SFT-PHYS-VALIDATION-ELECTROWEAK-MEASURED-VALUE-053

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The exact forced terminal on-shell share lies inside the complete PDG interval [22333,22351]/100000. Its exact complement lies inside the outward-propagated compatible-input W/Z squared interval. No uncertainty is widened and no inconsistent aggregate is converted into a passing or failing SFT law.

The exact forced terminal on-shell share lies inside the complete PDG interval [22333,22351]/100000. Its exact complement lies inside the outward-propagated compatible-input W/Z squared interval. No uncertainty is widened and no inconsistent aggregate is converted into a passing or failing SFT law.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of the sealed terminal electroweak value, like-typed direct measurements, exact interval transport, source consistency, custody, proof/measurement separation, successor closure and extension. The declared exact boundary is: The admitted target-free terminal on-shell value; the complete direct on-shell row; the complete compatible W and Z rows; the unchanged internally inconsistent all-input aggregation record; both source identities; and every measurement-method boundary. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-terminal-electroweak-value-versus-like-typed-direct-measurements__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-terminal-electroweak-value-versus-like-typed-direct-measurements	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:191e23e5365839ffb39393534fad4c68cbb3c69a26526969dd2a2aa70a15bcee`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3e4b8abb05c6d184cb34a5a25f61a48cc359196fa3dd7514ef58bd8faf631abb`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7b78e6fbad92bcdf6ed876234c371f9ed1c4a68f45d8bf116c90814ba7403638`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:76e5b62b331c12427c35dbdd6a447ce1f368b12099344a66614725bb637d339b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:522d95548ec25f2dc17eb43ebaff006b93c24978311ef817c611f7f1f2b8eb28`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0d160846dfd86800df046d3e306ab0801659da92682854572c47e80fff5724d3`. Independent certificate: `sha256:bdb4d5c3e0a39efc8e1b4da389915a147cc4ce1089435221e0618fded1b7cfce`. Engine external-validation hash: `sha256:18d3d16b8154e97a765842993bf67bae6d35a558ba3824aa5787478415a389f1`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-STANDARD-MODEL-REVIEW
- PDG-2024-W-BOSON-LISTING

Observed comparison records:

- The exact forced on-shell share 1930922298157999/8642477221479757 lies inside [22333,22351]/100000.
- Its exact One-complement lies inside the complete compatible-input W/Z squared interval.
- No uncertainty was widened and no measured row selected the formal relation.

- The all-input aggregate and its uncertainties remain unchanged, but its source-identified incompatible input prevents it from being treated as a second like-typed physical target.
- Tampered direct-angle and compatible-W controls reject.

Falsification condition: Reject if either source changes; the sealed exact share leaves the direct on-shell interval; its complement leaves the compatible W/Z squared interval; any uncertainty is widened; the inconsistent aggregate is deleted, used as a second law, or relabelled as an SFT result; or target data alter the formal survivor.

Measurement receipt: `sha256:fde304793a48fcfd4a37491fce37c4c1f7b254c39eecc9759ba83bd4fab871e7`. Isolation certificate: `sha256:c0357f9ab5def03b84b5aeda8ed2025cf948e92685b8c4d00e07b1152b38663c`. Custody certificate: `sha256:f5df2b4aa8c65fbe72832ec209c323fbe46f8ea1071271d17ae09ce89fb9ae7d`.

Registered source descriptions:

- Review of Particle Physics: Electroweak Model and Constraints on New Physics: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf (sha256:8642888a3408d8c57fc673b379325b07f02948135491f64a2e42320e8929320a)
- Review of Particle Physics: W-boson listing: https://pdg.lbl.gov/2024/listings/rpp2024-list-w-boson.pdf (sha256:91cb466bfea8fa49b53ae53d2168797189c14a5a114f30d7cc926f64c4c1e772)

Registered target identities:

- PDG-ELECTROWEAK-WITHHELD-COMPLETE-MEASURED-VALUE-RECORD from PDG-2025-STANDARD-MODEL-REVIEW

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured weak angle or boson mass in the formal relation
- no fitted rung, correction, coefficient, uncertainty multiplier or selected central value
- no mismatch relabelled as successful empirical closure
- no deletion or alteration of the all-input aggregation record
- no consensus classification used as proof; source consistency is checked from the published record
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:c6c5f370a54459c49dc300cfe78fe04ca6b1a2b9d3c23c6fecb1006e6631123f`; engine receipt `sha256:e93f04515944a1951b78463db3083350fbc77e6d8506edbcdc13178cedfbc852` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-ELECTROWEAK-MEASURED-VALUE-053-e93f04515944a195.json`; empirical-validation hash `sha256:4152cde723ac46d74d2053f6b43f8ac609c54133f27b27750e673223a7f8d7df`; measurement receipt `sha256:fde304793a48fcfd4a37491fce37c4c1f7b254c39eecc9759ba83bd4fab871e7`.

351. Common Fold scale measured-value correction

Claim identity: `SFT-PHYS-VALIDATION-COMMON-SCALE-MEASURED-VALUE-054`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The terminal on-shell share lies inside [22333,22351]/100000. The support-eight share 25/106 lies inside [2331,2367]/10000. The complete registered low-transfer vector preserves the sub-W direction. No mismatch, uncertainty multiplier, fitted rung or measurement-selected exception is admitted.

> *The terminal on-shell share lies inside [22333,22351]/100000. The support-eight share 25/106 lies inside [2331,2367]/10000. The complete registered low-transfer vector preserves the sub-W direction. No mismatch, uncertainty multiplier, fitted rung or measurement-selected exception is admitted.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030`
- `SFT-PHYS-VALIDATION-ELECTROWEAK-MEASURED-VALUE-053`
- `SFT-PHYS-COUPLING-RUNNING-CONVERGENCE-TERMINAL-006`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of common-scale carrier, target-free relation, provenance, custody, proof/measurement separation, row completeness, successor closure and absence of extra rules. The declared exact boundary is: The admitted common-axis receipt; terminal and support-eight exact shares; all four scheme rows; all four numeric low-transfer rows; all eight plotted measurement classes; threshold boundaries; complete strong/electromagnetic receipt; and exact source identities. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-common-scale-values-versus-like-typed-measurements__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-common-scale-values-versus-like-typed-measurements`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:dcdd173f7ca0731f1c4330774a7d9b935db9e655a6eecd78b139eaf107e6db12`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:85653d59c680816e42666db3be96a88a1a392189b7bef80bda4929e13ab36f54`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:57af3942a29ed8a3075b1b67012bceab6fc7696e61057e9bd160186ff7dfdfb3`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8bfda0144733672b4df92ccafcd2d8adb6a8eed355f33f69a2c90d98143251ac`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b9f8f831567c843db4b2268175a876079d6341e2f9ffd633b52ba46ce5fac0d6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c5a642b9a74d2c41c3e81b7cfd550e2dc1003d4e4d59c7500759e43760495805`. Independent certificate: `sha256:8f17c55fb5249ada38631b72e5d7d780f1e38575b3cd70eceea171406ede09fe`. Engine external-validation hash: `sha256:29fcb4125abefbdef3a90b5ae64f4d21af9460bacc7e0410b88eefda51d438ba`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2026-ELECTROWEAK-SCALE-VECTOR
- PDG-2025-2026-STRONG-EM-COMPLETE-RUNNING-VECTOR

Observed comparison records:

- The exact forced terminal weak share lies inside [22333,22351]/100000.
- The exact forced support-eight share 25/106 lies inside [2331,2367]/10000.
- The complete E158 and APV intervals preserve the sealed below-W direction relative to the MS-Z interval.
- NuTeV remains unchanged as a source-identified interpretation-sensitive DIS extraction; its displacement is not an SFT result or acceptance condition.
- No row was deleted and no uncertainty was widened; displaced terminal and APV controls reject.

Falsification condition: Reject if a source or dependency changes; either forced value leaves its like-typed measured interval; the below-W direction reverses; a row or threshold boundary is omitted; NuTeV displacement is made a passing result; any uncertainty is widened; or target data alter the formal survivor.

Measurement receipt: `sha256:4bc4c5859b4f63e25cc5a73d4814a0f0030c4668e4b01ac2b95e6ac075ee7661`. Isolation certificate: `sha256:ddf578f829cf128b80abcd8fddfdb2932301a9f31f95ea82e6d94b6f1868c44e`. Custody certificate: `sha256:30a93b5c0742b2693862b910e4cc205ecdc38d1e1b40a58954623ea3ca788efd`.

Registered source descriptions:

- Review of Particle Physics 2026: Electroweak Model and Constraints on New Physics: https://pdg.lbl.gov/2026/reviews/rpp2026-rev-standard-model.pdf (sha256:a102f6252b7190dc423200271dffa7c805cd15a50391b1c578853d2f777611cb)
- Complete admitted strong and electromagnetic running vector: repository-bound source record (sha256:b83331089d96c073fbd5101753ba5c4716ae1a8b1b891e068684b6f7246d9953)

Registered target identities:

- PDG-COMMON-SCALE-WITHHELD-COMPLETE-MEASURED-VALUE-RECORD from PDG-2026-ELECTROWEAK-SCALE-VECTOR

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured weak angle in the formal common-axis construction
- no fitted support, coefficient, correction, uncertainty multiplier or selected central value
- no NuTeV displacement relabelled as successful empirical closure
- no deletion, alteration or silent retyping of any registered measurement row
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:a4943e87cae5b12b391da771e8435eb70068faff98f29bea497e45577fc7a96f`; engine receipt `sha256:0514d2a2fe7ad505dfd99c494f0c6ac6214169791c09596d4657e32ee1c458d8` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-COMMON-SCALE-MEASURED-VALUE-054-0514d2a2fe7ad505.json`; empirical-validation hash `sha256:a27a65924cde81c28c15ac470e4e89221a82d44493ee31ae50e4e581f83eb326`; measurement receipt `sha256:4bc4c5859b4f63e25cc5a73d4814a0f0030c4668e4b01ac2b95e6ac075ee7661`.

352. Terminal cosmic transport measured-value correction

Claim identity: `SFT-PHYS-VALIDATION-COSMIC-TRANSPORT-MEASURED-VALUE-055`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The exact upper enclosure of the complete 32-row mean squared normalised residual is below the One after the fourth exact enclosure round (three forced bisection refinements). The exact 11/5, 22/5, 17/32 and tension-One values lie inside the complete registered Planck, Haridasu and constant-state intervals. No sigma multiplier, fitted component, uncertainty rescaling, deleted row or mismatch-as-success predicate remains.

> *The exact upper enclosure of the complete 32-row mean squared normalized residual is below the One after the fourth exact enclosure round (three forced bisection refinements). The exact 11/5, 22/5, 17/32 and tension-One values lie inside the complete registered Planck, Haridasu and constant-state intervals. No sigma multiplier, fitted component, uncertainty rescaling, deleted row or mismatch-as-success predicate remains.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-PROBABILITY-STATISTICS-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of exact transport carrier, unit-normalized residual relation, provenance, custody, proof/measurement separation, complete rows, successor closure and no extra rule. The declared exact boundary is: The admitted terminal transport receipt; all 32 CCH rows; H0 and complete Planck budget; both registered acceleration reconstructions; constant-w interval; DESI model-comparison record; exact rational root enclosures; and every source identity. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-cosmic-transport-versus-complete-unit-residual-and-like-typed-values__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-cosmic-transport-versus-complete-unit-residual-and-like-typed-values`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:686a44491ee46aac075aa4752daa819f6417f039614f1f361594560a3b246a25`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:624147300356701da75d1412c41d5a6fdfbcfd9dc40b727d4e7cf83350d4e4a6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1924ea5750e2ae67ee9bfacca7939eb2ecbed5b8312b9812998458a02b5758c1`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:57a3b73bd4e72c2475d11e25b201e6b86988e15fcdec4f5c1bcf6c3c248febd0`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e9198019efe4e3f15762d11adf9acd3777c4728d42e4a65b5062087f18936725`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:94a56325bcf7bab112806fef190159f248a5ebb19391bb72d6832f0ecedb6af6`. Independent certificate: `sha256:a4916f203d0a194a8fa9028afbcda359a26b81d0ddb6280f3bfde2ce0df8eaa6`. Engine external-validation hash: `sha256:c073fd4672a744a2fabb64a70d07e9cb00a6bc145bd0fd2a85f7e7afb46f3c58`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- GOMEZ-VALENT-2023-CCH-32
- PLANCK-2018-BAO-BASELINE-BUDGET
- HARIDASU-2018-LATE-EXPANSION
- GOMEZ-VALENT-2019-ACCELERATION
- ESCAMILLA-2024-DARK-ENERGY-STATE
- DESI-DR2-2025-COSMOLOGY

Observed comparison records:

- All 32 direct chronometer rows contribute once to a complete exact normalised-residual ledger; its upper enclosure is below the One on the fourth exact enclosure round, after three data-independent rational-root refinements.
- The exact equality 11/5 and onset 22/5 lie inside the complete Planck budget transports.
- The exact acceleration magnitude 17/32 and onset 22/5 lie inside the complete Haridasu q0 and transition intervals.
- Tension-One lies inside the registered constant-state interval; alternate reconstruction and DESI model-comparison records remain unchanged but are not acceptance conditions.

- No uncertainty multiplier was selected or widened; tampered chronometer and transition records reject.

Falsification condition: Reject if a source or dependency changes; the complete normalised-residual lower enclosure reaches the One; any forced threshold, acceleration or static-state value leaves its like-typed interval; any row is omitted; an uncertainty is rescaled; a method/model-comparison displacement is rewarded; or target data alter the formal survivor.

Measurement receipt: `sha256:49342e727ece7edb6b0a68b4d001c15b772ae3d10898be217b458b03ede64b2b`. Isolation certificate: `sha256:0c8e6b029ffb29a423ec31f2af85827dbb7aa0e416d4efd4a4d53aeedc658673`. Custody certificate: `sha256:072be24ecd90149ee7303d23dba00c628b498d56d4bafee71b1ce1a0f30db2d1`.

Registered source descriptions:

- Cosmic chronometers to calibrate the ladders and measure the curvature of the Universe: https://doi.org/10.1093/mnras/stad1511 (sha256:cbd38055c3c9b27b2cbe1fb7456c60dfd7df3eeb366182932b7899d25207b128)
- Planck 2018 plus BAO baseline parameter table: https://doi.org/10.1051/0004-6361/201833880 (sha256:a8525585688e7ba818f8650fc0f8b73a449823d5e199a8be36e37b8e73a0e612)
- An improved model-independent assessment of the late-time cosmic expansion: https://doi.org/10.1088/1475-7516/2018/10/015 (sha256:4ac5d80b39e3dc963d5cc7a0c1b8996b0b895a974f08fa736803186bd100f74c)
- Quantifying the evidence for the current speed-up of the Universe: https://arxiv.org/abs/1810.02278 (sha256:55100ed889f8aad5ada957f33759d954466b944d21b461b4c63c62f0b0fb45d0)
- The state of the dark energy equation of state circa 2023: https://arxiv.org/abs/2307.14802 (sha256:67c8d75e79fd85d727012e72e8838651de6720a2e8898e225e251601814c20a5)
- DESI DR2 Results II: BAO measurements and cosmological constraints: https://arxiv.org/abs/2503.14738 (sha256:1e82f26e4cc3901b16168cd147f252bfa804f9c3caad3f4f7e3532640d237841)

Registered target identities:

- COSMIC-TRANSPORT-WITHHELD-COMPLETE-MEASURED-VALUE-RECORD from GOMEZ-VALENT-2023-CCH-32

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no external H, density, q, transition or state target in the formal transport law
- no fitted H0, density, exponent, residual coefficient or selected uncertainty multiplier
- no rowwise two-standard-uncertainty shortcut
- no DESI model preference or alternate reconstruction relabelled as an SFT result
- no deleted measurement row or rescaled uncertainty
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:0fe774dfc5e9e0d08a491f543b4c3012dc0bd07ed2a9e715f47636f519c84c64`; engine receipt `sha256:e3e00a1c023f76cd0817c7eaec86c0243ab10d338e552f6bae1f90a94a959380` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-COSMIC-TRANSPORT-MEASURED-VALUE-055-e3e00a1c023f76cd.json`; empirical-validation hash `sha256:b050727f8732043890a71a6cb17e10e5581e06453e7f5c374ad632a2aa562c73`; measurement receipt `sha256:49342e727ece7edb6b0a68b4d001c15b772ae3d10898be217b458b03ede64b2b`.

353. Criticality and turbulence measured-value correction

Claim identity: `SFT-PHYS-VALIDATION-CRITICALITY-MEASURED-VALUE-056`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. All five manganite structural keys identify the binary local-order mean-field class. The complete fifteen-value mean squared normalised residual is exactly 5286961/10584000, below the One. The erbium beta/gamma/nu intervals contain 1/2, One and 1/2; the turbulence interval contains 2/3; and both physical routes exhibit the registered falling five-thirds plateau. No material mismatch is an acceptance condition.

> *All five manganite structural keys identify the binary local-order mean-field class. The complete fifteen-value mean squared normalized residual is exactly 5286961/10584000, below the One. The erbium beta/gamma/nu intervals contain 1/2, One and 1/2; the turbulence interval contains 2/3; and both physical routes exhibit the registered falling five-thirds plateau. No material mismatch is an acceptance condition.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-CRITICALITY-UNIVERSALITY-TURBULENCE-TERMINAL-047`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-PROBABILITY-STATISTICS-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of class carrier, complete structural-key relation, provenance, custody, proof/measurement separation, all-row retention, successor closure and no extra rule. The declared exact boundary is: The admitted two-class formal census; complete structural key for each of five manganites; all fifteen manganite exponent values; complete erbium vector; complete turbulence structure interval; both physical spectrum routes; limitations; and every source identity. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-class-keys-versus-complete-measured-vectors__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-class-keys-versus-complete-measured-vectors`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:e1d06bfae39d2baff3af1ee014becfb762a4b6c7b83b9d53d33f73dc5b647353`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1719b267af4323421d187741a62fca2f5e1213bb21ece4856f4b3b95d445c11d`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:4eafee2cb6316ccb3f6e636fb0c19676c7053c57e782ed0653bda8898ac0d21a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f3db1e1d3ac6e5656325c9d1a8aa679172df0a5fe985f6b0f652e110c0b3c3fb`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:82c45b89bef04fe802501c9d86a6ede46331807acc9e22c09efb48fe0d14b6bd`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:abecd535b035d40744922b3b96b09e6fbb2a89d4ccd2fb516f0531445269b187`. Independent certificate: `sha256:b056e6b7a7cba98ff7975a1e6b5fdf01d22ab2a91bbb18a36a9269d25a30c965`. Engine external-validation hash: `sha256:299fa83470a7c8596c73ec61feda3a7e046673df23abd28923bc07be8918cd31`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- SPRINGER-CETIN-MANGANITE-CRITICAL-EXPONENTS-2026
- MCMASTER-LIN-ERBIUM-CRITICAL-SCATTERING-1993
- APS-MCCOMB-TURBULENCE-STRUCTURE-2014
- APS-HUANG-TURBULENCE-SPECTRUM-2010

Observed comparison records:

- All five complete manganite structural class keys identify the registered mean-field class.
- All fifteen beta/gamma/delta measurements enter once; their exact mean squared normalised residual is 5286961/10584000, below the One.
- The complete erbium vector contains one-half, One and one-half; the independent turbulence interval contains two-thirds.
- Both physical spectral routes exhibit the falling five-thirds compensated plateau with all limitations retained.
- La02 remains visible without its displacement becoming a result; tampered vector and structural-key controls reject.

Falsification condition: Reject if a source or dependency changes; any material structural key is incomplete; the complete fifteen-value mean squared residual reaches the One; erbium or turbulence excludes a sealed exponent; either spectrum route is absent; any row is omitted or uncertainty rescaled; La02 displacement is rewarded; or targets alter the survivor.

Measurement receipt: `sha256:27d3a2750e7db5e6204fbeb1b647aa84ac9a4dfad6b9de6a383638e56f9a3402`. Isolation certificate: `sha256:9535981b71f46414bfc904845396564a93e81667af7b4faf6410228639f0798a`. Custody certificate: `sha256:be2305b343aaa6b76070795f86a948a65d6d7f04dffec2428a216149f722d20e`.

Registered source descriptions:

- Structural, magnetic, and magnetocaloric properties of La and Sm co-doped CaMnO3 manganites: https://doi.org/10.1007/s10854-026-17400-y (sha256:afd0f783bd40b8ed376238eda9b254387c75c0c26ecc57488ab9b81b33d1d455)
- Critical magnetic scattering in erbium: https://doi.org/10.1063/1.353720 (sha256:d8b6495b7defb46daa4d1162967d11b4680012ebd4a8e6d60e8892b83b9b3564)

- Spectral analysis of structure functions and their scaling exponents in forced isotropic turbulence: https://doi.org/10.1103/PhysRevE.90.053010 (sha256:0c9f4321b52f78fe64b02684b4139a450955c12092c4faf8487f49bf6ff1d0f5)
- Second-order structure function in fully developed turbulence: https://doi.org/10.1103/PhysRevE.82.026319 (sha256:1bff196a24bc3f852dd384469c5f8e4112fcdf717c2416190ad44b0012ac06cb)

Registered target identities:

- CRITICALITY-WITHHELD-COMPLETE-MEASURED-VALUE-RECORD from SPRINGER-CETIN-MANGANITE-CRITICAL-EXPONENTS-2026

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured exponent or material label in the formal class census
- no selected matching subset or La02 mismatch-as-success predicate
- no fitted exponent, uncertainty multiplier, intermittency correction or measurement-selected class
- no row deletion or uncertainty rescaling
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:972770658891b5371be10884db72f00aacd524b07d5fe190dc434a5567c25f4f`; engine receipt `sha256:7e437d81bee9d4c88b304a013c5c980fa9d0390f95acaf30c2dfb45b30a1602a` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CRITICALITY-MEASURED-VALUE-056-7e437d81bee9d4c8.json`; empirical-validation hash `sha256:51ddbbffc9736ecc850a4828f962f1499b2fdd51809d6e252688eebdeaf7828c`; measurement receipt `sha256:27d3a2750e7db5e6204fbeb1b647aa84ac9a4dfad6b9de6a383638e56f9a3402`.

354. Thermal history and physical helium measured-value correction

Claim identity: `SFT-PHYS-VALIDATION-THERMAL-HISTORY-MEASURED-VALUE-058`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Exact exponent One lies inside [49/50,517/500]. Exact physical helium-isotope share 59/240 lies inside [489/2000,2471/10000]. The one-sixth to one-seventh freezeout sequence, positive deuterium, finite recombination support, eighteen extrema and all seven TT rows remain complete. No excluded quarter or angular noninteger record is admitted as a successful result.

Exact exponent One lies inside [49/50,517/500]. Exact physical helium-isotope share 59/240 lies inside [489/2000,2471/10000]. The one-sixth to one-seventh freezeout sequence, positive deuterium, finite recombination support, eighteen extrema and all seven TT rows remain complete. No excluded quarter or angular noninteger record is admitted as a successful result.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037`
- `SFT-PHYS-THERMAL-HELIUM-ISOTOPE-TERMINAL-057`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of physical thermal carrier, isotope relation, provenance, custody, proof/measurement separation, complete rows, successor closure and no extra rule. The declared exact boundary is: The admitted analytic and physical-isotope receipts; complete temperature, helium, deuterium, freezeout, recombination and acoustic records; all uncertainty endpoints; all seven TT rows; all eighteen extrema; projection boundary; and every source identity. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-physical-thermal-values-versus-complete-measurements__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-physical-thermal-values-versus-complete-measurements`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:71d5b821c08b24d0a200db067d1344d662e111fd6b5b28faad0fb8bc19cc9dd5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d8bb99bae704fa0f9359dba4a0077ebb8e4a6ac52415f747430e8fe64fef82a2`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a96d9010099338a297ce51e80473e184dd3f77cf24a73a0b87f2751a0407e06e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bd8b792a7864e21a1369417e13b923b95dee61035cb784af21ec4fd3bff2d139`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:93b71dfa37d98bc1b88e9b4ab0e4dd780330e0080ff83b251d8f83bb9f56eabb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d3afd654004d34d24ba55451eda4328e7b249a86254e778902bcfc98e0dfe7e6`. Independent certificate: `sha256:61be22939d2ea95b2cc9e77c9d966e7dd3d30860ba7f9caeae64d805a13069e5`. Engine external-validation hash: `sha256:e0a2835a1ae33c589af028c1579f4e7a0ef3b3d20d95f099031507e2a749f214`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NOTERDAEME-ET-AL-2011-CMB-TEMPERATURE
- AVER-ET-AL-2026-PRIMORDIAL-HELIUM
- COOKE-PETTINI-STEIDEL-2018-DEUTERIUM
- PLANCK-2018-OVERVIEW-PEAKS
- PLANCK-2018-COSMOLOGICAL-PARAMETERS
- PDG-2025-BBN-REVIEW

Observed comparison records:

- Exact CMB temperature exponent One lies inside [49/50,517/500].
- Exact physical primordial helium share 59/240 lies inside [489/2000,2471/10000].
- The one-sixth to one-seventh freezeout sequence and positive deuterium channel remain complete.
- Finite recombination support, eighteen extrema and all seven TT rows remain source-bound and positive; angular projection records are not mismatch results.
- No uncertainty was widened; displaced helium and incomplete-peak controls reject.

Falsification condition: Reject if a source or dependency changes; exponent One or 59/240 leaves its complete measured interval; the freezeout sequence, positive deuterium, finite recombination support, extrema or TT rows are missing; an uncertainty is widened; a mismatch is rewarded; or targets alter either formal survivor.

Measurement receipt: `sha256:8663e550fdb13a666d3c2b1d099e1b9f44b175879d5652b7909bc1e4ee0cbbcf`. Isolation certificate: `sha256:c2d36c150cf297b75ea33c3be9994f9032dc76d36def77c065ed4bba8e0f36d2`. Custody certificate: `sha256:c0e4760965d8d0c0f6e4e11a1bf5a19d2ddcc3404e752855a9a3c46dba600899`.

Registered source descriptions:

- The evolution of the Cosmic Microwave Background Temperature: https://arxiv.org/abs/1012.3164 (sha256:aa79c65170e84d6a8dc71dbd0876538c19619a1c926453ce5a54bd96d3f1efb7)
- The LBT Yp Project IV: A New Value of the Primordial Helium Abundance: https://arxiv.org/abs/2601.22238 (sha256:72f646a345c36b77211e6035d3a78345498f0766cf9fc0a54932ebe0dd670c6a)
- One percent determination of the primordial deuterium abundance: https://arxiv.org/abs/1710.11129 (sha256:1a6ec19d6568b8854a40e9dd587906f99a8ea94ecd2d5c712ad86aa506b93f4a)
- Planck 2018 results I: overview and cosmological legacy: https://arxiv.org/abs/1807.06205 (sha256:dca932893b7d2724aa2b8d33170fc1b5682c425dd56fcd3c5d7b2098d377db5c)
- Planck 2018 results VI: cosmological parameters: https://arxiv.org/abs/1807.06209 (sha256:8e172730faf07c9f4ff3fdcc7043f76ed67df6f76066d47df30d693025b6ce77)
- Big-Bang Nucleosynthesis: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-bbang-nucleosynthesis.pdf (sha256:1843c1d9025aa33c059700708f1041c4462364cdb582325f0c685a6ba1b38484)

Registered target identities:

- THERMAL-HISTORY-WITHHELD-COMPLETE-MEASURED-VALUE-RECORD from AVER-ET-AL-2026-PRIMORDIAL-HELIUM

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured helium central value or uncertainty in either formal survivor
- no fitted decay, capture, binding, reaction, projection or abundance coefficient
- no exact-quarter-exclusion or noninteger-angular-position mismatch rewarded as closure
- no deleted row or rescaled uncertainty

- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:03d3d89369145a006af94d8a419a408810d74d099a771168fec92d1b766194af`; engine receipt `sha256:7339aa07e1a9f52e8ef1fede03cbe17d38834362f1c9d63bb00b05deb6d49021` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-THERMAL-HISTORY-MEASURED-VALUE-058-7339aa07e1a9f52e.json`; empirical-validation hash `sha256:2376571e5468f9eadcadf80482ad61e6a43a9b0bd50a3812c903ee5f114eaa47`; measurement receipt `sha256:8663e550fdb13a666d3c2b1d099e1b9f44b175879d5652b7909bc1e4ee0cbbcf`.

355. Complete hadron Regge measured-value correction

Claim identity: `SFT-PHYS-VALIDATION-HADRON-REGGE-MEASURED-VALUE-060`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The forced squared carriers 3/5, 9/5, 3, 21/5 and 27/5 all lie inside their corresponding complete measured resonance-support intervals. All five rows and the rho5 omission status are retained. No fitted slope, intercept, residual, selected row, uncertainty widening or mismatch-as-success remains.

> *The forced squared carriers 3/5, 9/5, 3, 21/5 and 27/5 all lie inside their corresponding complete measured resonance-support intervals. All five rows and the rho5 omission status are retained. No fitted slope, intercept, residual, selected row, uncertainty widening or mismatch-as-success remains.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-HADRON-REGGE-TERMINAL-005`
- `SFT-PHYS-HADRON-REGGE-DIMENSIONAL-TERMINAL-059`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed Regge carrier, complete resonance support, provenance, custody, proof/measurement separation, complete rows, successor closure and no extra rule. The declared exact boundary is: The formal 059 receipt; all five registered masses, uncertainties, widths, width uncertainties and statuses; the exact most-restrictive support construction; common post-seal GeV-squared translation; and all source identities. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-exact-regge-carriers-versus-complete-measured-resonance-supports__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-exact-regge-carriers-versus-complete-measured-resonance-supports`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9576dd4d668afb3fa2951c8e13c59ec3779a02432720bd2a996d06280a35fc15`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:909b4ce12e13596c9c1aa2b479c79c5d9bd49154e27dba348aa3bfbd6d11a683`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:af80693aeb595bd40d2593d435871114812faa4839bae161891105d654f06b58`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3cff8caf802fccc5e3d071067220be16da18b1ada81fcd6e29ddb5345415346d`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d5d152764842943169dc695f4c58d24fe26ace16bb8e830a768f0adfb194a7ac`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0c8bc06ef57ebb6e35a54fb76c53932605eebaea3d9d70ffdbde3261c5aee370`. Independent certificate: `sha256:92422c7be03302267a28cc17db31e79b1e7a1236591bbd40866a1b6cb8e39dfe`. Engine external-validation hash: `sha256:f6b749a0f22d24de38b5f7e9d0743d1420f8dd540efcf7f885c378d5eebeb1bb`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-RHO-770
- PDG-2025-A2-1320
- PDG-2025-RHO3-1690
- PDG-2025-A4-1970
- PDG-2025-RHO5-2350

Observed comparison records:

- Exact zero-parameter Regge base is 3/5 GeV^2 and exact successor step is 6/5 GeV^2.
- All five exact squared carriers lie inside the corresponding most-restrictive measured resonance-support intervals.
- Every mass, mass uncertainty, width, width uncertainty and listing status is retained once.
- The rho5 summary omission and single-measurement status remain explicit.
- No slope, intercept, residual or correction is fitted; displaced and missing-row controls reject.

Falsification condition: Reject if a source or dependency changes; any exact carrier leaves its most restrictive measured resonance support; a row or status is missing; a width or uncertainty is widened; a coefficient or correction is fitted; a mismatch is rewarded; or targets alter the formal survivor.

Measurement receipt:
`sha256:26f6d86b2e87b8eccb5e37b1685aec25dfdc5d151d16c80a04217dd6d85502b0`. Isolation certificate: `sha256:c62628600e61bebf70c8ee49456ef12c27caf8e1ab3c52887a2fc68acdb7996f`. Custody certificate: `sha256:295f755df18b9ca5edde5d8478bf4e238a5754e38b9fb1f54d3bb3b9917011e6`.

Registered source descriptions:

- Particle Data Group: https://pdg.lbl.gov/2025/listings/rpp2025-list-rho-770.pdf (sha256:6877a2e12365b035d3391a1a28af2f49693d8a749df276be3c772491a50ca356)
- Particle Data Group: https://pdg.lbl.gov/2025/listings/rpp2025-list-a2-1320.pdf (sha256:6877a2e12365b035d3391a1a28af2f49693d8a749df276be3c772491a50ca356)
- Particle Data Group: https://pdg.lbl.gov/2025/listings/rpp2025-list-rho3-1690.pdf (sha256:6877a2e12365b035d3391a1a28af2f49693d8a749df276be3c772491a50ca356)
- Particle Data Group: https://pdg.lbl.gov/2025/listings/rpp2025-list-a4-1970.pdf (sha256:6877a2e12365b035d3391a1a28af2f49693d8a749df276be3c772491a50ca356)
- Particle Data Group: https://pdg.lbl.gov/2025/listings/rpp2025-list-rho5-2350.pdf (sha256:6877a2e12365b035d3391a1a28af2f49693d8a749df276be3c772491a50ca356)

Registered target identities:

- HADRON-REGGE-WITHHELD-COMPLETE-MEASURED-VALUE-RECORD from PDG-2025-RHO-770

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no pole-mass standard uncertainty mislabelled as the support of an unstable resonance
- no fitted slope, intercept, string tension, residual, mass correction or selected trajectory subset
- no widened mass uncertainty or resonance width and no omission of rho5 status
- no target access before the formal survivor and prediction label seal
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof scalar

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest
`sha256:a0c377dd985d02964b48a385b4625fa59a3208339d8f60503314f04a7fced0ff`; engine receipt `sha256:15c9daaa90d88c5193e5e3af4a5ca64a75c0418ee85797a2b90607c37a21cb06` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-HADRON-REGGE-MEASURED-VALUE-060-15c9daaa90d88c51.json`; empirical-validation hash
`sha256:b66c9c9da0cfb84a24085e0b1d1bdb2ec00e7fd64a02bdbc0210b9403aca1f1e`; measurement receipt `sha256:26f6d86b2e87b8eccb5e37b1685aec25dfdc5d151d16c80a04217dd6d85502b0`.

356. Complete particle-generation, mass-pattern, mixing and longevity comparison

Claim identity: `SFT-PHYS-VALIDATION-PARTICLE-MODE-GENERATION-052`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The forced count three lies in the PDG LEP-SLC precision interval [2989,3003]/1000. The independent direct determination 292/100 with stated uncertainty 5/100 is preserved exactly and has central-value displacement 8/5 stated uncertainties from three; it is neither adjusted nor used to select the result. Both terminal charged-lepton mass-ratio intervals, the available s/d and b/s quark ratio intervals, all four terminal CKM intervals and the positive-neutrino mass/mixing support pass their registered comparisons; t/c remains without an exact scheme-matched direct comparator. The charged-lepton records give m_e<m_mu<m_tau and lifetime_e>lifetime_mu>lifetime_tau. These measurements test the terminal polynomial and ordering consequences, not the superseded site-as-mass, universal rate-ratio, dimensional lifetime-equality or extra-spatial-dimension claims.

> *The forced count three lies in the PDG LEP-SLC precision interval [2989,3003]/1000. The independent direct determination 292/100 with stated uncertainty 5/100 is preserved exactly and has central-value displacement 8/5 stated uncertainties from three; it is neither adjusted nor used to select the result. Both terminal charged-lepton mass-ratio intervals, the available s/d and b/s quark ratio intervals, all four terminal CKM intervals and the positive-neutrino mass/mixing support pass their registered comparisons; t/c remains without an exact scheme-matched direct comparator. The charged-lepton records give m_e<m_mu<m_tau and lifetime_e>lifetime_mu>lifetime_tau. These measurements test the terminal polynomial and ordering consequences, not the superseded site-as-mass, universal rate-ratio, dimensional lifetime-equality or extra-spatial-dimension claims.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051`
- `SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002`
- `SFT-PHYS-VALIDATION-QUARK-CKM-003`
- `SFT-PHYS-MATTER-CKM-TERMINAL-004`
- `SFT-PHYS-VALIDATION-NEUTRINO-MASS-MIXING-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed particle-mode law, full generation/mass/mixing/lifetime vector, provenance, target isolation, proof/measurement separation, complete-row retention, successor closure and extension. The declared exact boundary is: The admitted Claim 051; the precision fitted neutrino-count measurement and the unchanged independent direct determination; both charged-lepton ratios; all available quark ratios and the missing-comparator boundary; all terminal CKM rows; the positive-neutrino mass/mixing vector; all three charged-lepton masses and lifetimes; heavy-lepton search limits; and every interpretive limitation. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-particle-mode-generation-law-versus-complete-registered-spectrum-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-particle-mode-generation-law-versus-complete-registered-spectrum-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:234198c4dedadb99627b00ac01684e59ec98279b3dafe9b154ec25a1a9131ace`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9fe79983d3b6a4b8bc543dfd8e9dc5f53b1908f3855e6cc7aa49b06f9a01b32b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:206dd3eb485875b622471c84fb7661aff32d95c18749effd964e6c52ee7ae1f7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:31ba6457edc32db6d1ad37a5d913d2d3b023e56034a79ac890b44265cb17ac22`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6f167c1bbb175fa1a86526e6de7ff54306a535b9c92573ad855a4822e3d8dd37`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:698b9367acec9c6f40c5463c18f1c0c8f682c4f00ec78afdac8de48b65050da1`. Independent certificate: `sha256:8fe3ed5259739b150bbef45c19438bd41f267c21bfb732c49a37c87eef94be14`. Engine external-validation hash: `sha256:8572aa491dcd0973dcb9e6e6bf5c163ec07f8b93df9736332fbaaa19f1bbcc9f`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2025-LEPTON-SUMMARY
- NIST-CODATA-2022-ALL-CONSTANTS
- PDG-2025-QUARK-MASSES
- PDG-2025-CKM-MATRIX
- PDG-2025-NEUTRINO-MIXING

Observed comparison records:

- The precision fitted neutrino-type interval [2989,3003]/1000 contains the forced count three.
- The independent direct determination 292/100 with stated uncertainty 5/100 is retained without adjustment.
- Its exact central-value displacement from three is 8/5 stated uncertainties; it does not select or alter the forced law.
- Both terminal charged-lepton ratios, available s/d and b/s ratios, all terminal CKM rows and the positive-neutrino vector pass their registered comparisons.
- No exact scheme-matched direct t/c comparator is claimed.
- Charged-lepton mass intervals increase electron-muon-tau and lifetimes decrease in the same label order.
- Heavy-lepton search limits, measurement-model provenance and nondecisive neutrino-ordering boundaries are retained.
- No coordinate fraction is relabelled as mass and no structural reach count is relabelled as seconds.
- Tampered fit, altered direct record and reversed-lifetime controls reject.

Falsification condition: Reject if any source changes; the precision generation interval excludes three; the independent direct row is omitted, altered, rescaled or used to select the law; a registered terminal mass or mixing comparison fails; the charged-lepton mass/lifetime orders reverse; the missing t/c comparator is presented as measured; any search or model boundary is omitted; an old site coordinate is relabelled as mass; or target data alter the formal survivor.

Measurement receipt: `sha256:69150d847fbccce53bef6578047d3afceb4cd159eb05e20b02549956ac43a03d`. Isolation certificate: `sha256:bc0344002addffce79dec59fadf802a036be32f799a4e6f5ebf47b4e55bd75f8`. Custody certificate: `sha256:26d5c53f2b8b66cd7d153272f3f414edd69eb0243963b7704a255786369ca026`.

Registered source descriptions:

- Review of Particle Physics 2025 update: Leptons summary tables: https://pdg.lbl.gov/2025/tables/rpp2025-sum-leptons.pdf (sha256:5aa9a3a33b554204056cb04c42319bb32c9664ae518907652ff5cdc1cb87e5bb)
- 2022 CODATA recommended values: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)
- Review of Particle Physics: Quark Masses: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf (sha256:d544d099aa15739ec83d87711bd4c5b1e0a1032d6f70aaa4847ec78601f7aeae)
- Review of Particle Physics: CKM Quark-Mixing Matrix: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf (sha256:a0a78578971f38ff89c6fc5579bc608de41ec383a205dc25cba1d26f7145610a)
- Review of Particle Physics: Neutrino Masses, Mixing, and Oscillations: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf (sha256:d7067e2e3c9098cc924f10ffbca579c557fb8e848bf3acc17f9815598cdda7a6)

Registered target identities:

- PARTICLE-MODE-GENERATION-WITHHELD-COMPLETE-RECORD from PDG-2025-LEPTON-SUMMARY

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no external particle mass, lifetime, generation count, mixing entry or search limit in the formal survivor
- no omission, rescaling or reinterpretation of the independent direct neutrino-count determination or the absent exact t/c comparator
- no use of generation coordinate fractions as measured mass values
- no claim that structural subtraction reach equals a dimensional lifetime ratio
- no claim that quarter-mode 1:1:4 is a universal observed mixing or decay-rate ratio
- no inference that heavy-particle search limits prove absolute nonexistence
- no extra spatial dimension or compactification rule selected from particle data
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:7b1248718fba75f0e23a4f31f0b8fbf98cdd266cfdf8dfed9cc4c8e2d8660b55`; engine receipt `sha256:0a2ca2749eed1cdaf6b92ab645ace007b3190ac5e5bb952e3a0cbdedff1c0804` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-PARTICLE-MODE-GENERATION-052-0a2ca2749eed1cda.json`; empirical-validation hash `sha256:0bae7a0db26bde3b738ec6c24f82bfc74856941136042da60d0971c4ef3feb7b`; measurement receipt `sha256:69150d847fbccce53bef6578047d3afceb4cd159eb05e20b02549956ac43a03d`.

357. Measured dark abundance and complete Smithion/LFV standing-test record

Claim identity: `SFT-PHYS-VALIDATION-DARK-SMITHION-LFV-062`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The exact 27/5 ratio lies inside the complete Planck dark/baryon interval, and 27/5 times 0.0224 gives 0.12096 inside the reported cold-dark interval [0.119,0.121]. The 175-galaxy SPARC record rejects finite baryon-only inverse-square asymptotics and supports additional gravitating matter under the admitted law. MEG II and BaBar provide three positive upper limits after null searches, not three measured rates; the 3:5:20 and 4:1 LFV predictions remain standing tests. No measured Smithion mass exists, so all twelve mass-ratio enclosures remain standing predictions rather than fabricated measured-value matches.

> *The exact 27/5 ratio lies inside the complete Planck dark/baryon interval, and 27/5 times 0.0224 gives 0.12096 inside the reported cold-dark interval [0.119,0.121]. The 175-galaxy SPARC record rejects finite baryon-only inverse-square asymptotics and supports additional gravitating matter under the admitted law. MEG II and BaBar provide three positive upper limits after null searches, not three measured rates; the 3:5:20 and 4:1 LFV predictions remain standing tests. No measured Smithion mass exists, so all twelve mass-ratio enclosures remain standing predictions rather than fabricated measured-value matches.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061`
- `SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001`
- `SFT-PHYS-FIELD-INVERSE-SQUARE-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of the sealed law, Planck density rows, galaxy discriminator, LFV search rows, prediction status, target isolation, complete-row custody and no-extra-rule. The declared exact boundary is: The admitted Claim 061; both Planck density rows with uncertainties; the complete SPARC sample statement; all three registered LFV radiative upper-limit rows; every current Smithion measurement-status row; and no target access before seal. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-dark-smithion-lfv-law-versus-complete-current-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-dark-smithion-lfv-law-versus-complete-current-record`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:e37c8e8a0fb85346cac4c921b758a9b0adbee5b5a358a99d1be8f27ef73116a4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8b245e9d86ac5517361d422e8a43f2cee3d25a6c2e1db291e6617d70683e1605`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:eb6f888f446824a0cab8d1c97cc6d52652a7ffaed86670f240e31fe6b7493453`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7f053e393fab0a7adaae778ff2b7613165b852a1e23f2d3b9cd1a3d1d0c8cb02`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b28ad019c6371ee4d5bcebd9bcf0bae85ff05997cd206c01e9067efc1f4de628`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e4a059747073e52276d7db230a328c84cfc618a73a9adbf29066b563e3de9b45`. Independent certificate: `sha256:edc9f8101bfff4a5025bb403ee6d1f854e6c149f73fdc87c80c3b78c1fc12731`. Engine external-validation hash: `sha256:c16302fba31483b63cb4c01692f1bda069ba8f75e09da2798a935bb2324a087b`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PLANCK-2018-VI-ABSTRACT-DENSITIES
- SPARC-2016-175-GALAXY-MASS-MODELS
- MEGII-2025-MU-E-GAMMA
- BABAR-2010-TAU-LFV-GAMMA

Observed comparison records:

- The exact dark/baryon ratio 27/5 lies inside the complete Planck density-ratio interval.
- Comparison-side transport 27/5 times 0.0224 equals 0.12096 and lies inside [0.119,0.121].
- The complete 175-galaxy SPARC record requires additional gravitating support under the admitted inverse-square law; it does not alone identify a particle.
- MEG II and BaBar retain all three null-search upper limits. They do not constitute measured nonzero LFV rates, so no 3:5:20 match is fabricated.
- No registered Smithion mass measurement exists; all twelve spectra remain exact standing predictions.
- Tampered-density and fabricated-rate controls reject.

Falsification condition: Reject if any source changes; 27/5 leaves the complete Planck ratio interval; 0.12096 leaves the complete cold-dark interval; the galaxy discrepancy row is omitted; an LFV limit is altered or relabelled as a rate; a Smithion is relabelled as measured without a source; target data select the formal law; or any hostile control passes incorrectly.

Measurement receipt: `sha256:2da4d4ef1c11b3dec958dac4f13b5bac740eea49b084d8131d511b81b59ed11a`. Isolation certificate: `sha256:bd444dda787a62b1b01e8969c0d1e8aac8245ad3fc1740c14133b8c32c67bd8d`. Custody certificate: `sha256:f5dd79b438273ad54e593d3009e4b590d9c8e3656e2cf0117cc0502032081cc0`.

Registered source descriptions:

- Planck 2018 results. VI. Cosmological parameters: https://arxiv.org/abs/1807.06209 (sha256:274e0189de5846ce0b8c2d7b83ae06c72587cf8325d4d2b2338e88dd0a74a88f)
- SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves: https://arxiv.org/abs/1606.09251 (sha256:d089215877213661e40965543ee7e05736619082ad16d95e65ec059029588c63)
- New limit on the muon-to-electron-gamma decay with the MEG II experiment: https://arxiv.org/abs/2504.15711 (sha256:0efbf99543c92d340eb7b07f40e6fea580b7746caab26c9233d3e30e8b71a5b6)
- Searches for Lepton Flavor Violation in the Decays tau to e gamma and tau to mu gamma: https://arxiv.org/abs/0908.2381 (sha256:ba3e1352f4c14b1867e06f6436b4a6b1e1980cd55ce52d9a9454bd618ed70015)

Registered target identities:

- DARK-SMITHION-LFV-WITHHELD-COMPLETE-RECORD from PLANCK-2018-VI-ABSTRACT-DENSITIES

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no density, galaxy datum, search limit or particle mass in formal survivor selection
- no upper limit relabelled as an observed rate or relative-rate measurement
- no null search relabelled as proof of particle identity
- no unobserved Smithion relabelled as a measured discovery
- no fitted abundance, cross-section, mediator mass, branching fraction or uncertainty multiplier
- no negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:0111d63af2037414d6394950c563e819550f34d1196e42a6ee1a4614e0c787f9`; engine receipt `sha256:51e623e6512223c23a0f0cae604ca556ed30fd15173236ffd1c889dd6424e27d` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-DARK-SMITHION-LFV-062-51e623e6512223c2.json`; empirical-validation hash `sha256:8eb364d2768268207e98259bfe3f0269ddf1685006fb1d491750d3163521c832`; measurement receipt `sha256:2da4d4ef1c11b3dec958dac4f13b5bac740eea49b084d8131d511b81b59ed11a`.

358. Blind neutron-EDM and multi-channel proton-stability validation

Claim identity: `SFT-PHYS-VALIDATION-STRONG-CP-BARYON-STABILITY-064`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The PSI result retains its null central status, positive 1.1 and 0.2 times ten-to-the-minus-26 uncertainty magnitudes and positive 1.8 times ten-to-the-minus-26 90-percent upper bound. The six proton modes retain 90-percent partial-lifetime lower limits 2.4e34, 1.6e34, 1.4e34, 7.3e33, 7.2e33 and 4.5e33 years. No source reports a statistically significant proton-decay signal; the explicitly retained candidates are background compatible. Thus no registered row violates the sealed aligned-One or fibre-preserving baryon-One predictions.

The PSI result retains its null central status, positive 1.1 and 0.2 times ten-to-the-minus-26 uncertainty magnitudes and positive 1.8 times ten-to-the-minus-26 90-percent upper bound. The six proton modes retain 90-percent partial-lifetime lower limits 2.4e34, 1.6e34, 1.4e34, 7.3e33, 7.2e33 and 4.5e33 years. No source reports a statistically significant proton-decay signal; the explicitly retained candidates are background compatible. Thus no registered row violates the sealed aligned-One or fibre-preserving baryon-One predictions.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STRONG-CP-BARYON-STABILITY-TERMINAL-063`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`

- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed formal carrier, all registered EDM and proton-search rows, source binding, capability closure, proof/measurement separation, row completeness, successor closure and no-extra-rule. The declared exact boundary is: The sealed Claim 063; the direct neutron-EDM central-status, statistical uncertainty, systematic uncertainty and upper-limit rows; all six registered Super-Kamiokande mode limits and candidate interpretations; all four primary source snapshots; and no target access before the prediction seal. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-strong-alignment-and-baryon-stability-versus-complete-registered-direct-search-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-strong-alignment-and-baryon-stability-versus-complete-registered-direct-search-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:38100d6833e49ccffbea934b617894201371284cc9db173c1944618d174e17b9`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:576d04f2cc2e35de96d1f95fd64b1543e032a14fbc27c4ffa7bcb81c18a74c40`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:636aa152173b656c5c090c6023503779b90c47d91a6c35b0fa26756a8bef684f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ad6e2347a557026397acb3574713a1d49b7b2d0a4ae2ed9a7b9c97e3be0f3d68`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:afb52c0be23620df9d200e86ca2d5a08174b3678f1b571358534c5b77bea02c0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4d0799c837a396410e90962d9354a8b4a675a30f8bb6093077312c1c394f39eb`. Independent certificate: `sha256:219d673b42b16ff5a7b5ab93deca735ccae9d63fa7962177f2710df976e6eda0`. Engine external-validation hash: `sha256:1d3374be2ed6753471294e271f5bfb58cba46e4f0213f2cee5a7f84ba71f6a5b`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NEDM-2020-PSI-PERMANENT-EDM
- SUPERK-2020-PROTON-E-MU-PI0
- SUPERK-2024-PROTON-E-MU-ETA
- SUPERK-2026-PROTON-E-MU-TWO-PI0

Observed comparison records:

- PSI reports a null neutron-EDM central displacement with both positive uncertainty magnitudes and a finite 90-percent upper limit of 1.8e-26 e cm.
- The neutron-EDM row is consistent with the sealed empty-One strong-dipole prediction; it is not relabelled as numerical proof zero.
- All six registered Super-Kamiokande proton modes retain their finite 90-percent partial-lifetime lower limits.
- No registered mode reports a statistically significant proton-decay signal; every candidate interpretation, including background-compatible events, is retained.
- Finite lifetime limits are not called infinity and do not select the already sealed proton-stability law.
- Confirmed-EDM and confirmed-proton-decay hostile controls both reject.

Falsification condition: Reject if any source changes; the direct EDM record reports a confirmed nonempty displacement; any proton mode reports a statistically significant decay signal; a candidate or uncertainty is omitted or relabelled; a finite limit is called exact absence/infinity; target data select the formal law; or any hostile control passes incorrectly.

Measurement receipt: `sha256:13d218828f910869fcffd434f0367d39e0bbd287fb203e61f56303f6b7cf369c`. Isolation certificate: `sha256:7a7bcee294d5be0d191c5ffefa7728d992a8bdaa3535aea6976af11a81bc8ce3`. Custody certificate: `sha256:0f2c58711f4c0cc427856875accad7747aee8abbe00cda684166fe0a3c628cd0`.

Registered source descriptions:

- Measurement of the permanent electric dipole moment of the neutron: https://arxiv.org/abs/2001.11966 (sha256:1afeb830f372c7725cca9fd3d004333fe19cade698fd9798af9dfac7adc5c180)
- Search for proton decay via p to e-plus pi-zero and p to mu-plus pi-zero: https://arxiv.org/abs/2010.16098 (sha256:6fa532b3b2671e8d6e028308297b95837567a6e3608b471aaa8af00fff6a9679)
- Search for proton decay via p to e-plus eta and p to mu-plus eta: https://arxiv.org/abs/2409.19633 (sha256:5b4bf694378930d7e38935546fea9c5deced9531137b0da276778e7b67fb4937)
- Search for proton decay via p to e-plus two-pi-zero and p to mu-plus two-pi-zero: https://arxiv.org/abs/2604.10975 (sha256:06d10f843d39b54a14465628d72a9af18920976369326b1c717e3621b2109c98)

Registered target identities:

- STRONG-CP-BARYON-STABILITY-WITHHELD-COMPLETE-RECORD from NEDM-2020-PSI-PERMANENT-EDM

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no neutron-EDM value, uncertainty, proton candidate or lifetime limit in formal survivor selection
- no null central estimate relabelled as conventional numerical proof zero
- no finite experimental upper or lower limit relabelled as an exact absence or completed infinity

- no background-compatible event relabelled as proton decay and no candidate row omitted
- no fitted theta, axion coefficient, decay rate, lifetime, confidence multiplier or tolerance
- no negative, irrational, imaginary, floating, NaN, continuum or infinite Fold proof scalar

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:8892bb86644f083a8401009a9263a9eb545ef11baf0ad4641c7247d412119385`; engine receipt `sha256:48e9dbebdb29b3902876a09cde2154eea3700904898930c1f12ab0a5e5ed9a09` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-STRONG-CP-BARYON-STABILITY-064-48e9dbebdb29b390.json`; empirical-validation hash `sha256:725517ecde0475192e96b7c9f79419b33afebe5a5aacebf1dbae29ed89b05673`; measurement receipt `sha256:13d218828f910869fcffd434f0367d39e0bbd287fb203e61f56303f6b7cf369c`.

359. Blind terminal Higgs mass and self-coupling validation

Claim identity: `SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Using the post-seal PDG dimensional reference v = 12311/50 GeV, the sealed ratio predicts exactly m_H = 31557437733819647/251923197734500 GeV = 125.266104978... GeV. The PDG 2025 listed average is 626/5 +/- 11/100 GeV, whose reported-uncertainty interval [12509/100,12531/100] contains the prediction; the exact displacement from its central value is 16653377460247/251923197734500 GeV. ATLAS reports 12511/100 +/- 11/100 GeV and CMS reports 3126/25 +/- 3/25 GeV; their exact positive offsets remain in the certificate and are not called aggregate failures. The exact native lambda is 657077816569574182495 9729/50772238045420788745992200. The 2026 ATLAS-CMS 95-percent normalised kappa-lambda interval, recorded as a below-reference direction of magnitude 71/100 through 61/10, contains the SFT normalised unity correspondence but does not yet resolve native lambda precisely.

> *Using the post-seal PDG dimensional reference v = 12311/50 GeV, the sealed ratio predicts exactly m_H = 31557437733819647/251923197734500 GeV = 125.266104978... GeV. The PDG 2025 listed average is 626/5 +/- 11/100 GeV, whose reported-uncertainty interval [12509/100,12531/100] contains the prediction; the exact displacement from its central value is 16653377460247/251923197734500 GeV. ATLAS reports 12511/100 +/- 11/100 GeV and CMS reports 3126/25 +/- 3/25 GeV; their exact positive offsets remain in the certificate and are not called aggregate failures. The exact native lambda is 6570778165695741824959729/50772238045420788745992200. The 2026 ATLAS-CMS 95-percent normalized kappa-lambda interval, recorded as a below-reference direction of magnitude 71/100 through 61/10, contains the SFT normalized unity correspondence but does not yet resolve native lambda precisely.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-HIGGS-SYMMETRY-TERMINAL-065`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed Higgs carrier, dimensional reference, aggregate mass, individual mass rows, source custody, self-coupling boundary, measurement separation and no-extra-rule. The declared exact boundary is: The sealed Claim 065 ratio and coupling; the PDG 246.22 GeV dimensional reference; the complete PDG listed average and uncertainty; the direct ATLAS and CMS mass rows with every reported uncertainty; the latest combined ATLAS-CMS kappa-lambda interval; all five source snapshots; and no target access before seal. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-terminal-Higgs-ratio-and-self-coupling-versus-complete-current-record__source-bound-proof-trace__capability-closed-prediction_` _separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-terminal-Higgs-ratio-and-self-coupling-versus-complete-current-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:c9f03004accc73fcf2f616a607413669e9bb114672dda69254dc4dfb5765f1b2. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:7cb3166cd4916a6c62c86c5c2adc3ff09b42876cb10d5a8006aedbc1a51c6130.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:f94882b3dd5c935411d6ebaaeaa5fbbd9e7bd7589dee0a084614ccc76ef67cb1.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:569660b6c17442aad7ef9beabf9e8a81dcb02d4126315d8add78afa41d59b5f6.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:e6aa6fdc968a856053a7eb789a45daa916263e18008360f5727bfd6dec31037f.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:b9feec25ed76098c2f850434d0ccb1fd68e8de7db024e6f98697faebca453233. Independent certificate: sha256:d2fdff3ecd3577f12218d6ff2ecd49bcb7035ccd7170de7b4b0d05a954d1f956. Engine external-validation hash: sha256:7a680e6dcacaa8f558fdffd086fada8181356528069b1adae675f94c7b2a7191.

Shared claim-record clause PHYS-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-2024-ELECTROWEAK-VACUUM-SCALE
- PDG-2025-HIGGS-MASS-AVERAGE
- ATLAS-2023-COMBINED-HIGGS-MASS
- CMS-2024-FOUR-LEPTON-HIGGS-MASS

- ATLAS-CMS-2026-HIGGS-PAIR-COMBINATION

Observed comparison records:

- The sealed exact ratio and post-seal PDG 246.22 GeV dimensional reference give 125.266104978... GeV.
- The prediction lies inside the PDG 2025 125.20 +/- 0.11 GeV listed-average interval.
- ATLAS and CMS individual offsets are retained exactly and are not falsely described as one-uncertainty agreements.
- The exact native coupling is 0.1294167525...; normalised unity lies inside the 2026 combined direct-search interval.
- The direct self-coupling constraint remains broad and is not called a precision measurement.
- Outside-mass, outside-coupling and erased-offset hostile controls all reject.

Falsification condition: Reject if any source changes; the sealed mass prediction lies outside the complete authoritative aggregate interval; normalised unity lies outside the direct self-coupling interval; an individual offset or uncertainty is omitted; a broad limit is called a precision measurement; target data select or modify the formal law; a signed external coordinate becomes a Fold proof scalar; or any hostile control passes incorrectly.

Measurement receipt: `sha256:f86cdc91c32c1dae5d3677b2b4b0de33d01ac3f3cddbb717b8a379be425d5a29`. Isolation certificate: `sha256:477f2c064986183a80e924e90f8fd3cc398a221df87eb8e9cbe83db1a23b78c1`. Custody certificate: `sha256:628b1386b9dd4a7f87bf260ec9dd7cabacef458c49ea06bc754f825c1cdeb5c9`.

Registered source descriptions:

- Electroweak Model and Constraints on New Physics: https://pdg.lbl.gov/2024/reviews/rpp2024-rev-standard-model.pdf (sha256:4689caae925ee279228684249333a8395a40dda74a94863549b48f758d3fdd3f)
- Particle Data Group Higgs-boson listing: https://pdg.lbl.gov/2025/listings/rpp2025-list-higgs-boson.pdf (sha256:228835096e2003bb332f456ad412cbfbd628a6973be715309da7d2f419ac77bc)
- Combined measurement of the Higgs boson mass from H to gamma-gamma and H to ZZ-star to four leptons: https://arxiv.org/abs/2308.04775 (sha256:b3d2c09226e54f4553a381c70e41095f8f1c18aec3aabbc845e0481aa9207e4e)
- Measurement of the Higgs boson mass and width using the four-lepton final state: https://arxiv.org/abs/2409.13663 (sha256:8dfaa73b5df3a958998a1a95dc123dc02013ef22c0a4fc3f82253550fc1c1807)
- Combination of ATLAS and CMS searches for Higgs boson pair production at 13 TeV: https://arxiv.org/abs/2602.23991 (sha256:eaf44d724e33f500781a964a2918b5d2062c55be760704736346233ed02ff678)

Registered target identities:

- HIGGS-TERMINAL-WITHHELD-COMPLETE-RECORD from PDG-2025-HIGGS-MASS-AVERAGE

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no Higgs mass, VEV, self-coupling interval, uncertainty or aggregate in formal survivor selection
- no coefficient, correction term, target-selected tolerance, uncertainty inflation or fitted series
- no omission of the ATLAS or CMS individual offsets and no claim that both lie within one reported uncertainty
- no direct-search interval relabelled as a precision measurement of native lambda
- no signed external coordinate imported as a negative Fold proof scalar
- no negative, irrational, imaginary, floating, NaN, continuum or infinite Fold proof scalar

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:cb014d09798e1ef2e597b0c9d2d9081635d84d9aa8e8eb6ee91c05176b5a63a4`; engine receipt `sha256:f69729204538a9c0949fa5c4e3dfce51beece2ebe2d4e43163910d6192f9b523` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066-f69729204538a9c0.json`; empirical-validation hash sha256:56c919c8c0a8107a97ead48490ea16b9ae64ae334208a7ae8530cdfcd15c24a4; measurement receipt sha256:f86cdc91c32c1dae5d3677b2b4b0de33d01ac3f3cddbb717b8a379be425d5a29.

360. Blind stellar, galactic and tidal terminal validation

Claim identity: SFT-PHYS-VALIDATION-STELLAR-GALACTIC-TIDAL-068

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The NASA solar record reports sound-speed and adiabatic-index inference at the 1/10000 precision order and density at 1/1000; its reference-model convective boundary 1427/2000 lies inside the observed 713/1000 +/- 1/1000 interval and is retained only as external structure evidence. The complete six stellar slopes are 507/250, 1143/250, 5743/1000, 4329/1000, 3967/1000 and 573/200 with their six reported uncertainties; the 3967/1000 row contains four and the 573/200 row contains three, while the other four do not and remain explicit. SPARC contributes all 175 galaxies. The 153-galaxy baryonic Tully-Fisher result is 77/20 +/- 9/100 with intrinsic scatter 3/50: its central interval excludes four, while the reported systematic interval [7/2,4] contains the sealed endpoint. The Bullet Cluster records an eight-sigma mass/plasma separation. The Moon records equal approximately-655-hour orbital and rotation periods. MESSENGER records Mercury's 29323073/500000 +/- 11/1000000 day rotation and experimentally verifies its 3:2 resonance, preserving the separately generated eccentric-resonance boundary.

The NASA solar record reports sound-speed and adiabatic-index inference at the 1/10000 precision order and density at 1/1000; its reference-model convective boundary 1427/2000 lies inside the observed 713/1000 +/- 1/1000 interval and is retained only as external structure evidence. The complete six stellar slopes are 507/250, 1143/250, 5743/1000, 4329/1000, 3967/1000 and 573/200 with their six reported uncertainties; the 3967/1000 row contains four and the 573/200 row contains three, while the other four do not and remain explicit. SPARC contributes all 175 galaxies. The 153-galaxy baryonic Tully-Fisher result is 77/20 +/- 9/100 with intrinsic scatter 3/50: its central interval excludes four, while the reported systematic interval [7/2,4] contains the sealed endpoint. The Bullet Cluster records an eight-sigma mass/plasma separation. The Moon records equal approximately-655-hour orbital and rotation periods. MESSENGER records Mercury's 29323073/500000 +/- 11/1000000 day rotation and experimentally verifies its 3:2 resonance, preserving the separately generated eccentric-resonance boundary.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067
- SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009
- SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed physical carrier, seven-source custody, solar structure, complete six-row stellar relation, complete galaxy vector, central-plus-systematic Tully-Fisher comparison, Moon-plus-Mercury tidal boundary and no-extra-rule. The declared exact boundary is: The sealed Claim 067 receipt; every row from the seven immutable source snapshots; exact positive whole and fraction values; reported uncertainties and systematic intervals; all six stellar regimes; the complete SPARC population; the Bullet Cluster separation; the lunar 1:1 observation; the Mercury 3:2 boundary; and structural target denial before formal sealing. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-stellar-galactic-tidal-law-versus-complete-seven-source-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
`relation`	`sealed-stellar-galactic-tidal-law-versus-complete-seven-source-record`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:8a2aca0e7bf45cac58fc4a88bbbda934681d60af816702d9301a0b3ea68d5a48`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:af2210735a54649ca8ae52dedc43b60b5a6261827642aac4479775c9bac293f5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1a6b661784eeac5d9900d5e3b1fbdd96cd7c68088554a407e3c1172d48ec787c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8302a4bb44f6f038ed5326e59b92adf075ac0e67b439a3c5e0d9ac0ad40dd933`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:cc352e780d6815d7afd54c64c6066adb0d67ebefded9b213b6894867a920eded`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:eb56f2d40414512405ccbac57899a615c098e7af11df3ff0b9190561a4e09f94`. Independent certificate: `sha256:a91da4b516a074ba15fcca4787da29800ccda7fcb0e1e281d4991c757c63858f`. Engine external-validation hash: `sha256:72e7b0069871be252494e39a49b820076d3a41a76cf531298ab2cc6a84760f7d`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NASA-NTRS-HELIOSEISMOLOGY-2001
- EKER-2018-MAIN-SEQUENCE-RELATIONS
- SPARC-2016-ROTATION-CURVES
- LELLI-2019-BARYONIC-TULLY-FISHER
- CLOWE-2006-BULLET-CLUSTER
- NASA-GRC-MOON-SYNCHRONOUS-ROTATION
- NASA-NTRS-MESSENGER-MERCURY-RESONANCE

Observed comparison records:

- Helioseismology reports stable solar structure at parts-per-ten-thousand sound-speed precision; its conventional reference value is not relabelled as an SFT prediction.
- All six piecewise stellar slopes remain; only the high and very-high mass rows contain terminal powers four and three.
- All 175 SPARC galaxies and all 153 Tully-Fisher galaxies remain in the registered population records.
- The Tully-Fisher central interval excludes four while the complete reported systematic interval reaches four; both are retained.
- The Bullet Cluster records an eight-sigma separation between the mass and dominant plasma peaks.
- The Moon records 1:1 synchronous rotation; Mercury records the separately declared 3:2 eccentric-resonance boundary.
- Erased-row, central-only, false-lunar-lock and erased-Mercury hostile controls reject.

Falsification condition: Reject if any source identity changes; any of the six stellar regimes, both Tully-Fisher interval classes, the full galaxy counts, the Bullet separation, the Moon terminal or the Mercury boundary is omitted; the formal law is changed after target release; a central offset is concealed; a conventional reference model is called an SFT prediction; a fitted parameter enters; or any hostile control passes incorrectly.

Measurement receipt: `sha256:313655e8d2faf9fae7c637f0139713d9ddda3e18b70b3360397ac6adcca7c673`. Isolation certificate: `sha256:16c31cc82998c61691f15702aa12fcd5f61e912741ee9878d0f06a68ac97078c`. Custody certificate: `sha256:d914349f61c1421ff2bcd03fcde8b1a263ff788a5da65e88eb4c0993961cc4c5`.

Registered source descriptions:

- Our Sun IV: The Standard Model and Helioseismology: Consequences of Uncertainties in Input Physics and in Observed Solar Parameters: https://ntrs.nasa.gov/citations/20010111092 (sha256:e523f76ba101f8becc7c18e593d9dce2ce999e4393f3efbace74bada0a51e42e)
- Interrelated main-sequence mass-luminosity, mass-radius and mass-effective temperature relations: https://arxiv.org/abs/1807.02568 (sha256:38f82514c2bcc9f3732581232d86d0798fb758f84018ddce1ad5e5941a359ff3)
- SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves: https://arxiv.org/abs/1606.09251 (sha256:d089215877213661e40965543ee7e05736619082ad16d95e65ec059029588c63)
- The baryonic Tully-Fisher relation for different velocity definitions and implications for galaxy angular momentum: https://arxiv.org/abs/1901.05966 (sha256:a35dafd7d01967b64bbd78be5d337c0011f53fe3c3f6e224b32c23a7a7ba4e3f)
- A direct empirical proof of the existence of dark matter: https://arxiv.org/abs/astro-ph/0608407 (sha256:59f2fb580797de5e5007fcea490a0f6c36036364546dbbb6a319a6e2fb5d0209)
- Moon: https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/moon/ (sha256:8591513db27d380612afdf5d4c34e3fa1502fdf89100f745b88a578c49135df4)
- The Gravity Field, Orientation, and Ephemeris of Mercury from MESSENGER Observations After Three Years in Orbit: https://ntrs.nasa.gov/citations/20150000346 (sha256:836e77ddea47e52734be957e4f41bfcddfc247c8b9beec3a9b0eb900756cfd4e)

Registered target identities:

- STELLAR-GALACTIC-TIDAL-WITHHELD-COMPLETE-RECORD from COMPLETE-SEVEN-SOURCE-VECTOR

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no source row, stellar slope, galaxy fit, tidal period or uncertainty in formal survivor selection
- no fitted exponent, normalization, mass-to-light choice, halo profile, tolerance or correction term
- no deletion of four non-endpoint stellar regimes and no universal single stellar exponent
- no replacement of the Tully-Fisher central interval by its systematic interval or concealment of either

- no relabelling of a conventional solar reference-model value as an SFT dimensional prediction
- no universal lunar 1:1 rule extended across the explicitly separate Mercury eccentric resonance
- no negative, irrational, imaginary, floating, NaN, continuum or infinite Fold proof scalar

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d6683300dc5935bb78a3cc5c38421bab652cc3dc42f3bcb7353dca988b63be3d`; engine receipt `sha256:2410fe8ac97e22ffea7303b50155129dfe6c914d84e549c3c9bb6d24b0672ad6` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-STELLAR-GALACTIC-TIDAL-068-2410fe8ac97e22ff.json`; empirical-validation hash `sha256:f13a517866bb232d464d74547d87605e5b470146d0f89eacbddf5b33880a04c9`; measurement receipt `sha256:313655e8d2faf9fae7c637f0139713d9ddda3e18b70b3360397ac6adcca7c673`.

361. Blind stellar nuclear chain, collapse and heavy-element validation

Claim identity: `SFT-PHYS-VALIDATION-STELLAR-NUCLEAR-COLLAPSE-070`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete six NASA stage temperatures are (3/100,1/5,4/5,3/2,2,33/10) billion kelvin and strictly increase in the formal stage order; all durations (10000000,1000000,1000,1/10,2,1/100 years) remain and are not falsely called monotonic. Borexino reports 36/5 +3 -17/10 cpd per 100 tonnes, minimum significance five sigma at 99/100 confidence. SN1987A neutrinos were recorded by three detectors; the reported two-component preference exceeds 100:1 and the sub-One-second accretion component remains model-assisted. SN2014J records 847 and 1238 keV cobalt lines with fluxes 73/20 +/- 121/100 and 227/100 +/- 69/100 in 10^-4 photon units; the measured ratio 31/50 +/- 7/25 contains 17/25 and the spectrum significance is 113/10 sigma. GW170817 records the 810 nm strontium feature from 3/2 to 10 days, with 1/5-c broadening and 23/100-c blueshift.

The complete six NASA stage temperatures are (3/100,1/5,4/5,3/2,2,33/10) billion kelvin and strictly increase in the formal stage order; all durations (10000000,1000000,1000,1/10,2,1/100 years) remain and are not falsely called monotonic. Borexino reports 36/5 +3 -17/10 cpd per 100 tonnes, minimum significance five sigma at 99/100 confidence. SN1987A neutrinos were recorded by three detectors; the reported two-component preference exceeds 100:1 and the sub-One-second accretion component remains model-assisted. SN2014J records 847 and 1238 keV cobalt lines with fluxes 73/20 +/- 121/100 and 227/100 +/- 69/100 in 10^-4 photon units; the measured ratio 31/50 +/- 7/25 contains 17/25 and the spectrum significance is 113/10 sigma. GW170817 records the 810 nm strontium feature from 3/2 to 10 days, with 1/5-c broadening and 23/100-c blueshift.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069`
- `SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006`
- `SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005`
- `SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed carrier, source custody, complete stage table, direct fusion-neutrino row, collapse-neutrino vector, thermonuclear gamma vector, neutron-capture spectrum and no-extra-rule. The declared exact boundary is: The sealed Claim 069 receipt; all five immutable sources; all six stage temperatures and durations; Borexino rate and asymmetric errors; all three SN1987A detectors and model boundary; both SN2014J lines, fluxes, errors, ratio and significance; the complete registered GW170817 strontium row; and no target access before seal. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions.

Exactly one candidate survived: `complete-fold-carrier__sealed-stellar-nuclear-collapse-law-versus-complete-five-source-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-stellar-nuclear-collapse-law-versus-complete-five-source-record`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:606e96edc6791a89f2d03d4f09069c448275102eb50e8c6da23938285869906c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:295b2763cd15381d7792c51086f1bcf1631ba50e58ff10583c034d31fd31134c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9902e6720af71e081ceaffc4de9f1e7c20be7cc9bf946ce88e3e51b87111a98e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:38f5f64a33f4ef3e9174cdf3758984a1f3e16e343f2c18be29da2b902ef7fe3c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2b1785820c991b9e7f22c226823f597a44d41f30bf453d92d94891545b866950`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2c72d85ead6719362cd8b783b9b26a127a2e605061d0253b155fbb76a561a5c5`. Independent certificate: `sha256:d0078b02f631aec18fb80d091764b7e01e1011b73167cbec14781bff981b52fa`. Engine external-validation hash: `sha256:99883a85a8fc27fd03f1c3bedfebb1464b4722a7a15a2fb0898dc07f0197352c`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NASA-GSFC-COSMIC-ELEMENTS-STELLAR-STAGES
- BOREXINO-2020-CNO-NEUTRINOS
- LOREDO-LAMB-2001-SN1987A-NEUTRINOS

- DIEHL-2014-SN2014J-GAMMA-LINES
- WATSON-2019-GW170817-STRONTIUM

Observed comparison records:

- All six stellar stages have strictly increasing measured temperatures; every duration, including the nonmonotonic neon/oxygen pair, is retained.
- Borexino directly records CNO fusion neutrinos at 36/5 +3 -17/10 cpd per 100 tonnes and at least five-sigma significance.
- The complete three-detector SN1987A neutrino record is retained, with model preference explicitly identified as model-assisted.
- Both SN2014J cobalt lines, flux errors and ratio are retained; the measured ratio interval contains 17/25.
- GW170817 spectroscopy identifies strontium while retaining every spectral-modelling boundary.
- Erased-stage, hidden-irregularity, erased-detector, outside-ratio and erased-capture controls reject.

Falsification condition: Reject if any source changes; any stage, duration, detector, line, flux, uncertainty, confidence, feature or modelling boundary is omitted; the stage temperatures are not strictly ordered; the direct channel observations are absent; target data alter the sealed law; a fit enters; or any hostile control passes incorrectly.

Measurement receipt: `sha256:a7a660979d997ee220d888616b4cb0893150f610c7a11cfae2f04f62787b2370`. Isolation certificate: `sha256:a60aa94729e3f966a44a65c625162ae9a300b75ac462023625fec2a19707aa7a`. Custody certificate: `sha256:76ab98508c7a77d326f0211169d6bd8d4dc8bed11b0474a668d37a9fe0c7787a`.

Registered source descriptions:

- What is Your Cosmic Connection to the Elements?: https://imagine.gsfc.nasa.gov/educators/elements/imagine/05.html (sha256:156c6ce406b0ec49ceac6921ad36fccf5d8778fb5cec7539e24ee8cce961a794)
- Experimental evidence of neutrinos produced in the CNO fusion cycle in the Sun: https://arxiv.org/abs/2006.15115 (sha256:d61241252a22a4c9614d5a29a05cf7d211c4a8dde9dec77fe21e242238ed736d)
- Bayesian analysis of neutrinos observed from supernova SN 1987A: https://arxiv.org/abs/astro-ph/0107260 (sha256:72b8acc99ba37cf8557492669ce53851bf154b95950c7fc34e53a6681599b2d2)
- SN2014J gamma-rays from the 56Ni decay chain: https://arxiv.org/abs/1409.5477 (sha256:aaf0bc90867cdc5e13a8ee2dfd83a23def499e5d5be9bf3f7506e56e6afdaae1)
- Identification of strontium in the merger of two neutron stars: https://arxiv.org/abs/1910.10510 (sha256:34b7f1a277959937f868e6987ec8ce39d481165fccbea8cbab6cfa27457e77c1)

Registered target identities:

- STELLAR-NUCLEAR-COLLAPSE-WITHHELD-COMPLETE-RECORD from COMPLETE-FIVE-SOURCE-VECTOR

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no stage temperature, neutrino event, gamma line, spectral feature, abundance or source value in formal survivor selection
- no fitted reaction rate, explosion energy, nickel mass, standard-candle calibration, ejecta model or target-selected tolerance
- no deletion of the oxygen-duration irregularity, any SN1987A detector, either cobalt line or any GW170817 modelling boundary
- no model-assisted interpretation relabelled as a direct SFT dimensional prediction
- no negative, irrational, imaginary, floating, NaN, continuum or infinite Fold proof scalar

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:d37b32552b807987d11a36d7c667f15d25f1c45a6286b2e9a30fa9c218403a77`; engine receipt `sha256:d439aaf47e94dcc7baba1031cf0ac05c446d03fbd089ed86654618d815c2032c` at `receipts/eng` ine/model_admitted/SFT-PHYS-VALIDATION-STELLAR-NUCLEAR-COLLAPSE-070-d439aaf47e94dcc7.json; empirical-validation hash sha256:92967ad9cb187ae466cda94c34ca26c4c0571526b5e70ec403e4de0db3db117a; measurement receipt sha256:a7a660979d997ee220d888616b4cb0893150f610c7a11cfae2f04f62787b2370.

362. Blind compact-object and horizon-thermodynamics validation

Claim identity: SFT-PHYS-VALIDATION-COMPACT-HORIZON-THERMODYNAMICS-072

Shared claim-record clause PHYS-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The external dimensional record contains the 7/5-solar-mass electron-degenerate limit. The independently measured neutral-fermion-supported PSR J0740 mass interval is [201/100,215/100] solar masses, wholly above 7/5. The conditional merger-remnant upper records 267/100 and 61/20 solar masses are both above that measured interval and remain explicitly model-assisted rather than direct TOV measurements. Together with the earlier horizon-scale shadow/ringdown receipt, the post-seal record supports the ordered finite white-dwarf, neutron-star and horizon-endpoint classes. No direct astrophysical Hawking-radiation or temperature measurement exists in the registered vector, so the exact mT=1/16 law remains an unconfirmed, falsifiable prediction rather than a rewarded absence.

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Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071
- SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed compact carrier, complete object-family vector, source custody, dimensional-role separation, conditional-limit retention, Hawking standing-test status, target inaccessibility and no-extra-rule. The declared exact boundary is: The sealed Claim 071 receipt; all NASA white-dwarf rows; the full PSR J0740 mass, uncertainty, credibility and method record; both conditional LIGO-Virgo remnant limits; the immutable prior horizon-validation boundary; all three source snapshots; and no target access before seal. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-compact-horizon-law-versus-complete-current-object-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-compact-horizon-law-versus-complete-current-object-boundary	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:bfd7bc7756485a0dafece5f0e8f659d62f55132d13aa52d371010ecda340abc5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6bf6814835ef4eeb2f511d887bef6e8221922fe9e99c26b4a47af7ce909505ec`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:47197b8de4600a7aaad77e5868980b543cc3f3239b787d2fe6d7be881916958a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c0d38852e5d941098fdbbe8f5dada47c679aac539a404f1fdae54d21aaf1d1e0`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:0f770d778da96c076a83132c7d57e64499d4a9905fb4cd8bf31e828d99ed5d96`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:bce83e05cc4d335ba8b1bf4166592e2cf61e9eedceee0891844903b0ed01ba85`. Independent certificate: `sha256:74bc925ce6fa49912d235061d924e290c8362727f2593b5525e9bbbcc7630406`. Engine external-validation hash: `sha256:785891cb94b1f22d4c67da47f895069919dc319709bde25853120526159113fd`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NASA-GSFC-WHITE-DWARF-CHANDRASEKHAR
- FONSECA-2021-PSR-J0740-MASS
- LIGO-VIRGO-2019-GW170817-REMNANT

Observed comparison records:

- NASA records electron degeneracy and the 7/5-solar-mass white-dwarf limit.
- PSR J0740+6620 is measured at 52/25 +/- 7/100 solar masses; its complete interval lies above 7/5.
- Both LIGO-Virgo conditional remnant limits are retained with their model conditions and are not called direct TOV measurements.
- The prior horizon receipt is retained unchanged; direct Hawking radiation and temperature remain unmeasured standing tests.
- Hawking non-observation is explicitly not counted as confirmation of the formal mT=1/16 prediction.

- Reversed-order, collapsed-upper and erased-condition hostile controls all reject.

Falsification condition: Reject if any source changes; the measured neutral-fermion object no longer lies above the complete white-dwarf limit; either conditional remnant row is omitted or stripped of its condition; Hawking non-observation is counted as confirmation; target data select or modify Claim 071; an external signed coordinate becomes a Fold proof scalar; or any hostile control passes incorrectly.

Measurement receipt: `sha256:f6d5b035886daa8148cf24c06998e4f498ab93478bfd6af5988a21552f75299c`. Isolation certificate: `sha256:7d65d02a2f387d360082c171f4f6d10fe3548d315c86445fe47b15e1acf072c0`. Custody certificate: `sha256:cf5f250e6766c2a38fe66c7f8bb8e7daac18393d150e46819efa78efed40a9af`.

Registered source descriptions:

- NASA GSFC Imagine the Universe: White Dwarfs: https://imagine.gsfc.nasa.gov/science/objects/dwarfs2.html (sha256:29d25e42773e2b6c5c5f292ef97b6abb28e430f4b6af6336720ee5e3d41aa2a5)
- Refined Mass and Geometric Measurements of the High-Mass PSR J0740+6620: https://arxiv.org/abs/2104.00880 (sha256:66c7fbaacc06a5fa3513a7d963d2b300b05a0a52552e2358e9f257940f37b47a)
- Model comparison from LIGO-Virgo data on GW170817 binary components and consequences for the merger remnant: https://dcc.ligo.org/LIGO-P1800379/public (sha256:86d4efa4b4c5153799edf278bcd99b10753540337c876fbf505855b8bb949122)

Registered target identities:

- COMPACT-HORIZON-WITHHELD-COMPLETE-RECORD from COMPLETE-COMPACT-HORIZON-VECTOR

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no compact-object mass, uncertainty, merger condition, horizon image or Hawking status in formal survivor selection
- no fitted mass coefficient, equation-of-state choice, tolerance, temperature scale or target-selected correction
- no white-dwarf theory summary relabelled as direct mass measurement
- no observed neutron-star mass relabelled as the unknown maximum mass
- no conditional merger bound relabelled as a direct TOV measurement
- no Hawking non-observation rewarded as confirmation and no signed external coordinate imported as a Fold proof scalar

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:7e8c74ff06b33668cd0baec4d0ea0812645956b4f0d5f5904f3ea162ef1f56c1`; engine receipt `sha256:3df00caeaae7eb06d7b17539e9eb6282737e70c3a8c9a7208c83009b97acd124` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-COMPACT-HORIZON-THERMODYNAMICS-072-3df00caeaae7eb06.json`; empirical-validation hash `sha256:cc4c30b739d2862f07b85dc43938f0e80374ec5331adcff998cd5507863f64f2`; measurement receipt `sha256:f6d5b035886daa8148cf24c06998e4f498ab93478bfd6af5988a21552f75299c`.

363. Complete empirical gravitational-wave chirp, merger and ringdown comparison

Claim identity: `SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-074`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete empirical record supports all three forced waveform stages. GW151226 rises from 35 to 450 Hz and in amplitude over 55 cycles, retains positive radiated energy and records two initial components joining one final remnant. GW190521 retains a decaying least-damped l=m=2 remnant mode compatible with its full waveform analysis. Its short duration, conditional quasicircular interpretation and alternative-source boundary remain explicit. GW150914 is retained as observational development evidence, not counted as a blind target. No external value selects or modifies the 4096-form Claim 073 survivor, and no dimensional frequency or fitted damping value becomes a Fold proof scalar.

The complete empirical record supports all three forced waveform stages. GW151226 rises from 35 to 450 Hz and in amplitude over 55 cycles, retains positive radiated energy and records two initial components joining one final remnant. GW190521 retains a decaying least-damped l=m=2 remnant mode compatible with its full waveform analysis. Its short duration, conditional quasicircular interpretation and alternative-source boundary remain explicit. GW150914 is retained as observational development evidence, not counted as a blind target. No external value selects or modifies the 4096-form Claim 073 survivor, and no dimensional frequency or fitted damping value becomes a Fold proof scalar.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-TERMINAL-073`
- `SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVES-003`
- `SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed Fold sequence, complete post-seal event vector, source custody, rising-chirp ordering, two-to-one remnant transition, damped-mode support, scope retention and no-extra-rule. The declared exact boundary is: The immutable Claim 073 receipt; disclosed GW150914 development context; every GW151226 frequency, amplitude, cycle, source, remnant, energy, uncertainty and model-role row; every GW190521 duration, cycle, frequency, source, remnant, ringdown and conditional-interpretation row; and all hostile controls. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-Fold-chirp-merger-ringdown-sequence-versus-complete-postseal-LIGO-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-Fold-chirp-merger-ringdown-sequence-versus-complete-postseal-LIGO-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:92bdc306ff6274441d2cb013f7365da9d6177cbd4dc537e79abadc2fe70950f8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

`sha256:8d6ad202adf8a70915d4cead12907a1296914626f921ea9a01600d578bfc7520`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:329e5d4db94a3fa35b0cf093a42c00037843fff6d581493f77f15add230e698f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:14b95e26cf8f0a19780927b2e7c1fd82b6f912719570584d74226ee90573e6bd`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:45dece7ae7177ccf5e3e8cdfeca815fdc131af0943ffc70f5ffbc6d1a7b0bece`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:62c52df9671103756f013b7c2525757b1d2652c3e1149840f8f1bd0eca087d15`. Independent certificate: `sha256:5f5d4fa9d496a8064f43df7450856bc28c9e5333db1656d8560a041bf631e5b1`. Engine external-validation hash: `sha256:149d5efb655201049e1c3fcf2cd34337a83ca45684e321bd0240259b58079e1d`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- LIGO-VIRGO-GW151226-DISCOVERY-2016
- LIGO-VIRGO-GW190521-DISCOVERY-2020

Observed comparison records:

- GW150914 is retained as pre-seal observational context and is not counted as blind validation.
- GW151226 rises in frequency from 35 to 450 Hz and in amplitude over 55 cycles.
- GW151226 retains positive radiated energy and the two-component-to-one-remnant transition with full uncertainties and model role.
- GW190521 retains one remnant, a decaying least-damped quadrupolar mode and compatibility with the full waveform analysis.
- GW190521's conditional quasicircular interpretation, short-signal limit and alternative interpretations remain explicit.
- No dimensional frequency or exact half-One damping value is relabelled as a Fold proof scalar or direct universal measurement.

Falsification condition: Reject if a post-seal source changes; rising frequency/amplitude, positive radiation, two-to-one merger or decaying quadrupolar remnant support fails; any uncertainty or conditional/model boundary is omitted; GW150914 is relabelled as blind; a dimensional target selects Claim 073; or any hostile control passes.

Measurement receipt: `sha256:1e267de913c50323223c4e548c9b10fab5ccf279081db59446b78f1684309f7e`. Isolation certificate: `sha256:ed8c3a7d3d2a84f2d0ea64a6c3af2c31a2caa04320a413b092583784623e7870`. Custody certificate: `sha256:21fb28c3798ac2479fb847753b0a04b6e5bdffe39464ba5e502c59886c77d996`.

Registered source descriptions:

- LIGO Scientific Collaboration and Virgo Collaboration: https://dcc.ligo.org/LIGO-P151226/public (sha256:a59bb954e9850a8800e2b62e1808553012e791b879e00f108b0bf061bf530e72)
- LIGO Scientific Collaboration and Virgo Collaboration: https://dcc.ligo.org/P2000020/public (sha256:4d6661f18168267249b3ca043b0692d88376c9db9b3d9146a9500763f7b438a7)

Registered target identities:

- GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-WITHHELD-POSTSEAL-RECORD from COMPLETE-POSTSEAL-GRAVITATIONAL-WAVE-VECTOR

Shared claim-record clause PHYS-S028 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no claim that GW150914 was unseen before Claim 073
- no use of GW151226 or GW190521 values to select or alter the formal survivor
- no fitted chirp mass, waveform, merger time, damping coefficient, tone or tolerance
- no model-assisted remnant property relabelled as direct model-free measurement
- no claim that half-One damping or normalised Fold frequency was directly measured
- no omitted uncertainty, short-signal, conditional-interpretation or alternative-source row

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:76cd8e8bf521eeadf34de36bf9951e70cdd1578454b268cceb84a67a2da8d470; engine receipt sha256:80a9f13dc620d3cb2bf16597d54b63562aeac60ad1bab25d1327aabafb8bf576 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-074-80a9f13dc620d3cb.json; empirical-validation hash sha256:a197d901695d69375e3d5adc571616d5324340f82ae7539dd190aaff9b26b361; measurement receipt sha256:1e267de913c50323223c4e548c9b10fab5ccf279081db59446b78f1684309f7e.

364. Terminal Physics Grand Lock and extension-open no-omission certificate

Claim identity: SFT-PHYS-GRAND-LOCK-TERMINAL-075

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The frozen pre-lock Physics surface contains exactly 347 uniquely owned admitted claims and 534 transitive dependency nodes. Every Physics claim reaches SFT-FOUNDATION-ONE-001 in one acyclic dependency dictionary. Exactly 234 Physics claims carry empirical-test receipts, and every recorded unfavorable result or scope boundary remains bound. The current generator-three headline vector recomputes exactly, including inverse alpha 503846395469/3676744786, charged-lepton invariants 1, 1/6, 1/485 and 3/1454 plus the terminal product, quark products 1/383 and 1/3071, dark/baryon 27/5 with 27/32 and 5/32, Hubble 13/12 and 3305/3048, the terminal squared Planck/proton hierarchy, local vacuum floors 1/2^10 and 1/2^20, and normalized cosmological value 33/16. Under the complete generator successor three-to-four, every declared generator-dependent value changes while half-One, spatial rank three and boundary rank two remain fixed.

The frozen pre-lock Physics surface contains exactly 347 uniquely owned admitted claims and 534 transitive dependency nodes. Every Physics claim reaches SFT-FOUNDATION-ONE-001 in one acyclic dependency dictionary. Exactly 234 Physics claims carry empirical-test receipts, and every recorded unfavorable result or scope boundary remains bound. The current generator-three headline vector recomputes exactly, including inverse alpha 503846395469/3676744786, charged-lepton invariants 1, 1/6, 1/485 and 3/1454 plus the terminal product, quark products 1/383 and 1/3071, dark/baryon 27/5 with 27/32 and 5/32, Hubble 13/12 and 3305/3048, the terminal squared Planck/proton hierarchy, local vacuum floors 1/2^10 and 1/2^20, and normalized cosmological value 33/16. Under the complete generator successor three-to-four, every declared generator-dependent value changes while half-One, spatial rank three and boundary rank two remain fixed.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001
- SFT-PHYS-ATOMIC-CORRECTION-HIERARCHY-004
- SFT-PHYS-ATOMIC-CUBIC-SUPPORT-004
- SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001
- SFT-PHYS-ATOMIC-FIELD-SPLITTING-TERMINAL-005
- SFT-PHYS-ATOMIC-FINE-STRUCTURE-TERMINAL-005
- SFT-PHYS-ATOMIC-HYDROGEN-RYDBERG-TERMINAL-005

- SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-ATOMIC-HYPERFINE-TERMINAL-005
- SFT-PHYS-ATOMIC-LAMB-SHIFT-TERMINAL-005
- SFT-PHYS-ATOMIC-SHELL-PERIODICITY-TERMINAL-005
- SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005
- SFT-PHYS-ATOMIC-TRANSITION-SELECTION-004
- SFT-PHYS-BARYOGENESIS-DEPENDENCY-TERMINAL-021
- SFT-PHYS-COLLECTIVE-RADIATION-RESPONSE-TERMINAL-041
- SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071
- SFT-PHYS-COMPOSITE-CONFINING-SECTOR-TERMINAL-031
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001
- SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001
- SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-CONTINUUM-COARSE-GRAIN-001
- SFT-PHYS-COSMO-BACKGROUND-001
- SFT-PHYS-COSMO-BRANCH-BOUNDARY-001
- SFT-PHYS-COSMO-COMPLETE-BUDGET-001
- SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032
- SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001
- SFT-PHYS-COSMO-DISTANCE-001
- SFT-PHYS-COSMO-EXPANSION-001
- SFT-PHYS-COSMO-HUBBLE-CALIBRATION-001
- SFT-PHYS-COSMO-LENSING-001
- SFT-PHYS-COSMO-REDSHIFT-001
- SFT-PHYS-COSMO-SPATIAL-FLATNESS-001
- SFT-PHYS-COSMO-STRUCTURE-GROWTH-001
- SFT-PHYS-COUPLED-ENSEMBLE-SYNCHRONIZATION-TERMINAL-007
- SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008
- SFT-PHYS-COUPLING-ACCUMULATED-SEPARATION-TERMINAL-015
- SFT-PHYS-COUPLING-RUNNING-CONVERGENCE-TERMINAL-006
- SFT-PHYS-CRITICALITY-UNIVERSALITY-TURBULENCE-TERMINAL-047
- SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061
- SFT-PHYS-DECAY-WIDTH-BRANCHING-LIFETIME-TERMINAL-006

- SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003
- SFT-PHYS-DYNAMICS-POTENTIAL-EVOLUTION-003
- SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003
- SFT-PHYS-DYNAMICS-SYMMETRY-ACTION-TERMINAL-016
- SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003
- SFT-PHYS-ELECTROWEAK-WZ-RATIO-002
- SFT-PHYS-FIELD-ACTION-REACTION-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003
- SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001
- SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-INDUCTION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-FIELD-LORENTZ-TRANSFER-003
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003
- SFT-PHYS-FIELD-MAXWELL-PLANAR-CLOSURE-003
- SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-FIELD-SOURCE-RESPONSE-001
- SFT-PHYS-FINITE-QUANTUM-GRAVITY-TERMINAL-023
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-FLUID-DENSITY-001
- SFT-PHYS-FLUID-INVISCID-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-FLUID-TURBULENCE-001
- SFT-PHYS-FLUID-VISCOSITY-001
- SFT-PHYS-FOLD-UNIVERSE-TRANSPORT-TERMINAL-024
- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003

- SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002
- SFT-PHYS-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-TERMINAL-073
- SFT-PHYS-GRAVITY-CURVATURE-001
- SFT-PHYS-GRAVITY-EQUIVALENCE-001
- SFT-PHYS-GRAVITY-FIELD-SOURCE-001
- SFT-PHYS-GRAVITY-GEODESIC-001
- SFT-PHYS-GRAVITY-GRAVITON-POLARIZATION-003
- SFT-PHYS-GRAVITY-HORIZON-001
- SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003
- SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003
- SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003
- SFT-PHYS-GRAVITY-REDSHIFT-EQUIVALENCE-003
- SFT-PHYS-GRAVITY-STATIC-CLOCK-003
- SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003
- SFT-PHYS-GRAVITY-WEAK-FIELD-FLUX-003
- SFT-PHYS-HADRON-REGGE-DIMENSIONAL-TERMINAL-059
- SFT-PHYS-HADRON-REGGE-TERMINAL-005
- SFT-PHYS-HIGGS-SYMMETRY-TERMINAL-065
- SFT-PHYS-INFLATION-GROWTH-TERMINAL-039
- SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025
- SFT-PHYS-LATTICE-OPERATOR-TERMINAL-022
- SFT-PHYS-LYAPUNOV-KS-CORRESPONDENCE-TERMINAL-008
- SFT-PHYS-MATTER-BARYON-PHOTON-003
- SFT-PHYS-MATTER-BARYON-PHOTON-TERMINAL-004
- SFT-PHYS-MATTER-CKM-FIBRE-003
- SFT-PHYS-MATTER-CKM-PHYSICAL-003
- SFT-PHYS-MATTER-CKM-TERMINAL-004
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-PHYS-MATTER-CONFINEMENT-LIFT-003
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MATTER-DECAY-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-MATTER-GENERATION-DEPTH-003
- SFT-PHYS-MATTER-INTER-ENTRY-COUPLING-003
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003

- SFT-PHYS-MATTER-MIRROR-MASS-CLOSURE-003
- SFT-PHYS-MATTER-MIXING-001
- SFT-PHYS-MATTER-MIXING-CORRESPONDENCE-003
- SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-MATTER-PROTON-ELECTRON-003
- SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004
- SFT-PHYS-MATTER-QUARK-CUBICS-003
- SFT-PHYS-MATTER-QUARK-DRESSING-003
- SFT-PHYS-MATTER-QUARK-INVARIANTS-003
- SFT-PHYS-MATTER-SCATTERING-001
- SFT-PHYS-MEAS-BOUNDARY-GROWTH-001
- SFT-PHYS-MEAS-CALIBRATION-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-REFERENCE-REALIZATION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MEAS-UNIT-COMPARISON-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MECH-ACCELERATION-001
- SFT-PHYS-MECH-ANGULAR-MOMENTUM-001
- SFT-PHYS-MECH-ANGULAR-MOTION-001
- SFT-PHYS-MECH-COMPOSITE-CENTRE-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-MECH-DURATION-001
- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-FORCE-001
- SFT-PHYS-MECH-INERTIA-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-MECH-MOMENTUM-001
- SFT-PHYS-MECH-POWER-001
- SFT-PHYS-MECH-SPEED-VELOCITY-001

- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEDIATOR-RANGE-CHANNEL-TERMINAL-020
- SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005
- SFT-PHYS-MOLECULAR-SPECTRUM-HIERARCHY-004
- SFT-PHYS-NEUTRINO-CP-PHASE-002
- SFT-PHYS-NEUTRINO-MAJORANA-003
- SFT-PHYS-NEUTRINO-PMNS-ANGLES-002
- SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003
- SFT-PHYS-NEUTRINO-POSITIVE-MASS-003
- SFT-PHYS-NEUTRINO-SPLITTING-003
- SFT-PHYS-NEUTRINO-ZERO-NU-BETA-BETA-003
- SFT-PHYS-NUCLEAR-BINDING-001
- SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005
- SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001
- SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001
- SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006
- SFT-PHYS-NUCLEAR-FISSION-001
- SFT-PHYS-NUCLEAR-FUSION-001
- SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005
- SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006
- SFT-PHYS-NUCLEAR-LEVELS-001
- SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005
- SFT-PHYS-NUCLEAR-RADIOACTIVITY-001
- SFT-PHYS-NUCLEAR-REACTIONS-001
- SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005
- SFT-PHYS-NUCLEON-BINDING-TERMINAL-005
- SFT-PHYS-ODD-LATTICE-ALL-REGION-OCCUPANCY-TERMINAL-007
- SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009
- SFT-PHYS-PARKER-PROTON-ENERGY-TERMINAL-028
- SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051
- SFT-PHYS-PLASMA-COLLECTIVE-001
- SFT-PHYS-PLASMA-MHD-001
- SFT-PHYS-PLASMA-OSCILLATION-001
- SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009
- SFT-PHYS-PROTON-RADIUS-TERMINAL-029
- SFT-PHYS-QED-ELECTRON-MAGNETIC-ANOMALY-004
- SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003
- SFT-PHYS-QED-MUON-MAGNETIC-ANOMALY-004

- SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004
- SFT-PHYS-QUADRUPOLE-RADIATED-POWER-TERMINAL-012
- SFT-PHYS-QUANTUM-BELL-001
- SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001
- SFT-PHYS-QUANTUM-CONTEXTUALITY-001
- SFT-PHYS-QUANTUM-DECOHERENCE-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-QUANTUM-EVOLUTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-INCOMPATIBILITY-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-QUANTUM-OBSERVABLE-001
- SFT-PHYS-QUANTUM-PHYSICAL-STATE-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-SUPPORT-UNCERTAINTY-TERMINAL-049
- SFT-PHYS-QUANTUM-TUNNELLING-001
- SFT-PHYS-QUANTUM-WEIGHT-001
- SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003
- SFT-PHYS-RELATIVITY-TWO-HAND-DIRAC-SQUARE-003
- SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030
- SFT-PHYS-SCALE-PROTON-PLANCK-HIERARCHY-002
- SFT-PHYS-SCALE-PROTON-PLANCK-TERMINAL-003
- SFT-PHYS-SCATTERING-PARTITION-PATH-TERMINAL-017
- SFT-PHYS-SCATTERING-RUTHERFORD-COMPTON-TERMINAL-006
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001
- SFT-PHYS-SPACETIME-CLOCK-RATE-001
- SFT-PHYS-SPACETIME-CLOSED-TIMELIKE-ADMISSIBILITY-003
- SFT-PHYS-SPACETIME-EVENT-RELATION-001
- SFT-PHYS-SPACETIME-EXACT-INTERVAL-003
- SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001
- SFT-PHYS-SPACETIME-INTERVAL-001
- SFT-PHYS-SPACETIME-LENGTH-RELATION-001
- SFT-PHYS-SPACETIME-LIMIT-SPEED-001
- SFT-PHYS-SPACETIME-VELOCITY-COMPOSITION-TERMINAL-033

- SFT-PHYS-SPACETIME-WARP-ADMISSIBILITY-003
- SFT-PHYS-SPACETIME-WORMHOLE-ADMISSIBILITY-003
- SFT-PHYS-SPIN-STATISTICS-CONDENSATION-TERMINAL-045
- SFT-PHYS-STATIC-EXTERIOR-CLOCK-TERMINAL-011
- SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067
- SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069
- SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013
- SFT-PHYS-STRONG-CP-BARYON-STABILITY-TERMINAL-063
- SFT-PHYS-STRONG-FIELD-NONLINEAR-FIXED-POINT-TERMINAL-014
- SFT-PHYS-STRONG-RUNNING-DIRECTION-002
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010
- SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043
- SFT-PHYS-THERMAL-HELIUM-ISOTOPE-TERMINAL-057
- SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037
- SFT-PHYS-THERMO-ENTROPY-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-FLUCTUATION-001
- SFT-PHYS-THERMO-HEAT-WORK-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-LANDAUER-DEMON-TERMINAL-018
- SFT-PHYS-THERMO-LANDAUER-EMPIRICAL-019
- SFT-PHYS-THERMO-MICRO-MACRO-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-THERMO-STATE-RELATION-001
- SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001
- SFT-PHYS-THERMO-TEMPERATURE-001
- SFT-PHYS-THERMO-THIRD-LAW-001
- SFT-PHYS-VACUUM-ASYMMETRIC-BEAT-EXTRACTION-003
- SFT-PHYS-VACUUM-COMPLETE-CYCLE-LEDGER-003
- SFT-PHYS-VACUUM-DENSITY-SCALE-TERMINAL-035
- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-PHYS-VACUUM-INERTIA-UNITY-003
- SFT-PHYS-VACUUM-ODD-RECURRENCE-003

- SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003
- SFT-PHYS-VALIDATION-ATOMIC-CUBIC-SUPPORT-004
- SFT-PHYS-VALIDATION-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-001
- SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-002
- SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002
- SFT-PHYS-VALIDATION-COLLECTIVE-RADIATION-RESPONSE-042
- SFT-PHYS-VALIDATION-COMMON-SCALE-MEASURED-VALUE-054
- SFT-PHYS-VALIDATION-COMPACT-HORIZON-THERMODYNAMICS-072
- SFT-PHYS-VALIDATION-COSMIC-TRANSPORT-MEASURED-VALUE-055
- SFT-PHYS-VALIDATION-CRITICALITY-MEASURED-VALUE-056
- SFT-PHYS-VALIDATION-CRITICALITY-UNIVERSALITY-TURBULENCE-048
- SFT-PHYS-VALIDATION-DARK-SMITHION-LFV-062
- SFT-PHYS-VALIDATION-DYNAMICS-SPECTRA-003
- SFT-PHYS-VALIDATION-ELECTROWEAK-MEASURED-VALUE-053
- SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003
- SFT-PHYS-VALIDATION-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-VALIDATION-FIELD-ELECTRIC-STRENGTH-001
- SFT-PHYS-VALIDATION-FINITE-LOOPS-003
- SFT-PHYS-VALIDATION-FORCE-SECTOR-ANCHORS-003
- SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-074
- SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVES-003
- SFT-PHYS-VALIDATION-GRAVITY-CLOCK-EQUIVALENCE-003
- SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003
- SFT-PHYS-VALIDATION-HADRON-REGGE-MEASURED-VALUE-060
- SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066
- SFT-PHYS-VALIDATION-INFLATION-GROWTH-040
- SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-VALIDATION-INVERSE-SQUARE-001
- SFT-PHYS-VALIDATION-MAGNETIC-ANOMALIES-003
- SFT-PHYS-VALIDATION-MAJORANA-ZERO-NU-003
- SFT-PHYS-VALIDATION-MECHANICS-FORCE-001
- SFT-PHYS-VALIDATION-MECHANICS-MOMENTUM-001
- SFT-PHYS-VALIDATION-METROLOGY-MOLAR-PLANCK-001
- SFT-PHYS-VALIDATION-NEUTRINO-MASS-MIXING-003
- SFT-PHYS-VALIDATION-NONSTANDARD-SPACETIME-003
- SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001
- SFT-PHYS-VALIDATION-OPTICAL-OPERATIONS-003

- SFT-PHYS-VALIDATION-PARTICLE-MODE-GENERATION-052
- SFT-PHYS-VALIDATION-PROTON-ELECTRON-003
- SFT-PHYS-VALIDATION-PROTON-PLANCK-TERMINAL-003
- SFT-PHYS-VALIDATION-QUANTUM-SUPPORT-UNCERTAINTY-050
- SFT-PHYS-VALIDATION-QUARK-CKM-003
- SFT-PHYS-VALIDATION-RELATIVISTIC-FIELDS-003
- SFT-PHYS-VALIDATION-SPIN-STATISTICS-CONDENSATION-046
- SFT-PHYS-VALIDATION-STELLAR-GALACTIC-TIDAL-068
- SFT-PHYS-VALIDATION-STELLAR-NUCLEAR-COLLAPSE-070
- SFT-PHYS-VALIDATION-STRONG-CP-BARYON-STABILITY-064
- SFT-PHYS-VALIDATION-THERMAL-EQUILIBRIUM-044
- SFT-PHYS-VALIDATION-THERMAL-HISTORY-MEASURED-VALUE-058
- SFT-PHYS-VALIDATION-THERMAL-HISTORY-RECOMBINATION-038
- SFT-PHYS-VALIDATION-THERMO-MOLAR-ENERGY-001
- SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036
- SFT-PHYS-VALIDATION-VACUUM-EXTRACTION-003
- SFT-PHYS-VALIDATION-VACUUM-FLOOR-003
- SFT-PHYS-VALIDATION-VACUUM-INERTIA-003
- SFT-PHYS-VALIDATION-VACUUM-POLARIZATION-003
- SFT-PHYS-VALIDATION-VELOCITY-COMPOSITION-034
- SFT-PHYS-VALIDATION-WAVE-ENERGY-001
- SFT-PHYS-VALIDATION-WAVE-FREQUENCY-001
- SFT-PHYS-WAVE-DIFFRACTION-001
- SFT-PHYS-WAVE-DISPERSION-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-PERIOD-FREQUENCY-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-WAVE-RESONANCE-001
- SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001
- SFT-PHYS-WAVE-SUPERPOSITION-001
- SFT-PHYS-WEAK-PARITY-FIBRE-002
- SFT-PHYS-YANG-MILLS-SINGLET-GAP-EMPIRICAL-027
- SFT-PHYS-YANG-MILLS-SINGLET-GAP-TERMINAL-026

Generated grammar and closure boundary. Generate the complete twelve-axis product of ownership, receipt identity, registration identity, dependency closure, root reachability, value vector, generator perturbation, invariant control, cross-domain identity, empirical reconciliation, extension status and extra-rule forms. The declared exact boundary is: All 347 admitted pre-lock Physics claims, all 534 nodes in their dependency closure, every bound exact-result and empirical certificate row, the complete generator-three to generator-four adverse perturbation, all shared structural carriers and all 4096 alternatives. The generator produced `4096` named candidates and the decision support contains `4096` one-for-one decisions. Exactly one candidate survived: `unique-complete-physics-owner__receipt-and-certificate-hash-bound__registration-and-dependencies-hash-bound__complete-acyclic-dictionary__foundational-One-root__all-claim-exact-result-vector__every-and-only-declared-dependent-value-moves__binary-space-boundary-held__complete-cross-domain-identity-graph__complete-receipts-with-adverse-boundaries__current-evidence-closed-extension-open__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`ownership`	`unique-complete-physics-owner`	`omitted-or-duplicate-owner`	That is not a complete branch census.
`receipt`	`receipt-and-certificate-hash-bound`	`unbound-or-current-only-result`	That cannot detect replacement.
`registration`	`registration-and-dependencies-hash-bound`	`unstated-or-mutable-premises`	That erases the proof route.
`dependency`	`complete-acyclic-dictionary`	`orphan-or-cycle`	An orphan or cycle cannot derive from the One.
`root`	`foundational-One-root`	`multiple-or-missing-roots`	That adds an axiom or loses derivation.
`values`	`all-claim-exact-result-vector`	`headline-selection-only`	Selecting favourable headlines permits omission.
`perturbation`	`every-and-only-declared-dependent-value-moves`	`selected-values-move`	A selected subset cannot prove structural dependence.
`invariants`	`binary-space-boundary-held`	`silent-global-retuning`	Global retuning cannot localize cause.
`identity`	`complete-cross-domain-identity-graph`	`duplicate-unrelated-numbers`	Coincidence without provenance is not identity.
`empirical`	`complete-receipts-with-adverse-boundaries`	`favourable-rows-only`	Cherry-picking violates the empirical record.
`extension`	`current-evidence-closed-extension-open`	`permanently-locked-theory`	That would exclude lawful discovery.
`rule`	`no-extra-rule`	`free-extra-rule`	An extra rule is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:cf92b9cf88639e40fc0cbbe37f7361b252f6839e6a64cdbe7c0e48d47857a592`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:f4786d8a1868ac9dda94eacd530e2f7ee6b2edd640697d9918d5b1bb6ceb3819`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:4c0f5ac74cf9fe44c88c8e6ec55de9f9cb5ea1bf87dd112e081f86b6667d1afe`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:deb3e5409ab49a544b77cf151557f3e24de8e44a6784215a4d3349e493725803`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:10640f437fe38173ec3f5697342bb57007e4c75ed6169e83dac6e47d1b8c4210`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash:

sha256:bff4a8792ccae133d39c7661cf47ba22f3ea3330d4f8b3dc0c29228ff36077ef. Independent certificate: sha256:0599cd41db3228cf2f60d71a77582a1ed85a272cde0e16b19645ace592648898. Engine external-validation hash: sha256:65110685b0d6c88d72cdb3f2704a305b540d1c011e92b5a5ef87bb00c9dd5392.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause PHYS-S028 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no edit to the canonical engine or any admitted receipt
- no measurement, V1/V2 result or external equation selecting a formal survivor
- no fitted parameter, tolerance, candidate neighborhood or omitted adverse result
- no numerical-zero, negative, irrational, imaginary, floating, NaN or completed-infinite proof scalar
- no assertion that current completion prohibits a later lawfully forced extension

Shared claim-record clause PHYS-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest sha256:91770400e9861ef159a95209bd52e675bc08f4e1cd81bd02de93dba586758cb0; engine receipt sha256:ae18f67371c8e7054430935d6b5e5f3162f24cf9cba073769384bf7ba467d817 at receipts/engine/model_admitted/SFT-PHYS-GRAND-LOCK-TERMINAL-075-ae18f67371c8e705.json; empirical-validation hash None; measurement receipt None.

365. Complete Physics empirical-vector and adverse-result Grand Lock

Claim identity: SFT-PHYS-VALIDATION-GRAND-LOCK-076

Shared claim-record clause PHYS-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. Exactly 234 pre-lock Physics claims carry independently replicated empirical validation; their complete vector binds 147 distinct registered external source identities. All 234 empirical and external validation hashes are retained. Every available separate measurement receipt is retained, while six older claim packages lacking that later materialization field are explicitly identified rather than silently upgraded. All fourteen detected unfavourable-result or scope-boundary claims remain inside the same vector. The formal Grand Lock receipt predates this aggregate reconciliation; observations validate or challenge their declared claims but never select or alter a formal survivor.

> *Exactly 234 pre-lock Physics claims carry independently replicated empirical validation; their complete vector binds 147 distinct registered external source identities. All 234 empirical and external validation hashes are retained. Every available separate measurement receipt is retained, while six older claim packages lacking that later materialization field are explicitly identified rather than silently upgraded. All fourteen detected unfavorable-result or scope-boundary claims remain inside the same vector. The formal Grand Lock receipt predates this aggregate reconciliation; observations validate or challenge their declared claims but never select or alter a formal survivor.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAND-LOCK-TERMINAL-075
- SFT-PHYS-ATOMIC-FIELD-SPLITTING-TERMINAL-005
- SFT-PHYS-ATOMIC-FINE-STRUCTURE-TERMINAL-005
- SFT-PHYS-ATOMIC-HYDROGEN-RYDBERG-TERMINAL-005
- SFT-PHYS-ATOMIC-HYPERFINE-TERMINAL-005

- SFT-PHYS-ATOMIC-LAMB-SHIFT-TERMINAL-005
- SFT-PHYS-ATOMIC-SHELL-PERIODICITY-TERMINAL-005
- SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001
- SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001
- SFT-PHYS-CONTINUUM-COARSE-GRAIN-001
- SFT-PHYS-COSMO-BACKGROUND-001
- SFT-PHYS-COSMO-BRANCH-BOUNDARY-001
- SFT-PHYS-COSMO-COMPLETE-BUDGET-001
- SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032
- SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001
- SFT-PHYS-COSMO-DISTANCE-001
- SFT-PHYS-COSMO-EXPANSION-001
- SFT-PHYS-COSMO-HUBBLE-CALIBRATION-001
- SFT-PHYS-COSMO-LENSING-001
- SFT-PHYS-COSMO-REDSHIFT-001
- SFT-PHYS-COSMO-SPATIAL-FLATNESS-001
- SFT-PHYS-COSMO-STRUCTURE-GROWTH-001
- SFT-PHYS-COUPLED-ENSEMBLE-SYNCHRONIZATION-TERMINAL-007
- SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008
- SFT-PHYS-COUPLING-RUNNING-CONVERGENCE-TERMINAL-006
- SFT-PHYS-DECAY-WIDTH-BRANCHING-LIFETIME-TERMINAL-006
- SFT-PHYS-FIELD-ACTION-REACTION-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001
- SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-INDUCTION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-FIELD-MAGNETIC-001

- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-FIELD-SOURCE-RESPONSE-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-FLUID-DENSITY-001
- SFT-PHYS-FLUID-INVISCID-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-FLUID-TURBULENCE-001
- SFT-PHYS-FLUID-VISCOSITY-001
- SFT-PHYS-GRAVITY-CURVATURE-001
- SFT-PHYS-GRAVITY-EQUIVALENCE-001
- SFT-PHYS-GRAVITY-FIELD-SOURCE-001
- SFT-PHYS-GRAVITY-GEODESIC-001
- SFT-PHYS-GRAVITY-HORIZON-001
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-HADRON-REGGE-TERMINAL-005
- SFT-PHYS-LYAPUNOV-KS-CORRESPONDENCE-TERMINAL-008
- SFT-PHYS-MATTER-BARYON-PHOTON-TERMINAL-004
- SFT-PHYS-MATTER-CKM-TERMINAL-004
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MATTER-DECAY-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-MATTER-MIXING-001
- SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004
- SFT-PHYS-MATTER-SCATTERING-001
- SFT-PHYS-MEAS-CALIBRATION-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-REFERENCE-REALIZATION-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MEAS-UNIT-COMPARISON-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MECH-ACCELERATION-001

- SFT-PHYS-MECH-ANGULAR-MOMENTUM-001
- SFT-PHYS-MECH-ANGULAR-MOTION-001
- SFT-PHYS-MECH-COMPOSITE-CENTRE-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-MECH-DURATION-001
- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-FORCE-001
- SFT-PHYS-MECH-INERTIA-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-MECH-MOMENTUM-001
- SFT-PHYS-MECH-POWER-001
- SFT-PHYS-MECH-SPEED-VELOCITY-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005
- SFT-PHYS-NUCLEAR-BINDING-001
- SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005
- SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006
- SFT-PHYS-NUCLEAR-FISSION-001
- SFT-PHYS-NUCLEAR-FUSION-001
- SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005
- SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006
- SFT-PHYS-NUCLEAR-LEVELS-001
- SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005
- SFT-PHYS-NUCLEAR-RADIOACTIVITY-001
- SFT-PHYS-NUCLEAR-REACTIONS-001
- SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005
- SFT-PHYS-NUCLEON-BINDING-TERMINAL-005
- SFT-PHYS-ODD-LATTICE-ALL-REGION-OCCUPANCY-TERMINAL-007
- SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009
- SFT-PHYS-PARKER-PROTON-ENERGY-TERMINAL-028
- SFT-PHYS-PLASMA-COLLECTIVE-001
- SFT-PHYS-PLASMA-MHD-001
- SFT-PHYS-PLASMA-OSCILLATION-001
- SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009
- SFT-PHYS-PROTON-RADIUS-TERMINAL-029
- SFT-PHYS-QED-ELECTRON-MAGNETIC-ANOMALY-004
- SFT-PHYS-QED-MUON-MAGNETIC-ANOMALY-004

- SFT-PHYS-QUADRUPOLE-RADIATED-POWER-TERMINAL-012
- SFT-PHYS-QUANTUM-BELL-001
- SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001
- SFT-PHYS-QUANTUM-CONTEXTUALITY-001
- SFT-PHYS-QUANTUM-DECOHERENCE-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-QUANTUM-EVOLUTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-INCOMPATIBILITY-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-QUANTUM-OBSERVABLE-001
- SFT-PHYS-QUANTUM-PHYSICAL-STATE-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-TUNNELLING-001
- SFT-PHYS-QUANTUM-WEIGHT-001
- SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030
- SFT-PHYS-SCATTERING-RUTHERFORD-COMPTON-TERMINAL-006
- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001
- SFT-PHYS-SPACETIME-CLOCK-RATE-001
- SFT-PHYS-SPACETIME-EVENT-RELATION-001
- SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001
- SFT-PHYS-SPACETIME-INTERVAL-001
- SFT-PHYS-SPACETIME-LENGTH-RELATION-001
- SFT-PHYS-SPACETIME-LIMIT-SPEED-001
- SFT-PHYS-STATIC-EXTERIOR-CLOCK-TERMINAL-011
- SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013
- SFT-PHYS-STRONG-FIELD-NONLINEAR-FIXED-POINT-TERMINAL-014
- SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010
- SFT-PHYS-THERMO-ENTROPY-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
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- SFT-PHYS-THERMO-FLUCTUATION-001
- SFT-PHYS-THERMO-HEAT-WORK-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-LANDAUER-EMPIRICAL-019

- SFT-PHYS-THERMO-MICRO-MACRO-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-THERMO-STATE-RELATION-001
- SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001
- SFT-PHYS-THERMO-TEMPERATURE-001
- SFT-PHYS-THERMO-THIRD-LAW-001
- SFT-PHYS-VALIDATION-ATOMIC-CUBIC-SUPPORT-004
- SFT-PHYS-VALIDATION-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-001
- SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-002
- SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002
- SFT-PHYS-VALIDATION-COLLECTIVE-RADIATION-RESPONSE-042
- SFT-PHYS-VALIDATION-COMMON-SCALE-MEASURED-VALUE-054
- SFT-PHYS-VALIDATION-COMPACT-HORIZON-THERMODYNAMICS-072
- SFT-PHYS-VALIDATION-COSMIC-TRANSPORT-MEASURED-VALUE-055
- SFT-PHYS-VALIDATION-CRITICALITY-MEASURED-VALUE-056
- SFT-PHYS-VALIDATION-CRITICALITY-UNIVERSALITY-TURBULENCE-048
- SFT-PHYS-VALIDATION-DARK-SMITHION-LFV-062
- SFT-PHYS-VALIDATION-DYNAMICS-SPECTRA-003
- SFT-PHYS-VALIDATION-ELECTROWEAK-MEASURED-VALUE-053
- SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003
- SFT-PHYS-VALIDATION-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-VALIDATION-FIELD-ELECTRIC-STRENGTH-001
- SFT-PHYS-VALIDATION-FINITE-LOOPS-003
- SFT-PHYS-VALIDATION-FORCE-SECTOR-ANCHORS-003
- SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-074
- SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVES-003
- SFT-PHYS-VALIDATION-GRAVITY-CLOCK-EQUIVALENCE-003
- SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003
- SFT-PHYS-VALIDATION-HADRON-REGGE-MEASURED-VALUE-060
- SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066
- SFT-PHYS-VALIDATION-INFLATION-GROWTH-040
- SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-VALIDATION-INVERSE-SQUARE-001
- SFT-PHYS-VALIDATION-MAGNETIC-ANOMALIES-003
- SFT-PHYS-VALIDATION-MAJORANA-ZERO-NU-003

- SFT-PHYS-VALIDATION-MECHANICS-FORCE-001
- SFT-PHYS-VALIDATION-MECHANICS-MOMENTUM-001
- SFT-PHYS-VALIDATION-METROLOGY-MOLAR-PLANCK-001
- SFT-PHYS-VALIDATION-NEUTRINO-MASS-MIXING-003
- SFT-PHYS-VALIDATION-NONSTANDARD-SPACETIME-003
- SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001
- SFT-PHYS-VALIDATION-OPTICAL-OPERATIONS-003
- SFT-PHYS-VALIDATION-PARTICLE-MODE-GENERATION-052
- SFT-PHYS-VALIDATION-PROTON-ELECTRON-003
- SFT-PHYS-VALIDATION-PROTON-PLANCK-TERMINAL-003
- SFT-PHYS-VALIDATION-QUANTUM-SUPPORT-UNCERTAINTY-050
- SFT-PHYS-VALIDATION-QUARK-CKM-003
- SFT-PHYS-VALIDATION-RELATIVISTIC-FIELDS-003
- SFT-PHYS-VALIDATION-SPIN-STATISTICS-CONDENSATION-046
- SFT-PHYS-VALIDATION-STELLAR-GALACTIC-TIDAL-068
- SFT-PHYS-VALIDATION-STELLAR-NUCLEAR-COLLAPSE-070
- SFT-PHYS-VALIDATION-STRONG-CP-BARYON-STABILITY-064
- SFT-PHYS-VALIDATION-THERMAL-EQUILIBRIUM-044
- SFT-PHYS-VALIDATION-THERMAL-HISTORY-MEASURED-VALUE-058
- SFT-PHYS-VALIDATION-THERMAL-HISTORY-RECOMBINATION-038
- SFT-PHYS-VALIDATION-THERMO-MOLAR-ENERGY-001
- SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036
- SFT-PHYS-VALIDATION-VACUUM-EXTRACTION-003
- SFT-PHYS-VALIDATION-VACUUM-FLOOR-003
- SFT-PHYS-VALIDATION-VACUUM-INERTIA-003
- SFT-PHYS-VALIDATION-VACUUM-POLARIZATION-003
- SFT-PHYS-VALIDATION-VELOCITY-COMPOSITION-034
- SFT-PHYS-VALIDATION-WAVE-ENERGY-001
- SFT-PHYS-VALIDATION-WAVE-FREQUENCY-001
- SFT-PHYS-WAVE-DIFFRACTION-001
- SFT-PHYS-WAVE-DISPERSION-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-PERIOD-FREQUENCY-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-WAVE-RESONANCE-001
- SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001

- SFT-PHYS-WAVE-SUPERPOSITION-001
- SFT-PHYS-YANG-MILLS-SINGLET-GAP-EMPIRICAL-027
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of full empirical carrier, reconciliation relation, immutable provenance, sealed execution, separate observation record, complete favorable/adverse rows, successor closure and no-extra-rule. The declared exact boundary is: Every empirically tested pre-lock Physics claim; every source, validation and available measurement-receipt identity; every explicit legacy receipt shape; every retained unfavorable result and scope boundary; and all 256 alternatives. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__complete-Physics-empirical-vector-with-all-adverse-and-scope-boundary-records-retained__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	complete-Physics-empirical-vector-with-all-adverse-and-scope-boundary-records-retained	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:34282851b5cb8020dc5448ec744c2bb0eff495beb17951ae20f8b2d631410cc2`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2166f3eb725f8e397e29da86404a23c6c6beabadbe1c1653149666d38a885966`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:768689bc7417602c40fbf5c009cabdb2944d3b93cb862e6a5a459f097fbef4a6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f6b377ac20aa317940cf02063001fbf392bea2ce910b263150ce921e32411cf4`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

`sha256:604e0ad1d0c52955bad892346c3d808138fcbe75a74497118b5b6f694a919d4a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2c1c76e69ce41d87d211866b32d87a5d932767a610da975592feac6608c021ba`. Independent certificate: `sha256:7b0c9ee0781ae529abf368b8c34477f02a06b1adae84bb07d7921a77306defd9`. Engine external-validation hash: `sha256:dad245e3012d7c9dd22c6599a0f049ae623f49895dbafea6fe234fb3ed4183cb`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- AMDC-AME2020-MASS-1-2021
- AMDC-AME2020-MASS-1-BINDING-2021
- AMDC-NUBASE2020-2021
- AMDC-NUBASE2020-DECAY-HALFLIFE-2021
- APS-HUANG-TURBULENCE-SPECTRUM-2010
- APS-JILA-BEC-GROUND-OCCUPATION-1996
- APS-MCCOMB-TURBULENCE-STRUCTURE-2014
- APS-WILLIAMS-FALLER-HILL-1971-COULOMB
- ARXIV-1201.0501-FIZEAU-WATER-AIR
- ATLAS-2023-COMBINED-HIGGS-MASS
- ATLAS-CMS-2026-HIGGS-PAIR-COMBINATION
- ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026
- AVER-ET-AL-2026-PRIMORDIAL-HELIUM
- BABAR-2010-TAU-LFV-GAMMA
- BERUT-NATURE-2012
- BICEP-PLANCK-2022-TENSOR
- BIPM-SI-BROCHURE-9-2026
- BOREXINO-2020-CNO-NEUTRINOS
- CERN-OPEN-DATA-DATASET-API-2026-07-23
- CLOWE-2006-BULLET-CLUSTER
- CMS-2024-FOUR-LEPTON-HIGGS-MASS
- COOKE-PETTINI-STEIDEL-2018-DEUTERIUM
- D10E-STRONG-FIELD-NONLINEAR-FIXED-POINT-COMPARISON
- D10F-STRONG-CARRIER-MASSLESS-CONFINED-COMPARISON
- D9F-EXTERNAL-ORBITAL-DIMENSION-STABILITY
- D9M-POST-NEWTONIAN-FIXED-POINT-COMPARISON
- D9N-SYMMETRIC-SOURCE-CONSERVATION-COMPARISON
- D9O-STATIC-EXTERIOR-CLOCK-COMPARISON
- D9Q-QUADRUPOLE-RADIATED-POWER-COMPARISON
- DESAI-ET-AL-2025-PARKER-HCS-PROTONS
- DESI-DR2-2025-COSMOLOGY

- DIEHL-2014-SN2014J-GAMMA-LINES
- EKER-2018-MAIN-SEQUENCE-RELATIONS
- ESCAMILLA-2024-DARK-ENERGY-STATE
- FONSECA-2021-PSR-J0740-MASS
- GOMEZ-VALENT-2019-ACCELERATION
- GOMEZ-VALENT-2023-CCH-32
- GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026
- GWOSC-GW231123-135430-V3-2026-07-23
- GWOSC-V2-CATALOGUES-2026-07-23
- HARIDASU-2018-LATE-EXPANSION
- IAEA-ENDF-102-DATA-FORMATS-PROCEDURES
- IAEA-ENDF-CURRENT-2026-07-23
- IAEA-ENSDF-CURRENT-2026-07-23
- IAEA-INDC-NDS-0452-MAGIC-NUMBERS
- IAEA-NDS-JENDL1-INDC-JAP-45
- IAEA-NDS-LEXFOR-FISSION-2006
- IAEA-STI-PUB-1196-RADIATION-PHYSICS
- IAEA-TECDOC-2024-NUCLEAR-COGENERATION
- IAEA-THEORY-NUCLEAR-STRUCTURE-STI-PUB-249
- IAEA-WORLD-FUSION-OUTLOOK-2023
- IAPWS-RELEASES-2026-07-23
- IUPAC-NIST-ATOMIC-SHELL-PERIODICITY-2022-2026
- LELLI-2019-BARYONIC-TULLY-FISHER
- LIGO-VIRGO-2019-GW170817-REMNANT
- LIGO-VIRGO-GW151226-DISCOVERY-2016
- LIGO-VIRGO-GW190521-DISCOVERY-2020
- LOREDO-LAMB-2001-SN1987A-NEUTRINOS
- MATTER-FLAVOUR-AUTHORITATIVE-2022-2026
- MATTER-FLAVOUR-TERMINAL-ANOMALY-AUTHORITATIVE-2022-2026
- MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026
- MATTER-FLAVOUR-TERMINAL-PROTON-AUTHORITATIVE-2022-2026
- MCMASTER-LIN-ERBIUM-CRITICAL-SCATTERING-1993
- MEGII-2025-MU-E-GAMMA
- NASA-GRC-MOON-SYNCHRONOUS-ROTATION
- NASA-GSFC-COSMIC-ELEMENTS-STELLAR-STAGES
- NASA-GSFC-WHITE-DWARF-CHANDRASEKHAR
- NASA-LAMBDA-2026-07-23
- NASA-LAMBDA-COBE-FIRAS-2026-07-23

- NASA-LAMBDA-COBE-FIRAS-MONOPOLE-2026-07-23
- NASA-NTRS-HELIOSEISMOLOGY-2001
- NASA-NTRS-MESSENGER-MERCURY-RESONANCE
- NASA-PLASMA-FREQUENCY-PROBE-1992
- NASA-STEREO-ALFVEN-LIST
- NATURE-2019-PRAD-PROTON-RADIUS
- NATURE-2026-ELECTRONIC-HYDROGEN-PROTON-RADIUS
- NATURE-STORZ-LOOPHOLE-FREE-BELL-2023
- NBS-ACOUSTIC-CAVITY-1986
- NEDM-2020-PSI-PERMANENT-EDM
- NIM-NIST-ACOUSTIC-BOLTZMANN-2017
- NIST-ASD-5.12
- NIST-ATOMIC-TRANSITION-RATE-LIFETIME-2026
- NIST-CHEMISTRY-WEBBOOK-2026-07-23
- NIST-COBLENTZ-BLACKBODY
- NIST-CODATA-2014-HISTORICAL
- NIST-CODATA-2022
- NIST-CODATA-2022-ALL-CONSTANTS
- NIST-CODATA-2022-BOLTZMANN
- NIST-CODATA-2022-EXACT-C
- NIST-DIODE-LASER-LINEWIDTH
- NIST-HYDROGEN-RYDBERG-SUCCESSOR-2022-2026
- NIST-JILA-PAULI-BLOCKING-2001
- NIST-JOHNSON-NOISE-BOLTZMANN-2011
- NIST-KRAMIDA-HYDROGEN-CRITICAL-COMPILATION-2010-2019
- NIST-MOLECULAR-SPECTROSCOPY-H2-D2-SUCCESSOR
- NIST-NBS-ATOMIC-FIELD-SPLITTING-2026
- NIST-NCNR-DCS-SUPERFLUID-HELIUM
- NIST-SRD-INDEX-2026-07-23
- NOTERDAEME-ET-AL-2011-CMB-TEMPERATURE
- PDG-2024-ELECTROWEAK-VACUUM-SCALE
- PDG-2024-W-BOSON-LISTING
- PDG-2025-2026-STRONG-EM-COMPLETE-RUNNING-VECTOR
- PDG-2025-A2-1320
- PDG-2025-A4-1970
- PDG-2025-BBN-REVIEW
- PDG-2025-CKM-MATRIX
- PDG-2025-ELECTROMAGNETIC-RUNNING-ENDPOINT

- PDG-2025-HIGGS-MASS-AVERAGE
- PDG-2025-LEPTON-SUMMARY
- PDG-2025-NEUTRINO-MIXING
- PDG-2025-QUARK-MASSES
- PDG-2025-RHO-770
- PDG-2025-RHO3-1690
- PDG-2025-RHO5-2350
- PDG-2025-STANDARD-MODEL-REVIEW
- PDG-2025-SUMMARY-TABLES
- PDG-2026-ELECTROWEAK-REVIEW
- PDG-2026-ELECTROWEAK-SCALE-VECTOR
- PDG-2026-QCD-REVIEW
- PDG-2026-QUARK-MODEL-GLUEBALL-BOUNDARY
- PDG-2026-W-BOSON-LISTING
- PDG-ELECTROWEAK-PRECISION-2024-2025
- PDG-FORCE-SECTOR-ANCHORS-2025
- PDG-HADRON-MULTIPLETS-REGGE-2025
- PDG-NIST-NUCLEAR-RESIDUAL-RANGE-2025-2026
- PDG-NIST-NUCLEON-BINDING-2022-2026
- PH2-EXTERNAL-LYAPUNOV-KS-CORRESPONDENCE
- PH3-EXTERNAL-COUPLED-MAP-CRITICALITY
- PIECHOCINSKA-PRA-2000
- PLANCK-2018-BAO-BASELINE-BUDGET
- PLANCK-2018-COSMOLOGICAL-PARAMETERS
- PLANCK-2018-INFLATION
- PLANCK-2018-OVERVIEW-PEAKS
- PLANCK-2018-VI-ABSTRACT-DENSITIES
- PLANCK-PLA-BASE-NRUN-TTTEEE-BAO-LENSING-68PC
- RAUCH-COHERENT-SPINOR-ROTATION-1975
- RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026
- SFTOM-V1-E3-EXACT-OBSERVATION
- SFTOM-V1-E4-EXACT-OBSERVATION
- SPARC-2016-175-GALAXY-MASS-MODELS
- SPARC-2016-ROTATION-CURVES
- SPRINGER-CETIN-MANGANITE-CRITICAL-EXPONENTS-2026
- SUPERK-2020-PROTON-E-MU-PI0
- SUPERK-2024-PROTON-E-MU-ETA
- SUPERK-2026-PROTON-E-MU-TWO-PI0

- VACUUM-LINEAGE-AUTHORITATIVE-2015-2026
- WATSON-2019-GW170817-STRONTIUM

Observed comparison records:

- Complete immutable reconciliation of 234 empirically tested Physics claims.
- All 147 distinct registered external source identities remain present.
- Every empirical and external validation hash and every available separate measurement receipt remains bound.
- Six older materialization shapes without a separate measurement-receipt field are explicitly retained, not silently upgraded.
- All fourteen detected unfavourable-result or scope-boundary claims remain in the same evidence vector.
- Observation remains empirical evidence; prior observations are not relabelled as unseen predictions and never select a formal survivor.

Falsification condition: Reject if any empirical claim, validation hash, available measurement receipt, source identity, legacy receipt shape, unfavourable result or scope boundary is omitted or changed; if the formal receipt does not predate this reconciliation; if prior observations are relabelled as unseen; if measurement selects a formal survivor; or if any hostile control passes.

Measurement receipt: `sha256:6837b0a6ba85b94ed9a282a974e7f0b11514ab82e441e01587ae8db871f71bd1`. Isolation certificate: `sha256:4872b1dd81f1ece18a811463bf455907c2aa6a709b2dca12480fe968faee6613`. Custody certificate: `sha256:09c27a514eb50e5e0fc87a5e8b1ad256d070da51745fed7eeb7e5010a6a9efea`.

Registered source descriptions:

- complete immutable pre-lock Physics evidence surface: experiments/external_sources/physics/snapshots/physics-grand-lock-empirical-reconciliation-record.json (sha256:e233cd761aa874893d2c2a4e2b09f071297aee1204d531fce3d93429948177a3)

Registered target identities:

- SFT-V3-PHYSICS-COMPLETE-EMPIRICAL-RECONCILIATION from SFT-V3-PHYSICS-IMMUTABLE-EMPIRICAL-EVIDENCE-SURFACE

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no favourable-only selection or deletion of mismatch, non-observation, tension, uncertainty or scope boundary
- no prior observation relabelled as an unseen prediction
- no legacy certificate silently represented as having a later separate measurement-receipt field
- no measurement, consensus model, fit, parameter or tolerance selecting a formal survivor
- no edit to the canonical engine or any admitted receipt

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:3ccc728167a1e7a63da295820a4061a1e60d4c985824b86eb9305cd1f097be18`; engine receipt `sha256:93f4497d6f7ef3c477246079f62c21f96a7ae27fd9516fa876e9d413bbab569e` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-GRAND-LOCK-076-93f4497d6f7ef3c4.json`; empirical-validation hash `sha256:7ec55dbb12f906e009c994473b79791759328778eefc4d19b59a83ba32d3f3f2`; measurement receipt `sha256:6837b0a6ba85b94ed9a282a974e7f0b11514ab82e441e01587ae8db871f71bd1`.

366. Complete measured Tesla-resonance family and power-boundary comparison

Claim identity: `SFT-PHYS-VALIDATION-TESLA-RESONANCE-FAMILY-082`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete captured record supports bounded discrete recurrence, measured quarter-wave third and other odd harmonics, one longitudinal plus two transverse rank-three roles, and resonant connected-endpoint transfer with cavity storage. It simultaneously forces the empirical boundary: material mode speeds differ; transfer has decay, loss and fidelity limits; the Earth spherical-cavity sequence is not the quarter-wave odd sequence; and no captured experiment demonstrates source-free or unlimited power extraction.

The complete captured record supports bounded discrete recurrence, measured quarter-wave third and other odd harmonics, one longitudinal plus two transverse rank-three roles, and resonant connected-endpoint transfer with cavity storage. It simultaneously forces the empirical boundary: material mode speeds differ; transfer has decay, loss and fidelity limits; the Earth spherical-cavity sequence is not the quarter-wave odd sequence; and no captured experiment demonstrates source-free or unlimited power extraction.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-TESLA-BOUNDED-CAVITY-078`
- `SFT-PHYS-TESLA-ODD-QUARTER-WAVE-079`
- `SFT-PHYS-TESLA-LONGITUDINAL-TRANSVERSE-080`
- `SFT-PHYS-TESLA-RESONANT-TRANSFER-081`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of the four sealed resonance laws, five source-bound external records, complete favorable/adverse row retention, observational provenance, measurement separation and no-extra-rule closure. The declared exact boundary is: Formal Claims 078-081; all five frozen institutional source identities and captured artifacts; every bounded, odd-quarter, orientation, transfer, loss, Earth-cavity, speculative-power and delivery-type row; and all 256 registered comparison forms. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-tesla-resonance-family-versus-complete-five-source-observation-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-tesla-resonance-family-versus-complete-five-source-observation-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favourable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:73b63e008331ce3e86b445eda56e551abc008b19fe37a505f5a0a387753cad65`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for

every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:cf67c88a3c739f6b4d99aa476e4c88661001fde501f469ab594bbde94307a8aa`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8d452e6b3fad24f2d5160be591710bef3d2a11ac4ec4817d3081a91b0a306a54`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b35656f2a02d4e0316d3e77a93bbfe5f3bf5e48cf444f62d8ff32ec61ee7fa23`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:798bd5a4a08b09675a292e2f129f2222be34c5f21d48b1226bb3fb49729d729a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8e23b86ed48ff510e216029683d9069c22ab055089f83777b98e2e1299336683`. Independent certificate: `sha256:2cfb7f08c5fecfc0da31b43b28ad811e33391de1a9cc71c3d3cfcf62f68e6dec`. Engine external-validation hash: `sha256:210a086734b2cc0494960ab7d218cce3e0b59cc950949a60e2cd005fc6832051`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NIST-JRES-QUARTER-WAVE-1935
- NIST-SRDATA-DIAMOND-MODES-Z00435
- NIST-COHERENT-RESONANT-TRANSFER-2007
- NASA-EARTH-IONOSPHERE-RESONANCE-2003
- NASA-SCHUMANN-OBSERVATION-2011

Observed comparison records:

- QUARTER-WAVE-ODD-HARMONIC-STRUCTURE: predicted complete-measured-resonance-family__bounded-and-odd-quarter-modes__one-longitudinal-two-transverse__resonant-transfer-with-loss__earth-cavity-without-source-free-power; observed complete-measured-resonance-family__bounded-and-odd-quarter-modes__one-longitudinal-two-transverse__resonant-transfer-with-loss__earth-cavity-without-source-free-power; exact match True
- LONGITUDINAL-AND-TWO-TRANSVERSE-MODES: predicted complete-measured-resonance-family__bounded-and-odd-quarter-modes__one-longitudinal-two-transverse__resonant-transfer-with-loss__earth-cavity-without-source-free-power; observed complete-measured-resonance-family__bounded-and-odd-quarter-modes__one-longitudinal-two-transverse__resonant-transfer-with-loss__earth-cavity-without-source-free-power; exact match True
- RESONANT-CAVITY-TRANSFER-AND-STORAGE: predicted complete-measured-resonance-family__bounded-and-odd-quarter-modes__one-longitudinal-two-transverse__resonant-transfer-with-loss__earth-cavity-without-source-free-power; observed complete-measured-resonance-family__bounded-and-odd-quarter-modes__one-longitudinal-two-transverse__resonant-transfer-with-loss__earth-cavity-without-source-free-power; exact match True
- EARTH-CAVITY-MODES-AND-POWER-BOUNDARY: predicted complete-measured-resonance-family__bounded-and-odd-quarter-modes__one-longitudinal-two-transverse__resonant-transfer-with-loss__earth-cavity-without-source-free-power; observed complete-measured-resonance-family__bounded-and-odd-quarter-modes__one-longitudinal-two-transverse__resonant-transfer-with-loss__earth-cavity-without-source-free-power; exact match True
- EARTH-CAVITY-SATELLITE-DETECTION: predicted complete-measured-resonance-family__bounded-and-odd-quarter-modes__one-longitudinal-two-transverse__resonant-transfer-with-loss__earth-cavity-without-source-free-power;

observed complete-measured-resonance-family__bounded-and-odd-quarter-modes__one-longitudinal-two-transverse__resonant-transfer-with-loss__earth-cavity-without-source-free-power; exact match True

- deliberately tampered unfavourable control rejected
- direct source check five_source_rows_retained: True
- direct source check all_registered_classifications_retained: True
- direct source check quarter_wave_source_measured: True
- direct source check orientation_rows_measured: True
- direct source check resonant_transfer_measured: True
- direct source check earth_cavity_modes_retained: True
- direct source check power_claim_remains_proposed_not_measured: True
- direct source check satellite_detection_and_delivery_mismatch_retained: True
- direct source check formal_and_empirical_boundaries_complete: True
- all five source records and every limitation/adverse row retained
- source-free or unlimited power inference rejected as unsupported by the captured measurements

Falsification condition: Reject if any formal receipt, preregistration, source artifact or row changes; if quarter-wave odd response, the complete orientation inventory, resonant transfer/storage, or Earth-cavity detection is absent; if any loss, speed distinction, non-odd Earth sequence, speculative-power status or delivery mismatch is concealed; or if observation is allowed to alter a formal survivor.

Measurement receipt: `sha256:1fd0a5c0afa7d4cd90a932d78a30660711a2ef23aececcaa60a34b605adb181b`. Isolation certificate: `sha256:fbb659ef25078e452512c395ad665af6390b7ce8b307adc7e33259b8ca8be3d8`. Custody certificate: `sha256:34cff4fda9b96194edc7a02fb61ba32eaa3d22d8da6195ce3b88c484cae49ac0`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no historical Tesla reputation or V1/V2 result as a proof premise
- no measured frequency, apparatus length, gain or fidelity selecting a formal survivor
- no omission of the non-odd Earth sequence, material-speed distinctions, loss, decay, fidelity or delivery-type mismatch
- no inference from a proposed application to a measured source-free-energy result
- no universal efficiency, unlimited Earth-power or identity-of-substances claim
- no numerical-zero, negative, irrational, imaginary, floating or completed-infinite magnitude in the formal derivation

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:bbe4969d3376381da15284830d6a4d86f9cb8cbcc7dec77399c42e51fe5b51ed`; engine receipt `sha256:270786783cb68b4737d35faf5385079767c22036b03dc8f83965bb1eb2467b92` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-TESLA-RESONANCE-FAMILY-082-270786783cb68b47.json`; empirical-validation hash `sha256:55a5a993cab6ca208e83b73cd9112595650414648641 1ebf942b4a8cfca9a472`; measurement receipt `sha256:1fd0a5c0afa7d4cd90a932d78a30660711a2ef23aececcaa60a34b605adb181b`.

367. Complete measured vacuum/inertia-drive correspondence and apparatus boundary

Claim identity: `SFT-PHYS-VALIDATION-VACUUM-INERTIA-DRIVE-FAMILY-087`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The complete official record supports the external correspondence of the sealed structural family: a driven local-vacuum/inertia mechanism is explicitly described; nonempty vacuum response and boundary-dependent vacuum interaction are measured; and ordinary inertial/gravitational unity is constrained. It simultaneously forces the apparatus boundary: the captured Navy records call the device a theoretical concept, state that no prototype yet existed and propose later experiment; CNES does not test driven inertia; NIST retains coherent pump energy and forbids removal of real work from ground fluctuations. Therefore the exact structural channel, positive finite-depth bound and complete ledger remain admitted, while a useful device magnitude, completed public inertial-mass-reduction measurement and source-free cyclic gain are not claimed by this comparison.

> *The complete official record supports the external correspondence of the sealed structural family: a driven local-vacuum/inertia mechanism is explicitly described; nonempty vacuum response and boundary-dependent vacuum interaction are measured; and ordinary inertial/gravitational unity is constrained. It simultaneously forces the apparatus boundary: the captured Navy records call the device a theoretical concept, state that no prototype yet existed and propose later experiment; CNES does not test driven inertia; NIST retains coherent pump energy and forbids removal of real work from ground fluctuations. Therefore the exact structural channel, positive finite-depth bound and complete ledger remain admitted, while a useful device magnitude, completed public inertial-mass-reduction measurement and source-free cyclic gain are not claimed by this comparison.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-VACUUM-LOCAL-RESONANT-DRIVE-083`
- `SFT-PHYS-VACUUM-INERTIA-COVARIATION-084`
- `SFT-PHYS-VACUUM-INERTIA-POSITIVE-FLOOR-085`
- `SFT-PHYS-VACUUM-INERTIA-COMPLETE-LEDGER-086`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar and closure boundary. Generate the complete eight-axis product of the four sealed vacuum/inertia-drive laws, five source-bound official records, complete favorable/adverse/absent/untested retention, observational provenance, measurement separation and no-extra-rule closure. The declared exact boundary is: Formal Claims 083-086; all five frozen institutional source identities and six captured artifacts; every mechanism, theoretical-status, prototype-status, nonempty-vacuum, Casimir, pump-ledger, ordinary-unity and engineered-control-scope row; and all 256 registered comparison forms. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-vacuum-inertia-drive-family-versus-complete-five-source-observation-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Shared claim-record clause `PHYS-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-vacuum-inertia-drive-family-versus-complete-five-source-observation-vector`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3aa54cb18536810425e3c50a98cbdac29de48367ec7c15701dc9fdfdbc2eb5c9`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9486147e466f4c7a7e0d7b4370a2714b2fac5b42a659296944998e1dc2e54220`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:96c4cfb51268dd3eed79a38972dc935d466907b1ad3d3acc29177cc44f29f860`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f1f6bcc9aacf81a4ea9da010950461cb223130bba20bd4a737e73787ac4a57b1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:71130f9a86dfa61cb7f261fba68e46dd30a846c81ebae2fccf6d7a8517faacb4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e914ce00fe6e530defcb6c299f9937777b8d50c344c9853441f0902900307444`. Independent certificate: `sha256:43617226e5120b8c074573e69428171b09fb96f47402f517fad8de6b96d4ee98`. Engine external-validation hash: `sha256:f44058c5d559c76ec101bb0a6fd068a04f058cdf28ffc34f281999818a8b5bae`.

Shared claim-record clause `PHYS-S005` applies. The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- NAVAIR-PAX205-INERTIAL-MASS-REDUCTION-DEVICE
- NAVAIR-PAX205-PATENT-PROSECUTION-RECORD
- NIST-ZERO-POINT-FLUCTUATIONS-2015
- NIST-CASIMIR-VACUUM-FORCES-2013
- CNES-MICROSCOPE-FINAL-2022

Observed comparison records:

- PATENT-MECHANISM-AND-APPARATUS-CLAIM: predicted complete-vacuum-inertia-drive-record__official-mechanism-described__no-public-prototype-measurement-in-record__nonempty-vacuum-and-casimir-response-measured__ordinary-unity-constrained__pump-and-restoration-ledger-retained; observed complete-vacuum-inertia-drive-record__official-mechanism-described__no-public-prototype-measurement-in-record__nonempty-vacuum-and-casimir-response-measured__ordinary-unity-constrained__pump-and-restoration-ledger-retained; exact match True

- PUBLIC-DEMONSTRATOR-AND-TEST-STATUS: predicted complete-vacuum-inertia-drive-record__official-mechanism-described__no-public-prototype-measurement-in-record__nonempty-vacuum-and-casimir-response-measured__ordinary-unity-constrained__pump-and-restoration-ledger-retained; observed complete-vacuum-inertia-drive-record__official-mechanism-described__no-public-prototype-measurement-in-record__nonempty-vacuum-and-casimir-response-measured__ordinary-unity-constrained__pump-and-restoration-ledger-retained; exact match True
- NONEMPTY-VACUUM-AND-PUMP-BOUNDARY: predicted complete-vacuum-inertia-drive-record__official-mechanism-described__no-public-prototype-measurement-in-record__nonempty-vacuum-and-casimir-response-measured__ordinary-unity-constrained__pump-and-restoration-ledger-retained; observed complete-vacuum-inertia-drive-record__official-mechanism-described__no-public-prototype-measurement-in-record__nonempty-vacuum-and-casimir-response-measured__ordinary-unity-constrained__pump-and-restoration-ledger-retained; exact match True
- BOUNDARY-DEPENDENT-VACUUM-INTERACTION: predicted complete-vacuum-inertia-drive-record__official-mechanism-described__no-public-prototype-measurement-in-record__nonempty-vacuum-and-casimir-response-measured__ordinary-unity-constrained__pump-and-restoration-ledger-retained; observed complete-vacuum-inertia-drive-record__official-mechanism-described__no-public-prototype-measurement-in-record__nonempty-vacuum-and-casimir-response-measured__ordinary-unity-constrained__pump-and-restoration-ledger-retained; exact match True
- INERTIAL-GRAVITATIONAL-UNITY-AND-SCOPE-CONTROL: predicted complete-vacuum-inertia-drive-record__official-mechanism-described__no-public-prototype-measurement-in-record__nonempty-vacuum-and-casimir-response-measured__ordinary-unity-constrained__pump-and-restoration-ledger-retained; observed complete-vacuum-inertia-drive-record__official-mechanism-described__no-public-prototype-measurement-in-record__nonempty-vacuum-and-casimir-response-measured__ordinary-unity-constrained__pump-and-restoration-ledger-retained; exact match True
- deliberately tampered unfavorable control rejected
- direct source check five_source_identities_retained: True
- direct source check all_classifications_retained: True
- direct source check navy_mechanism_description_retained: True
- direct source check navy_no_prototype_status_retained: True
- direct source check prosecution_mechanism_retained: True
- direct source check prosecution_proposed_experiment_status_retained: True
- direct source check nist_ground_and_pump_boundary_retained: True
- direct source check nist_casimir_boundary_response_retained: True
- direct source check cnes_unity_and_scope_record_retained: True
- direct source check complete_power_and_scope_boundary_retained: True
- all official source identities and every favorable, adverse, absent, untested and scope-limiting row retained
- no public apparatus measurement invented and no formal channel falsification inferred from its absence

Falsification condition: Reject if any formal receipt, preregistration, source artifact or row changes; if the official mechanism description, no-prototype/proposed-experiment status, measured nonempty vacuum response, Casimir interaction, ordinary unity constraint or pump/restoration boundary is omitted; if patent status is relabelled as measurement; if lack of public device data is relabelled as formal falsification; or if observation alters a formal survivor.

Measurement receipt: `sha256:c93adfb85742c28ab95048e895b1f3276c34144a3c3ba79ea8434053c635de25`. Isolation certificate: `sha256:e1705dc2a889767101ad50fe7ff1c8a35c9fc30101cc37806952bcaa176a2731`. Custody certificate: `sha256:0da359028d4f38cf91b8342ce39d67b90b882f013c45e8a173fe5a20cc5a6995`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, patent formula, target magnitude or desired apparatus result as a proof premise
- no patent issuance, proposed experiment or concept paper relabelled as a measured device effect
- no absence of a public prototype measurement relabelled as falsification of the sealed exact structural channel
- no MICROSCOPE equivalence result relabelled as a driven local-inertia-control test
- no omission of pump energy, restoration cost, no-ground-state-work or apparatus-status rows
- no source-free energy, universal efficiency, useful propulsion or dimensional minimum-mass claim
- no numerical-zero, negative, irrational, imaginary, floating or completed-infinite magnitude in the formal derivation

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:1f1896f6dbf0b85b020e052b7da01dfc0c0e84109319b5d97df95ff5247b347b`; engine receipt `sha256:ae878e3fe8666a5471b9d19a2f67ab235b05f2187f3e9934523cc0e1661f4ca1` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VACUUM-INERTIA-DRIVE-FAMILY-087-ae878e3fe8666a54.json`; empirical-validation hash `sha256:b1010b89fc82039aca76f69977233104924e3e3bab5b77d7490787e6d6a26bca`; measurement receipt `sha256:c93adfb85742c28ab95048e895b1f3276c34144a3c3ba79ea8434053c635de25`.

368. Complete measured new-sector, particle-census and Smithion search record

Claim identity: `SFT-PHYS-VALIDATION-NEW-SECTOR-COMPLETE-FAMILY-095`

Shared claim-record clause `PHYS-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Theorem. The known external anchors reproduce three weak carriers, three colour charges and eight gluons and retain the known lepton/quark categories used in the 110-kind census. PDG lists axion-like, heavy-neutral-lepton and supersymmetric classes as search subjects, not confirmed additions in this record. ATLAS searches semi-visible/emerging confining dark jets and missing momentum and reports model-dependent exclusions, establishing the relevance of the sealed signature classes but not the specific 24/48 inventories, slopes 4/6 or a Smithion discovery. The existing dark record supports exact 27/5 abundance correspondence and additional gravitating support while retaining no registered Smithion mass. Thus every penta/hepta quantity remains a precise standing test; current non-observation neither confirms nor retires it.

> *The known external anchors reproduce three weak carriers, three colour charges and eight gluons and retain the known lepton/quark categories used in the 110-kind census. PDG lists axion-like, heavy-neutral-lepton and supersymmetric classes as search subjects, not confirmed additions in this record. ATLAS searches semi-visible/emerging confining dark jets and missing momentum and reports model-dependent exclusions, establishing the relevance of the sealed signature classes but not the specific 24/48 inventories, slopes 4/6 or a Smithion discovery. The existing dark record supports exact 27/5 abundance correspondence and additional gravitating support while retaining no registered Smithion mass. Thus every penta/hepta quantity remains a precise standing test; current non-observation neither confirms nor retires it.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- `SFT-PHYS-PENTA-COMPLETE-PHENOTYPE-088`
- `SFT-PHYS-HEPTA-COMPLETE-PHENOTYPE-089`
- `SFT-PHYS-PENTA-BETA-SLOPE-090`
- `SFT-PHYS-HEPTA-BETA-SLOPE-091`
- `SFT-PHYS-CATEGORY-CLEAN-PARTICLE-CENSUS-092`
- `SFT-PHYS-NO-EXTRA-SECTOR-PARTICLE-BOUNDARY-093`
- `SFT-PHYS-SMITHION-INTERACTION-SEARCH-SIGNATURES-094`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`

- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of the seven sealed new-sector laws, five source identities, complete favorable/adverse/absent/searched/unresolved retention, observational provenance, measurement separation and no-extra-rule closure. The declared exact boundary is: Formal Claims 088-094; all five registered source identities and eleven captured artifacts; every known-sector, known-particle, outside-list-search, collider-signature, abundance, gravitating-support, null and unmeasured-standing-prediction row; and all 256 comparison forms. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-new-sector-complete-family-versus-five-source-observation-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Shared claim-record clause PHYS-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-new-sector-complete-family-versus-five-source-observation-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favourable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:d732f2a78aeaa28003aa95656f1f334e9e9bc1bdf40f61d84880541627516429. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:7ea0adab0f9c5031e88c05164fd2b091327d684fd4e25baf45b4cad03505ffba.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:300e3098b5ceb014c5421419963b345857d34b6f9d01eac3ba481fb1924f25d2.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:bd613e67145557379ea1e130e3778561927d66ce1ca2af10703afced1a9f456a.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:8c6b69523646df69c790aff4dfe3f28df25d01e74821002b23a71445011b08f6.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e78b3dc5dd67d7ec1dc01d6f1598281795f56c01791cad71ed9f594c47c33b8e. Independent

certificate: `sha256:d446fcb696a9642d5f5bce65c9bcb9546c39c4c59d16ae8681b0d0a50029c60c`. Engine external-validation hash:
`sha256:223aca77589d0b3e16592732b551bcdc43e158de3819e9292c3bb721e6baa5ab`.

Shared claim-record clause `PHYS-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

External source identities:

- PDG-FORCE-SECTOR-ANCHORS-2025
- PDG-SUMMARY-TABLES-2025
- PDG-SEARCHES-HYPOTHETICAL-PARTICLES-2025
- ATLAS-DARK-SECTOR-JETS-2025
- DARK-SMITHION-STANDING-RECORD-2026

Observed comparison records:

- KNOWN-SECTOR-CARRIER-ANCHORS: predicted complete-new-sector-record__known-three-eight-anchors__known-fermion-categories__outside-list-classes-searched-not-confirmed__dark-jets-and-missing-momentum-searched__penta-hepta-and-smithions-remain-standing; observed complete-new-sector-record__known-three-eight-anchors__known-fermion-categories__outside-list-classes-searched-not-confirmed__dark-jets-and-missing-momentum-searched__penta-hepta-and-smithions-remain-standing; exact match True
- KNOWN-FERMION-KIND-INVENTORY: predicted complete-new-sector-record__known-three-eight-anchors__known-fermion-categories__outside-list-classes-searched-not-confirmed__dark-jets-and-missing-momentum-searched__penta-hepta-and-smithions-remain-standing; observed complete-new-sector-record__known-three-eight-anchors__known-fermion-categories__outside-list-classes-searched-not-confirmed__dark-jets-and-missing-momentum-searched__penta-hepta-and-smithions-remain-standing; exact match True
- OUTSIDE-LIST-SEARCH-STATUS: predicted complete-new-sector-record__known-three-eight-anchors__known-fermion-categories__outside-list-classes-searched-not-confirmed__dark-jets-and-missing-momentum-searched__penta-hepta-and-smithions-remain-standing; observed complete-new-sector-record__known-three-eight-anchors__known-fermion-categories__outside-list-classes-searched-not-confirmed__dark-jets-and-missing-momentum-searched__penta-hepta-and-smithions-remain-standing; exact match True
- CONFINING-JET-AND-MISSING-MOMENTUM-SEARCH: predicted complete-new-sector-record__known-three-eight-anchors__known-fermion-categories__outside-list-classes-searched-not-confirmed__dark-jets-and-missing-momentum-searched__penta-hepta-and-smithions-remain-standing; observed complete-new-sector-record__known-three-eight-anchors__known-fermion-categories__outside-list-classes-searched-not-confirmed__dark-jets-and-missing-momentum-searched__penta-hepta-and-smithions-remain-standing; exact match True
- SMITHION-ABUNDANCE-GRAVITY-AND-MEASUREMENT-STATUS: predicted complete-new-sector-record__known-three-eight-anchors__known-fermion-categories__outside-list-classes-searched-not-confirmed__dark-jets-and-missing-momentum-searched__penta-hepta-and-smithions-remain-standing; observed complete-new-sector-record__known-three-eight-anchors__known-fermion-categories__outside-list-classes-searched-not-confirmed__dark-jets-and-missing-momentum-searched__penta-hepta-and-smithions-remain-standing; exact match True
- deliberately tampered unfavorable control rejected
- direct source check five_source_identities_retained: True
- direct source check all_classifications_retained: True
- direct source check known_sector_anchors_and_predictions_retained: True
- direct source check known_particle_categories_retained: True
- direct source check outside_list_search_categories_retained: True
- direct source check atlas_confining_dark_jet_classes_retained: True
- direct source check atlas_limits_remain_model_dependent: True
- direct source check dark_abundance_and_gravity_rows_retained: True

- direct source check smithion_measurement_not_invented: True
- direct source check complete_scope_boundary_retained: True
- all standing predictions retained without fabricated measurement or retirement

Falsification condition: Reject if any formal receipt, preregistration, artifact or row changes; if a known count/category, searched hypothesis, collider signature/limit, abundance/gravity row or unmeasured-standing status is omitted; if a search is relabelled as discovery; if non-observation is relabelled as retirement; or if observation alters a formal survivor.

Measurement receipt: `sha256:193639c324b366998aa5ffded5ddc212277a9ee16fa90ed9aecf803f480c168f`. Isolation certificate: `sha256:be20ceb2c50d8b56867f25ed622b7e52bc7fc649bd1d7fac9775487dd5a6f072`. Custody certificate: `sha256:b2abd9c0bf13c9bfb2298afd65917dc5240e59aa5f85a6a83b65c6bc55829627`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Shared claim-record clause `PHYS-S028` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Boundary and non-claim. The following forms are explicitly excluded:

- no PDG, ATLAS, cosmology or null-search row visible to the formal survivor decision
- no known-sector count imported to select p-squared-less-One
- no searched hypothetical class relabelled as a confirmed fundamental kind
- no model-dependent collider exclusion generalised to every SFT signature
- no unobserved penta/hepta carrier, beta slope or Smithion mass represented as measured
- no current non-observation represented as retirement of a standing prediction
- no numerical-zero, negative, irrational, imaginary, floating or completed-infinite magnitude in the formal derivation

Shared claim-record clause `PHYS-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Immutable evidence identities. Source manifest `sha256:6edc3b3a0a14398e7a22e3fd791214a4adffe58809c4e8294fa3d2bdbba49e5e`; engine receipt `sha256:2f13335a3bfdd7a395689ff113676a6c3287d90c7b4c8e49eac53730d5c5f2bd` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-NEW-SECTOR-COMPLETE-FAMILY-095-2f13335a3bfdd7a3.json`; empirical-validation hash `sha256:12c193e25d9c75fcaf57f0f52dcb6e647cdc2dbb5747d6c069f7123ec53ea058`; measurement receipt `sha256:193639c324b366998aa5ffded5ddc212277a9ee16fa90ed9aecf803f480c168f`.

25. Cross-branch synthesis

The completed branch has one compositional arc. Measurement defines what an observation can retain. Mechanics defines relational change and transfer. Fields extend source-response over local support. Waves close recurrence and propagation. Thermodynamics identifies macro-observation fibres and information-retention costs. Physical quantum theory retains complete support, phase/action and joint composition. Matter identifies recurrent constituent and channel classes. Spacetime orders events by limiting causal paths; gravitation changes their source-linked relational closure. Collective matter takes finite cell networks to scale-declared macro-observations. The cosmological boundary exports universal propagation laws while refusing to disguise historical state as a law.

This unification is operational: the same Fold objects - exact counts, ratios, words, held labels, tables, transition traces and observation fibres - appear in every subbranch. The claim is not that established terminology selected those objects. Correspondence is registered only after each Fold form survives its generated alternatives.

26. Limitations and falsification

The paper makes no claim of an actually completed physical infinity, exact continuum substrate, derivation of contingent initial conditions, unbounded astronomical census or replacement of later Chemistry, Materials, Astronomy and Cosmology branches. It does not treat a simulation as empirical proof, an exact SI definition as an independent natural observation, or portal availability as numerical agreement. Categorical external rows test structural correspondence; the dedicated value adapters test numerical consequences. A new observation outside the registered boundary requires a new preregistration and cannot silently broaden an old receipt.

Accordingly, complete means complete to the current 29 July 2026 project-version investigation. It is not permanent closure or the end of Physics or science.

Any admitted empirical claim is falsified within its boundary by a preserved external row that does not match the sealed prediction, by target access before sealing, by omitted adverse evidence, by a changed source snapshot, by failure of its independent reconstruction or by an engine-constitution violation. The correct response is a halted new claim and investigation of the derivation or adapter. Existing immutable receipts remain identities of the artifacts they certify; new evidence cannot be retroactively written into them.

27. Reproducibility

The repository uses Python's standard library for the core engine and supports macOS, Windows and Linux without Docker. Researchers may inspect every claim package directly. The one-command repository verifier remains available for release validation, but this branch build deliberately used proportionate admission-time checks and a read-only evidence census rather than repeatedly replaying historical derivations. The Physics inventory is `publications/inventories/physics.json`; this manuscript's exact evidence map is `publications/current/physics/evidence_map.json`.

28. Conclusion

The paper's lead result is the exact derivation $\alpha^{-1} = 503846395469/3676744786$, not an isolated numerical fit but the first terminal constant in a connected Physics reconstruction. The same root and admission constitution force and validate the charged-lepton relations, precision anomalies, electroweak share, Higgs sector, Planck/proton hierarchy, proton radius, vacuum and cosmological ratios, spatial rank, inverse-square force law, nuclear and hadronic structure, quantum and relativistic correspondence, thermodynamics, collective matter and gravitation documented in the complete ledger.

At the declared derivational boundary, all 368 current V3 Physics claims and all 488 categorically Physics-owned V1/V2 atoms are engine-admitted and current-evidence closed, with no open Physics gap family. Grand Locks 075/076 preserve the pre-extension validation snapshot; the Unified Constants, Tesla resonance, vacuum/inertia and new-sector completion records reconcile the nineteen later claims. The 238 external-validation records retain their claim-specific chronology and evidence classes; they are not collapsed into a claim of universal empirical confirmation. Adverse, unavailable, unresolved and standing-prediction records remain visible, including the unobserved penta/hepta/Smithion family.

This candidate remains subject to the publication gates and Maria Smith's sole release authority. Physics remains open to lawful versioned extension, correction and falsification after publication. The result is an open, inspectable tree of physical laws whose authority rests on exact derivation, complete enumeration, sealed measurement, adverse evidence and reproducible traces rather than credentials, institutional permission, opaque prediction or consensus selection.

Current authoritative statement and receipt reconciliation

The current model-admitted census is authoritative at claim level. The entries below expose exact current statement or receipt strings that the earlier successor prose omitted or paraphrased. Historical wording remains preserved elsewhere in the paper; this reconciliation does not retroactively rewrite a prediction, chronology, evidence class, result or receipt.

`SFT-PHYS-MEAS-UNCERTAINTY-001`

Earlier display gap: exact statement.

Current exact statement: Measurement uncertainty is forced as the explicitly retained support of observation/reference distinctions not closed by the measurement process; it is not an irrational proof value or permission to choose a favorable result.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:9aef08a488c0a15fc4dd2132a1afd9c32d45bb0da6e6ed5c76f90c541f5ffc01`.

Receipt path:
`receipts/engine/model_admitted/SFT-PHYS-MEAS-UNCERTAINTY-001-9aef08a488c0a15f.json`.

SFT-PHYS-THERMO-HEAT-WORK-001

Earlier display gap: exact statement.

Current exact statement: Heat is energy transfer whose microscopic carrier label is closed by the receiving macro-observation; work is energy transfer whose organized source-to-response label remains observable.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:de7c30a2d5ab28ca55a30d35e4973d6197a3db788ad47d8604dca2b49d98e57c`.

Receipt path:
`receipts/engine/model_admitted/SFT-PHYS-THERMO-HEAT-WORK-001-de7c30a2d5ab28ca.json`.

SFT-PHYS-PLASMA-MHD-001

Earlier display gap: exact statement.

Current exact statement: Magnetofluid behavior is the compositional closure of conserved fluid flow with oriented charge transport and magnetic source-response on the same generated cell network.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:6cca90cf9ab0c0a753f0aac31257648c0164daf92d57347119bf09406431f404`.

Receipt path:
`receipts/engine/model_admitted/SFT-PHYS-PLASMA-MHD-001-6cca90cf9ab0c0a7.json`.

SFT-PHYS-MEAS-BOUNDARY-GROWTH-001

Earlier display gap: exact statement.

Current exact statement: Within a complete source-boundary dilution experiment, physical boundary-growth dimension is measured without a fitted exponent or logarithm by comparing exact positive distance and response ratios, generating every positive exponent by repeated composition, retaining the complete compatible family and closing all later candidates only after a monotonic strict-excess certificate.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:b2d8360be10ba4be5664b59ee4624d91d909962f83927b3bd98d4caf5b02e43b`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-MEAS-BOUNDARY-GROWTH-001-b2d8360be10ba4be.json`.

SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001

Earlier display gap: exact statement.

Current exact statement: For a complete boundary surrounding a localized source in the forced three-direction spatial carrier, holding one source-normal direction leaves exactly the unique positive predecessor rank two as free boundary organization; scaled equivalent boundary support is therefore the complete pair-cell product.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:10f70a4a463a28ed545c101dcf7528b6d5a7b04355e3c89f3911f8b95514717d`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001-10f70a4a463a28ed.json`.

SFT-PHYS-VALIDATION-INVERSE-SQUARE-001

Earlier display gap: exact statement.

Current exact statement: After the Fold structure independently forces and seals positive integer exponent two, the Williams, Faller and Hill Coulomb-law experiment reports q = (2.7 +/- 3.1) x 10^-16 in the tested form 1/r^(2+q). Converting only the reported finite interval to exact positive rational exponent bounds and enumerating every positive integer inside them leaves two as the sole compatible exponent.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:e70905b63729a3bf5619b26115f7fe8b78eb04aba53eba78f6697405d78a220b`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-INVERSE-SQUARE-001-e70905b63729a3bf.json`.

SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001

Earlier display gap: exact statement.

Current exact statement: The forced generator-three partition supplies one exact three-root mass carrier whose symmetric invariants are root sum One, pair sum one over the two Fold fibres paired with generator three, leading product one over the positive predecessor of twice the colour fifth-power support, and sharpened product obtained by the unique generator-three channel part. The resulting exact invariants are 1, 1/6, 1/485 and 3/1454; no irrational root is formed.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:9c7202057f7735d46e9b355f8611e63e9043d99b1a3ced16cadb6a468d09e598`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001-9c7202057f7735d4.json`.

SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001

Earlier display gap: exact statement.

Current exact statement: The admitted charged-lepton cubic is refined by the complete terminal electromagnetic self-coupling of its three-root product: alpha to the generator-three order is distributed over the complete three-cubed carrier and weighted once by the forced cover-depth pair five and seven, with the up-depth transported through one alpha and one colour channel. The exact correction is held from 3/1454.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:720b443c7267c7dc09cf7ecb5ffa226b9a0d6a745136689ef2e42f17f13b2d2f`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001-720b443c7267c7dc.json`.

SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003

Earlier display gap: exact statement.

Current exact statement: In forced three-space, three spatial axes each carry binary second-difference curvature while one temporal axis carries the same binary curvature; dimension-normalized balance again fixes speed-square at the One.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:e9314b2725d52aa3136325aee8ba08bca5fbc41e0ee3ee19db391ff98b7c1dac`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003-e9314b2725d52aa3.json`.

SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003

Earlier display gap: exact statement.

Current exact statement: Binary, generator-three and complete depth-three support force the squared PMNS triple and its normalized electron-flavour weights, while the unique self-antipodal Fold position forces the maximal phase carrier.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:6cc0b2828e8a304fd80c274f6e9a2e722cd1bebd68cff071f4c394bf25c872bd`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003-6cc0b2828e8a304f.json`.

SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003

Earlier display gap: exact statement.

Current exact statement: The bare gyromagnetic value is the binary count. The sole terminal electromagnetic self-return forces a strictly positive leading phase-normalized anomaly alpha/2, and any additional mass-scale channel has muon/electron sensitivity equal to the squared sealed mass ratio.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:e0a6844ac70a6e929c5e23ed576df2ea5c13eb4a05d33b148b2b3ad75a4c4f6f`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003-e0a6844ac70a6e92.json`.

SFT-PHYS-VALIDATION-PROTON-ELECTRON-003

Earlier display gap: exact statement.

Current exact statement: The sealed algebraic proton/electron interval is compared with the complete CODATA value and uncertainty; the non-overlap is retained as an unfavorable result.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:c9c81674c4671965244d499e7846cfd6154e3b256db1aaf0e14ef32e58972292`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-PROTON-ELECTRON-003-c9c81674c4671965.json`.

SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004

Earlier display gap: exact statement.

Current exact statement: A diameter-normalized finite turn has three complete generator sectors. Its sole remaining return passes through the forced up-depth seven and the complete binary support sixteen of the four-rung terminal promotion object, forcing the exact continued ratio 3 + 1/(7 + 1/16) = 355/113. This is an exact rational Fold projection, not an imported irrational proof value.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:95e58749591f916244b2fa09fc71e9f8fb13f9db848dc4c2c78ed1d74bd403f0`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004-95e58749591f9162.json`.

SFT-PHYS-HADRON-REGGE-TERMINAL-005

Earlier display gap: exact statement.

Current exact statement: Generator-three flavour support forces nine ordered flavour/antiflavour cells, partitioned as one invariant cell class and its eight-class positive predecessor. Complete three-place flavour support forces twenty-seven cells; direct exchange enumeration partitions them as ten symmetric, two eight-cell mixed hands and one antisymmetric class. Independently, one retained fixed-tube act per positive spin successor forces exact affine squared support. The exact theorem governs normalized fixed-carrier support; whether physical resonance masses realize literal equal spacing is a separate post-seal measurement.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:66bc011c6612641467f7deb8554279abbc55311f86f34efab51ac3fa2f718b29`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-HADRON-REGGE-TERMINAL-005-66bc011c66126414.json`.

SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005

Earlier display gap: exact statement.

Current exact statement: Short-range quarter-order residual recurrence and exact electromagnetic opposition force one positive bulk/surface/Coulomb/asymmetry/pairing ledger with no fitted coefficient. The surface share is exactly One over the binary-plus-generator interface carrier five; Coulomb loss is the admitted exact alpha over every ordered distinct proton path; asymmetry and pairing carry the admitted quarter-order. Exact rational radius enclosures, complete concave integer/parity enumeration below the forced mass 2^(5+7), and a two-case tail induction uniquely maximize normalized binding at mass count 62, charge count 28 and neutron count 34. No nuclide name or measured binding value enters execution.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:08d480f86053d84de80bb82879ba57fc11d0dbcba02557045244e7998cfa65a5`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005-08d480f86053d84d.json`.

SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005

Earlier display gap: exact statement.

Current exact statement: The least nonidentity nuclear junction and decomposition are binary. Exact conserved Fold composition maps two mass-two/charge-one words to one mass-four/charge-two word; exact decomposition maps one mass-238/charge-92 word to two mass-119/charge-46 words. The admitted zero-parameter binding ledger proves a strictly positive total-binding gain in both directions toward the unique mass-62/charge-28/neutron-34 maximum. Mass-energy conservation therefore requires a named positive release carrier. The normalized reaction barrier is the forced Half-One. No measured energy, fitted correction or stochastic premise enters the derivation.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:fbbdba722e6fb27d738ad3cd1df1f62e4caa979038b212147120ca814f0e25fa`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005-fbbdba722e6fb27d.json`.

SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006

Earlier display gap: exact statement.

Current exact statement: For the least light binary fusion representative and the registered symmetric heavy fission representative, exact zero-parameter binding-score enclosures force the fusion gain per conserved nucleon strictly above the fission gain per conserved nucleon, while the heavy reaction's larger nucleon count forces its total release strictly above the light reaction's total. Fusion requires two charged incident boundaries to approach through Coulomb opposition before short-range residual closure; fission has one parent and therefore proceeds through a neutral-capture or internal-surface carrier. Half-One and quarter-order structures remain normalized relations, not one universal dimensional threshold. Dimensional reaction energies and thresholds are opened only after sealing.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:1b13169930d9f4c22d75008ef7debe615a8c1dfdedee7a5f495b85c467444e50`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006-1b13169930d9f4c2.json`.

SFT-PHYS-LYAPUNOV-KS-CORRESPONDENCE-TERMINAL-008

Earlier display gap: exact statement.

Current exact statement: A complete m-label Fold step multiplies every uncast local separation by the positive whole m and multiplies complete word support by the same m. At depth d the support is exactly m^d and one further depth contributes one complete m-label distinction. Thus expansion and information growth have one forced exact carrier m. Conventional Lyapunov and entropy logarithms are correspondence labels at the external analytic boundary, not imported SFT proof values. Under the separately stated hypotheses of an entropy formula, the common external rate label corresponds to this same exact carrier.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:f0b74a8639f9e01be7b9adc31f83b4b24ed82f085e72c17736c38f030b5cae40`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-LYAPUNOV-KS-CORRESPONDENCE-TERMINAL-008-f0b74a8639f9e01b.json`.

SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008

Earlier display gap: exact statement.

Current exact statement: For a complete m-label Fold in the normalized drive-response transverse channel, one coupling step retains the exact transverse multiplier m(One-g). The unique neutral boundary is g=(m-1)/m; every smaller proper coupling expands the distinction and every larger proper coupling contracts it. For m=2 the boundary is the forced half-One. This threshold is not assigned to arbitrary network topologies, whose distinct transverse eigenvalue factors remain explicit external scope.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:845ceed3f993a0523e1f01a5ce9375d469e426be12539941c5e5fd5130b9534f`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008-845ceed3f993a052.json`.

SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009

Earlier display gap: exact statement.

Current exact statement: At a circular balance, dimension-forced gravity and centrifugal response are equal positive magnitudes. After every exact outward displacement q greater than the One, gravity has positive denominator q^(d-1) and centrifugal response q^3. Denominator order alone forces restoring gravity for d=2,3, equality at d=4, and nonrestoring response for every d at least five. Combined with independently forced three-space, orbital stability supplies a distinct discriminator selecting the stable side without signed forces, negative exponents or fitted parameters.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:9a067d10ece75b8300bec3c77193649313dbc2fcc116f721bd0fef54544a433c`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009-9a067d10ece75b83.json`.

SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009

Earlier display gap: exact statement.

Current exact statement: Binary support and generator-three colour force matter source seven-sixteenths; half-One is the coupling; quadratic field energy supplies self-source feedback. The exact map f-next=half-One times (seven-sixteenths plus f squared) has one admissible Fold-part fixed point, quarter-One. Starting from the linear field seven-thirty-seconds, every iterate remains below and approaches quarter-One; both its positive gap and each added correction contract at every successor by an exact factor below quarter-One. This depth-independent Fold theorem corresponds post-seal to weak-field post-Newtonian iteration without claiming universal numerical convergence outside its registered scalar channel.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:b0e30a2cb8fa9d9f2232df8f8790ab82d8fc2f8e6555326543e15fce2fb48bba`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009-b0e30a2cb8fa9d9f.json`.

SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010

Earlier display gap: exact statement.

Current exact statement: Four generated coordinate labels force ten symmetric rank-two slots paired one-to-one with one energy, three momentum and six stress source carriers. Generated coordinate shifts commute, so each of four curvature-divergence directions expands into identical carried and opposed derivative ledgers. Every componentwise source equation therefore preserves an exact local source ledger at every finite depth; removing one flow term breaks balance. No signed numerical proof scalar, imported derivative, target-selected component or free correction enters.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:79b12fc32760832dc52e87fcf8d743575ebe2f944c18e7cb24cbb41a6d8a5b22`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010-79b12fc32760832d.json`.

SFT-PHYS-STATIC-EXTERIOR-CLOCK-TERMINAL-011

Earlier display gap: exact statement.

Current exact statement: The generated horizon radius is twice the positive mass carrier. Rank-two boundary conservation forces field equal to horizon radius over radius paired with radius; one retained radius converts this to the well share horizon radius over radius; its positive complement completes the One and is the unique exterior clock-square coefficient. Flux is identical at every exact exterior radius, every outward binary successor halves the well and quarters the field, the boundary closes the clock record, and exact-square exterior pairs force the farther clock to run faster without irrational proof values.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:fca4c68a06026daea19121fd5b71d52bc98e2410be1f83eaef13ff62600d880c`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-STATIC-EXTERIOR-CLOCK-TERMINAL-011-fca4c68a06026dae.json`.

SFT-PHYS-COUPLING-ACCUMULATED-SEPARATION-TERMINAL-015

Earlier display gap: exact statement.

Current exact statement: For the admitted binary and generator-three running sectors on the common Fold support R, the exact adjacent gap at positive level n is 1/((R_n+2)(R_n+3)), with R_1=One and R_(n+1)=2R_n. The first gap is 1/12 and the second is 1/20. Every successor from the second gap onward is strictly below half its predecessor. Consequently every positive finite partial accumulation S_N satisfies 1/12 <= S_N < 11/60, and for every positive finite tolerance a generated level supplies a smaller exact remaining-tail envelope. This is constructive finite convergence, not a completed infinite sum.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:723ff785cf91a40c287d2f2b38394a63fd8bfaa7a0be7a34be2e9e973202f4bf`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-COUPLING-ACCUMULATED-SEPARATION-TERMINAL-015-723ff785cf91a40c.json`.

SFT-PHYS-DYNAMICS-SYMMETRY-ACTION-TERMINAL-016

Earlier display gap: exact statement.

Current exact statement: Exact Fold double-and-cast preserves the reduced odd denominator core of every positive rational part, giving a depth-independent native conserved charge. In every finite generated transition support, a symmetry is the complete bijection that preserves transition incidence, exact positive step carriers and held observation fibres. Its fibre identity is therefore conserved. Conversely, complete enumeration of the bijections preserving any registered conserved partition generates that partition's symmetry class. In the exact dyadic basin, every Fold act removes one binary denominator layer and the unique trajectory reaches the One, where the identity act is empty One. For every finite ranked descent path, action is the complete sum of exact positive oriented step magnitudes. Every monotone descent telescopes exactly to the endpoint separation; any detour retains an additional positive up/down carrier and cannot lower the action. Thus the local dyadic descent and global minimum-variation reading are one exact Fold structure at the declared boundary.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:3c158a5e6f5ad82f89435d0b00c6542d1230e20706f5cdadad0db202864d4075`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-DYNAMICS-SYMMETRY-ACTION-TERMINAL-016-3c158a5e6f5ad82f.json`.

SFT-PHYS-SCATTERING-PARTITION-PATH-TERMINAL-017

Earlier display gap: exact statement.

Current exact statement: For the registered one-target Fold geometry, the complete mutually exclusive outcome support is the equipotent pair scatter/pass, so each cell carries half-One and the pair reassembles the One. Cross section is the exact positive successful support relative to incident support. At positive target-number density n, encounter support per path is n times the section and the exact mean-free-path carrier is its reciprocal. At unit density, section One gives path One and section half-One gives path two; increasing section strictly shortens path. The partition is an exact support measure over held deterministic outcomes, not a random transition cause.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:b4b42d9e9e0a52a703a059c265cc94325c4f56907da3422c4cb83b0facd9efdf`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-SCATTERING-PARTITION-PATH-TERMINAL-017-b4b42d9e9e0a52a7.json`.

SFT-PHYS-THERMO-LANDAUER-DEMON-TERMINAL-018

Earlier display gap: exact statement.

Current exact statement: The Fold reset maps the two held preimages quarter-One and three-quarter-One to the common ready image half-One. It therefore closes exactly one binary distinction and has native separation half-One. Reversal is possible exactly when one of the two predecessor labels is retained. A Maxwell sorting cycle gains one gas distinction, stores it in the demon's two-label memory, and on reset must export one reverse label to the environment. The complete closed ledger therefore contains no free unrecorded gain. This is the exact model-native erasure law; dimensional thermal-energy correspondence is tested only after sealing.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:b130990c39c87ac65a1f5728a4c471bc05f8f5720d0413feebffb5b03760f88b`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-THERMO-LANDAUER-DEMON-TERMINAL-018-b130990c39c87ac6.json`.

SFT-PHYS-THERMO-LANDAUER-EMPIRICAL-019

Earlier display gap: exact statement.

Current exact statement: The independently admitted Fold reset closes one predecessor distinction and requires one environment reverse label. Before any external target is opened, this predicts that a physical two-state memory reset to one state has finite positive environmental dissipation and that a reliable slow erasure approaches a nonvanishing thermal bound. Post-seal primary theory and experiment match that structure: the conventional symmetric thermal-reservoir bound is average heat at least $k_B T \ln 2$, and measured long-cycle mean heat saturates it. The external logarithmic dimensional coefficient remains a measurement/theory inscription and is not equated numerically with the native half-One Fold separation.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:eb264a10c1fe3666cf25a5e18beb9bab271534333bfdee25fadc0ffafcc3a93e`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-THERMO-LANDAUER-EMPIRICAL-019-eb264a10c1fe3666.json`.

SFT-PHYS-MEDIATOR-RANGE-CHANNEL-TERMINAL-020

Earlier display gap: exact statement.

Current exact statement: A complete channel combination that reassembles the One has the structural empty mass record and retains its forward carrier at every generated finite step. Every proper broken channel has a positive mass shortfall. Repeated positive transfer of that share from forward to co-located rest support conserves the One and leaves a finite positive forward trace; the following act is the empty boundary rather than numerical zero. Independently, the characteristic normalized range scale is One divided by mass, so every larger positive mediator mass has shorter range. A massless inverse-square carrier remains positive at every finite radius. This forces the exact preserved/massless versus broken/massive discriminator without importing a Yukawa exponential or asserting a hard physical zero beyond the range scale.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:3ad2016a90d7adc2c1c5c5afdfdaf5510f73ade146e2ea2e0a097d9158621f54`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-MEDIATOR-RANGE-CHANNEL-TERMINAL-020-3ad2016a90d7adc2.json`.

SFT-PHYS-BARYOGENESIS-DEPENDENCY-TERMINAL-021

Earlier display gap: exact statement.

Current exact statement: Beginning from one exactly paired particle/antiparticle carrier, a positive retained baryon residue occurs in the complete minimal Fold process grammar if and only if three independently held distinctions co-occur: a transition changes the baryon tally, conjugate transition paths are distinguished by the admitted CP carrier, and reverse completion is held by a nonequilibrium process record. Complete enumeration of all eight presence/absence combinations leaves exactly one residue-bearing process. The already sealed terminal CKM and baryon-to-photon claims supply the post-derivation physical carrier and abundance comparison; they do not select this dependency law.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:738dd11360b3cb43280ce6b595023c6f6408166ab6ef96ad4f8e9a6d4592c7b4`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-BARYOGENESIS-DEPENDENCY-TERMINAL-021-738dd11360b3cb43.json`.

SFT-PHYS-LATTICE-OPERATOR-TERMINAL-022

Earlier display gap: exact statement.

Current exact statement: One conservative Fold lattice family covers one, two and three generated directions. Every site retains half-One and distributes the other half equally over its complete two-per-axis neighbour ring; this gives the one-direction half/quarter/quarter stencil and the planar four-neighbour eighth shares. A point source has positive peak magnitude two times the dimension and an equally counted opposition ring. Complete oriented taxicab enumeration forces causal-ball counts 5,13,25 in two directions and 7,25,63 in three. A cycle of N positive sites has exactly N phase mode carriers with update relation half-One centre plus quarter forward phase plus quarter reverse phase; the identity phase is the unique stationary flat mode. No irrational mode value is evaluated.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:2d54a071b1a994f26b3a05083d0d57ec33659b9d5430b17467f5dfd03d525fb8`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-LATTICE-OPERATOR-TERMINAL-022-2d54a071b1a994f2.json`.

SFT-PHYS-FINITE-QUANTUM-GRAVITY-TERMINAL-023

Earlier display gap: exact statement.

Current exact statement: The admitted quantum branch support and gravitational curvature-source law compose on one and the same finite Fold lattice. Three forced spatial directions plus one process direction give four coordinate roles and ten symmetric rank-two slots. Four exact source-conservation constraints and four generated coordinate-shift redundancies retain exactly two physical polarizations. The detached carrier has an empty mass/rest record and advances one causal cell per tick. At every positive depth the quantum support, positive distance floor, finite rational loop sum and quarter-area horizon ledger are held together, while the unique three-space theorem leaves the extra-dimension record empty.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:a627c3db6aff944a007b53df0a06975454cec52c113e672da62a440ebfd191e7`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-FINITE-QUANTUM-GRAVITY-TERMINAL-023-a627c3db6aff944a.json`.

SFT-PHYS-FOLD-UNIVERSE-TRANSPORT-TERMINAL-024

Earlier display gap: exact statement.

Current exact statement: Every reduced positive odd denominator defines an exact periodic Fold-orbit component. For coprime components m and n, the units of mn map bijectively to ordered component pairs, Fold doubling commutes with that map, and the composite period is the least common multiple of the component periods. This forces exact joint support and lockstep correlation. It does not force directed signalling or literal physical travel: Fold dynamics preserves the denominator, and one component trajectory is independent of the other once the composite initial state is fixed. Composition is a global preparation or description map, not a transition moving a pre-existing state between components. The physical transport record therefore remains empty unless a separately admitted causal operation is supplied.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:b03563b484c95269254dd2bb409ae42efbd2a71989eba3ac1b3dd8171c7e467d`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-FOLD-UNIVERSE-TRANSPORT-TERMINAL-024-b03563b484c95269.json`.

SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025

Earlier display gap: exact statement.

Current exact statement: The complete prime-sector ladder two, three, five and seven is governed by one exact table. Sector m has holding coupling (m-1)/m, positive mass shortfall 1/m, mixing share 1/(m-1), m-squared take One mediators and interior standing modes j/(m-1). Half-One is the unique self-antipodal standing mode common to every odd prime sector and is absent from the fundamental two-sector interior. On common positive support R, every sector has holding share (m+R-1)/(m+R); exact positive pair gaps preserve strict sector order and forbid finite triple coincidence while shrinking under every support successor. The Fold period dictionary is gravity one, electromagnetism two, strong three and joint recurrence six. The prior bundle claiming both slope m-1 and an abelian slope of zero is internally inconsistent and is rejected; the admitted terminal running laws replace it.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:464f20ad41d716a52836a9d5952a62991e76b488a2ea70539ea6f4c0ee8c8f0e`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025-464f20ad41d716a5.json`.

SFT-PHYS-YANG-MILLS-SINGLET-GAP-TERMINAL-026

Earlier display gap: exact statement.

Current exact statement: The generated colour-three sector holds two-thirds of the One and leaves the exact positive one-third complement. The two parts form a complete period-two Fold orbit. Colour observation admits only a nonempty closed singlet, with an antipodal pair as the least support, while the isolated colour record is empty and separation work increases by two-thirds at every positive finite successor. Consequently the normalized least physical colour-singlet excitation has the depth-invariant positive gap one-third. The locally propagating colour carrier nevertheless retains an empty mass label and One-cell causal motion: local carrier masslessness and physical colour-singlet gaplessness are not the same proposition.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:899e904600c1018041286e947de6acefe6c1afb50f1f8da202cfda587fa9568a`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-YANG-MILLS-SINGLET-GAP-TERMINAL-026-899e904600c10180.json`.

SFT-PHYS-YANG-MILLS-SINGLET-GAP-EMPIRICAL-027

Earlier display gap: exact statement.

Current exact statement: The independently admitted Fold theorem predicts only that the least physical colour-singlet excitation is positively separated from the empty excitation record; it does not predict a dimensionful glueball mass. After the prediction is sealed, the complete registered PDG 2026 pure-gauge lattice spectrum is opened. Its scalar ground-state interval and every retained tensor/pseudoscalar excitation interval lie strictly above the empty boundary and remain strictly ordered. The quenched-approximation, mixing, full-QCD and unresolved experimental-identification limitations are mandatory parts of the result.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:d85984ba94789e15d9fdb97588e63d318b014b3e84315ace56cb50c16cbeb2fe`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-YANG-MILLS-SINGLET-GAP-EMPIRICAL-027-d85984ba94789e15.json`.

SFT-PHYS-PARKER-PROTON-ENERGY-TERMINAL-028

Earlier display gap: exact statement.

Current exact statement: The admitted proton colour sector has exactly three charge labels and three-squared Take One non-return channels, hence eight carrier channels. The admitted field-energy composition pairs the electromagnetic amplitude with itself, so the dimensionless proton energy share is eight times alpha squared. At the historical leading alpha rung this is exactly 500000/1173679081; at the complete terminal alpha rung it is exactly 108147617771025486368/253861190227103943729961. No proton rest-energy value or Parker observation enters this derivation.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:41f17af1f94905b0e93d67d7dfc7a399fd9063a53e6db4d21d486e1a8c5f1673`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-PARKER-PROTON-ENERGY-TERMINAL-028-41f17af1f94905b0.json`.

SFT-PHYS-PROTON-RADIUS-TERMINAL-029

Earlier display gap: exact statement.

Current exact statement: The admitted colour-three proton has inner tripling site one-third and outer complement two-thirds. Every one of its three colour labels carries both Fold fibre labels, so complete edge transport is six times two-thirds, exactly four. Internal ordered colour support has nine pair cells and the proton carries one external unit charge, forcing ten complete charge-support cells. A spatial charge boundary receives one terminal electromagnetic traversal, not the two traversals of an energy self-composition; the inward held share is therefore alpha_terminal/10. The resulting dimensionless rms-radius coefficient is exactly 4(One-alpha_terminal/10), independent of the probe.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:77a31b069cc1cdd4bdf84f3ca08fb4849ae33fddb02945727f1809d9e4c2373d`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-PROTON-RADIUS-TERMINAL-029-77a31b069cc1cdd4.json`.

SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030

Earlier display gap: exact statement.

Current exact statement: The One-based Fold scale support is generated uniquely by binary succession R_next=2R, with exact relative spacing One/R and one local carrier act per support cell. The complete prime sectors share this one axis. In the binary electroweak sector, charged holding support (R+1)/(R+2) and neutral half-One force the squared share (R+2)^2/[4(R+1)^2+(R+2)^2], which strictly descends at every binary successor. Complete terminal promotion has support sixteen; holding the three generator directions forces active level thirteen, base 225/1009 and the single terminal alpha return over seventeen already admitted as the exact on-shell result. The source 2+R is a binary Fold power only at support two. Every common exact rational unit rescaling preserves like-dimension ratios, while the admitted proton-Planck relation fixes the physical scale ratio before any dimensional unit name or measured reference is applied post-seal.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:44d542f87d12844c02c321ac651008ae7ddd738a3b16e7d2c3d8700ec89eca55`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030-44d542f87d12844c.json`.

SFT-PHYS-COMPOSITE-CONFINING-SECTOR-TERMINAL-031

Earlier display gap: exact statement.

Current exact statement: Every generated even Fold sector s beyond the binary fibre holds the odd positive predecessor d=s-1. Its complete nonempty standing modes are the d-1 labels k/d. Fold doubling has the exact inverse (d+1)/2 and therefore permutes this support without changing denominator, forcing a finite disjoint first-return orbit partition. The modes form exactly (d-1)/2=(s-2)/2 positive antipodal pairs whose members reassemble the One, and the unique all-but-one holding coupling is d/s. Consequently sectors 8, 12, 18, 24 and 30 have respectively 6, 10, 16, 22 and 28 modes; orbit-size partitions (3,3), (10), (8,8), (11,11) and (28); pair counts 3, 5, 8, 11 and 14; and couplings 7/8, 11/12, 17/18, 23/24 and 29/30.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:eb8b53e1662db23eff80d371b5b6db094e3c7d4665b5ea0a6882ba3072e29ec8`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-COMPOSITE-CONFINING-SECTOR-TERMINAL-031-eb8b53e1662db23e.json`.

SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032

Earlier display gap: exact statement.

Current exact statement: Generator-three space forces matter transport by the third power of exact past stretch; one additional wave-recurrence stretch forces radiation by the fourth power; and the vacuum Fold invariant retains the One. With the admitted terminal present shares matter 5/16 and vacuum 11/16, the exact late squared-rate carrier is E2(r)=(11+5r^3)/16, the matter fraction is 5r^3/(11+5r^3), equality is r^3=11/5, acceleration onset is r^3=22/5, and today's accelerating separation has magnitude 17/32. Tension and acceleration are typed orientations, never negative proof scalars.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:504257d441fad45a8f173fad45015c7d723caa7636c517e71d27fddc43a2c42d`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032-504257d441fad45a.json`.

SFT-PHYS-SPACETIME-VELOCITY-COMPOSITION-TERMINAL-033

Earlier display gap: exact statement.

Current exact statement: Typed rest, exact positive sublimit parts and the complete present/absent bilinear grammar force one same-direction composition: u Fold-plus v=(u+v)/(One+uv). It is commutative, associative, closed below the limiting One, fixes the limiting One, and is exactly reconstructed by multiplying the forward/held pairs (One+u, One Take u) and (One+v, One Take v). Rest and a vanished held part are typed structures, never numerical zero; direction is held as a label, never a negative proof scalar.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:337fccb78afd2207d9976e8034d89d1db75ba9fed2f63c44a3b36ae0e90935a0`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-SPACETIME-VELOCITY-COMPOSITION-TERMINAL-033-337fccb78afd2207.json`.

SFT-PHYS-VALIDATION-VELOCITY-COMPOSITION-034

Earlier display gap: exact statement.

Current exact statement: After the exact Fold operation (u+v)/(One+uv) is sealed, the complete registered Lahaye-Labastie-Mathevet Fizeau record is opened. The measured water phase-slope 274/1000 with stated 3/1000 statistical uncertainty lies between the reported raw and profile/dispersion-corrected relativistic predictions 248/1000 and 299/1000, while ordinary addition predicts 563/1000. The best relativistic-row distance is 25/1000, against 289/1000 for ordinary addition. The air row also rejects ordinary addition and remains compatible with relativistic composition. Every calibration, profile and path-length limitation is retained.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:54dce92f903c6f102e4d39f3bdb826a89c05e69e80b6d6df66728711dbc03cf4`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VELOCITY-COMPOSITION-034-54dce92f903c6f10.json`.

SFT-PHYS-VACUUM-DENSITY-SCALE-TERMINAL-035

Earlier display gap: exact statement.

Current exact statement: Generator-three space has volume twenty-seven and unique least binary covering depth five. Both held Fold labels at every covering level force a complete boundary-record depth ten. Its least nonempty local vacuum amplitude is One/2^10; energy is the complete two-leg self-composition, uniquely forcing the local energy floor One/2^20. This local dimensionless cell floor is not the cosmological density fraction. The terminal cosmic vacuum share is 11/16, and three-space rate geometry forces the separately typed normalized cosmological magnitude Lambda(c/H)^2=3(11/16)=33/16. A dimensional Lambda follows only after a held external rate and speed reference: Lambda=(33/16)H^2/c^2.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:c7b4777b12fc70628b0fd9a2f7d957274b61ede7f3624a665781364c8c9f7723`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VACUUM-DENSITY-SCALE-TERMINAL-035-c7b4777b12fc7062.json`.

SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036

Earlier display gap: exact statement.

Current exact statement: After the formal vacuum law is admitted, the complete Planck page-225 68-percent rows and exact CODATA speed are opened. The exact terminal vacuum share 11/16 lies inside Planck's 0.6889 +/- 0.0056 interval. The separately forced normalized magnitude 33/16 lies inside the exact transported interval 3(0.6889 +/- 0.0056). The Hubble interval and exact limiting speed transport the sealed coefficient to a positive exact dimensional Lambda interval in inverse square megaparsecs. The local One/2^20 floor lies outside the global vacuum-fraction interval, confirming that direct unscaled identification is a type error.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:6379f0d35a606d217e2351a35c42ae258aea71c52d8d20d2c4abeef2fdd8c202`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036-6379f0d35a606d21.json`.

SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037

Earlier display gap: exact statement.

Current exact statement: Exact Fold temperature is transported inversely to exact scale growth, so temperature times scale is held. Distinct binding thresholds therefore impose one descending thermal order without a named chronology selecting it. The complete binary orbit on depth seven is (One, two, four)/seven. Its least live part fixes the freeze-out neutron share One/seven, not the old shorthand neutron/proton ratio. The proton share is six/seven and the freeze-out ratio is One/six. Transport into the least complete binary cover gives capture shares One/eight and seven/eight, hence neutron/proton One/seven; complete neutron pairing forces helium-family share One/four and hydrogen-family share three/four. Recombination is classified at the half-One bound/free midpoint, while every physical visibility record remains finite and positive rather than an instantaneous collapse. Pre-decoupling acoustics have exact positive-whole internal modes with alternating compression/rarefaction parity; observed angular peaks retain projection and driving records and are not asserted to be exact integer multiples.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:25dda78644ebc04ad43cd3e416950ad76dae6b124ed023f41364f0661eae9c00`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037-25dda78644ebc04a.json`.

SFT-PHYS-VALIDATION-THERMAL-HISTORY-RECOMBINATION-038

Earlier display gap: exact statement.

Current exact statement: After formal Claim 037 is admitted, direct high-redshift CMB thermometry, primordial helium and deuterium, Planck recombination and acoustic extrema, and the PDG freeze-out synthesis are opened. The measured temperature exponent contains exact One. The independently derived One/six to One/seven neutron/proton transport matches the external analytic sequence. Current direct helium measurement lies below and excludes exact One/four at one standard interval, so the old claim that the analytic family partition is the exact measured isotope abundance is rejected rather than fitted. Deuterium remains a positive minor channel. Planck reports finite recombination support and eighteen acoustic extrema; its angular multipoles reject exact integer multiples of the first peak, validating the formal separation between internal whole modes and observation.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:7440c78415e834656217aca301aaa8b6f987d6f795544eb5c03b0679970321f4`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-THERMAL-HISTORY-RECOMBINATION-038-7440c78415e83465.json`.

SFT-PHYS-INFLATION-GROWTH-TERMINAL-039

Earlier display gap: exact statement.

Current exact statement: Generator-three stable space has volume twenty-seven and is first completely covered by binary depth five, support thirty-two. Inflation is therefore an exact five-doubling cover process, not a conventional logarithmic duration imported from a potential. One held boundary distinction leaves thirty-one scalar support records and one least tensor-pair record, forcing scalar share 31/32 and tensor share 1/32. The exact quarter-One perturbation grows by two Fold steps through half-One to the One. Under later exact scale growth g, matter retention is One/g^3 and radiation retention One/g^4, so matter gains the exact relative factor g and supplies the earlier gravitational scaffold without a stochastic seed or fitted transfer function.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:00aa52b47f8ebfe57fb5a90530a54614ba175d87f6708c69e454fcf9fd888aa7`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-INFLATION-GROWTH-TERMINAL-039-00aa52b47f8ebfe5.json`.

SFT-PHYS-VALIDATION-INFLATION-GROWTH-040

Earlier display gap: exact statement.

Current exact statement: After formal Claim 039 is admitted, Planck's complete scalar spectral-index estimate and the improved BICEP/Keck-Planck tensor-to-scalar upper bound are opened. Exact scalar support 31/32 lies inside the registered Planck one-standard interval. Exact least tensor support 1/32 lies strictly below the registered 95 percent upper bound 4/125. Neither row selects or modifies the exact formal partition.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:8e06d75a25206d5cce129a9afd2583a54cb1e070a0daf6eeeff92ece99a8d01f`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-INFLATION-GROWTH-040-8e06d75a25206d5c.json`.

SFT-PHYS-COLLECTIVE-RADIATION-RESPONSE-TERMINAL-041

Earlier display gap: exact statement.

Current exact statement: Finite bosonic cavity support is the complete exact occupation-word census at held total quanta; modes costing more than the support carry only an empty record, so the ultraviolet tail closes without a continuum exponential. Common exact scaling of energy and mode costs preserves every occupation word, forcing peak-frequency covariance with temperature. Three spatial mode-count powers plus one energy power force radiated power ratio g^4. Fixed boundaries force acoustic frequencies n times the fundamental. Stimulated emission duplicates the held photon mode; inversion is strictly above half-One, lasing threshold is exact gain/loss equality, and linewidth times coherence ticks is One. Charge stiffness forces plasma-frequency-squared=nq^2/(m epsilon) and Debye-length-squared=epsilon T/(nq^2). Magnetic tension over inertia forces Alfvén-speed-squared=B^2/(mu rho). Squared carriers retain exact Fold arithmetic without importing irrational roots.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:50974ab152438d8b84fe8220281e0d98737df0880f9a6384d50bac6060987b07`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-COLLECTIVE-RADIATION-RESPONSE-TERMINAL-041-50974ab152438d8b.json`.

SFT-PHYS-VALIDATION-COLLECTIVE-RADIATION-RESPONSE-042

Earlier display gap: exact statement.

Current exact statement: After Claim 041 is sealed, NIST/NBS blackbody, laser and acoustic records and NASA plasma and Alfvén records are opened. The externally measured blackbody exponent is four, Coblentz's coefficient was within one percent and the spectrum shape agreed. Measured cavity resonances, strictly positive finite laser linewidths with feedback narrowing, direct density tracking by plasma frequency, and the registered 2007-2014 Alfvén-wave list all retain the formal relation directions without selecting their laws.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:1d7dd8e6373027c10bd83aa9ff128ebabe3fdc60569b24ad597f9c22abad2455`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-COLLECTIVE-RADIATION-RESPONSE-042-1d7dd8e6373027c1.json`.

SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043

Earlier display gap: exact statement.

Current exact statement: Temperature is the exact positive mean of a finite population's held throw records, so total throw equals population count times temperature. For every complete paired binary population, exhaustive multiplicity counting has one maximum at equal labels, forcing half-One equilibrium and detailed balance. For L generated levels the Fold recurrence uniquely supplies integer populations (2^(L-1),...,2,1), normalized by 2^L-1; complete enumeration at the same population and throw gives this dyadic ladder as the unique multinomial maximum. Equilibrium fluctuation three-quarter-One and response quarter-One are complementary and have the same quarter-One departure from half-One. A finite deterministic four-record cycle has mean half-One, making noise and response two readings of one retained orbit.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:cf536f7516eb1c85bb075500d9c1a0ebd64bdddb544878f26a27424d08de25bf`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043-cf536f7516eb1c85.json`.

SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002

Earlier display gap: exact statement.

Current exact statement: After the exact terminal charged-lepton invariant is sealed, exact rational root brackets predict both adjacent-root squared mass ratios and compare them with both complete NIST CODATA 2022 charged-lepton ratio intervals.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:1df007d68f197e5a8a8a58dad28190bbaf10229ab2f8136365a25622eba5d5d8`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002-1df007d68f197e5a.json`.

SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-002

Earlier display gap: exact statement.

Current exact statement: The exact Fold value two-thirds is sealed before both complete CODATA mass-ratio intervals are opened and propagated, with exact rational square-root enclosures, into a conservative Koide interval.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:4e7a64b6751e5622c827612840751a26b40b61ee657bb011eded7ff7914aa776`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-002-4e7a64b6751e5622.json`.

SFT-PHYS-VALIDATION-THERMAL-EQUILIBRIUM-044

Earlier display gap: exact statement.

Current exact statement: After the complete temperature, finite equilibrium and fluctuation-response law is sealed, the exact SI Boltzmann carrier and two physically distinct thermometry determinations are opened. Both complete measured intervals contain the exact carrier. Acoustic thermometry independently ties temperature to mean kinetic energy, while Johnson-noise thermometry measures thermal noise response to temperature and resistance. The external records do not report the internal dyadic population ladder or the three-quarter/quarter Fold coordinates as universal measured values, and that limitation is retained.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:60366227e00c50a3529d98538a3ca42d9620f4da30a174b9cff96f1122097dbd`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-THERMAL-EQUILIBRIUM-044-60366227e00c50a3.json`.

SFT-PHYS-SPIN-STATISTICS-CONDENSATION-TERMINAL-045

Earlier display gap: exact statement.

Current exact statement: The preserving and alternating exchange classes force the two finite quantum statistics. For N occupants over L levels, complete occupation enumeration gives C(N+L-1,N) preserving words with no finite per-level ceiling and C(L,N) alternating words with at most one occupant per level; when N>L the latter support is the empty One form. Exact dyadic Fold weighting normalizes every admitted finite occupation word without a continuum distribution. The complete two-label spin composition has three preserving readings and one alternating reading. An alternating constituent changes held fibre after one complete turn and first returns after two, whereas an alternating pair is preserving and returns after one. For every finite preserving population, repeated cold Fold weighting uniquely raises the shared ground-orbit weight through the already forced lock share (m-1)/m at a first finite depth; its exact canonical mean throw is the critical-temperature carrier. At minimum total throw, exhaustive occupation enumeration leaves exactly one word: the whole population in the shared ground orbit.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:f71da5b86f99d6569a1f33dc6fc37024cc5d458b12e625c5dbc4faf3c33ccda7`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-SPIN-STATISTICS-CONDENSATION-TERMINAL-045-f71da5b86f99d656.json`.

SFT-PHYS-VALIDATION-SPIN-STATISTICS-CONDENSATION-046

Earlier display gap: exact statement.

Current exact statement: After the complete finite spin-statistics and condensation law was officially sealed, three primary physical records were opened. A finite rubidium-87 Bose gas exhibits measured ground-state occupation that rises as temperature is lowered and a sharp transition at (94 +/- 5)/100 of its independently defined trap scale. A two-spin-state potassium-40 Fermi gas directly exhibits Pauli blocking, including a reported factor-two collision-cross-section reduction. Neutron interferometry measures a fermion return of 704 +/- 38 degrees, whose complete interval contains the sealed two-turn 720-degree correspondence. The trap-specific BEC temperature and collision-response magnitude remain external scale records; they are not used to manufacture the formal occupation or lock law.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:6c37132ae33b8f23b06e69b7034c5acdd39144d51b2661d7e65f441b400604eb`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-SPIN-STATISTICS-CONDENSATION-046-6c37132ae33b8f23.json`.

SFT-PHYS-CRITICALITY-UNIVERSALITY-TURBULENCE-TERMINAL-047

Earlier display gap: exact statement.

Current exact statement: The admitted binary self-antipodal lock fixes the continuous local-order threshold at half-One. Complete positive perfect-power scaling then forces beta=1/2 for the order carrier, nu=1/2 for the inverse-correlation carrier, gamma=One for reciprocal response, and delta=three for the critical-field relation; alpha and eta are empty One exponent records. Widom, Rushbrooke and Fisher close exactly without a numerical-zero proof scalar. Systems share this exponent law exactly when they share the same generated threshold, order-power and field-power key. Separately, the forced three-space branch count fixes a conserved cascade class: cube refinement carries a squared second-order structure record, forcing 2/3, while shell succession adds the One and forces a falling spectrum magnitude 5/3. These are two distinct generated universality classes, not one measurement-selected class.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:0601d19640943c4b99eb8cccf061e3115c520773eac5e762bf7c5b7440339b25`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-CRITICALITY-UNIVERSALITY-TURBULENCE-TERMINAL-047-0601d19640943c4b.json`.

SFT-PHYS-VALIDATION-CRITICALITY-UNIVERSALITY-TURBULENCE-048

Earlier display gap: exact statement.

Current exact statement: After Claim 047 and its complete 256-form derivation were officially sealed, four independent records were bound. Neutron scattering in erbium measures beta, gamma and nu intervals containing the sealed 1/2, One and 1/2 vector. Five manganite compositions report beta, gamma and delta: La00, La04, La06 and La08 contain the complete sealed 1/2, One and three vector, while La02 does not and is retained as an unfavorable class-membership control. An independent turbulence computation measures zeta_2=(679 +/- 13)/1000, whose exact interval contains 2/3, and a physical turbulent-channel experiment observes five-thirds compensated spectral plateaux by both Fourier and Hilbert routes. Finite-Reynolds and structure-function-range limitations remain explicit; no measured value selected or altered the formal law.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:075d304c85f48d6886ba7cb1fd44d5db25815cdf3a5d016502da1435654f1bdc`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CRITICALITY-UNIVERSALITY-TURBULENCE-048-075d304c85f48d68.json`.

SFT-PHYS-QUANTUM-SUPPORT-UNCERTAINTY-TERMINAL-049

Earlier display gap: exact statement.

Current exact statement: A complete binary preparation of N=2^k branches uniquely fixes its observation depth k and indivisible branch unit 1/N; depth three and eighth-One outcomes occur exactly for an eight-branch preparation, not universally. The Fold-held/returned parity table generates the complete Walsh observation without an imported signed, imaginary or irrational scalar. Exact phase-count orthogonality forces Parseval counting and therefore s_t*s_f>=N for every nonempty dyadic support. Weighting support counts by reciprocal exact grid spacings cancels the spacing and forces unit-free product (s_t*s_f)/N>=One; the squared normalized support-spread product is at least 1/N^2 and gives 1/16 at depth two. These are support-spread laws, not an unproved identification with statistical moment variance. Coprime 2-by-3 and 3-by-5 preparations visit every product cell once, but the complete product is factorable and is not entanglement by itself. Exhaustive joint-subset enumeration forces entanglement exactly at nonfactorability, preserves complete projections and makes remote relabelling locally invariant. Every factorized deterministic two-setting Bell record wins at most three of four setting classes; complete setting-inclusive joint support exceeds that factorization boundary while preserving exact local marginals and no signalling.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:1560f2e0de3870abac2bdc6575aa9811c4dc013d5a6d705e729a832dc451b79f`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-QUANTUM-SUPPORT-UNCERTAINTY-TERMINAL-049-1560f2e0de3870ab.json`.

SFT-PHYS-VALIDATION-QUANTUM-SUPPORT-UNCERTAINTY-050

Earlier display gap: exact statement.

Current exact statement: After Claim 049 and its complete local-response census were officially sealed, the primary Storz et al. record was bound. More than one million trials measured S=(20747 +/- 33)/10000. Its complete interval [20714,20780]/10000 lies above the local-factorization bound two, with a reported P value below 10^-108. The timing record also preserves space-like separation and the memory-robust analysis. The paper's measurement-independence assumption is retained rather than hidden: this result tests local factorization under the declared experimental conditions; it does not exclude every deterministic model and it does not import random setting generation as ontic nondeterminism.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:92401779cd8580dbaf3962207ae25e0b7fc75787f91e27681d1ac290de403085`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-QUANTUM-SUPPORT-UNCERTAINTY-050-92401779cd8580db.json`.

SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051

Earlier display gap: exact statement.

Current exact statement: Every positive finite m-Fold with m at least two has exactly m positive preimages for each positive image and exactly m-2 interior fixed modes. The proof is generated directly from the complete offset and fixed-index ranges and extends by induction. The generator-three fibre supplies three ordered generation labels represented equivalently by the half-One fibre (1/6,1/2,5/6), the One fibre (1/3,2/3,One), the interior five-Fold modes (1/4,1/2,3/4) and the quarter transition ladder; these are order-isomorphic coordinates, not measured masses. The generator-three state volume in three forced spatial directions is 3^3=27, whose least binary cover is depth five and least colour cover is depth three. Every finite recurrent particle-trace census injects into the exact internal stationary-mode ladder, so mode depth adds no spatial dimension. Generation order transports to the separately admitted charged-lepton, up-quark, down-quark and positive-neutrino polynomial carriers; it never licenses the superseded claim that a site fraction itself is a mass. The colour/binary duals 1/95, 1/383 and 1/485, the depth-independent heavy/light equals subtraction-reach identity, and the 1:1:4 quarter-mode squared multiplicity close as exact structural records; physical masses, mixing entries and lifetimes remain separate post-seal comparisons.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:18e4aac0792a01c613d31f4d980951b0e2da69107155c4e50967ed80c9eaa700`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051-18e4aac0792a01c6.json`.

SFT-PHYS-VALIDATION-ELECTROWEAK-MEASURED-VALUE-053

Earlier display gap: exact statement.

Current exact statement: The already sealed zero-parameter terminal on-shell share 1930922298157999/8642477221479757 is tested against the direct PDG on-shell measurement 0.22342 +/- 0.00009 and an independent compatible-input W/Z reconstruction. Both measured-value comparisons pass without fitting, rescaling or row deletion. The separately published all-input W aggregate contains a source-identified incompatible input and is retained unchanged as a measurement-method record; it is not admitted as an SFT mismatch or as a second physical target.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:e93f04515944a1951b78463db3083350fbc77e6d8506edbcdc13178cedfbc852`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-ELECTROWEAK-MEASURED-VALUE-053-e93f04515944a195.json`.

SFT-PHYS-VALIDATION-COMMON-SCALE-MEASURED-VALUE-054

Earlier display gap: exact statement.

Current exact statement: The admitted common-axis law independently fixes the terminal weak share and the support-eight share 25/106. The terminal value lies in the direct on-shell interval and 25/106 lies in the complete cesium atomic-parity-violation interval. The registered below-W direction is preserved. The NuTeV DIS extraction is retained unchanged with its source-reported interpretation concerns, but its displacement is neither an SFT result nor an acceptance condition.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:0514d2a2fe7ad505dfd99c494f0c6ac6214169791c09596d4657e32ee1c458d8`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-COMMON-SCALE-MEASURED-VALUE-054-0514d2a2fe7ad505.json`.

SFT-PHYS-VALIDATION-COSMIC-TRANSPORT-MEASURED-VALUE-055

Earlier display gap: exact statement.

Current exact statement: The zero-parameter law E2(r)=(11+5r^3)/16 is tested against all 32 direct chronometer rows by one complete exact normalized-residual ledger. Rational root enclosures refine until its decision is exact, replacing the selected two-uncertainty row rule. The exact 11/5 equality, 22/5 onset, 17/32 acceleration magnitude and tension-One state meet like-typed external intervals. Alternate reconstruction and model-fit records remain unchanged but are not rewarded as SFT mismatches.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:e3e00a1c023f76cd0817c7eaec86c0243ab10d338e552f6bae1f90a94a959380`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-COSMIC-TRANSPORT-MEASURED-VALUE-055-e3e00a1c023f76cd.json`.

SFT-PHYS-VALIDATION-CRITICALITY-MEASURED-VALUE-056

Earlier display gap: exact statement.

Current exact statement: Each of the five manganite materials is assigned its empirical class key from the complete observed transition order, interaction-range classification, Widom relation and beta/gamma/delta vector. All five share the registered mean-field key, and all 15 exponent values enter one exact unit-normalized residual ledger whose mean is below the One. La02 remains fully visible but its individual displacement is not rewarded as a result. Independent erbium and both turbulence measurements contain the exact Fold exponents.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:7e437d81bee9d4c88b304a013c5c980fa9d0390f95acaf30c2dfb45b30a1602a`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CRITICALITY-MEASURED-VALUE-056-7e437d81bee9d4c8.json`.

SFT-PHYS-THERMAL-HELIUM-ISOTOPE-TERMINAL-057

Earlier display gap: exact statement.

Current exact statement: The admitted One/four helium-family share is an analytic baryon partition, not yet the physical isotope mass share. Helium-four has four retained constituent positions and fifteen nonempty constituent subwords. Their complete incidence product has sixty cells. Binding the four positions into one physical isotope introduces exactly one collective composite-identity record: no hold leaves four unbound labels, while more than one hold adds an ungenerated independent composite identity. The remaining fifty-nine cells give the unique isotope conversion 59/60. Therefore the physical primordial helium-isotope share is (One/four)(59/60)=59/240 and the complementary hydrogen-family share is 181/240.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:a3b3a44a0d3add7032680427c3b2b504147c0eb824d05fa5d044cef454c5ebc4`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-THERMAL-HELIUM-ISOTOPE-TERMINAL-057-a3b3a44a0d3add70.json`.

SFT-PHYS-VALIDATION-THERMAL-HISTORY-MEASURED-VALUE-058

Earlier display gap: exact statement.

Current exact statement: The analytic One/four family partition and the independently enumerated physical isotope share 59/240 remain separately typed. The forced physical share lies inside the complete direct helium interval. Exact temperature transport exponent One lies inside the high-redshift thermometry interval; the typed one-sixth to one-seventh freeze-out sequence, positive deuterium channel, finite recombination record, eighteen extrema and all seven TT peak rows are retained. Angular projection differences are method records, not rewarded mismatches.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:7339aa07e1a9f52e8ef1fede03cbe17d38834362f1c9d63bb00b05deb6d49021`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-THERMAL-HISTORY-MEASURED-VALUE-058-7339aa07e1a9f52e.json`.

SFT-PHYS-HADRON-REGGE-DIMENSIONAL-TERMINAL-059

Earlier display gap: exact statement.

Current exact statement: The admitted two-hand relativistic closure forces the exact positive three-motion share 3/5. A confined rotating tube has the two forced Fold hands, so one spin successor retains one motion share on each hand and adds 2(3/5)=6/5 squared-support units. The first positive spin rank carries the unsucceeded 3/5 motion share. Therefore every positive rank is forced depth-independently as Q(J)=3/5+(J-1)6/5=(6J-3)/5. No slope, intercept, resonance mass, width or residual is fitted or available to the formal execution.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:f82510d61a2e1472c8d090f2fd1f63b53491bab9f875ce4ae27b0e0b6b5fc54e`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-HADRON-REGGE-DIMENSIONAL-TERMINAL-059-f82510d61a2e1472.json`.

SFT-PHYS-VALIDATION-HADRON-REGGE-MEASURED-VALUE-060

Earlier display gap: exact statement.

Current exact statement: The independently sealed zero-parameter law Q(J)=(6J-3)/5 is tested against the complete five-row physical resonance vector. Each row retains its reported mass, mass uncertainty, width, width uncertainty and listing status. The exact squared carrier must lie inside the most restrictive measured resonance support formed from the reported width minus its lower uncertainty and the inward mass-uncertainty endpoints. All five carriers pass. No central mass, slope, intercept, residual, uncertainty or width selects or modifies the law.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:15c9daaa90d88c5193e5e3af4a5ca64a75c0418ee85797a2b90607c37a21cb06`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-HADRON-REGGE-MEASURED-VALUE-060-15c9daaa90d88c51.json`.

SFT-PHYS-VALIDATION-PARTICLE-MODE-GENERATION-052

Earlier display gap: exact statement.

Current exact statement: The target-free terminal particle-mode law is compared with the complete registered lepton, quark, CKM and neutrino records. The PDG LEP-SLC fit interval [2989,3003]/1000 contains the forced generation count three. The independent direct determination 292/100 with stated uncertainty 5/100 is retained unchanged: its central-value displacement from three is exactly 8/5 stated uncertainties, with no rescaling, fitting, or use in selecting the law. Both terminal charged-lepton ratios pass; the registered s/d and b/s quark ratios pass while the absence of an exact scheme-matched t/c comparator remains explicit; all terminal CKM rows and the positive-neutrino PMNS/CP support pass at their registered boundaries. Charged-lepton masses increase electron-to-muon-to-tau while measured lifetimes decrease in that order.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:0a2ca2749eed1cdaf6b92ab645ace007b3190ac5e5bb952e3a0cbdedff1c0804`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-PARTICLE-MODE-GENERATION-052-0a2ca2749eed1cda.json`.

SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061

Earlier display gap: exact statement.

Current exact statement: The closed prime-sector ladder transports the already admitted coloured mass construction without a new scale. For each penta and hepta sector and each generated down/up depth, one exact cubic has root sum One, the forced pair carrier and product p/[p(2p^d-1)-1]. Complete 1/1024 enumeration finds exactly three disjoint positive roots per cubic and sixty exact halvings retain rational enclosures, closing twelve Smithion mass-ratio predictions. The complete p-member fibre is neutral, the least root is unique in each spectrum, and the closed sector/generation grammar leaves no lower same-charge decay image, forcing the lightest neutral singlet as the stable relic class. The same generated lepton preimages force mass-weighted flavour carriers 1/32, 5/96 and 5/24, hence 3:5:20 and a tau-channel ratio 4:1. Generator volume 27 and its least binary cover force dark/baryon 27/5 and matter/baryon 32/5. Measurements and search outcomes are inaccessible to this formal execution.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:71a131119ffd152690ec772019177fb06599474cd5e89b60a62c1b9b69ec5762`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061-71a131119ffd1526.json`.

SFT-PHYS-VALIDATION-DARK-SMITHION-LFV-062

Earlier display gap: exact statement.

Current exact statement: After the zero-parameter Claim 061 seals, the exact dark/baryon ratio 27/5 is compared with the complete Planck density-ratio interval and lies inside it. Transport of the comparison-side baryon central value gives Omega_c h-squared=0.12096, inside Planck's complete reported [0.119,0.121] interval. SPARC's complete 175-galaxy record retains observed-to-baryonic velocity discrepancies; under the separately derived inverse-square law this rejects a finite baryon-only asymptote and requires an additional gravitating distribution. MEG II and BaBar report no signal and only upper limits for the three registered radiative LFV channels. Therefore those rows do not measure the sealed 3:5:20 relative weights and are retained without manufacturing a confirmation or contradiction. The twelve Smithion mass ratios remain exact standing predictions because no corresponding measured particles exist in the registered evidence.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:51e623e6512223c23a0f0cae604ca556ed30fd15173236ffd1c889dd6424e27d`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-DARK-SMITHION-LFV-062-51e623e6512223c2.json`.

SFT-PHYS-STRONG-CP-BARYON-STABILITY-TERMINAL-063

Earlier display gap: exact statement.

Current exact statement: The complete colour-three fibre is independent of the binary handed fibre. With no generated hand selector, complete colour-times-hand support retains both hands; enumeration of all three nonempty hand subsets leaves that vectorial form uniquely complete. Opposition composed with paired parity therefore returns the aligned One, whereas either one-hand restriction returns the already admitted self-antipodal half-One weak phase. Strong CP violation, its electric-dipole carrier and any compensating axion-like rule are consequently absent empty-One records, not tuned numerical values. Independently, all 83 generated mediator signatures preserve their prime fibre. Complete enumeration finds 202 cross-fibre label pairs, including six colour-to-binary pairs, but no generated carrier for any of them. Every one of the 27 colour-three proton images retains three one-third baryon parts, exactly the One; finite composition preserves the same fibre.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:6bd69d2eb78150c11fefc82f512bdfd775eaa2f3a2667127407d7a34bd695657`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-STRONG-CP-BARYON-STABILITY-TERMINAL-063-6bd69d2eb78150c1.json`.

SFT-PHYS-VALIDATION-STRONG-CP-BARYON-STABILITY-064

Earlier display gap: exact statement.

Current exact statement: Only after Claim 063 sealed, the PSI neutron-EDM measurement and six direct Super-Kamiokande proton-decay modes were released to the comparator. PSI reports a null central displacement with positive statistical and systematic uncertainties and a 90-percent upper limit of 1.8 times ten-to-the-minus-26 elementary-charge centimetres. This finite limit is consistent with the structurally absent strong electric-dipole carrier but is not relabelled as an exact proof scalar. Super-Kamiokande reports no decay indication across e/mu plus pi-zero, eta and two-pi-zero modes, with all six finite 90-percent partial-lifetime lower limits retained. Candidate events explicitly classified as atmospheric-neutrino-background compatible remain candidates, not fabricated decays. The complete registered vector therefore leaves the sealed aligned-One and proton-stability laws unviolated at current experimental sensitivity.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:48e9dbebdb29b3902876a09cde2154eea3700904898930c1f12ab0a5e5ed9a09`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-STRONG-CP-BARYON-STABILITY-064-48e9dbebdb29b390.json`.

SFT-PHYS-HIGGS-SYMMETRY-TERMINAL-065

Earlier display gap: exact statement.

Current exact statement: The complete minimal vacuum census contains structural empty-One, the half-One and the One. Empty-One is absence rather than a field value; the One is unison rather than a displaced proper part. Half-One alone is positive, proper, self-antipodal and folds to the One, forcing the displaced ground without an inserted mass or potential. Its first binary successors retain the V2 leading rungs 1/2, 1/4 and 1/8. Terminal completion then acts on the scalar's complete two-hand by generator-three directional product: six cells. Holding the unique invariant return leaves five active cells, independently cross-locked to the least binary cover depth of the generation volume 27. One terminal alpha return therefore contributes exactly (6/5) alpha to the half-One mass ratio. The complete two-fibre sharing of its squared excitation forces the self-coupling to one half of that exact squared ratio. No mass, VEV, coupling or measurement selects either result.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:3f84e4ede48a489bc78a8cc47eac935bbff3566d1e561c26ef0059d7bbcbfb1e`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-HIGGS-SYMMETRY-TERMINAL-065-3f84e4ede48a489b.json`.

SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066

Earlier display gap: exact statement.

Current exact statement: Claim 065 sealed the exact ratio m_H/v = 2563352914777/5038463954690 and native self-coupling 6570778165695741824959729/50772238045420788745992200 before any Higgs target was released. The post-seal comparator uses the PDG electroweak scale 246.22 GeV only as a dimensional reference, giving the exact mass prediction 125.266104978... GeV. That prediction lies inside the complete PDG 2025 listed-average interval 125.09--125.31 GeV. The individual ATLAS and CMS offsets are retained rather than erased: they are approximately 1.42 and 1.88 of their reported combined uncertainties. The direct ATLAS-CMS pair-production constraint remains broad and contains the normalized unity self-coupling correspondence; it is not misrepresented as a precision measurement of native lambda.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:f69729204538a9c0949fa5c4e3dfce51beece2ebe2d4e43163910d6192f9b523`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066-f69729204538a9c0.json`.

SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067

Earlier display gap: exact statement.

Current exact statement: Half-One is the unique complete two-share balance, so normalized stellar pressure and gravity meet at one-half each. In three generated spatial directions, complete pressure response retains the three directions plus both Fold fibres, while gravity retains the three directions plus the One source: exact exponents 5/3 and 4/3. Cubing removes every root; compression gives q^5>q^4 and expansion gives 1/q^4>1/q^5, forcing radial restoration. Three-space volume and its one radiative recurrence force the two terminal luminosity classes M^3 and M^4; one fuel carrier then forces lifetime divisors M^2 and M^3. Under the admitted inverse-square law, circular balance is v^2=M(r)/r: a finite visible-mass asymptote must fall, while a flat curve forces enclosed support to grow exactly with radius. Three-space plus the orbital recurrence forces the fourth-power baryonic Tully-Fisher carrier. Finally, every finite rational spin-orbit mismatch loses one generated surplus cell per dissipative step and terminates at 1:1 when no separate eccentric or external forcing channel remains.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:24674dfa019c208012414ab3c587656ee404512f20d00ab44e26ee137be22ae2`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067-24674dfa019c2080.json`.

SFT-PHYS-VALIDATION-STELLAR-GALACTIC-TIDAL-068

Earlier display gap: exact statement.

Current exact statement: Claim 067 sealed hydrostatic restoration, the complete three/four terminal stellar luminosity classes, the flat-curve additional-support discriminator, the fourth-power baryonic rotation carrier and the isolated one-to-one tidal terminal before the seven-source target vector opened. Helioseismology records a stable solar structure at parts-per-ten-thousand sound-speed precision. All six measured piecewise stellar slopes remain present: only the high and very-high mass endpoints contain the sealed powers four and three, so no universal single-power claim is manufactured. The 153-galaxy baryonic Tully-Fisher central interval excludes four while its full reported systematic interval reaches four; both facts remain in the receipt. SPARC and the Bullet Cluster independently retain the extended-support discriminator. The Moon records 1:1 synchronous rotation, while Mercury's measured 3:2 resonance confirms the formal eccentric-forcing boundary rather than being erased as an exception.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:2410fe8ac97e22ffea7303b50155129dfe6c914d84e549c3c9bb6d24b0672ad6`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-STELLAR-GALACTIC-TIDAL-068-2410fe8ac97e22ff.json`.

SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069

Earlier display gap: exact statement.

Current exact statement: Every charged fusion-stage successor adds a complete positive source-to-target boundary carrier, so its access requirement is strictly greater than its predecessor at every finite depth. The corresponding supports are strictly nested. While a junction moves toward the unique all-mass binding maximum it retains positive release; at the globally closed mass-62, charge-28 terminal, ordinary fusion beyond the maximum has the empty One release form. With inward gravity retained, loss of this outward carrier forces collapse unless a distinct compressed nuclear support closes first; otherwise the horizon closes the path. A separate complete-fuel recurrence has no interior fixed point and terminates only when finite fuel is exhausted, forcing the thermonuclear-unbinding channel. Beyond the binding maximum, a neutral capture carries no inter-boundary charge path; repeated capture followed by radioactive label rebalance is therefore the unique generated heavy-element path that does not falsely demand exothermic charged fusion past the maximum.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:a41b28a935f54de5daabdbe77b6c1aba9286f174526b8feaa60829ec848e08af`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069-a41b28a935f54de5.json`.

SFT-PHYS-VALIDATION-STELLAR-NUCLEAR-COLLAPSE-070

Earlier display gap: exact statement.

Current exact statement: Claim 069 sealed the stage-successor, global binding terminal, support-loss collapse, thermonuclear and neutral-capture laws before the five-source vector opened. NASA records all six major burning stages at strictly increasing temperatures from 3/100 to 33/10 billion kelvin and retains every duration, including the nonmonotonic neon/oxygen duration pair. Borexino directly detects stellar CNO fusion neutrinos at 36/5 with +3 and -17/10 counts per day per 100 tonnes and better than five-sigma significance. The complete SN1987A three-detector neutrino record supports the collapse channel while its delayed-mechanism preference remains labelled model-assisted. SN2014J directly supplies both cobalt decay lines and their uncertainties; the measured ratio interval contains the registered branch ratio. GW170817 spectroscopy identifies strontium, directly locating a neutron-capture element in neutron-rich merger ejecta while retaining the source's spectral-modelling dependence.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:d439aaf47e94dcc7baba1031cf0ac05c446d03fbd089ed86654618d815c2032c`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-STELLAR-NUCLEAR-COLLAPSE-070-d439aaf47e94dcc7.json`.

SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071

Earlier display gap: exact statement.

Current exact statement: Complete three-space exclusion packing gives q^3 singly occupied cells, momentum depth q and relativistic support numerator q^4, while paired gravitational source support is q^6; gravity therefore overtakes the exclusion scaling after the exact base. The loaded upper preimage is three-quarter-One and Folds to the half-One balance. The binary fibre supplies exactly two pre-horizon exclusion families and no third. The already admitted horizon Fold doubles mass to radius, rank-two support makes area proportional to radius paired with itself, and two binary halvings force entropy to one quarter of that boundary support. The live vacuum recurrence and admitted quarter-One horizon witness fix temperature times mass to one-sixteenth, so temperature varies exactly inversely with mass. Every finite emission successor lowers positive mass, raises positive temperature and lowers area while the finite floor prevents a numerical-zero or infinite proof form.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:69152da01b20c6b1a7ab6ecfab4e44cf836d588cba4ea6f3b02d5827cb5a0b3f`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071-69152da01b20c6b1.json`.

SFT-PHYS-VALIDATION-COMPACT-HORIZON-THERMODYNAMICS-072

Earlier display gap: exact statement.

Current exact statement: Claim 071 sealed q^4 exclusion support against q^6 gravity, the exact two-family pre-horizon census, quarter-area information support and mT=1/16 before the compact-object targets opened. NASA's complete white-dwarf summary identifies electron degeneracy and the 7/5-solar-mass dimensional limit. The primary PSR J0740+6620 timing record measures 52/25 +/- 7/100 solar masses, placing a neutral-fermion-supported object strictly above the white-dwarf limit. The complete LIGO-Virgo GW170817 remnant analysis retains conditional upper boundaries 267/100 and 61/20 solar masses and explicitly identifies their model conditions. The earlier immutable horizon validation remains unchanged: horizon-scale shadow and ringdown are observed, while direct Hawking radiation, temperature and information reconstruction are not. That non-observation is a standing test boundary and is never counted as confirmation of mT=1/16.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:3df00caeaae7eb06d7b17539e9eb6282737e70c3a8c9a7208c83009b97acd124`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-COMPACT-HORIZON-THERMODYNAMICS-072-3df00caeaae7eb06.json`.

SFT-PHYS-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-TERMINAL-073

Earlier display gap: exact statement.

Current exact statement: Positive quadrupole radiation transfers a retained take from a restoring binary orbit at every dynamic successor. Source conservation therefore lowers the positive orbital remainder while the exact binding support increases and separation contracts. Inverse-square circular balance forces period squared equal to separation cubed, so shrinking separation uniquely raises the exact squared orbital frequency. The quadrupole repeats twice per orbit and therefore raises the gravitational-wave frequency by the same ordered chirp. Horizon contact joins two source supports into one remnant while retaining the radiation and component-label ledger. The remnant mass/turn labels hold one tone class; the unique nonempty equal binary contraction halves its amplitude at each finite return, forcing a damped ringdown without a negative exponent, irrational frequency or numerical-zero endpoint.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:7add0b09e795aee79758286268637f442b40155f4af5c83a4a123f90e2150e19`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-TERMINAL-073-7add0b09e795aee7.json`.

SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-074

Earlier display gap: exact statement.

Current exact statement: Claim 073 independently enumerated and sealed the exact Fold sequence. GW150914 remains disclosed as pre-seal observational context and is not relabelled as a blind prediction. The separately bound post-seal GW151226 record directly reports frequency and amplitude increasing across 55 cycles from 35 to 450 Hz, positive radiated energy and a two-component-to-one-remnant transition. The post-seal GW190521 record separately supports a decaying least-damped quadrupolar remnant mode while retaining the conditional quasicircular interpretation, short-signal boundary and alternative interpretations. Together the complete record validates the rising-chirp, merger and damped-ringdown classes without claiming that dimensional frequencies or half-One damping are universal measured Fold coordinates.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:80a9f13dc620d3cb2bf16597d54b63562aeac60ad1bab25d1327aabafb8bf576`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-074-80a9f13dc620d3cb.json`.

SFT-PHYS-GRAND-LOCK-TERMINAL-075

Earlier display gap: exact statement.

Current exact statement: Every Physics claim admitted before this lock is uniquely owned once, hash-bound to its receipt, certificate and registration, and included in a complete acyclic dependency dictionary whose every Physics node reaches the foundational One. The current exact headline vector independently recomputes from generator three. In the complete generator-successor adverse census, every value declared generator-dependent changes under three-to-four while the binary fibre, stable spatial rank and boundary rank remain held. Shared carriers form an explicit cross-domain identity graph. Every admitted empirical receipt, measurement-row preservation certificate, unfavorable result and scope boundary remains present. This closes the declared V3 Physics surface at this version without preventing a later lawfully forced extension.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:ae18f67371c8e7054430935d6b5e5f3162f24cf9cba073769384bf7ba467d817`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-GRAND-LOCK-TERMINAL-075-ae18f67371c8e705.json`.

SFT-PHYS-VALIDATION-GRAND-LOCK-076

Earlier display gap: exact statement.

Current exact statement: After formal Grand Lock 075 sealed, the complete pre-lock Physics empirical surface was reconciled as one immutable vector. All 234 empirically tested claim receipts, all 147 distinct registered external source identities, every empirical and external validation hash, every available measurement receipt, all six explicit legacy materialization shapes, and all fourteen detected unfavorable-result or scope-boundary claims remain present together. Observation is retained as empirical evidence; prior observations are not mislabelled as unseen predictions, and no measurement is allowed to select a formal survivor.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:93f4497d6f7ef3c477246079f62c21f96a7ae27fd9516fa876e9d413bbab569e`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-GRAND-LOCK-076-93f4497d6f7ef3c4.json`.

SFT-PHYS-UNIFIED-CONSTANTS-OBJECT-077

Earlier display gap: exact statement.

Current exact statement: The admitted electromagnetic, lepton, quark, cosmological, hierarchy and vacuum constants are not independent numerical dials. They are typed readings of one rooted Fold geometry: the One forces the Fold; the Fold forces binary two and generator three; three-space and boundary rank two force the shared cover depths five and seven; and those same retained carriers generate the complete registered exact cross-sector vector. A generator-successor dependency probe moves every and only generator-dependent carrier, while the binary-only half-One and independently held ranks remain fixed.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:e6112bcfa663b3d24727bd03688491f55fdb5fc32359542b92acee7e551fbac8`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-UNIFIED-CONSTANTS-OBJECT-077-e6112bcfa663b3d2.json`.

SFT-PHYS-TESLA-BOUNDED-CAVITY-078

Earlier display gap: exact statement.

Current exact statement: A positive finite connected propagation path bounded at both ends has one exact outward-return period and a whole-counted discrete mode family.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:52164b45a1cd9eac61d3fd2850d10dd144a0fc6d2699520bd10d166325f3126e`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-TESLA-BOUNDED-CAVITY-078-52164b45a1cd9eac.json`.

SFT-PHYS-TESLA-ODD-QUARTER-WAVE-079

Earlier display gap: exact statement.

Current exact statement: A standing recurrence joining opposed held endpoint roles closes only after an odd positive number of quarter-cycle acts, forcing the odd harmonic family.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:3f1341a5b993e042ffcea6cc7220100e8ee8ab98c59aafc3694426381e30c83c`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-TESLA-ODD-QUARTER-WAVE-079-3f1341a5b993e042.json`.

SFT-PHYS-TESLA-LONGITUDINAL-TRANSVERSE-080

Earlier display gap: exact statement.

Current exact statement: One exact source-return recurrence supports a held displacement along propagation and two held displacements across propagation in forced rank-three space.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:3c7be179aa4e991f5bf66fcb926a51b3660f9d11e9c70216ea0cf4e75fe91a09`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-TESLA-LONGITUDINAL-TRANSVERSE-080-3c7be179aa4e991f.json`.

SFT-PHYS-TESLA-RESONANT-TRANSFER-081

Earlier display gap: exact statement.

Current exact statement: A phase-matched drive on a finite connected resonant path reaches every path cell and transfers support while retaining cavity, delivered and loss carriers in one exact conserved ledger.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:33dbb3b402878cfd1553f2eaf0d25d1df3194bd29cfe390d43e7a13facde069c`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-TESLA-RESONANT-TRANSFER-081-33dbb3b402878cfd.json`.

SFT-PHYS-VALIDATION-TESLA-RESONANCE-FAMILY-082

Earlier display gap: exact statement.

Current exact statement: The four sealed V3 laws are compared with five complete institutional records. A measured NBS quarter-wave line responds near its third and other odd harmonics; the NIST diamond record retains round-trip echoes and one longitudinal plus two separately measured transverse modes; a NIST resonant cavity transfers, stores and retrieves a prepared state while retaining decay and fidelity limits; and NASA records the bounded Earth-ionosphere mode family. The Earth modes do not equal the quarter-wave odd sequence, and proposed wireless or free-energy applications are not measurements of source-free extraction. Every favorable, adverse, limitation and delivery-type row remains held.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:270786783cb68b4737d35faf5385079767c22036b03dc8f83965bb1eb2467b92`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-TESLA-RESONANCE-FAMILY-082-270786783cb68b47.json`.

SFT-PHYS-VACUUM-LOCAL-RESONANT-DRIVE-083

Earlier display gap: exact statement.

Current exact statement: A live exact local vacuum carrier can undergo a source-bound resonant transition while retaining its initial state, driven state and complete transferred support.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:3ffdf3606dd8f27146c3e0bb203c3758e85780049e8c60e074841d567bfc1e0a`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VACUUM-LOCAL-RESONANT-DRIVE-083-3ffdf3606dd8f271.json`.

SFT-PHYS-VACUUM-INERTIA-COVARIATION-084

Earlier display gap: exact statement.

Current exact statement: Because the admitted vacuum-to-inertia exchange ratio is the One, every lawful local vacuum-carrier change is held identically by its inertial carrier.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:a956088e2454d1594a7a04ca4f6050816bca30f59a32aae52dcafd1cb9617474`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VACUUM-INERTIA-COVARIATION-084-a956088e2454d159.json`.

SFT-PHYS-VACUUM-INERTIA-POSITIVE-FLOOR-085

Earlier display gap: exact statement.

Current exact statement: At every registered positive finite Fold depth, complete binary support forces an exact positive least carrier, so neither a driven vacuum carrier nor its unity-related inertia carrier can become structural absence.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:0a7552d2128afa000bc701c2199f3e7efdce1c5091bd09d612b3826c900c6e23`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VACUUM-INERTIA-POSITIVE-FLOOR-085-0a7552d2128afa00.json`.

SFT-PHYS-VACUUM-INERTIA-COMPLETE-LEDGER-086

Earlier display gap: exact statement.

Current exact statement: A lawful local vacuum/inertia drive retains the source, paired response, outward transfer, restoration transfer and information trace, forbidding unrecorded cyclic gain while preserving the outward event.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:3fc114314e2065848eca08536e13323688be410b9693a064921e2c8c62cb8b57`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VACUUM-INERTIA-COMPLETE-LEDGER-086-3fc114314e206584.json`.

SFT-PHYS-VALIDATION-VACUUM-INERTIA-DRIVE-FAMILY-087

Earlier display gap: exact statement.

Current exact statement: The four sealed V3 laws are compared with five official source identities. Navy records describe a driven local-vacuum/inertia mechanism while explicitly retaining theoretical, proposed-experiment and no-prototype status in the captured record. NIST measurements establish nonempty ground-state response and boundary-dependent vacuum interaction while retaining coherent-pump and no-ground-state-work boundaries. CNES constrains ordinary inertial/gravitational unity but is not a driven-inertia apparatus test. Every favorable, adverse, absent, untested and scope-limiting row remains held.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt:
`sha256:ae878e3fe8666a5471b9d19a2f67ab235b05f2187f3e9934523cc0e1661f4ca1`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VACUUM-INERTIA-DRIVE-FAMILY-087-ae878e3fe8666a54.json`.

SFT-PHYS-PENTA-COMPLETE-PHENOTYPE-088

Earlier display gap: exact statement.

Current exact statement: The generated sector-5 force retains its complete charge, mediator, antipodal-pair, carrier and confinement phenotype without importing a known-force template as a premise.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:11362854f921cda3962cfed7578380b32cca2663c85de8528d4aa17895fe20c4`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-PENTA-COMPLETE-PHENOTYPE-088-11362854f921cda3.json`.

SFT-PHYS-HEPTA-COMPLETE-PHENOTYPE-089

Earlier display gap: exact statement.

Current exact statement: The generated sector-7 force retains its complete charge, mediator, antipodal-pair, carrier and confinement phenotype without importing a known-force template as a premise.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt:
`sha256:57d1a7b380211d15d035e2853e74fa81dc7cd150f4acadef9a55aa1d8efb7a1a`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-HEPTA-COMPLETE-PHENOTYPE-089-57d1a7b380211d15.json`.

SFT-PHYS-PENTA-BETA-SLOPE-090

Earlier display gap: exact statement.

Current exact statement: The sector-5 coupling-to-shortfall relation independently forces beta slope 4 and its own sector-specific falsifier.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:df6eeb2137767de6ae27e4dde9a6f4655629411248f30136c5cbdf82c3df9198`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-PENTA-BETA-SLOPE-090-df6eeb2137767de6.json`.

SFT-PHYS-HEPTA-BETA-SLOPE-091

Earlier display gap: exact statement.

Current exact statement: The sector-7 coupling-to-shortfall relation independently forces beta slope 6 and its own sector-specific falsifier.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:bcdbca84f9778bcf437fc0fe253f13f2ab4672db1414d5e1cccf78c7cc38607e`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-HEPTA-BETA-SLOPE-091-bcdbca84f9778bcf.json`.

SFT-PHYS-CATEGORY-CLEAN-PARTICLE-CENSUS-092

Earlier display gap: exact statement.

Current exact statement: The finite prime-sector carrier inventory, named nonsector bosons, twelve known fermion kinds and twelve Smithion kinds form one category-clean census without omission or double counting.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:3d96ec5a6eb16daa8bc8971bfdb34d876fa76fb082da1737583c7eb209ec1e06`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-CATEGORY-CLEAN-PARTICLE-CENSUS-092-3d96ec5a6eb16daa.json`.

SFT-PHYS-NO-EXTRA-SECTOR-PARTICLE-BOUNDARY-093

Earlier display gap: exact statement.

Current exact statement: The counted prime-sector ceiling and category-clean census force the first excluded prime and an explicit two-sided outside-list falsification boundary.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:aee65ca87af0ec74235a3e62c475d6254301ef623dbbebef5c1582aab0bb9cd9`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-NO-EXTRA-SECTOR-PARTICLE-BOUNDARY-093-aee65ca87af0ec74.json`.

SFT-PHYS-SMITHION-INTERACTION-SEARCH-SIGNATURES-094

Earlier display gap: exact statement.

Current exact statement: The closed penta/hepta fibres and absence of a generated cross-fibre mediator force exact electromagnetic, nuclear-recoil, gravitational and collider search classes for Smithions.

Formal closure status: `depth_independent`.

Current external status: `independently_replicated`.

Model-admitted receipt: `sha256:5e1ffc9a06c962f4d8301714cf57d5a6e306884cf9c4c4fcfed4b43365c101b7`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-SMITHION-INTERACTION-SEARCH-SIGNATURES-094-5e1ffc9a06c962f4.json`.

SFT-PHYS-VALIDATION-NEW-SECTOR-COMPLETE-FAMILY-095

Earlier display gap: exact statement.

Current exact statement: Seven sealed V3 laws are compared with the complete registered PDG, ATLAS, Planck, SPARC, MEG II and BaBar record. Known sector counts and known particle categories are retained; outside-list classes remain searched hypotheses; collider dark-jet and missing-momentum signatures are active with model-dependent exclusions; and all penta/hepta carrier, slope and Smithion measurements remain explicit standing tests rather than fabricated discoveries or retired claims.

Formal closure status: `depth_independent`.

Current external status: `empirically_tested_and_independently_replicated`.

Model-admitted receipt: `sha256:2f13335a3bfdd7a395689ff113676a6c3287d90c7b4c8e49eac53730d5c5f2bd`.

Receipt path: `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-NEW-SECTOR-COMPLETE-FAMILY-095-2f13335a3bfdd7a3.json`.

Data and code availability

The open repository is `MettaMazza/ernos-labs-sft-platform`. The live model-admitted index is `census/claims.json`; the Physics field boundary is `output/release/physics-1.3.0/05_Physics-Current-Categorical-Inventory.json`. Every represented claim has its own package under `claims/<claim-id>/` and its immutable model-admitted receipt under `receipts/engine/model_admitted/`.

The publication evidence map is `output/release/physics-1.3.0/02_Physics-Paper-001-v1.3-Evidence-Map.json` and the candidate manifest is `output/release/physics-1.3.0/03_Physics-Paper-001-v1.3-Manifest.json`. Registered external records and source captures are retained under `experiments/external_sources/physics/` together with their source identities, transport outcomes and hashes. Files too large for ordinary Git remain checksum-bound and are distributed through the applicable existing Zenodo record rather than omitted or replaced.

Build, render, replay and verification programs are retained under `tools/`, with branch implementations under `sft/`, independent reconstructions under `generated/` and tests under `tests/`. The final approved Markdown, rendered PDF, evidence map, manifest, metadata and checksums must agree before deposit. No successful test is treated as empirical confirmation unless the claim-specific evidence protocol separately authorises that status.

Shared claim-record clauses

The clauses below previously appeared verbatim in multiple claim records. They are preserved once here to reduce mechanical repetition. Each in-text clause reference applies this exact wording at that claim location; no scientific statement, status, condition or qualification has been removed.

`PHYS-S001` - applied at 368 claim locations

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

`PHYS-S002` - applied at 368 claim locations

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

`PHYS-S003` - applied at 13 claim locations

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

`PHYS-S004` - applied at 368 claim locations

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

`PHYS-S005` - applied at 238 claim locations

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

`PHYS-S006` - applied at 9 claim locations

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

`PHYS-S007` - applied at 15 claim locations

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

`PHYS-S008` - applied at 24 claim locations

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

`PHYS-S009` - applied at 14 claim locations

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavourable row is accepted.

`PHYS-S010` - applied at 41 claim locations

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

`PHYS-S011` - applied at 10 claim locations

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

`PHYS-S012` - applied at 15 claim locations

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

`PHYS-S013` - applied at 31 claim locations

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

`PHYS-S014` - applied at 15 claim locations

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

`PHYS-S015` - applied at 24 claim locations

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

`PHYS-S016` - applied at 9 claim locations

Meaning. Dimensionless constants, mass scales and precision values are kept in Physics. Each reported value is tied to its zero-parameter Fold relation, sealed prediction, exact rational measurement adapter and immutable receipt. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

`PHYS-S017` - applied at 62 claim locations

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

`PHYS-S018` - applied at 14 claim locations

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

`PHYS-S019` - applied at 15 claim locations

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

`PHYS-S020` - applied at 17 claim locations

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

`PHYS-S021` - applied at 12 claim locations

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

`PHYS-S022` - applied at 34 claim locations

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

`PHYS-S023` - applied at 13 claim locations

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

`PHYS-S024` - applied at 18 claim locations

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

`PHYS-S025` - applied at 16 claim locations

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

`PHYS-S026` - applied at 8 claim locations

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced `256` named candidates and the decision support contains `256` one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

`PHYS-S027` - applied at 8 claim locations

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavourable interval is accepted.

`PHYS-S028` - applied at 64 claim locations

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

References and official data bodies

- Bureau International des Poids et Mesures, *The International System of Units (SI), 9th edition*.
- National Institute of Standards and Technology and CODATA, *2022 Fundamental Physical Constants - Complete Listing*.
- National Institute of Standards and Technology, Atomic Spectra Database, version 5.12.
- Particle Data Group, *2025 Review and Summary Tables*.
- International Atomic Energy Agency, ENSDF and ENDF evaluated nuclear-data systems.
- International Association for the Properties of Water and Steam, official releases.
- Gravitational Wave Open Science Center, GWTC catalog and event APIs.
- CERN Open Data Portal, public dataset API.
- NASA LAMBDA and COBE FIRAS public data products.
- Lundh, A. et al. (2018), *Industry sponsorship and research outcome*, systematic review and meta-analysis, https://doi.org/10.1007/s00134-018-5293-7.

- Fabbri, A. et al. (2018), *The influence of industry sponsorship on the research agenda*, https://pubmed.ncbi.nlm.nih.gov/30252531/.
- Gallo, S. A. et al. (2023), empirical study of grant-review ranking, rating and applicant effects, https://pmc.ncbi.nlm.nih.gov/articles/PMC10553257/.
- Demicheli, V. and Di Pietrantonj, C., Cochrane review of grant peer review, https://pmc.ncbi.nlm.nih.gov/articles/PMC8973940/.
- UNESCO (2021), *Recommendation on Open Science*, https://www.unesco.org/en/legal-affairs/recommendation-open-science.
- National Institute of Standards and Technology, *AI Risk Management Framework 1.0*, transparency, explainability and interpretability characteristics.
- Nature Communications (2025), consensus statement on null-result publication and research waste, https://pmc.ncbi.nlm.nih.gov/articles/PMC12459790/.
- Smith, Maria, *There Is No Nothing*, Ernos Labs methods and foundation paper.
- Smith, Maria, *From One to Fold*, Foundation branch evidence corpus.
- Smith, Maria, *From Fold to Mathematics*, Mathematics branch paper.
- Smith, Maria, *From Distinction to Information*, Information Science branch paper.
- Smith, Maria, *After Turing: The Fold Machine*, Classical Computation branch paper.
- Smith, Maria, *The Quantum Fold Machine*, Reversible and Quantum Computation branch paper.

Data, code, rights and participation

The canonical open repository is https://github.com/MettaMazza/ernos-labs-sft-platform. The release package contains this Markdown manuscript, the archival PDF, the complete evidence map, publication receipt, categorical inventory, claim packages, generated candidate censuses, elimination decisions, controls, independent validators, empirical source records and immutable engine receipts.

Independent review, replication, lawful extension and attempted invalidation are invited. Contact Maria.Smith.Sftoe@gmail.com or https://discord.gg/ucwGryVxGr. Scientific admission requires the complete derivation chain and an unchanged-engine receipt; neither reputation nor criticism alone changes the model census.

Suggested citation: Smith, Maria (2026), *From Fold to Physics: An Exact, Parameter-Free and Machine-Closed Reconstruction of Physical Science from Smithian Fold Theory*, Ernos Labs Physics Branch Paper 001, version 1.3.0.

Complete current-census successor supplement

The authoritative current scope of this paper is 368 live model-admitted claims (`physics` 368). Counts retained inside the preceding-version narrative describe the dated freeze at which those passages were written; this successor table and the machine census control the current total.

Every current claim identifier for this paper's branch surface already occurs in the inherited scientific body. No claim-level prose delta is required.

Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 1.3.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

This paper reports a zero-parameter reconstruction of physical science from the single foundational Smithian One theorem, without imported axioms, fitted coefficients, numerical zero, negative proof magnitudes, irrational or imaginary proof values, floating-point proof equality, or continuum premises. Its lead numerical result is an exact first-principles derivation of the fine-structure constant:

> $\alpha^{-1} = 503846395469/3676744786 = 137.035999177180855...$; $\alpha = 3676744786/503846395469 = 0.007297352564321794...$

The value was sealed before the registered CODATA 2022 target was released and lies inside the complete interval $137.035999177 \pm 0.000000021$. The same derivational constitution produces exact charged-lepton cubic and Koide relations, electron and muon magnetic anomalies, the on-shell electroweak share, Higgs mass and self-coupling, the Planck/proton hierarchy, proton radius, vacuum and cosmological ratios, nuclear closure numbers, hadron trajectories, inverse-square dilution, relativistic and quantum laws, thermodynamic and extraction boundaries, gravitational-wave ordering, the Unified Constants Object, Tesla resonance laws, vacuum/inertia co-variation and restoration, and the penta/hepta/Smithion standing-prediction family documented here.

The publication contains all 368 current engine-admitted Physics claims. Their generated grammars contain 257,776 candidates, 257,776 one-for-one decisions, 368 unique survivors and 1,472 passing mandatory adverse controls. Of those claims, 238 contain sealed post-derivation external-validation records; formal claims are connected to their registered empirical successors or retain their exact structural test boundary. The categorical clean-room audit closes all 488 Physics-owned V1/V2 atoms at the declared current-evidence boundary, with no open Physics atom or gap family. Grand Locks 075/076 preserve the 349-claim pre-extension snapshot; the separately admitted Unified Constants, Tesla resonance, vacuum/inertia and new-sector families add nineteen current claims with their own completion audits and terminal empirical receipts. The current dated inventory is therefore a 368-claim full-field projection, complete to known registered scope and permanently open to lawful extension, correction and falsification.

Scope and ownership boundary

This paper owns the `physics` surface comprising 368 current claims. Ownership identifies responsibility for derivation and falsification; it does not prevent criticism or lawful cross-branch use. Application performance, consensus, credentials and Lean acceptance cannot transfer ownership or select a law.

Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

Successor machine-identity appendix

- Preceding source: `publications/successors/physics/FROM_FOLD_TO_PHYSICS_PAPER_001_V1_3.md`
- Preceding source SHA-256: `sha256:3e0c2136752e3683ee6f599f234072b12a6ec01e08a4aa287d267717150c4883`
- Lean report: `generated/lean4_validation/reports/whole_model_validation.json`
- Lean report SHA-256: `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`
- Ordered census SHA-256: `sha256:f3f540f77a9b677bf7ddb9f5c6ea1ea41f08e57bb07163e25f385fd4d1dcde71`
- Execution manifest SHA-256: `sha256:517fd288f4484f43fdc0343b17fcabd06ea9e12c977a4512d4a891dd038fce41`
- Protected engine seal: `sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a`

- Verification-authority seal: `sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8`
- Publication authorised: `false`
- Remote actions performed: `false`

Lean 4 independent verification update

Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned `PASS`. Its immutable report identity for this successor is `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`.

Verified surface	Result
Ordered census claims	2,777
Proof-bearing accepted claim gates	2,777
Source-bound claims	2,777
Candidate records	898,902
Decision records	898,902
Passed controls	11,108
Registered branches	17
Issues	0

This paper's registered branch surface contributes **368** of the accepted claims.

Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar: `unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter. Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors. `#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted gate` only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

Successor conclusion

Version 1.4.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes 368 current claim(s) within `physics` to the total dependency spine.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.

Exact current claim-record preservation addendum

The records below are reproduced byte-for-byte from the current census and claim packages because literal-aware house-style editing must never alter an authoritative statement or receipt identity.

Post-seal gravitational-wave speed, polarization and quadrupole comparison

Claim ID: `SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVES-003`

Exact statement: The GW170817/GRB170817A measurement bounds relative gravity/light speed between -3 x 10^-15 and +7 x 10^-16; LIGO-Virgo catalog tests support tensorial polarization and quadrupolar strong-field modes. Frequency/source coverage remains finite and explicit.

Dependency route: `SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003` -> `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001` -> `SFT-PHYS-MEAS-TARGET-CUSTODY-001` -> `SFT-PHYS-MEAS-UNCERTAINTY-001` -> `SFT-MATH-EXACT-ARITHMETIC-001`

Candidate generator: Generate the complete eight-axis post-seal gravitational-wave comparison product.

Grammar boundary: Every registered multimessenger speed interval, polarization, quadrupolar ringdown and finite-coverage control row.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `complete-fold-carrier__sealed-gravity-wave-law-versus-complete-LIGO-GWOSC-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`

Exact result: Gravity/light common speed is confirmed at the reported multimessenger bound; tensor polarization and quadrupolar strong-field support are confirmed within the registered catalog analyses.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `empirically_tested_and_independently_replicated`

Engine receipt: `sha256:627976ac082693a34548353be3a26eae93d6804108d549a4ab6411c0575f70ea` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVES-003-627976ac082693a3.json`

Control	Passed	Expected	Observed
false_premise	True	reject the generated form lacking the complete physical carrier	reject the generated form lacking the complete physical carrier
tampered_source	True	reject a changed registered source identity	reject a changed registered source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject target access, forbidden proof values, imported laws and free parameters	reject target access, forbidden proof values, imported laws and free parameters

Post-seal nonstandard-spacetime realization boundary

Claim ID: `SFT-PHYS-VALIDATION-NONSTANDARD-SPACETIME-003`

Exact statement: The sealed wormhole, warp and closed-timelike results are admissibility theorems. The complete registered authority vector contains no confirmed physical realization of any of the three, and current catalog tests report no nonstandard strong-field signal in their stated analyses.

Dependency route: `SFT-PHYS-SPACETIME-WORMHOLE-ADMISSIBILITY-003` -> `SFT-PHYS-SPACETIME-WARP-ADMISSIBILITY-003` -> `SFT-PHYS-SPACETIME-CLOSED-TIMELIKE-ADMISSIBILITY-003` -> `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001` -> `SFT-PHYS-MEAS-TARGET-CUSTODY-001` -> `SFT-PHYS-MEAS-UNCERTAINTY-001` -> `SFT-MATH-EXACT-ARITHMETIC-001`

Candidate generator: Generate the complete eight-axis post-seal nonstandard-spacetime status product.

Grammar boundary: Every formal admissibility result, registered authority search row, non-detection scope and explicit no-realization statement.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `complete-fold-carrier__sealed-admissibility-laws-versus-complete-authority-status-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`

Exact result: Wormhole, warp and exact closed-timelike admissibility boundaries are formally sealed; no authoritative realization appears in the registered source vector, so all remain unconfirmed physical frontier predictions.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `empirically_tested_and_independently_replicated`

Engine receipt: `sha256:58e0992e82b5379cb948eeed83b8ea8babc833386599c5c8ab93eafbb4705fb3` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-NONSTANDARD-SPACETIME-003-58e0992e82b5379c.json`

Control	Passed	Expected	Observed
false_premise	True	reject the generated form lacking the complete physical carrier	reject the generated form lacking the complete physical carrier
tampered_source	True	reject a changed registered source identity	reject a changed registered source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject target access, forbidden proof values, imported laws and free parameters	reject target access, forbidden proof values, imported laws and free parameters